Semiclassical and Stochastic Gravity

Quantum Field Effects on Curved Spacetime

BEI-LOK B. HU AND
ENRIC VERDAGUER

CAMBRIDGE MONOGRAPHS
ON MATHEMATICAL PHYSICS

SEMICLASSICAL AND STOCHASTIC GRAVITY

The two pillars of modern physics are general relativity and quantum field theory, the former describes the large scale structure and dynamics of space-time, the latter, the microscopic constituents of matter. Combining the two yields quantum field theory in curved space-time, which is needed to understand quantum field processes in the early universe and black holes, such as the well-known Hawking effect. This book examines the effects of quantum field processes back-reacting on the background space-time which become important near the Planck time (10^{-43} sec). It explores the self-consistent description of both space-time and matter via the semiclassical Einstein equation of semiclassical gravity theory, exemplified by the inflationary cosmology, and fluctuations of quantum fields which underpin stochastic gravity, necessary for the description of metric fluctuations (space-time foams). Covering over four decades of thematic development, this book is a valuable resource for researchers interested in quantum field theory, gravitation and cosmology.

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Dedicated to

The loving memory of my parents, I-Ping Hu and Pie Wang
and my brother, Professor Bambi Hu – BLH

My mother Maria, *my daughter* Lena, *and*
the loving memory of my father Joan *and uncle* Angel – EV

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Preface

Research on the topics covered in this book began around 1974, after Hawking's epochal discovery, when effective action methods were introduced, and regularization schemes established, for curved spacetimes. This book could be viewed as v2 and v3 of "*quantum field theory in curved spacetime*" (QFTCST), established half a century ago, in two senses: as versions 2 and 3, the continued development of this field to "*semiclassical gravity*" in the early 1980s and "*stochastic gravity*" established in the mid-90s, and their many implications and applications in the ensuing years. It can also be viewed as Volumes 2 and 3 of the many well-written books on QFTCST, listed in Chapter 1, comprising the chapters in Parts I–II and Parts III–V, respectively. We see little need to explain the relevance of this subject matter to theoretical physics since it is well indicated in these earlier monographs. Suffice it to say that it is drawn by the allure of quantum gravity, theories for the microscopic structures of spacetime, but is built on the firm and weathered foundation of two well-established theories in the past century: general relativity for the large-scale structures of spacetime and quantum field theory for the description of matter, both valid through experimental and observational tests to an amazingly high degree of accuracy.

This is the long-awaited time to give thanks to those who have influenced our intellectual growth, shaped our professional paths and helped in the writing and editing of this book. BLH wishes to express this: I am deeply indebted to my Ph.D. advisor, the late Professor John A. Wheeler for his guidance, inspiration, patience and understanding, in the tumultuous late 1960s and early 70s, when not only theoretical physics, but also humanistic values and societal priorities were undergoing fundamental changes; to the late Professor Tulio Regge in showing how mathematics can be enjoyed like magic, especially when shown in the ambiance of music from his whistling of Italian opera arias. The highest-caliber scholarship of the late Professors S. Chandrasekhar and Bryce S. DeWitt, and of Professors Stephen L. Adler and Steven Weinberg, has continued to serve as a living standard of perfection for me. Professor Charles W. Misner, from whom I learned differential geometry and whose universe was given to me by Wheeler as a first exercise in theoretical cosmology, and Professor James B. Hartle, from whom I learned the versatile effective action method and many aspects of quantum field theory, are prime examples of how presumably self-absorbed researchers can also be very warm, modest and caring human beings.

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The raw materials of this book are largely based on papers co-authored with colleagues: James B. Hartle, Stephen A. Fulling, Leonard Parker, Hing-Tong Cho, Paul Anderson and his former Ph.D. student Jason Bates, and work we did with our former Ph.D. students and postdoctorals. We gladly acknowledge their essential contributions: Charis Anastopoulos, Daniel Arteaga, Esteban Calzetta, Roberto Camporesi, Antonio Campos, Ardeshir Eftekharzadeh, Markus Fröb, Chad Galley, Philip Johnson, Rosario Martín, Andrew Matacz, Denjoe O'Connor, Guillem Pérez-Nadal, Juan Pablo Paz, Nicholas Phillips, Stephen Ramsey, Alpan Raval, Albert Roura, Tsung-Chen Shen, Kazutomo Shiokawa, Sukanya Sinha, Chris Stephens, Aris Stylianopoulos and Yuhong Zhang.

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On a personal note, BLH is grateful for the unfailing support of his brother, Professor Shiu-Lok Hu, his cousin Kuen-Wai Lau and wife Alice Cho, and lifelong friend Dr. Shun-Ming Chung and his endearing family. He misses even more deeply his children Tung-Hui and Tung-Fei, his love in an eternal universe, even after all vital signals have disappeared in a black hole.

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1

Overview: Main Themes. Key Issues. Reader's Guide

In this overview we wish to place the body of work described in this book in perspective, connecting it with subject matters in other works sharing similar goals but pursued in different ways or operating at different levels of inquiry. There are many such subjects, but two groups stand out as quite obvious or timely. The obvious subject is quantum gravity (QG) (see, e.g., [1, 2, 3, 4, 5] and [6, 7, 8] for a sampling of the different schools of thought), pursued in the last seventy years, while the timely one is gravitational quantum physics (GQP) (e.g., [9]) a recently minted term presenting a new emphasis on two old disciplines. Quantum gravity refers to theories of the microscopic structures of spacetime, where micro refers to the Planck scale 10^{-33} cm or below, energy scale of 10^{19} GeV or above. Familiar representatives are string theory [10, 11], canonical loop quantum gravity [12, 13, 14, 15, 16, 17], spin network [18, 19, 20], group field theory [19, 21], asymptotic safety [5, 22, 23, 24, 25, 26], simplicies [27, 28, 29, 30, 31], causal dynamical triangulation [32, 33, 34], causal sets [35, 36, 37], etc. We shall comment in the last chapter of this book on how stochastic gravity is linked to quantum gravity, and how it can assist in unraveling the microscopic structures of spacetime.

These nonperturbative theories operative at the Planck scale are what quantum gravity entails. Their structures and contents are fundamentally different from *perturbative* quantum gravity [38, 39] built upon the quantized weak perturbations of classical background spacetimes – the spin-two gravitons, which should exist in nature and are in principle detectable [40, 41] by laboratory experiments at energies much lower than the Planck energy. First explored in the early 1960s [42] in the realm of particle physics [43], graviton physics shares many similar features with photon physics in quantum electrodynamics (QED) and stands on the same footing as the physics of intermediate bosons such as gluons in quantum chromodynamics (QCD) of strong interaction. It is the latter, perturbative quantum gravity, centered on graviton interactions,

which falls in the realm of gravitational quantum physics. Therefore, it deals with gravitational effects on quantum systems readily accessible at today's low energy in experiments on Earth or in space. In Sec. 1.3 we shall describe the relation of semiclassical gravity with gravitational quantum physics, point out the non-relation with the Newton–Schrödinger equation [44], and mention the role of stochastic gravity in addressing quantum information issues.

In this section we discuss two key issues, (1) self-consistent backreaction and non-Markovian dynamics; (2) coarse-graining, fluctuations and colored noise. Using an example from semiclassical gravity we point out the necessity of self-consistency in seeking simultaneous solutions of the equations of motion for the quantum matter field and the Einstein equation for the background spacetime, and the importance of including fluctuations in this consideration. In Sec. 1.2 we discuss two main themes: (1) the existence of a stochastic regime in relation to the quantum and the semiclassical regimes; (2) how the conceptual framework of open quantum systems and the influence functional, or its close kin, the 'in-in' or closed-time-path (CTP) or Schwinger–Keldysh effective action, are useful to connect these three levels of theoretical structures and the description of a physical system at each of these three levels. We use the more familiar moving charge quantum field system to illustrate how a correctly formulated approach to self-consistent backreaction leads to a modified Abraham–Lorentz–Dirac (ALD) equation for the motion of a charge with radiation-reaction which is pathology-free, while including the noise from the quantum field we obtain an ALD-Langevin equation describing the stochastic dynamics of the moving charge. In Sec. 1.4 we describe our approach and emphasis, then provide a guide to the readers.

1.1 From QFT in Curved Spacetime to Semiclassical and Stochastic Gravity

We begin with the two solid foundations of modern physics, both having stood the test of time: quantum field theory (QFT) for the description of matter, and general relativity for the description of the large-scale structure and dynamics of spacetime. Placing these two together for the description of quantum matter in a classical gravitational field yields quantum field theory (QFT) in curved spacetime (CST) – let us call this the Level 1 structure. This theory, largely accomplished in the 1970s, is now blessed with many excellent reviews and monographs [45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58]. The next two levels of structure, semiclassical gravity (SCG) at Level 2, established in the 1980s and stochastic gravity (StoG) at Level 3, commenced from the 1990s, are the target of investigation of this book.

At the first structural level, quantum field theory in curved spacetime describes the behavior of a quantum matter field, treated as a test field, propagating in a specified classical background spacetime. Important processes described

by this theory range from the *Casimir effect* of quantum fields in spacetimes with boundaries [59, 60, 61, 62] or non-trivial topology [63, 64, 65], to effects of vacuum polarization and vacuum fluctuations such as *particle creation* in the early universe [66, 67, 68, 69, 70, 71, 72, 73, 74, 46, 75, 76], and Hawking radiation of black holes [77, 78, 79, 80, 81, 82, 83], all in the first decade of its development.

The second structural level towards understanding the interaction of quantum fields with gravity is *backreaction*, i.e., the effects of quantum matter fields exerted on the spacetime, impacting on its structure and dynamics. Since the background spacetime remains classical the quantum matter acting as the source would have to come from the expectation value of the stress-energy tensor operator for the quantum fields with respect to some quantum state of symmetries commensurate with the spacetime. Since this object is quadratic in the field operators, which are only well defined as distributions on the spacetime, it contains ultraviolet divergences. Finding viable ways to regularize or renormalize this quantity defined the task of the second stage, in the mid-70s, in the theoretical development of QFTCST. Major regularization methods include adiabatic [84, 85, 86, 87], or 'n-wave' [69, 71, 88] regularization of quantum fields in dynamical spacetimes, suitable for cosmological particle creation processes, dimensional regularization [89, 90, 91, 92] which was successfully applied earlier to proving the renormalizability of QCD [93, 94, 95], the elegant zeta function method [96, 97, 98, 99] for quantum fields in spacetimes with Euclidean sections and the covariant point-separation method [46, 100, 101]. This period ended in 1978, when different regularization methods converge in producing (almost) the same results. The essential uniqueness (modulo some terms quadratic in the spacetime curvature which are independent of the quantum state) in the expectation value of the stress-energy operator via reasonable regularization techniques was proved by Wald [102, 103]. The criteria that a physically meaningful expectation value of the stress-energy tensor ought to satisfy are known as Wald's axioms.

The theory obtained from a self-consistent solution of the geometry and dynamics of the spacetime and the quantum matter field is known as *semiclassical gravity*. Determining the dynamics of spacetime with self-consistent backreaction of the quantum matter field is thus its central task: one assumes a general class of spacetime where the quantum fields live in and act on, and seeks solutions which satisfy simultaneously the Einstein equation for the spacetime and the field equations for the quantum fields. The Einstein equation which has the expectation value of the stress-energy operator of the quantum matter field as the source is known as the *semiclassical Einstein equation*. Semiclassical gravity was first investigated in cosmological backreaction problems in [104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114]. An example is the damping of anisotropy in Bianchi universes by the backreaction of particles created from the vacuum. Using the effect of quantum field processes such as particle creation to explain why the universe became isotropic in the context of chaotic cosmology [115, 116, 117] was investigated in the late seventies.

A well-known example of semiclassical gravity is the inflationary cosmology proposed in the early eighties by Guth [118] and others [119, 120, 121, 122, 342] where the vacuum expectation value of a gauge or Higgs field acts as source in the Einstein equation. It is easy to see that a constant energy density in a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) universe gives rise to exponential expansion, the case of eternal inflation described by a de Sitter universe. Such a solution is disallowed in classical cosmology because it corresponds to an unphysical equation of state where the pressure $p = -\rho$ energy density. A quantum source makes this solution of the semiclassical Einstein equation not only possible, but, as later development of inflationary cosmology showed, desirable.

Extending semiclassical gravity to *stochastic semiclassical gravity* is a Level 3 theoretical structure developed in the nineties. While semiclassical gravity is based on the semiclassical Einstein equation with the source given by the expectation value of the stress-energy tensor of quantum fields, stochastic semiclassical gravity includes also its fluctuations in a new stochastic semiclassical Einstein–Langevin equation. We will often use the shortened term *stochastic gravity* as there is no confusion as to the nature and source of stochasticity in gravity, here being induced by the quantum matter fields and not from classical sources (e.g., Moffett's theory [123]) or residing *ab initio* in the classical spacetime.

If the centerpiece of semiclassical gravity is the vacuum expectation value of the stress-energy tensor of a quantum field, the centerpiece in stochastic semiclassical gravity is the symmetrized stress-energy bitensor and its expectation value known as the *noise kernel*. The mathematical properties of this quantity, its physical contents in relation to the behavior of fluctuations of quantum fields in curved spacetimes and their backreaction in the spacetime dynamics engendering induced metric fluctuations are the main focus of this theory. How the noises associated with the fluctuations of quantum matter fields seed the structures of the universe, how they affect fluctuations of the black hole horizon and the backreaction of Hawking radiation on the black hole dynamics, as well as the implications on trans-Planckian physics, are new horizons to explore. With regard to the theoretical issues, stochastic gravity is the necessary foundation to investigate the validity of semiclassical gravity. It is also a useful platform supported by well-established low energy (sub-Planckian) physics to explore the connection with high energy (Planckian) physics in the realm of quantum gravity.

If the main issues in QFTCST are finding physically meaningful definitions of different vacua and how the divergences in the expectation values of the stress-energy tensor can be controlled by regularization and renormalization, then the main issues of semiclassical gravity are the self-consistent backreaction of matter fields and their effects on the structure and dynamics of spacetime. For stochastic gravity, the main issues are coarse-graining, noise and fluctuations. Let us select a sample problem (with details extracted from a later chapter) to illustrate the qualitative features of a backreaction problem in semiclassical gravity.

1.1.1 Self-Consistent Backreaction: Nonlocal Dissipation in Open Systems

Backreaction of quantum processes refers to the effects these processes have on the source which engenders them. The source could be a strong or time-dependent background field and the process may entail vacuum polarization or amplified fluctuation effects manifesting as particle creation. A well known classic example is the Schwinger process [124] of particle production in strong static fields. Particle pairs back-reacting on the field engenders dissipative dynamics, which tends to weaken the field [125, 126, 127]. For time-dependent fields, an active topic of current research is the dynamical Casimir effect, where vacuum fluctuations are parametrically amplified into real particles. Indeed, the mechanism is the same as in cosmological particle creation, and the backreaction problem is of interest there because it can significantly alter the dynamics of the early universe near the Planck time. In like manner particle creation from a moving mirror is an analog of Hawking radiation from black holes (or Unruh radiation from an accelerated detector), though the physics is different from cosmological particle creation, as spacetimes in this class (including the de Sitter universe) possess event horizons and share the same characteristic thermal distribution of particles created as a result of exponential red-shifting of the wave modes between the 'in' and 'out' states. The backreaction of Hawking radiation is of interest because it can alter the fate of an evaporating black hole emitting radiation and impacts on the related issues of black hole end state and information loss.

In semiclassical gravity one considers the effects of quantum matter field processes such as vacuum polarization (e.g., trace anomaly) and vacuum fluctuation (e.g., particle creation) exerted on the classical background spacetime. At the equation of motion level, the backreaction problem entails solving in a self-consistent manner both the matter field equations and the Einstein equations with the expectation values of the matter field as source (e.g., [107, 108]). Alternatively one can take the functional approach by integrating over the radiative contributions of the quantum matter field to obtain a one-loop effective action of the background gravitational field. From the variation of this effective action one obtains the equation of motion for the background spacetime now with the backreaction of matter field incorporated therein.

Two aspects in a backreaction problem at the semiclassical level stand out: one, related to backreaction, is the importance of self-consistency in the semiclassical Einstein equation; the other, related to semiclassicality, is decoherence in the quantum to classical transition and the appearance of dissipative dynamics.

1. Self-consistency in semiclassical gravity The necessity of self-consistency in the semiclassical backreaction problem is shown by Flanagan and Wald [128] who used the averaged null energy condition (ANEC) as a criterion to

verify this requirement. The expectation value $\langle T_{ab} \rangle$ of the renormalized stress-energy tensor of quantum fields generically violates the classical, local positive energy conditions of general relativity. Nevertheless, it is possible that $\langle T_{ab} \rangle$ may still satisfy some nonlocal positive energy conditions, ANEC being the most prominent. It states that $\int \langle T_{ab} \rangle k^a k^b \, d\lambda \geq 0$ along any complete null geodesic, where k^a denotes the geodesic tangent, with affine parameter λ. If ANEC holds, then traversable wormholes cannot occur. However, although ANEC holds in Minkowski spacetime, it is known that ANEC can be violated in curved spacetimes if one is allowed to choose the spacetime and quantum state arbitrarily, without imposition of the semiclassical Einstein equation, $G_{ab} = 8\pi G_{\text{N}} \langle T_{ab} \rangle$. Flanagan and Wald study a free, linear, massless scalar field with arbitrary curvature coupling in the context of perturbation theory about the flat spacetime/vacuum solution. At first order in the metric and state perturbations, and for pure states of the scalar field, they find that the ANEC integral vanishes, as it must for any positivity result to hold. For mixed states, the ANEC integral can be negative. However, they proved that if the ANEC integral transverse to the geodesic is averaged, using a suitable Planck scale smearing function, a strictly positive result is obtained in all cases except for the trivial flat spacetime/vacuum solution. These results suggest that if traversable wormholes do exist as self-consistent solutions of the semiclassical equations, they can only be Planck size. Their finding is in agreement with conclusions drawn by Ford and Roman [129, 130] from different arguments. (See also [131, 132, 133].)

2. Coarse-graining, decoherence and dissipation The procedure of integrating over fluctuations of the quantum field to obtain an effective action, and from there an effective equation of motion, is a form of coarse-graining, which is arguably the most important element in an open-system way of thinking [134, 135, 136, 137, 138]. There, a closed system C is divided into a system S of interest, in this case, the gravitational field, and its environment E, the quantum matter field. One can actually aim higher, and begin with a closed quantum system made up of a quantum gravity sector and a quantum matter sector. This would have been the case if we had a viable theory of quantum gravity – a theory for the microscopic constituents of spacetime and matter. One can then ask what conditions are necessary for the gravity sector to become classical. This was indeed explored in the early 90s, in the realm of quantum cosmology and semiclassical gravity. A necessary ingredient is decoherence, which can be understood in several ways, such as by way of decoherent or consistent histories [139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164] or via environment-induced decoherence [165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179]. We shall describe this aspect later in brief. The study of semiclassical and stochastic gravity begins with the stage where the gravity sector has already

been decohered sufficiently that it can be treated as a classical entity. In the next section we shall illustrate this procedure with the simple example of a particle–quantum field interaction, where we know the quantum theory for both the particle and the field. Here we wish to first highlight the main themes so we can have a better grasp of the key issues in semiclassical and stochastic gravity. In doing so we also hope to provide a good motivation for adopting the open quantum system conceptual framework for deeper inquires.

An Example

Let us examine a typical backreaction problem to highlight the second feature above, in two parts: (i) backreaction in the form of dissipation in the open system; (ii) coarse-graining of the environment, noise and fluctuations. We will extract the results of the calculations and spare the reader of the details, which are to be presented in Chapter 3.

Consider a massless conformally coupled quantum scalar field in 4 dimensions obeying the wave equation (2.31) in a classical radiation-filled Bianchi Type I universe with line element (3.55). (When the anisotropy β_{ij} in (3.55) or Q in (2.31) goes to zero, one recovers the radiation-dominated FLRW universe with scale factor a.) For spatially flat cosmology, only the expansion rate, i.e. the derivatives of a, β_{ij} are physically meaningful. The one-loop effective action incorporates the effects of the quantum field on the background geometry. In the Feynman diagram depiction, the loops account for quantum contributions of matter fields and the external legs attached to the loops represent the classical contributions of the gravitational field. The order of the vertices corresponds to the order of the coupling parameter (in this case β') in the perturbative expansion. The one-loop 'in-out' effective action for this problem was calculated in [110, 111]. But for an initial value problem where the evolutionary history of the system (rather than the transition amplitude) is desired one should use the 'in-in' (Schwinger–Keldysh or closed-time-path) effective action [180, 181] (expounded in Chapter 3) because only it can produce equations of motion which are real and causal [182]. The equation of motion for β_{ij} with backreaction from the created particles incorporated therein is an example of the semiclassical Einstein (SCE) equation.

It is convenient to work with a first integral of the SCE equation (3.107), where J_{ij} is an external source for switching on the anisotropy in the distant past and $c_{ij} = \int J_{ij}(\eta) d\eta$ (over conformal time η related to cosmic time t by $\eta = \int dt/a(\eta)$) is an integration constant which sets the magnitude and orientation of the initial anisotropy.

We can write this equation in a schematic form;

$$\frac{d}{d\eta}\left(\tilde{M}\frac{dq_{ij}}{d\eta}\right) + \mathcal{K}\frac{dq_{ij}}{d\eta} + kq_{ij} = c_{ij}, \tag{1.1}$$

where

$$\tilde{M} = \frac{1}{30(4\pi)^2} ln(\tilde{\mu}a), \tag{1.2}$$

$$k = -\frac{a^2}{8\pi G_{\rm N}} + \frac{1}{90(4\pi)^2}\left[\left(\frac{a'}{a}\right)^2 + \left(\frac{a''}{a}\right)\right], \tag{1.3}$$

$$\mathcal{K} q_{ij} = \int d\eta_2 \int d\eta_1 f(\eta_2 - \eta_1)\frac{dq_{ij}}{d\eta_1}, \tag{1.4}$$

where $\tilde{\mu}$ is the renormalization scale, a prime denotes $d/d\eta$ and $\mathcal{K}$ is a non-local operator acting on the function q_{ij}.

This equation is in the form of a damped driven harmonic oscillator where the "generalized coordinates" q_{ij} are the rate of anisotropic expansion β'_{ij} and the "generalized driving force" is c_{ij}. The "spring consant" k is time-dependent, so is the real mass $\tilde{M}$ (strictly speaking the damped harmonic oscillator analogy applies only when these quantities are positive), and the viscous force is velocity-dependent. The non-local kernel $\mathcal{K}$ links the "velocity" q_{ij} at different times, giving rise to a viscosity function γ encapsulating the effects of particle creation which is history-dependent. This memory effect reflects the non-Markovian nature of the resultant semiclassical geometrodynamics, and as we shall see, is a rather generic feature of backreaction processes. This is easy to understand from an open system viewpoint, the time scale of the natural dynamics of the system is different from that of the environment. When one incorporates the dynamics of the environment into that of the system – mathematically it entails turning two ordinary differential equations, one for each party, into an integro-differential equation for just one of them – the dynamics of the open system now contains two time scales reflected in the nonlocal kernel of the integro-differential equation [134].

With a real and causal equation of motion for $q_{ij} = \beta'_{ij}$ one can take the Fourier transform and identify from the dissipative term $i\omega\gamma q(\omega)$ (where $q \equiv q_{ij}q^{ij}$), i.e. the "resistance" component in a LCR circuit, the viscosity function $\gamma(\omega)$:

$$\gamma(\omega) = \frac{|\omega|^3}{60(4\pi)^2}. \tag{1.5}$$

The damping of anisotropy going like ω^4 translates to a dependence on the quadrature of the second derivative of β_{ij}, which can be identified as the lowest order terms of the Weyl curvature tensor. This leads to the result that the rate of particle production in anisotropic or inhomogeneous cosmological spacetimes is proportional to the Weyl curvature-squared $C_{abcd}C^{abcd}$ of the background geometry.

To check if it is correct to associate this viscosity function for the damping of anisotropy of spacetime with particle creation of the scalar field, one can calculate the energy dissipated in the spacetime dynamics within the history of

the universe and the total energy of particles created in the process. If one chooses to look at the geometrodynamics (the left-hand side of the SCE equation) one can obtain the (spectral) power $P(\omega)$ dissipated by a velocity-dependent viscous force $\mathbf{F}$ acting on the background spacetime simply from $P(\omega) = \mathbf{F} \cdot \mathbf{v}$. The dissipated energy density $\rho(\omega)$ is obtained by integrating this '(spectral) braking power' $P(\omega)$ over all frequencies.

$$\rho_{dissipation} = \int_0^\infty \frac{d\omega}{2\pi} [\omega \beta_{ij}(\omega)^*][\gamma(\omega)\omega\beta_{ij}(\omega)]. \tag{1.6}$$

Alternatively, focusing on the matter field sector (the right-hand side of the SCE equation) one can calculate the energy density of particles created from the vacuum. The power spectrum of particle pairs created by a given anisotropy history is given by

$$\mathcal{P}(\omega) = \frac{1}{30\pi^2}\omega^4 Tr\beta^*(2\omega)\beta(2\omega). \tag{1.7}$$

Integrating over the full spectrum $\int_0^\infty d\omega(2\omega)\mathcal{P}(\omega)$ produces the total energy density of particle pairs created, which is seen to be precisely equal to the energy density dissipated in the dynamics of spacetime.

This example illustrates that particle creation indeed exerts a dissipative effect on the background gravitational field. In particular, we have given a field-theoretical derivation of the viscosity function of the anisotropy damping process. One can perform similar calculations of particle creation of non-conformal fields in isotropic universes and obtain the viscosity function from its backreaction on the background spacetime. The rate of particle production is proportional to the scalar curvature-squared ξR^2 in FLRW spacetimes. For massive particles there will be a delta-function threshold.

This is all very nice, one may say, but it is only half of the story. An additional term of a stochastic nature has a reserved seat on the right hand side of (1.1) but has escaped our attention so far. The identification of it makes up the second part of this story, highlighting the second key issue, that of noise and fluctuations.

1.1.2 Fluctuations: Colored Noise from Coarse-Graining the Environment

The above example brings out three main themes of relevance to the subject matter we shall develop in this book. (1) Surpassing the theoretical structure of a prescribed curved spacetime exerting a one-way influence on the quantum fields living in it, the backreaction problem demands an account of the mutual influence of quantum matter fields present and the background spacetime in a self-consistent manner as embodied in the semiclassical Einstein equation with the expectation value of the stress-energy tensor of quantum fields acting as

its source, which defines semiclassical gravity. (2) The backreaction of quantum field processes such as particle creation alters the state and evolution of the background spacetime, generally resulting in the appearance of *nonlocal dissipative dynamics*. The physical meaning of dissipation can be seen more clearly in the conceptual framework of *open quantum systems*: A coarse-grained environment backreacting on a system results in an open system whose dynamics is no longer necessarily unitary. (3) In an open system perspective, where the classical spacetime is viewed as the system and the quantum matter field is viewed as its environment, a coarse-grained environment can under some rather general conditions (Gaussian systems for certain) be represented by *a noise term, usually colored*. This stochastic forcing term represents the fluctuations in the environment variables. The inclusion of the fluctuations of the stress-energy tensor entering as noise turns the semiclassical Einstein equation into an Einstein–Langevin (E–L) equation which defines stochastic gravity. With a noise of zero mean under stochastic average, taking the stochastic average of the E–L eqn reproduces the semiclassical Einstein equation. It is in this sense that the semiclassical gravity is regarded as a mean field theory.

Historically, the development of stochastic gravity took three stages. Stage 1 began around 1977, when different regularization schemes came to agreement enough to facilitate a proper treatment of the backreaction problems. By 1987 this task was largely completed, which led to the establishment of semiclassical gravity. An important step is the realization that the 'in-out' effective action must be replaced by the 'in-in' effective action to ensure a real and causal equation of motion for the spacetime dynamics. Stage 3 began in 1994 when the Einstein–Langevin equation was first proposed, followed by several worked-out examples. Why should there be a stochastic term and why was it not appearing earlier in the semiclassical Einstein equation – these were the questions asked and answered in the intervening years which marked Stage 2. By 1996 the basic elements of stochastic gravity were in place and the theoretical structure largely completed by 2000 [183]. The ensuing years saw applications of stochastic gravity to the structure formation problem in cosmology and the backreaction and fluctuations problems in black holes, as well as continual developments in the formulation of a validity criterion for semiclassical gravity and the calculation of the noise kernels or the stress-energy tensor correlations for spacetimes of importance to cosmology and black hole physics.

Let us see what this entails with the example described above. The statement is that a stochastic term s_{ij} can be accommodated on the right-hand side of (1.1)

$$\frac{d}{d\eta}\left(\tilde{M}\frac{dq_{ij}}{d\eta}\right) + \mathcal{K}\frac{dq_{ij}}{d\eta} + kq_{ij} = c_{ij} + s_{ij}, \tag{1.8}$$

with $s_{ij}(\eta) = \int d\eta' \xi_{ij}(\eta')$ where $\xi_{ij}(\eta)$ is a Gaussian type noise, which is completely characterized by its second moment

$$\langle \xi_{ij}(\eta) \rangle_\xi = 0$$
$$\langle \xi_{ij}(\eta_1)\xi_{kl}(\eta_2) \rangle_\xi = \nu_{ijkl}(\eta_1 - \eta_2). \tag{1.9}$$

Here $\nu_{ijkl}(\eta_1 - \eta_2)$ is known as the noise kernel. It is the two-point time-correlation function of the external stochastic source $\xi_{ij}(\eta)$. Since this correlation function is non-local, this noise is colored. In the above the angled brackets denote an average with respect to the stochastic functional distribution $\mathcal{P}[\xi]$ of $\xi^{(k)}(s)$, allowing the noise to be of different kinds (k), (see, e.g., [172])

$$\mathcal{P}[\xi^{(k)}] = P^{(k)} \exp\left\{ -\frac{1}{2\hbar} \int d^4y \int d^4x \xi^{(k)}(x) \left[\nu^{(k)}(x-y)\right]^{-1} \xi^{(k)}(y) \right\} \tag{1.10}$$

where $P^{(k)}$ is a normalization factor and we have suppressed the tensor indices on ξ, ν. With this noise, Eq. (1.8) now assumes the form of a generalized damped harmonic oscillator driven by a stochastic force s_{ij}. Inspirations leading to the discovery of this stochastic term in the Stage 2 activities described above, where open quantum system concepts and techniques were put to good use [184], came all the way from quantum Brownian motion studies of quantum stochastic processes. The details of how the noise term came to be recognized in the anisotropy damping problem are spelled out in [185]. We only show the results here to complete the story.

The noise kernel for the spatial anisotropy is given by

$$\nu(\eta) = \frac{1}{30(4\pi)^2} \int_0^{+\infty} \frac{d\omega}{2\pi} \pi\omega^4 \cos\omega\eta. \tag{1.11}$$

The verification of a noise term was shown in detailed calculations [185, 186], but the prediction of its existence [184] was aided by the fact that the dissipation and noise kernels for stochastic processes satisfy a generic fluctuation-dissipation relation (FDR), in analogy with the quantum Brownian model [187, 171, 172], namely,

$$\nu(\eta) = \int_0^\infty d\eta' K(\eta - \eta')\gamma(\eta') \tag{1.12}$$

where the fluctuation-dissipation kernel $K(\eta)$ is given by

$$K(\eta) = \int_0^\infty \frac{d\omega}{\pi}\ \omega \cos\omega\eta. \tag{1.13}$$

That a fluctuation-dissipation relation exists which encapsulates the backreaction problem of quantum field processes such as particle creation in cosmological spacetimes was shown in [185].

To summarize what we have learned in this example, whereas semiclassical gravity is based on the semiclassical Einstein equation with sources given by the expectation value of the stress-energy tensor of quantum fields, stochastic

semiclassical gravity is based on the Einstein–Langevin equation, which has in addition sources due to the noise kernel. The noise kernel is the vacuum expectation value of the stress-energy bitensor which describes the fluctuations of quantum matter fields in curved spacetimes.

1.1.3 Three Routes to Stochastic Gravity

Functional Approach

The above description highlighting the importance of including fluctuations over the mean value invoking open quantum system concepts and methods constitutes the functional approach to stochastic gravity. We use it as a physical motivation for the need of stochastic gravity theory. This route in the derivation of the Einstein–Langevin equation by way of the influence functional method is discussed in Chapter 10. The Feynman–Vernon influence functional is closely related to the Schwinger–Keldysh closed-time-path (CTP) or 'in-in' effective action methods. This will be introduced in Chapter 3 after a discussion in Chapter 2 of the traditional ('in-out') effective action for quantum fields in curved spacetime.

Axiomatic Approach

An alternative derivation of the Einstein–Langevin equation provided by Martin and Verdaguer [188] is based on the formulation of a self-consistent dynamical equation for a perturbative extension of semiclassical gravity able to account for the lowest order stress-energy fluctuations of matter fields. This axiomatic approach [183, 189] is useful to see the link in the structure of the theory from the mean value of the stress-energy tensor to the correlation functions. This is also presented in Chapter 10, where the full equivalence of the stochastic correlations and the quantum correlations is discussed. Their exact equivalence on the Minkowski background is proved in [190] and discussed in Chapter 11. For metric fluctuations in inflationary cosmology the equivalence of stochastic and the (standard) quantum correlations to linear order is shown in Chapter 13.

Semiclassical Stochastic Gravity as Effective Theory

In a broader perspective, from the viewpoint of effective field theory [191], it has long been proposed that general relativity can be regarded as the low energy limit of some more complete theory of quantum gravity, with the Planck length as the cut-off length scale (e.g., [192, 193]). This way of thinking not only mitigates the objection that quantum gravity is non-renormalizable or that the metric may not be a fundamental variable which must be quantized, it may actually lead to a realistic description of nature – a physically viable theory can offer a sound description of reality only within a specific range of length or energy scales

because of the limitation in the observer's ability to carry out measurements within a certain degree of precision.

For semiclassical gravity, there are at least two ways to justify the semiclassical Einstein equations [190]. (1) Axiomatic: It is the only consistent way to couple quantum matter to a classical metric, *a la* Wald's axioms mentioned above. (2) Large N: When the quantum gravity sector is coupled to a large number N of matter (free) fields [194], the lowest-order contribution in a $1/N$ expansion produces the semiclassical Einstein equations. When using large N the cut-off scale is shifted to the rescaled length. At the lowest order the fluctuations of the metric are suppressed but all matter loops are included.

In the same vein, stochastic gravity results from taking the next-to-leading order in the $1/N$ expansion of quantum gravity coupled to N matter fields. Whereas in semiclassical gravity the fluctuations of the metric are suppressed, here metric fluctuations are of linear order while graviton loops or internal graviton lines are sub-leading in comparison to matter loops. Semiclassical and stochastic gravity as effective theories in the large N context are discussed in Chapters 9 and 10.

1.1.4 The Importance of Including Fluctuations, in More General Terms

For a free quantum field semiclassical gravity is fairly well understood. The theory is in some sense unique, since the only reasonable c-number stress-energy tensor that one may construct with the stress-energy operator is a renormalized expectation value [102, 103]. However, the scope and limitations of the theory are not so well understood. The advent of stochastic gravity points to the importance of including fluctuations in the criterion for the stability of semiclassical gravity. In general terms when fluctuations become large compared to the mean, a theory based on the mean value can become unreliable. Semiclassical gravity being a mean-field theory, it is conceivable that it would break down when the fluctuations of the stress-energy operator are large [195, 196].

Calculations of the fluctuations of the energy density for Minkowski, Casimir and hot flat spaces as well as Einstein and de Sitter universes are available [196, 197, 198, 199, 200, 201, 183, 189, 202, 203, 204, 205]. It is less clear, however, how to quantify what a large fluctuation is, and different criteria have been proposed [196, 206, 207, 198, 199, 208, 209, 210, 211]. Including quantum fluctuations in the consideration of the validity of semiclassical gravity was discussed in [212]. A new criterion based on the quantum fluctuations of the semiclassical metric which also incorporates previous criteria [213, 209, 195, 196] in a unified and self-consistent way was proposed by Hu, Roura and Verdaguer [190, 214]. This will be discussed in Part III of this book.

One can see the perils of a mean field theory from the following example suggested by Ford [195]. Consider an isolated system in a quantum state which

is the superposition of one configuration of mass M with the center of mass at X_1 and, with equal amplitude, another configuration of the same mass with the center of mass at X_2. The semiclassical theory as described by the semiclassical Einstein equation predicts that the center of mass of the gravitational field of the system is centered at $(X_1+X_2)/2$. However, one would expect that if a succession of test particles are sent to probe the gravitational field of the above system half of the time they would react to a gravitational field of mass M centered at X_1 and half of the time to the field centered at X_2. The two predictions are clearly different; note that the fluctuation in the position of the center of masses is proportional to $(X_1 - X_2)^2$. Although this example raises the issue of the importance of fluctuations over the mean, it should not be taken too literally. In fact, if the previous masses are macroscopic the quantum system decoheres very quickly and instead of being described by a pure quantum state it is described by a density matrix which diagonalizes in a certain pointer basis; see [215, 175]. For observables attached to such a pointer basis the density matrix description is equivalent to that provided by a statistical ensemble. The results will differ, in any case, from the semiclassical prediction. This strongly suggests that a statistical term that describes the quantum fluctuations must be included in the semiclassical equations.

A systematic study of the connection between semiclassical gravity and open quantum systems resulted in the development of a new conceptual and technical framework where *(semiclassical) Einstein–Langevin equations*[1] were derived, as exemplified in the above anisotropy damping problem (see *e.g.*, [216, 217, 185, 186, 218, 219, 220]). Most of these results use the *influence functional method* of Feynman and Vernon [187], which is suitable when the primary interest rests in obtaining the coarse-grained effect of the environment on the system.

In the language of the *consistent histories* formulation of quantum mechanics the existence of a semiclassical regime for the dynamics of a quantum system has two requirements. The first is *decoherence*, which guarantees that probabilities can be consistently assigned to histories describing the evolution of the system, and the second is that these probabilities should peak near histories which correspond to solutions of classical equations of motion. The effect of the environment is crucial, on two counts: (i) to facilitate decoherence from the noise in it, and (ii) to induce dissipation in the system dynamics by the backreaction, resulting in a semiclassical stochastic dynamics of the open system. See, e.g., [147] for a lucid description of this point. Stochastic semiclassical equations are obtained in an open quantum system after a coarse-graining of the environmental degrees of freedom and a further coarse-graining in the system variables. Applying an

[1] The word semiclassical put in parentheses here refers to the fact that the noise source in the Einstein–Langevin equation arises from the quantum field, while the background spacetime is classical; generally we will not carry this word since there is no confusion that the source which contributes to the stochastic features of this theory comes from quantum fields.

open quantum system conceptual framework to explore issues in quantum cosmology was suggested in, e.g., [184, 221]. It is believed that classical spacetimes emerged after decoherence of the quantum wave function [222] of the universe in interaction with quantum matter acting as its environment. The analogy could be made formally [223] under certain assumptions. Minisuperspace and Born–Oppenheimer approximations are often invoked in the consideration of transition from quantum to semiclassical cosmology. This was an active topic of research in the mid-80s to early 90s. See, e.g., [224, 225, 226, 227, 228, 229, 230, 231, 232]. How quantum geometry and matter existed and interacted before the semiclassical epoch belongs to the tantalizing realm of quantum gravity.

1.2 Quantum, Stochastic, Semiclassical

In the above we use the example of a quantum matter field in a classical gravitational background and work our way up to the semiclassical and then the stochastic regimes. Establishment of the semiclassical theory demands a self-consistent treatment of backreaction; and of the stochastic theory, a suitable inclusion of fluctuations. The existence of a stochastic regime between the quantum and the semiclassical is a rather generic feature. Here we show the same structure appearing by going the other way, 'top-down'; i.e., start with a closed quantum system C comprised of two interacting subsystems, call one the system S, and the other the environment E, which contains a much larger number of degrees of freedom. We examine under what physical conditions and using what technical tools one can capture the emergence of a stochastic theory, which, upon suitable stochastic averaging, produces a good semiclassical theory, perhaps even ameliorating possible pathologies in the classical equation which cause lingering uneasiness. We will use a commonplace textbook system as example, that of a charged particle moving in a quantum field. The quantum particle acting in the place of a quantum sector for gravity can serve to illustrate some general features of the issues of importance to us, in the absence of an agreed-upon viable theory of quantum gravity.

A second, no smaller, purpose is to debunk the misconception that pathology exists in the equations obtained from backreaction. The well-known case is the Abraham–Lorentz–Dirac (ALD) equation [233, 234, 235] describing the motion of a classical charge with radiation-reaction [236, 237] plagued by problems such as pre-acceleration and runaway solutions. We shall show that when the proper conceptual order is followed and the correct technical procedures taken, there is no pathology. This applies also to the semiclassical Einstein equation, as we shall see in a later chapter. It is important to bring out these issues up front and show the proper way to deal with them, if only to dispel any mistaken conclusions drawn from improper treatments, that semiclassical theories are 'sick', and stochastic theories contain ingredients of 'arbitrariness'. On the first issue, assurance is aided by an order reduction scheme properly applied to the

semiclassical Einstein equation (e.g., [238]). Regarding the second issue, for Gaussian systems at least, the Feynman–Vernon functional identity provides a rigorous definition of noise for quantum fields. These points are expounded in later chapters.

1.2.1 Coarse-Graining a Quantum Closed System: Coarse-Grained and Stochastic Effective Action

We begin with the closed system of a quantum particle in a quantum field. A *closed system* can be meaningfully partitioned into subsystems if there exist discrepancies in the scales describing each subsystem, or in accordance to the physical scales present in their interaction strengths (measured by energy or time scales) in relation to the probing scale or resolution accuracy. If we are interested in the details of one such subsystem S (e.g., particle dynamics) and decide to ignore certain details of the other subsystems (e.g., details in the quantum field configurations, such as correlations and phase information) comprising the *environment* E, the distinguished subsystem is thereby rendered an *open system*. The overall effect of the coarse-grained environment on the open-system can be captured by the influence functional of Feynman and Vernon [187], or the closely related closed-time-path (CTP) effective action of Schwinger and Keldysh [239, 240]. These are the initial value, or 'in-in', formulations suitable for following the time evolution of a system, in contradistinction to the usual 'in-out' (Schwinger–DeWitt) effective action formulation useful for S-matrix transition amplitude calculations.

For the model of particle-field interactions under study, coarse-graining the quantum field yields a nonlocal *coarse-grained effective action* (CGEA) for the particle motion [241, 242]. An exact expression may be found in the special case that the particle and field are initially uncorrelated, and the field state is Gaussian. The CGEA may be used to treat the fully nonequilibrium quantum dynamics of interacting particles. Variation of the CGEA yields the equations of motion for the system which in the leading order approximation is the semiclassical limit, followed by higher-order quantum corrections. When decoherence due to interactions with the field is efficient enough and the particle trajectory becomes sufficiently well-defined, with some degree of stochasticity caused by noise [147], the CGEA can be meaningfully transcribed into a *stochastic effective action* (SEA) [243, 244, 245, 246], describing stochastic particle motion. Variation of this stochastic effective action leads to a Langevin equation.

At the semiclassical level, where a classical particle is treated self-consistently with backreaction from the quantum field, an equation of motion for the mean coordinates of the particle trajectory is obtained. For the particle–field system it is the ALD equation governing the dynamics of the particle. At the stochastic level when self-consistent backreaction of the fluctuations in the quantum field is included in the consideration, the ALD–Langevin (ALDL) equation is obtained.

Below, our discussion follows the pathway presented in [246]. We shall first examine the ALD equation and then identify its problems before discussing a sound approach to mitigate them.

1.2.2 Backreaction: The ALD Equation and Its Problems

The generally accepted classical equation of motion including electromagnetic (EM) radiation reaction for the velocity $v \equiv dz/d\tau$, where $z(\tau)$ is the trajectory and τ is the proper time, of a charged (e), spinless point particle (mass m) under the influence of an external force F_{ext} is the celebrated Abraham–Lorentz–Dirac (ALD) equation:

$$\dot{v}^\mu = \Gamma^\mu + F^\mu_L, \tag{1.14}$$

where the radiation reaction force $\Gamma^\mu = \frac{2e^2}{3c^3}\,(\dot{v}^2 - \ddot{v}^\mu)$ with $v^2 = v^\alpha v_\alpha$, and the Lorentz force is $F^\mu_L = \frac{e}{c} F^{\mu\nu} v_\nu$, where $F^{\mu\nu}$ is the EM field tensor. The timescale $\tau_0 = (2e^2/3mc^3)$ determines the relative importance of radiation-reaction. For electrons $\tau_0 \sim 10^{-24}$ secs is the time for light to traverse the classical electron radius $r_e \sim 10^{-15}$m.

Because the ALD equation is a third-order differential equation, seeking solutions to it requires the specification of the initial acceleration in addition to the usual position and velocity required by second-order differential equations. This opens the door to the appearance of runaway solutions. Physical (e.g. non-runaway) solutions may be enforced by transforming Eq. (1.14) to a second-order integral equation with boundary condition such that the final energy of the particle is finite and consistent with the total work done on it by external forces. But the removal of runaway solutions yields acausal solutions that pre-accelerate on timescales τ_0. This raises the question of causality in the classical theory of point particles and fields.

Much effort has been spent in the past century since Lorentz [234] in understanding charged particle radiation-reaction (e.g., [247]). To satisfy the self-consistency and causality requirements, different measures are introduced. Examples include imposing a high-energy cut-off for the field, an extended charge distribution, special boundary conditions, particle spin, and the use of perturbation theory or order reduction techniques (e.g.,[248, 249, 250, 251]). An alternative close to the route we are presenting here is by way of quantum Langevin equations [252, 253, 254]. We shall describe the approach of Johnson and Hu [245, 246, 255] which takes the same pathway as we do for semiclassical and stochastic gravity.

1.2.3 Decoherent Histories, Semiclassicality and Stochasticity

A consequence of coarse-graining the environment – quantum field – is the appearance of noise which is instrumental to the decoherence of the system and

the emergence of a classical particle trajectory. As described earlier decoherence refers to the loss of phase coherence in the quantum open system arising from the interaction of the system with the environment. When the environment is a quantum field, under certain conditions (e.g., Gaussian) or approximations, quantum fluctuations can act effectively as a classical stochastic source, or noise. Under reasonable physical conditions the evolution propagator for the reduced density matrix of the open system is dominated via the stationary phase approximation by the particle trajectory as an extremal solution of the real part of the coarse-grained effective action (CGEA). Because the CGEA is derived by summing over all histories of the quantum field, this extremal solution – in the case of an electron moving in an electromagnetic field, the ALD equation with self-consistent backreaction – incorporates the average radiation reaction force, and hence gives the correct semiclassical trajectory.

In this emergent picture of quantum to classical transition, there is always some degree of residual stochasticity in the system dynamics. Stochastic fluctuations around decoherent semiclassical trajectories are described by the imaginary part of the CGEA. The stochastic effective action encodes much of the quantum statistical information of the field and the state of motion of the system in a noise correlator for the particle. Variation of the stochastic effective action yields classical stochastic equations of motion for the system which embodies a *quantum* dissipation effect *over and above* the classical radiation reaction. It is this quantum dissipation, not the classical radiation reaction force, which balances the quantum fluctuations via a fluctuation-dissipation relation (FDR).[2]

1.2.4 Pathology-Free Modified ALD Equations

This view of the emergence of semiclassical solutions as decoherent histories also suggests a new way to look at the paradoxes of charged particle radiation reaction in the ALD equation. Since the semiclassical limit describes the equations of motion for the expectation value corresponding to the quantum-averaged particle trajectory after sufficient coarse-graining we are led to ask these questions: (1) Are the decoherent histories describing particle trajectories the solutions to a possibly modified ALD equation? (2) Are these solutions unique and runaway-free? (3) Are they causal without pre-acceleration on the coarse-grained scale in which they decohere?

These questions were answered in the affirmative by Johnson and Hu [246] with results obtained from the formulation described above, namely, via the coarse-grained and the stochastic effective actions. They showed that the semiclassical solutions are indeed described by a *modified (semiclassical) ALD equation* with time-dependent coefficients satisfying these criteria.

[2] Note this result clears up the misconception that the radiation reaction force is equivalent to the force due to vacuum fluctuations, e.g., [256, 237]. They are not; the former is a classical effect, whereas the latter is ostensibly quantum.

$$m_0 \ddot{z}_\mu = \frac{e^2}{c^3} \int d\tau' w_\mu G^R \left(z\left(\tau\right), z\left(\tau'\right)\right) + e w_\mu \varphi_{ext}\left(z\right), \tag{1.15}$$

where G^R is the field's retarded Green's function at two proper times on the trajectory and

$$w_\mu \equiv \dot{z}^\nu \dot{z}_{[\nu} \partial_{\mu]} - \ddot{z}_\mu. \tag{1.16}$$

The external potential φ_{ext} provides an external force $F^{ext}_\mu = e w_\mu \varphi_{ext}$ with two main components: a gradient force $e \dot{z}^\nu \dot{z}_{[\nu} \partial_{\mu]} \varphi_{ext} = e\left(\dot{z}_\mu \dot{z}_\nu - \eta_{\mu\nu}\right) \partial^\nu \varphi_{ext}$, and an effective contribution to the particle mass, $e \varphi_{ext} \ddot{z}_\mu$. Note that $\dot{z}^\mu w_\mu = \dot{z}^\mu \ddot{z}_\mu = 0$ identically, and therefore $\dot{z}^\mu F^{ext}_\mu = 0$, as required of a relativistic force that preserves the mass-shell constraint $\dot{z}^2 = -1$. The scalar field force F^{ext}_μ is analogous to the electromagnetic force $F^{EM}_\mu = \dot{z}^\nu F_{\mu\nu}$ which satisfies $\dot{z}^\mu F^{EM}_\mu = 0$.

The retarded Green's function is singular when $\tau = \tau'$, and so requires regularization. After proper treatments (see [245, 246, 255] for details), for sufficiently point-like particles,

$$m \ddot{z}_\mu\left(\tau\right) = \frac{\tau_0}{2}\left(\dot{z}_\mu \ddot{z}^2 + \dddot{z}_\mu\right) + e w_\mu \varphi_{ext}\left(z\right). \tag{1.17}$$

Thus, at low energies and long times, the radiation reaction force is of the usual ALD form, with no runaway trajectories. (Note that the coefficient $\tau_0/2$ for a conformal scalar field is half of the value for radiation reaction in the EM field.) In this modified ALD equation after a proper treatment of backreaction the time-dependent effects act to preserve causality, occurring in the short time after the field begins to dress the initial particle state. A conceptual way to understand this is that fine-grained histories operationally are never observed; physical observables are coarse-grained decohered histories. The coarse-graining scale is set by the measurement resolution in both time and space. With coarse-graining, the quantum fluctuations giving highly nonclassical trajectories are suppressed, and we therefore expect that a *causal and consistent* quantum theory should yield, upon suitable coarse-graining, a classical or semiclassical limit that is pathology free.

Whether the semiclassical solution is unique depends on both the initial particle and field states. If the field begins in a superposition of macroscopically distinct configurations, each configuration may lead to macroscopically distinct particle trajectories. This happens because there will, in that case, no longer be a single extremal solution of the CGEA around which the evolution propagator can be expanded. If these distinct coarse-grained trajectories decohere they are identified as a set of semiclassical solutions. For initial field states that are simple functional Gaussians like vacuum, squeezed, coherent or thermal field states, this problem does not arise. Finally, the particle's initial state may involve a superposition of distinguishable wave packets that, after coarse-graining and decoherence, lead to multiple semiclassical solutions. Thus, only for initially localized particle states in quantum fields that are sufficiently classical, i.e. not

involving macroscopically distinguishable superpositions, should one expect to find a unique single semiclassical solution.

The emergent nature of both the modified ALD equation and the ALD–Langevin equation in the semiclassical and stochastic regimes are the consequences of the effects of coarse-graining of the field environment and its backreaction on the particle system. Because the semiclassical limit describes the quantum average of coarse-grained histories, the ALD equation loses its meaning in the finest-grained quantum limit. Of interest to us here in the study of the effects of quantum fields on a curved spacetime are the corresponding semiclassical Einstein equation and the Einstein–Langevin equation.

1.3 Low Energy Limit: Relation to Gravitational Quantum Physics

We now turn our attention to the relation of semiclassical and stochastic gravity with the newly enacted field of gravitational quantum physics (GQP) [9]. GQP deals mainly with low energy (many orders of magnitude lower than the Planck energy) quantum phenomena in the presence of gravity, under present-day laboratory conditions, often with atomic-optical, molecular spectroscopy, superconductivity, and condensed matter ways and means. When quantum gravity is mentioned in this context it should be understood as referring to graviton physics which carries the quantum features of (perturbative) gravity or quantum aspects of atom-light-matter systems in a weak gravitational field.

We shall highlight the main subject here, the so-called "*alternative quantum theories*" (AQTs), and mention later another subject in passing, namely, quantum information in the presence of gravity, specifically, gravitational decoherence and gravitational entanglement ('gravitational cats'). (For an overview of these two aspects with references, see, e.g., [257].) AQTs often contain proposals at variance with standard quantum mechanics, while sharing the goal of overcoming the difficulties in quantum measurement, but drawing on gravity for assistance. Their hope is that gravity may play a pivotal role in the so-called 'collapse of the wave function', and provide an explanation to the question why objects with masses above a certain scale behave classically and those below quantum mechanically. Because of the involvement of gravity, practitioners in AQT also work with a theory they call semiclassical gravity, but as we shall see, it has basic differences from what is discussed in this book. For one, the types of Newton–Schrödinger equations (NSEs) [44] invoked in many of the AQTs are, contrary to common belief, *not* derivable from taking the weak field limit of general relativity and the nonrelativistic limit of quantum field theory. The second subject we want to discuss under GQP bears on two foundational issues of quantum information, namely, *quantum decoherence* and *quantum entanglement*. Quantum entanglement is the welcomed resource of quantum computing while quantum decoherence is the unwelcome factor to mitigate. One needs to find ways to keep a system's quantum nature long enough for quantum information processing to

be implemented. Here, both issues are framed in the context or in the settings of a gravitational field. Many laboratory quantum experiments involve gravity or micro-gravity. Experiments of quantum information and communication carried out in space certainly need to factor in the effects of the Earth's or even the solar system's gravitational fields. We shall focus on the cases where gravity is the context, not the setting. An example is the Diosi [258, 259, 260, 261, 262] - Penrose [3, 263, 264, 265] theory where *gravitational decoherence* is assumed to be responsible for a quantum object to acquire classical attributes. Another example is the gravitational cat state, an entangled state of a massive object being in two separate spatial locations at the same time, an issue at the very base and starting point of gravitational or relativistic quantum information [266]. (For issues and critiques on decoherence in quantum gravity, see e.g., [267, 268] and references therein.)

1.3.1 Alternative Quantum Theories and the Role of Gravity

General relativists have investigated the interplay between gravity and quantum largely from the angle of how quantum matter affects spacetime (Q $\rightarrow$ G). This is the pathway followed in this book, namely, from quantum field theory in curved spacetime to semiclassical to stochastic gravity. Asking the question in the other direction (G $\rightarrow$ Q), namely, how gravity enters in quantum phenomena, has been going on for just as long (e.g., [269]) mainly by quantum foundation theorists. A forefront issue is why macroscopic objects are found sharply localized in space (their wave functions 'collapsed' on definite locales) while those of microscopic objects extend over space. This contradiction is captured in the celebrated Schrödinger's cat.[3] One can very coarsely place these theories in three groups: the Diosi–Penrose theories invoking gravitational decoherence mentioned earlier; the Ghirardi–Remini–Weber (GRW) [270, 271, 272, 273] Pearle [274, 275, 276] models of continuous spontaneous localization (CSL), and the recent trace dynamics theory of Adler [277] which attempts to provide a sub-stratum theory from which quantum mechanics emerges. The review by Bassi et al. [278] contains a good summary of these theories and recent experiments designed to put them to tests.

Semiclassical gravity from the perspective of the AQTs The validity of semiclassical gravity in the form first proposed by Møller [279] and

[3] Note that 'cat-states' have been found for atoms whereas entangled 'dead' and 'alive' state for real cats, somewhat bigger than an atom, have not. The difference between micro and macro objects is crucial insofar as their quantum behaviors are concerned. A missing basic ingredient is nonequilibrium statistical mechanics which is needed to interpolate between micro/few-body effects and macro/many-body phenomena. Here lies also the importance of macroscopic quantum phenomena (MQP), a subject which has evaded serious theory efforts but is essential in understanding why cat-states can never be found for real cats.

Rosenfeld [280] in the 60s is often questioned by authors of Newton–Schrödinger equation, referring to the arguments of Eppley and Hannah [281], Page and Geilker [282], et al. Leaving aside the bigger question of whether gravity should be quantized, which had seen broader discussions since then [283, 284, 285, 286], the internal consistency of relativistic semiclassical gravity (RSCG)[4] by itself had also been investigated further and there are better responses to the challenges posed in the early 80s (read, e.g., papers by Kibble, Randjbar-Daemi and Duff in [2]). We refer to the following papers: Simon [287] on the stability of semiclassical gravity, Flanagan and Wald [128] on the integrity of relativistic semiclassical gravity when self-consistency is imposed in the choice of the classical spacetime and the quantum state of matter fields by the semiclassical Einstein equation, Anderson, Molina-Paris and Mottola [209], who showed the stability of semiclassical gravity based on linear response, and Hu, Roura and Verdaguer [190], who went further, showing the validity of SCG in a fully nonequilibrium setting, when induced metric fluctuations driven by quantum matter field fluctuations are included via self-consistent solutions of the Einstein–Langevin equation.

1.3.2 Newton–Schrödinger Equation Does Not Represent Semiclassical Gravity

The Newton–Schrödinger equations (NSEs) [44, 288] play a prominent role in alternative quantum theories (AQT) [278], emergent quantum mechanics [277], macroscopic quantum mechanics [289, 290] and gravitational decoherence [291, 292] such as invoked in the Diosi–Penrose theories. The class of theories built upon these equations has drawn increased attention because experimentalists often use it as the conceptual framework and technical platform for understanding the interaction of quantum matter with classical gravity at today's low energy (compared to Planck energy) and to compare their laboratory results. It is thus relevant and timely to explore the assumptions entering into the construction of these equations and the soundness of the theories built upon them, especially in relation to general relativity (GR) and quantum field theory (QFT), the two well-tested theories governing the dynamics of classical spacetimes and quantum matter.

Since NSEs are often conjured as the weak field (WF) limit of GR and the nonrelativistic (NR) limit of QFT, their viability is usually conveniently assumed by proxy with their well-accepted progenitor theories. This was put into question in a paper by Anastopoulos and Hu [293]. They show that NSEs do not follow from general relativity (GR) and quantum field theory (QFT), and there are no

[4] We add the adjective 'relativistic' when it becomes necessary to distinguish SCG from the usage by AQTs. Also, the SCE equation is sometimes called the Møller–Rosenfeld equation by AQTs. We prefer to stay with the name semiclassical Einstein equation, because, after all, it is Einstein's equation, no matter what enters as the source.

'many-particle' NSEs. They come to this conclusion from two considerations: (1) taking the NR limit of the semiclassical Einstein (SCE) equation, the central equation of relativistic semiclassical gravity, a fully covariant theory based on GR+QFT with self-consistent backreaction of quantum matter on the spacetime dynamics [294]. (2) working out a model (see [291]) with matter described by a scalar field interacting with weak gravity with a procedure the same as in deriving the NR limit of quantum electrodynamics (QED). Let us first present the NSE, highlight its features and problems, then show that it cannot be derived from GR+QFT.

The Newton–Schrödinger Equations (NSEs)

The NSE governing the wave function of a single particle $\psi(\mathbf{r}, t)$ is of the form

$$i\frac{\partial}{\partial t}\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + m^2 V_N[\psi]\psi, \tag{1.18}$$

where $V_N(\mathbf{r})$ is the (normalized) gravitational (Newtonian) potential given by

$$V_N(\mathbf{r}, t) = -\int d\mathbf{r}' \frac{|\psi(\mathbf{r}', t)|^2}{|\mathbf{r} - \mathbf{r}'|}. \tag{1.19}$$

It satisfies the Poisson equation

$$\nabla^2 V_N = 4\pi G_{\rm N}\mu, \tag{1.20}$$

where $\mu = m|\psi(\mathbf{r}, t)|^2$ is the mass density, the nonrelativistic (slow motion) limit of energy density ε.

The Newton–Schrödinger equation predicts spatial localization of the wave-function, and decoherence only as a consequence of spatial localization. This "collapse of the wave function" in space for macroscopic objects is the most desired feature of NSEs in many AQTs. However, as cautioned in [293] and explained below, on the GR side, one should not identify the gravitational potential as a dynamical variable, and on the QFT side, not mistake a field as a collection of particles described by single particle wave functions.

Problems with the Newton–Schrödinger Equations

A. In NSE the gravitational self-energy gives rise to the non-linear terms in Schrödinger's equation. In Diosi's theory the gravitational self-energy defines a stochastic term in the master equation. With GR+QFT gravitational self-energy only contributes to mass renormalization, at least in the weak field (WF) limit, and the Newtonian interaction term at the field level induces a divergent self-energy contribution to the single-particle Hamiltonian. It does not induce nonlinear terms to the Schrödinger equation for any number of particles.

B. The *one-particle NSE* appears as the Hartree approximation for N particle states as $N \to \infty$. Consider the ansatz $|\Psi\rangle = |\chi\rangle \otimes |\chi\rangle \ldots \otimes |\chi\rangle$ for an N-particle system. (See Eq. (1.25) for the QED analog). At the limit $N \to \infty$

the generation of particle correlations in time is suppressed and one gets an equation which reduces to the NS equation for χ [295, 296]. However, in the Hartree approximation $\chi(\mathbf{r})$ is *not* the wave-function $\psi(\mathbf{r})$ of a single particle, but a *collective variable* that describes a system of N particles under a mean field approximation.[5]

C. A severe problem of the NSE when applied to a single-particle wave function is its *probabilistic interpretation*. Consider two statistical ensembles of particles one of which is described by the wave-function $\psi_1(\mathbf{r})$ and the other by the wave function $\psi_2(\mathbf{r})$. The ensemble obtained from mixing these ensembles with equal weight is described in standard quantum theory by the density matrix $\rho(\mathbf{r},\mathbf{r}') = \frac{1}{2}[\psi_1(\mathbf{r})\psi_1^*(\mathbf{r}') + \psi_2(\mathbf{r})\psi_2^*(\mathbf{r}')]$. The usual Schrödinger evolution guarantees that the probabilistic interpretation of the density matrix remains consistent under time evolution $\rho_t(\mathbf{r},\mathbf{r}') = \frac{1}{2}[\psi_1(\mathbf{r},t)\psi_1^*(\mathbf{r}',t)] + \frac{1}{2}[\psi_2(\mathbf{r},t)\psi_2^*(\mathbf{r}',t)]$. This property does not apply for non-linear evolutions of the wave-functions. The problem of nonlinearity in quantum mechanics is an old issue which many AQTs are woefully aware of, so we will just mention it here without pressing the case.

In what follows we will show that the only meaningful description of quantum matter interacting with classical gravity is if the matter degrees of freedom are described in terms of quantum fields, *not* in terms of single-particle wave functions in quantum mechanics.

Non-Relativistic Weak Field Limit of SCE Equation

We now examine the nonrelativistic limit of the semiclassical Einstein (SCE) equation and show that it is qualitatively different from the 'many-particle' Newton–Schrödinger equation derived in, e.g., [297]. The SCE equation is in the form

$$G_{\mu\nu} = 8\pi G_{\rm N}\langle\Psi|\hat{T}_{\mu\nu}|\Psi\rangle,$$

where $G_{\mu\nu}$ is the Einstein tensor and $\langle\hat{T}_{\mu\nu}\rangle$ is the expectation value of the stress-energy density operator $\hat{T}_{\mu\nu}$ with respect to a given (Heisenberg-picture) quantum state $|\Psi\rangle$ of the field.

In the weak field limit the spacetime metric has the form $ds^2 = -(1-2V)dt^2 + d\mathbf{r}^2$, and the non-relativistic limit of the semiclassical Einstein equation takes the form

$$\nabla^2 V = 4\pi G_{\rm N}\langle\hat{\varepsilon}\rangle, \tag{1.21}$$

where $\hat{\varepsilon} = \hat{T}_{00}$ is the energy density operator. This can be solved to yield

$$V(\mathbf{r}) = -G\int d\mathbf{r}'\,\frac{\langle\hat{\varepsilon}(\mathbf{r}')\rangle}{|\mathbf{r}-\mathbf{r}'|}. \tag{1.22}$$

[5] It is long known [194] that RSCG is the theory obtained as the leading-order large N limit of N component quantum fields living in a curved spacetime. In Part III of this book we will show that the next-to-leading-order large N expansion produces stochastic semiclassical gravity.

The expectation value of the stress-energy tensor in general has ultraviolet divergences and need be regularized. The procedures have been established since the mid-70s, selectively described in Chapters 2, 4, 5 and 8.

Two key differences between the NR limit of SCE eqn and NSE are: (i) The energy density $\hat{\varepsilon}(\mathbf{r})$ is an operator, not a c-number. The Newtonian potential is not a dynamical object in GR, but subject to constraint conditions. (ii) The state $|\Psi\rangle$ of a field is a N-particle wave function. Quantum matter is coupled to classical gravity as a mean-field theory, well defined only when N becomes sufficiently large.

Starting from SCE the (misplaced) procedure leading one astray to an NSE is the treatment of $m|\psi(\mathbf{r},t)|^2$ as a mass density for a single particle, while in fact it is a quantum observable that corresponds to an operator $\hat{\epsilon}(\mathbf{r}) = \hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r})$ in the Hilbert space of the quantum field theory when the matter degrees of freedom are treated as quantum fields $\hat{\psi}(\mathbf{r})$ and $\hat{\psi}^\dagger(\mathbf{r})$, as they must be. Not treating these quantities as operators bears the problematic consequences A and B of NSE stated above.

Analogue to the Nonrelativistic Limit of QED

To cross check these observations Anastopoulos and Hu [293] carried out an independent calculation for matter described by a scalar field interacting with weak gravity, following the same procedures laid out in [291], namely, solve the constraint, canonically quantize the system, then take the nonrelativistic limit. This procedure is the same as taken in deriving the non-relativistic limit of QED. A Schrödinger equation of the form

$$i\partial|\psi\rangle/\partial t = \hat{H}|\psi\rangle, \tag{1.23}$$

with

$$\hat{H} = -\frac{\hbar^2}{2m}\int d\mathbf{r}\hat{\psi}^\dagger(\mathbf{r})\nabla^2\hat{\psi}(\mathbf{r}) - C\int\int d\mathbf{r}d\mathbf{r}'\frac{(\hat{\psi}^\dagger\hat{\psi})(\mathbf{r})(\hat{\psi}^\dagger\hat{\psi})(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} \tag{1.24}$$

is obtained, with the Coulomb potential replacing the Newtonian gravitational potential (C here accordingly contains the Coulomb constant). This equation is widely used in condensed matter physics.

The matrix elements of the operator (1.24) on the single-particle states $|\chi\rangle$ define the single-particle Hamiltonian:

$$\langle\chi_2|\hat{H}|\chi_1\rangle = -\frac{\hbar^2}{2m}\int d\mathbf{r}\chi_2^*(\mathbf{r})\nabla^2\chi_1(\mathbf{r}) - C\int d\mathbf{r}d\mathbf{r}'\frac{\chi_2^*(\mathbf{r}')\chi_1(\mathbf{r})\delta(\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}. \tag{1.25}$$

It is clear that the form of nonrelativistic QED, Eq. (1.24), is very different from that of the NSE (1.18) for gravity when considering a single particle state. For single-particle states the gravitational interaction leads only to a mass-renormalization term (similar to mass renormalization in QED). This is point A made earlier. Using the Hartree approximation to Eq. (1.24) leads to the same

result as the NR WF limit of SCE, not NSE. This is Point B made earlier. Details of this calculation can be found in [293].

This analysis shows that NSEs are not derivable from GR+QFT. Coupling of classical gravity with quantum matter at the semiclassical level can only be via mean fields. There are no N-particle NSEs. One can therefore conclude that theories based on Newton–Schrödinger equations assume unknown physics.

1.3.3 Nonrelativistic Weak Field Limit of the Einstein–Langevin Equation

To complete this task, it is instructive to also seek the NR WF limit of the Einstein–Langevin equation (ELEq). The merit of doing this is that it provides a natural origin of stochastic source in the Schrödinger equation, needed by many AQTs. It is natural because the noise arises from fluctuations of the quantum matter field, not put in by hand, as is done in many AQTs. The equations derived from taking the NR WF limit of the ELEqs have as base those obtained from the semiclassical Einstein equation, and the features we highlighted above carry over. These equations are not for a single or even many particles; they are the large N limit of an N component field. They are represented by a Hartree wave function which is a collective variable of the matter sector, only now accompanied by a noise term representing the quantum fluctuations of matter fields. (In this way, as stressed in [298], the NR WF limit of stochastic gravity differs from any NSEs with noise added in by hand, e.g., [299, 300].)

1.3.4 Stochastic Gravity needed for Gravity-Related Issues of Quantum Information

Schrödinger's cat state is perhaps the simplest entity embodying the uniquely quantum feature, namely, entanglement, also the most important resource in quantum information. It has been studied and tested out for numerous non-gravitational systems (e.g., in ion traps [301, 302, 303, 304]). Experimental schemes for testing quantum superposition of mechanical objects (e.g., mirrors, cantilevers) by their interaction with light and electric devices have also been explored [305, 306, 307, 308, 309] in opto- and nano-electro-mechanics.

In a quantum description of matter a single motionless massive particle can in principle be in a superposition state of two spatially separated locations. This superposition state in gravity, or gravitational cat ('gravcat') state, can lead to fluctuations in the Newtonian force exerted on a nearby test particle. As shown in [310], this effect lends a way for the detection of 'gravcat' state in the simplest setup in Newtonian gravity.

As an illustration of a gravitational cat state, consider the quantum description of a stationary point mass m localized around $\mathbf{x} = 0$ with spread σ, described by a Gaussian wave function with zero mean momentum:

$$\psi_0(\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{3/4}} e^{-\frac{\mathbf{x}^2}{4\sigma^2}}. \tag{1.26}$$

The position of the particle is described by the probability distribution $|\psi_0(\mathbf{x})|^2$. A particle of mass m_0 located at $\mathbf{R}$ will experience a Newtonian force acted on it by m given by

$$\mathbf{F} = -\frac{Gmm_0}{|\mathbf{R}-\mathbf{x}|^3}(\mathbf{R}-\mathbf{x}). \tag{1.27}$$

For $|\mathbf{R}| >> \sigma$ the fluctuations of the Newtonian force are negligible, which leads one to view it as a deterministic variable.

Now consider two Gaussian states peaked at $\pm\frac{1}{2}\mathbf{L}$ with zero mean momentum. A cat state can be constructed from their superposition,

$$\psi(\mathbf{x}) = \frac{1}{\sqrt{2}} \frac{1}{(2\pi\sigma^2)^{3/4}} \left[e^{-\frac{(\mathbf{x}+\mathbf{L}/2)^2}{4\sigma^2}} + e^{-\frac{(\mathbf{x}-\mathbf{L}/2)^2}{4\sigma^2}} \right]. \tag{1.28}$$

If $\mathbf{L}$ is of the order of magnitude of $\mathbf{R}$, the fluctuations of the Newtonian force (1.27) are non-negligible. Since the force is a function of $\mathbf{x}$, and $\mathbf{x}$ is described by an operator in quantum mechanics, the Newtonian force should also be described as an operator. But then, so would be the gravitational potential. In this sense, the cat state for the point mass has generated a cat state for the gravitational field.

What Is the Right Theory for the Description of a Gravitational Cat State?

This simple question touches on three aspects, quantum, gravity and information. Combining the first two ingredients, namely quantum and gravitational physics, the theory to use is quantum field theory in curved spacetime, albeit in the nonrelativisitc weak field limit suitable for tabletop experiments. Demanding self-consistency in the solutions of the dynamical equations for both classical gravity and for quantum matter, we need solutions of the semiclassical Einstein (SCE) equation in semiclassical gravity (SCG). How about the quantum information aspect? Specifically, how do we accommodate entangled states? If we stay within SCG, we run into trouble immediately.

Consider a single particle. For a superposition of two states each localized at $\mathbf{x} = \pm\frac{1}{2}\mathbf{L}$, the SCE equation predicts that the particle is most likely to be found at the center $\mathbf{x} = 0$, rather than at either side. This inadequacy of SCG has been pointed out in, e.g., [195, 2]. Some authors may use this as an indication that gravity needs be quantized. However, to capture the physics of the superposition state, what is needed here is to include considerations of the quantum correlations, or equivalently, quantum fluctuations of the mass density of the system. To second correlation order, stochastic gravity theory would suffice.

Gravitational entangled states have been on the minds of quantum cosmologists since the 1960s, in the context of the wave function of the universe.

The generic solution to the Wheeler–DeWitt equation [311, 312, 313, 314, 315] is a cat state, i.e., a superposition of two distinct cosmological spacetimes. Everett's 'many-world' interpretation of quantum mechanics [316] is in that spirit. Here, albeit in the weak field nonrelativistic limit, the quantum superposition nature of the theory remains.

It may come as a small surprise that even in an entry-level inquiry into the quantum informational properties of gravity such as the gravitational cat state, emphatically not at the high curvature realms of the very early universe or the late stages of black hole collapse, but in the fully accessible laboratory environment at low energy ranges, one needs to invoke stochastic gravity. The central quantity of importance for the problem described above is the energy density correlation. This corresponds to the noise kernel in stochastic gravity theory, evaluated in the weak field nonrelativistic limit. In this limit quantum fluctuations of the stress-energy tensor manifest as the fluctuations of the Newtonian force.

For the point mass what is needed is the stress-energy tensor and its fluctuations at the level of a single non-relativistic particle. In the non-relativistic limit, the dominant contribution to the stress-energy tensor comes from the mass density $\mu(\mathbf{r})$. The corresponding quantum operator is proportional to the number density operator defined in terms of the particle creation and annihilation operators, or equivalently, the non-relativistic quantum fields. Thus, expectation values and correlations of the mass density can be defined as expectation values of products of $\hat{\mu}(\mathbf{r}, t)$. Gravitational quantum information is an exciting aspect of current research in gravitational quantum physics [9] and relativistic quantum information [266]. For further investigation, see, e.g., [310, 317] and references therein.

1.4 Emphasis and Approach. Guide to the Reader

Here we give a brief description of the contents of this book, our emphasis and approach, the preparation needed, and a guide to readers of different backgrounds.

(1) *Background and preparation.* Because the subject matter discussed in this book spans over four decades, from around 1974 to 2018, and because there already exist excellent monographs on the subject of quantum field theory in curved spacetime (QFTCST), space limitation deters us from attempting even a brief summary of it, except for those topics essential to the development of semiclassical gravity, mainly, the effective action approach and the major regularization methods, namely, dimensional, zeta function, point-separation and adiabatic regularization, treated in Chapters 2, 4, 5 and 8 respectively. Thus we suggest those readers not entirely familiar with QFTCST to read one or two of these monographs (listed in Sec. 1.1.1) as

preparation, focusing on the basics, such as how the vacua in curved and dynamical spacetimes are defined [318] (and the ambiguities therein), how vacuum polarization and fluctuations of quantum fields manifest (e.g., in the Casimir effect [59, 60, 61]) in spacetimes with boundary or with nontrivial topology, and particle creation in two classes of spacetimes: (A) *dynamical spacetimes*, for understanding the early universe, from the FLRW universe [66] to the Bianchi universes [69] and the de Sitter (dS) universe (the $k = 0$ 'flat' coordinate representation – the 'Poincarè patch' – covering half of de Sitter spacetime in the conformal diagram; and the $k = 1$ 'closed' coordinate, a hyperbola of one sheet, covering the full dS spacetime) [319, 320]; and separately, (B) thermal particle creation in the class of *spacetimes with event horizons*, such as the Unruh effect [321] for uniformly accelerated (Rindler) observers, the Hawking effect [77] in eternal (Schwarzschild) [82] and collapsing black holes [78], in 'static' de Sitter spacetime [322], and the moving mirror analogues [323, 324].

(2) *Emphasis and approach.* When introducing a new subject it makes sense logically to think of first presenting the formal structures, and then illustrate them with applications. Here, from considerations of both space and motivation, we opted for a merged way: introduce a new formalism or methodology by showing how it works explicitly when implemented in a generic (another word for commonplace) example. E.g., we use the scalar field and the $\lambda\Phi^4$ theory as our workhorse, thus sacrificing gauge theories and spinors all together even though they play essential roles in the description of nature.[6] The generic spacetimes we shall study are the FLRW, Bianchi and de Sitter universe for cosmology and the Schwarzschild spacetime for black hole physics. The aim is to help the reader get familiarized with the new topics and the necessary techniques through working out in detail some concrete representative problems, such as the 'in-in' effective action (Chapter 2), the large N expansion and the two-particle-irreducible (2PI) effective action (Chapters 6 and 7), to learn the new concepts, such as the open quantum system paradigm, to gain new perspectives, such as the stochastic gravity approach to structure formation (Chapter 13), and to meet new challenges, such as the calculation of the noise kernel (the expectation value of the symmetrized stress-energy bitensor), all in the hope of lessening the reader's possible anxiety in trying to master the mathematical formalisms in abstract. This is a viable choice for us precisely because there exist excellent monographs

[6] The readers may find the following references helpful. For effective actions of *gauge fields* in curved spacetimes: DeWitt [325] Volume 1 p. 680; Parker & Toms [53] Chapter 7 on one-particle-irreducible (1PI)-effective action (EA) in curved spacetime; Calzetta & Hu [326] Chapter 7 on 'in-in' or closed-time-path (CTP) two-particle-irreducible (2PI) effective action of gauge fields in flat spacetime, also work of Berges [327] and Carrington [328]. For the treatment of gauge theories via the Vilkovisky–DeWitt effective action, see [329, 330] and chapters in [53]. For *Spinors* in curved space: Isham [331, 332], Penrose & Rindler [333], Birrell & Davies [47] (Sec. 3.8), Parker & Toms (Sec. 5.6–Sec. 5.9); Holland & Wald [54].

and reviews with the latter emphasis, notably authored by DeWitt, Fulling, Wald, Fredenhagen et al. If we use the familiar textbooks in quantum field theory for comparison, our approach would probably be considered to be closer to Ramond [334] than Itzykson and Zuber [335], to Amit [336] than Zinn-Justin [337], the latter classics are admirably more formal, complete and thorough. For monographs in QFTCST, loosely speaking, stylistically this book is closer to, probably a tad more formal than, Birrell and Davies. In fact our book begins where theirs ends, with interacting quantum field theory, but develops further the essential topics, such as the 'in-in' (Schwinger–Keldysh) or closed-time-path (CTP) effective action in Chapter 3 as an important extension of the 'in-out' (Schwinger–DeWitt) effective action discussed in Chapter 2, the infrared behavior of interacting quantum fields in inflationary cosmology in Chapter 6, etc., while adding another level of complexity to topics seen before, such as the renormalization group equations following the dimensional regularization of Φ^4 theory, the second variation of the effective action derived via the zeta-function method for calculating the noise kernel in Chapter 4, the treatment of point-separation in Chapter 5 not just as a technical convenience in identifying the ultraviolet divergences but respecting the two-point functions of the stress-energy tensor as physically important quantities.

(3) *Contents summary and guide.* This beginning Chapter 1 depicts the main themes and key issues in the contents of this book. We explain the mean field nature of semiclassical theory with respect to fluctuations and explain why a stochastic regime often exists between the quantum and semiclassical regimes in a physical system. This can be seen from the perspective of two key issues, the *backreaction* problem treated self-consistently, and the importance of *fluctuations*. We then shift our attention to the low energy end, the realm of *gravitational quantum physics*, where laboratory experiments are being designed to test the effects of gravity in atomic-molecular-optical, electromechanical, superconducting and condensed matter systems. We point out how *semiclassical gravity* defined here is different from those invoked in alternative quantum theories, and why *stochastic gravity* is needed in the discussion of gravitationally entangled states in quantum information. In the Epilogue we look at how stochastic gravity is linked to *quantum gravity*. We demarcate the nature and missions of quantum gravity from *emergent gravity*, the former entailing a bottom-up (in terms of energy scale) approach in the search for theories of the microscopic constituents of spacetime, and the latter, while positing such a microscopic theory, having to explain how the familiar low energy phenomena emerge from it. Stochastic gravity joins the former endeavor, serving to illuminate the mesoscopic domain in two senses. From the perspective of large N expansion, knowing that stochastic gravity corresponds to the next-to-leading order in a theory with a large number of matter field components, one can continue with the next order

up, which brings us one step closer to quantum gravity, albeit still in an effective theory sense. A more physical picture is to view stochastic gravity in a broader sense as encompassing the hierarchy of correlation functions of the stress-energy tensor of quantum matter fields *and* their induced metric correlations. At each correlation order of the matter field there is a corresponding order of metric correlations, connected by the Einstein–Langevin equations at that order. With quantum gravity possessing the full hierarchy of metric correlations (similar to the Schwinger–Dyson or the Boltzmann–BBGKY hierarchy), stochastic gravity realizes a 'kinetic theory' approach to quantum gravity.

The inner chapters are divided into five parts. The four chapters in **Part I** contain some foundational materials for QFTCST, focusing on the effective action, both the 'in-out' (in Chapter 2) and the 'in-in' kinds (in Chapter 3), which prepares us to tackle the backreaction problems. It also contains descriptions of three important regularization methods, dimensional (Chapter 2), zeta function (Chapter 4) and point-separation (Chapter 5). The second halves of both Chapter 4 and 5 contain advanced materials using these methods, in the derivations of the noise kernels. For readers already familiar with the core subjects of QFTCST, say, at the level of Birrell and Davies or Parker and Toms, they can skip the first halves of Chapter 2, 4 and 5 and focus on their second halves.

Part II contains more advanced material. Chapter 6 discusses the infrared behavior of interacting quantum fields in de Sitter spacetime, especially for massless minimally coupled scalar fields obeying an equation of motion of the same form as weak gravitational perturbations. This theoretical topic is of observational cosmological significance. A proper treatment of this problem by identifying the zero mode in a Euclidean field theory or by examining the late time behavior in a stochastic field theory already existed in the mid-80s and mid-90s, but the last decade saw a welcoming surge of interest in and investigations of this subject. In that context we also present the two-particle-irreducible (2PI) 'in-out' effective action in Chapter 6 for the treatment of infrared problems. This is followed in Chapter 7 by a presentation of the closed-time-path (CTP) 'in-in' 2PI effective action formalism and the large N expansion technique, applied to the $O(N)$ $\lambda\Phi^4$ theory. The equations of motion derived from these methods are needed for backreaction considerations which define semiclassical gravity. Chapter 8 presents the backreaction of quantum field processes in the early universe, enabling us to address these questions: Can the backreaction effect of the trace anomaly lead to singularity avoidance? Where does Starobinsky inflation, the model which has survived observational scrutiny thus far, enter? How does vacuum particle creation, believed to be one of the most significant quantum processes near the Planck time, alter the background spacetime dynamics? Chapter 8 also contains an example in inflationary cosmology, specifically,

how the fluctuations of the inflaton field get parametrically amplified and how particles created in this process heat up the supercooled universe after inflation, when the dynamics of all three parties, the background spacetime, the inflaton field and its fluctuations, are determined self-consistently.

Part III contains three chapters. It begins with Chapter 9, which prepares for the establishment of stochastic gravity theory. Starting with the consideration of general relativity as an effective field theory, it explains the effective action for quantum metric perturbations when they are coupled to N matter fields, for N large. The key ingredients are an explanation of how to select the adiabatic vacuum of the interacting gravitational field-matter field theory and the use of the effective action to generate two-point metric correlations. These are used later in Chapter 16 for conformal fields in de Sitter backgrounds. Chapter 10 shows two ways to derive the Einstein–Langevin equation (ELEq), the centerpiece of stochastic gravity: the axiomatic approach and the functional approach. There, the large N expansion is used to discuss the relation between stochastic and quantum correlations. In Chapter 11 an explicit illustration of how the ELEq is derived for quantum fields in the Minkowski spacetime is given. Solutions of the ELEq incorporating the backreaction of matter field fluctuations give rise to the induced metric fluctuations. From this and the intrinsic metric fluctuations one can draw the expected yet important conclusion that the Minkowski spacetime is stable against intrinsic gravitational and metric fluctuations induced by quantum matter fields.

Once the reader successfully masters the core structure of stochastic gravity from the materials in Part III, the ensuing chapters in Part IV and V will be understood as its logical derivatives with applications. **Part IV** contains two chapters on the backreaction and fluctuations in cosmology and one for black holes. In Chapter 12 we consider a conformally coupled scalar fields in an isotropic but spatially (weakly) inhomogeneous cosmology to illustrate the two levels of theoretical structure in the title of the book: (A) how the quantum matter fields back-react on the background spacetime, from solutions of the semiclassical Einstein equation and (B) how the fluctuations of the quantum matter field backreact on the background spacetime producing induced metric fluctuations from solutions of the Einstein–Langevin equation. The first part is in exact parallel to finding the backreaction effects of particle creation in a weakly anisotropic but homogeneous cosmology (Bianchi Type I), a problem discussed in Chapter 2 as a generic example of semiclassical gravity. From the CTP effective action we give the stress-energy tensor for that quantum field up to first order in the metric perturbations, the zeroth order equation being the semiclassical equation that gives rise to the trace-anomaly-driven inflation studied in Chapter 8. In the second part we go further to show the derivation of the ELEq short of solving it.

Chapter 13 shows how gravitational perturbation theory in the presence of quantum fields works for the cosmological structure formation problem. This

is an important topic in both traditional cosmology (starting with Lifshitz's classic 1946 paper [338]), where structures are believed to have arisen from classical white noise sources amplified by the expansion of the universe, and in modern cosmology with quantum fields as source (since the advent of Guth's inflationary cosmology [118]), where structures are seeded by quantum fluctuations of the inflaton field. Chapter 13 shows that the stochastic correlations produce results equivalent to the quantum correlations of quantized first (linear) order perturbations of the gravitational field. Stochastic gravity can deal very naturally with the fluctuations of the inflaton field beyond the linear approximation as in the trace-anomaly-induced inflation and the Starobinsky type II inflation.

Chapter 14 discusses the backreaction of Hawking radiation on black holes under quasi-static conditions, and metric fluctuations of an evaporating black hole. Already in his seminal paper Hawking [78] noticed the importance of metric fluctuations in the black hole evaporation process: "the area decrease as resulting from the fact that quantum fluctuations of the metric will cause the position and the very concept of the event horizon to be somewhat indeterminate." The black hole end state and information loss issue [339, 340] has dominated the discourse of particle/string and quantum gravity communities for over four decades. Thus we preface our main discussions in this chapter with a short summary of the current status. Despite the strong disagreement and diversity of thoughts, fortunately there is one solid ground which almost everyone agrees on: semiclassical gravity.

The theoretical structure of the backreaction and fluctuations problem in black hole spacetimes is the same as for cosmological spacetimes, and a strategy to pursue this problem in Schwarzschild black holes had been outlined in [341]. A major factor retarding the progress of this project is that the radial function in Schwarzschild spacetime does not have an analytic form and numerical solutions are very involved.

In contrast, the high symmetry of the de Sitter spacetime makes it possible to push forward a thorough investigation of the correlation functions of covariant geometric objects, as detailed in the three chapters in **Part V**. There is also a shift of emphasis: the 'in-in' effective action for the gravitational perturbations are used for computing correlations of *quantum* metric perturbations. In stochastic gravity we do not quantize the metric perturbations; in fact, the stochastic correlations are shown to agree with the quantum correlations in the large N approximation. Here, the separation of induced and intrinsic correlations is no longer useful, because both the induced and the intrinsic parts diverge but their sum is finite. Thus we return to our starting point, the 'in-in' effective action. Chapter 15 shows a calculation of the fluctuations in the stress-energy tensor in de Sitter spacetime, in parallel to the derivations of the noise kernel by the zeta function method in Chapter 4 and the point separation method in Chapter 5. Chapter 16 presents the computation of the two point scalar, vector and tensor

perturbations. Chapter 17 presents the computation of the correlation functions of the Riemann tensor, which naturally contains the Weyl-Weyl correlators, and ends with a description of how these quantities are useful for the calculation of quantum fluctuation-induced phenomena in early universe cosmology and black hole physics.

Part I

Effective Action and Regularization, Stress Tensor and Fluctuations

2

'In-Out' Effective Action. Dimensional Regularization

The effective potential governing a quantum field in a background spacetime is an important and useful quantity in quantum field theory, as its variation gives the equation of motion for the field with quantum corrections. However, it is well defined only for flat or static spacetimes and for a constant field. Effective action is for more general conditions, allowing for dynamical background spacetimes and background fields, its variation producing dynamical equations for both the field and the background spacetime in a self-consistent manner.

Effective actions have immediate relevance for the study of quantum processes in black holes and the early universe. The period of interest when quantum field and particle physics effects take a center stage is from the grand unification theory (GUT) time ($t_{GUT} \approx 10^{-35}$ sec), when the strong and electroweak forces are believed to be unified, to the Planck time ($t_p \approx 10^{-43}$ sec) [the Planck length $\ell_p \equiv \sqrt{\hbar G_{\rm N}/c^3} = 1.6 \times 10^{-33} cm$], when quantum gravitational effects become important. This epoch may have seen one or more phase transitions or symmetry-breaking processes associated with possible quantum gravity, induced gravity, Kaluza–Klein theory, string theory or supersymmetry ($\approx 10^{17}$ GeV) theories.

A dynamical background geometry dictates a corresponding change in the background field through the wave equation in curved spacetimes (e.g., for scalar fields, involving the Laplace–Beltrami operator). Take the inflationary cosmology [118, 342] active at the grand unified theory (GUT) time as an example. A GUT-epoch phase transition based on an effective potential, such as the Coleman-Weinberg (CW) [343] potential invoked in the "new" inflationary cosmological model [120, 119] derived from a flat-space quantum field theory, is, strictly speaking, structurally inconsistent because it presumes a constant background Higgs field while the background cosmological spacetime is dynamical. An effective action is needed in its place. One should look into the details of different epochs and make assumptions or approximations accordingly. For example, during the *supercooling stage* the Higgs field is almost constant. This could justify the use

of an effective potential. In the following *slow-roll* stage one can use a quasilocal effective action as the background field is slowly varying. In the *reheating* stage the rapid oscillation of the background field (the inflaton) as it makes its way toward the true vacuum produces abundant particles from the quantum fluctuations via parametric amplification. Particle creation and interaction provide the fuel for reheating, and their backreaction on the background inflaton field ushers the background spacetime into a post-inflation radiation-dominated phase. In the reheating epoch a quasilocal approximation is no longer valid. One needs to use a method which respects the full dynamics of the background field in a dynamical spacetime, namely, the full effective action.

Strong backreaction effects from quantum field processes prevail also at the Planck energy [the Planck mass is defined as $m_p \equiv \sqrt{\hbar c/G_{\rm N}} = 2.2 \times 10^{-5} g$], where rapid changes in the background spacetime and background field bring forth copious particle creation from the vacuum, and the created particles in turn act to alter the dynamics of the background spacetime in a major way, such as the isotropization and homogenization of the universe at the Planck time. We will treat these processes in these cases in later chapters. In this chapter we will present the familiar Schwinger–DeWitt effective action, the so-called 'in-out' formalism discussed in most textbooks, suitable for the computation of S matrix scattering or transition amplitudes. In the next chapter we will present the Schwinger–Keldysh effective action in the so-called 'in-in', or 'closed-time-path' (CTP) formalism necessary for the derivation of the dynamics of expectation values.

We begin in the first part with a quick refreshment of basic field theory and quantum fields in curved spacetime, just to remind the readers of some key concepts, such as under what conditions the vacuum states are well-defined, and to introduce some necessary expressions such as the conformal stress-energy tensor. In the second part we construct the 'in-out' effective action of an interacting quantum field and apply it to study the effects of particle creation and interaction in the early universe. We illustrate how dimensional regularization is implemented. The third part of this chapter treats the case where changes in the background spacetime and fields are gradual enough that one can perform a derivative expansion beyond the constant background, introduce momentum space representation for the propagators and obtain quasi-local effective actions in a closed form. The fourth part of this chapter is devoted to dimensional regularization and the derivation of renormalization group equations, using the $\lambda\Phi^4$ theory as an example.

2.1 Quantum Field Theory in Dynamical Spacetimes: Key Points

Let us begin with a free massive m scalar field ϕ in Minkowski spacetime whose dynamics are described by the classical action

$$S[\phi] = \int d^4x \left\{ -\frac{1}{2} (\nabla\phi)^2 - V[\phi(x)] \right\}, \tag{2.1}$$

where $(\nabla\phi)^2$ in Minkowski space is equal to $\partial_\mu\phi\partial^\mu\phi = (\partial\phi)^2$ and the potential $V(\phi)$ is a real functional of the field variable ϕ. For a massive scalar field with $\lambda\phi^4$ self interaction the potential is

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4, \tag{2.2}$$

where λ is the coupling constant. The equations of motion are given by the variational principle $\delta S/\delta\phi = S_{,\phi} = 0$. They read

$$\Box\phi - V'(\phi) = 0, \tag{2.3}$$

where $\Box = \partial_\mu\partial^\mu$ in Minkowski space and $V'(\phi) = dV(\phi)/d\phi$.

2.1.1 Some Basics of Quantum Field Theory (QFT)

The theory is quantized by replacing the field variable ϕ by an operator-valued distribution Φ. In the *Heisenberg picture*, for each event x there is an operator $\Phi(x)$ acting on some Hilbert space $\mathcal{H}$ of states. The conjugate momentum π goes over to the momentum operator Π, and the Poisson brackets become the *equal-time canonical commutation relations*. A free field corresponds to a quadratic potential $V(\Phi)$. A generic example is a free massive scalar field with $V(\Phi) = \frac{1}{2}m^2\Phi^2$. The Heisenberg equation of motion for this field becomes the *Klein–Gordon equation* $\Box\Phi(x) - m^2\Phi(x) = 0$.

Assume that the field lives in a finite large spatial volume[1] V. Expanding the scalar field operator in (spatial) Fourier modes, we have

$$\Phi(t,\mathbf{x}) = \frac{1}{\sqrt{V}}\sum_{\mathbf{k}}\mathbf{f}_{\mathbf{k}}(t)u_{\mathbf{k}}(\mathbf{x}), \tag{2.4}$$

where $\mathbf{k} = 2\pi\mathbf{n}/L$, and $\mathbf{n} = (n_1, n_2, n_3)$ in general consists of a triplet of integers. In Minkowski space the spatial mode functions are simply $u_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{x}}$. In the infinite volume continuum limit this becomes

$$\Phi(t,\mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}}\, e^{i\mathbf{k}\cdot\mathbf{x}}\mathbf{f}_{\mathbf{k}}(t)\,. \tag{2.5}$$

The (operator-valued) amplitude function $\mathbf{f}_{\mathbf{k}}(t)$ for each mode $\mathbf{k}$ obeys a harmonic oscillator equation

$$\ddot{\mathbf{f}}_{\mathbf{k}} + \omega_{\mathbf{k}}^2\mathbf{f}_{\mathbf{k}} = 0, \tag{2.6}$$

where an overdot denotes d/dt and $\omega_{\mathbf{k}}^2 = |\mathbf{k}|^2 + m^2$ in Minkowski space.

Given two complex independent solutions $f_{\mathbf{k}}$, $f_{\mathbf{k}}^*$ of Eq. (2.6), we may write

$$\mathbf{f}_{\mathbf{k}}(t) = f_{\mathbf{k}}(t)\,\hat{a}_{\mathbf{k}} + f_{\mathbf{k}}^*(t)\,\hat{a}_{-\mathbf{k}}^\dagger. \tag{2.7}$$

[1] The notation for spatial volume V can be distinguished from the potential $V(\Phi)$, which carries an argument, the 1-loop effective potential $V^{(1)}$ which has a superscript, and the spacetime volume Vol.

Let us introduce the Wronskian $W(f,g) = f\dot{g} - g\dot{f}$, which is conserved by Eq. (2.6), and impose the normalization

$$(f_{\mathbf{k}}, f^*_{\mathbf{k}}) = i\hbar. \tag{2.8}$$

The Heisenberg picture Φ and its conjugate momentum $\Pi = \delta\mathcal{L}/\delta(\partial_0\Phi)$ obey the equal-time commutation relation. Note that the scalar delta function $\delta(\mathbf{x},\mathbf{x}')$ in curved spacetime is defined by $\int d^4\mathrm{x}\sqrt{-g}\delta(\mathbf{x},\mathbf{x}')h(\mathbf{x}) = h(\mathbf{x}')$, where h is any test function. The equal-time commutation relations between quantum fields and their conjugate momenta induce these relations between the particle creation $\hat{a}^\dagger_{\mathbf{k}}$ and annihilation $\hat{a}_{\mathbf{k}}$ operators:

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] = \left[\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{k}'}\right] = 0;\ \left[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{k}'}\right] = \delta(\mathbf{k}-\mathbf{k}'). \tag{2.9}$$

We say that each choice of the basis functions $f_{\mathbf{k}}$ constitutes a *particle model*, where $f_{\mathbf{k}}$ is the positive frequency component and $f^*_{\mathbf{k}}$ is the negative frequency component of the $\mathbf{k}$ th mode; the state which is destroyed by all the $\hat{a}_{\mathbf{k}}$'s is the vacuum of the particle model. The vacua of different particle models are in general inequivalent.

Now consider a free massive quantum scalar field in a dynamical background spacetime. When a mode decomposition is possible the (c-number) amplitude function of the $\mathbf{k}$ th mode obeys, from Eq. (2.6), the wave equation

$$\frac{d^2 f_{\mathbf{k}}}{dt^2} + \omega^2_{\mathbf{k}}(t) f_{\mathbf{k}}(t) = 0, \tag{2.10}$$

where the natural frequency $\omega_{\mathbf{k}}$ now acquires an explicit time dependence and the mode equation generally possesses time-dependent solutions. Unlike for QFT in constant background field or static spacetimes, a consistent separation into positive and negative energy solutions of the wave equation is not always possible. The definition of a vacuum state becomes a fundamental challenge in the construction of QFT in time-dependent backgrounds [318].

There are a few situations where vacuum states in QFT in dynamical spacetimes are well-defined: (1) the so-called statically bounded or asymptotically stationary spacetimes, where it is assumed that at $t = \pm\infty$ the background spacetime becomes stationary and the background fields become constant; (2) conformally-invariant fields in conformally static spacetimes. In both cases the Fock spaces are well defined and one can calculate the amplitude for particle creation in an S-matrix sense. (3) If the background spacetime does not change too rapidly (to be quantified later by a nonadiabaticity parameter) there is a conceptually clear and technically simple method in defining the so-called (nth order) *adiabatic vacuum or number state*. We shall treat cases (1) and (2) below and case (3) in Chapter 8.

2.1.2 Particle Number and Energy-Momentum Tensor

We begin with a formal rendition to the parametric oscillator equation (2.10) describing the amplitude function of the **k**th normal mode. Following [69, 71] we seek a solution of $f_{\mathbf{k}}(t)$ in the form:

$$f_{\mathbf{k}}(t) = \sqrt{\frac{\hbar}{2\omega_{\mathbf{k}}}}\,\{\alpha_{\mathbf{k}} e_{\mathbf{k}}^{-} + \beta_{\mathbf{k}} e_{\mathbf{k}}^{+}\}\,; \quad e_{\mathbf{k}}^{\pm} \equiv \exp\left\{\pm i \int \omega_{\mathbf{k}} dt\right\} \tag{2.11}$$

in terms of a positive-frequency component and a negative-frequency component $e_{\mathbf{k}}^{\mp}$, with coefficients $\alpha_{\mathbf{k}}, \beta_{\mathbf{k}}$ respectively. Only with a well-defined vacuum will they convey the meaning of particles and antiparticles. Transformation from one particle representation to another, sometimes referred to as a 'quasi-particle' basis, is facilitated by a Bogoliubov transformation on the functions $\alpha_{\mathbf{k}}, \beta_{\mathbf{k}}$.

Since the single equation (2.11) does not determine the coefficients $\alpha_{\mathbf{k}}$ and $\beta_{\mathbf{k}}$ uniquely, we need another condition, which is chosen so that the Wronskian condition Eq. (2.8) is satisfied. The auxiliary condition imposed on $\dot{f}_{\mathbf{k}}$ is

$$\dot{f}_{\mathbf{k}}(t) = -i\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2}}\,(\alpha_{\mathbf{k}} e_{\mathbf{k}}^{-} - \beta_{\mathbf{k}} e_{\mathbf{k}}^{+})\,. \tag{2.12}$$

Inverting these two equations we can express the complex functions $\alpha_{\mathbf{k}}, \beta_{\mathbf{k}}$ in terms of $f_{\mathbf{k}}, \dot{f}_{\mathbf{k}}$ as follows:

$$\alpha_{\mathbf{k}} = \sqrt{\frac{\omega_{\mathbf{k}}}{2\hbar}}\left(f_{\mathbf{k}} + \frac{i}{\omega_{\mathbf{k}}}\dot{f}_{\mathbf{k}}\right) e_{\mathbf{k}}^{+}\,, \qquad \beta_{\mathbf{k}} = \sqrt{\frac{\omega_{\mathbf{k}}}{2\hbar}}\left(f_{\mathbf{k}} - \frac{i}{\omega_{\mathbf{k}}}\dot{f}_{\mathbf{k}}\right) e_{\mathbf{k}}^{-}. \tag{2.13}$$

Making use of the Wronskian condition we obtain

$$s_{\mathbf{k}} \equiv |\,\beta_{\mathbf{k}}\,|^2 = \frac{1}{2\hbar\omega_{\mathbf{k}}}\left(\left|\dot{f}_{\mathbf{k}}\right|^2 + \omega_{\mathbf{k}}^2\,|f_{\mathbf{k}}|^2\right) - \frac{1}{2}. \tag{2.14}$$

It is tempting to regard $s_{\mathbf{k}} = |\beta_{\mathbf{k}}|^2$ as the amount of particle production. However, that is only true when the vacuum state is well-defined, to make sense of particles.

The stress-energy tensor of a massive scalar field in Minkowski space with metric $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$ is

$$T_{\mu\nu}^{Mink} = \nabla_\mu\phi\nabla_\nu\phi - \frac{1}{2}\eta_{\mu\nu}\nabla^\rho\phi\nabla_\rho\phi - \frac{1}{2}\eta_{\mu\nu}m^2\phi^2. \tag{2.15}$$

The energy density is given by the expectation value of the 00 component of $T_{\mu\nu}$ with respect to the Minkowski vacuum $|\,0\rangle$, i.e.,

$$\rho_0^{Mink} \equiv \langle 0\,|\,T_{00}\,|\,0\rangle = \int \frac{d^3\mathbf{k}}{2(2\pi)^3}(|\,\dot{f}_{\mathbf{k}}\,|^2 + \omega_{\mathbf{k}}^2\,|\,f_{\mathbf{k}}\,|^2) = \int \frac{d^3\mathbf{k}}{(2\pi)^3}(2s_{\mathbf{k}} + 1)\frac{\hbar\omega_{\mathbf{k}}}{2}. \tag{2.16}$$

The second equality comes from using (2.5) for the field decomposition in terms of $\mathbf{f}_{\mathbf{k}}(t)$ and then (2.7) in terms of the creation and annihilation operators for the stress-energy tensor (2.15). The third equality follows from using (2.14).

Comparing this to the Hamiltonian of a finite system of parametric oscillators with canonical coordinates $q_{\mathbf{k}}$ and their conjugate momenta $\pi_{\mathbf{k}}$

$$H^{Mink}(t) = \frac{1}{2}\sum_k (\pi_{\mathbf{k}}^2 + \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2) = \sum_k (N_{\mathbf{k}} + \frac{1}{2})\hbar\omega_{\mathbf{k}}, \tag{2.17}$$

one can identify $\mid f_{\mathbf{k}} \mid^2$ and $\mid \dot{f}_{\mathbf{k}} \mid^2$ with the canonical coordinates $q_{\mathbf{k}}^2$ and momentum $\pi_{\mathbf{k}}^2$, and the eigenvalue of H_0 is the energy $E_{\mathbf{k}} = (N_{\mathbf{k}} + \frac{1}{2})\hbar\omega_{\mathbf{k}}$. The analogy of particle creation with parametric amplification is formally clear: (2.14) defines the number operator

$$N_{\mathbf{k}}(t) = \frac{1}{2\hbar\omega_{\mathbf{k}}}(\pi_{\mathbf{k}}^2 + \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2) - \frac{1}{2} = s_{\mathbf{k}}, \tag{2.18}$$

and (2.16) says that the energy density of vacuum particle creation comes from the amplification of that from vacuum fluctuations $\hbar\omega_{\mathbf{k}}/2$ by the factor $\mathcal{A}_{\mathbf{k}} = 2s_{\mathbf{k}} + 1$. For free quantum fields in Minkowski space the vacuum remains the same throughout its evolution; there is no vacuum instability, no particle creation. Thus $s_{\mathbf{k}} = 0$ and $\mathcal{A}_{\mathbf{k}} = 1$, i.e., there is no amplification of the vacuum fluctuations. Note again these relations are formal, awaiting a physical definition of the vacuum state, which varies for different circumstances. We shall return to this point in Chapter 8 when we introduce the adiabatic vacuum and explain adiabatic regularization.

2.1.3 QFT in Curved Spacetime

The action for a free massive (m) scalar field $\Phi(x)$ coupled to an n-dimensional background spacetime with metric $g_{\mu\nu}$ and scalar curvature R is given by

$$S_f[\Phi, g] = \int d^n x\,(-g)^{-1/2}\left\{\frac{1}{2}\Phi(x)\Box_x\Phi(x) - \frac{1}{2}\Big[m^2 + (1-\xi)\,\xi_n R\Big]\Phi^2(x)\right\}, \tag{2.19}$$

where $x = (t, \mathbf{x}), \xi_n = \frac{(n-2)}{4(n-1)}$, and ξ is the field-curvature coupling constant, which is equal to 0 for conformal coupling and 1 for minimal coupling. (The g under the square root sign denotes the determinant of the metric tensor, while that in the argument denotes dependence on the metric tensor.) The equation of motion for the scalar field $\Phi(x)$ is obtained from

$$\frac{\delta}{\delta\Phi(x)} S[\Phi, g] = 0, \tag{2.20}$$

namely,

$$\Big[-\Box_x + m^2 + (1-\xi)\xi_n R\Big]\Phi(x) = 0. \tag{2.21}$$

Here

$$\Box \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu = \frac{1}{\sqrt{-g}}\frac{\partial}{\partial x^\mu}\left(\sqrt{-g}\, g^{\mu\nu}\,\frac{\partial}{\partial x^\nu}\right) \tag{2.22}$$

is the Laplace–Beltrami operator defined on the background spacetime with metric $g_{\mu\nu}$. It is a generalization of the d'Alambertian wave operator in Minkowski space to curved spacetime.

In the canonical quantization approach, one assumes a foliation of spacetime into dynamically evolving, time-ordered, spacelike hypersurfaces Σ. If the three-dimensional space Σ possesses some symmetry, such as a homogeneous space, possessing a group of motion, a separation of variables is usually possible, which enables a normal mode decomposition of the field. (The spacetimes considered in this book, e.g., Friedmann–Lemaître–Robertson–Walker (FLRW), de Sitter (dS) and Schwarzschild spacetimes all possess these properties.) One can then impose canonical commutation relations on the creation and annihilation operators corresponding to the (time-dependent) amplitude functions of each normal mode, define the vacuum and number states, and then construct the Fock space. For Minkowski space, a unique global Killing vector ∂_t orthogonal to all constant-time spacelike hypersurfaces facilitates an unambiguous separation of the positive- and negative-frequency modes, and thus a well-defined and unique vacuum. A unique global Killing vector ∂_t exists also for curved yet static spacetimes. In a general curved dynamical spacetime, general covariance precludes any such privileged choice of time and slicing. There is no natural mode decomposition and no unique vacuum [318, 48]. We assume however that the background spacetime under consideration has at least enough symmetry to allow for a normal mode decomposition of the invariant operator at any constant-time slice.

The classical field theory is quantized by replacing the field variable ϕ by the operator-valued distribution Φ. Consider the field Φ in a coordinate volume $V = L^3$ with coordinate length L. We can expand the field Φ in terms of a complete set of (spatial) orthonormal modes $u_{\mathbf{k}}(\mathbf{x})$ as in Eq. (2.4). We use $\mathbf{x}$ as a generic notation for the spatial coordinates, also applicable for spatially non-flat spacetimes, e.g., in S^3 with radius a, $V = 2\pi^2 a^3$, one can use the hyperspherical coordinates $\mathbf{x} = (\chi, \theta, \phi)$, and the wave numbers are then labeled by the corresponding quantum numbers $\mathbf{k} = (n, l, m)$, with n being the principal quantum number and (l, m) that of S^2. As before, we write the operator-valued amplitude function $\mathbf{f}_{\mathbf{k}}(t)$ in terms of the time-independent annihilation operators $a_{\mathbf{k}}$ and the (c-number) amplitude functions $f_{\mathbf{k}}(t)$ as in (2.7). The canonical commutation rules on Φ then imply the conditions on $\hat{a}_{\mathbf{k}}$ and $\hat{a}^\dagger_{\mathbf{k}'}$ as in (2.9).

In curved space the inequivalence of Fock space representation due to the lack of a global time-like Killing vector makes the constant separation of positive and negative-frequency components in general impossible. The mixing of positive- and negative-frequency components is the source of particle creation in the

second quantization description. Particle creation may arise from topological, geometrical, or dynamical causes. Cosmological particle creation is by nature a dynamically induced effect; beware that in dynamical spacetimes the inequivalence of vacua exists at all times throughout the evolution. We have restricted our attention to a free-field thus far: particles are not produced from interactions, but rather from the excitation (parametric amplification [68]) of vacuum fluctuations (or quantum noise) by the changing background gravitational field. Particle creation and interaction described by an interacting quantum field theory will be treated in Sec. 4.

2.1.4 Friedmann–Lemaître–Robertson–Walker (FLRW) Universes

The class of Friedmann–Lemaître–Robertson–Walker (FLRW) universes plays an important role in the establishment of the so-called Standard Model in cosmology. The line element has the form:

$$ds^2_{RW} = -dt^2 + a^2(t)d\Omega^2_3, \tag{2.23}$$

where $a(t)$ is the time-dependent scale factor and $d\Omega^2_3$ is the line element of the 3D hypersurface, which comes in three types: spatially-closed (k=+1) with $d\Omega^2_{S3}$ of a 3-sphere S^3, spatially flat (k=0) with $d\Omega^2_{E^3}$ the 3D Euclidean space, and spatially open (k = −1) with $d\Omega^2_{H^3}$ of a 3D hyperbolic space. The line element $d\Omega^2_{S^3}$ of a 3-sphere is given by

$$d\Omega^2_{S^3} = d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2), \tag{2.24}$$

where χ, θ, ϕ are the angular coordinates on S^3 with ranges $0 \leq \chi, \theta \leq \pi$, $0 \leq \phi \leq 2\pi$. The line element of the a 3D hyperbolic space H^3 is

$$d\Omega^2_{H^3} = d\chi^2 + \sinh^2\chi(d\theta^2 + \sin^2\theta\, d\phi^2). \tag{2.25}$$

where $\chi \geq 0$. The three corresponding static cases, when a is a constant, are the (closed) Einstein universe, the Minkowski space and the open Einstein universe, respectively.

For the spatially-flat FLRW spacetime with line element

$$ds^2 = -dt^2 + a^2(t)d\Omega_{E^3} = -dt^2 + a^2(t)\sum_{i=1}^{3} dx_i^2 \tag{2.26}$$

the spatial mode functions are simply $u_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{x}}$ and the wave equation for the amplitude function of the $\mathbf{k}$th mode in cosmic time t becomes

$$\ddot{f}_k(t) + 3H\ \dot{f}_k(t) + [\omega_k^2(t) + q(t)]f_k(t) = 0, \tag{2.27}$$

where, as before, an overdot denotes taking the derivative with respect to cosmic time, $\cdot = d/dt$. Because of spatial isotropy, f depends only on $k \equiv |\mathbf{k}|$. Here

$$\omega_k^2(t) = \frac{k^2}{a^2} + m^2; \quad q = (1-\xi)\,\xi_n R, \tag{2.28}$$

and

$$R = 6\left[\dot{H}(t) + 2H^2(t)\right], \tag{2.29}$$

with $H(t) \equiv \frac{\dot{a}}{a}$ being the expansion (Hubble) rate of the background space. We have grouped terms containing two time derivatives of a (second derivative or first derivative squared) and call them q. As we will see in Chapter 8, they are of second adiabatic order while ω_k is of zero adiabatic order.

Using the conformal time η which is related to the cosmic time t by $\eta = \int^t dt'/a(t')$, the metric of the FLRW spacetime can be written as

$$ds^2_{RW} = -dt^2 + a(t)^2 d\Omega_3^2 = a^2(\eta)(-d\eta^2 + d\Omega_3^2). \tag{2.30}$$

Thus in conformal time η the closed, flat and open RW metrics are conformally related by the conformal factor $a(\eta)$ to the static Einstein universe, Minkowski space and the open Einstein universe respectively. They are conformally static.

Now consider a massive scalar field in a spatially-flat FLRW universe. The conformal amplitude function $\chi_{\mathbf{k}}(\eta) \equiv a(\eta) f_{\mathbf{k}}(\eta)$ related to the c-number amplitude function $f_{\mathbf{k}}$ for the $\mathbf{k}$th normal mode satisfies, from Eq. (2.27), the following wave equation:

$$\chi''_{\mathbf{k}}(\eta) + [\Omega^2_{\mathbf{k}}(\eta) + Q]\chi_{\mathbf{k}}(\eta) = 0, \tag{2.31}$$

where a prime denotes $' \equiv d/d\eta$ and

$$\Omega_k^2(\eta) \equiv \omega_k^2(t)a^2 = k^2 + m^2 a^2 \tag{2.32}$$

is the time-dependent natural frequency. The term Q comes from field-curvature coupling which for the FLRW universe is given by $Q = Q_\xi = \xi R a^2$. It takes on a different form $Q = Q_\beta$ for conformal fields in anisotropic spacetimes such as Bianchi universes, to be discussed in Chapter 3.

One readily observes that, for massless ($m = 0$) conformally coupled ($\xi = 0$) fields in a spatially-flat FLRW universe, the conformal wave equation admits simple solutions

$$\chi_{\mathbf{k}}(\eta) = Ae^{i\omega_{\mathbf{k}}\eta} + Be^{-i\omega_{\mathbf{k}}\eta}, \tag{2.33}$$

which are of the same form as traveling waves in flat space. Since $\omega_{\mathbf{k}}(m = 0, \xi = 0) = k = \text{const}$, the positive- and negative-frequency components remain separated and there is no particle production. This is an observation made by Parker [344].

In this connection, Grishchuk [345] showed that there is no production of gravitons in a radiation-dominated FLRW universe. This is easily seen as follows: The gravitons are quantized linear perturbations off a background spacetime. In a FLRW universe, just as in Minkowski spacetime, there are two polarizations, each obeying an equation – the Lifshitz equation [338] – which has the same form as a massless minimally-coupled ($\xi = 1$) scalar field [346] (See Sec. 6.1.1.). In FLRW universe $R = 6a''/a^3$, the wave equation (2.31) reads, in conformal time,

$$\chi''_{\mathbf{k}}(\eta) + (k^2 - a''/a)\chi_{\mathbf{k}}(\eta) = 0. \tag{2.34}$$

For a radiation-dominated FLRW universe, $a \sim t^{\frac{1}{2}} \sim \eta$, and thus $R = 0$. The natural frequency is a constant and there is no production of massless minimally coupled scalar particles or gravitons in the conformal vacuum. We refer to these two simple inductions about particle creation in FLRW universe from field and spacetime conformal transformation considerations as the Parker–Grishchuk lemma [344, 345].

More generally, the wave equation for each normal mode contains a term proportional to the time-dependent natural frequency. The negative-frequency modes can thus be excited by the dynamics of the background through $a(\eta)$ and $R(\eta)$. In analogy with the time-dependent Schrödinger equation, one can view the $\omega_k^2 + Q$ term in (2.31) as a time-dependent potential $V(\eta)$ which can induce backscattering of waves [68, 71], thus mixing the positive and negative frequency components in each mode. This, as we have learned, in the second quantization language, signifies particle creation.

Note that the basic mechanism for cosmological particle creation is different from thermal particle creation in black holes [78], accelerated detectors [321] or moving mirrors [323, 324], the latter cases involving the presence of an event horizon or the exponential red-shifting of outgoing modes [347, 348, 349].

We shall return to this subject in Sec. 3, in connection with the conformal anomaly. We now turn to a path-integral description of particle creation, in terms of the instability of the vacuum as it evolves from past infinity to future infinity. The key figure is the vacuum persistence amplitude.

2.2 The Schwinger–DeWitt ('In-Out') Effective Action

The vacuum persistence amplitude $\langle 0_+|0_-\rangle$ is the transition amplitude that the vacuum $|0_-\rangle$ (the 'in' vacuum) at an initial time, say, $t = -\infty$ remains the vacuum $|0_+\rangle$ (the 'out' vacuum) at late times, say, $t = +\infty$. For a system of matter field Φ and gravitational field g, the vacuum persistence amplitude has a functional representation

$$\langle 0_+|0_-\rangle = \int D\Phi Dg \exp\{iS[\Phi, g]\}, \tag{2.35}$$

where the integral is over all fields Φ and gauge-equivalent 4-geometries g and with the specified 3-geometries and field configurations on the initial and final spacelike hypersurfaces. These slices at $t = \pm\infty$ are usually chosen such that a vacuum or adiabatic vacuum state is well-defined, corresponding to vanishing graviton and particle number. The values of 3-geometries and fields on these 'in' and 'out' states define the boundary conditions. The action $S[g, \Phi]$ for this system consists of two parts,

$$S[\Phi, g] = S_f(\Phi) + S_g^{(0)}[g], \tag{2.36}$$

where $S_g^{(0)}[g]$ is the classical gravitational (Einstein–Hilbert) action

$$S_g^{(0)}[g] = \frac{1}{16\pi G_{\rm N}} \int_{\mathcal{D}} d^4x\sqrt{-g}\; R, \tag{2.37}$$

where $G_{\rm N}$ is the Newton constant, R is the curvature scalar of the background spacetime and $\mathcal{D}$ is the spacetime region enclosed by the initial and final hypersurfaces. The quantum matter field action $S_f(\Phi)$ is

$$S_f[\Phi] = \int_{\mathcal{D}} d^4x\sqrt{-g}\; L, \tag{2.38}$$

where L is the matter field Lagrangian density.

Let us consider first the case where there is only a scalar field $\Phi(x)$ present, coupled linearly to an external source $J(x)$. We will add on the consideration of a graviton field later. We shall also work in a n-dimensional spacetime, in preparation for dimensional regularization. The vacuum persistence amplitude $\langle 0_+|0_-\rangle$ is given by the generating functional Z obtained by functionally integrating the actions over the scalar field Φ in a background metric g:

$$\begin{aligned}\langle 0_+|0_-\rangle = Z[J, g] &= N \int D\Phi\; \exp\left\{ i\, S[\Phi, g] + i \int d^n x\, \sqrt{-g(x)}\; J(x)\Phi(x) \right\} \\ &= e^{i\, W[J,g]}.\end{aligned} \tag{2.39}$$

To incorporate the quantum contributions, we make a background field decomposition $\Phi = \phi + \varphi$ and expand S around a saddle point $\phi(x)$ up to $\delta^2 S/\delta\phi(x)\phi(x')$ term,

$$\begin{aligned}S[\Phi, g] = S[\phi(x), g] &+ \int d^n x\, \sqrt{-g}\; \phi(x) J(x) \\ &+ \int d^n x\, \left\{ \frac{\delta S[\phi(x), g]}{\delta\phi(x)} + \sqrt{-g}\; J(x) \right\} \big[\Phi(x) - \phi(x)\big] \\ &+ \frac{1}{2} \int d^n x\, d^n x'\; \big[\Phi(x) - \phi(x)\big] \frac{\delta^2 S[\phi, g]}{\delta\phi(x)\delta\phi(x')} \big[\Phi(x') - \phi(x')\big] + \cdots .\end{aligned} \tag{2.40}$$

The saddle point field $\phi(x)$ satisfies

$$\frac{\delta}{\delta\phi} S[\phi, g] + \sqrt{-g}\; J = 0, \tag{2.41}$$

which is identical to the classical field in the source-free limit. After performing the Gaussian functional integral over the fluctuation fields $\varphi = \Phi - \phi$ we obtain for W, defined in (2.39),

$$W[J,g] = -i \ln Z[J,g] = S[\phi,g] + \int d^n x \sqrt{-g}\; J(x)\phi(x) - \frac{i}{2} \ln \mathrm{Det} G, \quad (2.42)$$

where G is the Green function

$$G^{-1}(x,x') = \frac{\delta^2 S[\phi,g]}{\delta\phi(x)\delta\phi(x')}. \quad (2.43)$$

The effective field $\bar{\phi}$ is defined as[2]

$$\begin{aligned} \bar{\phi} = \langle 0_+|\Phi|0_-\rangle &= Z^{-1}[J,g] \int D\Phi\; \Phi\; \exp\Big\{ i\, S[\Phi,g] + \int d^n x \sqrt{-g}\; J\phi \Big\} \\ &= \frac{\delta}{\delta J} W[J,g] \end{aligned} \quad (2.44)$$

We now perform a functional Legendre transformation. Since at one loop ϕ and $\bar{\phi}$ are indistinguishable we have

$$\Gamma[\bar{\phi},g] = W[J,g] - \int d^n x \sqrt{-g}\; J\bar{\phi}(x). \quad (2.45)$$

The effective action $\Gamma[\bar{\phi},g]$ is equal to W at vanishing source, whence, by virtue of (2.39),

$$\langle 0_+|0_-\rangle = \exp\Big\{ i\, \Gamma[\bar{\phi},g] \Big\}. \quad (2.46)$$

It yields the one-particle irreducible (1PI) Green function.

In the above we have only considered quantum contributions of the scalar field in a given geometry g. Quantum contributions of the gravitational field – gravitons – can be incorporated similarly. Gravitons are quantized weak linear perturbations $h_{\alpha\beta}$ off of a background spacetime with metric $g^{(0)}_{\alpha\beta}$, i.e., the metric of the weakly perturbed spacetime is $g_{\alpha\beta} = g^{(0)}_{\alpha\beta} + h_{\alpha\beta}$. Adding in the action a term of the form $\int d^n x \sqrt{-g}\; g_{\alpha\beta} T^{\alpha\beta}$, where $T^{\alpha\beta}$ is the source driving the graviton field $h_{\alpha\beta}$, one can then integrate over both the scalar field and the graviton field configurations to get the generating functional of both fields.

Therefore, in the presence of external sources J and $T^{\alpha\beta}$ of the matter and gravitational fields respectively, the generating functional for a quantum field becomes

[2] Note in the 'in-out' setup the effective field is in general complex. It is not the mean field, which is properly defined only for expectation values obtainable from the 'in-in' formulation to be discussed in the next chapter. To avoid overladen use of symbols both quantities are denoted by an overbar; however, their differences can be distinguished by the context.

$$Z[J,T^{\alpha\beta}] = \langle 0_+|0_-\rangle_{J,T^{\alpha\beta}} = \int D\Phi Dg\ \exp\Big\{ i\,S[\Phi,g] + i\,J\Phi + i\,T^{\alpha\beta}g_{\alpha\beta}\Big\}$$
$$= e^{iW[J,T^{\alpha\beta}]}. \tag{2.47}$$

The effective action $\Gamma[\bar\phi,\bar g]$ is defined as the Legendre transform of $W[J,T^{\alpha\beta}]$

$$\Gamma[\bar\phi,\bar g] = W[J,T^{\alpha\beta}] - J\Phi - T^{\alpha\beta}g_{\alpha\beta} \tag{2.48}$$

where

$$\bar\phi = \frac{\delta W[J,T^{\alpha\beta}]}{\delta J}, \qquad\qquad \bar g_{\alpha\beta} = \frac{\delta W[J,T^{\alpha\beta}]}{\delta T^{\alpha\beta}}. \tag{2.49}$$

When $J=0$ and $T^{\alpha\beta}=0$ we can write

$$\langle 0_+|0_-\rangle = e^{i\Gamma[\bar\phi,\bar g]}, \tag{2.50}$$

where the effective action is evaluated at the mean field $\bar\phi$ and the particular geometry $\bar g$ with vanishing sources $J[\bar\phi,\bar g]=0$ and $T^{\alpha\beta}[\bar\phi,\bar g]=0$. This geometry $\bar g$ is called the *effective geometry*, which in the 'in-out' formalism is complex. To get a real geometry governed by a causal equation of motion one needs to use the 'in-in' formalism treated in the next chapter. Since

$$\frac{\delta\Gamma[\bar\phi,\bar g]}{\delta\bar g_{\alpha\beta}(x)} = -T^{\alpha\beta}[\bar g], \tag{2.51}$$

this condition is equivalent to

$$\frac{\delta\Gamma[\bar\phi,\bar g]}{\delta\bar g_{\alpha\beta}} = 0, \tag{2.52}$$

which can be regarded as the defining condition for the effective geometry $\bar g$. The effective geometry which extremizes the effective action is the normalized matrix element of the metric field operator $\hat g_{\alpha\beta}(x)$ between the initial and final vacua:

$$\bar g_{\alpha\beta}(x) = \frac{\langle 0_+|\hat g_{\alpha\beta}(x)|0_-\rangle}{\langle 0_+|0_-\rangle}. \tag{2.53}$$

If the initial vacuum is stable so that $|0_+\rangle = |0_-\rangle$, then the effective geometry and the effective action are real, and $|\langle 0_+|0_-\rangle|^2 = 1$. If particles are produced, then $|0_+\rangle$ will not be the same as $|0_-\rangle$ and both the effective geometry and effective action will be complex. Indeed,

$$|\langle 0_+|0_-\rangle|^2 = \exp\Big\{-2\,\mathrm{Im}\,\Gamma[\bar\phi,\bar g]\Big\} \equiv 1 - P. \tag{2.54}$$

Thus the imaginary part of the effective action evaluated at the mean geometry is directly related to the total probability of producing a pair of particles from the initial vacuum in the history of the universe. If $\mathrm{Im}\,\Gamma[\bar\phi,\bar g]$ is small, then

$$P \simeq 2\,\mathrm{Im}\,\Gamma[\bar\phi,\bar g]. \tag{2.55}$$

Except in very special cases, the effective action cannot easily be evaluated in closed form. A systematic approximation is to reinstate the $\hbar$ in S (i.e., as $S/\hbar$) and expand $\Gamma[\bar{\phi}, \bar{g}]$ in powers of $\hbar$, i.e.

$$\Gamma[\bar{\phi}, \bar{g}] = \Gamma^{(0)}[\bar{\phi}, \bar{g}]/\hbar + \Gamma^{(1)}[\bar{\phi}, \bar{g}] + \hbar\Gamma^{(2)}[\bar{\phi}, \bar{g}] + \cdots \tag{2.56}$$

and keep only the first few terms. This is the familiar loop expansion. In the lowest order, the effective action is just the classical action, i.e., $\Gamma^{(0)}[\bar{\phi}, \bar{g}] = S_g^{(0)}[\bar{g}]$. In the one-loop ($\hbar^0$) order $\Gamma^{(1)}$ is obtained by evaluating the defining functional integral by the method of stationary phase. Rewriting the 'ln Det' in (2.42) to 'Tr ln', we have

$$\Gamma^{(1)}[\bar{\phi}, \bar{g}] = -\frac{i}{2} \sum_f \epsilon_f \Big[\mathrm{Tr} \ln G^f(x, x') \Big]_{reg}. \tag{2.57}$$

The sum is over all the fields present in the original action including gravity and the ghost fields. ϵ_f is 1 for bosons and -1 for fermions. The function $G^f(x, x')$ is the Green function for the field f propagating in the background metric $\bar{g}$. The trace Tr includes spacetime labels and the subscript *reg* indicates that it is suitably regularized. We will discuss dimensional regularization in a later section and zeta-function and point-separation regularization in Chapters 4 and 5.

In the following we shall consider the case where only quantum fluctuations of the scalar field, and not the gravitational field, is included. The effective action which incorporates such contributions yields an effective geometry $\bar{g}$ and an effective field $\bar{\phi}$, in general complex, satisfying the equations of motion

$$\frac{\delta\Gamma[\bar{\phi}, \bar{g}]}{\delta \bar{g}_{\alpha\beta}} = 0, \qquad\qquad \frac{\delta\Gamma[\bar{\phi}, \bar{g}]}{\delta \bar{\phi}} = 0. \tag{2.58}$$

They are the Einstein equation and the Klein–Gordon equation modified by quantum fluctuations of the scalar field. (Since the difference of $\Gamma^{(1)}[\bar{\phi}]$ and $\Gamma^{(1)}[\phi]$ are in higher-loop terms, we may sometimes use ϕ and $\bar{\phi}$ interchangeably as functional variables.)

2.2.1 Conformal Stress-Energy Tensor and Conformal Anomaly

The stress-energy tensor obtained from the variation of the action (2.19) with respect to the metric $g_{\mu\nu}$ is given by,

$$\begin{aligned} \Lambda_{\mu\nu} = {} & \nabla_\mu \Phi \nabla_\nu \Phi - \frac{1}{2} g_{\mu\nu} \nabla^\rho \Phi \nabla_\rho \Phi - \frac{1}{2} g_{\mu\nu} m^2 \Phi^2 \\ & + (1 - \xi)\xi_n \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \Phi^2 \\ & + (1 - \xi)\xi_n [g_{\mu\nu} \Box(\Phi^2) - \nabla_\mu \nabla_\nu (\Phi^2)]. \end{aligned} \tag{2.59}$$

This is what Callen Coleman and Jackiw [350] called the 'new, improved' energy-momentum tensor. For conformal fields in curved spacetimes it is conformally

related to the flat space counterpart. Note the appearance of ξ, the curvature scalar R, and the $\Box(\Phi^2)$ terms. Recall also from (2.19) that we define ξ such that $\xi = 0$ corresponds to conformal coupling. The vacuum energy density of a massless conformal field is given by the expectation value of the 00 component of $\Lambda_{\mu\nu}$with respect to the conformal vacuum.

Conformal Anomaly

Earlier we learned that there is no particle creation from a conformally coupled field in a conformally static spacetime. This would be the case for free massless conformal fields ($m = \xi = 0$) in a FLRW universe. One needs to break the conformal symmetry to see particle creation, such as for massive, or non-conformal fields, for interacting fields, or for anisotropic or inhomogeneous perturbations of the FLRW universe. We shall consider the first three cases in this chapter, the case of a weakly anisotropic universe in the next chapter and a weakly inhomogeneous universe in Chapter 12.

An interesting fact is, a quantity of different quantum field origin known as the trace anomaly exists even in cases where there is no particle creation, such as for conformally invariant fields in a conformally flat spacetime. Its physical origin can be attributed to vacuum polarization rather than vacuum fluctuations which seed particle creation upon parametric amplification in cosmology or due to event horizons as in black holes. 'Trace' here refers to the trace of the stress-energy tensor. For classical conformal fields such as the photon, the trace of its stress-energy tensor is zero, but for a quantum conformal field, its trace is non-zero, thus referred to as an anomaly. In fact there is much deeper meaning (showing up in the regularization of the stress-energy tensor) and implications (as a likely candidate for primordial inflation) associated with this term. Let us recount the origin of the conformal anomaly from the regularization of quantum fields in curved space.

For a conformally invariant field theory, such as the massless conformal-coupled scalar field under consideration, the counter-action S^c introduced to cancel the ultraviolet divergences must be constructed from curvature quantities which are conformally invariant or pure divergences in the limit of four dimensions. This requirement leads to a counteraction of the general form, written in the context of dimensional regularization:

$$S_c = \frac{\mu_c^{n-4}}{n-4} \int d^n x \sqrt{-g}\, \big(A\, G + B\, F\big), \tag{2.60}$$

where μ_c is a renormalization mass scale with the dimension of an inverse length, and the constants $A = -(5760\pi^2)^{-1}, B = (1920\pi^2)^{-1}$ for a scalar field. Also,

$$\begin{aligned} G &= R_{abcd}R^{abcd} - 4R_{ab}R^{ab} + R^2, \\ F &= R_{abcd}R^{abcd} - 2R_{ab}R^{ab} + \frac{1}{3}R^2, \end{aligned} \tag{2.61}$$

where G is the argument of the Gauss–Bonnet identity in four dimensions and thus a pure divergence. F in four dimensions becomes the square of the Weyl tensor and thus is conformally-invariant. This counteraction will lead to a trace anomaly of the form

$$T_A = \alpha \Box R + \beta (R_{ab} R^{ab} - \frac{1}{3} R^2) + \gamma C_{abcd} C^{abcd}, \tag{2.62}$$

where α, β and γ are constants related by $3\alpha - \beta - 2\gamma = 0$. Their values are

$$\alpha = \frac{1}{2880\pi^2} \begin{cases} 1 \\ 6 \\ 12 \end{cases} \qquad \beta = \frac{1}{2880\pi^2} \begin{cases} 1 \\ 11 \\ 62 \end{cases} \tag{2.63}$$

for scalar, four-component neutrino and Maxwell fields respectively [351, 352]. They are related to A, B by

$$\alpha = \frac{2}{3} B, \qquad \beta = -2A, \qquad \gamma = A + B. \tag{2.64}$$

This results in a regularized action for the matter field stress-energy tensor and three additional terms in curvature quadratures. Since these terms are conserved geometric objects and purely local, i.e., it is constructed purely from the metric, the curvature, and a finite number of its covariant derivatives, they can be moved to the left-hand side of the Einstein equation, and combined with existing curvature quadratures in the theory, with renormalized coupling constants. Doing so results in an extended theory of general relativity, known as semiclassical gravity, which encompasses the dynamics of quantum matter fields in a dynamical classical spacetime in a self-consistent way, embodied in the semiclassical Einstein equation.

The trace anomaly T_A can be formally written as the functional derivative of the one-loop effective action $\Gamma^{(1)}_{TA}$ with respect to the metric

$$T_A = \frac{2}{\sqrt{-g}} g_{\alpha\beta} \frac{\delta \Gamma^{(1)}_{TA}}{\delta g_{\alpha\beta}}. \tag{2.65}$$

The action $\Gamma^{(1)}_{TA}$ which gives rise to the trace anomaly T_A cannot in general be expressed as an integral of a polynomial in the curvature, though local non-polynomial expressions are possible.

For massless conformal scalar fields in the FLRW spacetime the variation in (2.65) is with respect to the scale factor $a(\eta)$. It can be shown that the following action generates the trace anomaly T_A [353]:

$$\Gamma^{(1)}_{TA}[a] = V \int_0^\infty d\eta \left[-3\alpha \left(\frac{a''}{a} \right)^2 + \beta \left(\frac{a'}{a} \right)^4 \right], \tag{2.66}$$

There are many ways to see how the trace anomaly arises. Derivation via the zeta function [96, 97] and the point separation methods [46, 100] will be discussed in Chapters 4 and 5 respectively. In a path-integral representation Fujikawa [354]

showed that it can be identified with the Jacobian factor for the functional measure under the conformal transformation. A perturbative method is used in [110] to derive the trace anomaly. We shall use this way to illustrate the renormalization of an interacting field theory in the following section. Conformal anomalies for interacting scalar fields are treated in [355]. The semiclassical Einstein equation will be discussed further in Chapter 7 and later chapters. The effect of the trace anomaly on the cosmological singularity is treated in Chapter 8. Here we focus on the effects of particle creation and interaction.

2.2.2 Semiclassical Einstein equation

Our introduction of the counter-action (2.60) to explain the origin of the conformal anomaly is catered to a conformal field. But the general expression in the form of 'squared' curvature tensors in the counteraction, such as contained in the F and G terms, holds for general nonconformal, massive fields. The semiclassical action for gravity has the generic form

$$S^{\rm G}[g^{\mu\nu}] = \frac{1}{16\pi G_{\rm N}} \int d^4x\sqrt{-g}\left[R - 2\Lambda + \left(aR^2 + bR^{ab}R_{ab} + cR^{abcd}R_{abcd}\right)\right], \tag{2.67}$$

where a, b, and c are constants with dimensions of length squared, R_{abcd} is the Riemann tensor, R_{ab} is the Ricci tensor, Λ is the cosmological constant (with dimension of inverse length-squared), $\sqrt{-g}$ is the square root of the determinant of $g_{\mu\nu}$, and $G_{\rm N}$ (with dimension of length divided by mass) is Newton's constant. The inclusion of arbitrary coupling ξ to the quantum field and the higher-order curvature terms R^2, $R^{ab}R_{ab}$ and $R^{abcd}R_{abcd}$ is needed for the cancellation of the ultraviolet divergences which arise in the matter quantum fields resulting in a renormalization of a, b, c, $G_{\rm N}$, Λ, m, ξ, and λ. These quantities before renormalization are bare; their observable physical counterparts are renormalized. Semiclassical gravity theory entails finding self-consistent solutions of the quantum matter field equations and the Einstein equation with these quantum fields as sources. In four spacetime dimensions the generalized Gauss–Bonnet theorem provides a relation between these three constants a, b, and c. Thus they are not all independent. Classical Einstein gravity is obtained by setting $a = 0$ and $b = 0$.

In the counteraction, instead of the Riemann curvature tensor-squared term, one can also use the Weyl curvature tensor-squared term. Thus an alternative form of $S^{\rm G}[g^{\mu\nu}]$ can be

$$S^{\rm G}[g^{\mu\nu}] = \frac{1}{16\pi G_{\rm N}} \int d^4x\sqrt{-g}\left[R - 2\Lambda + \left(a\,R^2 + \alpha_2\,R^{ab}R_{ab} + \gamma\,C^{abcd}C_{abcd}\right)\right], \tag{2.68}$$

where, comparing with (2.67), $\alpha_2 = b+c$ and $\gamma = 3c/2$. The stress-energy tensors are obtained by taking the functional derivative of this action with respect to g_{ab}. The three different curvature-squared terms $R^2, R^{ab}R_{ab}, C_{abcd}C^{abcd}$ in the counteraction give rise to the following three types of tensors

$$\begin{aligned} {}^{(1)}H_{ab} &= -\frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{ab}}\int d^4x\sqrt{-g}\; R^2 \\ &= -2g_{ab}\Box R + 2\nabla_a\nabla_b R - 2RR_{ab} + \frac{1}{2}g_{ab}R^2, \end{aligned} \tag{2.69}$$

$$\begin{aligned} {}^{(2)}H_{ab} &= -\frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{ab}}\int d^4x\sqrt{-g}\; R^{ab}R_{ab} \\ &= \frac{1}{2}\, g_{ab}R^{cd}R_{cd} - \Box R_{ab} - \frac{1}{2}\Box R g_{ab} + \nabla_a\nabla_b R - R^{cd}R_{cadb}, \end{aligned} \tag{2.70}$$

$$\begin{aligned} {}^{(C)}H_{ab} &= -\frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{ab}}\int d^4x\sqrt{-g}\; C_{abcd}C^{abcd} \\ &= -4\nabla^c\nabla^d C_{abcd} + 2R^{cd}C_{abcd}. \end{aligned} \tag{2.71}$$

Grouping these three terms together we call their sum H_{ab} (with no superscript):

$$\mathrm{H}_{ab} = a\,{}^{(1)}H_{ab} + \alpha_2\,{}^{(2)}H_{ab} + \gamma\,{}^{(C)}H_{ab} \tag{2.72}$$

In a semiclassical approximation only terms that are first order in $\hbar$ will be considered. Remember also in 4D owing to the existence of the Gauss–Bonnet theorem, only two of these terms are independent [356].

In conformally flat, four-dimensional spacetimes where the Weyl curvature tensor $C^c{}_{def}$ vanishes, ${}^{(1)}H_{ab}$ and ${}^{(2)}H_{ab}$ no longer remain linearly independent. In this case ${}^{(1)}H_{ab} = 3\;{}^{(2)}H_{ab}$. A useful new quantity appears instead,

$$\begin{aligned} {}^{(3)}H_{ab} &= -\frac{1}{12}R^2 g_{ab} + R^{cd}R_{cadb} \\ &= -R_a{}^c R_{cb} + \frac{2}{3}RR_{ab} + \frac{1}{2}R_{cd}R^{cd}g_{ab} - \frac{1}{4}R^2 g_{ab}, \end{aligned} \tag{2.73}$$

which is conserved only in conformally flat spacetimes, but not as a result of a variational derivation, as are ${}^{(1)}H_{ab}$, ${}^{(2)}H_{ab}$, nor as the limit of a conserved quantity in non-conformally flat spacetimes [357]. It is second order in derivatives of the metric, unlike ${}^{(1)}H_{ab}$ and ${}^{(2)}H_{ab}$, which are of fourth order. Nevertheless, it is allowed by Wald's axioms [103], and, in general, contributes to the conformal anomaly. Under the preconditions for the existence of ${}^{(3)}H_{ab}$ one can also write H_{ab} as

$$\mathrm{H}_{ab} = a\,\hbar^{(1)}H_{ab} + \alpha_2\,\hbar^{(2)}H_{ab} + \beta\,\hbar^{(3)}H_{ab} + O(\hbar^2), \tag{2.74}$$

where it should be understood that the ${}^{(3)}H_{ab}$ term is only present when ${}^{(1)}H_{ab} = 3{}^{(2)}H_{ab}$ which means that only two of these terms are independent. One can choose to fold ${}^{(2)}H_{ab}$ into ${}^{(1)}H_{ab}$ and shorten this to

$$\mathrm{H}_{ab} = \alpha\,\hbar{}^{(1)}H_{ab} + \beta\,\hbar{}^{(3)}H_{ab} + O(\hbar^2), \tag{2.75}$$

hence, $\alpha = a + \alpha_2/3$. Taking the trace of this expression yields the conformal anomaly (2.62). When the Weyl curvature vanishes because of conformality, only two terms with constants α, β remain.[3]

The semiclassical Einstein equation has the generic form

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{N}}\,[\langle T_{ab}\rangle + \mathrm{H}_{ab}] \tag{2.76}$$

where $\langle T_{ab}\rangle = \mathcal{O}(\hbar)$ is the expectation value of the matter stress-energy tensor. For convenience, we consider only massless, conformally coupled fields (of arbitrary spin). We may reasonably restrict the form of $\langle T_{ab}\rangle$ to obey Wald's physical axioms: (1) covariant conservation, (2) causality, (3) standard results for 'off-diagonal' matrix elements, and (4) standard results in Minkowski space [102, 103, 362]. Wald showed that any $\langle T_{ab}\rangle$ that obeys the first three axioms is unique up to the addition of a local, conserved tensor. Furthermore, any local, conserved tensor can reasonably be considered part of the geometrodynamics and so be written on the left-hand side of the field equations. In the spirit of renormalization, the three new terms of curvature quadratures, called collectively H_{ab}, arising from the counteractions introduced to cancel the ultraviolet divergences in the quantum matter fields, can now be moved to the LHS of the Einstein equation. Giving the term H_{ab} a new name $\Xi_{ab} \equiv -8\pi G_{\mathrm{N}}\mathrm{H}_{ab}$ so it can now be regarded as belonging to the family of geometric objects, we obtain the semiclassical Einstein equation

$$R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab} + \Xi_{ab} = 8\pi G_{\mathrm{N}}(T^m_{ab} + \langle T^f_{ab}\rangle), \tag{2.77}$$

where we have included the stress-energy tensor of any classical matter T^m_{ab} present, and the vacuum expectation value of the stress-energy tensor of the quantum fields present $\langle T^f_{ab}\rangle$, to one loop $\hbar$ order.

In the next section we show the example of a $\lambda\Phi^4$ quantum field, carry out the perturbation treatment, work out the regularization, derive the SCE equation and discuss its solutions with bearings on how quantum field processes impact on the early universe.

[3] The ${}^{(1)}H_{ab}$, ${}^{(2)}H_{ab}$ tensors were introduced in [358] and [85]. The $\alpha_1, \alpha_2, \alpha_3$ in [357] correspond to a, α_2, β here. The k_1, k_3 in [359, 360] correspond to our α, β. Finally, the (k_3, k_2, k_1) in [361] correspond to our $(2880\pi^2)(\alpha, \beta, \gamma)$ respectively.

2.3 Effective Action of an Interacting Field: Particle Creation and Interaction

We now apply the 'in-out' effective action formalism discussed in the last section to an interacting field,[4] using a background field decomposition.[5] Consider a massive (m), self-interacting $\lambda\Phi^4$ scalar field coupled (ξ) to a radiation-filled spatially flat FLRW universe with metric $g_{\mu\nu}$ and scalar curvature R. We shall calculate the particle creation probability and derive the coupled equations of motion for the metric and the background field.[6]

Let us begin with the simplest case of a free ($\lambda = 0$), massless ($m = 0$), conformally coupled ($\xi = 0$) scalar field and build up from there. The classical radiation with energy density $\rho_r = \tilde{\rho}_r\, a^{-4}$ is described by the classical action

$$S_{rad} = \int d^4x \, \Big[-\tilde{\rho}_r\Big], \tag{2.78}$$

where $\tilde{\rho}_{\rm r}$ is a constant which measures the total number of photons (entropy content) in the universe. The classical gravitational action for the spatially flat FLRW universe is

$$S_g^{(0)}[a] = -\frac{V}{16\pi G_{\rm N}} \int_0^\infty d\eta \, \Big[6(a')^2\Big] = -V \int_0^\infty d\eta \, \Big[6\left(\frac{a'}{\kappa}\right)^2\Big], \tag{2.79}$$

where $\kappa \equiv \sqrt{16\pi G_{\rm N}}$, $V = a^3 V_0$, V_0 being the volume at a reference time t_0, and $R = 6a''/a^3$ where a prime denotes taking the derivative with respect to the conformal time η, related to cosmic time t by $t \equiv \int a(\eta) d\eta$.

The classical action which depicts a radiation-filled spatially flat FLRW universe is then given by

$$S^{(0)} = S_g^{(0)}[a] + S_{rad} = -V \int_0^\infty d\eta \, \Big[6\frac{(a')^2}{\kappa^2} + \tilde{\rho}_r\Big]. \tag{2.80}$$

A superscript (0) is placed on S to denote the zeroth order, or at the background level, in relation to perturbative calculations performed later. We see that at this order $\delta S^{(0)}/\delta a = 0$ gives the familiar classical Einstein equation in the Standard Model of cosmology whose G_{00} component reads

$$3\left(\frac{\dot{a}}{a}\right)^2 = 8\pi G_{\rm N}\rho. \tag{2.81}$$

with ρ given by ρ_r.

[4] For treatments of interacting quantum fields in curved spacetime, see, e.g., [363, 92, 364, 365, 366, 367, 368] in its initial stage of development and Chapter 10 of [47] for further discussions and references.

[5] For the general formalism of background field method in curved spacetime and its application, see, e.g., [369, 370] for scalar fields and pure gauge fields, [371, 372] for fermion fields interacting with gauge fields, and [373] for non-abelian gauge fields.

[6] These equations being complex, the 'in-out' effective action is subject to the same criticism we issued earlier: a proper treatment should be by way of the 'in-in' effective action which gives real and causal equations of motion.

We now consider the cases when m, ξ, λ are non-zero. We shall work in n-dimensions in anticipation of dimensional regularization. The discussion below follows the work of [374].

The action for a massive (m), self-interacting $\lambda\Phi^4$ scalar field coupled (ξ) to a radiation-filled spatially flat FLRW universe with metric $g_{\mu\nu}$ and scalar curvature R is given by

$$S^f[\Phi, g] = \int d^n x\sqrt{-g}\left\{\frac{1}{2}\Phi(x)\Box_x\Phi(x) - \frac{1}{2}\Big[m^2 + (1-\xi)\,\xi_n R\Big]\Phi^2(x) - \frac{\lambda}{4!}\,\Phi^4(x)\right\}. \tag{2.82}$$

The scalar field $\Phi(x)$ satisfies the equation of motion

$$\frac{\delta}{\delta\Phi(x)}S^f[\Phi, g] = 0, \tag{2.83}$$

namely,

$$\Big[-\Box_x + m^2 + (1-\xi)\xi_n R\Big]\Phi(x) + \frac{\lambda}{6}\,\Phi^3(x) = 0. \tag{2.84}$$

At 1-loop the Green function $G(x, x')$ for a scalar field [given by Eq. (2.43) with $G^{-1}(x, x') = \delta^2 S_f/\delta\bar{\phi}(x)\,\delta\bar{\phi}(x')$] satisfies the equation (in n-dimensions)

$$\sqrt{-g(x)}\Big[-\Box_x + m^2 + (1-\xi)\,\xi_n R + \frac{\lambda}{2}\,\bar{\phi}^2\Big]G(x, x') = -\delta^n(x - x'). \tag{2.85}$$

For free, massless, conformally coupled fields ($\lambda = m = \xi = 0$) in an FLRW universe, the Green function G_0

$$G_0(x, x') = [a(\eta)]^{1-\frac{n}{2}}\,G_F(x, x')\,[a(\eta')]^{1-\frac{n}{2}} \tag{2.86}$$

is conformally related to the flat-space Feynman Green function:

$$G_F(x, x') = -\frac{1}{(2\pi)^n}\int d^n k\frac{e^{ik(x-x')}}{k^2 - i\epsilon}. \tag{2.87}$$

For small m^2, ξ and λ, one can solve this equation perturbatively. Defining

$$V(\bar{\phi}, g) = m^2 - \xi\xi_n R(x) + \frac{1}{2}\lambda\bar{\phi}^2(x) \equiv m_\xi^2(x) + \frac{1}{2}\lambda\bar{\phi}^2(x) \tag{2.88}$$

as the perturbation potential, G satisfies an integral equation

$$G(x, x') = G_0(x, x') + \int d^n y\; G_0(x, y)V(y)G(y, x'), \tag{2.89}$$

or, in operator form,

$$\begin{aligned} G = G_0 + G_0VG &= G_0 + G_0VG_0 + G_0VG_0VG_0 + \cdots \\ &= G_0(1 - VG_0)^{-1}. \end{aligned} \tag{2.90}$$

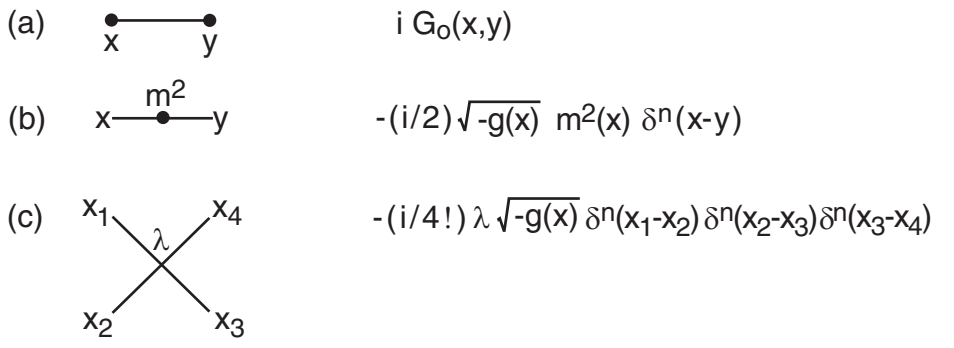


Figure 2.1 A diagrammatic representation of Feynman rules in configuration space. (a) A line denotes the propagator of massless free field. (b) A black dot denotes a two-point vertex $m^2(x) = m^2a^2(\eta) - \xi R/6$, where $\xi = 0$ denotes conformal coupling and R is the scalar 4-curvature. (c) denotes the four-point vertex of Φ^4 self-interaction with coupling constant λ.

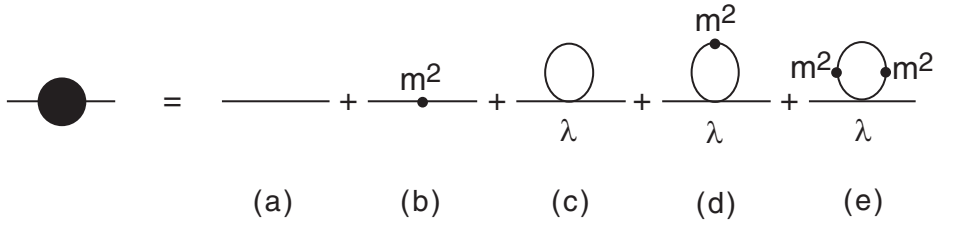


Figure 2.2 One-loop expansion of a two-point dressed Green function up to second order in λ and $m^2(x)$.

Substituting this into the 1-loop effective action yields

$$\Gamma[\bar{\phi}, g] - \frac{i}{2}\,\mathrm{Tr}\ln G_0 - \frac{i}{2}\sum_{n=1}^{\infty}\frac{1}{n}\int d^n x_1 \cdots \int d^n x_n\; V(x_1)G_0(x_1,x_2)\cdots V(x_n)G_0(x_n,x_1) + O(\hbar^2) \tag{2.91}$$

If we adopt the Feynman rules for the propagator and the vertex functions as depicted in Fig. 2.1, then up to 1-loop we need to sum over all diagrams in Fig. 2.2 and Fig. 2.3 for contributions to the first and second order of λ, respectively.

We observe by power-counting that

$$G_F(x_1 - x_2)G_F(x_2 - x_3)\cdots G_F(x_l - x_1) \sim \int d^n k\; k^{-2l} \tag{2.92}$$

which contains ultraviolet divergences only for $l = 1$ and 2 (in 1-loop graphs) at $n = 4$. We see that the lowest order bubble diagram Fig. 2.2(c) does not

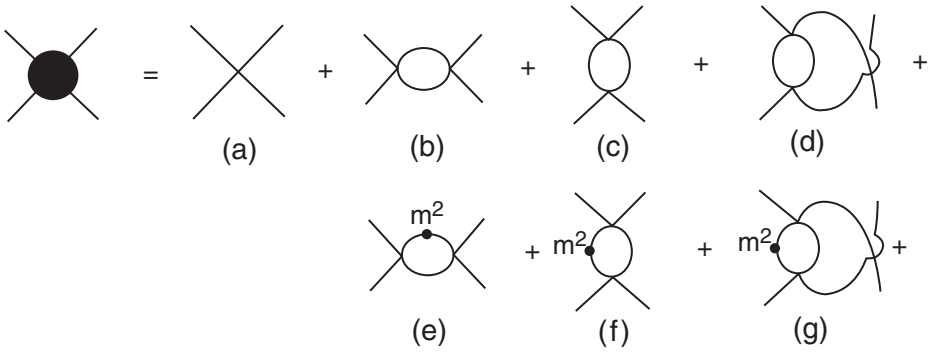


Figure 2.3 One-loop expansion of a dressed four-point vertex up to second order in λ and $m^2(x)$.

contribute because it is proportional to $G_F(0) \sim \int d^n k\ (k^2 - i\varepsilon)^{-1}$, which gives zero regularized value in dimensional regularization. This leaves only diagrams Fig. 2.2(d) to $\mathrm{O}(\lambda)$ and Fig. 2.3(b), (c) and (d) to $\mathrm{O}(\lambda^2)$, which must be regularized. They all contain products of propagators in the form $G_F(x-x')G_F(x'-x)$, which in momentum space representation is given by

$$(2\pi)^n \delta^n(p_1 + p_2) \int \frac{d^n q}{(2\pi)^n}\ \frac{1}{q^2 - i\varepsilon} \frac{1}{(q+p)^2 - i\varepsilon}. \tag{2.93}$$

This can be evaluated by rotating both p^0 and q^0 by $\pi/2$ in the complex plane and rotating p^0 back $-\pi/2$ after integration, yielding

$$\begin{aligned} G_F(x-x')G_F(x'-x) = {} & \frac{i}{8\pi^2(n-4)}\, \delta^n(x-x') \\ & - \frac{i}{16\pi^2}\Big[2 + \psi(1) + \ln 4\pi - i\pi\Big]\delta^n(x-x') \\ & + \frac{i}{16\pi^2} \int \frac{d^n p}{(2\pi)^n}\ e^{ip(x-x')} \ln \frac{p^2}{\mu_c^2} + \mathcal{O}(n-4), \end{aligned} \tag{2.94}$$

where $\psi(z)$ is the digamma function and μ_c is a constant with mass dimension. The first term is the singular part. From this we can easily identify the singular part of Fig. 2.2(d) to be

$$-i\sqrt{-g(x)}\ \frac{-1}{16\pi^2(n-4)} \Big(m^2 - \frac{1}{6}\xi R\Big)\delta^n(x_1 - x_2), \tag{2.95}$$

which requires a term in the counter Lagrangian

$$L_1^{CT} = L_{m,\xi}^{CT} = -\frac{1}{2}\Big(\delta m^2 - \frac{1}{6} R\delta\xi\Big)\Phi^2, \tag{2.96}$$

$$\delta m^2 = \frac{\lambda m^2}{16\pi^2(n-4)}\,, \qquad \delta\xi = \frac{\lambda\xi}{16\pi^2(n-4)}, \tag{2.97}$$

while the singular part from each of the three diagrams Fig. 2.2(b), (c), and (d) is

$$i\,\lambda\mu_c^{2(n-4)}\sqrt{-g(x)}\;\frac{\lambda}{16\pi^2(n-4)}\,\delta^n(x_1-x_2)\delta^n(x_2-x_3)\delta^n(x_3-x_4), \tag{2.98}$$

which requires a combined counterterm (hence a factor of 3)

$$L_2^{CT}=L_\lambda^{CT}=-\frac{\delta\lambda}{4!}\,\Phi^4,\qquad\text{where}\qquad \delta\lambda=\frac{3\lambda^2}{16\pi^2(n-4)}. \tag{2.99}$$

Using these results we can compute the 1-loop effective action up to the second order, namely, in λ^2, m^4 and ξ^2

$$\Gamma[\Phi,a]=S[\Phi,a]-\frac{i}{4}\int d^nx d^nx'\,\Big\{a^2(x)\Big[m_\xi^2(x)+\frac{\lambda}{2}\,\mu_c^{2(n-4)}\bar\phi^2(x)\Big]G_F(x-x')$$
$$\times\,G_F(x'-x)a^2(x')\Big[m_\xi^2(x')+\frac{\lambda}{2}\,\mu_c^{2(n-4)}\bar\phi^2(x')\Big]\Big\}. \tag{2.100}$$

We see that the term $a^4m_\xi^2(x)G_F(x-x')G_F(x'-x)m_\xi^2(x')$ contains a divergence

$$\frac{\sqrt{-g}}{32\pi^2(n-4)}\Big(m^2-\frac{1}{6}\xi R\Big)^2, \tag{2.101}$$

which requires a counter Lagrangian

$$L_3^{CT}=L^{CT}_{\bar\Lambda,\epsilon_0,\epsilon_1}=\delta\bar\Lambda+\delta\epsilon_0 R+\frac{1}{2}\delta\epsilon_1 R^2, \tag{2.102}$$

where $\epsilon_0=(16\pi G_{\rm N})^{-1}=\kappa^{-2}$, with $G_{\rm N}$ being Newton's constant and $\bar\Lambda=2\epsilon_0\Lambda$, with Λ being the cosmological constant. (Compared to the coefficients a,b in (2.67), we note that $\frac{1}{2}\epsilon_1=\epsilon_0 a$, also, later in (2.170), $\frac{1}{2}\epsilon_2=\epsilon_0 b$.) We have

$$\delta\bar\Lambda=\frac{m^4}{32\pi^2(n-4)}\,,\quad \delta\epsilon_0=-\frac{m^2}{16\pi^2(n-4)}\Big(\frac{\xi}{6}\Big),\quad \delta\epsilon_1=\frac{\xi^2}{36\times16\pi^2(n-4)}. \tag{2.103}$$

This comes from the free field two-point vacuum bubble diagram and the counterterms correspond to the renormalization of the cosmological constant Λ, Newton's constant (contained in ϵ_0) and the coefficient of the quadratic curvature ϵ_1, respectively. By adding the three counteraction terms from (2.96), (2.99) and (2.100), $S_i^{CT}=\int d^4x\;a^4L_i^{CT}$, and replacing the bare coupling constants and geometric parameters with the renormalized ones, we get, upon taking the limit $n\to4$, the effective action for the scalar field up to one-loop order

$$\Gamma_{ren}[\Phi,\alpha]=\Gamma[\Phi,\alpha]+S_1^{CT}+S_2^{CT}+S_3^{CT}=S_{f0}[\Phi,a]+\Gamma^{(1)}_{ren}[\Phi,a], \tag{2.104}$$

where

$$\Gamma^{(1)}_{ren} = \int d^4x\, a^4(\eta) M^2(\bar\phi)\, \frac{1}{32\pi^2}\Big(\ln a + \frac{i\pi}{2}\Big) - \int d^4x \int d^4x'\; a^2(\eta) a^2(\eta') M^2(\bar\phi) K(x,x') M^2(\bar\phi). \tag{2.105}$$

Here,

$$M^2 \equiv m^2_\xi + \frac{1}{2}\,\lambda\bar\phi^2 = m^2 - \frac{1}{6}\,\xi R + \frac{1}{2}\,\lambda\bar\phi^2, \tag{2.106}$$

and

$$K(x,x') = \frac{1}{64\pi^2}\int \frac{d^4p}{(2\pi)^4}\; e^{ip(x-x')} \ln\frac{p^2}{\mu^2}. \tag{2.107}$$

(We have redefined the renormalization constant μ to absorb the constant $2 + \psi(1) + \ln 4\pi$). For further discussions on the renormalization of the $\lambda\Phi^4$ theory in curved spacetime see, e.g., [366, 367, 368].

This regularized effective action is the starting point for deriving the equation of motion and calculating the particle production and backreaction effects. Two remarks are in order:

1. The divergences we identified and the counter-terms we introduced from the Feynman diagrams are exactly the same as those obtained from more general considerations for arbitrary curved-spacetimes, such as from the work of Bunch and Parker [364], Toms [368], Hu and O'Connor [375] and others. As shown in [111], one can also deduce the Gauss–Bonnet and quadratic curvature terms from the higher-order Feynman diagrams. This is the source of the trace anomaly.
2. One can almost guess the form of the effective Lagrangian from the results of the free field case whose role is now played by the background field. In place of the $-\xi\xi_n R$ factor in the free field case we have the effective mass $M^2 = m^2 - \frac{1}{6}\xi R + \frac{\lambda}{2}\bar\phi^2$. (The $\frac{1}{2}\lambda\bar\phi^2$ term comes from the 1-loop approximation of the $\frac{\lambda}{4!}\Phi^4$ potential in the saddle-point expansion.) Except for the additional counterterm for λ which needs be deduced from the vertex diagrams, the rest of the procedure is similar to that for the free field case. The derivation of the non-local operator $K(x-x')$ in the finite contribution is exactly identical.

We can now add up all the terms to get the total effective action up to one-loop

$$S = S_{rad} + S_{g0} + S_{f0} + \Gamma^{(1)}_{ren} + \Gamma^{(1)}_{TA}, \tag{2.108}$$

where the five terms are given by (2.78), (2.37), (2.38), (2.105) and (2.66) respectively. Written explicitly in terms of the FLRW spacetime and using the conformally related field quantities $\chi = a\bar\phi$, they become

$$S_{rad} = V\int d\eta\,(-\tilde{\rho}_r), \quad S_{g0} = -\frac{3}{8\pi G}V\int d\eta\,(a')^2, \tag{2.109}$$

$$S_{f0} = \int d^4x\,\Big\{-\frac{1}{2}\eta^{\mu\nu}\partial_\mu\chi(x)\partial_\nu\chi(x) - \frac{\xi}{2}\Big(\frac{a''}{a}\Big)\chi^2 - \frac{1}{2}\,m^2a^2\chi^2 - \frac{\lambda}{4!}\chi^4\Big\}, \tag{2.110}$$

$$\Gamma^{(1)}_{ren} = \int d^4x\,\mathcal{M}^2(x)\frac{1}{32\pi^2}\Big(\ln a + \frac{i\pi}{2}\Big) - \int d^4xd^4x'\,\mathcal{M}^2(x)K(x-x')\mathcal{M}^2(x'), \tag{2.111}$$

where

$$\mathcal{M}^2 \equiv M^2a^2 = m^2a^2 - \xi\,\frac{a''}{a} + \frac{\lambda}{2}\,\chi^2. \tag{2.112}$$

We shall consider only homogeneous fields $\bar{\phi}(t)$ or $(\chi(\eta))$ and for computational simplicity take the local truncation for $K(x-x')$, i.e. $K(x-x') \simeq \delta(x-x')$. (The non-local effect for free fields has been studied by Hartle [376].) Under these assumptions, the total action is

$$\begin{aligned} S[a,\chi] = V\int d\eta\,\Big\{&-\tilde{\rho}_r - \frac{1}{2}\tilde{a}'^2 + \frac{1}{2}\chi'^2 - \frac{1}{2}\,\tilde{m}^2\tilde{a}^2\chi^2 + \frac{\xi}{2}\,\Big(\frac{\tilde{a}''}{\tilde{a}}\Big)\chi^2 - \frac{\lambda}{4!}\,\chi^4 \\ &+ \Big[\tilde{m}^2\tilde{a}^2 - \xi\Big(\frac{\tilde{a}''}{\tilde{a}}\Big) + \frac{\lambda}{2}\chi^2\Big]^2\,\frac{1}{32\pi^2}\Big(\ln\bar{\ell}_p\tilde{a} + \frac{i\pi}{2}\Big)\Big\}. \end{aligned} \tag{2.113}$$

The rescaled (tilde) parameters are measured in ratio to the Planck length $\ell_p \equiv \sqrt{G_{\rm N}}$ in geometric units.

$$\tilde{a} = \bar{\ell}_p^{-1}\,a, \quad \tilde{m} = \bar{\ell}_p\,m, \quad \tilde{\mu} = \bar{\ell}_p\,\mu. \tag{2.114}$$

where we have defined $\bar{\ell}_p \equiv \sqrt{\frac{4\pi}{3}}\ell_p$ for convenience. Particle production can be calculated from the effective action and its backreaction obtained by solving the equations for the effective geometry. We divide into cases distinguished in accordance to whether the perturbative parameters m (mass), ξ (field coupling), χ (background field) and λ (self-interaction) are zero or not. From the form of this Lagrangian we see that the effect of interaction (λ) always manifests through the background field (χ). Hence the case of zero background field leads to results equivalent to free fields ($\chi = 0 \Rightarrow \lambda = 0$) but the converse is not true, because the background field is also coupled to m, ξ, etc. For the treatment of massive fields the present perturbative method fails at late times because the relevant parameter ma becomes large at large a. One knows however that the classical behavior of the universe containing massive fields at late times is given by that of a matter-dominated solution. We will derive the equations of motion for massless conformal fields only: this case singles out the effects of non-zero background field χ and its interaction λ on particle creation and backreaction.

For non-zero conformal background fields, the mass term which breaks the conformal invariance provides a coupling between the geometric scale factor a and the background scalar field χ [see Eq. (2.113)]. In the massless case a

and χ will each evolve independently as a, $\chi \sim \eta$. For self-interacting, nonzero background field (λ, $\chi \neq 0$) the results are non-trivial. Note that the generalization of the Parker–Grishchuk lemma [344, 345] (namely, no particle production of conformally invariant fields in conformally flat spacetimes) to $\lambda\phi^4$ theory refers only to the unsplit field Φ satisfying (2.84). The lemma does not apply to nonzero background fields order by order, under a background field decomposition. From the action

$$S[\tilde{a},\phi] = V\int d\eta\,\left\{-\frac{1}{2}\tilde{a}'^2 - \tilde{\rho}_r + \frac{1}{2}\chi'^2 - \frac{\lambda}{4!}\chi^4 + \frac{1}{32\pi^2}\frac{\lambda^2}{4}\chi^4\Big(\ln\bar{\ell}_p\tilde{a} + \frac{i\pi}{2}\Big)\right\}, \tag{2.115}$$

we get the dynamic equations

$$\tilde{a}'' + \frac{1}{32\pi^2}\frac{\lambda^2}{4}\,\chi^4\tilde{a}^{-1} = 0, \tag{2.116}$$

$$\chi'' + \frac{\lambda}{3!}\,\chi^3 - \frac{\lambda^2}{32\pi^2}\chi^3\Big(\ln\bar{\ell}_p\tilde{a} + \frac{i\pi}{2}\Big) = 0. \tag{2.117}$$

This has a first integral given by

$$-\frac{1}{2}\tilde{a}'^2 + \rho_r + \frac{1}{2}\,\chi'^2 + \frac{\lambda}{4!}\,\chi^4 - \frac{1}{32\pi^2}\frac{\lambda^2}{4}\,\chi^4\Big(\ln\bar{\ell}_p\tilde{a} + \frac{i\pi}{2}\Big) = E. \tag{2.118}$$

At early times $\eta \to 0$ the only asymptotic solution for $\tilde{a} \to 0$ is

$$\begin{aligned}\tilde{a} &\sim A\eta + A'\eta^5 + \cdots\\ \tilde{\chi} &\sim B\eta + B'\eta^5\ln(\bar{\ell}_p A\eta) + \cdot\cdot,\cdot\end{aligned} \tag{2.119}$$

where we have assumed that the contribution from the logarithmic terms dominates, and A, B are constants satisfying $A^2 - B^2 = 2\tilde{\rho}_r$ and

$$A' = -\frac{\lambda^2}{2560\pi^2}\frac{B^4}{A}, \qquad\qquad B' = \frac{\lambda}{640}B^3. \tag{2.120}$$

At late times $\eta \to \infty$, several characteristic solutions are allowed. (See [374].) Out of these possibilities, at least one case is a physically acceptable solution:

$$\tilde{a} \sim C\,\eta + C'\eta^{-3}(\ln\bar{\ell}_p A\eta)^{-4} + \cdots, \qquad \chi \sim D'(\eta\ln\bar{\ell}_p A\eta)^{-1} + \cdots \tag{2.121}$$

where

$$C = \sqrt{2\tilde{\rho}_r}, \qquad\qquad C' = -\frac{8\pi^2}{3\lambda^2 C}, \qquad\qquad D' = 8\pi/\lambda.$$

Here $\tilde{a} \to \infty$ while $\chi \to 0$ at $\eta \to \infty$. Note that the $(\ln\bar{\ell}_p A\eta)^{-1}$ factor always enter for massless fields. The magnitude of the correction term is proportional to C', which contains the scale $(\lambda^2\tilde{\rho}_r)^{-1/4}$. Here in contradistinction to the massive zero-field case, it is the coupling constant λ which enters in determining the scale where deviation from the classical solutions appears.

The particle production probability density is given by $P = 2\,\mathrm{Im}\,\Gamma$. From it we obtain for this case the rate of particle production as

$$\begin{aligned} P &= (256\pi^2)^{-1}\lambda^2 B^4 \eta^4 \quad \text{at early time,} \\ &= (256\pi^2)\lambda^2 D^4 \frac{1}{\eta^4 (\ln \bar{\ell}_p A\eta)^4} \quad \text{at late times,} \end{aligned} \tag{2.122}$$

where $D = 8\pi/[\lambda(\ln \bar{\ell}_p^2 A + i\pi/2)^{1/2}]$. This sample calculation shows how to obtain the particle creation probability from the in-out effective action. It also provides the equations of motion describing the backreaction effects of particle creation and interaction in influencing the background geometry, albeit complex. We next turn to the case when the background spacetime and field vary slowly and describe a way to obtain the effective action in closed form.

2.4 Quasilocal Effective Action for Slowly Varying Background

Effective actions are not easily derivable in closed forms, except perhaps for Gaussian systems in spacetimes with suitable symmetries. For spacetimes with high enough symmetries one can use the well-developed mathematical tools of harmonic analysis in homogeneous or symmetric spaces [377] to obtain globally well-defined harmonic functions (see, e.g., the review of Camporesi [378]) and from which to construct the effective actions. These properties are put to good use in the zeta function regularization techniques described in Chapter 4. For general curved spacetimes one often appeals to perturbative methods, such as a Riemann normal coordinate expansion for the metric, which enables a momentum space representation of the Green function on a local coordinate patch, as done in [364].[7]

In this section we show how to construct the effective Lagrangian for a $\lambda\Phi^4$ theory in curved spacetime which takes into account the lowest-order radiative correction (one-loop approximation, first order in $\hbar$) up to the second order in the variation of the background field $L(\phi, \dot{\phi}, \ddot{\phi})$, but *exact to all orders in the coupling constant* λ. The result is generally valid for slowly varying background spacetimes and background fields. The exact range of validity in realistic conditions will depend on the functional form of the background metric and the observer's resolution capability which tracks the variational orders of the background field. Our treatment here follows [375] where the Schwinger–DeWitt proper-time formalism

[7] More general than the method used in [364] on a local Minkowski patch, group theoretical methods and quasilocal expansion have earlier been devised in homogeneous spaces to rendering partial differential equations to ordinary differential equations [379], and applied to the study of tensor perturbations in homogeneous cosmologies [380]. For the connection of this group theoretical method with the standard tensor harmonics method used in Lifshitz's derivation of the wave equation [338] governing the weak gravitational perturbations of the FLRW universe, see [381]. The group-theoretical method of [379] was also used to analyze the stability of instanton solutions in quantum gravity and Kaluza–Klein theories [382, 383].

[124, 384, 46, 96] is used for the derivation of the effective Lagrangian. The field is expanded in a Taylor series around an arbitrary spacetime point and the Green function is sought from a solution of the inhomogeneous wave equation. In this expansion if only field variations up to the second order are kept, the effective one-loop Lagrangian can be cast in an exact analytic form. This is made possible by a reduction-to-quadrature procedure for the momentum variables in the functional integral. First used by Brown and Duff [385] and Iliopoulos, Itzykson, and Martin [386] in flat-space field theory, this procedure was extended to curved space by Hu and O'Connor [375] using the momentum-space representation for the Green function of Bunch and Parker [364].

Using a small proper time expansion[8] of the effective action one can identify and remove the ultraviolet divergences by the standard method of dimensional regularization. From this one can derive a generalized expression for the 'HaMiDeW' (Hadamard–Minakshisundaram–Pleijel–DeWitt) coefficient [389, 390, 384] a_2 of the Laplace–Beltrami operator with a spacetime-dependent background scalar field. (We will see more of this in Chapters 4 and 5.) From the counter-terms a set of renormalization group (RG) equations for the coupling constants and various parameters of the theory can be obtained [375].

2.4.1 Background and Fluctuation Field

Consider a massive (m) self-interacting (λ) single component scalar field Φ coupled (ξ) to a general n-dimensional curved spacetime with metric $g_{\mu\nu}$ of Lorentzian signature $(-,+,+,+,\cdots)$ and scalar curvature R. It is described by the Lagrangian density

$$L_\phi^{(0)}[\Phi, g_{\mu\nu}] = -\frac{1}{2}\,\Phi\Big[-\Box + (1-\xi)\xi_n R + m^2\Big]\Phi - \frac{\lambda}{4!}\,\Phi^4, \tag{2.123}$$

where

$$\Box \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu = \frac{1}{\sqrt{-g}}\frac{\partial}{\partial x^\mu}\left(\sqrt{-g}\,g^{\mu\nu}\,\frac{\partial}{\partial x^\nu}\right) \tag{2.124}$$

is the Laplace–Beltrami operator in a spacetime with metric $g_{\mu\nu}$. It is a generalization of the d'Alambertian wave operator in Minkowski space to curved spacetime. Note here we use $\xi = 0, 1$ to denote conformal and minimal coupling, respectively and $\xi_n = \frac{1}{4}\frac{(n-2)}{(n-1)}$ is the conformal coefficient in n dimensions.

[8] Even though the small proper time expansion is often used to identify ultraviolet divergences, more general usage is facilitated by a form suggested by Jack, Parker and Toms [387, 388], namely, the expansion of the kernel of the Feynman propagator in curved spacetime can be written in a form in which all the terms containing the scalar curvature R are generated by a simple overall exponential factor which sums all terms containing R, including those with nonconstant coefficients, in the proper-time series. This result is valid for an arbitrary spacetime and for any spin. It also applies to the heat kernel. This form of the expansion is useful for exploring nonperturbative effects in quantum field theory.

The action has a minimum at $\Phi = \bar{\phi}$, which satisfies the classical equation of motion

$$A_2\,\bar{\phi}(x) = \left(-\Box + M_2^2\right)\bar{\phi}(x) = 0, \tag{2.125}$$

where

$$M_2^2 = m^2 + (1-\xi)\xi_n R + \frac{\lambda}{6}\,\bar{\phi}^2. \tag{2.126}$$

Fluctuations $\varphi = \Phi - \bar{\phi}$ around the classical background field $\bar{\phi}$ satisfy an equation (to lowest order)

$$A_1\,\varphi(x) = \left(-\Box + M_1^2\right)\varphi(x) = 0, \tag{2.127}$$

where

$$M_1^2 = m^2 + (1-\xi)\xi_n R + \frac{\lambda}{2}\,\bar{\phi}^2. \tag{2.128}$$

At times these quantities may come in handy:

$$M_\xi^2 \equiv m^2 + (1-\xi)\xi_n R\,, \quad M_\lambda^2 \equiv m^2 + \frac{\lambda}{2}\phi^2, \tag{2.129}$$

but note the difference from $m_\xi^2 \equiv m^2 - \xi\xi_n R$ defined in (2.88) earlier. The M_1, M_2, M_ξ, M_λ are effective masses which depend on the coupling ξ, background curvature R, and the background field $\bar{\phi}$. Contributions of the fluctuation field to the equation of motion for ϕ enter through the vacuum expectation value and the thermal average of its variance $\frac{1}{2}\lambda\,\langle\phi^2\rangle$, which acts as additional terms in the effective mass M_2.

The effective action Γ related to $L_{\rm eff}$ by

$$\Gamma[\phi, g_{\mu\nu}] = \int d^n x\,\sqrt{-g}\,L_{\rm eff} \tag{2.130}$$

is expanded perturbatively in powers of $\hbar$ as

$$\Gamma[\phi] = S[\phi] + \Gamma^{(1)} + \Gamma', \tag{2.131}$$

where $S[\phi]$ is the classical action,

$$S = \int d^d x\,\sqrt{-g}\,L_\phi^{(0)}, \qquad L_\phi^{(0)} = L[\phi, g_{\mu\nu}] \tag{2.132}$$

and $\Gamma^{(1)}$ and Γ' are the one loop and higher loop effective actions:

$$\Gamma^{(1)} = \int d^n x\,\sqrt{-g}\,L^{(1)} = -\frac{i\hbar}{2}\ln \mathrm{D}etG. \tag{2.133}$$

Here G is the bare Feynman Green's function satisfying

$$A_1 G(x,x') = -(-g)^{-1/2}\delta(x,x') \tag{2.134}$$

or

$$A_1 \bar{G}(x,x') = -\delta(x,x'), \tag{2.135}$$

where

$$\bar{G}(x,x') = (-g)^{-1/4}(x)\, G(x,x')\, (-g)^{-1/4}(x'). \tag{2.136}$$

(See, e.g., [46].) In a static homogeneous spacetime ϕ is a constant field, in which case one can define an effective potential V as

$$V(\phi) = -(Vol)^{-1}\Gamma(\phi), \tag{2.137}$$

where Vol denotes the spacetime volume. In general ϕ has temporal and spatial dependence, which renders $V(\phi)$ ill-defined.

2.4.2 Slowly Varying Background Field in Spacetimes with Small Curvature

Under circumstances where the background spacetime and the background field change slowly one can carry out a quasilocal expansion of the field and the metric around any spacetime point x^μ, including their derivatives up to a certain order. E.g., to second derivative order the background field assumes the form,

$$\phi^2(x') = \phi^2(x) + \phi^2_{,\mu}(x)(x'-x)^\mu + \frac{1}{2}\,\phi^2_{,\mu\nu}(x'-x)^\mu(x'-x)^\nu + \cdots, \tag{2.138}$$

and the effective Lagrangian we seek will then be a functional of ϕ and its derivatives up to this order.

The effect of a slowly varying background field on quantum processes such as symmetry breaking can be derived from such an effective Lagrangian. For spatially homogeneous backgrounds as in homogeneous cosmology, where the field has trivial spatial dependence and varies only in time, this Lagrangian is related but not equivalent to the adiabatic expansion method.[9] The range of validity of these expansions in realistic calculations will of course depend on the choice of time and the functional dependence of the background metric and the dynamics of the background field.

[9] The Taylor expansion used here and the adiabatic expansion are not always equivalent in that the definition of an adiabatic vacuum (or adiabatic n-particle state in finite temperature field theory [391, 392]) is devised such that there will be no particle present to within a definite adiabatic order. Although one can loosely say that terms with nth derivatives are of nth adiabatic order, the reverse is not always true. Terms of a definite adiabatic order, say in the energy density, usually involve combinations of terms of the same derivative order and products of terms of lower derivative orders. The definition of particle states associated with adiabatic expansion has a precise physical meaning but not so for the simple Taylor expansion beyond incorporating the nonlocal characteristics of the background field.

In a local Riemann normal coordinate expansion around a spacetime point x, the metric can be written as

$$g_{\mu\nu} = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\alpha\nu\beta}y^\alpha y^\beta - \frac{1}{6}R_{\mu\alpha\nu\beta;\gamma}y^\alpha y^\beta y^\gamma - \left(\frac{1}{20}R_{\mu\alpha\nu\beta;\gamma\delta} - \frac{2}{45}R_{\alpha\mu\beta\lambda}R^\lambda{}_{\gamma\nu\delta}\right)y^\alpha y^\beta y^\gamma y^\delta + \cdots \tag{2.139}$$

where $\eta_{\mu\nu}$ is the Minkowski metric with signature $(-,+,+,+,\ldots), y = x' - x$, and the coefficients in this expansion are evaluated at $y = 0$. Using this expansion for the wave operator A_1 in (2.127) we get an equation for the Green function $\bar{G}(x,x')$ in Eq. (2.136) up to quadratic order in y

$$\left(\eta^{\mu\nu}\partial_\mu\partial_\nu - \alpha^2 - \beta_\mu y^\mu - \frac{1}{4}\gamma^2_{\mu\nu}y^\mu y^\nu\right)\bar{G}(y) = \delta^d(y), \tag{2.140}$$

where

$$\begin{aligned}
\alpha^2 &= m^2 - \xi\,\xi_n R + \frac{1}{2}\lambda\phi^2 \equiv m_\xi^2 + \frac{1}{2}\lambda\phi^2 \equiv m^2 + U,\\
\beta_\mu &= U_{;\mu},\\
\frac{1}{4}\gamma^2_{\mu\nu} &= \frac{1}{2}U_{;\mu\nu} + a_{\mu\nu},
\end{aligned} \tag{2.141}$$

with

$$a_{\mu\nu} = \frac{1}{120}R_{;\mu\nu} - \frac{1}{40}\Box R_{\mu\nu} + \frac{1}{30}R_\mu{}^\lambda R_{\lambda\nu} - \frac{1}{60}R^\kappa{}_\mu{}^\lambda R_{\kappa\lambda} - \frac{1}{60}R^{\lambda\rho\kappa}{}_\mu R_{\lambda\rho\kappa\nu}. \tag{2.142}$$

Equation (2.140) for the Green function $\bar{G}$ will reduce to that in Minkowski space if U were a constant. This is the case for constant background fields ϕ and for either conformal coupling ($\xi = 0$) or for spacetimes of constant four curvature (e.g. de Sitter universe).

For simplicity we work with the Euclidean representation obtained from a Wick rotation to imaginary time $t \to -i\,\tau$. The Euclidean version has meaning, e.g, representing a finite-temperature quantum field theory, only when the effective mass is a constant. The Minkowskian version is the physically meaningful one. For a treatment of generalizing the usual effective potential to include the kinetic terms in the effective action, see [393], where results for the leading-order gradient terms in the one-loop effective action for scalar fields valid for both zero and finite temperatures are given when quantum corrections arise from scalar, spinor, or gauge fields.

The Euclidean Green function G_E defined by

$$\bar{G}_E(\tau,\mathbf{x},\tau',\mathbf{x}') = -i\,\bar{G}(-i\tau,\mathbf{x};\,-i\tau',\mathbf{x}') \tag{2.143}$$

satisfies the equation (note here and henceforth, y^μ and p^μ are the Euclideanized form of the original y^μ and p^μ, e.g., $\tau = i\,t$.)

$$\Big(\partial_E^2 + \partial_x^2 - \alpha^2 - \hat{\beta}_\mu y^\mu - \frac{1}{4}\,\hat{\gamma}^2_{\mu\nu} y^\mu y^\nu\Big)\bar{G}_E = -\delta(\tau, \mathbf{x}, \tau', \mathbf{x}'), \tag{2.144}$$

where (with $i = 1, 2, \ldots, n-1$)

$$\hat{\beta}_0 = -i\,\beta_0, \quad \hat{\beta}_i = \beta_i, \quad \hat{\gamma}^2_{00} = -\gamma^2_{00}, \quad \hat{\gamma}^2_{0i} = -i\gamma^2_{0i}, \quad \hat{\gamma}^2_{ij} = \gamma^2_{ij}. \tag{2.145}$$

In the coordinate patch defined in (2.139) one can introduce a momentum space centered at point x and define the Fourier transform $G_E(p)$ by

$$\bar{G}_E(y) = \int \frac{d^n x}{(2\pi)^n}\, e^{ipy}\, \bar{G}_E(p), \tag{2.146}$$

where $py = p_\mu y^\mu = \eta^{\mu\nu} p_\mu y_\nu$.

This momentum representation is well-defined only in a local neighborhood of $y = 0$ and does not carry the global properties of $G(x, x)$. However, for the study of ultraviolet divergences in two-point functions (like the Feynman propagator) or bivector quantities (like the energy-momentum tensor) in the coincidence limit, or for the study of systems involving low-order quasilocal variations of the background field, momentum-space representation is still applicable. Keeping the so-called 'kinetic terms', the quasi-local effective action which includes gradual changes in the background field is an improvement over the effective potential which is defined only for a constant background field. This is useful for the description of 'slow-roll' inflationary cosmology. We will pick up this topic again in Chapter 7. Note, however, quasi-local effective actions are not suitable for the consideration of processes involving rapid changes of the background metric, as in cosmological particle production, or processes depending on the global properties of spacetime, e.g., topological effects on phase transitions.

The momentum space Green function $G_E(p)$ satisfies

$$\Big(p^2 + \alpha^2 + i\,\hat{\beta}_\mu \partial_p^\mu - \frac{1}{4}\,\hat{\gamma}^2_{\mu\nu}\partial_p^\mu \partial_p^\nu\Big)\bar{G}_E(p) = 1, \tag{2.147}$$

where $\partial_p^\mu = \frac{\partial}{\partial p_\mu}$. From here on we will drop the subscript E and the overbar on G_E, with the understanding that we are dealing with Euclidean Green functions. For constant effective mass $\alpha = \text{constant}, \hat{\beta} = \hat{\gamma} = 0$ the Euclidean Green function has the flat space form

$$G(p) = (p^2 + \alpha^2)^{-1} = \int_0^\infty ds\; e^{-\alpha^2 s} e^{-p^2 s}, \tag{2.148}$$

where we have expressed it in a proper time (s) integral representation. The configuration space Green function $G(x, x')$ in the coincidence limit ($y \to 0$) becomes

$$G(x, x) = \int \frac{d^n p}{(2\pi)^n} \int_0^\infty ds\; e^{-\alpha^2 s} e^{-p^2 s}. \tag{2.149}$$

In the general case of variable effective mass, when $\phi(x)$ and $R(x)$ have space and time dependence, we want to find solutions of the momentum space Green function equation with $\hat{\beta}_\mu, \hat{\gamma}_{\mu\nu} \neq 0$. If we only include up to second derivatives of $\phi(x)$, we can write $G(p)$ in the same form as in Eq. (2.148) but with p^2 replaced by a general quadratic polynomial in p^μ

$$G(p) = \int_0^\infty ds\, e^{-\alpha^2 s} \exp\Big[-p^\mu A_{\mu\nu}(s) p^\nu + i\, B_\mu(s) p^\mu + C(s)\Big], \tag{2.150}$$

where A, B, and C are functions of $\hat{\beta}$ and $\hat{\gamma}$. The configuration space Euclidean Green function now has the general form

$$G(x,x) = \frac{1}{(2\pi)^n} \int_0^\infty ds\, e^{-\alpha^2 s + C} \int d^n p\, e^{ip\cdot y} e^{-(p\cdot A\cdot p - iB\cdot p)}. \tag{2.151}$$

Under a change of variables $p \to p' = p - \frac{i}{2} A^{-1}\cdot B$, the integral becomes Gaussian, which can easily be evaluated. Taking the coincidence limit $y \to 0$ of $G(x,x')$ gives

$$G(x,x) = \int_0^\infty \frac{ds}{(4\pi s)^{n/2}}\ \exp\Big[-\alpha^2 + C - \frac{1}{4}\, B\cdot A^{-1}\cdot B - \frac{1}{2}\ \mathrm{Tr}\ln(As^{-1})\Big], \tag{2.152}$$

where

$$\begin{aligned} A &= \hat{\gamma}^{-1}\cdot \tanh\hat{\gamma} s, \\ B &= 2\,\hat{\gamma}^{-2}\cdot(1 - \mathrm{sech}\hat{\gamma} s)\cdot\hat{\beta}, \\ C &= -\frac{1}{2}\ \mathrm{Tr}\ln(\cosh\hat{\gamma} s) - \hat{\beta}\cdot\hat{\gamma}^{-3}\cdot(\tanh\hat{\gamma} s - \hat{\gamma} s)\cdot\hat{\beta}. \end{aligned} \tag{2.153}$$

The one-loop contribution $L^{(1)}$ to the effective Lagrangian L_{eff} [10] which is related to the Green's function by

$$\frac{\partial L^{(1)}}{\partial \alpha^2} = -\frac{\hbar}{2}\, G(x,x) \tag{2.154}$$

can be obtained by integrating $G(x,x)$ with respect to α^2. (The constant of integration is accounted for by the renormalization of the parameters in the gravitational action. See (2.172) below.) We finally obtain the effective Lagrangian

$$\begin{aligned} L^{(1)} &= \frac{\hbar}{2(4\pi)^{n/2}} \int_0^\infty \frac{ds}{s^{1+n/2}}\ e^{-\alpha^2(\phi)s - f(s)} \\ &\equiv \frac{\hbar}{2(4\pi)^{n/2}} \int_0^\infty \frac{ds}{s^{1+n/2}}\ e^{-m^2 s} \bar{K}(x,x,s), \end{aligned} \tag{2.155}$$

[10] The Euclidean L_{eff} differs in sign from that of the Minkowskian L_{eff} defined in (2.130). To recover expressions in Lorentzian form, use the inverse transformations in (2.145) for converting $\hat{\beta}$ and $\hat{\gamma}$ back to β and γ by $\tau \to it$ and $\sqrt{-g_E}\, L_E \to -\sqrt{-g}\, L$.

where $\bar{K}(x,x,s)$ is a heat kernel, the object of attention in Chapter 4, and

$$f(s) = -C + \frac{1}{4}B\cdot A^{-1}\cdot B + \frac{1}{2}\,\mathrm{Tr}\ln(As^{-1})$$
$$= \frac{1}{2}\,\mathrm{Tr}\ln\Big[(\hat{\gamma}s)^{-1}\cdot\sinh(\hat{\gamma}s)\Big] + \hat{\beta}\cdot\hat{\gamma}^{-3}\cdot\left[2\tanh\left(\frac{1}{2}\hat{\gamma}s\right) - \hat{\gamma}s\right]\cdot\hat{\beta}, \quad (2.156)$$

with α, $\hat{\beta}$, $\hat{\gamma}$ given by (2.141) and (2.145).[11] This is, in the characterization of [385], an "exact" result for the effective Lagragian in curved space, modulo the obvious limitations of quasilocal approximations described above.

The expansion of the integrand of $L^{(1)}$ in (2.155) for small s can be performed in different ways. Factoring out the constant m^2 term as is done here and grouping the field variables with the curvature terms is closest in spirit and form to flat-space field theory. The curvature of spacetime and the fields enter only through the a_l coefficients. However, when dealing with problems involving interacting fields in dynamic spacetimes, the mass acquires additional contributions from ϕ, and R, and it is the modified mass $\alpha^2 = m^2+\frac{1}{2}\lambda\phi^2-\xi R/6$ or its related quantities M_1^2 or M_2^2 which are the more relevant physical parameters. For example, in symmetry-breaking problems, at the classical level, the phase transition occurs near $M_2^2 = m^2 + \frac{1}{6}\lambda\phi^2 + (1-\xi)R/6 = 0$, while at the quantum level it occurs near $M_2^2 + \frac{1}{2}\lambda\langle\varphi^2\rangle_0 = 0$, where the effective mass acquires a radiative correction. In dynamic spacetimes, further contributions can arise from the kinetic terms of the wave operator. One should seek approximations according to the correct range of the effective mass instead of the intrinsic mass. This difference shows up more distinctly in problems like symmetry breaking when one is interested in the long-range or infrared (IR) behavior of the theory. On the other hand, the identification of ultraviolet (UV) divergences, corresponding to the large mass, small proper time or small curvature domain, only picks up the local structure of spacetime. In the same vein, the UV behavior should also be identical to that obtained from exact models, as long as the spacetime is locally Minkowskian.

2.5 Regularization of Quasilocal Lagrangian for $\lambda\Phi^4$ Field

The expression for $L^{(1)}$ derived above is formal and divergent. In the Schwinger proper time representation small s pertains to the ultraviolet range [46, 96]. We can thus perform a small proper time series expansion of the heat kernel $\bar{K}(x,x,s) \equiv e^{-Us-f(s)}$ in its integrand (2.155) and identify the UV divergent terms.

[11] Note that except for the Euclideanized variables, its form is identical to the flat-space result of Ref. [385], as it should, since it is based on a local (momentum-space) technique. The quasilocal expansion (2.138) of the fields and the Riemann normal coordinate expansion (2.139) of the metric result in expressions of the same form as the Taylor expansion of fields in flat space. Spacetime curvature and background-field variations appear through the expansion coefficients $\alpha, \hat{\beta}, \hat{\gamma}$ in the Taylor series.

HaMiDeW Coefficients and Trace Anomaly

Expanding separately $e^{-f(s)}$ and e^{-Us} in series sums, we have

$$e^{-f(s)} = 1 - \frac{\hat{\gamma}^2}{12}\, s^2 + \frac{\hat{\beta}^2}{12}\, s^3 + \frac{\hat{\gamma}^4}{160}\, s^4 + \cdots \equiv \sum_{l=0}^{\infty} b_l s^l, \tag{2.157}$$

where $\hat{\gamma}^2 \equiv \operatorname{Tr} \hat{\gamma}_{\mu\nu}^2$ whence

$$\begin{aligned} \bar{K}(x,x,s) &\equiv e^{-Us-f(s)} \equiv \sum_{l=0}^{\infty} a_l\, s^l \\ &= \Big(1 - Us + \frac{U^2 s^2}{2} - \frac{U^3 s^3}{3!} - \frac{\hat{\gamma}^2 s^2}{12} + \frac{\hat{\beta}^2 s^3}{12} + \cdots\Big). \end{aligned} \tag{2.158}$$

The coefficients a_l are the HaMiDeW coefficients, here generalized to include changing background curvature and fields contained in $U(\phi) = \frac{1}{2}\lambda\phi^2 - \frac{1}{6}\xi R$. They correspond to the operator

$$H = -(-g)^{-1/4}\Box_E(-g)^{-1/4} + m^2 + \frac{\lambda}{2}\phi^2(x) + \frac{1}{6}\,(1-\xi)R \tag{2.159}$$

of an inhomogeneous scalar field $\bar{\phi}(x)$ in a background with curvature R. The first three coefficients are given by

$$a_0 = 1, \qquad a_1 = -U, \qquad a_2 = \frac{U^2}{2} - \frac{\hat{\gamma}^2}{12}. \tag{2.160}$$

While our derivation follows the pathway of Schwinger–DeWitt, this generalized form of a_l can also be obtained using differential geometric methods, see, e.g., Gilkey [394].

Recalling the definition of $\hat{\gamma}_{\mu\nu}^2$ from (2.141) and (2.142) we see that

$$\frac{1}{4}\hat{\gamma}^2 = -\frac{1}{2}\Box_E U - a^{\lambda}{}_{\lambda} \tag{2.161}$$

and

$$a^{\lambda}{}_{\lambda} = -\frac{1}{60}\Big(\Box_E R - R^{ab}R_{ab} + R^{abcd}R_{abcd}\Big), \tag{2.162}$$

where R, R_{ab}, and R_{abcd} are the scalar, Ricci, and Riemann curvature tensors, respectively. Equivalently, $a^{\lambda}{}_{\lambda}$ can be expressed in terms of the two covariant quantities F and G, introduced earlier (2.61) in the context of the trace anomaly. The a_2 coefficient enters in determining the trace anomaly T_A

$$T_A = \langle T^{\mu}{}_{\mu}\rangle_{\rm ren} = -\frac{a_2(x)}{16\pi^2}. \tag{2.163}$$

Eqs (2.160) and (2.161) show that the trace anomaly of an inhomogeneous scalar field in curved spacetime acquires an additional nonlocal contribution from the second variation of the scalar field. (Note the absence of first derivative terms

proportional to β, as expected.) Their contribution to the trace anomaly (analogous to the $\Box_E R$ term) may have theoretical and cosmological consequences for models involving time-dependent Higgs fields (e.g., symmetry breaking in GUT, induced gravity or higher-order curvature gravitational theories). The backreaction of trace anomaly and other vacuum polarization quantum effects on the early universe will be discussed in Chapter 8.

Dimensional Regularization

The one-loop effective Lagrangian is given by

$$L^{(1)} = \frac{\hbar}{2(4\pi)^{n/2}} \sum_{l=0}^{\infty} a_l \, m^{n-2l} \, \Gamma(l - n/2), \tag{2.164}$$

where

$$\Gamma(x) \equiv \int_0^{\infty} ds \, s^{x-1} e^{-s} \tag{2.165}$$

is the Gamma function. In order to keep this in the same dimension as $L^{(1)}$, it is customary to introduce a renormalization mass parameter such that

$$L^{(1)} = \frac{\hbar}{2(4\pi)^{n/2}} \left(\frac{m}{\mu_c}\right)^{n-4} \sum_{l=0}^{\infty} a_l \, m^{4-2l} \, \Gamma(l - n/2). \tag{2.166}$$

To proceed with dimensional regularization[12], we expand the first three terms in the series near $n = 4$. Making use of the well-known expansion relations for $(m/\mu_c)^{n-4}$ and $\Gamma(l - n/2)$, we get

$$\begin{aligned} L^{(1)} = &-\frac{\hbar m^4}{32\pi^2} \left\{ \frac{1}{n-4} + \frac{1}{2} \left[\ln\left(\frac{m^2}{4\pi {\mu_c}^2}\right) + \gamma_\epsilon - \frac{3}{2} \right] \right\} \\ &+ \frac{\hbar a_1 m^2}{16\pi^2} \left\{ \frac{1}{n-4} + \frac{1}{2} \left[\ln\left(\frac{m^2}{4\pi {\mu_c}^2}\right) + \gamma_\epsilon - \frac{1}{2} \right] \right\} \\ &- \frac{\hbar a_2}{16\pi^2} \left\{ \frac{1}{n-4} + \frac{1}{2} \left[\ln\left(\frac{m^2}{4\pi {\mu_c}^2}\right) + \gamma_\epsilon \right] \right\} + \frac{\hbar}{2(4\pi)^2} \sum_{l=0}^{\infty} a_l \, m^{4-2l} \, \Gamma(l-2). \end{aligned} \tag{2.167}$$

Here and only here $\gamma_\epsilon = 0.5772$ is the Euler constant.

In dimensional regularization the divergent terms [$\sim (n-4)^{-1}$] are removed by the introduction of counter-terms L_c in the effective Lagrangian, which in their most general form contain linear combinations of F and G, defined in (2.61).

[12] Dimensional regularization was invented by 't Hooft and Veltman [93, 94] and Bollini and Giambiagi [395] and applied successfully to proving the renormalizability of gauge theories. A good introduction to renormalization and the renormalization group can be found in the book by Collins [396]. Dimensional regularization was introduced to curved space by Drummond and Shore [397, 92], Brown and Cassidy [398, 399], Bunch and Parker [364] and many others. For further discussions and more references, see [47, 53].

The regularized Lagrangian then consists of terms in (2.167) together with the counter-terms. In a renormalization scheme, the divergent terms are absorbed into redefined coupling constants of the theory, namely, the cosmological constant Λ, Newton's constant $G_{\rm N}$ [contained in $\epsilon_0 \equiv (16\pi G_{\rm N})^{-1}$], the constants ϵ_1, ϵ_2, ϵ_3, and ϵ_4 of the curvature squared, the $\Box_E R$ terms [see (2.170) below] for the geometry, and m, θ, ξ, and λ for the fields.

Carrying out a small-s expansion of $L^{(1)}$ using α^2 as the mass parameter, we obtain the regularized one-loop effective Lagrangian

$$L^{(1)} = -\frac{\hbar}{64\pi^2}\left[\alpha^4\left(\ln\frac{\alpha^2}{\mu^2} - \frac{3}{2}\right) - \frac{\hat{\gamma}^2}{6}\ln\frac{\alpha^2}{\mu^2}\right] + \sum_{l=3}^{\infty} b_l\,\alpha^{4-2l}\,\Gamma(l-2), \qquad (2.168)$$

where $b_0 = a_0 = 1,\ b_1 = 0,\ b_2 = -\hat{\gamma}^2/12$. Here we have redefined μ related to the μ_c in (2.167) by

$$\ln\mu^2 = \ln 4\pi{\mu_c}^2 - \gamma - \frac{2}{n-4}, \qquad (2.169)$$

making it agree with the μ_H used in [97]. Note the disappearance of U terms in b_l but the presence of nonlocal term $\Box_E U$ in b_2. The terms in (2.168) proportional to $\hat{\gamma}^2$ come from the changing background field.

In anticipation of the appearance of higher-order curvature terms in the renormalization procedure, we write the classical Lagrangian for the background spacetime in the most general allowed form:

$${L_g}^{(0)} = \epsilon_0 R - \bar{\Lambda} + \frac{1}{2}\epsilon_1 R^2 + \frac{1}{2}\epsilon_2 C^2 + \epsilon_3 G + \epsilon_4 \Box_E R, \qquad (2.170)$$

where $\epsilon_0 = (16\pi G_{\rm N})^{-1}$, with $G_{\rm N}$ being Newton's constant. To compare with the coefficients in (2.67), $\bar{\Lambda} = 2\epsilon_0\Lambda$, the cosmological constant, $\frac{1}{2}\epsilon_1 = \epsilon_0 a,\ \frac{1}{2}\epsilon_2 = \epsilon_0 b$.

We have also added the total divergence terms G and $\Box_E R$ with coupling constants ϵ_3 and ϵ_4 for completeness, but they only enter in spacetimes with boundaries. The complete Lagrangian to one-loop is given by the sum of (2.170), (2.82) and (2.168):

$$L = {L_g}^{(0)} + {L_\Phi}^{(0)} + L^{(1)}. \qquad (2.171)$$

All parameters of the theory prior to this point are bare quantities.

Renormalization Parameters

We now proceed to define the renormalized parameters by suitable renormalization conditions:

$$
\begin{aligned}
m^2 &= -\left[\mathrm{Re}\frac{\partial^2 L}{\partial\phi^2}\right]_{\phi=0,\ R=0}, & \lambda &= -\left[\mathrm{Re}\frac{\partial^4 L}{\partial\phi^4}\right]_{\phi=\phi_0,\ R=0},\\
\theta &= \left[\mathrm{Re}\frac{\partial L}{\partial\Box_E\phi^2}\right]_{\phi=\phi_\theta,\ R=0}, & \frac{1-\xi}{6} &= -\left[\mathrm{Re}\frac{\partial^3 L}{\partial R\partial\phi^2}\right]_{\phi=0,\ R=R_0},\\
0 &= \left[\mathrm{Re}L\right]_{\phi=\bar{\phi},\ R=0}, & \epsilon_0 &= \left[\mathrm{Re}\frac{\partial L}{\partial R}\right]_{\phi=0,\ R=0},\\
\epsilon_1 &= \left[\mathrm{Re}\frac{\partial^2 L}{\partial R^2}\right]_{\phi=0,\ R=R_1}, & \epsilon_2 &= 2\left[\mathrm{Re}\frac{\partial L}{\partial C^2}\right]_{\phi=0,\ R=R_2},\\
\epsilon_3 &= \left[\mathrm{Re}\frac{\partial L}{\partial G}\right]_{\phi=0,\ R=R_3}, & \epsilon_4 &= \left[\mathrm{Re}\frac{\partial L}{\partial\Box_E R}\right]_{\phi=0,\ R=R_4}.
\end{aligned}
\tag{2.172}
$$

The real part in the above equations should be taken, as the renormalized terms are required to be real quantities. The coupling constants λ, θ, ξ, ϵ_1, ϵ_2, ϵ_3 and ϵ_4 are defined at the energy scales corresponding to the values $\phi = \phi_0$, ϕ_θ, $R = R_0$, R_1, R_2, R_3 and R_4, respectively, which may in general be all different as the energy scales at which they are measured need not be the same. How these coupling constants behave under changes of scales is determined by their associated renormalization-group equations.

2.6 Renormalization Group Equations

The renormalized parameters are related to the bare ones by $m^2 = {m_B}^2 + \delta m^2$, etc., where the counter-terms δm^2 are obtained from (2.172). The complete result is as follows:

$$\delta m^2 = \tilde{h}\lambda m^2\left[\ln\frac{m^2}{\mu^2} - 1\right] \equiv \tilde{h}\lambda m^2(\tau - \tau_m), \tag{2.173}$$

$$\delta\lambda = 3\tilde{h}\lambda^2\left[\ln\frac{m^2 + \frac{1}{2}\lambda{\phi_0}^2}{\mu^2} + \frac{8}{3} - \frac{8(m^2 + \frac{1}{4}\lambda{\phi_0}^2)m^2}{3(m^2 + \frac{1}{2}\lambda{\phi_0}^2)^2}\right] \equiv 3\tilde{h}\lambda^2(\tau - \tau_\lambda), \tag{2.174}$$

$$\delta\theta = -\frac{\tilde{h}\lambda}{12}\left[\ln\frac{m^2 + \frac{1}{2}\lambda{\phi_\theta}^2}{\mu^2}\right] \equiv -\frac{\tilde{h}\lambda}{12}(\tau - \tau_\theta), \tag{2.175}$$

$$\delta\xi = \tilde{h}\lambda\xi\left[\ln\frac{m^2 - \frac{1}{6}\xi R_0}{\mu^2}\right] \equiv \tilde{h}\lambda\xi(\tau - \tau_0), \tag{2.176}$$

$$\delta\bar{\Lambda} = \frac{\tilde{h}m^4}{2}\left[\ln\frac{m^2}{\mu^2} - \frac{3}{2}\right] + \delta\bar{\Lambda}(\langle\phi\rangle) \equiv \frac{\tilde{h}m^4}{2}(\tau - \tau_{\bar{\Lambda}}) + \delta\bar{\Lambda}(\langle\phi\rangle), \tag{2.177}$$

$$\delta\epsilon_0 = \tilde{h}m^2\frac{\xi}{6}\left[\ln\frac{m^2}{\mu^2} - 1\right] \equiv \tilde{h}m^2\frac{\xi}{6}(\tau - \tau_{\epsilon_0}), \tag{2.178}$$

$$\delta\epsilon_1 = -\frac{\tilde{h}\xi^2}{36}\left[\ln\frac{m^2-\frac{1}{6}\xi R_1}{\mu^2}\right] \equiv -\frac{\tilde{h}\xi^2}{36}(\tau-\tau_1), \tag{2.179}$$

$$\delta\epsilon_2 = \frac{\tilde{h}}{60}\left[\ln\frac{m^2-\frac{1}{6}\xi R_2}{\mu^2}\right] \equiv \frac{\tilde{h}}{60}(\tau-\tau_2), \tag{2.180}$$

$$\delta\epsilon_3 = -\frac{\tilde{h}}{360}\left[\ln\frac{m^2-\frac{1}{6}\xi R_3}{\mu^2}\right] \equiv -\frac{\tilde{h}}{360}(\tau-\tau_3), \tag{2.181}$$

$$\delta\epsilon_4 = \frac{\tilde{h}}{36}\left(\xi+\frac{1}{5}\right)\left[\ln\frac{m^2-\frac{1}{6}\xi R_4}{\mu^2}\right] \equiv \frac{\tilde{h}}{36}\left(\xi+\frac{1}{5}\right)(\tau-\tau_4), \tag{2.182}$$

where $\tau = \ln(m^2/\mu^2)$, $\tilde{h} = \hbar/32\pi^2$. We include the terms $\epsilon_4\Box_E R$ and $\theta\,\Box_E\phi^2$ in our classical actions, ${L_g}^{(0)}$ and ${L_\phi}^{(0)}$ in (2.170), (2.123) even though these are purely surface terms, since divergent terms proportional to them occur at the one-loop level. In the spirit of renormalization these divergences are absorbed into the appropriate bare parameters in the Lagrangian, and in this way they are transformed into the corresponding renormalized parameters. The Gauss–Bonnet density G is introduced for similar reasons; its integral over the manifold is a topological invariant $32\pi^2\chi$, χ being the Euler characteristic. We thus expect χ to be present in the action for all manifolds, with or without boundary.

From the counter-terms (2.173)–(2.182) one can obtain the renormalization-group equations for each of the parameters:

$$\begin{aligned}
&\frac{dm^2}{d\tau} = \tilde{h}\lambda m^2, \qquad \frac{d\lambda}{d\tau} = 3\tilde{h}\lambda^2, \qquad \frac{d\theta}{d\tau} = -\frac{\tilde{h}\lambda}{12}, \qquad \frac{d\xi}{d\tau} = \tilde{h}\lambda\xi,\\
&\frac{d\bar{\Lambda}}{d\tau} = \frac{\tilde{h}m^4}{2}, \qquad \frac{d\epsilon_0}{d\tau} = \tilde{h}m^2\frac{\xi}{6}, \qquad \frac{d\epsilon_1}{d\tau} = -\frac{\tilde{h}\xi^2}{36},\\
&\frac{d\epsilon_2}{d\tau} = \frac{\tilde{h}}{60}, \qquad \frac{d\epsilon_3}{d\tau} = -\frac{\tilde{h}}{360}, \qquad \frac{d\epsilon_4}{d\tau} = \frac{\tilde{h}}{36}\left(\xi+\frac{1}{5}\right).
\end{aligned} \tag{2.183}$$

Solutions to these equations can be obtained with the appropriate boundary conditions which preserve Eqs. (2.173)–(2.182) at the specific renormalization point:

$$m^2(\tau) = m^2(\tau_m)\left[\frac{L(\tau_m)}{L(\tau)}\right], \tag{2.184}$$

$$\lambda(\tau) = \frac{\lambda(\tau_\lambda)}{L^3(\tau)}, \tag{2.185}$$

$$\theta(\tau) = \theta(\tau_\theta) - \frac{1}{12}\left[\ln\frac{L(\tau_\theta)}{L(\tau)}\right], \tag{2.186}$$

$$\xi(\tau) = \xi(\tau_0)\left[\frac{L(\tau_0)}{L(\tau)}\right], \tag{2.187}$$

$$\bar{\Lambda}(\tau) = \bar{\Lambda}(\tau_{\bar{\Lambda}}) - \frac{m^4(\tau_m)}{2\lambda(\tau_\lambda)}L^2(\tau_m)[L(\tau) - L(\tau_{\bar{\Lambda}})], \tag{2.188}$$

$$\epsilon_0(\tau) = \epsilon_0(\tau_{\epsilon_0}) - \frac{\xi(\tau_0) m^2(\tau_m)}{6\lambda(\tau_\lambda)} L(\tau_m) L(\tau_0)[L(\tau) - L(\tau_{\epsilon_0})], \tag{2.189}$$

$$\epsilon_1(\tau) = \epsilon_1(\tau_1) + \frac{\xi^2(\tau_0)}{36\lambda(\tau_\lambda)} L^2(\tau_0)[L(\tau) - L(\tau_1)], \tag{2.190}$$

$$\epsilon_2(\tau) = \epsilon_2(\tau_2) + \frac{\tilde{h}}{60}(\tau - \tau_2), \tag{2.191}$$

$$\epsilon_3(\tau) = \epsilon_3(\tau_3) - \frac{\tilde{h}}{360}(\tau - \tau_3), \tag{2.192}$$

$$\epsilon_4(\tau) = \epsilon_4(\tau_4) - \frac{\xi(\tau_0) L(\tau_0)}{72\lambda(\tau_\lambda)}[L^2(\tau) - L^2(\tau_4)] + \frac{\tilde{h}}{180}(\tau - \tau_4), \tag{2.193}$$

where $L(\tau) = [1 - 3\tilde{h}\lambda(\tau_\lambda)(\tau - \tau_\lambda)]^{1/3}$ and τ_m to τ_4 are defined in Eqs. (2.173)–(2.182). Henceforth, whenever the explicit functional dependence of the parameters on τ is not displayed they should be regarded as assuming their values at their respective renormalization points.

A subset of these equations have also be obtained by other regularization methods, such as via zeta-function regularization [400] discussed in Chapter 4. As the background spacetime considered here is completely general we have the RG equation for the coupling constant of the conformal-tensor-squared term, which is not present, e.g., for the Einstein universe.

Since the RG equations are derived from the counter-terms in the regularization of ultraviolet divergences, they are insensitive to the large-scale properties of the background field, and thus can in principle also be obtained by other local expansion methods. Solutions to these RG equations which govern the strength of the coupling constants as a function of energy and curvature should be useful for the analysis of spacetime-curvature effects on elementary field interactions.

The renormalized one-loop effective Lagrangian is given by

$$\begin{aligned}
L_{\rm eff} = {} & \epsilon_0 R - \bar{\Lambda} + \frac{\epsilon_1}{2} R^2 + \frac{\epsilon_2}{2} C^2 + \epsilon_3 G + \epsilon_4 \Box_E R - \frac{1}{12}(1-\xi) R\phi^2 + \theta \Box_E \phi^2 \\
& - \frac{\hbar}{64\pi^2}\left\{ \frac{\lambda}{6} \Box_E \phi^2 \ln\left|\frac{M_\lambda^2}{m^2 + \lambda {\phi_\theta}^2/2}\right| - \frac{\xi}{6} \lambda \phi^2 R \left(\ln\left|\frac{M_\lambda^2}{m^2 - \xi R_0/6}\right| - \frac{3}{2}\right) \right. \\
& - \frac{1}{3} m^2 \xi R \left(\ln\left|\frac{M_\lambda^2}{m^2}\right| - \frac{1}{2}\right) + \frac{\xi^2}{36} R^2 \left(\ln\left|\frac{M_\lambda^2}{m^2 - \xi R_1/6}\right| - \frac{3}{2}\right) \\
& - \frac{1}{60} C^2 \ln\left|\frac{M_\lambda^2}{m^2 - \xi R_2/6}\right| + \frac{1}{180} G \ln\left|\frac{M_\lambda^2}{m^2 - \xi R_3/6}\right| \\
& \left. - \frac{1}{18}\left(\xi + \frac{1}{5}\right) \Box_E R \ln\left|\frac{M_\lambda^2}{m^2 - \xi R_4/6}\right| \right\} + \delta\bar{\Lambda}(\langle\phi\rangle) + \frac{\hbar}{64\pi^2}\left[\alpha^4 - \frac{\hat{\gamma}^2}{6}\right] \ln\frac{\alpha^2}{M_\lambda^2} \\
& + \frac{\hbar}{32\pi^2} \int_0^\infty \frac{ds}{s^3} \left[e^{-\alpha^2 s - f(s)} - 1 + \frac{\hat{\gamma}^2 s^2}{12}\right] + L_{\rm CW},
\end{aligned} \tag{2.194}$$

where $\delta\bar{\Lambda}$ is determined by the vacuum energy of flat space and L_{CW} denotes the Coleman–Weinberg potential for $\lambda\phi^4$ theory [343] in flat space:

$$L_{\rm CW} = \frac{1}{2}\,\phi\,\Box_E\phi - \frac{1}{2}\,m^2\phi^2 - \frac{\lambda}{4!}\,\phi^4 - \frac{\hbar}{64\pi^2}\left\{ M_\lambda^4 \ln\left|\frac{M_\lambda^2}{m^2}\right| - \frac{m^2}{2}\lambda\phi^2 + \frac{\lambda\phi^4}{4}\left(\ln\left|\frac{m^2}{m^2+\lambda\phi_0^2/2}\right| - \frac{25}{6} + \frac{8}{3}\frac{m^2(m^2+\frac{1}{4}\lambda\phi_0^2)}{(m^2+\frac{1}{2}\lambda\phi_0^2)^2}\right)\right\} \tag{2.195}$$

where $M_\lambda^2 = m^2 + \frac{\lambda}{2}\,\phi^2$. The integral expression in (2.194) contains terms which were finite to start with. The analytic form of $f(s)$ is given by (2.156).

Combining the results obtained above for the effective action with the results of integrating the renormalization-group equations (2.184)–(2.193), we obtain the renormalization-group-improved effective Lagrangian

$$L_{\rm eff} = \epsilon_0(\tau)R - \bar{\Lambda}(\tau) + \frac{1}{2}\,\epsilon_1(\tau)R^2 + \frac{1}{2}\,\epsilon_2(\tau)C^2 + \epsilon_3(\tau)G + \epsilon_4(\tau)\Box_E R + \frac{1}{2}\phi\,\Box_E\phi + \theta(\tau)\,\Box_E\phi^2 - \frac{1}{2}\,m^2(\tau)\phi^2 - \frac{1}{12}\,[1-\xi(\tau)]\phi^2 - \frac{\lambda(\tau)}{4!}\,\phi^4 + \frac{\hbar}{32\pi^2}\int_0^\infty \frac{ds}{s^3}\,\left[e^{-\alpha^2 s - f(s)} - 1 + \frac{\hat{\gamma}^2 s^2}{12}\right]. \tag{2.196}$$

where $\tau = \ln(\alpha^2/m^2)$ and the parameters are as in (2.184)–(2.193).

The above restults for the self-interacting scalar field were first obtained by Hu and O'Connor [375]. Paz and Mazzitelli [401] later derived the renormalized equations for the quantum field and background spacetime using the adiabatic regularization method (discussed in Chapter 8). Kirsten *et al.* [402] and Elizalde *et al.* [403] derived the renormalization group-improved effective Lagrangian for $O(N)$ scalar theory in slowly varying gravitational fields using zeta-function regularization and heat-kernel techniques. See also Markkanen and Tranberg [404].

Renormalization group equations in curved spacetime was first investigated by Brown and Collins [399], Nelson and Panagaden [405], Hu, O'Connor and Shen [400, 375], Buchbinder [406]. Toms [373, 407] used a gauge-invariant background field method to study the behaviour of a non-abelian gauge theory with fermions in the high curvature limit, and Parker and Toms [408, 387] performed a renormalization-group analysis of grand unified theories in curved spacetimes. For later developments, see the work of Elizalde and Odintsov, [409, 410, 411] and, especially, Holland and Wald [412].

The usage of effective action for treating quantum fields is very broad. For further work covering spinor and gauge fields, see, e.g., [52, 58]. For a taste of the more sophisticated diffeomorphism-invariant Vilkovisky–DeWitt effective action, see, e.g., [413, 329, 414, 53].

3

'In-In' Effective Action. Stress Tensor. Thermal Fields

3.1 The 'In-In' Effective Action

The effective action method we introduced in the last chapter is seen to be well suited to the treatment of backreaction problems for quantum processes in dynamical background spacetimes, as it yields equations of motion for both the quantum field and the spacetime in a self-consistent way. There is, however, one outstanding issue in the *'in-out'* (Schwinger–DeWitt) formulation, in that the effective geometry it engenders is complex and the equation of motion is acausal, thus marring the physical identity of an effective geometry and obscuring the physical meaning of effects related to backreaction, such as dissipation. This stems from the particular way the conventional *in-out* effective action method is formulated, i.e. in terms of the vacuum persistence amplitude $Z[J] = \langle 0_+|0_-\rangle$ which is the transition amplitude between the *in* and *out* vacuum under the action of a single source. To obtain a real and causal equation of motion one should use the *in-in* (or Schwinger–Keldysh) effective action which describes the evolution of the same *in* vacuum $Z[J_+, J_-] = \sum_\Psi \langle 0_-|\Psi\rangle_{J_-}\langle\Psi|0_-\rangle_{J_+}$ under the action of two different sources: J_+ propagating forward and J_- backward in time. Because of this it is also called the closed-time-path (CTP) formalism. In the 'in-out' formulation one calculates the matrix elements $\langle 0_+|\hat{T}|0_-\rangle$ of an operator $\hat{T}$, which, allowing for the instability of the vacuum, is in general complex. In the 'in-in' formulation one obtains the expectation values $\langle 0_-|\hat{T}|0_-\rangle$ of a physical observable, which are real. One can in principle obtain the *'in-in'* expectation values by summing over the *'in-out'* matrix elements over a complete set of intermediate states. However, this is not easily found, except for spaces of high symmetry. Instead, this motivates seeking a 'doubling of the paths' representation – one can obtain the same result by integrating over a closed-time-path, as suggested by Schwinger [239] (also [415, 416]) and Keldysh [240], as we now explain.

3.1.1 'In-Out' for Transition Amplitudes, 'In-In' for Expectation Values

In essence the two formulations are suitable for different sets of problems. In the *in-out* formalism, since the *in* and *out* states are stipulated, it amounts to posing a boundary value problem. Since it gives the transition amplitude between the *in* and the *out* vacuum it is more suitable for scattering and transition rate problems, like quantum tunneling and particle creation. The *in-in* formalism amounts to posing an initial value problem for the system. It is therefore more suitable for calculating the evolution of the expectation values of physical variables, e.g. the causal and correlational properties of the system. Thus, for particle creation and backreaction problems, whereas the *in-out* formalism conveniently yields the transition probability of pair creation $P = 2\mathrm{Im}\Gamma$, where Γ is the 'in-out' effective action, as we saw in the last chapter, the *in-in* formalism yields a real and causal equation of motion for a real effective geometry including backreaction effects. As we shall see in the example below the *in-in* formalism also gives a clear physical interpretation of the statistical mechanical meaning of dissipation, noise and the non-Markovian (processes with memory) nature of all backreaction effects, here involving quantum fields *on* a curved background spacetime. We begin with a short introduction to the *in-in* formalism, first in flat spacetime and then extend it to curved spacetime. To avoid overladen discussion we will go into details only in those parts which differ from the *in-out* formulation.

Consider the field operator Φ and states for an interacting field theory with classical action

$$S[\Phi] = \int d^4x \, \left\{ -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2 - \frac{\lambda}{4!}\Phi^4 \right\}. \tag{3.1}$$

As the (self) coupling constant λ is adiabatically switched off at $t \to \pm\infty$, the theory becomes free in the distant past and future. The *in* vacuum $|0_-\rangle$ and the *out* vacuum $|0_+\rangle$ coincide with the free field vacuum in the past and future asymptotic regions respectively. If the field is coupled to an external, c-number source $J(x)$, the states will evolve in time according to

$$|0_-\rangle_{J(t)} = \mathcal{T} \exp\left[i \int_{-\infty}^{t} dt' \int d^3x \; J(\vec{x},t')\Phi(\vec{x},t') \right] |0_-\rangle, \tag{3.2}$$

where $\mathcal{T}$ denotes taking the temporal order. The vacuum persistence amplitude defined as

$$Z(J) = \langle 0_+|0_-\rangle_J = \langle 0_+|\mathcal{T} \exp\left[i \int d^4x \; J(x)\Phi(x) \right] |0_-\rangle \tag{3.3}$$

contains all the dynamical information about the theory. It has a path integral representation

$$Z[J] = \int D\Phi \; \exp\Big\{ i \Big(S[\Phi] + J\Phi \Big) \Big\}, \tag{3.4}$$

where the integral is over all field configurations. For massive field theories like (3.1) one usually adds an imaginary part $(-i\epsilon)$ to the mass to account for the boundary conditions. $Z[J]$ generates the time-ordered matrix elements of a product of n fields between the *in* and *out* states,

$$\langle 0_+|T\,\Phi(x_1)\cdots\Phi(x_n)|0_-\rangle = (-i)^n \frac{\delta^n}{\delta J(x_1)\cdots\delta J(x_n)} Z[J]\,\Big|_{J=0}. \tag{3.5}$$

To look for a functional which generates expectation values with respect to the *in*-state, following [239], we introduce two external sources $J_+(x)$ and $J_-(x)$ and consider the quantity

$$Z[J^+, J^-] = {}_{J_-}\langle 0_-|0_-\rangle_{J_+}. \tag{3.6}$$

In contrast to the *in-out* formalism, where one lets the *in* vacuum evolve under the influence of an external source and compares the result with the *out* vacuum, in the *in-in* formalism, one lets the *in* vacuum evolve independently under two sources J_+ and J_-, and compares the results at a future time t_*. We may rewrite (3.6) as

$$\begin{aligned} Z[J_+, J_-] = \int D\Phi\ \langle 0_-|\tilde{\mathcal{T}} \exp\left[-i\int_\infty^{t*} dt \int d^3x\ J_-(x)\Phi(x)\right]|\psi\rangle \\ \times \langle\psi|\mathcal{T} \exp\left[i\int_{-\infty}^{t*} dt \int d^3x\ J_+(x)\Phi(x)\right]|0_-\rangle, \end{aligned} \tag{3.7}$$

where $\tilde{\mathcal{T}}$ denotes taking the anti-temporal order. Here $|\psi\rangle$ is an element of a complete, orthonormal set of common eigenvectors of the field operators at some arbitrary time t^* in the future

$$\Phi(\vec{x}, t^*)|\psi\rangle = \psi(\vec{x}, t^*)\ |\psi\rangle. \tag{3.8}$$

For simplicity we will assume $t^* = +\infty$ for all practical purposes.

From the definition (3.6) and (3.7) we obtain the following useful relations

$$Z[J, J] = 1, \qquad Z[J_+, J_-] = (Z[J_-, J_+])^*, \tag{3.9}$$

$$(-i)^n \frac{\delta^n Z[J_+, J_-]}{\delta J_+(x_1)\cdots\delta J_+(x_n)}\bigg|_{J_+=J_-=0} = \langle 0_-|\mathcal{T}\Big[\Phi(x_1)\cdots\Phi(x_n)\Big]|0_-\rangle, \tag{3.10}$$

$$(+i)^m \frac{\delta^m Z[J_+, J_-]}{\delta J_-(x_1)\cdots\delta J_-(x_m)}\bigg|_{J_+=J_-=0} = \langle 0_-\ |\ \tilde{\mathcal{T}}\Big[\Phi(x_1)\cdots\Phi(x_m)\Big]|0_-\rangle, \tag{3.11}$$

$$\begin{aligned} (-i)^{n-m} \frac{\delta^{n+m} Z[J_+, J_-]}{\delta J_-(x_1)\cdots\delta J_-(x_m)\delta J_+(y_1)\cdots\delta J_+(y_n)}\bigg|_{J_+, J_-=0} \\ = \langle 0_-|\tilde{\mathcal{T}}\Big[\Phi(x_1)\cdots\Phi(x_m)\Big]\mathcal{T}\Big[\Phi(y_1)\cdots\Phi(y_n)\Big]|0_-\rangle. \end{aligned} \tag{3.12}$$

Observe that Z generates expectation values other than the time-ordered ones. $Z[J_+, J_-]$ has a functional integral representation

$$Z[J_+, J_-] = \int D\Phi_+(x) D\Phi_-(x)\ \exp\Big\{ i\left(S[\Phi_+] + J_+\Phi_+ - S^*[\Phi_-] - J_-\Phi_-\right)\Big\}, \tag{3.13}$$

where $J\Phi$ denotes $\int d^4x\ J(x)\Phi(x)$ and S^* indicates that in this functional, m^2 carries an $+\ i\epsilon$ term. The integral in (3.13) is over all field configurations which coincide at $t = t^*$ (in practice, $t = +\infty$). We do not require the fields to go to zero as $t \to \infty$ (eventually we may consider that $i\epsilon$ is switched off in this region), nor do we require their time variation at $t = t^*$ be equal $\delta_t\Phi_+ \mid_{t=t^*} = \delta_t\Phi_- \mid_{t=t^*}$.

As in the *in-out* formalism, sometimes it is easier to work with the generating functional $W[J_+, J_-]$

$$Z[J_+, J_-] \equiv e^{iW[J_+, J_-]}, \ \text{ or } \ W[J_+, J_-] = -i \ln Z[J_+, J_-], \tag{3.14}$$

which generates normalized expectation values. Let us consider the mean fields defined by

$$\bar\phi_+(x) = \frac{\delta W[J_+, J_-]}{\delta J_+(x)} \quad \text{and} \quad \bar\phi_-(x) = -\frac{\delta W[J_+, J_-]}{\delta J_-(x)}. \tag{3.15}$$

If $J_+ = J_- = J$, then $\bar\phi_+ = \bar\phi_- = \bar\phi$ is the expectation value of the field with respect to the state which evolved from $|0_-\rangle$ under the influence of the source J. The *in-in* effective action Γ is defined as the Legendre transform of W, i.e.,

$$\Gamma[\bar\phi_+, \bar\phi_-] = W[\bar J_+, \bar J_-] - \bar J_+\bar\phi_+ + \bar J_-\bar\phi_-, \tag{3.16}$$

where $\bar J_+, \bar J_-$ are the solutions of

$$\bar\phi_\pm = \pm\, \frac{\delta}{\delta J_\pm} W[\bar J_+, \bar J_-]. \tag{3.17}$$

We then have

$$\frac{\delta\Gamma[\bar\phi_+, \bar\phi_-]}{\delta\bar\phi_\pm} = \mp\, \bar J_\pm. \tag{3.18}$$

Equations (3.15) or their equivalent (3.18) are the equations of motion for $\bar\phi$. As different from the *in-out* formalism, $\bar\phi$, the common value of $\bar\phi_+$ and $\bar\phi_-$, is real and depends causally on J, the common value of J_+ and J_-. The reality of $\bar\phi$ follows directly from $\bar\phi$ being the expectation value of the fields with respect to some state. As for causality, let us compute the functional derivative $\delta\bar\phi(x)/\delta J(x')$ using (3.9),

$$\frac{\delta\bar\phi(x)}{\delta J(x')} = (-i)\left[\frac{\delta^2 Z}{\delta J_+(x')\delta J_+(x)} + \frac{\delta^2 Z}{\delta J_-(x')\delta J_+(x)}\right]_{(J_+ = J_- = J)}. \tag{3.19}$$

From the definition (3.17) it is easy to show that this is zero whenever $t' > t$. Thus $\phi(x)$ depends only on the values of J in the past of x (from Lorentz invariance, we may say "in the past light cone of x"). Jordan [181] has demonstrated these properties to the two-loop order. Observe that in practice only one of the equations (3.18) need be solved. Since $\Gamma[\bar{\phi}, \bar{\phi}]$ is identically zero, we have the identity

$$\frac{\delta\Gamma}{\delta\bar{\phi}_+}\bigg|_{(\bar{\phi}_+=\bar{\phi}_-=\bar{\phi})} = -\frac{\delta\Gamma}{\delta\bar{\phi}}\bigg|_{(\bar{\phi}_+=\bar{\phi}_-=\bar{\phi})}. \tag{3.20}$$

So if $J_+ = J_- = J$, (3.18) always has a solution with $\bar{\phi}_+ = \bar{\phi}_- = \bar{\phi}$ and each of the equations in (3.18) implies the other.

3.1.2 In-In (CTP) Green Functions and Perturbative Expansion

Exact computations of Z, W, or Γ are usually impossible except for special cases. In general one needs to resort to perturbative methods. Observe that the path integral representation (3.13) of Z has the same form as the *in-out* path integral representation of Z for a theory of two scalar fields, and therefore the perturbative evaluation of Z may proceed exactly as in the conventional case. For example, W will be the sum of all connected graphs of the two-component field theory, while Γ will be the sum of the one particle irreducible (1PI) graphs.

Let us consider an example, that of a $\lambda\Phi^4$ field (3.1). In the free field theory the path integral (3.13) is Gaussian and Z takes on the form

$$\begin{aligned} Z_{\rm free}[J_+, J_-] = \exp\Big\{ \frac{i}{2} \int d^4x\, d^4x' \, \Big[& J_+(x) G_{++}(x,x') J_+(x') \\ & - J_+(x) G_{+-}(x,x') J_-(x') - J_-(x) G_{-+}(x,x') J_+(x') \\ & + J_-(x) G_{--}(x,x') J_-(x') . \Big] \Big\} . \end{aligned} \tag{3.21}$$

The kernels G_{++} and G_{--} are symmetric, and $G_{+-}(x,x') = G_{-+}(x',x)$. They will later be identified with the familiar Green functions (3.29)–(3.32). They are related to the classical fields $\bar{\phi}_\pm(x)$ of Eq. (3.15) by

$$\bar{\phi}_+(x) = \int d^4x' \, \Big\{ G_{++}(x,x') J_+(x') - G_{+-}(x,x') J_-(x') \Big\}, \tag{3.22}$$

$$\bar{\phi}_-(x) = \int d^4x' \, \Big\{ G_{-+}(x,x') J_+(x') - G_{--}(x,x') J_-(x') \Big\}. \tag{3.23}$$

One can introduce an alternative representation for the fields as

$$\bar{\phi}_+(x) = Z^{-1}[J_+, J_-]\langle 0_-| \left\{\tilde{\mathcal{T}} \exp\left[-i\int d^4x'\ J_-\Phi\right]\right\} \times \left\{\mathcal{T}\exp\left[i\int_{t'>t} d^4x' J_+\Phi\right]\right\}\Phi(x)\left\{\mathcal{T}\exp\left[i\int_{t'<t} d^4x\ J_+\Phi\right]\right\}|0_-\rangle, \tag{3.24}$$

$$\bar{\phi}_-(x) = Z^{-1}[J_+, J_-]\langle 0_-| \left\{\tilde{\mathcal{T}} \exp\left[-i\int_{t'<t} d^4x'\ J_-\Phi\right]\right\}\Phi(x) \times \left\{\tilde{\mathcal{T}}\exp\left[-i\int_{t'>t} d^4x'\ J_-\Phi\right]\right\}\left\{\mathcal{T}\exp\left[i\int d^4x'\ J_+\Phi\right]\right\}|0_-\rangle. \tag{3.25}$$

From the Heisenberg equations of motion for a free field, the canonical commutation relations, and (3.25), it is easy to show that

$$(-\Box + m^2)\bar{\phi}_\pm(x) = J_\pm(x). \tag{3.26}$$

As $t \to -\infty, \bar{\phi}_+(\bar{\phi}_-)$ contains only negative (positive) frequencies, and as $t \to +\infty$, $\bar{\phi}_+(x)$ goes into $\bar{\phi}_-(x)$, and $\delta_t\bar{\phi}_+(x)$ goes into $\delta_t\bar{\phi}_-$. A positive frequency component is defined as possessing the form $\exp(-i\omega t)$, $\omega > 0$. The solution of (3.26) is

$$\bar{\phi}_+(x) = -\int d^4x'\ \left[\Delta_F(x-x')J_+(x') + \Delta_-(x-x')J_-(x')\right], \tag{3.27}$$

$$\bar{\phi}_-(x) = +\int d^4x'\ \left[\Delta_+(x-x')J_+(x') + \Delta_D(x-x')J_-(x')\right], \tag{3.28}$$

where

$$\Delta_F(x-x') = +\int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot(x-x')}(-p^2 - m^2 + i\epsilon)^{-1}, \tag{3.29}$$

$$\Delta_D(x-x') = -\int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot(x-x')}(-p^2 - m^2 - i\epsilon)^{-1}, \tag{3.30}$$

$$\Delta_+(x-x') = -\int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot(x-x')}\, 2\pi i\, \delta(p^2+m^2)\theta(+p^0), \tag{3.31}$$

$$\Delta_-(x-x') = -\int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot(x-x')}\, 2\pi i\, \delta(p^2+m^2)\theta(-p^0), \tag{3.32}$$

are the Feynman, Dyson and positive & negative frequency Wightman function, respectively. Comparing (3.23) and (3.27) we may write

$$Z_{\rm free}[J_+, J_-] = \exp\Big[-\frac{i}{2}\int d^4x d^4x' \Big\{J_+(x)\Delta_F(x-x')J_+(x') + J_+(x)\Delta_-(x-x')J_-(x') + J_-(x)\Delta_+(x-x')J_+(x') + J_-(x)\Delta_D(x-x')J_-(x')\Big\}\Big]. \tag{3.33}$$

Comparing with (3.21), we have, in the present convention, $G_{++}(x,x') = \Delta_F(x,x')$, $G_{--}(x,x') = \Delta_D(x,x')$, $G_{-+}(x,x') = \Delta_+(x,x')$, and $G_{+-}(x,x') = \Delta_-(x,x')$. From (3.33), (3.9), (3.10) and (3.11) we find, for a free field,

$$\langle 0_-|\mathcal{T}\ \Phi(x)\Phi(x')\ |0_-\rangle = i\,\Delta_F(x-x'), \tag{3.34}$$

$$\langle 0_-|\tilde{\mathcal{T}}\ \Phi(x)\Phi(x')\ |0_-\rangle = i\,\Delta_D(x-x'), \tag{3.35}$$

$$\langle 0_-|\ \Phi(x)\Phi(x')\ |0_-\rangle = i\,\Delta_+(x-x'). \tag{3.36}$$

For interacting fields, we write

$$Z[J_+,J_-] = \exp\left\{-i\,\frac{\lambda}{4!}\int d^4x\,\left[\frac{\delta^4}{\delta J_+^4(x)} - \frac{\delta^4}{\delta J_-^4(x)}\right]\right\} Z_{\rm free}[J_+,J_-] \tag{3.37}$$

with $Z_{\rm free}$ given by (3.33) and (3.28). From (3.37), the evaluation of Z, W or Γ proceeds as usual. There will be two kinds of vertices, "+" vertices coming from the $\delta^4/\delta J_+^4$ term, and "−" vertices coming from the $\delta^4/\delta J_-^4$ term. There are three kinds of internal legs, "$++$", "$--$" or "$+-$", distinguished by the signs of the vertices, the external source or the background fields to which they are attached. The corresponding Feynman rules are given by Eqs. (3.34), (3.35), (3.36), as can be seen by writing them as derivatives of $Z_{\rm free}$. Observe that, according to Eqs. (3.36) and (3.31), the momentum flowing through a "$+-$" line is always on the mass-shell. Also the "$+-$" lines never appear as external lines in the effective action, because they are eliminated by the Klein–Gordon operators which remove the propagators from these external legs.

3.1.3 One-Loop CTP Effective Action

The evaluation of the 'in-in' or CTP effective action $\Gamma_{\rm CTP}[\bar{\phi}_a]$ (we use subscript a,b to denote CTP indices $+$ or $-$) in loop order proceeds the same way as in the 'in-out' formulation. Loop order refers to an expansion of $W[J_a]$ in powers of $\hbar$. Now, solve (3.14) by the steepest descent method. Let $\bar{\phi}_a^{(0)}(x)$ be the solutions of the classical field equations; we expand the exponent in $Z[J_a]$ about these background fields up to the second derivative of $\mathcal{S}[\phi_a]$. The integration therein is Gaussian and we can write to one loop order,

$$e^{iW[J_a]} \simeq e^{iW^{(0)}[J_a]}\Big[\det A_{ab}(x,y)\Big]^{-1/2}, \tag{3.38}$$

where

$$W^{(0)}[J_a] = \mathcal{S}[\bar{\phi}_a^{(0)}] + \int d^nx\ J_a\bar{\phi}_a^{(0)}, \tag{3.39}$$

and $A_{ab}(x,y)$ is a 2×2 matrix defined by $A_{+-}(x,y)=A_{-+}(x,y)\equiv 0$, and

$$A_{++}(x,y)\equiv \left.\frac{\delta^2 S[\phi_+]}{\delta\phi_+(x)\delta\phi_+(y)}\right|_{\phi_+=\bar\phi_+^{(0)}}, \tag{3.40}$$

$$A_{--}(x,y)\equiv -\left.\frac{\delta^2 S[\phi_-]}{\delta\phi_-(x)\delta\phi_-(y)}\right|_{\phi_-=\bar\phi_-^{(0)}}. \tag{3.41}$$

In terms of the propagator $G(x,y)=A^{-1}(x,y)$ (both are 2×2 CTP matrices) which is a functional of the background fields $\bar\phi_a^{(0)}(x)$ we can write (3.38) as

$$W[J_a]\simeq W^{(0)}[J_a]-\frac{i}{2}\,\mathrm{Tr}\ln G. \tag{3.42}$$

The effective action which is a functional of $\bar\phi_a$ can now be explicitly found to the same order. Using the fact that $\bar\phi_a$ differs from $\bar\phi_a^{(0)}$ by a term of order $\hbar$ one can show from (3.39) that $W^{(0)}[J_a]\simeq \mathcal{S}[\bar\phi_a]+\int d^nx\; J_a\bar\phi_a$, and we have finally

$$\Gamma_{\mathrm{CTP}}[\bar\phi_a]\simeq \mathcal{S}[\bar\phi_a]-\frac{i}{2}\,\mathrm{Tr}\ln G. \tag{3.43}$$

Background Field Method

Background field techniques developed in the conventional 'in-out' framework carry over to the 'in-in' formalism easily. However, it is important to note that Φ_+ and Φ_- are not independent integration variables, but are linked through the boundary conditions on the hypersurface at a common time t^* in the future.

To adopt the background field method we can think of the pair (Φ_+,Φ_-) as a field doublet in some internal space. In particular, for the 1-loop effective action we will find

$$\Gamma[\bar\phi_+,\bar\phi_-]=S[\bar\phi_+]-S[\bar\phi_-]-\frac{i}{2}\ln \mathrm{D}et \begin{pmatrix} \left\{\Box-m^2+i\epsilon-\frac{\lambda}{2}\,\bar\phi_+^2\right\} & 0 \\ 0 & -\left\{\Box-m^2-i\epsilon-\frac{\lambda}{2}\,\bar\phi_-^2\right\} \end{pmatrix}^{-1}. \tag{3.44}$$

In computing the inverse, however, it is important to keep in sight the correct boundary conditions: $\bar\phi_+$ and $\bar\phi_-$ must satisfy the conditions stated following Eq. (3.26).

Infinities and Counteraction

The infinities which appear in the *in-in* effective action can be rendered finite by adding to the classical action the same set of counter terms used in the *in-out* formalism. The crucial observation here is that a primitively divergent graph in the *in-in* effective action contains only vertices of the same sign. If there were

vertices of different signs, at least two internal lines would be of "$+-$" type (the graph is 1PI), the corresponding momenta would be on-shell, the corresponding loop integral would be finite, and the graph would not be primitively divergent. Since the graphs of the *in-in* effective action with all vertices of the same sign are just the graphs of the *in-out* theory plus their complex conjugates, so the primitive divergences must be the same. Once the primitive divergences are controlled, it is only a matter of combinatorics to show that the overlapping divergences disappear as well.

An important particular case of the *in-in* vacuum formalism is that when either J_+ or J_- does not induce any transition from the vacuum to n-particle states. For example, if J_- does not create particles, the *in* vacuum evolves only by acquiring a phase, which is given by the vacuum persistence amplitude $Z[J]$ of Eq. (3.3). So we have, in this case only,

$$Z[J_+, J_-] = Z[J_-]^* Z[J_+], \tag{3.45}$$

$$W[J_+, J_-] = W[J_+] - W^*[J_-], \tag{3.46}$$

$$\Gamma[\bar{\phi}_+, \bar{\phi}_-] = \Gamma[\bar{\phi}_+] - \Gamma^*[\bar{\phi}_-]. \tag{3.47}$$

Here, functionals of a single argument refers to the conventional *in-out* approach. Eq. (3.47) shows that the conventional effective action may be considered as a particular case of the *in-in* one. For a massive $\lambda\Phi^4$ theory, a constant source (and in particular, a zero source) does not cause transitions from the vacuum.

3.2 'In-In' Formalism for Quantum Fields in Curved Spacetime

The extension of the *in-in* formalism to curved background geometry does not pose any new problem over and above those encountered in the conventional *in-out* approach. Of course, for a general curved spacetime there will not be a preferred "time," but still one can define *in* and *out* states on Cauchy surfaces $\Sigma_\pm$ in the distant past and future, and use an evolution operator of the form

$$U[\Sigma_+, \Sigma_-] = P \exp\left[i \int_{\mathcal{D}} d^4x \, \sqrt{-g} \, J(x)\Phi(x) \right], \tag{3.48}$$

where P stands for path ordering and $\mathcal{D}$ is the spacetime volume sandwiched in between these Cauchy surfaces. Surfaces of constant time now should be thought of as Cauchy surfaces, and all spacetime integrals should refer to the invariant measure $\sqrt{-g}\, d^4x$. In particular, conditions (3.26) on the background (classical) fields hold, although now the definition of positive frequency must be decided on a case-by-case basis in accordance with the appropriate normal mode decomposition. As in the usual approach, the definition of the vacuum states depends on whether there exist positive frequency modes in the asymptotic regions. The form of the propagators in (3.28) alters, but equations (3.27), (3.33) and (3.34), (3.35), (3.36) still hold.

The graphs which express the perturbative expansion of the *in-in* functionals are the same as in flat spacetime theory, although one loses the convenience of a globally defined momentum representation. For the few well-known field theories studied in curved space, those which are renormalizable in flat space-time are also renormalizable in curved background, and the counterterms in the *in-in* effective action are the same as those for the *in-out* effective action.

Semiclassical Theory of Gravity

The full *in-in* effective action is given by

$$\Gamma[g^+_{\mu\nu},\bar\phi_+,g^-_{\mu\nu},\bar\phi_-] = -i\ln\int Dh^+_{\mu\nu}Dh^-_{\mu\nu}D\varphi_+D\varphi_-D\psi_{\text{ghost fields}} \times\exp\Big[i\Big\{S[g^+_{\mu\nu}+h^+_{\mu\nu},\bar\phi_++\varphi_+]-S[g^-_{\mu\nu}+h^-_{\mu\nu},\bar\phi_-+\varphi_-] +(\text{gauge fixing, ghost and tadpole terms})\Big\}\Big] \tag{3.49}$$

where $g^\pm_{\mu\nu}$ is the gravitational background field, $h^\pm_{\mu\nu}$ the fluctuation field, and $\bar\phi_\pm,\varphi_\pm$ represent the matter background and fluctuation fields respectively. The field equations for gravity have the form

$$\frac{\delta\Gamma}{\delta g^\pm_{\mu\nu}} = \mp J^{\mu\nu}_\pm. \tag{3.50}$$

The physical gravitational field $g_{\mu\nu}$ in the presence of a source $J^{\mu\nu}$ and background matter fields $\bar\phi$ is obtained under the conditions $J^{\mu\nu}_+ = J^{\mu\nu}_- = J^{\mu\nu}$, $\bar\phi_+ = \bar\phi_- = \bar\phi$ and $g^+_{\mu\nu} = g^-_{\mu\nu} = g_{\mu\nu}$. Write $S[g_{\mu\nu},\bar\phi] = S_g[g_{\mu\nu}] + S_f[g_{\mu\nu},\bar\phi]$, where subscripts g and f denote gravitational and matter fields respectively. The semi-classical approximation amounts to neglecting the fluctuations of the gravitational fields and ghosts in (3.49), whereby one obtains

$$\Gamma[g^+_{\mu\nu},\bar\phi,g^-_{\mu\nu},\bar\phi^-] = S_g[g^+_{\mu\nu}] - S_g[g^-_{\mu\nu}] + \Gamma_f[g^+_{\mu\nu},\bar\phi_+,g^-_{\mu\nu},\bar\phi_-], \tag{3.51}$$

$$\Gamma_f[g^+_{\mu\nu},\bar\phi_+,g^-_{\mu\nu},\bar\phi_-] = i\ln\int D\varphi_+D\varphi_-\ \exp\Big\{i\,S_f[g^+_{\mu\nu},\bar\phi_++\varphi_+] - S_f[g^-_{\mu\nu},\bar\phi_-+\varphi_-]+\text{tadpoles}\Big\}. \tag{3.52}$$

The field equations become

$$\frac{\delta S_g}{\delta g^{\mu\nu}}[g_{\mu\nu}] = -J_{\mu\nu} - \frac{\delta\Gamma_f}{\delta g^{+\mu\nu}}[g_{\mu\nu},\bar\phi,g_{\mu\nu},\bar\phi] \tag{3.53}$$

where $g^+_{\mu\nu}$ and $g^-_{\mu\nu}$ must be identified only after the variation has been taken. Until that point, $g^-_{\mu\nu}$ is to be considered as just another matter field. This ensures that $\delta\Gamma_f/\delta g^+_{\mu\nu}$ is covariantly conserved, since $\Gamma_f - S_g[g^-_{\mu\nu}]$ is invariant under coordinate changes. $2\sqrt{-g}\,\delta\Gamma_f/\delta g^{+\mu\nu}$ yields the expectation value of the stress-energy tensor of the matter fields, taken with respect to the state which evolves

from the *in* vacuum under the influence of combined gravitational and matter fields $g_{\mu\nu}$ and $\bar{\phi}$.

In spacetimes where an unambiguously defined global time exists, such as in static spaces (e.g., Einstein universe or de Sitter universe in the static coordinate) or in conformally static spaces (e.g., the Friedmann–Lemaître–Robertson–Walker universes, see Chapter 2) it may be more convenient to adopt a preferred coordinate system chosen with these special properties to express the effective action. We will do this in many examples here and later.

For treating problems of statistical field theory in curved space, the closed-time-path functional formalism can be generalized by considering initial states described by a density matrix. This is related to the influence functional formalism [187] which has been applied to nonequilibrium quantum field-theoretical processes in gravitation, cosmology, particle nuclear and atomic condensed matter physics. (See, e.g., [326].)

3.3 In-In Effective Action in Bianchi Type I Universe

Let us illustrate the 'in-in' formalism described above with a calculation of the *in-in* effective action for scalar fields in a Bianchi Type I universe. To do this it is instructive to first work out the 'in-out' effective action discussed in the last chapter for particle creation backreaction considerations so we can compare the two formulations. We will see the distinct improvements the 'in-in' formalism brings forth.

3.3.1 Bianchi Universes

We provide some background description of the Bianchi universes. Bianchi classified all spatially homogeneous spacetimes into nine types (some types are further distinguished by a sub 0 or h, such as Type VII_h, etc.) [417, 418]. The class of Bianchi Types I–IX spaces, especially the Type I and Type IX spaces, encompasses many important classical cosmological models for the early universe. The 4D space-time maximally symmetric de Sitter universe is another important class, owing to its relevance to inflationary cosmology. The last part of this book is devoted to a discussion of quantum field theory in de Sitter space.

The line elements of the Bianchi universes are given by

$$ds^2 = -dt^2 + \sum_{a,b=1}^{3} \gamma_{ab}(t)\sigma^a(x)\sigma^b(x), \tag{3.54}$$

where $\gamma_{ab}(t)$ is the metric tensor and $\sigma^a(x)$ are the invariant basis one-forms on the homogeneous hypersurfaces, satisfying the structure condition $d\sigma^a = \frac{1}{2}C^a_{bc}\sigma^b \wedge \sigma^c$, where C^a_{bc} are the structure constants of the underlying Lie group. For a diagonal type-IX (mixmaster) universe, $C^a_{bc} = \epsilon_{abc}$ the totally

antisymmetric tensor, and $\gamma_{ab} = l_a^2 \delta_{ab}$, where l_a are the three principal radii of curvature. The case when two of the three l_a's are equal is called the Taub universe, which when all three are equal gives the closed Friedmann–Lemaître–Robertson–Walker (FLRW) universe.

The Bianchi Type I universe has zero intrinsic curvature but nonzero extrinsic curvature from its expansion and contraction, whose rates differ in three directions. The line element is given by

$$ds^2 = -dt^2 + \sum_{i,j=1}^{3} \ell_{ij}^2(t) dx^i dx^j = a^2(\eta)\Big[-d\eta^2 + \sum_{i,j=1}^{3} e^{2\beta(\eta)_{ij}} dx^i dx^j\Big] \tag{3.55}$$

where $\eta = \int dt/a$ is the conformal time and β_{ij} is a symmetric, traceless 3×3 matrix describing the anisotropy of the geometry. For Bianchi Type I one can choose a coordinate such that β_{ij} is diagonal[1] with the three elements β_i. The line element of Bianchi Type I universe is thus

$$ds^2 = -dt^2 + \sum_{i=1}^{3} \ell_i^2(t)(dx^i)^2 \tag{3.56}$$

where $\ell_i = a(t)e^{\beta_i}$ is the scale factor in the x_i direction. This is a generalization of the spatially flat isotropic FLRW universe to the case where the universe expands anisotropically. A useful quantity for this measure is the anisotropy parameter $Q_\beta \equiv -\frac{1}{2}\sum_{i>j}(\dot{\ell}_i/\ell_i - \dot{\ell}_j/\ell_j)^2$, where an overdot denotes d/dt. When there is no matter present a solution to the vacuum Einstein equation exists, called the Kasner universe:[2]

$$\ell_i(t) = t^{p_i}, \quad \text{where} \quad \sum_{i=1}^{3} p_i^2 = \sum_{i=1}^{3} p_i = 1. \tag{3.57}$$

We see from this relation that the universe expands when two of the p_is are positive and one negative, and contracts when two of the p_is are negative and one is positive. The Kasner solution is important because it is a generic behavior of the universe at every point in space near the singularity where the most general solution of the Einstein equation [116, 117] is found to be an inhomogeneous Kasner solution. It is also known as a 'velocity-dominated solution' [421] reflecting the fact that near the cosmological singularity the extrinsic curvature (measuring the time rate of change of the

[1] This diagonal form has full generality because there is no spatial curvature. In a Bianchi Type IX universe where the spatial curvature is present, spacetimes represented by the full matrix β_{ij} are more general than that of the diagonal metric which is Misner's mixmaster universe [419]; the off-diagonal components signify rotation [420].

[2] By contrast there is no vacuum FLRW solution. Thus in the FLRW class, "matter does matter."

scale factors) dominates over the intrinsic curvature (in the 3- geometry).[3] Defining the isotropic expansion rate $\alpha = \ln a$ then we see $p_i = \alpha + \beta_i$. The spatially flat FLRW universe corresponds to the case $\beta = 0$ with scale factor $a = e^{\alpha}$.

3.3.2 In-Out Effective Action

Let us first calculate the *in-out* effective action for a massless conformal scalar field in a spatially homogeneous universe with small anisotropy. This is an example of backreaction calculation on the effect of particle creation, leading to the damping of anisotropy in the early universe. Our discussion here follows [110].

Consider a free massless conformally coupled scalar field in a classical Bianchi Type I universe with line element (3.56) filled with classical radiation (with energy density $\rho_r = \tilde{\rho}_r/a^4$, where $\tilde{\rho}_r$ is a constant which measures the total number of photons in the universe). The classical action for radiation is given by (2.78).

For the problem at hand, in the limit of vanishing anisotropy, the universe will become a radiation-filled FLRW universe, depicted by the classical action (2.80). We shall make the small anisotropy approximation and calculate the effective action Γ up to the second order in the perturbation expansion in β, i.e.,

$$\Gamma[a,\beta] = \sum_n \Gamma_n(\beta^n), \quad n = 0, 1, 2 \cdots \tag{3.58}$$

Calculation of the third-order (β) contribution is necessary in order to determine the form of the counteraction completely.

The classical gravitational action expanded to $O(\beta^2)$ is given by

$$S_g^{(0)}[a,\beta] = V \int d\eta \left[-6(a')^2 + a^2 \beta'_{ij}\beta'^{ij}\right] = S_{g0} + S_{g2}. \tag{3.59}$$

Recall the superscript (0) denotes background spacetime, which, in the lowest order is the FLRW metric. The subscript number denotes the perturbation order in β. The classical scalar field action is

$$S_f[\Phi, g] = -\frac{1}{2}\int d^4x \sqrt{-g}\left[g^{\alpha\beta}\partial_a\Phi\partial_\beta\Phi + \frac{R}{6}\Phi^2\right]. \tag{3.60}$$

Expanding in powers of β, we have

$$S_f[\Phi, a, \beta] = S_0[\Phi, a] + \sum_{n=1}^{2} S_n[\Phi, a, \beta], \tag{3.61}$$

[3] These results were found by Belinsky, Khalatnikov and Lifshitz [116, 117] and Misner [419], thus often referred to as the BKL–M solutions. These are vacuum solutions, hence the saying "matter doesn't matter" in the BKL–M generalized Kasner class of solutions near the cosmological singularity, in contradistinction to the FLRW class.

where

$$S_0 = -\frac{1}{2}\int d^4x\, a^2\Big[\eta^{\alpha\beta}\partial_\alpha\Phi\partial_\beta\Phi + (a''/a)\Phi^2\Big], \tag{3.62}$$

$$S_1 = \int d^4x\, a^2\, \beta^{ij}\partial_i\Phi\partial_j\Phi, \tag{3.63}$$

$$S_2 = -\int d^4x\, a^2\Big[\frac{1}{12}\beta'_{ij}\beta'^{ij}\Phi^2 + \beta^{ik}\beta^j_k\,\partial_i\Phi\partial_j\Phi\Big], \tag{3.64}$$

where $\eta^{\alpha\beta}$ denotes the Minkowski metric. We now expand the Green's function in powers of β. The Green's function G_0 in the limit of exact isotropy ($\beta_{ij} = 0$ in (3.56)) is conformally related to the flat-space Green's function. If we denote symbolically the contribution to the wave equation of first-order action in (3.63) by V_1, and that of the second-order action in (3.64) by V_2 etc., then we can write

$$G = G_0 + G_0\,(V_1 + V_2 + \cdots)\,G = G_0 + G_0V_1G_0 + \Big[G_0V_2G_0 + G_0V_1G_0V_1G_1\Big] + \cdots \tag{3.65}$$

Inserting this into (2.57) we find,

$$\Gamma_0[a] = -6V\int d\eta\,\left(\frac{a'}{\kappa}\right)^2 - \frac{i}{2}\Big[\mathrm{Tr}\ln G_0\Big]_{\rm reg}, \tag{3.66}$$

$$(\Gamma_{00}) \qquad (\Gamma_{01})$$

$$\Gamma_1[a,\beta] = -\frac{i}{2}\Big[\mathrm{Tr}(V_1G_0)\Big]_{\rm reg} \tag{3.67}$$

$$(a)$$

$$\Gamma_2[a,\beta] = \frac{V}{16\pi G_{\rm N}}\int d\eta\, a^2\beta'_{ij}\beta'^{ij} - \frac{i}{2}\Big[\mathrm{Tr}(V_2G_0)\Big]_{\rm reg} - \frac{1}{4}\Big[\mathrm{Tr}(V_1G_0V_0G_0)\Big]_{\rm reg} \tag{3.68}$$

$$(\Gamma_{20}) \qquad (b1) \qquad (b2)$$

$$\Gamma_3[a,\beta] = -\frac{i}{2}\Big[\mathrm{Tr}(V_3G_0 + V_2G_0V_1G_0 + \frac{1}{3}V_1G_0V_1G_0V_1G_0)\Big]_{\rm reg}. \tag{3.69}$$

$$(c1) \qquad (c2) \qquad (c3)$$

(In the above, quantities with a subscript 'reg' indicates they are regularized.) Those terms of higher order than the lowest can be represented by a series of Feynman diagrams described below.

In the limit of exact isotropy one gets the result for a FLRW universe. As we saw in Chapter 2 the one-loop effective action $\Gamma_0[a]$ for a conformally invariant scalar field contains the trace anomaly, whose effect will be discussed in Chapter 8.

To calculate the regularized trace of G in its higher-order contributions to Γ_1 and Γ_2, we shall use the method of dimensional regularization in the number

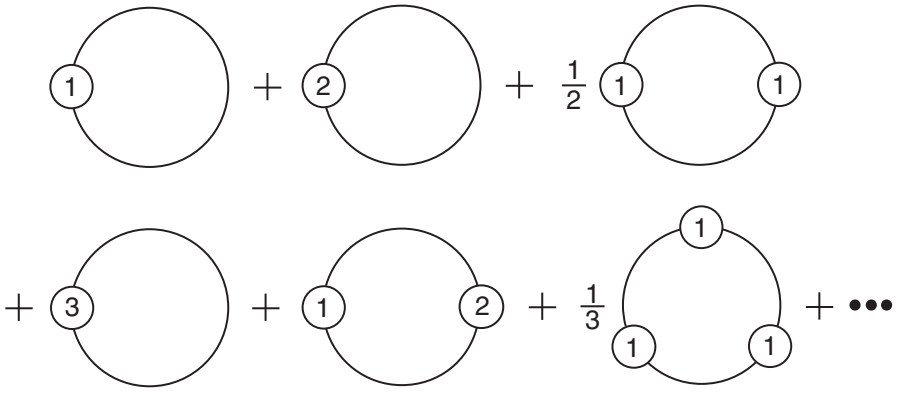


Figure 3.1 Feynman diagram expansion of the effective action. The circles containing the number n represent the interaction V_n.

of conformally related flat-space dimensions. In this prescription one first writes down the n-dimensional Green's function in the limit of exact isotropy as

$$G_0(x,x') = [a(\eta)]^{1-n/2} G_F(x,x') [a(\eta')]^{1-n/2}, \tag{3.70}$$

where

$$G_F(x,x') = -\int \frac{d^n k}{(2\pi)^n} \frac{e^{ik'(x-x')}}{k^2 - i\epsilon} \tag{3.71}$$

is the flat-space Feynman Green's function for a scalar field. Inserting these expressions into (3.67) and dimensionally regulating the Feynman integrals, we see that only terms (b2), (c2), and (c3) in (3.68) and (3.69) contribute to the one-loop effective action. Their divergent parts are to be canceled by the addition of a counteraction S_c in the effective action. The most general form of S_c which is conformally invariant (or a pure divergence in 4-dimension) is

$$S_c = \frac{\mu_c^{n-4}}{n-4} \int d^n x \sqrt{-\bar{g}}\, (A\, G + B\, F), \tag{3.72}$$

where $\bar{g}$ is the background metric, μ_c is the renormalization mass scale with dimensions of inverse length, and F, G were defined in (2.61) The numerical constants A and B are given by

$$B = (1920\pi^2)^{-1} = -3A, \qquad\qquad A = -(5760\pi^2)^{-1}. \tag{3.73}$$

There are a number of ways to determine these coefficients [352], one of which [111] is via a perturbative expansion from the second and third order (in β) divergent terms in Γ. This counteraction generates a trace anomaly of the form given in Eq. (2.62), where α, β, and γ are related to A, B by a general relation [422]

$$\alpha = 2B/3, \qquad \beta = -2A, \qquad \text{and} \qquad \gamma = A + B. \tag{3.74}$$

After adding the counteraction to cancel the ultraviolet divergences one finds the regularized ('in-out') one-loop effective action to be

$$\Gamma_2[a,\beta] = V\int_{-\infty}^{\infty} d\eta \left\{ \left[\left(\frac{a}{\kappa}\right)^2 - \varpi\left(\frac{a''}{a}\right) - \varpi\left(\frac{a'}{a}\right)^2 \right] \beta'_{ij}\beta'^{ij} + 3\varpi\left[i\,\frac{\pi}{2} + \ln a \right] \beta''_{ij}\beta''^{ij} \right\} - 3V\varpi \int_{-\infty}^{+\infty} d\eta \int_{-\infty}^{+\infty} d\eta'\ \beta''_{ij}(\eta)\tilde{K}(\eta-\eta')\beta''^{ij}(\eta'). \tag{3.75}$$

where $\varpi = (2880\pi^2)^{-1}$, and

$$\tilde{K}(\eta) = \frac{1}{\pi}\int_0^{\infty} d\omega\ \cos(\omega\eta)\ \ln\frac{\omega}{\mu}. \tag{3.76}$$

We have combined all scales which enter in the same way into a single regularization scale μ. Note that the expression is complex and nonlocal.

One can now determine the effective geometry by varying this effective action $\Gamma = \Gamma_0 + \Gamma_2$ with respect to a and to β_{ij}. Thus, the equation for $a(\eta)$

$$\delta\Gamma[a,\beta]/\delta a = 0 \tag{3.77}$$

is, to all orders in β, the modified Einstein equation

$$R = -8\pi G_{\rm N} T_A \tag{3.78}$$

where T_A is the trace anomaly for the Type I universe. To second order in β it reads,

$$\begin{aligned} &-\frac{a^2}{\varpi\kappa^2}\Big[6\frac{a''}{a} + \mathrm{Tr}(\beta'^2)\Big] \\ &\quad = -\frac{3a''''}{a} + \frac{12a'a'''}{a^2} + 9\Big(\frac{a''}{a}\Big)^2 - 24\Big(\frac{a'}{a}\Big)^2\Big(\frac{a''}{a}\Big) + 6\Big(\frac{a'}{a}\Big)^4 \\ &\qquad + \mathrm{Tr}\Big[\frac{3}{2}\Big((\beta')'\Big)^2 + 2\Big(\frac{a'}{a}\Big)\Big(\beta'^2\Big)' - \frac{1}{2}\Big(\beta'^2\Big)'' + 2\Big(\frac{a'}{a}\Big)'\beta'^2\Big], \end{aligned} \tag{3.79}$$

where $\beta'_{ij} \equiv d\beta_{ij}/d\eta$ and we have used β'^2 as a shorthand for $\beta'_{ij}\beta'^{ij}$. The equation governing the anisotropy function is

$$\delta\Gamma[a,\beta]/\delta\beta_{ij} = \tau^{ij}, \tag{3.80}$$

where the external source τ^{ij} which fixes the amount of anisotropy in the model vanishes everywhere except on the boundaries of the spacetime region under consideration. From this and the form of Γ_2 given by (3.68) one can easily check that the source term in (2.48) is given by $-2\Gamma_2$ and hence $W = \Gamma_0 - \Gamma_2$. In the interior region under consideration, $\tau^{ij} = 0$. Since Γ_0 and Γ_1 are independent of

β and to $O(\beta^2), \Gamma_2$ depends only on κ_{ij} and κ'_{ij}, the equation which governs the anisotropy can be reduced to

$$\delta\Gamma_2[a,\beta']/\delta{\beta'}^{ij} = -2c_{ij}, \tag{3.81}$$

where c_{ij} is a 3×3 matrix of constants. Written explicitly (3.81) reads

$$3\varpi\frac{d}{d\eta}\Big\{-\Big[\frac{i\pi}{2}+\ln a\Big]\frac{d{\beta'}_{ij}}{d\eta}+\int_0^\infty d\eta'\; K(\eta-\eta')\,\frac{d{\beta'}_{ij}}{d\eta'}\Big\}$$
$$+\Big[\Big(\frac{a}{\kappa}\Big)^2-\varpi\Big(\frac{a'}{a}\Big)^2-\varpi\Big(\frac{a''}{a}\Big)\Big]\,{\beta'}_{ij}=c_{ij}. \tag{3.82}$$

This is a linear integro-differential equation for the anisotropy function ${\beta'}_{ij}$ for any given scale factor $a(\eta)$. Eqs. (3.79) and (3.82) are two coupled equations for a and β_{ij} and thus the entire classical geometry to this order. With suitable boundary conditions they determine the vacuum persistence amplitude, the particle production probabilities, and the classical geometry with self-consistent backreaction of the produced particles. The self-consistency at all orders makes the effective action a better method for backreaction problems than the equations of motion approach for the geometry and the fields, as the latter usually involves staggered orders. However, note the appearance of the $\frac{i\pi}{2}$ term in (3.82) which results in a complex geometry. This is symptomatic of the 'in-out' formulation which yields complex transition amplitudes in general. It can produce correctly the probability of particle creation in the whole history of the universe, which we shall show below. But to obtain a real and causal equation of motion we need to use the 'in-in' effective action which produces physical expectation values. This we shall show in the next section.

One can formally expand the classical geometry in powers of the initial anisotropy. Thus, for the scale factor,

$$a(\eta) = a_0(\eta) + a_2(\eta) + \cdots . \tag{3.83}$$

Here $a_0(\eta)$ is the scale factor for the exactly isotropic universe, the behavior of which under the trace anomaly will be discussed in Chapter 8. It does not contribute to conformal field particle creation because, as explained in Chapter 2, the FLRW universe is conformally static. Particle production happens only when there is deviation from conformality, as for an anisotropic universe. The quantity a_2 represents the corrections to the scale factor including the backreaction of the produced particles due to anisotropic expansion. The lowest-order term in a similar expansion for the anisotropy function κ_{ij} satisfying (3.82) is linear in the initial anisotropy. (We shall understand by β_{ij} this solution in linear order.) It is this term which gives rise to particle production and contributes to the corrections a_2 in the scale factor by its backreaction. The total particle production probability P (when it is small) obtained by Hartle and Hu [110] is given (to second order in the initial anisotropy) by

$$P \simeq 2\,\mathrm{Im}\,\Gamma_2[a_0,\beta] = \frac{1}{1920\pi}\int d^4x\sqrt{-g}\;C^*{}_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta} \tag{3.84}$$

where $C_{\alpha\beta\gamma\delta}$ is the Weyl curvature tensor which measures the deviation of spacetime from conformal flatness, and a star indicates its complex conjugate. This expression is shared by particle creation in weakly inhomogeneous cosmologies as shown by Campos and Verdaguer [423], discussed in Chapter 12.

3.3.3 In-In Effective Action

We now proceed to derive the 'in-in' effective action for the same problem. We begin by doubling the matter field (Φ_+,Φ_-) and the anisotropy metric variables $(\beta^+_{ij},\beta^-_{ij})$ which we may at times denote simply by (β^+,β^-),

$$\Gamma_f[\beta^+,\beta^-] = -i\ln\int D\Phi_+ D\Phi_-\;\exp\Big\{iS_f[\beta^+,\Phi_+] - S_f[\beta^-,\Phi_-]\Big\}. \tag{3.85}$$

Note that only the equations which follow from the variation of Γ with respect to β^+_{ij} need to be considered (e.g., terms like $-S_g[\beta^-_{ij}]$ can be omitted). The scalar field contribution is, from (3.61),

$$\begin{aligned}\Gamma_f[\beta^+,\beta^-] = -i\ln\int D\Phi_+ D\Phi_-\;&\exp\Big\{i\Big(S_0[\Phi_+] - S_0[\Phi_-]\Big)\Big\}\\ &\times\Big\{1 + i\,S_1[\beta^+,\Phi_+] - i\,S_1[\beta^-,\Phi_-] + i\,S_2[\beta^+,\Phi_+] - i\,S_2[\beta^-,\Phi_-]\\ &-\frac{1}{2}\,S_1[\beta^+,\Phi_+]^2 - \frac{1}{2}\,S_1[\beta^-,\Phi_-]^2 + S_1[\beta^+,\Phi_+]S_1[\beta^-,\Phi_-] + \mathcal{O}(\beta^3)\Big\}\end{aligned} \tag{3.86}$$

where S_0, S_1, S_2 are terms in the scalar field action expanded in orders of β given by Eqs. (3.62), (3.63) and (3.64) respectively.

Computing the functional integrals with respect to the isotropic action and taking its logarithm are equivalent to computing each term as a sum of Feynman graphs and retaining only the connected ones. The two-point functions in the background FLRW space are given by

$$\langle\Phi_+(x)\Phi_+(x')\rangle = i\left[a(\eta)a(\eta')\right]^{1-n/2}\Delta_F(x-x'), \tag{3.87}$$

$$\langle\Phi_-(x)\Phi_-(x')\rangle = i\left[a(\eta)a(\eta')\right]^{1-n/2}\Delta_D(x-x'), \tag{3.88}$$

$$\langle\Phi_+(x)\Phi_-(x')\rangle = i\left[a(\eta)a(\eta')\right]^{1-n/2}\Delta_-(x-x'), \tag{3.89}$$

where $\Delta_{F,\,D}$ and Δ_- are the Feynman, Dyson and negative-frequency Wightman functions, respectively, with $m^2 = 0$. The terms in the *in-in* effective action which contribute to the variation with respect to β^+ are

$$\Gamma_f[\beta^+,\beta^-] = \frac{i}{2}\,\langle S_1[\beta^+,\Phi_+]^2\rangle - i\,\langle S_1[\beta^+,\Phi_+]S_1[\beta^-,\Phi_-]\rangle. \tag{3.90}$$

One can now begin to see the difference from the *in-out* approach, i.e., the second term is new. Using (3.60), Wick's theorem, (3.87) and the explicit forms of Δ_F, Δ_D and Δ_- we get

$$\begin{aligned}
&\frac{i}{2}\,\langle S_1[\beta^+,\Phi_+]^2\rangle \\
&\quad = -i\int d^n x\, d^n x'\ \beta^+_{ij}(\eta)\beta^+_{kl}(\eta')\int\frac{d^n p}{(2\pi)^n}\ e^{ip\cdot(x-x')}\int\frac{d^n q}{(2\pi)^n} \\
&\qquad q^i q^k (p-q)^j (p-q)^l [-q^2+i\varepsilon]^{-1}\times[-(p-q)^2+i\varepsilon]^{-1}, \qquad (3.91)\\
&-i\,\langle S_1[\beta^+,\Phi_+]S_1[\beta^-,\Phi_-]\rangle \\
&= 2i\int d^n x\, d^n x'\ \beta^+_{ij}(\eta)\beta^-_{kl}(\eta')\int\frac{d^n p}{(2\pi)^n}e^{ip\cdot(x-x')}\int\frac{d^n q}{(2\pi)^n} \\
&\quad q^i q^k (p-q)^j (p-q)^l\Big\{2\pi i\,\delta(q^2)\,\theta(-q^0)\Big\}\Big\{2\pi i\,\delta[(p-q)^2]\,\theta[-(p^0-q^0)]\Big\}. \qquad (3.92)
\end{aligned}$$

Since the anisotropy depends only on η we may integrate over the space variables to get (setting $V=1$)

$$\begin{aligned}
\frac{i}{2}\,\langle S_1[\beta^+,\Phi_+]^2\rangle &= \int d\eta\, d\eta'\ \beta^+_{ij}(\eta)\beta^+_{kl}(\eta')\int\frac{d\omega}{2\pi}\ e^{-i\omega(\eta-\eta')} \\
&\qquad\Big[-i\int\frac{d^n q}{(2\pi)^n}\ q^i q^j q^k q^l [-q^2+i\epsilon]^{-1}\times[(\omega-q^0)^2-\mathbf{q}^2+i\epsilon]^{-1}\Big], \qquad (3.93)\\
i\,\langle S_1[\beta^+,\Phi_+]S_1[\beta^-,\Phi_-]\rangle &= 2\int d\eta d\eta'\ \beta^+_{ij}(\eta)\beta^-_{kl}(\eta')\int\frac{d\omega}{2\pi}\ e^{-i\omega(\eta-\eta')} \\
&\qquad\Big[+i\int\frac{d^n q}{(2\pi)^n}\ q^i q^j q^k q^l\Big\{2\pi i\,\delta[q^2]\,\theta[-q^0]\Big\} \\
&\qquad\times\Big\{2\pi i\,\delta[(\omega-q^0)^2-\mathbf{q}^2]\,\theta[-(\omega-q^0)]\Big\}\Big]. \qquad (3.94)
\end{aligned}$$

If the argument of both δ-functions must vanish then $\omega=+2q^0$. Since q^0 must be negative, we may write

$$\begin{aligned}
&-i\,\langle S_1[\beta^+,\Phi_+]S_1[\beta^-,\Phi_-]\rangle \\
&\qquad = 2\int d\eta d\eta'\ \beta^+_{ij}(\eta)\beta^-_{kl}(\eta')\int\frac{d\omega}{2\pi}\ e^{-i\omega(\eta-\eta')}\theta(-\omega) \\
&\qquad\quad\times\Big\{+i\int\frac{d^n q}{(2\pi)^n}\ q^i q^j q^k q^l\Big\{2\pi i\,\delta(q^2)\,\theta(-q^0)\Big\}\Big\{2\pi i\,\delta[(\omega-q^0)^2-\mathbf{q}^2]\Big\}. \qquad (3.95)
\end{aligned}$$

One could compute this by brute force, but it is perhaps more suggestive to note that the q integral here (including the i factor) is simply twice the imaginary part of the q-integral in Eq. (3.93), computed according to the Cutkosky rules.

Now the "++ graph" in Eq. (3.93) is also the scalar field contribution to the *in-out* effective action. It can be written as

$$\frac{i}{2}\,\langle S_1[\beta^+,\Phi_+]^2\rangle = \int d\eta d\eta'\;\beta^+_{ij}(\eta)\beta^{+ij}(\eta')\int_{-\infty}^{+\infty}\frac{d\omega}{2\pi}\;e^{-i\omega(\eta-\eta')} \times\left\{\frac{-(-\omega^2)^2}{4(4\pi)^2(n^2-1)}\left[\epsilon^{-1}+\frac{1}{2}\ln\frac{-(\omega^2+i\epsilon)}{4\pi\mu^2}\right]\right\}. \tag{3.96}$$

The new +− contribution is given by

$$-i\,\langle S_1[\beta^+,\Phi_+]S_1[\beta^-,\Phi_-]\rangle = 2\int d\eta d\eta'\;\beta^+_{ij}(\eta)\beta^{-ij}(\eta')\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\;e^{-i\omega(\eta-\eta')} \times\left\{\frac{-(-\omega^2)^2}{4(4\pi)^2(n^2-1)}\,i\pi\,\theta(-\omega)\right\}. \tag{3.97}$$

Observe that only the ++ term is divergent, therefore the pole term is the same as in the *in-out* approach.

Introduce the notation

$$R_n(\eta)\equiv\int\frac{d\omega}{2\pi}\;e^{-i\omega\eta}\;\frac{(-\omega^2)^{\frac{n}{2}}}{2}\ln\left[\frac{-(\omega^2+i\varepsilon)}{4\pi\mu^2}\right], \tag{3.98}$$

$$I_n(\eta)\equiv\int\frac{d\omega}{2\pi}\;e^{-i\omega\eta}\;(-\omega^2)^{\frac{n}{2}}\;\pi\,\theta(-\omega), \tag{3.99}$$

and comparing (3.96) and (3.75), we observe that the pole parts cancel, leaving a finite residue. Finally we obtain the equation for β_{ij} :

$$\begin{aligned}&\frac{\delta}{\delta\beta^+_{ij}}\big(S_g+\Gamma_f\big)\Big|_{\beta^+=\beta^-=\beta}\\ &\quad=-2\epsilon_0\,\frac{d}{d\eta}\big(a^2\beta'_{ij}\big)+\frac{1}{30(4\pi)^2}\frac{d^2}{d\eta^2}\Big[\beta''_{ij}\ln(\mu a)\Big]+\frac{1}{90(4\pi)^2}\frac{d}{d\eta}\Big[\Big(\frac{a'^2}{a^2}+\frac{a''}{a}\Big)\beta'_{ij}\Big]\\ &\qquad-\frac{1}{30(4\pi)^2}\int_{-\infty}^{\infty}d\eta'\;\beta_{ij}(\eta')\,R_4(\eta-\eta')-\frac{1}{30(4\pi)^2}\int_{-\infty}^{\infty}d\eta'\;\beta_{ij}(\eta')\,i\,I_4(\eta-\eta')\\ &\quad=-J_{ij}(\eta).\end{aligned} \tag{3.100}$$

(Recall from Chapter 2 that $\epsilon_0\equiv\frac{1}{16\pi G_{\rm N}}$.) We have introduced an external source J_{ij} in order to be able to switch on the anisotropy in the distant past and redefined μ to absorb the nonessential constants.

The difference between Eq. (3.100) deduced from the *in-in* effective action and Eq. (3.82) deduced from the *in-out* approach lies exclusively in the term containing the I kernel. When this term is absent, the resulting equation (3.82), as we noticed before, is neither real nor causal. This becomes obvious if we write

$$\frac{1}{2}\ln\left[\frac{-(\omega^2+i\epsilon)}{4\pi\mu^2}\right]=\ln\frac{|\omega|}{\sqrt{4\pi\mu^2}}-i\,\frac{\pi}{2}, \tag{3.101}$$

and

$$R_4(\eta) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \left[\omega^4 \cos(\omega\eta) \ln \frac{|\omega|}{\sqrt{4\pi\mu^2}} \right] - i \, \frac{\pi}{2} \frac{d^4}{d\eta^4} \, \delta(\eta). \tag{3.102}$$

(Hereafter we choose the cut of the logarithm along the negative real axis.) The nonlocal part of (3.100) is given by the kernel

$$\begin{aligned} K_n(\eta) = R_n(\eta) + i \, I_n(\eta) &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \, e^{-i\omega\eta} \, (-\omega^2)^{\frac{n}{2}} \left[\ln \frac{|\omega|}{\sqrt{4\pi\mu^2}} + i \, \frac{\pi}{2} \, \mathrm{sgn}(-\omega) \right] \\ &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \, e^{-i\omega\eta} \, (-\omega^2)^{\frac{n}{2}} \, \ln \frac{-i(\omega + i\varepsilon)}{\sqrt{4\pi\mu^2}}. \end{aligned} \tag{3.103}$$

The third identity above follows from the second identity via (3.100),

$$\frac{\partial}{\partial\omega} \Big[\ln|\omega| + i \, \frac{\pi}{2} \, \mathrm{sgn}(-\omega) \Big] = \frac{1}{\omega} - i \, \pi \, \delta(\omega) = (\omega + i \, \varepsilon)^{-1}. \tag{3.104}$$

It is clear that for even n, $K(\eta)$ is real, because the real part of its Fourier transform is even and its imaginary part is odd. It is causal because all the singularities of the integral lie in the upper complex half plane. We observe that a causal and non-local equation could not have been derived from an action functional depending solely on β^+ as in the *in-out* approach. In fact one can eliminate the variable ω completely. Explicitly, the kernel $K_n(\tau)$ can be written as

$$K_n(\tau) = \theta(\tau) \, \frac{1}{(\sqrt{4\pi\mu^2} \, \tau)^{n+1}} \, \frac{1}{\Gamma(-n)} \Big[\psi(-n) - \ln(\sqrt{4\pi\mu^2} \, \tau) \Big], \tag{3.105}$$

for all n with $\psi(z) = \frac{d}{dz} \ln \Gamma(z)$ where $\Gamma(z)$ is the Gamma function. In principle one should use the formula for $n = 4$, but the resulting η' integral would not be well-defined (as the definition of the Fourier transform of a distribution is incorrectly applied). We will obtain a well-defined expression if we write

$$\begin{aligned} \int_{-\infty}^{\infty} d\eta' \, \beta_{ij}(\eta') K_4(\eta - \eta') &= \int_{-\infty}^{\infty} d\eta' \, \beta_{ij}(\eta') \Big[-\frac{\partial^5}{\partial\eta'^5} \, K_{-1}(\eta - \eta') \Big] \\ &= \int_{-\infty}^{\eta} d\eta' \, \Big[\frac{\partial^5}{\partial\eta'^5} \beta_{ij}(\eta') \Big] \Big[\ln(\eta - \eta') - \psi(1) \Big]. \end{aligned} \tag{3.106}$$

The term containing $\psi(1)$ is a local term which can be absorbed in a redefinition of the unit of mass μ in (3.100). In order to find the actual evolution of β_{ij} it is convenient to integrate both sides of (3.100) once with respect to η. The resulting equation is

$$\begin{aligned} &- 2\epsilon_0 \, a^2 \beta'_{ij} + \frac{1}{30(4\pi)^2} \frac{d}{d\eta} \Big\{ \beta''_{ij} \ln(\mu a) \Big\} + \frac{1}{90(4\pi)^2} \Big(\frac{a'^2}{a^2} + \frac{a''}{a} \Big) \beta'_{ij} \\ &\quad - \frac{1}{30(4\pi)^2} \int_{-\infty}^{\eta} d\eta' \Big[\frac{d^4}{d\eta'^4} \beta_{ij}(\eta') \Big] \ln(\eta - \eta') \\ &\quad = - \int_{-\infty}^{\eta} d\eta' J_{ij}(\eta') = -c_{ij}(\eta). \end{aligned} \tag{3.107}$$

If the source J_{ij} operates only in the distant past, we may take c_{ij} to be a constant for all finite values of η. Although Eq. (3.107) is different in form from those obtained from the *in-out* approach, its solution is close to (the real part of) the solution from the latter, as given in the previous section. In particular, (3.107) admits the conformally complete solution with β' going to a constant as $\eta \to -\infty$, while approaching the classical behavior $\beta'_{ij} \sim c_{ij}\eta^{-2}$ as $\eta \to +\infty$. This is the case because the behavior of β_{ij} is dominated by the local terms in both limits.

Particle Creation as a Dissipative Process

We mentioned in Chapter 1 that the backreaction of a system's environment often engenders dissipative dynamics in the open system. The equations of motion governing the open system derived from an *in-in* effective action being real and causal (in time domain), one can unambiguously identify in them the source of such dissipative behavior. In the present problem, the kernel K_4 acquires (in frequency domain) an imaginary part because (and only because) the conformal vacuum is unstable in the presence of anisotropy or inhomogeneity. This gives an unambiguous physical meaning to the characterization of particle creation as a dissipative process [182, 424].

We may obtain a quantitative check on the relationship between the imaginary part of K_4 and particle creation if we consider the energy density of the fields, $a^4T_0^0$. This can be found from the conservation law

$$(a^4T_0^0)' = a'\frac{\partial}{\partial a}\Gamma_f + \beta'_{ij}\frac{\partial}{\partial\beta^+_{ij}}\Gamma_f. \tag{3.108}$$

The contribution of the non-local terms in (3.100) to the energy density is

$$\frac{1}{30(4\pi)^2}\int_{-\infty}^{\eta} d\eta'\ \beta'_{ij}(\eta')\int_{-\infty}^{\infty} d\eta''\ \beta^{ij}(\eta'')K_4(\eta'-\eta''). \tag{3.109}$$

As $\eta \to \infty$ we may write this in terms of $\beta_{ij}(\omega) = \int d\eta\, e^{+i\omega\eta}\,\beta_{ij}(\eta)$, i.e.,

$$\frac{1}{30(4\pi)^2}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\,\omega^4\Big[\ln\frac{|\omega|}{\sqrt{4\pi\mu^2}} + i\,\frac{\pi}{2}\,\mathrm{sgn}(-\omega)\Big](+i\,\omega)\,\mathrm{Tr}\big\{\beta^*(\omega)\beta(\omega)\big\}. \tag{3.110}$$

We see that in the limit $\eta \to \infty$, the contribution of the real part cancels off, while the imaginary part gives a positive definite contribution. Thus the energy density associated with anisotropy dissipated is

$$\rho_{ad} = \frac{1}{120(4\pi)^2}\int_0^{\infty} d\omega\ (2\omega)\ \omega^4\,\mathrm{Tr}\big\{\beta^*(\omega)\beta(\omega)\big\}. \tag{3.111}$$

Since the cause of anisotropy dissipation arises from particle creation, this energy density in geometrodynamics must be balanced by the energy density of the

particles created. To see this, observe that the spectrum of particle pairs created by a given anisotropy history $\beta_{ij}(\eta)$ is

$$P(\omega) = \frac{\omega^4}{30(\pi)^2}\,\mathrm{Tr}\big\{\beta^*(2\omega)\beta(2\omega)\big\}. \tag{3.112}$$

Thus the total energy of particles created in the whole history is given by

$$\rho_{pc} = \int_0^\infty d\omega\ (2\omega)\,P(\omega) \tag{3.113}$$

which is equal to ρ_{ad}, as it should be.

To convince ourselves that the particles in (3.112) are really the created ones, it is enough to see that as $\beta \to +\infty$ all the vacuum polarization terms fade away and $a^4T_0^0$ reduces to $\rho_{pc} + constant$. In other words, the newly created particles are now behaving just like the classical radiation originally present. Moreover, K_4 is the only non-local kernel leading to a real causal equation with just the right amount of particles created. This is so because reality and (3.109) determine the imaginary part of K_4, and causality implies a Kramers–Kronig relation which then fixes the real part up to local terms.

We see from the above example of anisotropy dissipation that the *in-in* effective action is the appropriate formalism to use for treating backreaction problems of quantum process. It has the advantage that (a) the field equations are real and causal, so that the solutions can be identified with the dynamics of physical fields and geometry, and (b) the formalism provides an unambiguous description of the dissipative nature of the backreaction effect of quantum processes [184]. This method has also been successfully incorporated in statistical field theory for the treatment of a variety of nonequilibrium quantum processes. See, e.g., [326]. We shall continue the studies of cosmological backreaction problems in Chapter 8.

3.4 Expectation Value of the Stress Energy Tensor for Interacting Fields

In this section we discuss the *in-in* effective action for self-interacting quantum fields in curved spacetimes. We continue to use the $\lambda\Phi^4$ as an example to illustrate higher loop calculations, renormalization of the *in-in* effective action, and the calculation of vacuum expectation values of the stress-energy tensor.

The gravitational action has been introduced in (2.67). The action for a scalar $\lambda\Phi^4$ field in a general n-dimensional background spacetime is given by (2.82):

$$S_f[\Phi, g] = \int d^n x\,\sqrt{-g}\,\Big\{\frac{1}{2}\Phi(x)\Box_g\Phi(x) - \frac{1}{2}\Big[m^2 + (1-\xi)\,\xi_n R\Big]\Phi^2(x) - \frac{\lambda}{4!}\,\Phi^4(x)\Big\}. \tag{3.114}$$

where, as a reminder, $\Box_g$ denotes the Laplace–Beltrami operator in terms of the covariant derivative ∇_μ in a n-dimensional background spacetime with metric $g_{\mu\nu}$ and scalar curvature R. As before, $\xi_n = \frac{(n-2)}{4(n-1)}$ and we use $\xi = 0,\ 1$ for conformal and minimal coupling of the field to the spacetime respectively. From this action one can obtain the stress-energy tensor of the matter field

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_f}{\delta g^{\mu\nu}}, \tag{3.115}$$

which acts as source of the semiclasical Einstein equation.

We assume that in the infinite past m^2, ξ and λ are adiabatically switched off. This defines the 'in'-vacuum to be the conformal vacuum. We will make a perturbative expansion of the *in-in* effective action in powers of these parameters, specifically, to first order in λ and to second order in $M_\xi^2 = m^2 + (1-\xi)\xi_n R$. This means we will consider graphs with no more than two loops. We will need to know the two-loop counter terms in S_g^{ct}, and the one-loop counterterms in S_f^{ct} (the background scalar field is assumed to be zero). These have been computed in [374]. To this order there is neither wave function nor coupling constant renormalization.

We now specialize to the isotropic, spatially flat FLRW spacetime with line element (2.26). As λ and M_ξ are adiabatically switched on, conformal invariance is broken and the vacuum becomes unstable. We must now consider the full CTP effective action as a function of two variables a_+, a_-

$$\Gamma[a_+, a_-] = S_g[a_+] - S_g[a_-] - i\,\ln\int D\Phi_+ D\Phi_- \exp\Big\{i\Big[\Big(S_f + S_f^{ct}\Big)(\Phi_+, a_+) - \Big(S_f + S_f^{ct}\Big)(\Phi_-, a_-)\Big]\Big\}. \tag{3.116}$$

Introduce a change of variables $\Phi_\pm = (a_\pm)^{1-n/2}\chi_\pm$. The Jacobian of it is simply the path integral in the free conformally invariant case, which we know is a constant. After this transformation, we get

$$\Gamma[a_+, a_-] = S_g[a_+] - S_g[a_-] - i\,\ln\int D\chi^+ D\chi^- \exp\Big\{i\Big[\Big(S_f + S_f^{ct}\Big)(\chi_+, a_+) - \Big(S_f + S_f^{ct}\Big)(\chi_-, a_-)\Big]\Big\}, \tag{3.117}$$

where

$$S_f(\chi, a) = \int d^n x\,\Big\{-\frac{1}{2}\eta^{\mu\nu}\partial_\mu\chi\partial_\nu\chi - \frac{1}{2}\mathfrak{M}^2\chi^2 - \frac{\lambda a^{-\varepsilon}}{4!}\chi^4\Big\}, \tag{3.118}$$

$$S_f^{ct}(\chi, a) = \int d^4 x\,\Big\{\frac{\lambda\,\mathfrak{M}^2}{2(4\pi)^2\varepsilon}\chi^2 + \mathcal{O}(\lambda^2)\Big\}. \tag{3.119}$$

Here again, $\varepsilon = n - 4$ and $\mathfrak{M}^2 = M_\xi^2 a^2$. We have effectively reduced our problem to that of a theory in flat spacetime with position-dependent (when $n = 4$)

interactions. Expanding up to the desired orders we find (keeping only terms which contain a_+)

$$\Gamma = S_g + \Delta\Gamma \tag{3.120}$$

with

$$\begin{aligned}
\Delta\Gamma = \frac{i}{2}\Big\{ \frac{1}{4}\int d^4x d^4x' \; \Big[& -\frac{1}{2}\mathfrak{M}_+^2(x)\mathfrak{M}_+^2(x')\langle\chi_+^2(x)\chi_+^2(x')\rangle_c \\
& + \mathfrak{M}_+^2(x)\mathfrak{M}_-^2(x')\langle\chi_+^2(x)\chi_-^2(x')\rangle_c \\
& + \frac{1}{2}\,\mathfrak{M}_+^2(x)\frac{\lambda}{(4\pi)^2\varepsilon}\mathfrak{M}_+^2(x')\langle\chi_+^2(x)\chi_+^2(x')\rangle_c \\
& - \mathfrak{M}_+^2(x)\frac{\lambda}{(4\pi)^2\varepsilon}\mathfrak{M}_+^2(x')\langle\chi_+^2(x)\chi_-^2(x')\rangle_c\Big] \\
- \frac{\lambda}{6}\int d^4x d^4x' d^4x'' \; \Big[& \frac{-3}{4\times 4!}\,\mathfrak{M}_+^2(x)\mathfrak{M}_+^2(x')a_+^{-\varepsilon}(x'')\langle\chi_+^2(x)\chi_+^2(x')\chi_+^4(x'')\rangle_c \\
& + \frac{3}{4\times 4!}\,\mathfrak{M}_+^2(x)\mathfrak{M}_+^2(x')a_-^{-\varepsilon}(x'')\langle\chi_+^2(x)\chi_+^2(x')\chi_-^4(x'')\rangle_c \\
& + \frac{6}{4\times 4!}\,\mathfrak{M}_+^2(x)\mathfrak{M}_-^2(x')a_+^{-\varepsilon}(x'')\langle\chi_+^2(x)\chi_-^2(x')\chi_+^4(x'')\rangle_c \\
& - \frac{6}{4\times 4!}\,\mathfrak{M}_+^2(x)\mathfrak{M}_-^2(x')a_-^{-\varepsilon}(x'')\langle\chi_+^2(x)\chi_-^2(x')\chi_-^4(x'')\rangle_c\Big]\Big\}.
\end{aligned} \tag{3.121}$$

We observe that the graph $\langle\chi_-^2(x')\chi_-^2(x')\chi_+^4(x'')\rangle_c$ does not contribute because the corresponding integral is finite. Introducing the Feynman rules based on (3.29)–(3.32) and the kernels R_0, I_0 from (3.98) and (3.99), we get

$$\begin{aligned}
\Delta\Gamma = & -\frac{1}{2(4\pi)^2}\int d\eta d\eta'\; \mathfrak{M}_+^2(\eta)\Big\{\Big[\frac{\delta(\eta-\eta')}{\varepsilon} + R_0(\eta-\eta')\Big]\mathfrak{M}_+^2(\eta') \\
& \qquad + 2i\, I_0(\eta-\eta')\mathfrak{M}_-^2\Big\} \\
& + \frac{\lambda}{(4\pi)^2\varepsilon}\int d\eta d\eta'\; \mathfrak{M}_+^2(\eta)\Big\{\Big[\frac{\delta(\eta-\eta')}{\varepsilon} + R_0(\eta-\eta')\Big]\mathfrak{M}_+^2(\eta') \\
& \qquad + 2i\, I_0(\eta-\eta')\mathfrak{M}_-^2(\eta')\Big\} \\
& - \frac{\lambda}{2(4\pi)^4}\int d\eta d\eta' d\eta''\; \mathfrak{M}_+^2(\eta)\Big\{\mathfrak{M}_+^2(\eta')a_+^{-\varepsilon}(\eta'') \\
& \qquad \times \Big[\frac{\delta(\eta-\eta'')}{\varepsilon} + R_0(\eta-\eta'')\Big]\Big[\frac{\delta(\eta'-\eta'')}{\varepsilon} + R_0(\eta'-\eta'')\Big] \\
& \quad + \mathfrak{M}_+^2(\eta')I_0(\eta-\eta'')\times I_0(\eta'-\eta'')
\end{aligned}$$

$$+ 2i\,\mathfrak{M}_-^2(\eta')a_+^{-\varepsilon}(\eta'')\Big[\frac{\delta(\eta-\eta'')}{\varepsilon}+R_0(\eta-\eta'')\Big]\times I_0(\eta'-\eta'')$$
$$+ 2i\,\mathfrak{M}_-^2(\eta')a_-^{-\varepsilon}(\eta'')\Big[\frac{\delta(\eta'-\eta'')}{\varepsilon}+R_0^*(\eta'-\eta'')\Big]\times I_0(\eta-\eta'')\Big\}. \tag{3.122}$$

After some straightforward calculation we obtain

$$\begin{aligned}
\Delta\Gamma = &-\frac{1}{2(4\pi)^2\varepsilon}\int d\eta\;\mathfrak{M}_+^4 \\
&-\frac{1}{2(4\pi)^2}\int d\eta d\eta'\;\mathfrak{M}_+^2(\eta)\Big[R_0(\eta-\eta')\mathfrak{M}_+^2(\eta')+2i\,I_0(\eta-\eta')\,\mathfrak{M}_-^2(\eta')\Big] \\
&+\frac{\lambda}{2(4\pi)^4}\int d\eta\;\mathfrak{M}_+^4(\eta)\Big[\varepsilon^{-2}+\varepsilon^{-1}\ln a_+-\frac{1}{2}\big(\ln a_+\big)^2\Big] \\
&+\frac{\lambda}{(4\pi)^4}\int d\eta d\eta'\;\mathfrak{M}_+^2(\eta)\Big[R_0(\eta-\eta')\,\mathfrak{M}_+^2(\eta)\ln a_+(\eta) \\
&\qquad +i\,I_0(\eta-\eta')\,\mathfrak{M}_-^2(\eta')\ln a_+(\eta')+i\,I_0(\eta-\eta')\,\mathfrak{M}_-^2(\eta')\ln a_-(\eta')\Big] \\
&-\frac{\lambda}{2(4\pi)^4}\int d\eta d\eta'\;\mathfrak{M}_+^2(\eta) \\
&\times\Big\{\mathfrak{M}_+^2(\eta')\int d\eta''\;\Big[R_0(\eta-\eta'')R_0(\eta'-\eta'')+I_0(\eta-\eta'')I_0^*(\eta'-\eta'')\Big] \\
&+2i\,\mathfrak{M}_-^2(\eta')\int d\eta''\;\Big[R(\eta-\eta'')I_0(\eta'-\eta'')+R_0^*(\eta'-\eta'')I_0(\eta-\eta'')\Big]\Big\}.
\end{aligned} \tag{3.123}$$

Up to first order in λ and second order in M_ξ^2 we have

$$\begin{aligned}
S_g = \int d\eta\;a^n\Big\{&\epsilon_0\,R+\frac{1}{2(4\pi)^2\varepsilon}M_\xi^4-\frac{\lambda}{2(4\pi)^4\varepsilon^2}M_\xi^4 \\
&+\frac{1}{180(4\pi)^2\varepsilon}\Big(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}-R_{\mu\nu}R^{\mu\nu}\Big)\Big\},
\end{aligned} \tag{3.124}$$

where, as before, we assume $\Lambda = a = b = c = 0, \mu = 1$ in (2.57). The full effective action is therefore $\Gamma = S_E + \Gamma_f$, where S_E is Einstein's action, and

$$\Gamma_f = \frac{1}{180(4\pi)^2}\int d\eta\,\Big\{\Big(\frac{a'}{a}\Big)^4-3\Big(\frac{a''}{a}\Big)^2+90\mathfrak{M}^4\Big[\ln a-\frac{\lambda}{2}(\ln a)^2\Big]\Big\}, \tag{3.125}$$

plus non-local terms being the same as in Eq. (3.123).

We see that overlapping divergences have canceled out, as expected. To better appreciate the content of Eq. (3.123), let us compute the trace of the energy-momentum tensor

$$T = (a^{-3})\frac{\delta}{\delta a_+}\Gamma_f[a_+,a_-]\,\Big|_{a_+=a_-=a}. \tag{3.126}$$

The variation is to be taken with a^- held constant. We get, after identifying a_+ and a_-,

$$
\begin{aligned}
T(\eta) = T_A &+ \frac{1}{2(4\pi)^2 a^2}\int d\eta' \,\Big\{\frac{\partial}{\partial a}\big[\mathfrak{M}\ln a\big]\Big\}(\eta') \\
&- \frac{\lambda}{4(4\pi)^4 a^2}\int d\eta' \,\Big\{\frac{\partial}{\partial a}\big[\mathfrak{M}^4(\ln a)^2\big]\Big\}(\eta') \\
&- \frac{1}{(4\pi)^2 a^3}\int d\eta' \,\Big\{\frac{\partial}{\partial a}\mathfrak{M}^2\Big\}(\eta')\int d\eta''\,\mathfrak{M}^2(\eta'')K_0(\eta'-\eta'') \\
&+ \frac{\lambda}{(4\pi)^4 a^3}\int d\eta' \,\Big\{\frac{\partial}{\partial a}\mathfrak{M}^2\Big\}(\eta')\int d\eta''\,\mathfrak{M}^2(\eta'')\Big[a(\eta')\ln a(\eta)\Big]K_0(\eta'-\eta'') \\
&+ \frac{\lambda}{(4\pi)^4 a^3}\int d\eta' \,\Big\{\frac{\partial}{\partial a}\ln a\Big\}(\eta')\,\mathfrak{M}^2(\eta')\int d\eta''\,\mathfrak{M}^2(\eta'')K_0(\eta'-\eta'') \\
&- \frac{\lambda}{(4\pi)^4 a^3}\int d\eta' \,\Big\{\frac{\partial}{\partial a}\mathfrak{M}^2\Big\}(\eta')\int d\eta''\,\mathfrak{M}^2(\eta'')\int d\eta'''\,K_0(\eta'-\eta'')K_0(\eta'''-\eta''),
\end{aligned}
\tag{3.127}
$$

where T_A denotes the conformal anomaly terms and K_0 is the kernel defined in Eq. (3.103). We observe that T is real and depends causally on the evolution of the conformal factor.

In the massless, free field case ($m = \lambda = 0$), the nonlocal term in (3.127) reduces to the one computed in [425] (the local terms need not be identical since they depend on the renormalization prescription). This reconfirms that (3.127) represents the *in-in* expectation value in the state which reduces to the conformal vacuum in the conformal limit. The contribution of the nonlocal terms to the energy is, in the limit $\eta \to \infty$

$$
\begin{aligned}
\Delta(a^4 T^0{}_0) &= \int_{-\infty}^{\infty} d\eta\; a' a^3 T(\eta) \\
&= -\frac{1}{(4\pi)^2}\int d\eta\,\Big[\frac{d}{d\eta}\mathfrak{M}^2(\eta)\Big]\int d\eta'\,\mathfrak{M}^2(\eta')K_0(\eta-\eta') \\
&\quad - \frac{\lambda}{(4\pi)^4}\int d\eta\,\Big[\frac{d}{d\eta}\mathfrak{M}^2(\eta)\Big]\int d\eta'\,\mathfrak{M}^2(\eta')\int d\eta''\,K_0(\eta-\eta'')K_0(\eta''-\eta') \\
&\quad + \frac{\lambda}{(4\pi)^4}\int d\eta\,\Big[\frac{d}{d\eta}\mathfrak{M}^2(\eta)\Big]\int d\eta'\,\big(\mathfrak{M}^2\ln a\big)(\eta')K_0(\eta-\eta') \\
&\quad + \frac{\lambda}{(4\pi)^4}\int d\eta\,\Big[\frac{d}{d\eta}\big(\mathfrak{M}^2\ln a\big)(\eta)\Big]\int d\eta'\,\mathfrak{M}^2(\eta')K_0(\eta-\eta').
\end{aligned}
\tag{3.128}
$$

Introducing the Fourier transforms of K_0, $\mathfrak{M}^2$ and $\mathfrak{M}^2\ln a$, we obtain

$$
\begin{aligned}
\Delta(a^4 T^0{}_0) = \frac{1}{4(4\pi)^2}\int_0^\infty d\omega\;(2\omega)\Big\{\Big[1+\frac{\lambda}{(4\pi)^2}\ln\frac{|\omega|}{\sqrt{4\pi\mu^2}}\Big]\,|\mathfrak{M}^2(\omega)|^2 \\
- 2\frac{\lambda}{(4\pi)^2}\,\mathrm{Re}\Big[(\mathfrak{M}^2\ln a)(\omega)\mathfrak{M}^{2*}(\omega)\Big]\Big\}.
\end{aligned}
\tag{3.129}
$$

(As usual, the dependence on the renormalization scale μ is only apparent, since the full effective action is μ independent: a change in μ would be compensated by a change in the renormalized gravitational constants and M_ξ^2.) We find, as discussed in Section 3.3.2, that particle production depends only on the imaginary parts of the kernels in the frequency domain. This also reaffirms that self-interaction enhances particle production [363, 426], and its effect already appears to first order in the coupling constants.

In summary, we have computed the *in-in* effective action and the expectation values of the stress-energy tensor. Renormalization requires only counter terms for the *in-out* effective action. The equations of motion for the background metric are real, causal and nonlocal, the nonlocal terms being related to particle production processes.

Here, as in Chapter 2, we have chosen an easy enough interacting scalar field theory to illustrate the 'in-in' effective action. For treatment of 'in-in' effective action for gauge fields in flat space, see, e.g., Chapter 7 of [326]. For additional applications to cosmological problems see, e.g., [427] and [428], Chapter 15 of [326] and references therein. Since it is the basic tool of semiclassical and stochastic gravity it will appear in most later chapters of this book.

3.5 CTP Effective Action for Thermal Fields

Finite temperature quantum field theory [429, 430, 431] for systems in thermal equilibrium is usually constructed in the imaginary-time (Matsubara) formulation, where the many-body system is assumed to remain in thermal equilibrium throughout. Issues of acausality can arise when one forces the imaginary-time formalism on time-dependent systems. Real time methods are more suitable to treat systems under nonequilibrium conditions [432, 433, 434].

In this last section we consider the class of problems when the initial quantum state of the field is defined by a thermal density matrix operator ρ. We present an 'in-in' effective action [435, 436, 437] treatment of thermal field theory. For simplicity we shall work with a free scalar field in flat spacetime. This initial-value formulation is particularly suited to obtaining dynamical physical quantities under fully nonequilibrium conditions. Our presentation below follows [438] which feeds into a closed-time-path (CTP) field theory treatment of the nonequilibrium dynamics of interacting quantum systems such as presented in [326].

When an external classical source $J(x)$ is introduced to a system of scalar field $\phi(x)$ whose dynamics is described by a time-independent Hamiltonian $H[\phi(x)]$ the driven Hamiltonian assumes the form $H + J(x)\phi(x)$. If the initial quantum state of the field is defined by the density matrix operator ρ, we can define the CTP generating functional $Z_\rho[J]$ as a statistical average over the initial quantum state:

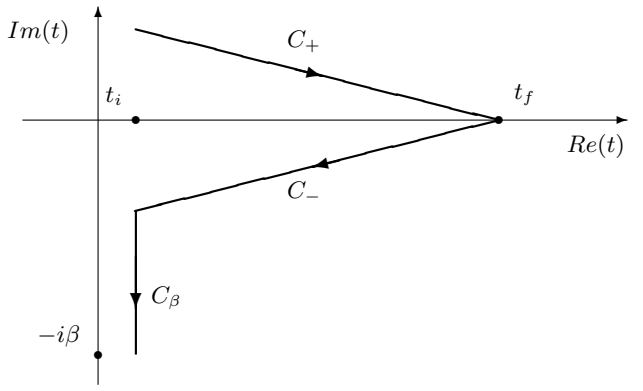


Figure 3.2 Integration contour in the complex time plane for the CTP path integral for a system with an initial state in thermal equilibrium.

$$Z_\rho[J] \equiv \mathrm{Tr}\Big\{\rho\, T_C \exp\Big[i\int_C d^n x\; J(x)\phi(x)\Big]\Big\}. \tag{3.130}$$

Here, $\phi(x)$ is the scalar field operator which evolves in time under the unperturbed Hamiltonian H, n is the number of spacetime dimensions, T_C is the time ordering operator defined along a contour C in complex time which goes forward in time and then backward to the initial point (see Fig. 3.2) and the integral symbol $\int_C$ refers to a complex integration along this path. Note that in this picture the time evolution of the initial state is determined by the external source $J(x)$ alone. We shall apply this formalism to a system initially in thermal equilibrium at temperature β^{-1}.

An integral representation of the generating functional can be obtained by introducing a complete basis of eigenstates $\{|\varphi, t_i\rangle\}$ of the field operator $\phi(x)$ at the initial time t_i, *i.e.* $\phi(t_i, \vec{x})|\varphi, t_i\rangle = \varphi(\vec{x})|\varphi, t_i\rangle$ where $\varphi(\vec{x})$ is the eigenvalue. This representation for $Z_\rho[J]$ is

$$Z_\rho[J] = \int d\varphi d\varphi'\; \langle\varphi, t_i|\,\rho\,|\varphi', t_i\rangle\langle\varphi', t_i|\,T_C \exp\Big[i\int_C d^n x\; J(x)\phi(x)\Big]\,|\varphi, t_i\rangle. \tag{3.131}$$

The last term in the integrand in the form of a transition amplitude can have a path integral presentation

$$\begin{aligned} Z_\rho[J] = &\int d\varphi d\varphi'\; \langle\varphi, t_i|\,\rho\,|\varphi', t_i\rangle \\ &\times \int_{\phi(t_i)=\varphi}^{\phi(t_i)=\varphi'} \mathcal{D}\phi\; \exp\Big[i\int_C d^n x\; \Big\{\mathcal{L}[\phi(x)] + J(x)\phi(x)\Big\}\Big] \end{aligned} \tag{3.132}$$

with the appropriate boundary conditions. Here, $\mathcal{L}[\phi(x)]$ is the Lagrangian density corresponding to the unperturbed Hamiltonian of the scalar field theory.

We follow the same procedure as before and introduce the generating functional $W_\rho[J] \equiv -i \ln Z_\rho[J]$ that generates the connected part of the n-point functions. The CTP effective action is its Legendre transform,

$$\Gamma_\rho^{\rm CTP}[\bar\phi] \equiv W_\rho[J] - \int_C d^n x \; J(x)\bar\phi(x), \tag{3.133}$$

where $\bar\phi(x)$ is the expectation value of the field $\phi(x)$ over the initial state in the presence of the external source $J(x)$,

$$\bar\phi(x) \equiv \frac{\delta W_\rho[J]}{\delta J(x)} \equiv \frac{{\rm Tr}\Big\{\rho\, T_C\, \phi(x) \exp\Big[i\int_C d^n x' \; J(x')\phi(x')\Big]\Big\}}{{\rm Tr}\Big\{\rho\, T_C\, \exp\Big[i\int_C d^n x' \; J(x')\phi(x')\Big]\Big\}}. \tag{3.134}$$

In a source-free situation $J = 0$, $\bar\phi$ takes on the mean value of the field. The CTP effective action is the generating functional of one-particle-irreducible (1PI) graphs and contains all the quantum corrections to the classical action. With a given initial state a real dynamical equation of motion with causal boundary conditions can be derived from this effective action.

3.5.1 Initial State in Thermal Equilibrium

Consider the Lagrangian density for a free massive scalar field

$$\mathcal{L}[\Phi(x)] = -\frac{1}{2}\left(\partial_\mu\Phi\partial^\mu\Phi + m^2\Phi^2\right). \tag{3.135}$$

The initial thermal state of this system is described by a normalized density matrix of the form $\rho_\beta = e^{-\beta H}/\,{\rm Tr}\{e^{-\beta H}\}$, where H is the Hamiltonian operator corresponding to (3.135) at some initial time t_i; *i.e.*, the unperturbed Hamiltonian of the system before the external source $J(x)$ is connected. In this case, the path integral representation of the generating functional given in Eq. (3.132) can be written as

$$\begin{aligned} Z_\beta[J] = \frac{1}{{\rm Tr}\{e^{-\beta H}\}} \int d\phi d\phi' \; \langle\phi, t_i - i\beta|\phi', t_i\rangle \\ \times \int_{\Phi(t_i)=\phi}^{\Phi(t_i)=\phi'} D\Phi \; \exp\Big\{i\int_C d^n x \; \mathcal{L}[\Phi(x)] + J(x)\Phi(x)\Big\}, \end{aligned} \tag{3.136}$$

where the integration is over a complex time path C composed of the usual forward C_+ and backward C_- real time branches plus an imaginary time branch C_β extending to $-i\beta$. (See Fig. 3.2.) In deriving this expression, we have used the fact that the density matrix operator ρ_β can be seen as a time translation operator in the complex plane because the unperturbed Hamiltonian is the generator of time evolution for the eigenstates of the field operator $\Phi(x)$. In other words, since the field transforms as $\Phi(t,\vec{x}) = e^{itH}\Phi(0,\vec{x})e^{-itH}$ and the eigenstates

evolve in time simply as $|\phi, t\rangle = e^{itH}|\phi, 0\rangle$, the density matrix generates a translation in complex time $t \to t - i\beta$. The above representation for $Z_\beta[J]$ may be written in a more compact form as

$$Z_\beta[J] = \mathcal{N} \int D\Phi \, \exp\left\{ i \int_C d^n x \, \mathcal{L}[\Phi(x)] + J(x)\Phi(x) \right\}. \tag{3.137}$$

The path integral is defined over all possible field configurations along the time path C with initial boundary condition $\Phi(t_i) = \Phi(t_i - i\beta)$ and the normalization term $\mathcal{N}$ includes all the factors which are independent of the external source $J(x)$.[4]

To perform the path integral of the generating functional it is convenient to make the change of field variables $\Phi(x) \to \Phi(x) - \int_C d^n x' \, G_\beta^C(x - x')J(x')$, where the propagator $G_\beta^C(x - x')$ satisfies the differential equation,

$$(\Box_C - m^2)\, G_\beta^C(x - x') = \delta_C(x - x'). \tag{3.138}$$

The subscript C in the d'Alambertian $\Box_C$ and the delta function δ_C indicates that these objects are defined along the complex path C. The subscript β in G_β^C is a reminder that this propagator has to be constructed with the appropriate thermal boundary conditions. If we redefine the normalization factor $\mathcal{N}$ by $Z_\beta[0]$ in order to include a new path integral independent of $J(x)$, the change of variable yields

$$Z_\beta[J] = Z_\beta[0] \, \exp\left\{ -\frac{i}{2} \int_C d^n x d^n x' \; J(x) G_\beta^C(x - x') J(x') \right\}. \tag{3.139}$$

It is clear from the thermal version of (3.130) and the above expression for the generating functional that the propagator $G_\beta^C(x - x')$ can be interpreted as the thermal average of two time-ordered (T_C) operators along C

$$G_\beta^C(x - x') = -i \, \langle T_C \, \Phi(x)\Phi(x') \rangle_\beta. \tag{3.140}$$

Here, angular brackets around an operator A denote taking the trace of A with respect to the thermal density operator, i.e., $\langle A \rangle_\beta \equiv \mathrm{Tr}\{\rho_\beta A\}$. One can show that the Wick theorem also applies at finite temperature [435, 434].

Let us find an explicit expression for the thermal propagator. Using the step function θ_C, defined also along the contour C, one can decompose the propagator as

$$\begin{aligned} G_\beta^C(\tau - \tau'; \vec{x} - \vec{x}') = {} & \theta_C(\tau - \tau') \; G_\beta^+(\tau - \tau'; \vec{x} - \vec{x}') \\ & + \theta_C(\tau' - \tau) \; G_\beta^-(\tau - \tau'; \vec{x} - \vec{x}'), \end{aligned} \tag{3.141}$$

[4] Note in the usual path integral approach to thermal field theory [432, 433, 434] the external source is, by design, also non-zero along C_β, whereas in our case [439, 440, 437] the contribution to the path integral along the segment C_β has no dynamical consequences because the external source is zero along this vertical segment. However, it can be shown that both procedures are consistent asymptotically, when $t_i \to -\infty$ and $t_f \to \infty$.

where we have denoted the time variable along the path C by τ and defined the propagators

$$G^{+}_{\beta}(\tau-\tau';\vec{x}-\vec{x}') = G^{-}_{\beta}(\tau'-\tau;\vec{x}'-\vec{x}) \equiv -i\langle\Phi(\tau;\vec{x})\Phi(\tau';\vec{x}')\rangle_{\beta}. \tag{3.142}$$

The trace nature of the generating functional and the Hamiltonian time evolution of the field operator $\Phi(x)$ impose some relations on certain propagators. For example, they lead to equalities like

$$G^{-}_{\beta}(\tau;\vec{x}) = G^{+}_{\beta}(\tau-i\beta;\vec{x}), \tag{3.143}$$

which are generally known as the Kubo–Martin–Schwinger (KMS) relations [441, 442, 443]. In fact, these conditions are direct consequences of thermal equilibrium and are necessary to preserve the periodic boundary condition in the change of field variable previously performed in order to simplify the path integral representation of the generating functional [432, 444]. The solution of (3.138), compatible with the KMS relation, may be expressed in terms of the thermal function $n_{\beta}(p^0) = (e^{\beta p^0}-1)^{-1}$ as

$$\begin{aligned} &G^{C}_{\beta}(\tau-\tau';\vec{x}-\vec{x}') \\ &\quad = \int \frac{d^n p}{(2\pi)^n}\, e^{ip\cdot(x-x')}\,[\theta_C(\tau-\tau') + n_{\beta}(p^0)]\,(-2\pi i)\,\mathrm{sgn}(p^0)\;\delta(p^2+m^2). \end{aligned} \tag{3.144}$$

In actual computations, it is convenient to introduce a matrix notation for the thermal propagator G^{C}_{β} according to the segment of the path, C_+ or C_-, the time variable falls under. There are four different possibilities:

- if $\tau,\,\tau' \in\ C_+$ or $\tau,\,\tau' \in\ C_-$ we may write, respectively,

$$G^{\beta}_{\pm\pm}(x-x') = \int \frac{d^n p}{(2\pi)^n}\, e^{ip\cdot(x-x')}\left[\frac{\mp 1}{p^2+m^2\mp i\epsilon} - 2\pi i\; n_{\beta}(|p^0|)\delta(p^2+m^2)\right], \tag{3.145}$$

- if $\tau \in\ C_+$ and $\tau' \in\ C_-$, or vice versa, then respectively,

$$G^{\beta}_{\pm\mp}(x-x') = \int \frac{d^n p}{(2\pi)^n}\, e^{ip\cdot(x-x')}(-2\pi i)\Big[\theta(\mp p^0) + n_{\beta}(|p^0|)\Big]\delta(p^2+m^2). \tag{3.146}$$

Since the time variable in the above equations takes values on the real axis alone, these propagator components may be expressed as thermal averages

$$\begin{aligned} G^{\beta}_{++}(x-x') &= -i\,\langle\mathcal{T}\,\Phi(x)\Phi(x')\rangle_{\beta}, \\ G^{\beta}_{--}(x-x') &= -i\,\langle\tilde{\mathcal{T}}\,\Phi(x)\Phi(x')\rangle_{\beta} \\ G^{\beta}_{-+}(x-x') &= -i\,\langle\Phi(x)\Phi(x')\rangle_{\beta} = G^{\beta}_{+-}(x'-x), \end{aligned} \tag{3.147}$$

where $\mathcal{T}$ and $\tilde{\mathcal{T}}$ are the usual time and anti-time ordering operators, respectively. If we define in the same way the external sources $J_\pm(t,\vec{x}) \equiv J(\tau,\vec{x})$ for $\tau \in C_\pm$, we can write the generating functional in matrix form

$$Z_\beta[J_a] = Z_\beta[0]\exp\left\{-\frac{i}{2}\int_{-\infty}^{\infty} d^n x d^n x'\; J_a^T(x) G_{ab}^\beta(x-x') J_b(x')\right\}. \tag{3.148}$$

Here, the subindices a, b take on values $+,-$ and the transposed source vector is defined by $J_a^T(x) \equiv (+J_+(x),\, -J_-(x))$. The minus sign on J_- reflects the fact that the segment C_- runs from $+\infty$ to $-\infty$.

If we wish to specify initial *vacuum* boundary conditions $\rho_0 \equiv |0,\mathrm{in}\rangle\langle 0,\mathrm{in}|$ instead of thermal equilibrium conditions, we could still use (3.148). Just substitute the thermal propagator G_{ab}^β by the vacuum propagator $G_{ab}^0 \equiv \lim_{\beta\to\infty} G_{ab}^\beta$ (see, e.g., [182, 423, 437]). The derivation of the CTP effective action in the case with an initial pure vacuum state can be found in [423]. To obtain the CTP effective action at finite temperature in terms of the propagators we only need to impose the appropriate thermal boundary conditions. The result is (see also [437, 440])

$$\Gamma_{\mathrm{CTP}}^\beta[\bar{\phi}_a] = S[\bar{\phi}_a] - \frac{i}{2}\,\mathrm{Tr}\{\ln G_{ab}^\beta\}, \tag{3.149}$$

where $S[\bar{\phi}_a]$ is the classical action for the scalar field evaluated at the expectation value of $\Phi_a(x)$ in the presence of the external source $J_a(x)$.

3.5.2 CTP Effective Action for Two Fields

We now derive the CTP effective action at finite temperature for two fields in the semiclassical theory. This generalization of Eq. (3.149) is useful for the consideration of problems which can be cast into open quantum systems. An example is the analysis of gravitational perturbations $g_{\mu\nu}(x)$ in a thermal scalar field $\Psi(x)$ acting as an environment [438]. For illustrative purpose, consider for simplicity two different scalar fields $\Psi(x), \Phi(x)$ described by the general Lagrangian density

$$\mathcal{L}[\Phi(x)] + \mathcal{L}[\Psi(x),\Phi(x)], \tag{3.150}$$

where the second term is assumed to be quadratic in the environmental field $\Psi(x)$. If we are only interested in the dynamics of the system field $\Phi(x)$ here, we introduce an external source coupled only to $\Phi(x)$. The CTP generating functional for the system field Φ will be

$$e^{iW_\beta[J_\pm]} \equiv \int D\Phi\, D\Psi\; \exp\left\{i\int_C d^n x\; \mathcal{L}[\Phi(x)] + \mathcal{L}[\Psi(x),\Phi(x)] + J(x)\Phi(x)\right\}, \tag{3.151}$$

where the time integration contour C is shown in Fig. 3.2. while appropriate thermal boundary conditions are enforced on the environment field $\Psi(x)$. If the

system field $\Phi(x)$ is assumed to evolve classically, the above CTP generating functional may be approximated by

$$e^{iW_\beta[J_\pm]} \simeq \exp\left\{ i \int_C d^n x \; \mathcal{L}[\bar{\phi}(x)] + J(x)\bar{\phi}(x) \right\} \times \int D\Psi \; \exp\left\{ i \int_C d^n x \; \mathcal{L}[\Psi(x), \bar{\phi}(x)] \right\}, \tag{3.152}$$

where $\bar{\phi}(x)$ is the classical field written in terms of $J(x)$. The remaining path integral is in fact a functional of $\bar{\phi}(x)$ alone and can be identified with the influence functional of Feynman and Vernon [187]. If $\mathcal{L}[\Psi(x), \bar{\phi}(x)]$ is quadratic in $\Psi(x)$, the path integral is Gaussian and may be performed exactly. In matrix notation we have

$$\exp\left\{ -\frac{i}{2} \; \mathrm{Tr} \ln \bar{G}^\beta_{ab}[\bar{\phi}_\pm] \right\}, \tag{3.153}$$

where $\bar{G}^\beta_{ab}[\bar{\phi}_\pm]$ is the corresponding matrix propagator with thermal boundary conditions of the Lagrangian density $\mathcal{L}[\Psi(x), \bar{\phi}(x)]$. When this result is substituted in the CTP generating functional and the Legendre transform with respect to $J(x)$ is performed, we obtain the thermal CTP effective action as

$$\Gamma^\beta_{CTP}[\bar{\phi}_\pm] = S[\bar{\phi}_+] - S[\bar{\phi}_-] - \frac{i}{2} \; \mathrm{Tr} \ln \bar{G}^\beta_{ab}[\bar{\phi}_\pm]. \tag{3.154}$$

We have shown how to apply the CTP effective action to finite temperature quantum field theory. Among many possibilities we mention two directions where this can be extended to a broad scope. One down-to-earth application is the use of CTP effective action to obtain equations of motion for fully nonequilibrium quantum many-body systems. See e.g., [445, 326]. The other is using the AdS–CFT correspondence to compute real-time Green's functions for a class of finite temperature field theories from their AdS gravity duals as illustrated in [446, 447]. Their method can be applied to computing higher point Lorentzian signature correlators. A mathematically elegant formulation of the Schwinger–Keldysh formalism invoking BRST symmetries in superspace was recently proposed in [448, 449].

4

Stress-Energy Tensor and Correlators: Zeta-Function Method

In this and the following chapter we will discuss two important regularization methods for treating the ultraviolet divergences in the stress-energy tensors of quantum fields in curved spacetime. We have presented the popular dimensional regularization in Chapter 2, where we also used the results for the $\lambda\Phi^4$ theory to derive its renormalization group equations in curved spacetime. In Chapter 8 we shall invoke the adiabatic regularization method suitable for dynamical spacetimes in the reheating problem. In this chapter we introduce the zeta (ζ) -function regularization method and in the next chapter the point-splitting regularization method. The zeta function regularization is probably the most elegant of the four main methods used for quantum fields in curved spacetime, linked to the heat kernel and spectral theorems in mathematics. The only drawback is, the zeta-function method can only be applied to Riemannian spaces (often called Euclidean, whose metrics have a $++++$ signature) where the invariant operator is of the elliptic type, as opposed to the hyperbolic type in pseudo-Riemannian spaces (sometimes called Lorentzian, with a $-+++$ signature). Besides, the space needs to have sufficiently large symmetry that the spectrum of the invariant operator can be calculated in analytic form explicitly.

In the first part of this chapter we will define the zeta function, show how to calculate it in several representative spacetimes and how the zeta-function regularization scheme works. We will then relate it to the heat kernel and derive the effective Lagrangian from it via the Schwinger proper time formalism. In the second part of this chapter we show how to obtain the correlation function of the stress-energy bitensor, also known as the noise kernel, from the second metric variation of the effective action. Noise kernel plays a central role in stochastic gravity as much as the expectation values of stress-energy tensor does for semiclassical gravity. We shall see more of the physical meanings of both in later chapters.

Part I: Zeta Function, Heat Kernel and Effective Potential

4.1 Zeta Function Regularization of 1-loop Effective Potential

The 1-loop effective action $\Gamma^{(1)}[\phi]$ expressed in terms of the logarithm of the determinant of an invariant operator A is, from (2.42),

$$\Gamma^{(1)}[\phi] = \frac{i\hbar}{2} \ln(\mathrm{Det}\mu^{-2} iA). \tag{4.1}$$

A constant mass scale μ is introduced to make the measure $d[\phi]$ of the functional integral for the generating functional dimensionless. We learned from Chapter 2 that the effective potential $V^{(1)}_{eff}(\bar{\phi}) = \frac{1}{Vol}\Gamma[\bar{\phi}]$, (where Vol denotes the volume of spacetime) for a constant effective field $\bar{\phi}$ is the first term in an expansion of the effective action in powers of the derivatives of the background field. The zeta function method can only deal with situations where the background spacetime is static and the background field is a constant. This includes the important class of Schwarzschild or Reissner–Nordström spacetimes for static (charged) black holes, where the famous Hawking effect [78, 82] can be derived. However, it is not applicable for quantum fields in cosmological spacetimes with time-dependent scale factors, except for de Sitter space in (a) the static coordinate, where the Gibbons–Hawking thermal radiance [322], similar to the Unruh effect [321] in uniformly accelerated observers, can be derived, or (b) the Euclidean representation, as a 4-sphere S^4 embedded in a 5-dim Euclidean space E^5. We shall discuss this last case in a later section.

The determinant of A is formally divergent and needs to be regularized. We shall use the Schwinger proper time representation of the effective Lagrangian and present the zeta-function method of Dowker and Critchley [96], based on the zeta function defined by [450]. In a later section we shall introduce the closely related zeta-function method of Hawking [97].

Assume that the metric on an Euclidean section $\mathcal{M}_E$ (a subscript E denotes quantities in the Euclidean section) of a n-dim manifold $\mathcal{M}$ is real and positive so that the invariant operator A_E is real, elliptic and self-adjoint. Assume A_E has a complete set of eigenfunctions ϕ_n belonging to the eigenvalues λ_n,

$$A_E \phi_n = \lambda_n \phi_n. \tag{4.2}$$

If $\mathcal{M}_E$ is a direct product space $= S^1 \times B^b$ with one of the spacetime dimensions compactified into a circle, $\lambda_N = k_0^2 + \omega_N^2$, where $k_0 = 2\pi n_0/\beta$, and ω_N (N denoting a collection of quantum numbers) are the eigenvalues of the operator A on B^b. If the imaginary-time dimension is compactified, we obtain the Matsubara finite temperature ($k_B T = \beta$) theory. If one of the spatial dimensions is compactified we obtain a cylindrical spacetime. Kaluza–Klein theory for the unification of gravity and electromagnetism is built on this class of spacetimes, with the generalized Einstein equation in the compactified 5th dimension describing electromagnetism.

Consider a massive (m) $\lambda\Phi^4$ field. In a background-field decomposition $\Phi = \bar{\phi}+\varphi$ of Φ into a background field $\bar{\phi}$ and a fluctuation field φ, the small fluctuation operator is $A \equiv \Box + M_1^2$ (see Chapter 2 for explanations). Here, $\Box$ is the Laplace–Beltrami operator on an n-dimensional spacetime manifold and the effective mass M_1 was defined in (2.128).

The Feynman Green function $G(x, x')$ expressed in the Schwinger proper time basis $|x\rangle$

$$AG = 1, \quad G(x, x') = \langle x|G|x'\rangle \tag{4.3}$$

is related to the *heat kernel* $K(x, x', s)$ by

$$G(x, x') = \int_0^\infty ds\, e^{-m^2 s} K(x, x', s) \tag{4.4}$$

where

$$K(x, x', s) = \langle x|K|x'\rangle. \tag{4.5}$$

It is named such because it obeys a heat (or diffusion) equation. Now, introduce G^ν, the νth power of G. Using the semi-group property $K(s)K(s') = K(s + s')$ we see that G^ν is related to K by

$$G^\nu(x, x') = [\Gamma(\nu)]^{-1} \int ds\; s^{\nu-1} e^{-m^2 s} K(x, x', s), \tag{4.6}$$

where $\Gamma(\nu)$ is the gamma function with argument ν. It is defined for elliptic operators on compact manifolds. In the Schwinger proper time representation (x, x') the *generalized zeta function* (to distinguish it from the Riemann zeta function ζ_R) is defined as the diagonal elements of $G^\nu(x, x')$ obtainable as the Mellin transform of the trace of the heat kernel, namely,

$$\zeta(x, x, \nu, m^2) \equiv G^\nu(x, x). \tag{4.7}$$

Through the Green function the generalized ζ-function can be expressed in terms of the eigenvalues λ_N of the invariant operator in the compact manifold, where N denotes the set of its good quantum numbers. Thus,

$$\zeta(\nu) = \sum_N (\mu^{-2}\lambda_N)^{-\nu}, \tag{4.8}$$

where μ is the constant mass scale introduced in (4.1).

The zeta function regularization procedure consists of examining the coincidence limit $x' \to x$ of $G^\nu(x, x')$ and letting ν tend to 1. Referring back to (2.154), replacing α^2 there by m^2 leads to

$$\frac{\partial L^{(1)}}{\partial m^2} = \frac{\hbar}{2} \lim_{x'\to x} G(x, x'). \tag{4.9}$$

Integrating over m^2 we get

$$L^{(1)}_{reg} = -\frac{\hbar}{2} \int_{m^2}^\infty G^{reg}(x, x, m'^2) dm'^2, \tag{4.10}$$

where $L^{(1)}_{reg} = 0$ is assumed in the limit of infinite m^2. (For a finite m, setting the Schwinger proper time $s \to \infty$ corresponds to reaching the infrared limit. In that limit, for manifolds with boundaries, there will be extra contributions, see [62].) Here we are interested in identifying the ultraviolet divergences which happen (for a finite m) in the $s \to 0$ limit.

Zeta function regularization is implemented by replacing G_{reg} by G^{ν} and defining

$$L^{(1)}_{reg} = L^{(\nu)}, \quad L^{(1)} = \lim_{\nu \to 1} L^{(\nu)}, \tag{4.11}$$

with

$$L^{(\nu)} = -\frac{\hbar}{2} \int_{m^2}^{\infty} \zeta(\nu, m'^2) dm'^2 = -\frac{\hbar}{2} (\nu - 1)^{-1} \zeta(\nu - 1, m^2). \tag{4.12}$$

From this we get

$$L^{(1)} = -\frac{\hbar}{2} \big[\lim_{\nu \to 1} (\nu - 1)^{-1} \zeta(0, m^2) + \zeta'(0, m^2) \big], \tag{4.13}$$

where

$$\zeta'(\nu) \equiv (d/d\nu)\zeta(\nu). \tag{4.14}$$

The one-loop effective Lagrangian multiplied to the spacetime volume Vol of the compact manifold gives the one-loop effective potential $V^{(1)}$. Thus, in terms of the zeta function and its derivative,

$$V^{(1)}[\bar{\phi}] = -\frac{\hbar}{2} (Vol)^{-1} [\zeta(0)/(\nu - 1) + \zeta'(0)] \tag{4.15}$$

The divergence in $V^{(1)}$ is seen to arise from $\nu = 1$ and is to be removed by the addition of counter terms. The function of ν in this sense is similar to that of n in dimensional regularization, where an n-dimensional field theory is worked out to facilitate the identification of terms which diverge as $n \to 4$.

To compute the zeta function we need to know the spectrum of the invariant operators. For many-component fields in static spatially- homogeneous spaces $\lambda_{\mathbf{n}} = k_0^2 + \kappa_{\mathbf{n}}^2 + M_2^2$, where $k_0 = (-\infty, \infty)$, and $\kappa_{\mathbf{n}}$ are the eigenvalues with $\mathbf{n}$ denoting the collective spatial quantum numbers of the Laplace operator $\Box$ on the homogeneous three-space. M_2 defined in (2.126) is given by

$$M_2 = m^2 + (1 - \xi)\xi_n R + \frac{\lambda}{6}\phi^2. \tag{4.16}$$

We will show explicitly how to calculate the zeta function, and from it the effective potential. As an appetizer, consider thermal field theory in 2D. The imaginary-time (Euclidean) finite-temperature theory compactifies the time dimension and changes its topology from R^1 to S^1. The eigenvalues become

discrete: $k_0 = 2\pi n_0/\beta, n_0 = 0, \pm 1, \pm 2, \ldots$ The zeta function for a massless scalar field in $S^1 \times R^1$ spacetime is

$$\zeta(\nu) = \sum_{n=-\infty}^{+\infty} \int dk[(2\pi n/\beta)^2 + k^2]^{-\nu}. \tag{4.17}$$

The effective potential is obtained as $V^{(1)} = -\frac{\pi}{6\beta^2}$. This expression is used to compare with results from two important effects where thermal radiance appears. An observer moving with constant acceleration a in a 2D spacetime (represented by a 2D Rindler space) will see a thermal spectrum with an inverse Unruh temperature $\beta = a/2\pi$. An observer near the horizon of a two-dimensional black hole with mass M (in the Hartle–Hawking vacuum) sees an inverse Hawking temperature $\beta = 8\pi M$.

In the following sections we will provide four examples to illustrate the zeta function method: (1) $S^1 \times R^3$: *thermal fields* in three-dimensional flat space with compactified imaginary-time dimension and (2) the *Einstein Universe* with topology $R^1 \times S^3$ where the spatial dimension is a hypersphere S^3. The Einstein universe is a closed Friedmann–Lemaître–Robertson–Walker (FLRW) universe with constant radius. It is not a viable cosmological model because it is static and dynamically unstable. However, it provides a good platform to illustrate how to calculate certain physical quantities, such as the vacuum energy density in the Casimir effect. We shall use the zeta function method to derive the 1-loop effective potential for scalar fields, following [400]. The characteristic function on S^3 is the hyperspherical function with quantum numbers $\mathbf{n} = (n, l, m)$ with ranges $n = 0, 1, 2, \ldots; l = 0, 1, 2, \ldots, n$ and $m = -l, -l + 1 \ldots, l$.

For waves or fields defined on the spatial hypersurfaces of spatially-homogeneous but anisotropic spacetimes, take for example the Bianchi Type IX containing the mixmaster and the Taub universe. One can use the SO(3)-invariant representation function D^J_{KM} with quantum numbers $\mathbf{n} = (J, K, M)$ where $J = 0, 1, \ldots; K, M = -J$ to J [451] for the calculation of the zeta function. Since these spacetimes are dynamical and of cosmological relevance, the quantity of physical interest is the effective action of quantum fields, not the effective potential. However, zeta function method can be used to derive the effective potential for their static versions, useful for the study of the effect of intrinsic curvature and spatial topology on the symmetry behaviors of quantum fields, the subject matter of Chapter 6. An example is the calculation of the effective potential for the study of the effect of curvature anisotropy in the Taub universe [452].

Another important class of spacetimes where zeta function method can be effectively applied for the derivation of the effective potential is the maximally-symmetric spacetimes containing (3) the *de Sitter* (dS) and (4) the *anti-de Sitter* (AdS) space. For the Euclidean de Sitter (EdS) and AdS spaces, n are the principal quantum numbers of a 4-sphere S^4 of constant positive curvature and a 4-dim hyperbolic space H^4 of constant negative curvature, respectively. We will treat both cases in later sections of this chapter.

4.2 One-Loop Finite Temperature Effective Potential

For static spacetimes, because of the existence of Euclidean sections and the Kubo–Martin–Schwinger (KMS) condition [441, 442, 443], finite temperature quantum field theory on a manifold $\mathcal{M} = R^1 \times B^b$ is formally equivalent to an Euclidean quantum field theory on $\mathcal{M}_E = S^1 \times B^b$, where time t in $\mathcal{M}$ is replaced by imaginary time τ in $\mathcal{M}_E$, with periodicity $\beta = 1/k_B T$ corresponding to the inverse temperature. In this case the invariant operator has a simple spectrum, from which it is easy to calculate the finite temperature effective Lagrangian (potential) and free energy density.

In a finite-temperature quantum field theory the thermal average of an operator O is given by

$$< O >\equiv \left[\mathrm{Tr}\left(e^{-\beta H}\right)\right]^{-1} \mathrm{Tr}\left(e^{-\beta H} O\right), \tag{4.18}$$

where H is the Hamiltonian and the trace Tr is taken with respect to states in the Hilbert space of the quantum field. The finite temperature Feynman Green function is defined as

$$G_\beta(x, x') = i\langle \mathcal{T}\phi(x)\phi(x')\}\rangle, \tag{4.19}$$

where $\mathcal{T}$ is the time-ordering operator. The finite-temperature one-loop effective action is given by

$$\Gamma_\beta^{(1)} = -\frac{1}{2} i \,\mathrm{tr}\,(\ln G_\beta) = -\frac{1}{2} i \ln \mathrm{Det} G_\beta, \tag{4.20}$$

where the spacetime tr is defined as

$$\mathrm{tr}\ A = \int d^n x\ A(x, x). \tag{4.21}$$

It is known in flat space finite-temperature theory that the thermal Green function can be obtained as an image sum of zero-temperature Green functions [453]

$$G_\beta(x, x', \tau) = \sum_{m=-\infty}^{\infty} G_\infty(x, x' - im\lambda\beta, \tau), \tag{4.22}$$

where λ is the time-like unit vector $(1, 0, \ldots, 0)$. For static spacetimes, this relation can be used to construct finite-temperature field theories.

The Green function can be expressed in the Schwinger proper-time representation

$$G_\beta = i \int_0^\infty d\tau\, e^{-im^2\tau} K_\beta(\tau), \tag{4.23}$$

with a formal mass parameter m^2. Here $K_\beta(x,x',\tau)$ is the transition amplitude propagator which satisfies the Schrödinger equation

$$\left(-i\frac{\partial}{\partial\tau}+A\right)K_\beta(x,x',\tau)=-ig^{-1/2}\delta(x-x')\delta(\tau), \tag{4.24}$$

and has period β in imaginary time.

The one-loop effective Lagrangian $L_\beta^{(1)}$ can be obtained from G_β by using the relation

$$\frac{\partial L_\beta^{(1)}(x;m^2)}{\partial m^2}=\frac{1}{2}iG_\beta(x,x;m^2). \tag{4.25}$$

For a general static spacetime, one can conformally transform the metric to one where the g_{00} component is a function of time only. The associated metric where this function is equal to one is called the *optical metric*. For the effective Lagrangian which is time-independent (thus corresponds to an effective potential),

$$L_\beta^{(1)}=-\frac{1}{2}i\mathrm{tr}_3\left(\ln G_\beta\right). \tag{4.26}$$

The subscript 3 is used to denote operations or quantities over the three-space which is the spatial section of $\mathcal{M}_E$ [62]. For example,

$$\mathrm{tr}_3A=\int d\mathbf{x}\,g^{1/2}A(\mathbf{x},t;\mathbf{x},t), \tag{4.27}$$

where the integration is only over the spatial region. $L_\beta^{(1)}$ can be expressed in terms of the *thermal zeta- function* [454] $\zeta_\beta(\nu)$ by

$$L_\beta^{(1)}=-\frac{i}{2}\left[\frac{\mathrm{tr}_3\zeta_\beta(0)}{\nu-1}+\mathrm{tr}_3\zeta'_\beta(0)\right]. \tag{4.28}$$

In a finite-temperature field theory there are no new ultraviolet divergences beyond that which appear in the zero temperature theory. Absorbing the pole term in the renormalization of the parameters of the field theory, the renormalized free energy in the optical metric is given by [62]

$$\bar{F}_{\mathrm{ren}}=-\bar{L}^{(1)}_{\beta(\mathrm{ren})}=\frac{i}{2}\bar{\mathrm{tr}}_3\bar{\zeta}'_\beta(0), \tag{4.29}$$

where an overbar denotes the quantities in the optical metric.

Expressions for the renormalized free energy (or, equivalently, the negative of the effective Lagrangian), the internal energy and the entropy in the optical manifold have been obtained by Dowker and Kennedy [62]. We give here only the case for spacetimes without boundary. For the free energy,

$$\bar{F}_{\mathrm{ren}}=-\frac{\pi^2}{90}\frac{\bar{c}_0}{\beta^4}-\frac{1}{24}\frac{\bar{c}_1}{\beta^2}-\bar{\mathrm{tr}}_3\bar{\zeta}'_3(0,\infty)\frac{1}{2\beta}-\frac{\bar{c}_2}{16\pi^2}\left[\ln\left(\frac{\beta}{4\pi}\right)+\gamma\right]+O(\beta^2); \tag{4.30}$$

and for the thermal energy,

$$\bar{E}_{\rm ren} = \frac{\pi^2}{30}\frac{\bar{c}_0}{\beta^4} + \frac{1}{24}\frac{\bar{c}_1}{\beta^2} - \frac{\bar{c}_2}{16\pi^2}\left[\ln\left(\frac{\beta}{4\pi}\right) + \gamma + 1\right] + O(\beta^2), \tag{4.31}$$

where $\bar{c}_l = \bar{a}_l + \bar{b}_l$ are the generalized Minakshisundaram coefficients [390] with $\bar{a}_l$ containing only integral l being the 'HaMideW' coefficients (introduced in Chapter 2) for spaces without boundary, and $\bar{b}_l$ depending on the induced metric and extrinsic curvature of the boundary, where l can be half-integral. (Consult [62] for their values.) We see the first term is the usual Planck distribution with $\bar{c}_0 = \int d\mathbf{x}\, \bar{g}^{1/2}$ being the Riemannian volume of $\bar{\mathcal{M}}_3$. The remaining terms represent corrections to the Planck distribution due to the curvature of the manifold.

The zeta function or spectral function in Lie groups and symmetric spaces with application to finite temperature theory and other topics can be found in books and reviews, e.g., [455, 378, 456, 457]

We now show how to calculate the finite temperature energy density, as needed for later purposes. For the simplest case of a massless conformal scalar field ($m = \xi = 0$), the energy density is given by the sum of $\epsilon_N = \omega_N$ multiplied by the degeneracy d_N and weighted by a Bose–Einstein distribution $n(\epsilon)$:

$$\rho_T = \frac{1}{V}\Sigma_N \epsilon_N d_N/(e^{\epsilon_N \beta} - 1) = \rho_{Pl} + \rho_c, \tag{4.32}$$

where $V = 2\pi^2 a^3$ is the volume of S^3. The sum is a Riemann ζ_R function, which can be written as the Planck term ρ_{Pl} plus a correction term ρ_c [458]

$$\rho_{Pl} = \frac{\pi^2 T^4}{30}, \quad \rho_c = \frac{12T^4}{\pi^2}\Sigma_{n=1}^{\infty} Re\ \zeta_R(4, 1 - 2\pi i n\Theta), \tag{4.33}$$

where $\Theta = Ta$ measures the number of photons in the universe. For fixed a, large (small) Θ corresponds to high (low) temperature. In these two limits $\rho_c \to -\rho_{CA}$ the Casimir energy density as $T \gg \omega_N$ and $\rho_c \to -\rho_{Pl}$, the Planck energy density, as $T \ll \omega_N$. Combining this with the vacuum energy density

$$\rho_0 = \rho_{CA} + \rho_{TA}, \tag{4.34}$$

the total energy density $\rho \to \rho_0$ at zero temperature, and $\rho \to \rho_{TA} + \rho_r$ at high temperatures. This result is useful for the consideration of thermal quantum processes in the early universe [391].

To get the free energy density from the energy density, one can use the thermodynamic relations in the Gibbs canonical ensemble, namely,

$$F = -\partial(\beta E)/\partial\beta \equiv E - TS. \tag{4.35}$$

One easily verifies that this gives the same result as that obtained from the zeta function directly.

4.3 One-Loop Effective Potential in the Einstein Universe

In this section we shall use the zeta function method to derive the one-loop effective potential for a massive scalar field coupled arbitrarily to the Einstein universe which is a static snapshot of the closed FLRW universe. It is useful for the analysis of how curvature and topology of space affect the symmetry behavior of a quantum field, the theme of Chapter 6.

The metric of the Einstein universe is

$$ds^2_{Ein} = -dt^2 + a^2 d\Omega^2_{S3} \,, \tag{4.36}$$

where a is the radius of S^3. The line element $d\Omega^2_{S3}$ of a 3-sphere is given by

$$d\Omega^2_{S3} = d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2) \,, \tag{4.37}$$

where χ, θ, ϕ are the angular coordinates on S^3 with ranges $0 \leq \chi, \theta \leq \pi,\ 0 \leq \phi \leq 2\pi$.

We consider a massive (m) self-interacting $\lambda\Phi^4$ scalar field Φ coupled arbitrarily (ξ) to the curvature. The fluctuation field φ around a background field $\bar{\phi}$ obeys to linear order in φ the wave equation

$$\left[\Box + (1-\xi)\frac{R}{6} + m^2 + \frac{\lambda}{2}\bar{\phi}^2\right]\varphi = 0 \,, \tag{4.38}$$

where $\Box$ is defined on $S^3 \times R^1$ with scalar curvature $R = 6/a^2, \xi$ is the coupling constant between the field and the background spacetime, with $\xi = 0$ denoting conformal and $\xi = 1$ minimal coupling. At the classical level the curvature term is seen yielding a non-zero contribution to the effective mass whenever the coupling is non-minimal ($\xi \neq 1$), namely, $M_1^2 \equiv M_\lambda^2 + (1-\xi)\frac{R}{6}$ and $M_\lambda^2 \equiv m^2 + \frac{\lambda}{2}\bar{\phi}^2$ in a 4-dim spacetime. Suppose the system exists in a phase of broken symmetry ($m^2 < 0$); then by increasing the curvature of spacetime beyond a certain critical value R_c, symmetry can be restored. The effect of quantum corrections and finite temperature corrections show up as additional terms in the effective mass M_1^2 in the form $\frac{1}{2}\langle\varphi\rangle^2 = \frac{1}{2}\lambda(\langle\varphi^2\rangle_0 + \langle\varphi^2\rangle_T)$. In general the quantum correction term is smaller than the classical term by a factor of $\lambda\hbar/(64\pi^2)$, except for the case of minimal (or near-minimal $\xi \leq 1$) coupling, where quantum correction becomes the main source for symmetry restoration.

An effective potential V can be defined for constant background spacetime g and background field $\bar{\phi}$ from the effective action $\Gamma(\bar{\phi})$ by

$$V(\bar{\phi}) = -(Vol)^{-1}\Gamma(\Phi)|_{\Phi=\bar{\phi}}, \tag{4.39}$$

where Vol is the spacetime volume. In the functional integral perturbative approach, the effective action $\Gamma[\bar{\phi}]$ is given by:

$$\Gamma[\bar{\phi}] = \Gamma^{(0)}_B[\bar{\phi}] + \Gamma^{(1)} + O(\hbar^2), \tag{4.40}$$

where $\Gamma_B^{(0)}$ is the classical action

$$\Gamma_B^{(0)}[g,\bar{\phi}] = \int d^4x\sqrt{g}\left\{\epsilon_{0B}R - \bar{\Lambda}_B - \frac{1}{2}\bar{\phi}\,[\Box + m_B^2 + (1-\xi_B)R/6]\,\bar{\phi} - \frac{\lambda_B}{4!}\bar{\phi}^4\right\}. \tag{4.41}$$

The gravitational part is the same as in (2.170), where $\epsilon_0 \equiv (16\pi G_{\rm N})^{-1}$, with $G_{\rm N}$ the Newton constant and $\bar{\Lambda} = 2\epsilon_0\Lambda$, with Λ the cosmological constant. Subscripts B denote the quantities are bare. (We shall leave out the quadratic curvature terms for now until regularization consideration invites their presence.) The field part is the same as in (2.19) except for a sign difference on the $\Box$ operator because we are in Euclidean space. $\Gamma^{(1)}$ is the 1-loop effective action

$$\Gamma^{(1)} = \frac{1}{2}i\hbar\ln(\mathrm{Det}\mu^{-2}iA). \tag{4.42}$$

The operator A is given here by

$$A \equiv \Box + m^2 + (1-\xi)\frac{R}{6} + \frac{1}{2}\lambda\bar{\phi}^2 = \Box + M_1^2, \tag{4.43}$$

where M_1 was defined in (2.128). For the $O(N)\lambda\Phi^4$ theory treated in the next section and Chapter 6, in the large N limit, A will contain M_2 defined in (2.126) instead.

The eigenvalues of the operator A on the Euclideanized metric $S^1 \times S^3$ are given by $\lambda_{\mathbf{n}} = k_0^2 + \omega_{nlm}^2$, where $k_0 = 2\pi n_0/\beta$, and $\omega_{n\ell m}$ are the eigenvalues of the operators A on the 3-sphere. For the Einstein universe with radius a

$$\omega_{nlm}^2 = (q^2-1)/a^2 + M_1^2\,, \tag{4.44}$$

where $q \equiv n+1$. The label $\mathbf{n}$ in $\lambda_{\mathbf{n}}$ denotes the collective time and hyperspherical quantum numbers (n_0, n, l, m) on $S^1 \times S^3$, with ranges $n_0 = 0 \pm 1, \pm 2 \ldots, n = 0, 1, 2, \ldots, l = 0, 1, \ldots n$ and $m = -l, -l+1, \ldots l$.

Using the generalized ζ-function defined by (4.8) one can express the one-loop effective potential from (4.15) as

$$V^{(1)}[\bar{\phi}] = -\frac{1}{2}\hbar(Vol)^{-1}\left[\zeta'(0) + \zeta(0)/\nu\right]. \tag{4.45}$$

For the specific purpose of identifying where ultraviolet divergences arise, we have replaced the $\nu - 1$ factor in the denominator of the term with $\zeta(0)$ there by ν here and let $\nu \to 0$.

In the limit of large β (at very low temperature, in the continuum limit) one can approximate $\sum_{n_0}$ by $\beta \int dk_0/2\pi$ whence

$$\begin{aligned}\zeta(\nu) &= \mu^{2\nu}\frac{\beta}{2\pi}\int_{-\infty}^{\infty} dk_0 \sum_{n,l,m}\left(k_0^2+\omega_{nlm}^2\right)^{-\nu} \\ &= \mu^{2\nu}\frac{\beta}{\sqrt{4\pi}}\frac{\Gamma(\nu-\frac{1}{2})}{\Gamma(\nu)}\sum_{n,l,m}\left(\omega_{nlm}^2\right)^{-\nu+\frac{1}{2}} \\ &= \frac{\beta}{\sqrt{4\pi}}\frac{(\mu a)^{2\nu}}{a}\frac{\Gamma(\nu-\frac{1}{2})}{\Gamma(\nu)}Z\left(\nu-\frac{1}{2},x\right), \end{aligned} \tag{4.46}$$

where

$$Z(\nu,x)=\sum_{q=1}^{\infty}\frac{q^2}{(q^2+x)^{\nu}}, \tag{4.47}$$

and

$$x = M_1^2a^2-1 = M_\lambda^2 a^2-\xi. \tag{4.48}$$

To evaluate $\zeta'(0)$, one can expand $Z(\nu,x)$ in a Laurent series about their simple pole $Z(\nu,x)=Z_{-1}(\nu,x)/\nu+Z_0$, and express

$$\lim_{\nu\to 0}\left[\Gamma(\nu-\frac{1}{2})/\Gamma(\nu)\right]\frac{(a\mu)^{2\nu}}{a} = -\frac{\sqrt{4\pi}}{a}\left[\nu+\left(2+\ln\frac{1}{4}a^2\mu^2\right)\nu^2+O(\nu^3)\right]. \tag{4.49}$$

In this way, one obtains the 1-loop effective potential (4.45) as

$$V^{(1)}[\bar{\phi}] = \left(\frac{\hbar}{4\pi^2a^4}\right)\left[Z_0(-\frac{1}{2},x)+Z_{-1}(-\frac{1}{2},x)\left(2+\ln\frac{1}{4}a^2\mu^2\right)\right], \tag{4.50}$$

where μ^2 is redefined to include a divergent part.[1]

To calculate the counter terms we need to compute Z_0 and Z_{-1} in $V^{(1)}$. From (4.48) by means of the Plana sum formula one gets

$$Z_{-1}(-\frac{1}{2},x) = -\frac{1}{16}x^2 \tag{4.51}$$

[1] Equations obtained using the regularization method of Dowker and Critchley [96] can be converted to those obtained by the method of Hawking [97] via the transformation:

$$\ln\mu_H^2 = \ln 4\mu_{DC}^2+\frac{1}{\nu}-2.$$

For brevity, we have applied this transformation to $\mu^2(=\mu_{DC}^2)$. Thus thereafter, one should regard $\mu^2(=\mu_H^2)$ as carrying a hidden divergent part.

$$Z_0(-\frac{1}{2},x) = (1+x)^{\frac{1}{2}} - \frac{1}{2}x\left(1+\frac{1}{4}x\right)^{\frac{1}{2}} + \frac{1}{16}x^2\left\{\ln\left[\frac{1}{2}(2+\sqrt{4+x})\right]^2 + \frac{1}{2}\right\} + I_{\frac{1}{2}}(x) \tag{4.52}$$

where

$$I_a(x) = i\int_0^\infty dt \frac{\{(2+it)^2[(2+it)^2+x]^a - (2-it)^2[(2-it)^2+x]^a\}}{e^{2\pi t}-1}.$$

A compendium of formulas useful for calculating the zeta functions in terms of Plana sums and their asymptotic expansions for small and large values of their arguments can be found in Appendices B and C of [452].

Since the values of the coupling constants are physically determined at low energy or small curvature, and the properties of the theory are independent of the point of renormalization, we can choose to renormalize $V^{(1)}$ at the near-flat space limit (small R or large $a \Rightarrow M_1^2 a^2 \simeq M_\lambda^2 a^2 \gg 1$ or large x). The expression for Z_0 in this limit ($x \gg 1$) is given by

$$Z_0(-\frac{1}{2},x) = \frac{1}{16}x^2\left(\ln\frac{1}{4}x + \frac{1}{2}\right) \quad (x \gg 1). \tag{4.53}$$

The renormalized 1-loop effective potential is given by

$$\begin{aligned} V^{(1)}[\bar\phi] &= \epsilon_0 R - \bar\Lambda + \frac{1}{2}\epsilon_1 R^2 + (1-\xi)\frac{R}{12}\bar\phi^2 + V_{CW}[\bar\phi] \\ &+ \left(\frac{\hbar}{64\pi^2}\right)\left\{-\frac{1}{3}m^2\xi R\left[\ln\left|\frac{M_\lambda^2}{m^2}\right| - \frac{1}{2}\right]\right. \\ &- \frac{1}{6}\xi\lambda\bar\phi^2 R\left[\ln\left|\frac{M_\lambda^2}{(m^2-\frac{\xi R_0}{6})}\right| - \frac{3}{2}\right] + \frac{1}{36}\xi^2 R^2\left[\ln\left|\frac{M_\lambda^2}{(m^2-\frac{\xi R_1}{6})}\right| - \frac{3}{2}\right] \\ &- \left.\left(\frac{x^2}{a^4}\right)\left[\ln\left|\frac{M_\lambda^2 a^2}{4}\right| + \frac{1}{2}\right]\right\} + \left(\frac{\hbar}{4\pi^2 a^4}\right)Z_0 + \delta\bar\Lambda(\langle\varphi\rangle) \end{aligned} \tag{4.54}$$

We have reinstated the R^2 term as needed from renormalization considerations. Here $R_0, R_1, \delta\Lambda$ are the same quantities defined earlier in (2.194), namely, R_0, R_1 are the scalar curvatures at the renormalization scales of the constants ξ, R^2 respectively, and $\delta\bar\Lambda$ is determined by the vacuum energy in flat space. Note for Z_0 the exact non-flat form (4.52) should be used. Setting $\xi = 0$ gives the case for conformal fields. This result obtained by zeta-function regularization should be compared with the (more complete) expression of the effective Lagrangian (2.194) obtained by dimensional regularization.

The Coleman–Weinberg effective potential $V_{CW}[\bar{\phi}]$ for $\lambda\phi^4$ theories in flat space is given by

$$V_{CW}[\bar{\phi}] = \frac{1}{2}m^2\bar{\phi}^2 + \frac{\lambda}{4!}\bar{\phi}^4 + \frac{\hbar}{64\pi^2}\left\{M_\lambda^4 \ln\left|\frac{M_\lambda^2}{m^2}\right| - \frac{1}{2}m^2\lambda\bar{\phi}^2 + \frac{1}{4}\lambda\bar{\phi}^4\left[\ln\left|\frac{m^2}{(m^2+\frac{1}{2}\lambda\phi_0^2)}\right| - \frac{25}{6} + \frac{8m^2(m^2+\frac{1}{4}\lambda\phi_0^2)}{3(m^2+\frac{1}{2}\lambda\phi_0^2)^2}\right]\right\}, \quad (4.55)$$

where ϕ_0 is the value of the scalar field where λ is renormalized.

Note that for the determination of the counterterms of renormalization the near-flat space regime approximation ($M_1^2a^2 \gg 1$) can be used for Z_0. However, if one is interested in how spacetime curvature affects the symmetry behavior of interacting quantum fields one should consider the high curvature regime corresponding to small conformally-related effective mass $M_1^2a^2 \ll 1$. As a precursor of Chapter 6 on the theme of infrared behavior of quantum fields we provide also the results for the large curvature regime. With $y \ll 1$ (where $y \equiv x + 1 \equiv M_1^2a^2$), Z_0 can be approximated by

$$Z_0(-\frac{1}{2}, y-1) = -0.41150 + y^{\frac{1}{2}} - 0.60223y - 0.0031778y^2 + \sum_{n=3}^{\infty}\frac{(-1)^n\Gamma(n-\frac{1}{2})}{n!\Gamma(-\frac{1}{2})}\eta(n-\frac{1}{2})y^n \equiv b_{\frac{1}{2}}y^{\frac{1}{2}} + \sum_{n=0}^{\infty}b_n y^n \quad (y \ll 1) \quad (4.56)$$

where $\eta(\nu) = \sum_{s=2}^{\infty}\frac{s^2}{(s^2-1)^\nu}$. Notice the appearance of the $y^{\frac{1}{2}}$ term.

Whether and where a system's vacuum or ground state exists at the order parameter field $\bar{\phi}_{min}$ is determined by the condition $V^{(1)\prime}[\bar{\phi}_{min}] = 0$. If a second minimum of lower energy exists at $\bar{\phi}_{min} \neq 0$ the system will tend to evolve toward this 'true' vacuum. Symmetry restoration occurs when the effective potential assumes a global minimum at $\bar{\phi} = 0$. An important parameter for determining the nature of this phase transition is the effective mass defined by $V^{(1)\prime\prime}[\bar{\phi}_{min}]$ which measures the curvature of the effective potential at the minimum. For a quadratic potential near the minimum the condition $dV^{(1)}[\bar{\phi}]/d\bar{\phi}^2|_{\bar{\phi}=0} = 0$ yields an algebraic equation $y_0 \equiv y(\bar{\phi} = 0)$ for the determination of the critical curvature R_c when symmetry breaking occurs.

$$y_0 + \frac{\hbar\lambda}{8\pi^2}C(a,\xi) + \lambda y_0^{-\frac{1}{2}} = 0. \quad (4.57)$$

$$C(a,\xi) = 2b_1 - \frac{1}{4}m^2a^2(1 + \ln|\frac{1}{4}m^2a^2|) + \frac{1}{4}\xi(2 + \ln|\frac{1}{4}(m^2 - \frac{\xi}{6}R_0)a^2|). \quad (4.58)$$

Further details can be found in the original paper [400]. (The definitions of R_0, R_1 are reversed here.)

4.4 $O(N)$ Self-Interacting Scalar Field in Curved Spacetime

Here, we follow the presentation in [504]. Consider an N-component self-interacting scalar field Φ^a $(a = 1, \ldots, N)$ on an *Euclidean* manifold of dimension D (where the signature of the metric is $(+,+,+,\ldots)$, obtained by changing the Lorentzian time t to the Euclidean time $\tau = it$) coupled to the background spacetime with scalar curvature R described by the action

$$S_{\rm E}[\Phi] = i \int d^D x \sqrt{g} \left[-\frac{1}{2} \Phi^a \Box_{\rm E} \Phi^a + \frac{1}{2} M_\xi^2 \Phi^2 + \frac{\lambda}{4!} \Phi^4 \right], \tag{4.59}$$

where $M_\xi^2 = m^2 + (1-\xi)\xi_n R$. In the integral over the D-dim Euclidean space, $x^0 = \tau$ and $\Box_{\rm E} = \sqrt{g}\,\partial_\mu(\sqrt{g}\,g^{\mu\nu}\partial_\nu)$ is now an elliptic operator. We shall keep D general here until later when we consider it to be a direct product of a noncompact space of dimension c and a compact space of dimension b. E.g., the Euclidean de Sitter space is a 4-sphere in E^5, thus $c = 0, D = b = 4$.

Decompose the field Φ^a into a background field $\bar\phi^a$ and a fluctuation field φ^a, i.e., $\Phi^a = \bar\phi^a + \varphi^a$. The background field $\bar\phi^a$ is required to satisfy the classical equations of motion with an arbitrary external source. Such a shift eliminates the linear term in the fluctuation field, which is equivalent to performing a Legendre transform. The resultant action is

$$\begin{aligned} S_{\rm E}[\bar\phi, \varphi] = S_{\rm E}[\bar\phi] + \int d^D x \sqrt{g} \\ \times \left\{ \frac{1}{2} \varphi^a \left[\left(-\Box_{\rm E} + M_\xi^2 + \frac{\lambda}{6} \bar\phi^2 \right) \delta^{ab} + \frac{\lambda}{3} \bar\phi^a \bar\phi^b \right] \varphi^b + \frac{\lambda}{6} \bar\phi^a \varphi^a \varphi^2 + \frac{\lambda}{4!} \varphi^4 \right\}. \end{aligned} \tag{4.60}$$

Notice that there are two kinds of vertices: a three-point vertex proportional to $\lambda \bar\phi^a(x)$ and a four-point vertex proportional to λ.

The effective action $\Gamma_{\rm E}[\bar\phi]$ is obtained by functionally integrating over the fluctuation field φ :

$$e^{-\Gamma_{\rm E}[\bar\phi]} = \int [d\varphi]\, e^{-S_{\rm E}[\bar\phi, \varphi]}, \tag{4.61}$$

where $[d\varphi]$ denotes functional differential of φ. The wave operator A^{ab} for the fluctuating field is given by

$$A^{ab} = \left(-\Box_{\rm E} + M_1^2 \right) \frac{\bar\phi^a \bar\phi^b}{\bar\phi^2} + \left(-\Box_{\rm E} + M_2^2 \right) \left(\delta^{ab} - \frac{\bar\phi^a \bar\phi^b}{\bar\phi^2} \right), \tag{4.62}$$

where M_1, M_2 were defined earlier in (2.128) and (2.126):

$$M_1^2 = M_\xi^2 + \frac{\lambda}{2} \bar\phi^2, \quad M_2^2 = M_\xi^2 + \frac{\lambda}{6} \bar\phi^2. \tag{4.63}$$

Here $\bar\phi^a \bar\phi^b / \bar\phi^2$ and $\delta^{ab} - \bar\phi^a \bar\phi^b / \bar\phi^2$ are orthogonal projectors. The former projects along the direction in the internal space picked out by $\bar\phi^a$ and the latter projects

into an $(N-1)$-dimensional subspace orthogonal to the direction of $\bar{\phi}^a$. Note that the operator $\Box_{\rm E}$ does not commute with the projectors unless $\bar{\phi}^a$ is a constant.

When the direction in group space picked out by $\bar{\phi}^a$ does not vary from point to point around the manifold (this does not necessarily imply $\bar{\phi}^2$ is a constant) the Euclidean Green's function for A^{ab} is given by

$$G^{ab} = G_1 \frac{\bar{\phi}^a \bar{\phi}^b}{\bar{\phi}^2} + G_2 \left(\delta^{ab} - \frac{\bar{\phi}^a \bar{\phi}^b}{\bar{\phi}^2} \right) , \tag{4.64}$$

where the G_i $(i = 1, 2)$ are the Green's functions for the operators $-\Box_{\rm E} + M_i^2$, that is, $(-\Box_{\rm E} + M_i^2)G_i = -\delta$.

The one-loop contribution is given by the sum of the logarithms of the determinants of the fluctuation operators. So in this case, by using the projection operators, the one-loop effective action is given by

$$\begin{aligned} \Gamma_{\rm E}[\bar{\phi}] &= S_{\rm E}[\bar{\phi}] - \frac{1}{2}\mathrm{Tr}\ln G_1 - \frac{N-1}{2}\mathrm{Tr}\ln G_2 \\ &= S_{\rm E}[\bar{\phi}] + \frac{1}{2}\sum_l d_l \ln \lambda_{1l} + \frac{N-1}{2}\sum_l d_l \ln \lambda_{2l} \ , \end{aligned} \tag{4.65}$$

where $S_{\rm E}[\bar{\phi}]$ is the classical action, λ_{il} and d_l are the eigenvalues and degeneracies of the operators $-\Box_{\rm E} + M_i^2$.

The Euclidean de Sitter spacetime is a 4 sphere S^4 embedded in E^5 with volume $Vol = 8\pi^2 a^4/3$. The mode functions we use correspond to choosing the de Sitter-invariant or Euclidean vacuum. The 1-loop effective potential for an N component scalar field with $O(N)$ symmetry in de Sitter spacetime is given by

$$V^{(1)} = \frac{1}{2}\sum_{\ell=0}^{\infty} d_\ell \ln(\lambda_{1\ell}\mu^{-2}) + \frac{N-1}{2}\sum_{\ell=0}^{\infty} d_\ell \ln(\lambda_{2\ell}\mu^{-2}), \tag{4.66}$$

where

$$\lambda_{i\ell} = \frac{\ell(\ell+3)}{a^2} + M_i^2, \quad (i = 1, 2), \qquad d_\ell = \frac{(\ell+1)(\ell+2)}{6}(2\ell+3). \tag{4.67}$$

The summation leads to an ultraviolet divergence which must be regularized. For illustration purpose we shall apply the zeta-function regularization scheme of Hawking [97] here, which is a variant of Dowker & Critchley [96] used in earlier sections.

The generalized zeta function $\zeta(\nu)$ with the degeneracies d_ℓ of the eigenvalues written explicitly is given by

$$\zeta(\nu) = \sum_{\ell=0}^{\infty} d_\ell (\lambda_\ell \mu^{-2})^{-\nu}. \tag{4.68}$$

This function has an analytic continuation into the complex ν plane. After continuation the resulting function has a simple pole at $\nu = -1$ but is analytic in the region of $\nu = 0$, and for sufficiently negative ν, it agrees with the series

sum Eq. (4.68). The Taylor expansion in ν about zero of (4.68) enables one to identify the coefficients in powers of ν

$$\zeta(0) = \sum_{\ell=0}^{\infty} d_\ell \quad \text{and} \quad \zeta'(0) = -\sum_{\ell=0}^{\infty} d_\ell \ln(\lambda_\ell \mu^{-2}). \tag{4.69}$$

Making this identification yields a convenient definition of the divergent sums occurring in $\ln \mathrm{Det} = \mathrm{Tr} \ln$ of an operator with a discrete spectrum. The $\ln \mathrm{Det}$ of any operator A is related to $-\zeta_A'(0)$ by

$$\ln \mathrm{Det}[A\mu^{-2}] = -\zeta_A'(0). \tag{4.70}$$

The calculation of the 1-loop scalar contribution to the effective potential in de Sitter space is then reduced to an examination of the function

$$\zeta(x_i, \nu) = \sum_{\ell=0}^{\infty} \frac{1}{6}(\ell+1)(\ell+2)(2\ell+3)[\ell(\ell+3) + x_i]^{-\nu}\,, \tag{4.71}$$

where $x_i = M_i^2 a^2$ and the scaling property has been employed to extract an overall factor of $\mu^2 a^2$. The effective potential to 1-loop for an N component scalar field is then given by

$$\begin{aligned} V_{eff}(\bar{\phi}) = V(\bar{\phi}) - \frac{1}{2Vol}[\zeta'(x_1, 0) + \zeta(x_1, 0)\ln(\mu^2 a^2) \\ + (N-1)[\zeta'(x_2, 0) + \zeta(x_2, 0)\ln(\mu^2 a^2)]. \end{aligned} \tag{4.72}$$

Expressions for the zeta function (4.71) and its generalization to arbitrary spin have been obtained by Candelas and Raine [459], Dowker and Critchley [96], Allen [460] and others. Our derivation here follows the Ph.D. thesis work of O'Connor [461]. The resultant expressions are

$$\zeta(x_i, 0) = \frac{1}{12}x_i^2 - \frac{1}{3}x_i + \frac{29}{90} \tag{4.73}$$

and

$$\begin{aligned} \zeta'(x_i, 0) = \frac{1}{6}\int dx_i \left\{ \frac{2}{3} - (x_i - 2)\left[\psi\left(\frac{3}{2} + \left(\frac{9}{4} - x_i\right)^{1/2}\right)\right.\right. \\ \left.\left. + \psi\left(\frac{3}{2} - \left(\frac{9}{4} - x_i\right)^{1/2}\right) - 1\right]\right\} + const. \end{aligned} \tag{4.74}$$

The value of the constant is not important as it ultimately gets absorbed in the renormalization process. Since $\psi(z) = d\ln\Gamma(z)/dz$ is the logarithmic derivative of the gamma function it obeys the functional relationship $\psi(z+1) = \psi(z) + \frac{1}{z}$ derived from that of the gamma function. This relation allows us to shift the argument of the ψ function to values that yield more rapid convergence, since

we can transfer this shift to a shift in x_i. After these manipulations the following expression valid for large $x_i + 4$ is obtained

$$\zeta'(x_i, 0) = const - \ln(x_i) - (\zeta(x_i, 0) - 1)\ell n(x_i + 4) + \frac{1}{8}x^2 + \frac{4259}{945}(x_i + 4)^{-1} + \frac{298799}{50400}(x_i + 4)^{-2} + \cdots \tag{4.75}$$

Similarly, a small x_i expansion of Eq. (4.74) yields

$$\zeta'(x_i, 0) = const - \ln(x_i) + \frac{2}{9}(4 - 3\gamma)x_i - \frac{1}{12}\left[\frac{5}{6} - 2\gamma + \frac{2}{3}(\psi'(4) - \psi'(1))\right] x_i^2 + \cdots \tag{4.76}$$

and $\frac{5}{6} - 2\gamma + \frac{2}{3}(\psi'(4) - \psi'(1)) = -2(\gamma + \frac{1}{27})$.

With these expressions we can now write down the effective potential in de Sitter space by substituting (4.75) or (4.76) into (4.72). For small x_i

$$V^{(1)}(\bar{\phi}) = \frac{1}{2Vol}\{const + \ln(x_1) + (N-1)\ln(x_2) + \frac{1}{3}[\ln(\mu^2 a^2) + 2\gamma - \frac{8}{3})]\,[x_1 + (N-1)x_2] - \frac{1}{12}[\ln(\mu^2 a^2) + 2\gamma + \frac{2}{27})]\,[x_1^2 + (N-1)x_2^2] + \cdots\Big\}. \tag{4.77}$$

and for large $x_i + 4$

$$V^{(1)}(\bar{\phi}) = \frac{1}{2Vol}\Big\{const - N\ln n(\mu^2 a^2) + \ln(x_1) + (N-1)\ln(x_2) + [\zeta(x_1, 0) - 1]\ln(\frac{x_1 + 4}{\mu^2 a^2}) + (N-1)[\zeta(x_2, 0) - 1]\ln(\frac{x_2 + 4}{\mu^2 a^2}) - \frac{1}{8}[x_1^2 + (N-1)x_2^2] + \cdots\Big\}. \tag{4.78}$$

From Eqs (4.77) and (4.78) we see that quantum corrections to the classical potential are small except when $x_i \to 0$ where there is a logarithmic singularity. The potential also appears to become complex when $x_i < 0$, but only after passing through $-\infty$. Mean field theory predicts an asymmetric ground state when $x_i < 0$ and a second-order critical point when x_i, evaluated at $\bar{\phi} = 0$, is zero. Therefore, the appropriate expansion for an examination of symmetry breaking is Eq. (4.77). What this means is that the 1-loop calculation in de Sitter space is untrustworthy when the classical analysis predict a broken symmetry, or the presence of a second order transition. To overcome these infrared divergence problems one has to resort to some procedure which includes an infinite number of self energy graphs.

We shall see in Chapter 6 that the nonperturbative contribution to the infrared divergence can be captured by identifying the zero mode of the invariant operator of the massless minimally coupled interacting scalar field in S^4. This was first pointed out in [462, 461], rediscovered by [463] and further developed by others.

4.4.1 Scalar QED in de Sitter Space

Phase transitions at the grand unified theory scale involve gauge fields. Scalar quantum electrodynamics (SQED) is probably the simplest gauge theory model which is still computationally manageable. In this section we shall derive the one loop effective potential for SQED in S^4 following the work of O'Connor [461].

The Euclidean action functional is given by

$$S[\Phi, A] = \int d^D x \sqrt{g} \left(\frac{1}{4} F^2 - \frac{1}{2} \Phi^a (D_\mu D^\mu \Phi)^a + \frac{1}{2} \left[m^2 + \frac{(1-\xi)}{6} R \right] \Phi^2 + \frac{\lambda}{4!} \Phi^4 \right) \tag{4.79}$$

where $D_\mu^{ab} = \nabla_\mu \delta^{ab} + e \epsilon^{ab} A_\mu, e$ is the electric charge, $F^2 = F_{\mu\nu} F^{\mu\nu}$ and $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ with the usual notation [460]. Consider the case of a background scalar field $\Phi^a = \bar{\phi}^a + \varphi^a$ but no background gauge field $A_\mu = 0 + a_\mu$, since we are only interested in breaking the de Sitter invariance of the background. In this case the effective action $\Gamma[\bar{\phi}]$ is obtained from the functional integral

$$e^{-\Gamma[\bar{\phi}]} = \int [d\varphi][da] \mathrm{Det}'[-\nabla^2] \delta[\nabla \cdot a] e^{-S[\bar{\phi}, \varphi, a]}. \tag{4.80}$$

Here the gauge-fixing δ-functional projects out the longitudinal photon degrees of freedom, Det denotes the compensating gauge-fixing Faddeev–Popov determinant and a prime on it means the zero modes are left out. The action functional to be integrated over $S[\bar{\phi}, \varphi, a]$ is obtained by first splitting the scalar and gauge fields into background fields plus their fluctuation fields, then requiring the background fields to satisfy the equations of motion with an arbitrary background current. It is given by

$$\begin{aligned} S[\bar{\phi}, \varphi, a] = S[\bar{\phi}, 0] + \int d^D x \sqrt{g} [& \frac{1}{2} a_\mu (\Box^{\mu\nu} + e^2 \bar{\phi}^2 g^{\mu\nu}) a_\nu \\ & - \frac{1}{2} (\nabla \cdot a)^2 - e (\nabla \cdot a) \varphi^a \epsilon^{ab} \bar{\phi}^b - 2 e \varphi^a \epsilon^{ab} a_\mu \nabla^\mu \bar{\phi}^b \\ & + \frac{1}{2} \varphi^a A^{ab} \varphi^b - e \varphi^a \epsilon^{ab} a_\mu \nabla^\mu \varphi^b + e^2 \bar{\phi}^a a^2 \varphi^2 \\ & + \frac{1}{2} e^2 a^2 \varphi^2 + \frac{\lambda}{6} \bar{\phi}^a \varphi^a \varphi^2 + \frac{\lambda}{4!} \varphi^4] \end{aligned} \tag{4.81}$$

where $\Box_{\mu\nu} = -\nabla^2 g_{\mu\nu} + R_{\mu\nu}, R_{\mu\nu}$ being the Ricci tensor and A^{ab} is the small fluctuation operator (4.62) for a two component scalar field:

$$A^{ab} = \left[-\Box + \left(m^2 + \frac{(1-\xi)}{6}R\right) + \frac{\lambda}{6}\bar{\phi}^4\right]\delta^{ab} + \frac{\lambda}{3}\bar{\phi}^a\bar{\phi}^b\,. \tag{4.82}$$

We can use Hodge-decomposition to separate $a_\mu = a_\mu^T + a_\mu^L + a_\mu^H$ into a transverse (T), a longitudinal (L) and a harmonic (H) part. This is an orthogonal decomposition so that the action splits into a sum of the individual contributions

$$S[\bar{\phi}, \varphi, a] = S_T[\bar{\phi}, \varphi, a_T] + S_L[\bar{\phi}, \varphi, a_L] + S_H[\bar{\phi}, \varphi, a_H]. \tag{4.83}$$

For the d-sphere ($d > 1$) there is no harmonic part, since the number of harmonic one-forms is equal to the first Betti number of the manifold which is zero for the d-sphere ($d \neq 1$). The gauge-fixing delta-functional only applies to the longitudinal part, therefore the integration over the longitudinal degrees of freedom is trivial. The relevant functional integral is

$$\int [da^L] Det'[-\mu^{-2}\nabla^2]\delta[\nabla \cdot a^L] e^{-S_L[\bar{\phi}, \varphi, a^L]}. \tag{4.84}$$

Noting that $a_\mu^L = \nabla_\mu \chi$, a change of integration variable from a_L to χ is appropriate to yield

$$\begin{aligned}\int [d\chi] \mid Det'[\nabla] \mid Det'[-\mu^{-2}\nabla^2]\, \delta[\nabla^2\chi] e^{-S_L[\bar{\phi}, \varphi, \nabla\chi]} \\ = \int [d\chi] Det'[-\mu^{-2}\nabla^2]^{1/2}\delta[\chi] e^{-S_L[\bar{\phi}, \varphi, \nabla\chi]} = Det'[-\mu^{-2}\nabla^2]^{1/2}.\end{aligned} \tag{4.85}$$

We are left with the remaining functional integral over the transverse fluctuations of the gauge field and the scalar field.

$$e^{-\Gamma[\bar{\phi}]} = \int [d\varphi][da^T] Det'[-\mu^{-2}\nabla^2]^{1/2} e^{-S[\bar{\phi}, \varphi, a^T]} \tag{4.86}$$

and

$$\begin{aligned}S[\bar{\phi}, \varphi, a_T] = S[\bar{\phi}, 0] + \int d^D x\sqrt{g}\left\{\frac{1}{2}a_\mu^T\left[\Box^{\mu\nu} + e^2\bar{\phi}^2 g^{\mu\nu}\right]a_\nu^T + \frac{1}{2}\varphi^a A^{ab}\varphi^b\right. \\ \left. - 2e\varphi^a\epsilon^{ab}a_\mu^T\nabla^\mu\bar{\phi}^b - e\varphi^a\epsilon^{ab}a_\mu^T\nabla^\mu\varphi^b + \frac{1}{2}e^2 a_T^2\varphi^2 + \frac{\lambda}{6}\bar{\phi}^a\varphi^a\varphi^2 + \frac{\lambda}{4!}\varphi^4\right\}.\end{aligned} \tag{4.87}$$

These equations are suitable for calculating the effective action for the scalar field to all orders when there is no background gauge field. If we are satisfied with deriving the effective potential then the background scalar field $\bar{\phi}^a$ can be set to be a constant and the term $2e\varphi^a\epsilon^{ab}a_\mu^T\nabla^\mu\bar{\phi}^b$ drops out of the action Eq. (4.87).

The one-loop contribution to the effective action is therefore given by

$$\Gamma[\bar{\phi}] = S[\bar{\phi}] + \frac{1}{2}\mathrm{Tr}\ln[\mu^{-2}A^{ab}] - \frac{1}{2}\mathrm{Tr}'\ln[-\mu^{-2}\nabla^2] + \frac{1}{2}\mathrm{Tr}\ln[\mu^{-2}(\Box^{\alpha\beta} + e^2\bar{\phi}^2 g^{\alpha\beta})] + \cdots \tag{4.88}$$

where Tr' means that the zero-modes are left out, since these are genuine Faddeev-Popov zero modes. The first term in (4.88) is the contribution from a two component scalar field, therefore the remaining task is the calculation of the determinant arising from the fluctuations in the gauge field.

For de Sitter space, using the eigenvalues and their degeneracy for S^4 of the operator $\Box_{\mu\nu}$

$$\lambda_\ell = (\ell^2 + 5\ell + 6)a^{-2},$$
$$d_\ell = \frac{1}{2}(\ell+1)(\ell+4)(2\ell+5), \tag{4.89}$$

we get the one-loop contribution to Γ from the transverse field

$$\Gamma_T^{(1)}[\bar{\phi}] = \frac{1}{2}\mathrm{Tr}\ln[\mu^{-2}(\Box^{\alpha\beta} + e^2\bar{\phi}^2 g^{\alpha\beta})] = \frac{1}{4}\sum_{\ell=0}^{\infty}(\ell+1)(\ell+4)(2\ell+5)\ln[\mu^{-2}a^{-2}(\ell^2+5\ell+6+s^2)], \tag{4.90}$$

where $s^2 = e^2\bar{\phi}^2a^2$. We can evaluate this as in the scalar case by the zeta function method.

Defining the zeta function as

$$\zeta(s,\nu) = \frac{1}{2}\sum_{\ell=0}^{\infty}(\ell+1)(\ell+4)(2\ell+5)[\ell^2+5\ell+6+s^2]^{-\nu} \tag{4.91}$$

we have

$$\Gamma_T^{(1)}[\bar{\phi}] = -\frac{1}{2}[\zeta'(s,0) + \zeta(s,0)\ln(\mu^2a^2)]\ . \tag{4.92}$$

The effective potential for SQED in de Sitter space shown here, originally derived by O'Connor [461], was applied to the analysis of symmetry behavior of SQED with implications on phase transitions in the early universe [462]. See also [464]. We will further discuss the infrared behavior of quantum fields in de Sitter universe in Chapter 6.

Part II: Stress Energy Correlations

In the second part of this chapter we will show how the correlation functions for the quantum stress energy bitensor results from the second metric variation of the effective action. This parallels the definition of the expectation value of the quantum stress-energy tensor resulting from the first metric variation of the

effective action. Only from a regularized or renormalized effective action can a physically meaningful expectation value be derived. Likewise, the correlation function is defined in terms of the second variation of the *regularized* effective action. Note that the correlation of the stress-energy tensor computed for two distinct points is finite regardless of whether the effective action is regularized or not (excluding lightlike separated points for a massless quantum field). It is only when the autocorrelation is computed that the issue of regularization arises. The correlation function defined in terms of the regularized effective action is a consistent approach since it is defined as the second variation of the same object for which the expectation value is the first variation. We give a formal derivation in the section following. In the next section we use the second variation of the regularized effective action to derive the stress-energy tensor correlations, and provide two examples, that in hot flat space and in N-dimensional anti-de Sitter (AdS^N) space. Our presentation is based on the work of Cho, Hu and Phillips [197, 465, 466].

4.5 Stress-Energy 2-Pt Function from 2nd Variation of Effective Action

As we have shown in the first part of this chapter, the generalized zeta function of a quantum system is given as the Mellin transform of the trace of its heat kernel. The effective action is then given by the derivative of the zeta function. The key step in the zeta function regularization method is to control the divergence of the heat kernel present by following its behavior as the Schwinger proper time is set to zero. This is done by introducing positive powers of the proper time and then performing an analytic continuation in powers of the proper time. At the end of the calculation, the variable of analytic continuation is set to zero. This is a consistent procedure in that the introduced power can be relaxed to zero at any point of the calculation to recover the initial formal expression.

The second variation of the generalized zeta function is facilitated by the Schwinger–DeWitt proper time expansion, which shows how the trace of the heat kernel responses. For second variation we only need to know the trace of a pair of heat kernels. For each of these traces we introduce a power of the proper time variable. The stress-energy correlation function is expressed in terms of the traces of the system's heat kernel, regularized via the generalized zeta function method. A nice feature is that we can relax the introduced powers at any time and recover the formal expression for the correlation function.

For geometries admitting a mode decomposition of the invariant operator, we display the correlation function explicitly in terms of these modes. For homogeneous spaces or spacetimes the calculations are simpler because it is possible to obtain analytic expressions.

We start by considering the generating functional of a scalar field ϕ in the Euclidean section Σ of a spacetime manifold $\mathcal{M}$,

$$Z = \int \mathcal{D}\phi \, e^{-S[\phi]} = \langle 0_+ | 0_- \rangle \tag{4.93}$$

and its functional derivatives with respect to the metric:

$$\begin{aligned}
\frac{\delta Z}{\delta g^{ab}(x)} &= -\int \mathcal{D}\phi \frac{\delta S}{\delta g^{ab}(x)} e^{-S} = -\langle 0_+ | \frac{\delta S}{\delta g^{ab}(x)} | 0_- \rangle, \\
\frac{\delta^2 Z}{\delta g^{ab}(x) \delta g^{c'd'}(y)} &= \int \mathcal{D}\phi \left[\frac{\delta S}{\delta g^{ab}(x)} \frac{\delta S}{\delta g^{c'd'}(y)} - \frac{\delta^2 S}{\delta g^{ab}(x) \delta g^{c'd'}(y)} \right] e^{-S} \\
&= \langle 0_+ | \frac{\delta S}{\delta g^{ab}(x)} \frac{\delta S}{\delta g^{c'd'}(y)} | 0_- \rangle - \langle 0_+ | \frac{\delta^2 S}{\delta g^{ab}(x) \delta g^{c'd'}(y)} | 0_- \rangle,
\end{aligned} \tag{4.94}$$

where $|0_-\rangle, |0_+\rangle$ are the vacua defined at the in and out states respectively. Using $W = \ln Z$ from (2.39) but for Euclidean space, which is equal to the effective action Γ in the absence of external source, this becomes

$$\frac{\delta^2 W}{\delta g^{ab}(x) \delta g^{c'd'}(y)} = \frac{1}{Z} \frac{\delta^2 Z}{\delta g^{ab}(x) \delta g^{c'd'}(y)} - \frac{\delta W}{\delta g^{ab}(x)} \frac{\delta W}{\delta g^{c'd'}(y)}. \tag{4.95}$$

The ('in-out') transition amplitude of the quantum stress-energy tensor is given by

$$\langle T_{ab} \rangle = \frac{\langle 0_+ | T_{ab} | 0_- \rangle}{\langle 0_+ | 0_- \rangle} = \frac{2}{\sqrt{g(x)}} \frac{\langle 0_+ | \delta S / \delta g^{ab}(x) | 0_- \rangle}{\langle 0_+ | 0_- \rangle} = -\frac{2}{\sqrt{g(x)}} \frac{\delta W}{\delta g^{ab}(x)}. \tag{4.96}$$

In analogy with this, we define the ('in-out') correlation function for the stress-energy tensor as

$$\begin{aligned}
\langle T_{ab}(x) T_{c'd'}(y) \rangle &= \frac{\langle 0_+ | T_{ab}(x) T_{c'd'}(y) | 0_- \rangle}{\langle 0_+ | 0_- \rangle} \\
&= \frac{4}{\sqrt{g(x)g(y)}} \frac{1}{Z} \langle 0_+ | \frac{\delta S}{\delta g^{ab}(x)} \frac{\delta S}{\delta g^{c'd'}(y)} | 0_- \rangle \\
&= \frac{4}{\sqrt{g(x)g(y)}} \left[\frac{\delta^2 W}{\delta g^{ab}(x) \delta g^{c'd'}(y)} + \frac{\delta W}{\delta g^{ab}(x)} \frac{\delta W}{\delta g^{c'd'}(y)} \right. \\
&\qquad \left. + \langle \frac{\delta^2 S}{\delta g^{ab}(x) \delta g^{c'd'}(y)} \rangle \right].
\end{aligned} \tag{4.97}$$

For any local action the last term will not contribute to the final expression for the stress-energy correlation function. For such an action, this expression will depend on x, y as a delta function $\delta(x-y)$. Thus it need not be considered when computing the correlation function for $x \neq y$. The autocorrelation is obtained by taking the coincidence limit $y \to x$.

Recognizing the second term in the last line as a product of expectation values of the stress-energy tensor, we can define the bitensor

$$\begin{aligned}\Delta T^2_{abc'd'}(x,y) &\equiv \langle T_{ab}(x)T_{c'd'}(y)\rangle - \langle T_{ab}(x)\rangle\langle T_{c'd'}(y)\rangle \\ &= \frac{4}{\sqrt{g(x)g(y)}}\frac{\delta^2 W}{\delta g^{ab}(x)\delta g^{c'd'}(y)}.\end{aligned} \tag{4.98}$$

4.5.1 Second Variation of the Effective Action

The classical action of a scalar field $\phi(x)$ is

$$S[\phi] = \frac{1}{2}\int d\mathbf{x}\phi(x)H\phi(x), \tag{4.99}$$

where H is a second-order elliptic operator. From the spectral decomposition of this operator, $H = \sum_n \lambda_n |n\rangle\langle n|$, where n denotes the collective indices of the spectrum, the effective action can be expressed as

$$W = -\frac{1}{2}\ln \mathrm{Det}(H/\mu^2) = -\frac{1}{2}\mathrm{Tr}\ \ln\frac{H}{\mu^2} = -\frac{1}{2}\sum_n \ln\frac{\lambda_n}{\mu^2}, \tag{4.100}$$

where we assume the zero modes of H have been projected out and μ^2 is a normalization constant with dimensions of mass-squared. This expression for the effective action is only formal since H is not trace class. We regularize the effective action and the expressions derived from it via the zeta function method.

The zeta function for this system is defined as

$$\zeta_H(\nu) = \mathrm{Tr}\ e^{-\nu \ln H/\mu^2} = \mu^{2\nu}\sum_n \lambda_n^{-\nu}. \tag{4.101}$$

For $2\nu > \dim\ \mathcal{M}$, this sum is convergent. Then an analytic continuation in ν is found such that it includes a neighborhood of $\nu = 0$. From this analytic continuation the regularized effective action is given as

$$W_R = \frac{1}{2}\frac{d\zeta(\nu)}{d\nu}\bigg|_{\nu=0} = \frac{1}{2}\zeta'(0). \tag{4.102}$$

For positive H the definition of the gamma function yields

$$\zeta_H(\nu) = \frac{\mu^{2\nu}}{\Gamma(\nu)}\int_0^\infty s^{\nu-1}\mathrm{Tr}\ U_0(s)ds, \quad U_0(s) = e^{-sH}, \tag{4.103}$$

i.e. as seen before, the zeta function is given as the Mellin transform of the trace of the heat kernel $U_0(t)$. We know $\mathrm{Tr}\ U_0(s) \sim s^{-d/2}$ for $s \to 0$. Hence for (4.103) to be convergent, we have the condition $\nu > d/2$. This is most often the expression from which an analytic continuation is derived. In fact, one of the main points of the zeta function idea is that formal expressions such as $\mathrm{Tr}\ U_0(t)$ need be modified by the introduction of a factor s^ς. Then once the analytic continuation is found, one takes $\varsigma = 0$.

We now consider the effect of two small metric perturbations δ_1 and δ_2 on the effective action. (They are assumed to be independent of the order with which they act.) The response of the effective action to these perturbations is

$$\begin{aligned}\delta_2\delta_1 W_R[g] &= W_R[g+\delta_1+\delta_2]+W_R[g]-W_R[g+\delta_1]-W_R[g+\delta_2] \\ &= \frac{1}{2}\frac{d}{d\nu}\left(\delta_2\delta_1\zeta_H(\nu)\right)\Big|_{\nu=0}, \qquad (4.104)\\ \delta_2\delta_1\zeta_H(\nu) &= \frac{\mu^{2\nu}}{\Gamma(\nu)}\int_0^\infty ds\; s^{\nu-1} \\ &\quad \mathrm{Tr}\left\{e^{-s(H+\delta_1 H+\delta_2 H)}-e^{-s(H+\delta_1 H)}-e^{-s(H+\delta_2 H)}+e^{-sH}\right\}. \qquad (4.105)\end{aligned}$$

To evaluate this, we perform a Schwinger proper time expansion [124]. For $U(s)=e^{-s(H+H_1)}$, where $H_1=\delta_1 H \ll H$,

$$\begin{aligned}\mathrm{Tr}\; U(s) &= \mathrm{Tr}\; U_0(s) - s\mathrm{Tr}\;\left[H_1 U_0(s)\right] \\ &\quad + \frac{s^2}{2}\int_0^1 du_1 \mathrm{Tr}\;\left[H_1 U_0((1-u_1)s) H_1 U_0(u_1 s)\right]+\cdots \qquad (4.106)\end{aligned}$$

Using this, we can write the response of the zeta function to this expansion as

$$\begin{aligned}\delta_2\delta_1\zeta_H(\nu) &= \frac{\mu^{2\nu}}{2\Gamma(\nu)}\int_0^\infty ds\; s^{\nu+1}\int_0^1 du_1\left\{\mathrm{Tr}\;\left[(\delta_1 H)U_0((1-u_1)s)(\delta_2 H)U_0(u_1 s)\right]\right. \\ &\quad \left.+\mathrm{Tr}\;\left[(\delta_2 H)U_0((1-u_1)s)(\delta_1 H)U_0(u_1 s)\right]\right\}. \qquad (4.107)\end{aligned}$$

As it stands, it is not finite. When the traces are taken involving $U_0((1-u_1)s)$ and $U_0(u_1 s)$, the divergences at $(1-u_1)s, u_1 s \to 0$ are present. At this point, we modify the above expression for the second variation of the zeta function by introducing the factor $[u_1(1-u_1)s^2]^\varsigma$. At the end of the calculation, once the analytic continuation is found, we set $\varsigma=0$. We view the introduction of this factor as an extension of the zeta function method to include the second variation. The replacement of $U_0(s)\to s^\varsigma U_0(s)$ is in the spirit of the zeta function method which reproduces the known results when applied to familiar problems, such as finding the first variation, in which case it produces the expectation value of the quantum stress-energy tensor. After the change of variables

$$u=(1-u_1)s, \quad v=u_1 s \qquad (4.108)$$

the twice-varied zeta function transforms to

$$\begin{aligned}\delta_2\delta_1\zeta_H(\nu) &= \frac{\mu^{2\nu}}{2\Gamma(\nu)}\int_0^\infty du\int_0^\infty dv(u+v)^\nu u(uv)^\varsigma\left\{\mathrm{Tr}\;\left[(\delta_1 H)U_0(u)(\delta_2 H)U_0(v)\right]\right. \\ &\quad \left.+\mathrm{Tr}\;\left[(\delta_2 H)U_0(u)(\delta_1 H)U_0(v)\right]\right\}. \qquad (4.109)\end{aligned}$$

Considering the first trace in the above expression, we find

$$\begin{aligned}\text{Tr } [(\delta_1 H)U_0(u)(\delta_2 H)U_0(v)] &= \sum_{n,n'} \langle n'|(\delta_1 H)e^{-uH}|n\rangle\langle n|(\delta_2 H)e^{-vH}|n'\rangle \\ &= \sum_{n,n'} e^{-u\lambda_n - v\lambda'_m}\langle n'|(\delta_1 H)|n\rangle\langle n|(\delta_2 H)|n'\rangle. \quad (4.110)\end{aligned}$$

By defining

$$T_{ab}[\phi_n(x), \phi'^*_n(x)] \equiv \frac{2}{\sqrt{g(x)}}\langle n'|\frac{\delta H}{\delta g^{ab}(x)}|n\rangle = \frac{2}{\sqrt{g(x)}}\int d\mathbf{z}\phi'^*_n(z)\frac{\delta H}{\delta g^{ab}(x)}\phi_n(z), \quad (4.111)$$

we can now write the stress-energy correlation bitensor as

$$\begin{aligned}\Delta T^2_{abc'd'}(x,y) = \frac{1}{2}\frac{d}{d\nu}\Bigg[\frac{\mu^{2\nu}}{\Gamma(\nu)}\int_0^\infty du\int_0^\infty dv(u+v)^\nu(uv)^\varsigma \\ \sum_{n,n'} e^{-u\lambda_n - v\lambda_{n'}}T_{ab}[\phi_n(x),\phi^*_{n'}(x)]T_{c'd'}[\phi_{n'}(y),\phi^*_n(y)]\Bigg] \quad (4.112)\end{aligned}$$

where the $\nu, \varsigma \to 0$ limit is understood.

For homogeneous spacetimes this implies $\Delta T^2_{abcd}(x,y)$ will only depend on $r = x - y$. Thus we can average over all x and do this while leaving the points separated by r. Also, the homogeneity will usually lead to a degeneracy of the eigenvalues. Thus the collective quantum number n can be split into principal and degenerate parts $n \to n, m$ and the eigenvalues only depend on n: $\lambda_{nm} \to \lambda_n$. This allows the sum over the degenerate indices to be done before evaluation of the zeta function. Putting all these together the stress-energy two-point function becomes

$$\begin{aligned}\Delta T^2_{abc'd'}(r) = \frac{1}{2\Omega}\frac{d}{d\nu}\Bigg[\frac{\mu^{2\nu}}{\Gamma(\nu)}\int_0^\infty du\int_0^\infty dv(u+v)^\nu(uv)^\varsigma\Bigg(\sum_{n,n'} e^{-u\lambda_n - v\lambda_{n'}} \\ \times \sum_{mm'}\int_{\mathcal{M}} d\mathbf{x}\, T_{ab}\left[\phi_{nm}(x),\phi^*_{n'm'}(x)\right]T_{c'd'}\left[\phi_{n'm'}(x+r),\phi^*_{nm}(x+r)\right]\Bigg)\Bigg], \quad (4.113)\end{aligned}$$

where $\Omega = \int_{\mathcal{M}} d\mathbf{x}$, with $\mathbf{x}$ a 4-dim vector, is the volume of the compact (spatial) manifold, which is different from the spacetime volume Vol (unless if the time direction is compactified, like the S^4 example of de Sitter discussed in an earlier section). If the manifold is noncompact, it is understood to be the unit volume.

Note that in the prescription of [197], an additional regularization factor $(uv)^\varsigma$ has been introduced. This is because these authors were interested in the fluctuations of the stress-energy tensor in the coincident limit of $\Delta T^2_{\mu\nu\alpha'\beta'}(x,x')$ where, in this limit, further divergences occur which call for an additional regularization factor. However, if the goal is to obtain correlators in the non-coincident case,

i.e., with two points separated, as is set here, then keeping this factor is not necessary, albeit the expression in Eq. (4.113) is more symmetric with this factor, Here we can first take the $\nu \to 0$ limit without spoiling the regularization and the expression in Eq. (4.113) will become

$$\Delta T^2_{abc'd'}(x,x') = \frac{1}{2}\int_0^\infty du \int_0^\infty dv (uv)^\varsigma \sum_{n,n'} e^{-u\lambda_n - v\lambda_{n'}} \times T_{ab}\left[\phi_n(x), \phi^*_{n'}(x)\right] T_{c'd'}\left[\phi_{n'}(x'), \phi^*_n(x')\right]\Big|_{\varsigma\to 0}. \tag{4.114}$$

We shall see from the following that with this expression the integrations over u and v effectively separate. The calculations are therefore simplified considerably.

4.5.2 Second Variation of the Zeta Function

We now develop the general form for the second variation of the zeta function for the Klein–Gordon field. We assume the Euclidean metric of the manifold $\mathcal{M} = S^1 \times \Sigma$ is of the form

$$g_{ab} = \begin{pmatrix} 1 & 0 \\ 0 & h_{ij} \end{pmatrix}, \tag{4.115}$$

where h_{ij} is the metric for the spatial section Σ. We denote the time and spatial variables by $x = (\tau, \mathbf{x})$, the invariant spatial volume form by $d\mathbf{x}$. (Σ here referring only to spatial hypersurfaces is more specialized than B^b introduced beneath (4.2).)

The wave operator for the Klein–Gordon field is given by

$$H = -\Box + \xi R + m^2 = -\frac{\partial^2}{\partial\tau^2} - {}^\Sigma\Delta + \xi R + m^2. \tag{4.116}$$

Here, and for the rest of this chapter, relieved of the focus on the conformal properties of the field, we revert to the more commonly used definition of ξ, that is, $\xi = 0$ denotes minimal coupling and $\xi = 1/6$ (in 4 dimension spacetime) for conformal coupling. Let $u_n(\mathbf{x})$ be the eigenfunctions of ${}^\Sigma\Delta$: ${}^\Sigma\Delta u_n(\mathbf{x}) = -\kappa_n^2 u_n(\mathbf{x})$, where n denotes the (collective) quantum numbers for the spatial part of the spectrum. We assume the $u_n(\mathbf{x})$ are orthonormal. The Euclidean time is made periodic with a period of $\beta = 1/T$, where T can be interpreted as a temperature with the Boltzmann constant $k = 1$. The eigenfunctions are thus given by

$$\phi_{k_0,n}(x) = \frac{e^{-ik_0\tau}}{\sqrt{\beta}} u_n(\mathbf{x}), \quad k_0 = \frac{2\pi n_0}{\beta}, \quad n_0 = 0, \pm 1, \pm 2, \ldots \tag{4.117}$$

and the eigenvalues by (see e.g., [400])

$$\lambda_{k_0,n} = k_0^2 + \kappa_n^2 + \xi R + m^2. \tag{4.118}$$

From our definition of the stress-energy tensor (4.111) we find

$$\begin{aligned} T_{ab}[\phi,\psi](x) &\equiv \frac{2}{\sqrt{g(x)}}\int\sqrt{g(x')}\psi(x')\left(\frac{\delta H_{x'}\phi(x')}{\delta g^{ab}(x)}\right)dx' \\ &= -(\nabla_a\psi\nabla_b\phi+\nabla_b\psi\nabla_a\phi)+g_{ab}\left(\nabla_c\psi\nabla^c\phi+\psi\nabla_c\nabla^c\phi\right) \\ &\quad -2\xi(g_{ab}\Box(\psi\phi)-\nabla_a\nabla_b(\psi\phi)+\psi\phi R_{ab}). \end{aligned} \tag{4.119}$$

This differs not unexpectedly from the usual definition of $T_{ab}[\psi,\phi]$. The eigenvalues used are themselves different, since we have the extra τ-degree of freedom. This expression can be used to calculate the autocorrelation of the energy density $\rho = T_{00}$

$$\Delta\rho^2(x) \equiv \lim_{y\to x}\left(\langle\rho(x)\rho(y)\rangle - \langle\rho(x)\rangle\langle\rho(y)\rangle\right), \tag{4.120}$$

which provides a measure for the magnitude of the fluctuations of the stress-energy vacuum expectation value. This choice yields

$$\begin{aligned} \rho[\phi,\psi] &= -\left(\partial_\tau\phi\partial_\tau\psi-\psi\partial_\tau^2\phi\right)+\left(\nabla_i\psi\nabla^i\phi+\psi^\Sigma\Delta\phi\right) \\ &\quad -2\xi(\psi\Delta\phi+\phi\Delta\psi+2\nabla_i\psi\nabla^i\phi), \end{aligned} \tag{4.121}$$

where i is summed over the spatial degrees of freedom only. This becomes

$$\rho[\phi_n,\phi^*_{n'}] = -\left(k_0^2+k_0k_0'-(2\xi-1)\kappa_n^2-2\xi\kappa_{n'}^2\right)\phi_n\phi^*_{n'}-(4\xi-1)\left(\nabla_i\phi_n\right)\left(\nabla^i\phi^*_{n'}\right) \tag{4.122}$$

when we specialize to eigenmodes (4.117). When we consider $|\rho[\phi_n,\phi^*_{n'}]|^2$ the contribution from the $\phi^*_n\phi_{n'}\nabla_i\phi_n\nabla^i\phi^*_{n'}$ and its conjugate will vanish when summed over. This will be shown case by case. This assumption provides

$$\begin{aligned} |\rho[\phi_n,\phi^*_{n'}]|^2 &= \left(k_0^2+k_0k_0'-(2\xi-1)\kappa_n^2-2\xi\kappa_{n'}^2\right) \\ &\quad\times\left(k_0'^2+k_0k_0'-(2\xi-1)\kappa_{n'}^2-2\xi\kappa_n^2\right)|\phi_n|^2|\phi^*_{n'}|^2 \\ &\quad+(4\xi-1)^2\left|\left(\nabla_i\phi_n\right)\left(\nabla^i\phi^*_{n'}\right)\right|^2. \end{aligned} \tag{4.123}$$

4.5.3 Quantum Vacuum Energy Density Fluctuations

The expressions derived above are valid for finite temperature fields owing to the periodicity condition imposed on the imaginary time. To consider quantum vacuum effects we will for the rest of this subsection set the temperature at zero, i.e. $\beta\to\infty$ whereby $\sum_{n_0=-\infty}^{\infty}\to(\beta/2\pi)\int_{-\infty}^{\infty}dk_0$. We can now do these sums/integrals easily, since they are all moments of gaussians. We find

$$\sum_{n_0=-\infty}^{\infty} e^{-(2\pi n_0/\beta)^2 u} \to \frac{\beta}{2\pi}\int_{-\infty}^{\infty} e^{-uk_0^2}dk_0 = \frac{\beta}{2\sqrt{\pi}}u^{-1/2}, \tag{4.124}$$

$$\sum_{n_0=-\infty}^{\infty}\left(\frac{2\pi n_0}{\beta}\right)^2 e^{-(2\pi n_0/\beta)^2 u} \to \frac{\beta}{2\pi}\int_{-\infty}^{\infty} k_0^2 e^{-uk_0^2}dk_0 = \frac{\beta}{4\sqrt{\pi}}u^{-3/2},$$

and there is zero contribution for any odd power of k_0 or k_0'. The normalization of ϕ_n and $\phi_{n'}$ provides a factor of β^{-2} which cancels the powers of β introduced in approaching the $\beta\to\infty$ limit. Introducing the functions

$$\Xi_{nn'}(u,v) = \frac{1}{8\pi\Omega(uv)^{\frac{1}{2}}}\left\{\frac{1}{uv}+\frac{\kappa_n^2}{v}+\frac{\kappa_{n'}^2}{u}+2\kappa_n^2\kappa_{n'}^2\right\}\sum_{mm'}\int_\Sigma |u_{nm}|^2|u_{n'm'}|^2 d\mathbf{x} \tag{4.125}$$

$$\Theta_{nn'}(u,v) = \frac{1}{8\pi\Omega(uv)^{\frac{1}{2}}}\left(\frac{1}{u}+2\kappa_n^2\right)\sum_{mm'}\int_\Sigma u_{nm}\nabla_i u^*_{nm}u^*_{n'm'}\nabla^i u_{n'm'}d\mathbf{x} \tag{4.126}$$

$$\Pi_{nn'}(u,v) = \frac{1}{4\pi\Omega(uv)^{\frac{1}{2}}}\sum_{mm'}\int_\Sigma \left|\nabla_i u_{nm}\nabla^i u^*_{n'm'}\right|^2 d\mathbf{x} \tag{4.127}$$

and

$$\Psi_{nn'}(u,v) = \Xi_{nn'}(u,v)+\Theta_{nn'}(u,v)+\Theta_{n'n}(v,u)+\Pi_{nn'}(u,v), \tag{4.128}$$

we define the zeta function associated with Ψ to be

$$\zeta_\Psi(\nu,\varsigma) = \sum_{n,n'}\int_0^\infty\int_0^\infty (u+v)^\nu (uv)^\varsigma \Psi_{nn'}(u,v)e^{-\lambda_n u-\lambda_{n'}v}dvdu. \tag{4.129}$$

From it, we derive an expression for the regularized autocorrelation of the energy density as

$$\Delta\rho^2 = \frac{1}{2}\,\frac{d}{d\nu}\left[\frac{\mu^{2\nu}}{\Gamma(\nu)}\zeta_\Psi(\nu,\varsigma)\right]\Bigg|_{\nu=0,\varsigma=0} = \frac{1}{2}\lim_{\varsigma\to 0}\zeta_\Psi(0,\varsigma), \tag{4.130}$$

where we have used $1/\Gamma(\nu)\sim\nu+\gamma\nu^2+0(\nu^3)$ (γ is Euler's constant) for $\nu\sim 0$. We have assumed $\varsigma > d/2+1$, in which case both $\zeta_\Psi(0,\varsigma)$ and $d\zeta_\Psi(\nu,\varsigma)/d\nu|_{\nu=0}$ are finite. Thus to find the regularized expression for $\Delta\rho^2$, we need to find the analytic continuation of $\zeta_\Psi(0,\varsigma)$ in ς such that it is finite at $\varsigma=0$.

4.6 Energy Density Fluctuations in $\Sigma = R^d\times S^1$

As another useful illustration of the zeta function method we calculate the variance of the energy density for a massless minimally coupled scalar field on a $(d+2)$-dimensional flat spacetime which is periodic on one spatial dimension with period L. Zeta function method has been effectively applied to the calculation of the energy density of quantum fields and its fluctuations for the study of the Casimir effect. See, e.g., [197, 205].

4.6.1 Casimir Energy Density Fluctuations

For this geometry, the spatial eigenfunctions are

$$u_{\mathbf{k}n}(\mathbf{x}, z) = \frac{e^{i\mathbf{k}\cdot\mathbf{x}+ilz}}{\sqrt{(2\pi)^d L}}, \quad \mathbf{k} \in R^d, \quad n = 0, \pm 1, \pm 2, \ldots, \tag{4.131}$$

with $l = 2\pi/L$. Denoting by $\mathbf{x} = (x_1, \ldots, x_d)$ the coordinates for the open dimensions and z for the one compact dimension, we have ${}^{\Sigma}\Delta = \sum_{j=1}^{d} \partial_{x_j}^2 + \partial_z^2$ and hence $\kappa_{k,n}^2 = k^2 + l^2 n^2$, ($k^2 = |\mathbf{k}|^2$). From this we see we should take k and n as the principal indices and the angular degeneracy of $\mathbf{k}$ as the degenerate indices, i.e. $\sum_m \to \int d\Omega_{d-1}$, integration over the unit $d-1$ sphere. We denote the volume of the S^{d-1} sphere as $V_{S^{d-1}}$.

To evaluate (4.95) we sum over the degenerate indices and perform the volume average

$$\int d\Omega_{d-1} \int d\Omega'_{d-1} \frac{1}{\mathrm{Vol}(\Sigma)} \int d\mathbf{x} dz |u_{kn}|^2 |u_{k'n'}|^2 = \left(\frac{V_{S^{d-1}}}{(2\pi)^d L} \right)^2 \tag{4.132}$$

from which (4.125) becomes

$$\begin{aligned} \Xi_{knk'n'}(u, v) = \frac{1}{8\pi} \left(\frac{V_{S^{d-1}}}{(2\pi)^d L} \right)^2 \frac{1}{(uv)^{\frac{1}{2}}} \\ \times \left\{ \frac{1}{uv} + \frac{k^2 + l^2 n^2}{v} + \frac{k'^2 + l^2 n'^2}{u} + 2\left(k^2 + l^2 n^2\right)\left(k'^2 + l^2 n'^2\right) \right\}. \end{aligned} \tag{4.133}$$

Also, since $\nabla_i u_{kn} \nabla^i u^*_{k'n'} = (kk' \cos\gamma + l^2 nn') u_{kn} u^*_{k'n'}$ where $\cos\gamma = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'$, the momentum correlation term (4.127) is

$$\Pi_{knk'n'}(u, v) = \frac{1}{8\pi} \left(\frac{V_{S^{d-1}}}{(2\pi)^d L} \right)^2 \frac{2}{(uv)^{\frac{1}{2}}} \left\{ \frac{k^2 k'^2}{d} + l^4 n^2 n'^2 \right\}. \tag{4.134}$$

Here we have used $\int d\Omega_{d-1} \int d\Omega'_{d-1} \cos\gamma = 0$ and $\int d\Omega_{d-1} \int d\Omega'_{d-1} \cos^2\gamma = (V_{S^{d-1}})^2/d$. Note that when summed over n and n', $\Theta_{knk'n'}$ vanishes.

We now turn to the principal index sums for k and k'. First consider the case when $n = 0$ (or $n' = 0$). This leads us to evaluate

$$\int_0^\infty dk\, k^{d-1} \int_0^\infty du\, u^{\varsigma - a} k^b e^{-k^2 u}. \tag{4.135}$$

Here $(a, b) = (3/2, 0)$ or $(1/2, 2)$, and $2a + b - 3 = 0$. We assume a boundary exists on the non-periodic dimensions at large distance and move it to infinity. This is effected by having $\int_0^\infty dk \to \lim_{\epsilon \to 0} \int_\epsilon^\infty$. Then (4.135) becomes

$$\begin{aligned} \int_\epsilon^\infty dk\, k^{d-1} \int_0^\infty du\, u^{\varsigma - a} k^b e^{-k^2 u} = \Gamma(\varsigma - a + 1) \int_\epsilon^\infty dk\, k^{d - 2\varsigma} \\ \overset{\varsigma \to 0}{\longrightarrow} -\frac{\Gamma(1 - a)}{d + 1} \epsilon^{d+1} \overset{\epsilon \to 0}{\longrightarrow} 0, \end{aligned} \tag{4.136}$$

where we have used $\varsigma > d/2$ in evaluating the $k-$integration. Thus we find there is no contribution from the $n = 0$ or $n' = 0$ terms in ζ_Ψ.

Turning to the case where n and n' do not vanish, we consider the function

$$\Psi_{nn'}(u,v) = \int_0^\infty k^{d-1}dk \int_0^\infty k'^{d-1}dk'\ \Psi_{knk'n'}(u,v)\ e^{-uk^2-vk'^2}. \tag{4.137}$$

In the integrand, there are terms with either factors of k^{d-1} or k^{d+1}, similiarly for k'. Using

$$\begin{aligned}\int_0^\infty k^{d-1}dke^{-uk^2} &= \frac{1}{2}\Gamma\left(\frac{d}{2}\right)u^{-d/2},\\ \int_0^\infty k^{d+1}dke^{-uk^2} &= \frac{1}{2}\Gamma\left(\frac{d}{2}\right)u^{-d/2}\frac{d}{2u},\end{aligned} \tag{4.138}$$

we have the rule that upon doing the k, k' integrations, we get an overall factor of $(\Gamma\left(\frac{d}{2}\right)/2)^2(uv)^{-d/2}$ and each factor of k^2 becomes $d/2u$, while k'^2 becomes $d/2v$. Applying this rule, we get from (4.128)

$$\begin{aligned}\Psi_{nn'}\left(\frac{u}{l^2},\frac{v}{l^2}\right) = \frac{1}{8\pi}&\left[\frac{V_{Sd-1}l^{d+3}}{2(2\pi)^dL}\Gamma\left(\frac{d}{2}\right)\right]^2\left(\frac{1}{uv}\right)^{\frac{d+3}{2}}\\ &\times\left\{\frac{(d+1)(d+2)}{2}+(1+d)(un^2+vn'^2)+4uvn^2n'^2\right\}\end{aligned} \tag{4.139}$$

and the corresponding zeta function

$$\begin{aligned}\zeta_\Psi(0,\varsigma) = B\sum_{n,n'=1}^\infty\int_0^\infty du\int_0^\infty dv(uv)^{\varsigma-\frac{d+3}{2}}e^{-n^2u-n'^2v}\\ \times\left\{\frac{(d+1)(d+2)}{2}+(1+d)(un^2+vn'^2)+4uvn^2n'^2\right\},\end{aligned} \tag{4.140}$$

where

$$B = \frac{1}{8\pi}\left[\frac{V_{Sd-1}l^{d+1-2\varsigma}}{(2\pi)^dL}\Gamma\left(\frac{d}{2}\right)\right]^2. \tag{4.141}$$

This form of the zeta function now allows us to perform the needed analytic continuation. We make use of the relation

$$\sum_{n=1}^\infty n^a\int_0^\infty s^{\nu-1}e^{-n^2s} = \Gamma(\nu)\,\zeta_R(2\nu-a), \tag{4.142}$$

where $\zeta_R(\nu)$ is the Riemann zeta function. Recalling (4.130), the variance of the energy density is

$$\Delta\rho^2 = \frac{B}{2}\zeta_R(-(d+1))^2 \left\{ \frac{(d+1)(d+2)}{2}\Gamma\left(-\frac{d+1}{2}\right)^2 \right.$$
$$\left. +2(d+1)\Gamma\left(-\frac{d+1}{2}\right)\Gamma\left(-\frac{d-1}{2}\right) + 4\Gamma\left(-\frac{d-1}{2}\right)^2 \right\}. \qquad (4.143)$$

Since $\Gamma\left(-(d-1)/2\right) = -((d+1)/2)\Gamma\left(-(d+1)/2\right)$, the second and third terms in the above expression cancel, leaving

$$\Delta\rho^2 = \frac{(d+1)(d+2)}{2}\left[\frac{V_{S^{d-1}}}{2\pi^{d+1}L^{d+2}}\Gamma\left(\frac{d}{2}\right)\Gamma\left(\frac{d+2}{2}\right)\zeta_R(d+2)\right]^2 \qquad (4.144)$$

by way of the Riemann zeta function reflection formula:

$$\Gamma\left(\frac{\nu}{2}\right)\zeta_R(\nu) = \pi^{\nu-\frac{1}{2}}\Gamma\left(\frac{1-\nu}{2}\right)\zeta_R(1-\nu). \qquad (4.145)$$

This is the result for the variance of the vacuum energy density (at zero temperature) for a massless minimally coupled quantum scalar field in a $d+2$-dimensional spacetime periodic in one spatial dimension. To get a measure of the fluctuations, we consider the dimensionless quantity [196]

$$\Delta' = \frac{\langle\rho^2\rangle - \langle\rho\rangle^2}{\langle\rho\rangle^2}. \qquad (4.146)$$

For the specified system the energy density is

$$\langle\rho\rangle = -\frac{V_{S^{d-1}}}{2\pi^{d+1}L^{d+2}}\Gamma\left(\frac{d}{2}\right)\Gamma\left(\frac{d+2}{2}\right)\zeta_R(d+2). \qquad (4.147)$$

Using these two expressions we obtain

$$\Delta'(\Sigma = R^d \times S^1) = \frac{(d+1)(d+2)}{2}. \qquad (4.148)$$

Using a different method Kuo and Ford [196] obtained $\Delta' = 6$ for the case of $\Sigma = R^2 \times S^1$, same as the expression obtained here, with $d = 2$. It is noteworthy that the relative amount of fluctuation increases quadratically with the dimension of the spacetime. The fact that the variance of the energy density compared to the mean is of the order of unity – this been shown for a number of cases, such as in the Einstein universe by Phillips and Hu [197] – is an indicator that the fluctuations of the stress-energy tensor of quantum fields are just as important a factor to consider as its mean value. This is a central theme in the research of Fewster, Ford and Roman [467, 468] and an impetus for taking seriously the effects of quantum fluctuations, as addressed in the theory of stochastic gravity to be expounded in the second half of this book.

Stress-Energy Correlators in Hot Flat Spaces

Thermal fields in flat space is another simple case to illustrate the use of zeta function methods. We don't need to repeat much here with the understanding that in the imaginary time formulation the space for thermal fields in flat space has the same topology as the Casimir geometry considered above. While the former has a compact imaginary time dimension the latter has one compactified spatial dimension. The expression for the stress-energy bitensor correlator or the noise kernel of a conformal scalar field in hot flat space was derived by Phillips and Hu [201] using a Gaussian form for the Wightman Green function, which for an ultrastatic spacetime is exact. We will show this derivation in the next chapter via the point separation method. The stress-energy correlators of massless, minimally coupled quantum fields in hot flat spaces have been derived with the generalized zeta function method by Cho and Hu [466].

4.7 Correlations of the Stress-Energy Tensor in AdS Space

In this section we apply the ζ-function formalism to a more elaborate case, calculation of the expectation value and the correlation functions of the stress-energy tensor for scalar fields in N-dim anti-de Sitter (AdS) spaces. AdS space has taken center stage in the last two decades because of the wide ranging implications of the holography principle embodied in the anti-de Sitter (AdS) space - conformal field theory (AdS–CFT) and gravity-gauge duality correspondence [469, 470, 471, 472, 473, 474].

4.7.1 Expectation Value of the Stress-Energy Tensor

We first calculate the expectation value. The Euclideanized N-dim AdS space is the hyperbolic space H^N with the metric

$$ds^2 = d\sigma^2 + a^2 \sinh^2\left(\frac{\sigma}{a}\right) d\Omega^2_{N-1}, \tag{4.149}$$

where σ is the geodesic distance, a the radius of the AdS^N space, and $d\Omega^2_{N-1}$ the metric for the $(N-1)$-sphere. The eigenfunctions $\phi_{\kappa lm}$ obey the equations

$$\left(-\Box - \frac{\rho_N}{a^2}\right)\phi_{\kappa lm} = \left(\frac{\kappa^2}{a^2}\right)\phi_{\kappa lm}, \tag{4.150}$$

where $\rho_N = (N-1)/2$, and are given by

$$\begin{aligned}\phi_{\kappa lm} &= c_l(\kappa)\left(\sinh\frac{\sigma}{a}\right)^{1-\frac{N}{2}} P^{1-l-\frac{N}{2}}_{-\frac{1}{2}+i\kappa}\left(\cosh\frac{\sigma}{a}\right) Y_{lm}(\Omega) \\ &\equiv f_{\kappa l}(\sigma) Y_{lm}(\Omega),\end{aligned} \tag{4.151}$$

where $P_\nu^\mu(x)$ is the associated Legendre function and $Y_{lm}(\Omega)$ are the hyperspherical harmonics. The normalization constant is given by

$$c_l(\kappa) = \frac{|\Gamma(i\kappa + \rho_N)|}{|\Gamma(i\kappa)|}. \tag{4.152}$$

Later we shall need the normalization constant for $l = 0$. For odd N,

$$|c_0(\kappa)|^2 = \frac{1}{a^N} \prod_{j=0}^{\rho_N} (\kappa^2 + j^2) = \frac{1}{a^N} \sum_{n=1}^{\rho_N} c_{2n} \kappa^{2n}, \tag{4.153}$$

and for even N,

$$|c_0(\kappa)|^2 = \frac{1}{a^N} (\kappa \tanh \pi\kappa) \prod_{j=\frac{1}{2}}^{\rho_N} (\kappa^2 + j^2) = \frac{1}{a^N} \tanh \pi\kappa \sum_{n=0}^{\rho_N - \frac{1}{2}} c_{2n+1} \kappa^{2n+1}. \tag{4.154}$$

We now proceed to evaluating the regularized expectation value of the stress-energy tensor in AdS spaces. From Eq. (4.96),

$$\langle T_{\mu\nu}(x) \rangle = -\frac{1}{2} \frac{d}{d\nu} \left\{ \frac{\mu^{2\nu}}{\Gamma(\nu)} \int_0^\infty dt\, s^\nu \int_0^\infty d\kappa \right.$$
$$\left. \times \sum_{lm} e^{-\frac{s}{a^2}\left(\kappa^2 + \rho_N^2 + m^2 a^2 - \xi N(N-1)\right)} T_{\mu\nu} \left[\phi_{\kappa lm}(x), \phi^*_{\kappa lm}(x)\right] \right\}_{\nu \to 0}. \tag{4.155}$$

Note that κ is a continuous variable and l and m are discrete. H^N is a homogeneous space so $\langle T_{\mu\nu}(x) \rangle = F g_{\mu\nu}(x)$ where F is a constant. It is obvious that

$$F = \langle T_{\sigma\sigma}(x) \rangle|_{x \to 0} = -\frac{1}{2} \frac{d}{d\nu} \left\{ \frac{\mu^{2\nu}}{\Gamma(\nu)} \int_0^\infty ds\, s^\nu \int_0^\infty d\kappa \right.$$
$$\left. \sum_{lm} e^{-\frac{s}{a^2}\left(\kappa^2 + a^2 b\right)} T_{\sigma\sigma} \left[\phi_{\kappa lm}(x), \phi^*_{\kappa lm}(x)\right] \right\}_{x, \nu \to 0}, \tag{4.156}$$

where $a^2 b = \rho_N^2 + m^2 a^2 - \xi N(N-1)$. Also, we have taken the limit $x \to 0$ to simplify the evaluation. $T_{\sigma\sigma}\left[\phi_{\kappa lm}(x), \phi^*_{\kappa lm}(x)\right]$ can be further simplified as

$$T_{\sigma\sigma}\left[\phi_{\kappa lm}(x), \phi^*_{\kappa lm}(x)\right] = -2 \partial_\sigma \phi^*_{\kappa lm} \partial_\sigma \phi_{\kappa lm}$$
$$+ \left[-2 \left(\xi - \frac{1}{4} \right) \Box + 2\xi \nabla_\sigma \nabla_\sigma + 2\xi \left(\frac{N-1}{a^2} \right) \right] |\phi_{\kappa lm}|^2 . \tag{4.157}$$

Making use of the addition theorem of the hyperspherical harmonics,

$$\sum_m Y^*_{lm}(\Omega) Y_{lm}(\Omega') = \frac{(2l + N - 2)\Gamma\left(\frac{N-2}{2}\right)}{4\pi^{\frac{N}{2}}} C_l^{\frac{N-2}{2}} (\Omega \cdot \Omega'), \tag{4.158}$$

where C_l^n is the Gegenbauer polynomial, we have

$$\sum_m |\phi_{\kappa lm}|^2 = \frac{(2l+N-2)\Gamma(l+N-2)}{2^{N-1}\pi^{\frac{N-1}{2}}\Gamma\left(\frac{N-1}{2}\right)\Gamma(l+1)}|c_l(\kappa)|^2 \times \left[\left(\sinh\frac{\sigma}{a}\right)^{1-\frac{N}{2}} P^{1-l-\frac{N}{2}}_{-\frac{1}{2}+i\kappa}\left(\cosh\frac{\sigma}{a}\right)\right]^2. \tag{4.159}$$

Moreover, from the short distance expansion of $\phi_{\kappa lm}$,

$$\left(\sinh\frac{\sigma}{a}\right)^{1-\frac{N}{2}} P^{1-l-\frac{N}{2}}_{-\frac{1}{2}+i\kappa}\left(\cosh\frac{\sigma}{a}\right) = \frac{2^{1-l-\frac{N}{2}}}{\Gamma\left(l+\frac{N}{2}\right)}\left(\frac{\sigma}{a}\right)^l\left\{1-\left[\frac{l}{6}+\frac{\kappa^2+(\rho_N+l)^2}{2(2l+N)}\right]\left(\frac{\sigma}{a}\right)^2+\cdots\right\}, \tag{4.160}$$

only the $l=0$ and the $l=1$ terms will contribute to the l sum in the limit x or $\sigma\to 0$. In particular,

$$\left(\sum_m |\phi_{\kappa 0m}|^2\right)_{\sigma\to 0} = \frac{|c_0(\kappa)|^2}{2^{N-1}\pi^{\frac{N}{2}}\Gamma\left(\frac{N}{2}\right)} \tag{4.161}$$

$$\left(\sum_m \partial_\sigma\phi^*_{\kappa 1m}\partial_\sigma\phi_{\kappa 1m}\right)_{\sigma\to 0} = \frac{(\kappa^2+\rho_N^2)|c_0(\kappa)|^2}{2^{N-1}\pi^{\frac{N}{2}}N\Gamma\left(\frac{N}{2}\right)a^2} \tag{4.162}$$

$$\left(\Box\sum_m |\phi_{\kappa 0m}|^2\right)_{\sigma\to 0} = -\frac{(\kappa^2+\rho_N^2)|c_0(\kappa)|^2}{2^{N-2}\pi^{\frac{N}{2}}N\Gamma\left(\frac{N}{2}\right)a^2} \tag{4.163}$$

$$\left(\Box\sum_m |\phi_{\kappa 1m}|^2\right)_{\sigma\to 0} = \frac{(\kappa^2+\rho_N^2)|c_0(\kappa)|^2}{2^{N-2}\pi^{\frac{N}{2}}N\Gamma\left(\frac{N}{2}\right)a^2}. \tag{4.164}$$

Hence,

$$\left(\Box\sum_{lm}|\phi_{\kappa lm}|^2\right)_{x\to 0} = \left(\nabla_\sigma\nabla_\sigma\sum_{lm}|\phi_{\kappa lm}|^2\right)_{x\to 0} = 0 \tag{4.165}$$

and we have

$$\left(\sum_{lm} T_{\sigma\sigma}[\phi_{\kappa lm}(x),\phi^*_{\kappa lm}(x)]\right)_{x\to 0} = -\frac{|c_0(\kappa)|^2}{2^{N-2}\pi^{\frac{N}{2}}N\Gamma\left(\frac{N}{2}\right)a^2}[\kappa^2+\rho_N^2-\xi N(N-1)]. \tag{4.166}$$

The constant F in Eq. (4.156) becomes

$$F = \frac{1}{2^{N-1}\pi^{\frac{N}{2}}N\Gamma\left(\frac{N}{2}\right)}\frac{d}{d\nu} \times\left\{\frac{\mu^{2\nu}a^{2\nu}}{\Gamma(\nu)}\int_0^\infty ds\, s^\nu\int_0^\infty d\kappa\, e^{-s(\kappa^2+a^2b)}|c_0(\kappa)|^2(\kappa^2+a^2b-m^2a^2)\right\}_{\nu\to 0}. \tag{4.167}$$

Using the finite sum representation for $|c_0(\kappa)|^2$ in the case of odd dimensions in Eq. (4.153), one gets

$$F = \frac{m^2 a^{2-N}}{2^N \pi^{\frac{N}{2}-1} N\Gamma\left(\frac{N}{2}\right)} \sum_{n=1}^{\rho_N} (-1)^{n+1} c_{2n} (a^2 b)^{n-\frac{1}{2}}. \tag{4.168}$$

For even dimensions, $|c_0(\kappa)|^2$ involves the term $\tanh \pi\kappa = 1 - 2/(e^{2\pi\kappa}+1)$, and one gets

$$\begin{aligned} F = \frac{a^{-N}}{2^{N-1}\pi^{\frac{N}{2}} N\Gamma\left(\frac{N}{2}\right)} \sum_{n=0}^{\rho_N - \frac{1}{2}} c_{2n+1} \Bigg[& \frac{(-1)^{n+1}}{2(n+1)} (a^2 b)^{n+1} \\ & + \frac{(-1)^{n+1}}{2} (a^2 b)^n m^2 a^2 \left(d_n - \ln \frac{b}{\mu^2} \right) + 2m^2 a^2 H_n(1; \sqrt{a} b) - 2H_n(0) \Bigg], \end{aligned} \tag{4.169}$$

where

$$d_0 = 0, \qquad d_n = \sum_{k=1}^{n} \frac{1}{k} \quad (n \geq 1), \tag{4.170}$$

and the function $H_n(\nu; \mu)$ is defined by the integral

$$H_n(\nu; \mu) = \int_0^\infty \frac{\kappa^{2n+1} d\kappa}{(e^{2\pi\kappa}+1)(\kappa^2+\mu^2)^\nu}. \tag{4.171}$$

For a derivation using a different coordinate system, arriving at the same result, see [475].

4.7.2 Correlation Functions of the Stress-Energy Tensor

In a maximally symmetric space like the Euclidean space R^N or the hyperbolic space H^N, any bitensor[2] can be expressed in terms of a set of basis bitensors which carry the symmetry properties of these spaces [476]. The first basis bitensor is the bi-scalar function $\tau(x, x')$ which is the geodesic distance between x and x'. Applying the covariant derivative on it one could define

$$n_\mu = \nabla_\mu \tau(x, x') \quad ; \quad n_{\alpha'} = \nabla_{\alpha'} \tau(x, x'), \tag{4.172}$$

where $n_\mu(x, x')$ is a vector at x and a scalar at x', while $n_{\alpha'}(x, x')$ is a scalar at x and a vector at x'. Next, we have the parallel propagator $g_\mu{}^{\alpha'}(x, x')$ which parallel transports any vector v^μ from x to x'. The transported vector is $v^\mu g_\mu{}^{\alpha'}(x, x')$. It is easy to see that $n_\mu(x, x') = -g_\mu{}^{\alpha'} n_{\alpha'}(x, x')$. $\tau(x, x'), n_\mu(x, x'), n_{\alpha'}(x, x')$ and $g_\mu{}^{\alpha'}(x, x')$ constitute this set of basis bitensors. All other bitensors can be

[2] Readers not so familiar with bitensors may wish to first read Sec. 1.3 of Chapter 5, then return to this topic here.

expressed in terms of them with coefficients depending only on the geodesic distance between the two points. For example,

$$\nabla^{\mu} n_{\nu} = A(\tau)\left(g^{\mu}{}_{\nu} - n^{\mu} n_{\nu}\right) \tag{4.173}$$

$$\nabla^{\mu} n_{\alpha'} = B(\tau)\left(g^{\mu}{}_{\alpha'} + n^{\mu} n_{\alpha'}\right) \tag{4.174}$$

$$\nabla^{\mu} g_{\nu\alpha'} = -(A(\tau) + B(\tau))\left(g^{\mu}{}_{\nu} n_{\alpha'} + g^{\mu}{}_{\alpha'} n_{\nu}\right), \tag{4.175}$$

where $A(\tau)$ and $B(\tau)$ are function of τ only. For the Euclidean R^N spaces, $A = 1/\tau$ and $B = -1/\tau$. For hyperbolic H^N spaces, $A = \coth(\tau/a)/a$ and $B = -\mathrm{csch}(\tau/a)/a$.

Since $\Delta T^2_{\mu\nu\alpha'\beta'}(x, x')$ is a symmetric bitensor, one could express it in terms of these basis bitensors. Taking the symmetries of the indices into account, we have

$$\begin{aligned}\Delta T^2_{\mu\nu\alpha'\beta'}(x, x') = {} & C_1(\tau) n_{\mu} n_{\nu} n_{\alpha'} n_{\beta'} + C_2(\tau)\left(n_{\mu} n_{\nu} g_{\alpha'\beta'} + g_{\mu\nu} n_{\alpha'} n_{\beta'}\right) \\ & + C_3(\tau)\left(n_{\mu} g_{\nu\alpha'} n_{\beta'} + n_{\nu} g_{\mu\alpha'} n_{\beta'} + n_{\mu} g_{\nu\beta'} n_{\alpha'} + n_{\nu} g_{\mu\beta'} n_{\alpha'}\right) \\ & + C_4(\tau)\left(g_{\mu\alpha'} g_{\nu\beta'} + g_{\mu\beta'} g_{\nu\alpha'}\right) + C_5(\tau) g_{\mu\nu} g_{\alpha'\beta'}.\end{aligned} \tag{4.176}$$

Using the derivatives in Eqs. (4.173) to (4.175), the conservation conditions $\nabla^{\mu} \Delta T^2_{\mu\nu\alpha'\beta'}(x, x') = 0$ can be expressed as three equations on the coefficients,

$$\begin{aligned}&\frac{dC_1}{d\tau} + \frac{dC_2}{d\tau} - 2\frac{dC_3}{d\tau} + (N-1)AC_1 + 2BC_2 - 2\left((N-2)A + NB\right)C_3 = 0 \\ &\frac{dC_2}{d\tau} + \frac{dC_5}{d\tau} + (N-1)AC_2 + 2BC_3 - 2(A+B)C_4 = 0 \\ &\frac{dC_3}{d\tau} - \frac{dC_4}{d\tau} + BC_2 + NAC_3 - N(A+B)C_4 = 0.\end{aligned} \tag{4.177}$$

Moreover, the traceless condition $g^{\mu\nu} \Delta T^2_{\mu\nu\alpha'\beta'}(x, x') = 0$ can be written as

$$\begin{aligned}&C_1 + NC_2 - 4C_3 = 0 \\ &C_2 + 2C_4 + NC_5 = 0\end{aligned} \tag{4.178}$$

To obtain the five coefficients $C_1(\tau)$ to $C_5(\tau)$ on H^N, we just need to evaluate five components of $\Delta T^2_{\mu\nu\alpha'\beta'}(x, x')$. Here we shall choose $\Delta T^2_{\sigma\sigma\sigma'\sigma'}(x, x')$, $\Delta T^2_{\sigma\sigma\theta'\theta'}(x, x')$, $\Delta T^2_{\sigma\theta\sigma'\theta'}(x, x')$, $\Delta T^2_{\theta\theta\phi'_1\phi'_1}(x, x')$ and $\Delta T^2_{\theta\phi_1\theta'\phi'_1}(x, x')$. Since H^N is homogeneous, the correlator should only depend on the geodesic distance between the two points. Hence, it is possible to simplify the consideration by appropriately choosing x and x'. First, we shall set x and x' to have the same angular coordinates $\Omega' \to \Omega$. After that we shall take the limit $\sigma' \to 0$. In effect we shall take x' to be at the origin.

As $\Omega' \to \Omega$, various bitensors can be simplified as follows.

$$\tau(x,x') = |\sigma - \sigma'|, \tag{4.179}$$

$$n_\mu(x,x') = \delta_{\mu\sigma}, \tag{4.180}$$

$$n_{\alpha'}(x,x') = -\delta_{\alpha'\sigma'}, \tag{4.181}$$

$$g_{\sigma\sigma'}(x,x') = 1, \tag{4.182}$$

$$g_{\theta\theta'}(x,x') = a^2 \sinh\left(\frac{\sigma}{a}\right) \sinh\left(\frac{\sigma'}{a}\right), \tag{4.183}$$

$$g_{\phi_1\phi_1'}(x,x') = a^2 \sinh\left(\frac{\sigma}{a}\right) \sinh\left(\frac{\sigma'}{a}\right) \sin^2\theta, \tag{4.184}$$

and so on. Note that the non-diagonal elements of $g_{\mu\alpha'}$ vanish in this limit. Using this result the relationship between the various components of $\Delta T^2_{\mu\nu\alpha'\beta'}(x,x')$ and the coefficients C_i also simplify. Then we have, as $\Omega' \to \Omega$,

$$\Delta T^2_{\sigma\sigma\sigma'\sigma'}(x,x') = C_1 + 2C_2 - 4C_3 + 2C_4 + C_5, \tag{4.185}$$

$$\Delta T^2_{\sigma\sigma\theta'\theta'}(x,x') = a^2 \sinh^2\left(\frac{\sigma'}{a}\right)(C_2 + C_5), \tag{4.186}$$

$$\Delta T^2_{\sigma\theta\sigma'\theta'}(x,x') = -a^2 \sinh\left(\frac{\sigma}{a}\right) \sinh\left(\frac{\sigma'}{a}\right)(C_3 - C_4), \tag{4.187}$$

$$\Delta T^2_{\theta\theta\phi_1'\phi_1'}(x,x') = a^4 \sinh^2\left(\frac{\sigma}{a}\right) \sinh^2\left(\frac{\sigma'}{a}\right) \sin^2\theta C_5, \tag{4.188}$$

$$\Delta T^2_{\theta\phi_1\theta'\phi_1'}(x,x') = a^4 \sinh^2\left(\frac{\sigma}{a}\right) \sinh^2\left(\frac{\sigma'}{a}\right) \sin^2\theta C_4. \tag{4.189}$$

The evaluation of the various components of the correlator can be further simplified if we take $\sigma' \to 0$. As we have seen in the last section, only terms with low values of l and l' will contribute.

The procedure which leads to the derivation of the 5 coefficients expressed in terms of Legendre functions is explained in [465] where one can find explicit expressions for the small and large geodesic distance limits of C_1 to C_5 in Euclidean AdS^N for arbitrary N. These results are expected to be useful for issues which invoke the correlation of stress-energy tensor in AdS spaces.

5

Stress-Energy Tensor and Correlation. Point Separation

In this chapter we focus on the stress-energy bitensor and its symmetrized product with two goals: (1) to present the methodology of the *point-separation regularization* scheme, (2) to use it to calculate the *noise kernel* which is the correlation function of the stress-energy bitensor, and to explore its properties. The first goal is an important component of quantum field theory in curved spacetime established in the 1970s (the theory which the Hawking effect is based on) and useful for the backreaction problems in semiclassical gravity explored in the 1980s (the theory inflationary cosmology is based on); the second goal is a necessity for the stochastic gravity theory established in the 1990s.

The contents of this chapter can be divided into three parts. In the first part we will introduce the necessary properties and geometric tools for analyzing and understanding the bitensors of interest: a bitensor is a geometric object that has support at two separate spacetime points. In particular, the noise kernel is a rank two tensor in the tangent space at x and in the tangent space at y. The Green functions $G(x, y)$, whose importance in physics needs no belaboring, are biscalars: they transform as scalars at both points x and y. In the second part we will present the point-separation regularization method for treating the ultraviolet (UV) divergences of the stress-energy tensor for a scalar field in a general curved spacetime. We then present some simple field theory examples for calculating the regularized stress-energy tensor, one being finite temperature quantum fields in flat space, the other being quantum fields in the optical Schwarzschild metric. For these theories the Green functions have a Gaussian form which make it easy to calculate the regularized stress-energy tensor.

In the third part, we turn our attention to the correlation functions of the stress-energy bitensor, the noise kernel. We derive a formal expression for the noise kernel in terms of the higher-order covariant derivatives of the Green functions taken at two separate points. One simple fact we will show is that for a massless conformal field the trace of the noise kernel identically vanishes. We

then discuss how to calculate this quantity under the Gaussian approximation which gives exact results for ultrastatic metrics such as the Einstein universe [197], the hot flat space and the optical Schwarzschild spacetime [477]. In the last part we calculate the noise kernel for a conformal field in de Sitter (dS) space, both in the conformal Bunch–Davies vacuum and in the static Gibbons–Hawking vacuum [478]. Drawing on the close parallel between the Gibbons–Hawking state in static dS and the Hartle–Hawking state in the Schwarzschild spacetime where it is easy to derive Hawking radiation from an eternal black hole, these results may prove useful for the investigation of the backreaction and fluctuation effects of quantum fields in the early universe and in black holes, a practical application of semiclassical and stochastic gravity theories.

5.1 Stress-Energy Bitensors and Products

5.1.1 Point-Separation Regularization as a Mathematical Motivation

The point separation scheme introduced in the 1960s by DeWitt [384] was brought to more popular use in the 70s in the context of quantum field theory in curved spacetime [46, 100, 101] as a means for obtaining a finite quantum stress-energy tensor. Since the stress-energy tensor is built from the product of a pair of field operators evaluated at the same point, ultraviolet (UV) divergences appear unavoidably. Unlike regularization schemes based on coordinate-dependent methods, such as the introduction of a cut-off function, the Pauli–Villar regularization or the adiabatic regularization, point-separation method can be formulated in a manifestly covariant way. In this scheme, a single point x is split into a pair of closely spaced points x and x'. The terms involving field products which are UV-'problematic', such as $\hat{\phi}(x)^2$, become $\hat{\phi}(x)\hat{\phi}(x')$, whose expectation value is well-defined. One then brings the two points back (taking the coincidence limit) to identify the divergences present, which will then be removed (by regularization) or moved to be grouped with the geometric objects (by renormalization of their coupling constants), thereby obtaining a well-defined, finite stress-energy tensor at a single point. In this context point-separation was used as a technique (many practitioners may still view it as a trick, a rather unwieldy one) for the purpose of identifying the ultraviolet divergences.

In a different, perhaps more enlightened, viewpoint, ordinary quantum field theory defined on single points of spacetime is considered as a low energy limit of fields defined on extended objects like fundamental strings or loops. In this light, point-separated stress-energy bitensors may carry some fundamental physical meaning as they contain information on the fluctuations and correlation of quantum fields, and by consistency with the gravity sector, can provide a probe into the quantum properties of the microscopic features of spacetimes [479]. Taking this view, we may also gain a new perspective on ordinary quantum field theory defined on single points: the coincidence limit is an approximation when

spacetime structure is viewed from a much larger scale than the fundamental scale (the Planck scale) or at a much lower energy (than the Planck energy). Ordinary quantum field theory defined on single points merely depicts the low energy limit of the more complete theory of matter and spacetimes. Classical general relativity and semiclassical gravity are only low energy effective theories.

In this perspective we can transcend the traditional view of treating point-separation as a tool of convenience for regularization ('splitting a point' into two and then bringing them back together for the sake of identifying the divergences), instead viewing the bitensor as having its own identity and serving a grander purpose. As we learned earlier, *quantum fields in curved spacetime* is a test-field limit of *semiclassical gravity* where backreaction from the *mean* of the quantum matter field stress-energy tensor is included in a self-consistent manner. *Stochastic gravity* extends semiclassical gravity by including the backreaction of the *fluctuations* of the quantum matter field stress-energy tensor. Thus stochastic gravity (in a restricted sense) based on the two-point functions of the stress-energy tensor is the 'entrance' into the mesoscopic domain [480, 481], where induced metric fluctuations ('spacetime foams') begin to appear, while stochastic gravity (in a broader sense) built on the full tower of products of the stress-energy bitensor [482][1] play a role in connecting with the more complete theories for the microscopic structures of spacetime, aka quantum gravity, from the low energy upward (a 'bottom-up' approach to quantum gravity [483]). And, if the general structure of many physical theories where a stochastic regime lies between the quantum and the semiclassical is of any guidance, then features of stochastic gravity should appear in all viable theories of quantum gravity when they approach the mid- and low-energy regimes. We will develop this line of reasoning in the Epilogue.

5.1.2 Stochastic Gravity as a Physical Motivation

In semiclassical gravity the classical spacetime (with metric g_{ab}) is driven by the expectation value $\langle\rangle$ of the stress-energy tensor operator $\hat{T}_{ab}$ of a quantum field with respect to some quantum state. One main task in the 70s was to obtain a regularized $\langle\hat{T}_{ab}\rangle$ as a source of the semiclassical Einstein equation (SCE) (e.g.,[182, 423] and earlier work referred therein). In stochastic semiclassical gravity of the 90s [479, 294] fluctuations of the stress-energy tensor [196, 197, 199] produce the induced metric fluctuations in the classical spacetime via the Einstein–Langevin equation [216, 217, 186, 220]. This stochastic term measures the fluctuations of quantum sources (e.g., arising from the difference of particles created in neighboring histories [182]) and is intrinsically linked to the dissipation in the dynamics of spacetime by a fluctuation-dissipation relation

[1] The recent work of Fewster, Ford and Roman [467, 468] indicates that the contributions from the higher moments of the stress-energy tensor can be significant.

[185, 186], which embodies the full backreaction effects of quantum fields on classical spacetime.

Schematically the Einstein–Langevin equation takes the form

$$G_{ab}[g] + \Lambda g_{ab} = 8\pi G_{\rm N}(T_{ab}{}^{c} + \hat{T}^{qs}_{ab})$$
$$\hat{T}^{qs}_{ab} \equiv \langle \hat{T}_{ab} \rangle_q + T^{s}_{ab}, \tag{5.1}$$

where G_{ab} is the Einstein tensor associated with the metric of spacetime g_{ab}, and $\Lambda, G_{\rm N}$ are the cosmological and Newton constants respectively. Here we use the superscripts c, s, q to denote classical, stochastic and quantum respectively. The new term $T^{s}_{ab} = 2\tau_{ab}$ which is of classical stochastic nature measures the fluctuations of the stress-energy tensor of the quantum field. To see what τ_{ab} is, first define

$$\hat{t}_{ab}(x) \equiv \hat{T}_{ab}(x) - \langle \hat{T}_{ab}(x) \rangle \hat{I}, \tag{5.2}$$

which is a tensor operator measuring the deviations from the mean of the stress-energy tensor. We are interested in the correlation of these operators at different spacetime points. Here we focus on the stress-energy (operator-valued) bitensor $\hat{t}_{ab}(x)\hat{t}_{c'd'}(y)$ defined at nearby points (x, y).[2]

The noise kernel $N_{abc'd'}$ is the expectation value of the stress-energy bitensor, defined as

$$N_{abc'd'}(x, y) \equiv \frac{1}{2}\langle \{\hat{t}_{ab}(x), \hat{t}_{c'd'}(y)\} \rangle, \tag{5.3}$$

where $\{\}$ means taking the symmetric product.[3] The noise kernel defines a real classical Gaussian stochastic symmetric tensor field τ_{ab} which is characterized to lowest order by the following relations,

$$\langle \tau_{ab}(x) \rangle_\tau = 0, \quad \langle \tau_{ab}(x)\tau_{c'd'}(y) \rangle_\tau = N_{abc'd'}(x, y), \tag{5.4}$$

where $\langle\,\rangle_\tau$ means taking a statistical average with respect to the noise distribution τ (for simplicity we don't consider higher-order correlations). Since $\hat{T}_{ab}$ is self-adjoint, one can see that $N_{abc'd'}$ is symmetric, real, positive and semi-definite. Furthermore, as a consequence of (5.3) and the conservation law $\nabla^a \hat{T}_{ab} = 0$, this stochastic tensor τ_{ab} is divergenceless in the sense that $\nabla^a \tau_{ab} = 0$ is a deterministic zero field. Also $g^{ab}\tau_{ab}(x) = 0$, signifying that there is no stochastic correction to the trace anomaly (if T_{ab} is traceless). (See [189].) Here all covariant derivatives are taken with respect to the background metric g_{ab} which is a

[2] Because of the necessary elaborations on the properties and manipulations of bitensors in this chapter for clarity we shall use unprimed indices attached to x and primed indicies attached to x' on the bitensor.

[3] This picks out the real part of $\hat{t}_{ab}(x)\hat{t}_{c'd'}(y)$ which is the part that originates from the real part of the effective action. The derivation of noise kernel from the Feynman–Vernon influence functional approach [187] is well-known: the noise kernel appears in the real part of the influence action [216, 185, 186, 183, 189]. An equivalent way is through the closed-time-path coarse-grained effective action [241, 242].

solution of the semiclassical equations. Taking the statistical average of (5.1) with respect to the noise distribution τ, as a consequence of the noise correlation relation (5.4),

$$\langle \hat{T}^{qs}_{ab} \rangle_\tau = \langle \hat{T}_{ab} \rangle_q \tag{5.5}$$

we recover the semiclassical Einstein equation which is (5.1) without the T^s_{ab} term. It is in this sense that we view semiclassical gravity as a mean field theory.

The Noise Kernel

The definition (5.3) of the noise kernel immediately implies that it is symmetric under the interchange of the two spacetime points and the corresponding pairs of indices so that

$$N_{abc'd'}(x,x') = N_{c'd'ab}(x',x). \tag{5.6}$$

The noise kernel obeys other important properties [188, 223]:

I. For any real tensor field $f^{ab}(x)$ it is *positive semi-definite*, namely,

$$\int d^4x\sqrt{-g(x)} \int d^4x'\sqrt{-g(x')}\, f^{ab}(x) N_{abc'd'}(x,x') f^{c'd'}(x') \geq 0; \tag{5.7}$$

II. *Conservation laws*, which is clear from (5.3),

$$\nabla^a N_{abc'd'} = \nabla^b N_{abc'd'} = \nabla^{c'} N_{abc'd'} = \nabla^{d'} N_{abc'd'} = 0; \tag{5.8}$$

III. When the field is conformally invariant the *partial traces must vanish*,

$$N^a{}_{ac'd'} = N_{ab}{}^{c'}{}_{c'} = 0; \tag{5.9}$$

IV. For conformally-invariant fields the noise kernel has a simple *scaling behavior under conformal transformations*. In Sec. 5.3.3 we will show that under a conformal transformation between two conformally related D-dimensional spacetimes with metrics of the form $\tilde{g}_{ab} = \Omega^2(x)\, g_{ab}$ and conformally related states, the noise kernel transforms as

$$\tilde{N}_{abc'd'}(x,x') = \Omega^{2-D}(x)\, \Omega^{2-D}(x')\, N_{abc'd'}(x,x'). \tag{5.10}$$

5.1.3 Properties of Bitensors and Their Symmetrized Products

The geometric properties of bitensors are summarized below [200].

n-Tensors and End-Point Expansions

An object like the Green function $G(x,y)$ is an example of a *biscalar*: it transforms as scalar at both points x and y. We can also define a *bitensor* $T_{a_1\cdots a_n\, b'_1\cdots b'_m}(x,y)$: upon a coordinate transformation; this transforms as a rank n tensor at x and a rank m tensor at y. We will extend this up to a *quad-tensor* $T_{a_1\cdots a_{n_1}\, b'_1\cdots b'_{n_2}\, c''_1\cdots c''_{n_3}\, d'''_1\cdots d'''_{n_4}}$ which has support at four points x, y, x', y', transforming as rank n_1, n_2, n_3, n_4 tensors at each of the four points. This also sets the notation we will use: unprimed indices referring to the tangent space constructed above x, single primed indices to y, double (even) primed to x' and triple (odd) primed to y'. For each point, there is the covariant derivative ∇_a at that point. Covariant derivatives at different points commute and the covariant derivative at, say, point x', does not act on a bitensor defined at, say, x and y:

$$T_{ab';c;d'} = T_{ab';d';c} \quad \text{and} \quad T_{ab';c''} = 0. \tag{5.11}$$

To simplify notation, henceforth we will eliminate the semicolons after the first one for multiple covariant derivatives at multiple points.

Having objects defined at different points, the coincident limit is defined as evaluation "on the diagonal", in the sense of the spacetime support of the function or tensor, and the notation $[G(x,y)] \equiv G(x,x)$ is used. This extends to n-tensors as

$$\left[T_{a_1\cdots a_{n_1}\, b'_1\cdots b'_{n_2}\, c''_1\cdots c''_{n_3}\, d'''_1\cdots d'''_{n_4}}\right] = T_{a_1\cdots a_{n_1}\, b_1\cdots b_{n_2}\, c_1\cdots c_{n_3}\, d_1\cdots d_{n_4}}, \tag{5.12}$$

i.e., this becomes a rank $(n_1+n_2+n_3+n_4)$ tensor at x. The multi-variable chain rule relates covariant derivatives acting at different points, when we are interested in the coincident limit:

$$\left[T_{a_1\cdots a_m\, b'_1\cdots b'_n}\right]_{;c} = \left[T_{a_1\cdots a_m\, b'_1\cdots b'_n;c}\right] + \left[T_{a_1\cdots a_m\, b'_1\cdots b'_n;c'}\right]. \tag{5.13}$$

This result is referred to as *Synge's theorem* in this context [48].

The bitensor of *parallel transport* $g_a{}^{b'}$ is defined such that when it acts on a vector $v_{b'}$ at y, it parallel transports the vector along the geodesics connecting x and y. This allows us to add vectors and tensors defined at different points. We cannot directly add a vector v_a at x and vector $w_{a'}$ at y. But by using $g_a{}^{b'}$, we can construct the sum $v^a + g_a{}^{b'} w_{b'}$. We will also need the obvious property $\left[g_a{}^{b'}\right] = g_a{}^b$.

The main biscalar we need for this work is the *world function* $\sigma(x,y)$. This is defined as a half of the square of the geodesic distance between the points x and y. It satisfies the equation

$$\sigma = \frac{1}{2}\sigma^{;p}\sigma_{;p}. \tag{5.14}$$

A notational convention: when the world function appears with indices a covariant derivative is implied : $\sigma^a \equiv \sigma^{;a}$, i.e., taking the covariant derivative at x,

while $\sigma^{a'}$ means the covariant derivative at y. This is done since the vector $-\sigma^a$ is the tangent vector to the geodesic with length equal the distance between x and y. As σ^a records information about distance and direction for the two points it is useful for constructing series expansions of a biscalar. The *end-point expansion* of a biscalar $S(x,y)$ is of the form

$$S(x,y) = A^{(0)} + \sigma^p A^{(1)}_p + \sigma^p\sigma^q A^{(2)}_{pq} + \sigma^p\sigma^q\sigma^r A^{(3)}_{pqr} + \sigma^p\sigma^q\sigma^r\sigma^s A^{(4)}_{pqrs} + \cdots \tag{5.15}$$

where, following our convention, the expansion tensors $A^{(n)}_{a_1\cdots a_n}$ with unprimed indices have support at x and hence the name end-point expansion. Only the symmetric part of these tensors contribute to the expansion. For the purposes of multiplying series expansions it is convenient to separate the distance dependence from the direction dependence. This is done by introducing the unit vector $p^a = \sigma^a/\sqrt{2\sigma}$. Then the series expansion can be written

$$S(x,y) = A^{(0)} + \sigma^{\frac{1}{2}} A^{(1)} + \sigma A^{(2)} + \sigma^{\frac{3}{2}} A^{(3)} + \sigma^2 A^{(4)} + \cdots \tag{5.16}$$

The expansion scalars are related to the expansion tensors via $A^{(n)} = 2^{n/2} A^{(n)}_{p_1\cdots p_n} p^{p_1}\cdots p^{p_n}$.

The last object we need is the *Van Vleck–Morette* $D(x,y)$, defined as $D(x,y) \equiv -\det(-\sigma_{;ab'})$. The related biscalar

$$\Delta^{1/2} = \left(\frac{D(x,y)}{\sqrt{g(x)g(y)}}\right)^{\frac{1}{2}} \tag{5.17}$$

satisfies the equation

$$\Delta^{1/2}\left(4 - \sigma_{;p}{}^{p}\right) - 2\Delta^{1/2}{}_{;p}\sigma^{;p} = 0 \tag{5.18}$$

with the boundary condition $\left[\Delta^{1/2}\right] = 1$.

Further details on these objects and discussions of the definitions and properties are contained in [100, 484], where it is shown how the defining equations for σ and $\Delta^{1/2}$ are used to determine the coincident limit expression for the various covariant derivatives of the world function ($[\sigma_{;a}]$, $[\sigma_{;ab}]$, etc.) and how the defining differential equation for $\Delta^{1/2}$ can be used to determine the series expansion of $\Delta^{1/2}$.

For example,

$$\sigma_{;ab}(x,x') = g_{ab}(x) - \frac{1}{3}R_{acbd}(x)\,\sigma^c(x,x')\sigma^d(x,x') + \cdots . \tag{5.19}$$

Examination of this expansion shows that to zeroth order in σ^a

$$\sigma_{;abc} = 0. \tag{5.20}$$

One can also expand $\Delta^{1/2}$ in powers of σ^a with the result that [100, 101]

$$\Delta^{1/2}(x,x') = 1 + \frac{1}{12} R_{ab}\,\sigma^a\sigma^b - \frac{1}{24} R_{ab;c}\,\sigma^a\sigma^b\sigma^c \tag{5.21}$$
$$+ \frac{1}{1440}(18\,R_{ab;cd} + 5\,R_{ab}\,R_{cd} + 4\,R_{paqb}\,R_c{}^q{}_d{}^p)\;\sigma^a\sigma^b\sigma^c\sigma^d + O[(\sigma^a)^5].$$

Stress-Energy Bitensors

We start with products of field operators defined at one point in curved spacetime, which are the familiar objects in conventional quantum field theory. We will then use point separation to introduce the bi-objects.

For simplicity our analysis is carried out in Riemannian geometries (with metric signature $(+,+,+,+)$). The results can be continued back to the Lorentzian geometry (with metric signature $(-,+,+,+)$) while exercising some caution, e.g., when null directions are involved $\sigma = 0$, which we will mention duly. For a free classical scalar field with the action[4]

$$S[\phi] = -\frac{1}{2}\int (m^2\,\phi^2 + \phi^2\,R\,\xi + \phi_{;p}\,\phi^{;p})\,\sqrt{g}\,d^4x, \tag{5.22}$$

the classical stress-energy tensor is

$$T_{ab}(x) \equiv \frac{2}{\sqrt{g(x)}}\frac{\delta S[\phi]}{\delta g^{ab}(x)} = (1-2\,\xi)\;\phi_{;a}\,\phi_{;b} + \left(2\,\xi - \frac{1}{2}\right)\;\phi_{;p}\,\phi^{;p}\,g_{ab}$$
$$+ 2\xi\,\phi\;\;(\phi_{;p}{}^{p}\,g_{ab} - \phi_{;ab}) + \phi^2\,\xi\,\left(R_{ab} - \frac{1}{2}R\,g_{ab}\right) - \frac{1}{2}m^2\,\phi^2\,g_{ab}. \tag{5.23}$$

When we make the transition to quantum field theory, we promote the field $\phi(x)$ to a field operator $\hat{\phi}(x)$. The fundamental problem of defining a quantum operator for the stress-energy tensor is immediately visible: the field operator appears quadratically. Since $\hat{\phi}(x)$ is an operator-valued distribution, products at a single point are not well-defined. But if the product is point-separated $\hat{\phi}^2(x) \to \hat{\phi}(x)\hat{\phi}(x')$, they are finite and well-defined.

Let us first seek a point-separated extension of these classical quantities and then consider the corresponding quantum field operators. Point separation is symmetrically extended to products of covariant derivatives of the field according to

$$(\phi_{;a})\,(\phi_{;b}) \to \frac{1}{2}\left(g_a{}^{p'}\nabla_{p'}\nabla_b + g_b{}^{p'}\nabla_a\nabla_{p'}\right)\phi(x)\phi(x'), \tag{5.24}$$

$$\phi\;(\phi_{;ab}) \to \frac{1}{2}\left(\nabla_a\nabla_b + g_a{}^{p'}g_b{}^{q'}\nabla_{p'}\nabla_{q'}\right)\phi(x)\phi(x'). \tag{5.25}$$

[4] In this chapter we shall use the conventional definition of ξ, being 0 for minimal coupling and 1/6 for conformal coupling in 4D.

The bivector of parallel displacement $g_a{}^{a'}(x,x')$ is included so that we may have objects that are rank 2 tensors at x and scalars at x'.

To carry out point separation on (5.23), we first define the differential operator

$$\begin{aligned}\mathcal{T}_{ab} = {} & \frac{1}{2}(1-2\xi)\left(g_a{}^{a'}\nabla_{a'}\nabla_b + g_b{}^{b'}\nabla_a\nabla_{b'}\right) + \left(2\xi - \frac{1}{2}\right) g_{ab}g^{cd'}\nabla_c\nabla_{d'} \\ & - \xi\left(\nabla_a\nabla_b + g_a{}^{a'}g_b{}^{b'}\nabla_{a'}\nabla_{b'}\right) + \xi g_{ab}\left(\nabla_c\nabla^c + \nabla_{c'}\nabla^{c'}\right) \\ & + \xi\left(R_{ab} - \frac{1}{2}g_{ab}R\right) - \frac{1}{2}m^2 g_{ab}, \end{aligned} \tag{5.26}$$

from which we obtain the classical stress-energy tensor as

$$T_{ab}(x) = \lim_{x'\to x} \mathcal{T}_{ab}\phi(x)\phi(x'). \tag{5.27}$$

When promoting the classical c-number field to a quantum operator-valued field, since neither point should be favored, the product of field configurations is taken to be the symmetrized operator product, denoted by curly brackets:

$$\phi(x)\phi(y) \to \frac{1}{2}\left\{\hat{\phi}(x), \hat{\phi}(y)\right\} = \frac{1}{2}\left(\hat{\phi}(x)\hat{\phi}(y) + \hat{\phi}(y)\hat{\phi}(x)\right). \tag{5.28}$$

With this, the point-separated stress-energy tensor operator is defined as

$$\hat{T}_{ab}(x,x') \equiv \frac{1}{2}\mathcal{T}_{ab}\left\{\hat{\phi}(x), \hat{\phi}(x')\right\}. \tag{5.29}$$

While the classical stress-energy tensor was defined at the coincidence limit $x' \to x$, we cannot attach serious physical meaning to the quantum stress-energy tensor at one point until the issue of regularization is dealt with. Instead, consider taking the expectation values of the products of the field operators defined at two separate points. This biscalar quantity is called the Hadamard (or Schwinger) function

$$G^{(1)}(x,x') = \left\langle \left\{\hat{\phi}(x), \hat{\phi}(x')\right\}\right\rangle, \tag{5.30}$$

and the point-separated stress-energy tensor is defined as

$$\langle \hat{T}_{ab}(x,x')\rangle = \frac{1}{2}\mathcal{T}_{ab}G^{(1)}(x,x'), \tag{5.31}$$

where, since $\mathcal{T}_{ab}$ is a differential operator, it can be taken "outside" the expectation value. The expectation value of the point-separated quantum stress-energy tensor for a free, massless ($m = 0$) conformally coupled ($\xi = 1/6$) scalar field on a four-dimensional spacetime with scalar curvature R is given by

$$\langle \hat{T}_{ab}(x,x')\rangle = \frac{1}{6}\left(g^{p'}{}_{b}\, G^{(1)}{}_{;p'a} + g^{p'}{}_{a}\, G^{(1)}{}_{;p'b}\right) - \frac{1}{12} g^{p'}{}_{q}\, G^{(1)}{}_{;p'}{}^{q}\, g_{ab}$$
$$- \frac{1}{12}\left(g^{p'}{}_{a}\, g^{q'}{}_{b}\, G^{(1)}{}_{;p'q'} + G^{(1)}{}_{;ab}\right) + \frac{1}{12}\left(\left(G^{(1)}{}_{;p'}{}^{p'} + G^{(1)}{}_{;p}{}^{p}\right) g_{ab}\right)$$
$$+ \frac{1}{12} G^{(1)}\left(R_{ab} - \frac{1}{2} R\, g_{ab}\right). \tag{5.32}$$

Symmetrized Products of the Stress Energy Bi-Tensors

The symmetrized product of the (mean subtracted) stress-energy tensor operator

$$2N_{ab,c'd'}(x,y) = \left\langle \left\{ \hat{T}_{ab}(x) - \langle \hat{T}_{ab}(x)\rangle, \hat{T}_{c'd'}(y) - \langle \hat{T}_{c'd'}(y)\rangle \right\} \right\rangle$$
$$= \left\langle \left\{ \hat{T}_{ab}(x), \hat{T}_{c'd'}(y) \right\} \right\rangle - 2\langle \hat{T}_{ab}(x)\rangle\langle \hat{T}_{c'd'}(y)\rangle \tag{5.33}$$

is called the *noise kernel* in stochastic semiclassical gravity. Since $\hat{T}_{ab}(x)$ defined at one point can be ill-behaved as it is generally divergent, one can question the soundness of these quantities. But *the noise kernel is finite for* $y \neq x$. All field operator products present in the first expectation value that could be divergent are canceled by similar products in the second term. We will replace each of the stress-energy tensor operators in the above expression for the noise kernel by their point-separated versions, effectively separating the two points (x, y) into the four points (x, x', y, y'). This will allow us to express the noise kernel in terms of a pair of differential operators acting on a combination of four- and two-point functions. Wick's theorem will allow the four-point functions to be re-expressed in terms of two-point functions. From this we see that all possible divergences for $y \neq x$ will cancel.

To obtain the point-separated bitensor, one should replace each of the stress-energy tensor operators in (5.33) with the corresponding point-separated version (5.29), with $\mathcal{T}_{ab}$ acting at x and x' and $\mathcal{T}_{c'd'}$ acting at y and y'. The noise kernel is then defined as

$$2N_{ab,c'd'}(x,y) = \lim_{x'\to x}\lim_{y'\to y} \mathcal{T}_{ab}\mathcal{T}_{c'd'}\, G(x,x',y,y'), \tag{5.34}$$

where the four point-function is

$$G(x,x',y,y') = \frac{1}{4}\left[\left\langle \left\{ \left\{ \hat{\phi}(x), \hat{\phi}(x') \right\}, \left\{ \hat{\phi}(y), \hat{\phi}(y') \right\} \right\} \right\rangle \right.$$
$$\left. -2\left\langle \left\{ \hat{\phi}(x), \hat{\phi}(x') \right\} \right\rangle \left\langle \left\{ \hat{\phi}(y), \hat{\phi}(y') \right\} \right\rangle \right]. \tag{5.35}$$

We assume the pairs (x, x') and (y, y') are each within their respective Riemann normal coordinate neighborhoods so as to avoid problems that possible geodesic caustics might be present.

Wick's theorem, for the case of free fields which we are considering, gives the simple product four-point function in terms of a sum of products of Wightman functions $iG^+(x,y) = \langle\hat{\phi}(x)\,\hat{\phi}(y)\rangle$ which we denote as G_{xy}:

$$\langle\hat{\phi}(x)\,\hat{\phi}(y)\,\hat{\phi}(x')\,\hat{\phi}(y')\rangle = G_{xy'}\,G_{yx'} + G_{xx'}\,G_{yy'} + G_{xy}\,G_{x'y'}. \tag{5.36}$$

Expanding out the anti-commutators in (5.35) and applying Wick's theorem, the four-point function becomes

$$G(x,x',y,y') = G_{xy'}\,G_{x'y} + G_{xy}\,G_{x'y'} + G_{yx'}\,G_{y'x} - G_{yx}\,G_{y'x'}. \tag{5.37}$$

We can now easily see that the noise kernel defined via this function is indeed well defined for the limit $(x',y') \to (x,y)$:

$$G(x,x,y,y) = 2\,\left(G_{xy}^2 + G_{yx}^2\right), \tag{5.38}$$

From this we can see that the noise kernel is also well defined for $y \neq x$; any divergence present in the first expectation value of (5.35) has been canceled by those present in the pair of Green functions in the second term.

5.2 Point-Separation Regularization of the Stress-Energy Tensor

We pointed out earlier that geometric quantities which have supports at two spacetime points like the bitensor may possess valuable information about the mesoscopic structure of spacetime at higher energies. Moving in the direction toward lower energies or poorer resolution, when we can no longer clearly discern the separated points, relevant physical quantities like the stress-energy tensor defined at a single spacetime point in ordinary field theory would show divergences at high frequencies. The covariant point-separation regularization scheme helps to identify where the UV divergences occur so one can fix them in some reasonable way. This section is devoted to a description of this method.

We start with the stress-energy bitensor and expand it in two different forms, the Schwinger–DeWitt expansion and the Hadamard expansion. Each has its usage for slightly different purposes but they are interconnected (see, e.g. [485]). We can see from the point-separated form of the stress-energy tensor (5.31) what we need to regularize is the Green function $G^{(1)}(x,x')$. Once the Green function has been regularized such that it is smooth and has a well-defined $x' \to x$ limit, the stress-energy tensor will be well defined. In Minkowski space, a common practice is by way of 'normal ordering', which hinges on the existence of a unique vacuum. For a general curved spacetime, there is no unique vacuum, and hence no unique mode expansion with which to build a normal ordering prescription. But we can still ask if there is a way to identify the contributions to be subtracted to yield a unique quantum stress-energy tensor.

5.2.1 Wald–ALN Prescription

One such prescription for the regularization of stress-energy tensor in curved spacetime is that of Wald [81], and Adler, Lieberman, and Ng [486] (with corrections [102]) summarized in [49]. We give a short synopsis below.

The idea builds on the fact that for $G(x,x')_\omega = \langle\omega|\hat{\phi}(x)\hat{\phi}(x')|\omega\rangle$, the function

$$F(x,x') = G(x,x')_{\omega_1} - G(x,x')_{\omega_2} \tag{5.39}$$

is a smooth function of x and x', where ω_1 and ω_2 denote two different states. This means the difference between the stress-energy tensor for two states is well defined for the point-separation scheme, i.e.,

$$\mathcal{F}_{ab} = \frac{1}{2}\lim_{x'\to x}\mathcal{T}_{ab}\left(F(x,x') + F(x',x)\right) \tag{5.40}$$

is well defined. So a bi-distribution $G^L(x,x')$ might be useful for the vacuum subtraction. At first, it would seem unlikely we could find such a unique bi-distribution. Wald found that with the introduction of four axioms for the regularized stress-energy tensor

$$\langle\hat{T}_{ab}(x)\rangle_{\rm ren} = \lim_{x'\to x}\frac{1}{2}\mathcal{T}_{ab}\left(G^{(1)}(x,x') - G^L(x,x')\right) \tag{5.41}$$

$G^L(x,x')$ is uniquely determined, up to a local conserved curvature term. The Wald axioms are [102, 49]:

1. The difference between the stress-energy tensor for two states should agree with (5.40);
2. The stress-energy tensor should be local with respect to the state of the field;
3. For all states, the stress-energy tensor is conserved: $\nabla^a\langle T_{ab}\rangle = 0$;
4. In Minkowski space, the result $\langle 0|T_{ab}|0\rangle = 0$ is recovered.

We are still left with the problem of determining the form of such a bi-distribution. Hadamard [389] showed that the Green function for a large class of states takes the form (in four spacetime dimensions)

$$G^L(x,x') = \frac{1}{8\pi^2}\left(\frac{2U(x,x')}{\sigma} + V(x,x')\log\sigma + W(x,x')\right), \tag{5.42}$$

with $U(x,x')$, $V(x,x')$ and $W(x,x')$ being smooth functions.[5] We refer to Eqn (5.42) as the 'Hadamard expansion ansatz'.

Since the functions $V(x,x')$ and $W(x,x')$ are smooth functions, they can be expanded as

[5] When working in the Lorentz sector of a field theory, we must modify the above function to account for null geodesics, since $\sigma(x,x') = 0$ for null separated x and x'. This problem can be overcome by using $\sigma \to \sigma + 2i\epsilon(t-t') + \epsilon^2$. At any point in a Lorentian analysis the above replacement for σ can be implemented.

$$V(x,x') = \sum_{n=0}^{\infty} v_n(x,x')\sigma^n \tag{5.43}$$

$$W(x,x') = \sum_{n=0}^{\infty} w_n(x,x')\sigma^n \tag{5.44}$$

with the v_n's and w_n's themselves smooth functions. These functions and $U(x,x')$ are determined by substituting $G^L(x,x')$ in the wave equation $OG^L(x,x') = 0$, where O is some hyperbolic differential operator, and equating to zero the coefficients of the explicitly appearing powers of σ^n and $\sigma^n \log\sigma$. Doing so, we get the infinite set of equations

$$U(x,x') = \Delta^{1/2}; \tag{5.45}$$

$$2H_0 v_0 + O\Delta^{1/2} = 0; \tag{5.46}$$

$$2nH_n v_n + O v_{n-1} = 0, \quad n \geq 1; \tag{5.47}$$

$$2H_{2n} v_n + 2nH_n w_n + O w_{n-1} = 0, \quad n \geq 1, \tag{5.48}$$

with

$$H_n = \sigma^{ip}\nabla_p + \left(n - 1 + \frac{1}{2}(\Box\sigma)\right). \tag{5.49}$$

From Eqs (5.45)–(5.47), the functions v_n are completely determined. In fact, they are symmetric functions, and hence $V(x,x')$ is a symmetric function of x and x'. On the other hand, the field equations only determine w_n, $n \geq 1$, leaving $w_0(x,x')$ completely arbitrary. This reflects the state dependence of the Hadamard form above. Moreover, even if $w_0(x,x')$ is chosen to be symmetric, this does not guarantee that $W(x,x')$ will be. By using axiom (4) $w_0(x,x') \equiv 0$. With this choice, its Minkowski spacetime limit is $G^L = 1/(4\pi^2\sigma)$ where now $2\sigma = (t-t')^2 - (\mathbf{x}-\mathbf{x}')^2$ and this corresponds to the correct vacuum contribution that needs to be subtracted.

With this choice though, we are left with a $G^L(x,x')$ which is not symmetric and hence does not satisfy the field equation at x', for fixed x. Wald [102] showed this in turn implies axiom (3) is not satisfied. He resolved this problem by adding to the regularized stress-energy tensor a term which cancels that which breaks the conservation of the old stress-energy tensor:

$$\langle T_{ab}^{\text{new}} \rangle = \langle T_{ab}^{\text{old}} \rangle + \frac{1}{2(4\pi)^2} g_{ab}\,[v_1]\,, \tag{5.50}$$

where $[v_1] = v_1(x,x)$ is the coincident limit of the $n = 1$ solution of Eq (5.48). This yields a stress-energy tensor which satisfies all four axioms and produces the well known trace anomaly $\langle T_a{}^a \rangle = [v_1]/8\pi^2$. We can view this redefinition as taking place at the level of the stress-energy tensor operator via

$$\hat{T}_{ab} \to \hat{T}_{ab} + \frac{\hat{1}}{2(4\pi)^2} g_{ab}\,[v_1]\,. \tag{5.51}$$

Since this amounts to a constant shift of the stress-energy tensor operator, it will have no effect on the noise kernel or fluctuations, as they are the variance about the mean. This is further supported by the fact that there is no noise associated with the trace, which we shall show below. Since this result was derived by only assuming that the Green function satisfies the field equation in *one* of its variables, it is independent of the issue of the lack of symmetry in the Hadamard ansatz (5.42).

Using the above formalism we now derive the coincident limit expression for the stress-energy tensor. To get a meaningful result, we must work with the regularization of the Wightman function, obtained by following the same procedure outlined above for the Hadamard function:

$$G_{\rm ren}(x,y) \equiv G^{+}_{\rm ren}(x,y) = G^{+}(x,y) - G^{L}(x,y). \tag{5.52}$$

In doing this, we assume the singular structure of the Wightman function is the same as that for the Hadamard function. In all applications, this is indeed the case. Moreover, if one works in the Euclidean section there is no issue of operator ordering. For now we only consider spacetimes with no time dependence present in the final coincident limit result, so there is also no issue of Wick rotation back to a Lorentzian signature.

Since the point-separated stress-energy tensor $T_{ab'}(x,y)$ involves covariant derivatives at the two points at which it has support, when we take the coincident limit we can use Synge's theorem (5.13) to move the derivatives acting at y to ones acting at x.

When the Green function is available in closed analytic form one can carry out an end-point expansion according to (5.15), displaying the ultraviolet divergence. Subtraction of the Hadamard ansatz (5.42), expressed as a series expansion, (5.43) and (5.44), will render this Green function finite in the coincident limit. With this, one can calculate the regularized stress-energy tensor.

Finding the form of the Green function is easy for conformal fields, such as an electromagnetic or conformally coupled massless scalar field, in conformally flat or static spacetimes because one can apply a conformal transformation to both the field and the spacetime and render the problem to be one in flat space or a static space. For spacetimes of constant curvature, such as the de Sitter or anti-de Sitter spacetimes, owing to their maximally symmetric nature, the Green functions have exact analytic forms. We will show an example of this for the de Sitter space in the last section of this chapter.

Another way to approach this problem is to assume some approximations to the Green functions. A commonly used approximation is the Gaussian [487, 488] approximation. In spacetimes with ultra-static metrics, namely, with $g_{00} = 1$, e.g, for hot flat space, the Gaussian form is exact. In the Einstein universe, the exact Green function is easily calculable.

The class of optical metrics is conformally related to the ultrastatic metrics by $g^{op}_{\mu\nu} = \Omega^2 g^{us}_{\mu\nu}$ where Ω is a conformal factor. Thus for a conformally invariant field one can find exact analytic forms of its Green functions $G(x, y)$ by carrying out a conformal transformation $G^{op}(x, y) = \Omega^{-1}(x)G^{us}(x, y)\Omega^{-1}(y)$ on the Green functions $G^{us}(x, y)$ of the corresponding ultrastatic spaces. We will introduce the ultra-static spacetimes, the conformal transformations and the Gaussian approximation to the Green function in the next subsection.

5.2.2 Ultra-Static Spacetimes

Given any static metric, one can always transform it to an ultra-static one, called the *optical metric*, by a conformal transformation. This class of metric has the form

$$ds^2 = -dt^2 + g_{ij}(\vec{x})\, dx^i dx^j, \tag{5.53}$$

where the metric functions g_{ij} are independent of the time t. In this case Synge's world function is given by

$$\sigma(x, x') = \frac{1}{2}\left(-(t - t')^2 + \mathbf{r}^2\right). \tag{5.54}$$

Introduce

$$\varsigma \equiv \sqrt{2^{(3)}\sigma}, \tag{5.55}$$

where ${}^{(3)}\sigma$ is the part of σ that depends on the spatial coordinates. Some useful properties of ς (not related to the ςs in other chapters) are

$$\nabla_i \varsigma = \frac{{}^{(3)}\sigma_i}{\varsigma}, \tag{5.56}$$

$$\nabla^2 \varsigma = \frac{{}^{(3)}\sigma^i_i - 1}{\varsigma}, \tag{5.57}$$

$$\nabla_i \varsigma\ \nabla^i \varsigma = \frac{{}^{(3)}\sigma_i\, {}^{(3)}\sigma^i}{2^{(3)}\sigma} = 1, \tag{5.58}$$

$$2\left({}^{(3)}\Delta^{1/2}\right)_{,i}\ \ \nabla^i \varsigma = \left(\frac{4}{\varsigma} - \nabla^2 \varsigma\right) {}^{(3)}\Delta^{1/2}, \tag{5.59}$$

where $\nabla^2 = \nabla^i \nabla_i$. From Eqs. (5.17) and (5.54) one can easily see that for an ultra-static spacetime the Van Vleck determinant ${}^{(3)}\Delta$ for the spatial metric g_{ij} coincides with the Van Vleck determinant Δ for the full spacetime. (The advantage of using ${}^{(3)}\Delta$ rather than Δ is that, although noncovariant, it is valid for arbitrary time separations, and one only needs to expand in powers of ς.)

When we are interested in spacetimes for which we know the Green function in a conformally related space time, we can modify the above procedure. The simple conformal transformation property of Green functions allows us to get the Green function in the physical metric we are interested in from that in the conformally

related metric (for the Gaussian approximation, this is the conformally optical metric). The main obstacle to overcome is the subtraction of the Hadamard ansatz. The divergent Green function is defined in terms of the optical metric while the Hadamard ansatz in terms of the physical metric. We need to re-express the transformed optical metric in terms of the physical metric. The defining equations for the geometric objects (*e.g.* Eqns (5.14) and (5.18)) on the optical metric are transformed to the physical metric and recursively solved. Now the Green function series expansion can be written solely in terms of the physical metric.

5.2.3 Gaussian Approximation

In the Schwinger–DeWitt proper-time formalism [124, 384] the Green function is expressed in terms of the heat kernel $K(x, y, s)$, with s being the Schwinger proper time, via

$$G(x, y) = \int_0^\infty K(x, y, s) ds. \tag{5.60}$$

The heat kernel $K(x, y, s)$ (we have absorbed the $e^{-m^2 s}$ factor in (4.4) and call it K here) satisfies

$$\left[\frac{\partial}{\partial s} - \left(\Box - \frac{R}{6}\right)\right] K(x, y, s) = 0, \tag{5.61}$$

$$K(x, y, 0) = \delta(x - y). \tag{5.62}$$

The optical metric for an ultrastatic spacetime has the product form

$$ds^2 = g_{ab} dx^a dx^b = d\tau^2 + g_{ij} dx^i dx^j \tag{5.63}$$

We assume in the Euclidean sector the imaginary time dimension is periodic with period $\beta = 1/T$ with T the temperature (setting $k_B = 1$). For a black hole, $T = \kappa/(2\pi)$, where κ is the surface gravity, which can be regarded as a temperature parameter for our purpose here. This form of the metric allows the kernel to take on the product form

$$K(x, y, s) = K_1(\tau, \tau', s) K_3(\mathbf{x}, \mathbf{y}, s), \tag{5.64}$$

with the kernels K_1, K_3 satisfying

$$\left(\frac{\partial}{\partial s} - \frac{\partial^2}{\partial \tau^2}\right) K_1(\tau, \tau', s) = 0, \tag{5.65}$$

$$\left[\frac{\partial}{\partial s} - \left(\nabla_i \nabla^i - \frac{R}{6}\right)\right] K_3(\mathbf{x}, \mathbf{y}, s) = 0. \tag{5.66}$$

Equation (5.65) has the periodic solution

$$K_1(\tau,\tau',s)=\frac{\kappa}{2\pi}\sum_{n=-\infty}^{\infty}\exp\left(-\kappa^2n^2s+i\kappa n\Delta\tau\right),\quad \Delta\tau=\tau-\tau' \tag{5.67}$$

Equation (5.66) in general is difficult to solve, but a useful approximate solution [487] exists via the Gaussian approximation to the path integral representation. For K_3, it takes the form

$$K_{3\mathrm{Gauss}}(\mathbf{x},\mathbf{y},s)=\frac{{}^{(3)}\Delta^{1/2}}{(4\pi s)^{\frac{3}{2}}}\exp\left(-\frac{{}^{(3)}\sigma}{2s}\right). \tag{5.68}$$

The complete Hadamard–Minakshisundaram–Pleijel–DeWitt expansion [384] would be

$$K_3=K_{3\mathrm{Gauss}}\sum_{n=0}^{\infty}a_n(\mathbf{x},\mathbf{y})s^n, \tag{5.69}$$

where the constants a_n are the 'HaMiDeW' coefficients. The Gaussian approximation amounts to only taking the first term in this power series.

By putting (5.67) and (5.68) back into (5.64) and carrying out the integration (5.60), Page obtains

$$G_{\mathrm{Gauss}}(x,y)=\frac{\kappa\Delta^{1/2}}{8\pi^2\varsigma}\frac{\sinh\kappa\varsigma}{(\cosh\kappa\varsigma-\cos\kappa\tau)} \tag{5.70}$$

as the Gaussian approximation for the Green function.

5.2.4 Regularized Stress Tensor in Ultrastatic Spacetimes

To regularize the stress-energy tensor, we start by expanding the Green function (5.70) about the coincident limit. As for the order of expansion, since the noise kernel contains terms with four covariant derivatives of the Green function, shown in the next section, this expansion needs to be to fourth order in σ^a, or second order in $\sigma=\sigma^p\sigma_p/2$, and fourth order in $\Delta\tau$. Doing this expansion yields

$$\begin{aligned}G_{\mathrm{Gauss}}=&\frac{\Delta^{\frac{1}{2}}}{8\,\pi^2\,\sigma}+\frac{\Delta^{\frac{1}{2}}}{8\,\pi^2}\left\{\frac{\kappa^2}{6}+\frac{\kappa^4}{180}\left(2\,\Delta\tau^2-\sigma\right)\right.\\&\left.+\frac{\kappa^6}{3780}\left(4\,\Delta\tau^4-6\,\Delta\tau^2\,\sigma+\sigma^2\right)\right\}+O\left(\sigma^{\frac{5}{2}},\Delta\tau^5\right).\end{aligned} \tag{5.71}$$

By subtracting the Hadamard ansatz from the Gaussian Green function

$$S(x,y)=\frac{1}{16\,\pi^2}\left(\frac{2\,\Delta^{\frac{1}{2}}}{\sigma}+\sigma\,w_1+\sigma^2\,w_2\right)+O\left(\sigma^3\right), \tag{5.72}$$

and noting that the $V(x,x')$ term is absent, since there is no $\log\sigma$ divergence present in the expansion of the Gaussian approximation to the Green function, we get the renormalized Green function

$$G_{\rm ren} = G_{\rm gauss} - S. \tag{5.73}$$

The divergent term present (5.71) is canceled by the divergent term from the Hadamard ansatz.

We next turn to developing the series expansion

$$\begin{aligned} G_{\rm ren} = \frac{1}{(4\pi)^2}\big(&G^{(0)}_{\rm ren} + \sigma^{;p}\, G^{(1)}_{{\rm ren}p} + \sigma^{;p}\,\sigma^{;q}\, G^{(2)}_{{\rm ren}pq} + \sigma^{;p}\,\sigma^{;q}\,\sigma^{;r}\, G^{(3)}_{{\rm ren}pqr} \\ &+\sigma^{;p}\,\sigma^{;q}\,\sigma^{;r}\,\sigma^{;s}\, G^{(4)}_{{\rm ren}pqrs}\big) \end{aligned} \tag{5.74}$$

of the regularized Green function. With this, it will be straightforward to compute the coincident limits of the various covariant derivatives needed. We start by assuming the expansions

$$\Delta^{\frac{1}{2}} \approx 1 + \sigma^{;p}\,\sigma^{;q}\,\Delta^{(2)}{}_{pq} + \sigma^{;p}\,\sigma^{;q}\,\sigma^{;r}\,\Delta^{(3)}{}_{pqr} + \sigma^{;p}\,\sigma^{;q}\,\sigma^{;r}\,\sigma^{;s}\,\Delta^{(4)}{}_{pqrs} \tag{5.75}$$

$$\Delta\tau^2 \approx \sigma^{;p}\,\sigma^{;q}\,\delta\tau^{(2)}{}_{pq} + \sigma^{;p}\,\sigma^{;q}\,\sigma^{;r}\,\delta\tau^{(3)}{}_{pqr} + \sigma^{;p}\,\sigma^{;q}\,\sigma^{;r}\,\sigma^{;s}\,\delta\tau^{(4)}{}_{pqrs} \tag{5.76}$$

$$w_1 \approx w_1^{(0)} + \sigma^{;p}\, w_1^{(1)}{}_p + \sigma^{;p}\,\sigma^{;q}\, w_1^{(2)}{}_{pq} \tag{5.77}$$

$$w_2 \approx w_2^{(0)} \tag{5.78}$$

The specific values of the expansion tensors in these series are derived in the Appendices of [201].

The above expressions contain up to fourth-order terms but only terms up to the second order will be needed for the regularization of the stress-energy tensor. Carrying out the subtraction (5.73) and substituting the expansions (5.75)–(5.78), we find the expansions tensors in (5.74) to be

$$G^{(0)}_{\rm ren} = \frac{\kappa^2}{3} \tag{5.79}$$

$$G^{(1)}_{{\rm ren}a} = 0 \tag{5.80}$$

$$G^{(2)}_{{\rm ren}ab} = \frac{\kappa^2}{3}\Delta^{(2)}{}_{ab} + \frac{\kappa^4}{180}\left(4\,\delta\tau^{(2)}{}_{ab} - g_{ab}\right) - \frac{1}{2}w_1^{(0)}\, g_{ab}. \tag{5.81}$$

We show only up to second-order terms for the present purpose. Using the explicit forms of the expansion tensor values given in (the Appendices of) [201] we get

$$G^{(2)}_{{\rm ren}ab} = \frac{\kappa^2}{36}R_{ab} + \frac{\kappa^4}{180}\left(4\,\delta_a{}^{\tau}\,\delta_b{}^{\tau} - g_{ab}\right), \tag{5.82}$$

where we have also used

$$w_1^{(0)} = -\frac{1}{240}\left(R_{;p}{}^{p} - R_{pq}\,R^{pq} + R_{pqrs}\,R^{pqrs}\right) = 0, \tag{5.83}$$

which hold for ultra-static metrics.

For flat space, this reduces to

$$G^{(2)}_{\mathrm{ren}ab} = \frac{\kappa^4}{180}\left(-\eta_{ab} + 4\,\eta_a{}^\tau\,\eta_b{}^\tau\right). \tag{5.84}$$

Using the end-point series expansion (5.74) of G_{ren}, the coincident limit of terms with up to four covariant derivatives can be computed. We simply differentiate the series (5.74) and then use the results from the Appendix of [201] for the coincident limits of the covariant derivatives of the σ. The results are (again we show only terms up to second order):

$$16\,\pi^2\,[G_{\mathrm{ren}}] = G^{(0)}_{\mathrm{ren}} \tag{5.85}$$

$$16\,\pi^2\,[G_{\mathrm{ren};a}] = G^{(0)}_{\mathrm{ren};a} \tag{5.86}$$

$$16\,\pi^2\,[G_{\mathrm{ren};ab}] = G^{(0)}_{\mathrm{ren};(ab)} + 2\,G^{(2)}_{\mathrm{ren}(ab)}. \tag{5.87}$$

We now have all the information we need to compute the coincident limit of the expectation value of the stress-energy bitensor. Since the point separated expression $\langle T_{ab'}(x,y)\rangle$ involves covariant derivatives at the two points at which it has support, when we take the coincident limit we can use Synge's theorem to move the derivatives acting at the second point y to ones acting at the first point x.

For a massless, conformally coupled scalar field the point-separated stress-energy tensor is

$$\begin{aligned}\langle T_{ab}(x,y)\rangle = {} & \frac{1}{3}\left(g^{p'}{}_b\,G_{\mathrm{ren};p'a} + g^{p'}{}_a\,G_{\mathrm{ren};p'b}\right) - \frac{1}{6}g^{p'}{}_q\,G_{\mathrm{ren};p'}{}^q\,g_{ab} \\ & - \frac{1}{6}\left(g^{p'}{}_a\,g^{q'}{}_b\,G_{\mathrm{ren};p'q'} + G_{\mathrm{ren};ab}\right) + \frac{1}{6}\left(\left(G_{\mathrm{ren};p'}{}^{p'} + G_{\mathrm{ren};p}{}^{p}\right)\,g_{ab}\right) \\ & - \frac{1}{12}G_{\mathrm{ren}}\,\left(R\,g_{ab} - 2\,R_{ab}\right).\end{aligned} \tag{5.88}$$

We take the coincident limit and utilize Synge's theorem to obtain

$$\begin{aligned}\langle T_{ab}\rangle_{\mathrm{ren}} = {} & \frac{1}{6}\left(3\,[G_{\mathrm{ren};a}]_{;b} + 3\,[G_{\mathrm{ren};b}]_{;a} - [G_{\mathrm{ren}}]_{;ab} - 6\,[G_{\mathrm{ren};ab}]\right) \\ & - \frac{1}{6}\left(3\,[G_{\mathrm{ren};p}]^{;p} - [G_{\mathrm{ren}}]_{;p}{}^{p} - 3\,[G_{\mathrm{ren};p}{}^{p}]\right)\,g_{ab} \\ & - \frac{1}{12}[G_{\mathrm{ren}}]\,\left(R\,g_{ab} - 2\,R_{ab}\right) \\ = {} & \frac{1}{16\pi^2}\left\{-\frac{1}{6}\left(G^{(0)}_{\mathrm{ren};ab} + 12\,G^{(2)}_{\mathrm{ren}ab} - G^{(0)}_{\mathrm{ren}}\,R_{ab}\right)\right. \\ & \left. -\frac{1}{12}\left(G^{(0)}_{\mathrm{ren}}\,R - 2\,G^{(0)}_{\mathrm{ren};p}{}^{p} - 12\,G^{(2)}_{\mathrm{ren}p}{}^{p}\right)\,g_{ab}\right\}.\end{aligned} \tag{5.89}$$

With the explicit values (5.82) for the expansion tensors we obtain

$$\langle T_{ab}\rangle_{\mathrm{ren}} = \frac{\kappa^4}{1440\,\pi^2}\left(g_{ab} - 4\,\delta_a{}^\tau\,\delta_b{}^\tau\right). \tag{5.90}$$

This expression has been derived by several other methods in the literature, e.g., in [488] Eq. (58).

We now show how to derive the explicit forms of the regularized stress-energy tensor for two cases, the hot flat space and the optical Schwarzschild black hole spacetime.

Hot Flat Space

The imaginary-time (τ) finite temperature $T = \kappa/2\pi$ quantum field theory in Euclidean flat space (with $x^a = (\tau, x, y, z,)$) is a simple enough example which can be used to compare the different methods introduced. For a massless scalar field the non-zero coincident limits of the derivatives of $G_{\rm ren}$ are [from Eq. (5.87) using (5.84)],

$$[G_{\rm ren}] = \frac{T^2}{12} \tag{5.91}$$

$$[G_{{\rm ren};ab}] = -\frac{(\pi^2\, T^4)}{90}\,(\delta_{ab} - 4\,\delta_a{}^\tau\,\delta_b{}^\tau)\,, \tag{5.92}$$

from which we obtain readily the stress-energy tensor in the familiar form

$$\langle T_{ab}\rangle = {\rm diag}\left\{-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right\}\rho, \quad \rho = -\frac{\pi^2\, T^4}{30}. \tag{5.93}$$

Optical Schwarzschild Black Hole

The optical spacetime conformally related to the Schwarzschild black hole spacetime has the line element in the coordinates $x^a = (\tau, r, \theta, \phi)$

$$ds^2 = d\tau^2 + \left(1 - \frac{2M}{r}\right)^{-2} dr^2 + \left(1 - \frac{2M}{r}\right)^{-1} r^2\left(d\theta^2 + \sin^2\theta d\phi^2\right). \tag{5.94}$$

Taking $\kappa = 2\pi T$ and $T = 1/(8\pi M)$ we choose the quantum state corresponding to the Hartle–Hawking state in the conformally–related Schwarzschild spacetime. For a massless conformal scalar field ($m = 0$, $\xi = 1/6$) the stress-energy tensor is

$$\left\langle T_a^b\right\rangle = {\rm diag}\left\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1\right\}\frac{\pi^2\, T^4}{30}. \tag{5.95}$$

We recover the standard thermal result for the stress-energy tensor.

5.3 The Noise Kernel: Structure, Forms and Computations

In the last section we discussed how the Hadamard expansion and the Gaussian approximation work to obtain a finite expression for the Green functions, from which one can derive a regularized stress-energy tensor. We also showed that in the class of ultra-static spacetimes the Gaussian approximation gives an exact expression for the Green function. In this section we discuss how to

calculate the noise kernel, or the correlator of the stress-energy bitensor, with these methods. We first derive a formal expression of the noise kernel in terms of the fourth covariant derivative of the Green functions and then show how to use conformal transformations to obtain the noise kernels from a known form in a simpler spacetime. With the Green functions derived earlier under the Gaussian approximation we proceed to evaluate the noise kernel for a massless scalar field in hot flat space. We also mention results in the ultrastatic metric conformal to the Schwarzschild metric obtained in [201].

The Green function under the Gaussian approximation yields a fairly accurate result for the stress-energy tensor, i.e., to second order in the field products, as was shown by Page [488]. However, the Gaussian approximation breaks down at fourth order, and this error will show up in the noise kernel because it needs up to four covariant derivatives of the Green function. Indeed, the trace of the noise kernel computed under the Gaussian approximation fails to vanish, as shown in [201].

Finally we turn our attention to spacetimes of constant curvature, specifically the de Sitter space. The hope is that with maximal symmetry we may be able to obtain an exact analytic expression for the noise kernel. We show the derivation of the noise kernel in two coordinatizations of de Sitter, one the ($k = 0$) FLRW coordinatization which covers half of the space and the other the static coordinatization which covers a quarter of the space in the conformal diagram. In the former case the Bunch–Davies vacuum has been useful for treating quantum processes in inflationary cosmology, while in the latter case the Gibbons-Hawking vacuum [322] was introduced in analogy with the Hartle–Hawking vacuum [82] for treating thermal radiance in black hole spacetimes.

5.3.1 Noise Kernel in Terms of Covariant Derivatives of Green Functions

We begin by deriving a formal expression for the noise kernel in terms of up to the fourth-order covariant derivatives of the Green function taken at two separate points. This expression was obtained in [183, 200]. For this purpose we need to keep the two pairs of points (x, x') and (y, y') distinct so we can keep track of which covariant derivative acts on which arguments of which Wightman function. As an example (the complete calculation is quite long), consider the result of the first set of covariant derivative operators in the differential operator (5.26), from both $\mathcal{T}_{ab}$ and $\mathcal{T}_{c'd'}$, acting on $G(x, x', y, y')$:

$$\frac{1}{4}(1-2\xi)^2\left(g_a{}^{p''}\nabla_{p''}\nabla_b + g_b{}^{p''}\nabla_{p''}\nabla_a\right) \times\left(g_{c'}{}^{q'''}\nabla_{q'''}\nabla_{d'} + g_{d'}{}^{q'''}\nabla_{q'''}\nabla_{c'}\right)G(x,x',y,y') \tag{5.96}$$

with the notation: ∇_a acts at x, $\nabla_{c'}$ at y, $\nabla_{b''}$ at x' and $\nabla_{d'''}$ at y'. Expanding out the differential operator above, we can determine which derivatives act on which Wightman function:

$$\frac{(1-2\,\xi)^2}{4}\Big[g_{c'}{}^{p'''}\,g^{q''}{}_{a}\,(G_{xy';bp'''}\,G_{x'y;q''d'}+G_{xy;bd'}\,G_{x'y';q''p'''}$$
$$+\,G_{yx';q''d'}\,G_{y'x;bp'''}+G_{yx;bd'}\,G_{y'x';q''p'''})$$
$$+\,g_{d'}{}^{p'''}\,g^{q''}{}_{a}\,(G_{xy';bp'''}\,G_{x'y;q''c'}+G_{xy;bc'}\,G_{x'y';q''p'''}$$
$$+\,G_{yx';q''c'}\,G_{y'x;bp'''}+G_{yx;bc'}\,G_{y'x';q''p'''})$$
$$+\,g_{c'}{}^{p'''}\,g^{q''}{}_{b}\,(G_{xy';ap'''}\,G_{x'y;q''d'}+G_{xy;ad'}\,G_{x'y';q''p'''}$$
$$+\,G_{yx';q''d'}\,G_{y'x;ap'''}+G_{yx;ad'}\,G_{y'x';q''p'''})$$
$$+\,g_{d'}{}^{p'''}\,g^{q''}{}_{b}\,(G_{xy';ap'''}\,G_{x'y;q''c'}+G_{xy;ac'}\,G_{x'y';q''p'''}$$
$$+\;G_{yx';q''c'}\,G_{y'x;ap'''}+G_{yx;ac'}\,G_{y'x';q''p'''})\Big]\,. \tag{5.97}$$

If we now let $x' \to x$ and $y' \to y$ the contribution to the noise kernel is (including the factor of $\frac{1}{8}$ present in the definition of the noise kernel):

$$\frac{1}{8}\left\{(1-2\,\xi)^2\;(G_{xy;ad'}\,G_{xy;bc'}+G_{xy;ac'}\,G_{xy;bd'})\right.$$
$$\left.+(1-2\,\xi)^2\;(G_{yx;ad'}\,G_{yx;bc'}+G_{yx;ac'}\,G_{yx;bd'})\right\} \tag{5.98}$$

That this term can be written as the sum of a part involving G_{xy} and one involving G_{yx} is a general property of the entire noise kernel. It thus takes the form

$$N_{abc'd'}(x,y) = N_{abc'd'}\left[G_+(x,y)\right] + N_{abc'd'}\left[G_+(y,x)\right]. \tag{5.99}$$

We will present the form of the functional $N_{abc'd'}\left[G\right]$ shortly. First we note, for x and y time-like separated, the above split of the noise kernel allows us to express it in terms of the Feynman (time-ordered) Green function $G_F(x,y)$ and the Dyson (anti-time-ordered) Green function $G_D(x,y)$:

$$N_{abc'd'}(x,y) = N_{abc'd'}\left[G_F(x,y)\right] + N_{abc'd'}\left[G_D(x,y)\right]. \tag{5.100}$$

Recall when the quantum stress-energy tensor fluctuations determined in the Euclidean section is analytically continued back to Lorentzian signature ($\tau \to it$), the time-ordered product results. On the other hand, if the continuation is $\tau \to -it$, the anti-time-ordered product results. With this in mind, the noise kernel is seen to be related to the quantum stress-energy tensor fluctuations (derived from the second variation of the effective action in the last chapter) as

$$4N_{abc'd'} = \Delta T^2_{abc'd'}|_{t=-i\tau,t'=-i\tau'} + \Delta T^2_{abc'd'}|_{t=i\tau,t'=i\tau'}\,. \tag{5.101}$$

The complete form of the functional $N_{abc'd'}\left[G\right]$ is

$$N_{abc'd'}\left[G\right] = \tilde{N}_{abc'd'}\left[G\right] + g_{ab}\tilde{N}_{c'd'}\left[G\right] + g_{c'd'}\tilde{N}'_{ab}\left[G\right] + g_{ab}g_{c'd'}\tilde{N}\left[G\right] \tag{5.102}$$

with

$$\begin{aligned}\tilde{N}_{abc'd'}\,[G] &= (1-2\,\xi)^2\;(G_{;c'b}\,G_{;d'a}+G_{;c'a}\,G_{;d'b})+4\,\xi^2\;(G_{;c'd'}\,G_{;ab}+G\,G_{;abc'd'})\\ &\quad-2\,\xi\;(1-2\,\xi)\;(G_{;b}\,G_{;c'ad'}+G_{;a}\,G_{;c'bd'}+G_{;d'}\,G_{;abc'}+G_{;c'}\,G_{;abd'})\\ &\quad+2\,\xi\;(1-2\,\xi)\;(G_{;a}\,G_{;b}\,R_{c'd'}+G_{;c'}\,G_{;d'}\,R_{ab})\\ &\quad-4\,\xi^2\;(G_{;ab}\,R_{c'd'}+G_{;c'd'}\,R_{ab})\,G+2\,\xi^2\,R_{c'd'}\,R_{ab}G^2, \qquad (5.103)\\ \tilde{N}'_{ab}\,[G] &= 2\;(1-2\,\xi)\left[\left(2\,\xi-\frac{1}{2}\right)\;G_{;p'b}\,G_{;}^{\;\;p'}{}_{a}+\xi\;\left(G_{;b}\,G_{;p'a}{}^{p'}+G_{;a}\,G_{;p'b}{}^{p'}\right)\right]\\ &\quad-4\,\xi\left[\left(2\,\xi-\frac{1}{2}\right)\;G_{;}^{\;\;p'}\,G_{;abp'}+\xi\;\left(G_{;p'}{}^{p'}\,G_{;ab}+G\,G_{;abp'}{}^{p'}\right)\right]\\ &\quad-(m^2+\xi R')\;[(1-2\,\xi)\;G_{;a}\,G_{;b}-2\,G\,\xi\,G_{;ab}]\\ &\quad+2\,\xi\left[\left(2\,\xi-\frac{1}{2}\right)\;G_{;p'}\,G_{;}^{\;\;p'}+2\,G\,\xi\,G_{;p'}{}^{p'}\right]\;R_{ab}\\ &\quad-(m^2+\xi R')\;\xi\,R_{ab}G^2, \qquad (5.104)\\ \tilde{N}\,[G] &= 2\left(2\,\xi-\frac{1}{2}\right)^2\;G_{;p'q}\,G_{;}^{\;\;p'q}+4\,\xi^2\;\left(G_{;p'}{}^{p'}\,G_{;q}{}^{q}+G\,G_{;p}{}^{p}{}_{q'}{}^{q'}\right)\\ &\quad+4\,\xi\;\left(2\,\xi-\frac{1}{2}\right)\;\left(G_{;p}\,G_{;q'}{}^{pq'}+G_{;}^{\;\;p'}\,G_{;q}{}^{q}{}_{p'}\right)\\ &\quad-\left(2\,\xi-\frac{1}{2}\right)\;\left[(m^2+\xi R)\;G_{;p'}\,G_{;}^{\;\;p'}+(m^2+\xi R')\;G_{;p}\,G^{;p}\right]\\ &\quad-2\,\xi\;\left[(m^2+\xi R)\;G_{;p'}{}^{p'}+(m^2+\xi R')\;G_{;}^{\;\;p}{}_{p}\right]G\\ &\quad+\frac{1}{2}\,(m^2+\xi R)\;(m^2+\xi R')\,G^2. \qquad (5.105)\end{aligned}$$

For a massless, conformally-coupled scalar field ($\xi=\frac{1}{6}$ and $m=0$), the noise kernel functional is

$$\begin{aligned}9\tilde{N}_{abc'd'}\,[G] &= 4\;(G_{;c'b}\,G_{;d'a}+G_{;c'a}\,G_{;d'b})+G_{;c'd'}\,G_{;ab}+G\,G_{;abc'd'}\\ &\quad-2\;(G_{;b}\,G_{;c'ad'}+G_{;a}\,G_{;c'bd'}+G_{;d'}\,G_{;abc'}+G_{;c'}\,G_{;abd'})\\ &\quad+2\;(G_{;a}\,G_{;b}\,R_{c'd'}+G_{;c'}\,G_{;d'}\,R_{ab})\\ &\quad-(G_{;ab}\,R_{c'd'}+G_{;c'd'}\,R_{ab})\,G-\frac{1}{2}R_{c'd'}\,R_{ab}G^2, \qquad (5.106)\\ 36\tilde{N}'_{ab}\,[G] &= 8\left(-G_{;p'b}\,G_{;}^{\;\;p'}{}_{a}+G_{;b}\,G_{;p'a}{}^{p'}+G_{;a}\,G_{;p'b}{}^{p'}\right)]\\ &\quad+4\left(G_{;}^{\;\;p'}\,G_{;abp'}-G_{;p'}{}^{p'}\,G_{;ab}+G\,G_{;abp'}{}^{p'}\right)\\ &\quad-2\,R'\;(2\,G_{;a}\,G_{;b}-G\,G_{;ab})\\ &\quad-2\;\left(G_{;p'}\,G_{;}^{\;\;p'}-2\,G\,G_{;p'}{}^{p'}\right)\;R_{ab}-R'\,R_{ab}G^2, \qquad (5.107)\end{aligned}$$

$$
\begin{aligned}
36\tilde{N}\,[G] = {} & 2\,G_{;p'q}\,G_{;}^{\;p'q} + 4\,\left(G_{;p'}{}^{p'}\,G_{;q}{}^{q} + G\,G_{;p}{}^{p}{}_{q'}{}^{q'}\right) \\
& - 4\,\left(G_{;p}\,G_{;q'}{}^{pq'} + G_{;}^{\;p'}\,G_{;q}{}^{q'}{}_{p}\right) \\
& + R\,G_{;p'}\,G_{;}^{\;p'} + R'\,G_{;p}\,G^{;p} \\
& - 2\,\left(R\,G_{;p'}{}^{p'} + R'\,G_{;p}{}^{p}\right)G + \frac{1}{2}R\,R'G^2. \qquad (5.108)
\end{aligned}
$$

For the minimal coupling ($\xi = 0$) case:

$$\tilde{N}_{abc'd'}\,[G] = G_{;c'b}\,G_{;d'a} + G_{;c'a}\,G_{;d'b}, \qquad (5.109)$$

$$\tilde{N}'_{ab}\,[G] = -G_{;p'b}\,G^{p'}_{;a} - m^2\,G_{;a}\,G_{;b}, \qquad (5.110)$$

$$\tilde{N}\,[G] = \frac{1}{2}\left(G_{;p'q}\,G_{;}^{\;p'q} + m^2\,\left(G_{;p'}\,G_{;}^{\;p'} + G_{;p}\,G^{;p}\right) + m^4G^2\right). \qquad (5.111)$$

5.3.2 Trace of the Noise Kernel

One of the most interesting and surprising results to come out of the investigations undertaken in the 1970s of the quantum stress tensor was the discovery of the trace anomaly [489]. Note that when the trace of the stress-energy tensor $T = g^{ab}T_{ab}$ is evaluated for a field configuration that satisfies the field equation

$$\frac{\delta S[\phi]}{\delta\phi(x)} = 0 \Rightarrow (\Box - \xi R - m^2)\,\phi = 0, \qquad (5.112)$$

the trace is seen to vanish for the massless conformally coupled field. When this analysis is carried over to the renormalized expectation value of the quantum stress-energy tensor, the trace no longer vanishes. As we mentioned earlier, according to Wald [103], this is due to the failure of the renormalized Hadamard function $G_{\rm ren}(x, x')$ to be symmetric in x and x', implying it does not necessarily satisfy the field equation (5.112) in the variable x'. We can determine the noise associated with the trace in this way. Taking the trace at both points x and y of the noise kernel functional (5.100):

$$
\begin{aligned}
N\,[G] = {} & g^{ab}\,g^{c'd'}\,N_{abc'd'}\,[G] \\
= {} & -24\,G\,\xi\,\left\{\left(m^2 + \frac{1}{2}\xi R\right)\,G_{;p'}{}^{p'} + \left(m^2 + \frac{1}{2}\xi R'\right)\,G_{;p}{}^{p}\right\} \\
& + 36\,\xi^2\,\left\{G_{;p'}{}^{p'}\,G_{;p}{}^{p} + G\,G_{;p}{}^{p}{}_{p'}{}^{p'}\right\} + \left(m^2 + \frac{1}{2}\xi R\right)\,\left(m^2 + \frac{1}{2}\xi R'\right)G^2 \\
& + 24\left(\frac{1}{6} - \xi\right)\left\{3\left(\frac{1}{6} - \xi\right)G_{;p'p}\,G_{;}^{\;p'p} - 3\xi\,\left(G_{;p}\,G_{;p'}{}^{pp'} + G_{;p'}\,G_{;p}{}^{pp'}\right)\right. \\
& \left. + \left(m^2 + \frac{1}{2}\xi R\right)\,G_{;p'}\,G_{;}^{\;p'} + \left(m^2 + \frac{1}{2}\xi R'\right)\,G_{;p}\,G^{;p}\right\}. \qquad (5.113)
\end{aligned}
$$

For the massless conformal case, this reduces to

$$N[G] = \frac{1}{18}\left\{RR'G^2 - 6G\left(R\Box' + R'\Box\right)G + 18\left[\left(\Box G\right)\left(\Box' G\right) + \Box'\Box G\right]\right\}, \tag{5.114}$$

which holds for any function $G(x, y)$. For G being the Green function, it satisfies the field equation (5.112)

$$\Box G = (m^2 + \xi R)G. \tag{5.115}$$

We will only assume the Green function satisfies the field equation in its first variable. Using the fact $\Box' R = 0$ (because the covariant derivatives act at a different point than at which R is supported), it follows that

$$\Box'\Box G = (m^2 + \xi R)\Box' G. \tag{5.116}$$

With these results, the noise kernel trace becomes

$$\begin{aligned} N[G] = 4 &\left(m^2\ (1-3\,\xi) + \frac{1}{2}\,R\ (1-6\xi)\ \xi\right) \\ &\times \left\{G^2\ (2\,m^2 + R'\,\xi) + (1-6\,\xi)\ G_{;p'}\,G^{p'}_{;} - 6\,G\,\xi\,G_{;p'p'}\right\} \\ &+ 4\left(\frac{1}{6} - \xi\right)\left\{3\ (2\,m^2 + R'\,\xi)\ G_{;p}\,G^{;p} - 18\,\xi\,G_{;p}\,G_{;p'pp'}\right. \\ &\left. +18\left(\frac{1}{6} - \xi\right)\ G_{;p'p}\,G_{;p'p}\right\}. \end{aligned} \tag{5.117}$$

We see that it vanishes for the massless conformal case, thus there is no noise associated with the trace anomaly. This result derived from the point-separated noise kernel is completely general since we have assumed the Green function is only satisfying the field equations in its first variable. This condition holds not just for the classical field case, but also for the regularized quantum case, where we do not expect the Green function to satisfy the field equation in both variables. One can see this result from a simple observation. Since the trace anomaly is known to be locally determined and quantum state independent, whereas the noise present in the quantum field is non-local, it is hard to find a noise associated with it. This is in agreement with previous findings [216, 185, 186], derived from the Feynman–Vernon influence functional formalism [187].

In summary, we have derived a general expression for the noise kernel, or the correlator of the stress-energy bitensor, for a quantum scalar field in a general curved spacetime using the point separation method. The general form is expressed as products of covariant derivatives of the quantum field's Green function. It is finite when the noise kernel is evaluated for distinct pairs of points (and non-null points for a massless field). This expression (5.101)–(5.104) for the noise kernel of a scalar field is completely general and can be used with or without first considering issues of renormalization of the Green function. We have

shown that the trace of the noise kernel vanishes, confirming there is no noise associated with the trace anomaly.

5.3.3 Noise Kernel under Conformal Transformation

For the investigation of the conformal properties of the noise kernel such as in the proof we just provided on the vanishing of its trace, we derive in this section the form of the noise kernel under a conformal transformation. This is also useful for explicit calculations of the noise kernel as we have seen for the ultra-static spacetimes. For instance, the noise kernel can be derived easily in an optical Schwarzschild metric. One can then use a conformal transformation to obtain the expressions for the physical metric. We provide two alternative proofs based respectively on the use of quantum operators and on functional methods (see, e.g., Appendix of [477]).

We first show how the classical stress-energy tensor of a conformally invariant scalar field rescales under a conformal transformation $g_{ab} \to \tilde{g}_{ab} = \Omega^2(x)\, g_{ab}$. The key point is that the classical action of the field $S[\phi, g]$ remains invariant (up to surface terms) if one rescales appropriately the field: $\phi \to \tilde{\phi} = \Omega^{(2-D)/2}\phi$. Taking this into account, one easily gets the result from the definition of the stress-energy tensor as a functional derivative of the classical action:

$$\begin{aligned}\tilde{T}_{ab} &= \frac{2\tilde{g}_{ac}\tilde{g}_{bd}}{\sqrt{-\tilde{g}}}\frac{\delta S[\tilde{\phi},\tilde{g}]}{\delta \tilde{g}_{cd}} = \frac{2\tilde{g}_{ac}\tilde{g}_{bd}}{\sqrt{-\tilde{g}}}\frac{\delta S[\phi, g]}{\delta \tilde{g}_{cd}} \\ &= \Omega^{2-D}\frac{2g_{ac}g_{bd}}{\sqrt{-g}}\frac{\delta S[\phi,g]}{\delta g_{cd}} = \Omega^{2-D}\, T_{ab}. \end{aligned} \tag{5.118}$$

For conformally invariant fields the scaling behavior of the noise kernel Eq. (5.10) under conformal transformations can be shown as follows. Promoting the classical field ϕ in Eq. (5.118) to an operator in the Heisenberg picture, the operator $\hat{T}_{ab}(x)$ would be divergent because it involves products of the field operator at the same point. However, in order to calculate the noise kernel what one actually needs to consider is $\hat{t}_{ab}(x) = \hat{T}_{ab}(x) - \langle \hat{T}_{ab}(x)\rangle$ and this object is UV finite, i.e., its matrix elements $\langle\Phi|\hat{t}_{ab}(x)|\Psi\rangle$ for two arbitrary states $|\Psi\rangle$ and $|\Phi\rangle$ (not necessarily orthogonal) are UV finite because Wald's axioms [47] guarantee that $\langle\Phi|\hat{T}_{ab}(x)|\Psi\rangle$ and $\langle\Phi|\Psi\rangle\langle\hat{T}_{ab}(x)\rangle$ have the same UV divergences and they cancel out.

We proceed as follows: Start by introducing a UV regulator (it is useful to consider dimensional regularization since it is compatible with the conformal symmetry for scalar and fermionic fields, but this is not indispensable since we will remove the regulator at the end without making any subtraction of non-invariant counterterms). Next, apply the operator version of Eq. (5.118) to the operators $\hat{t}_{ab}(x)$ appearing in Eq. (5.33) defining the noise kernel. Since all UV divergences cancel out, as argued above, one can then safely remove the regulator and be left with Eq. (5.10).

We now make use of this conformal relation to calculate the noise kernel of several spacetimes of physical interest. As an immediate application of the conformal relation proven above we show how to derive the noise kernel for a massless conformally coupled scalar field in the class of conformally flat spacetimes which includes important physical examples such as the de Sitter/anti-de Sitter and the FLRW spacetimes. Note these results are useful for those metrics that are conformal to the full Minkowski metric (or at least a part of it that contains a Cauchy surface).

Let us first calculate the noise kernel for a massless conformally coupled scalar field in Minkowski spacetime. We begin with the Wightman function in flat space in the Minkowski vacuum

$$G^{+}(x,x') = \frac{1}{8\pi^2\sigma(x,x')} + \frac{i}{8\pi}\delta(\sigma(x,x'))\mathrm{sgn}(t-t'), \tag{5.119}$$

with

$$\sigma(x,x') = \frac{1}{2}\left[-(x^0 - x^{0'})^2 + (\vec{x} - \vec{x}')^2\right]. \tag{5.120}$$

In general σ satisfies the relationship

$$\sigma = \frac{1}{2}g_{ab}\sigma^a\sigma^b = \frac{1}{2}g_{a'b'}\sigma^{a'}\sigma^{b'}, \tag{5.121}$$

with

$$\begin{aligned}\sigma^a &\equiv \sigma^{;a},\\ \sigma^{a'} &\equiv \sigma^{;a'}.\end{aligned} \tag{5.122}$$

Recall primes on indices indicate that the indices are at the point x'. If the points are close together then

$$\begin{aligned}\sigma^a &= x^a - x^{'a},\\ \sigma^{a'} &= -(x^a - x^{'a}),\end{aligned} \tag{5.123}$$

and of course, in general,

$$\begin{aligned}\sigma_a &= g_{ab}\sigma^b\\ \sigma_{a'} &= g_{a'b'}\sigma^{b'}.\end{aligned} \tag{5.124}$$

Because the noise kernel is quadratic in the Wightman function, one would expect to see formally divergent terms in the coincidence limit and for null separations which go like derivatives of $i\delta(\sigma)/\sigma$ and $\delta(\sigma)^2$. Using the symmetries of the noise kernel it is easy to show that all of the $i\delta(\sigma)/\sigma$ terms must vanish identically; however, the meaning of the $\delta(\sigma)^2$ terms is more difficult to understand.

As discussed in [490] for flat space and in [491] for de Sitter, these divergences may be more easily analyzed in Fourier space by integration against a test

function; the resulting expressions are explicitly finite. The use of smearing functions in point-separated stress-energy bitensors for the calculation of vacuum energy density fluctuations in Minkowski and Casimir states is illustrated in [199].

In the present case the square of the Wightman function is

$$G^+(x,x')^2 = \frac{1}{64\pi^4\sigma^2} + \frac{i\delta(\sigma)\mathrm{sgn}(t-t')}{64\pi^3\sigma} - \frac{\delta(\sigma)^2}{64\pi^2}. \tag{5.125}$$

When integrated against a test function $f(x')$, we find that the divergent $f(x)\delta(0)$ and $if(x)/0$ terms arising from the integrals over $f(x')\delta(\sigma)^2$ and $if(x')\delta(\sigma)/\sigma$ are exactly the terms needed to cancel the divergence arising from the integral over $1/\sigma^2$ such that the total result is finite. Because of this, the total integral is just the Hadamard finite part,

$$\int d^4x' f(x')[G^+(x,x')]^2 = \mathcal{H}\left[\int d^4x' \frac{f(x')}{64\pi^4\sigma(x,x')^2}\right]. \tag{5.126}$$

Therefore, the square of the Wightman function is a well-defined distribution.

A similar cancellation happens for the noise kernel. The resulting expression for the noise kernel for the conformally invariant scalar field in the Minkowski vacuum state is

$$\begin{aligned} N_{abc'd'}(x,x') = \Bigg[& \frac{\sigma_a\sigma_b\sigma_{c'}\sigma_{d'}}{48\pi^4\sigma^6} + \frac{\sigma_{(a}\eta_{b)(c'}\sigma_{d')}}{24\pi^4\sigma^5} + \frac{4\eta_{a(c'}\eta_{d')b} - \eta_{ab}\eta_{c'd'}}{192\pi^4\sigma^4} \\ & - \frac{\sigma_a\sigma_b\sigma_{c'}\sigma_{d'}}{576\pi^2}\left(9\delta''(\sigma)^2 - 8\delta'(\sigma)\delta'''(\sigma) + \delta(\sigma)\delta''''(\sigma)\right) \\ & + \frac{\sigma_a\sigma_b\eta_{c'd'} + \eta_{ab}\sigma_{c'}\sigma_{d'}}{576\pi^2}\left(5\delta'(\sigma)\delta''(\sigma) - \delta(\sigma)\delta'''(\sigma)\right) \\ & + \frac{\sigma_{(a}\eta_{b)(c'}\sigma_{d')}}{144\pi^2}\delta(\sigma)\delta'''(\sigma) - \frac{\eta_{a(c'}\eta_{d')b}}{288\pi^2}\left(4\delta'(\sigma)^2 + \delta(\sigma)\delta''(\sigma)\right) \\ & + \frac{\eta_{ab}\eta_{c'd'}}{576\pi^2}\left(\delta'(\sigma)^2 - \delta(\sigma)\delta''(\sigma)\right)\Bigg] \\ = \mathcal{H}\Bigg[& \frac{\sigma_a\sigma_b\sigma_{c'}\sigma_{d'}}{48\pi^4\sigma^6} + \frac{\sigma_{(a}\eta_{b)(c'}\sigma_{d')}}{24\pi^4\sigma^5} + \frac{4\eta_{a(c'}\eta_{d')b} - \eta_{ab}\eta_{c'd'}}{192\pi^4\sigma^4}\Bigg], \end{aligned} \tag{5.127}$$

where the divergences arising from an integral over the delta functions are exactly those needed to cancel the divergences in the integration over powers of $1/\sigma$. Here $(\ldots)$ indicates symmetrization of the indices, η_{ab} is the Minkowski metric, and $\eta_{ac'} = \mathrm{diag}(-1,1,1,1)$ is the bivector of parallel transport in Minkowski space for Cartesian coordinates. With the transformation $g_{ab} \to \tilde{g}_{ab} = \Omega^2(x)\, g_{ab}$ the noise kernel in spacetimes conformally related to Minkowski is then given by

$$\begin{aligned} \widetilde{N}^{conf}_{abc'd'}(x,x') = \mathcal{H}\Big[& \Omega(x)^{-2}\Omega(x')^{-2}\left(\frac{\sigma_a\sigma_b\sigma_{c'}\sigma_{d'}}{48\pi^4\sigma^6} + \frac{\sigma_{(a}\eta_{b)(c'}\sigma_{d')}}{24\pi^4\sigma^5}\right. \\ & \left. + \frac{4\eta_{a(c'}\eta_{d')b} - \eta_{ab}\eta_{c'd'}}{192\pi^4\sigma^4}\right)\Big]. \end{aligned} \tag{5.128}$$

This result agrees with results for the noise kernel in the Minkowski vacuum state obtained by Martín and Verdaguer [183] [189, 490, 477], and with results for the noise kernel in dS and AdS spacetimes by Osborn and Shore [204] via conformal field theory, and by Cho and Hu [465] obtained by the generalized zeta function method described in the last Chapter. They also show [1018] how, by applying the appropriate conformal transformations, one can obtain the various noise kernels in Robertson-Walker spacetimes from the noise kernels in Minkowski and Einstein spacetimes.

5.3.4 Noise Kernels in Exact Forms

We now show two examples by this method, the hot flat space and the de Sitter space.

Hot Flat Space

Consider the quantum field in a thermal state at an arbitrary temperature T and let the points be separated in a non-null direction. This is an easy example because in flat space the function $U(x, x')$ is equal to one. Therefore, the expression for $G^+(x, x')$ in Eq. (5.70) is exact so long as the points are separated in a non-null direction. This expression can be substituted into Eqs. (5.102) and (5.106)–(5.108) to obtain an exact expression for the noise kernel. Here the quantity ς takes on the following simple form in Cartesian coordinates and components:

$$\varsigma = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2} = |\vec{r} - \vec{r}'|. \tag{5.129}$$

The only subtlety which one must be aware of is that the point separation must be arbitrary before the derivatives are computed. Once they are computed, then any point separation that one desires can be used.

All components of the noise kernel can be derived in exact analytic forms when the points are separated in a non-null direction. Both conservation and the vanishing of the partial traces have been checked in [477] where one component is displayed, as follows:

$$\begin{aligned} N_{ttt't'} = & \frac{\kappa^2 \sinh^2(\kappa\varsigma)}{192\pi^4\varsigma^6(\cosh(\kappa\Delta t) - \cosh(\kappa\varsigma))^2} \\ & + \frac{\kappa^3 \sinh(\kappa\varsigma)}{96\pi^4\varsigma^5(\cosh(\kappa\Delta t) - \cosh(\kappa\varsigma))^3}\,[1 - \cosh(\kappa\Delta t)\cosh(\kappa\varsigma)] \\ & + \frac{\kappa^4}{192\pi^4 r^4(\cosh(\kappa\Delta t) - \cosh(\kappa\varsigma))^4}\,[2 - 2\cosh(\kappa\Delta t)\cosh(\kappa\varsigma) \\ & - \cosh^2(\kappa\varsigma) + \cosh^2(\kappa\Delta t)\cosh(2\kappa\varsigma)] \end{aligned}$$

$$
\begin{aligned}
&+\frac{\kappa^5\sinh(\kappa\varsigma)}{288\pi^4\varsigma^3(\cosh(\kappa\Delta t)-\cosh(\kappa\varsigma))^4}\left[2\cosh(\kappa\Delta t)\right.\\
&\left.-\cosh(2\kappa\Delta t)\cosh(\kappa\varsigma)-\cosh(\kappa\Delta t)\cosh^2(\kappa\varsigma)\right]\\
&-\frac{\kappa^6}{576\pi^4\varsigma^2(\cosh(\kappa\Delta t)-\cosh(\kappa\varsigma))^6}\left[12-6\cosh^2(\kappa\Delta t)\right.\\
&+\cosh^4(\kappa\Delta t)-12\cosh(\kappa\Delta t)\cosh(\kappa\varsigma)+\cosh^3(\kappa\Delta t)\cosh(\kappa\varsigma)\\
&-18\cosh^2(\kappa\varsigma)+12\cosh^2(\kappa\Delta t)\cosh^2(\kappa\varsigma)-2\cosh^4(\kappa\Delta t)\cosh^2(\kappa\varsigma)\\
&+17\cosh(\kappa\Delta t)\cosh^3(\kappa\varsigma)+3\cosh^3(\kappa\Delta t)\cosh^3(\kappa\varsigma)-\cosh^4(\kappa\varsigma)\\
&\left.-6\cosh(2\kappa\Delta t)\cosh^4(\kappa\varsigma)-\cosh(\kappa\Delta t)\cosh^5(\kappa\varsigma)\right]. \qquad (5.130)
\end{aligned}
$$

De Sitter Spacetime

The de Sitter spacetime admits many coordinatizations. We show two of them here, one which covers half of the space (in a spatially flat or k = 0 FLRW coordinatization, sometimes referred to as the Poincaré patch) and one which covers only one quarter of the space (the static coordinate) in the conformal diagram. The former is used frequently for the description of inflationary cosmology. The latter, having the same causal structure as the Schwarzschild spacetime, has been used for the study of thermal radiance by an observer in de Sitter spacetime by Gibbons–Hawking [322] similar to the derivation of Hawking radiation in Schwarzschild spacetime by Hartle and Hawking [82]. For completeness we should add that a coordinate, the 'closed' dS, which covers the whole de Sitter space is in the form of a hyperbola of one sheet in an embedding diagram in E^5, and another, the Euclidean dS, is the Euclideanized coordinate of S^4 in E^5.

The de Sitter space in the $k = 0$ FLRW coordinatization when expressed in conformal time η $(-\infty < \eta < 0)$ with scale factor $a(\eta) = \alpha/(-\eta)$ $(0 < a < \infty)$ has the line element

$$ds^2 = \frac{\alpha^2}{(-\eta)^2}(-d\eta^2 + dx^2 + dy^2 + dz^2). \qquad (5.131)$$

The noise kernel for the conformally invariant scalar field in the conformal vacuum in de Sitter space can be obtained from Eq. (5.128) with the substitutions (where now the scale factor a is called Ω)

$$
\begin{aligned}
t \to -\eta, \qquad & t' \to -\eta'\\
\Omega(x) = \frac{\alpha}{(-\eta)}, \qquad & \Omega(x') = \frac{\alpha}{(-\eta')}. \qquad (5.132)
\end{aligned}
$$

The metric for the static de Sitter spacetime has a form similar to that of Schwarzschild spacetime. The coordinate transformation to the static coordinates is given by

$$
\begin{aligned}
x &\equiv \frac{e^{-T/\alpha}}{\sqrt{B}}\rho\sin\theta\cos\phi,\\
y &\equiv \frac{e^{-T/\alpha}}{\sqrt{B}}\rho\sin\theta\sin\phi,\\
z &\equiv \frac{e^{-T/\alpha}}{\sqrt{B}}\rho\cos\theta,\\
-\eta &\equiv \alpha\frac{e^{-T/\alpha}}{\sqrt{B}},
\end{aligned}
\tag{5.133}
$$

and the resulting line element is

$$
ds^2 = -BdT^2 + \frac{d\rho^2}{B} + \rho^2 d\theta^2 + \rho^2\sin^2\theta d\phi^2 \ , \tag{5.134}
$$

with $B = 1 - \frac{\rho^2}{\alpha^2}$. For an observer situated at the origin, $B = 0$ is a cosmological horizon which marks the boundary of his observable universe. We shall use this coordinate system for de Sitter to study the noise kernel near the horizon and compare its behavior with that found for the approximate noise kernel in Schwarzschild spacetime.

From the definitions given in Eq. (5.133), we transform Eq. (5.128) to the static coordinates using the relation

$$
N_{abc'd'}(x,x') = \frac{\partial x^A}{\partial x^a}\frac{\partial x^B}{\partial x^b}\frac{\partial x'^{C'}}{\partial x'^{c'}}\frac{\partial x'^{D'}}{\partial x'^{d'}}N_{ABC'D'}(x,x'), \tag{5.135}
$$

where we have used capital letters to represent indicies of the comoving coordinates and lower case to represent indicies of the static coordinates.

To avoid coordinate singularities, we express the noise kernel in terms of an orthonormal frame at each of the two points. We do this by introducing orthonormal basis vectors at each point which satisfy

$$
(e_{\hat{a}})^c(e_{\hat{b}})_c = \eta_{\hat{a}\hat{b}}, \tag{5.136}
$$

$$
(e^{\hat{a}})_c(e_{\hat{a}})_d = g_{cd}, \tag{5.137}
$$

where $\eta_{\hat{a}\hat{b}}$ is the Minkowski metric. The components of a vector may be written in the orthonormal basis as

$$
A_{\hat{a}} = (e_{\hat{a}})^a A_a. \tag{5.138}
$$

Similarly, the noise kernel in this basis is

$$
N_{\hat{a}\hat{b}\hat{c}'\hat{d}'}(x,x') = (e_{\hat{a}})^a(e_{\hat{b}})^b(e_{\hat{c}'})^{c'}(e_{\hat{d}'})^{d'}N_{abc'd'}(x,x'). \tag{5.139}
$$

For the static de Sitter coordinates, we choose basis vectors such that

$$
\begin{aligned}
(e_{\hat{T}})^T &= \sqrt{-g^{TT}}\\
(e_{\hat{\rho}})^\rho &= \sqrt{g^{\rho\rho}}\\
(e_{\hat{\theta}})^\theta &= \sqrt{g^{\theta\theta}}\\
(e_{\hat{\phi}})^\phi &= \sqrt{g^{\phi\phi}}.
\end{aligned}
\tag{5.140}
$$

All other components are zero.

The expressions resulting from this procedure are quite long. We present only two of them here (the rest can be obtained by consulting the authors of [477, 478]):

$$\begin{aligned} N_{\hat{T}\hat{T}\hat{T}'\hat{T}'}(x,x') &= (BB')^{-1} N_{TTT'T'}(x,x') \\ &= \frac{1}{12\pi^4 \left[\alpha^2\left(\sqrt{BB'}\,\tau - 2\right) + 2\rho\rho'\cos(\gamma)\right]^6} \\ &\quad \times \Big\{\alpha^4 \left[-12\sqrt{BB'}\,\tau + BB'\,(\tau^2+14)\right. \\ &\quad \left. -\,(2B+2B'-6)\,(\tau^2-1)\right] \\ &\quad +4\alpha^2\rho\rho'\cos(\gamma)\left(3\sqrt{BB'}\,\tau - 2\,(\tau^2-1)\right) \\ &\quad +2\rho^2\rho'^{\,2}\,(\tau^2-1)\cos(2\gamma)\Big\} \end{aligned} \tag{5.141}$$

$$\begin{aligned} N_{\hat{T}\hat{\rho}\hat{T}'\hat{\rho}'}(x,x') &= N_{T\rho T'\rho'}(x,x') \\ &= \frac{\alpha^2}{6\pi^4 \left[\alpha^2\left(\sqrt{BB'}\,\tau - 2\right) + 2\rho\rho'\cos(\gamma)\right]^6} \\ &\quad \times \Big\{\alpha^2\cos(\gamma)\left[-4\sqrt{BB'}\,\tau + BB'\,(\tau^2+4)\right. \\ &\quad \left. -\,(B+B'-4)\,(\tau^2-2)\right] \\ &\quad +\rho\rho'(\cos(2\gamma)+3)\left(\sqrt{BB'}\,\tau - \tau^2 + 2\right)\Big\}, \end{aligned} \tag{5.142}$$

where

$$\tau \equiv 2\cosh(\Delta T/\alpha), \tag{5.143}$$

$$\cos\gamma \equiv \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\phi-\phi'), \tag{5.144}$$

and $B' = 1-\rho'^{\,2}/\alpha^2$. The expressions shown in Eqs. (5.141) and (5.142) are valid only for non-null separations of the points.[6]

Behavior near the horizon When either of the two points approaches the cosmological horizon, we see that $\rho \to \alpha$ and $B \to 0$ (or $\rho' \to \alpha$ and $B' \to 0$, respectively). However, note that at the horizon T is a null coordinate and is infinite. Therefore, ΔT is ill-defined when one or both points are on the horizon. Nonetheless, if we fix the value of ΔT, then inspection of Eq. (5.99) shows that both of the components displayed therein remain bounded when either point is

[6] Note that we have neglected the delta function contributions in the above expressions. According to the authors of [477, 478] this is due to the fact that the initial computations of the delta function contributions each contained approximately 2,700 terms and they have not yet found a means by which to simplify the expressions into displayable forms.

arbitrarily close to the horizon, so long as the points are spacelike or timelike separated.[7]

For example, if we take ρ' to be arbitrarily close to the horizon, then

$$N_{\hat{T}\hat{T}\hat{T}'\hat{T}'}(x,x') \approx -\frac{\tau^2-1}{384\pi^4\alpha^4(\alpha-\rho\cos(\gamma))^6} \times \left[\alpha^2(B-3)+4\alpha\rho\cos(\gamma)-\rho^2\cos(2\gamma)\right] \tag{5.145}$$

$$N_{\hat{T}\hat{\rho}\hat{T}'\hat{\rho}'}(x,x') \approx -\frac{\tau^2-2}{384\pi^4\alpha^3(\alpha-\rho\cos(\gamma))^6} \times \left[(B-4)\alpha\cos(\gamma)+\rho(\cos(2\gamma)+3)\right]. \tag{5.146}$$

When $\rho=\rho'$ and both points are near the horizon, we have

$$N_{\hat{T}\hat{T}\hat{T}'\hat{T}'}(x,x') \approx \frac{\tau^2-1}{3072\pi^4\alpha^8}\csc^8\left(\frac{\gamma}{2}\right). \tag{5.147}$$

and

$$N_{\hat{T}\hat{\rho}\hat{T}'\hat{\rho}'} \approx -\frac{\tau^2-2}{3072\pi^4\alpha^8}\csc^8\left(\frac{\gamma}{2}\right). \tag{5.148}$$

Therefore, the noise kernel is bounded when either or both of the points are near the horizon so long as the separation in the T coordinate is fixed and either $\gamma\neq 0$, $\rho\neq\rho'$, or both. For $\gamma=0$ and $\rho=\rho'$, the above expression is not bounded as the two points approach the horizon. This is expected since, on the cosmological horizon, T is a null coordinate and ΔT is ill-defined.

[7] We find this behavior to be true for all components of the noise kernel when expressed in the orthonormal frame.

Part II

Infrared Behavior, 2PI, $1/N$, Backreaction and Semiclassical Gravity

6

Infrared Behavior of Interacting Quantum Fields

6.1 Overview: Relevance, Issues and Approaches

In Chapters 2, 4, and 5 we have discussed the ultraviolet (UV) divergences arising from the contributions of high frequency or short wavelength modes in quantum field theories (QFT) in curved spacetimes (CST) and introduced three methods – dimensional, zeta-function and point-separation – to regularize them. In this chapter we focus on the opposite domain, namely, infrared (IR) divergences arising from the long wavelength modes in the spectrum of the fluctuation wave operator. Traditionally, in flat space QFT, IR divergences are often fixed in a rather simplistic way, by introducing a low frequency cut-off in the integration over modes, and terms containing such cut-offs are cast away, with the justification that they represent processes at far distances beyond the realm of concern to the problem under study. This kind of argument obviously fails for field effects even in flat space with boundaries. There is nontrivial physics associated with the boundary and different boundary conditions imposed on the field also make a difference. (This is an understatement in the era when AdS–CFT correspondences show up in many areas of physics.) In the same vein, in curved spacetimes, the large-scale structure of spacetime is expected to play a nontrivial role in the IR physics, not to mention the need for covariance, where, like the regularization of UV divergences, cut-off function methods are often untenable. The curvature and topology of spacetime, and how the field is coupled to the spacetime, become important factors in the investigation of IR physics.

There was significant progress in the 70s with the invention and application of quantum field theory techniques to the treatment of infrared problems, notably in the works of Coleman, Cornwall, Jackiw, Politzer, and Tomboulis [492, 493, 494, 495] in the two-particle-irreducible effective action, the large N expansion of 'tHooft [496] and the decoupling theorem of Appelquist and Carazzone [497]. A well-known example is the treatment of divergences of QFT at high temperatures by Dolan and Jackiw [429] and Bernard [72].

Infrared physics is at the focus of critical phenomena, an important area of physical, biological, and social sciences containing many profound ideas. Near the critical points the correlation functions often show divergences. To understand how a system behaves physically near a critical point, one needs to calculate the critical exponents in order to place the system in the correct universality class. Significant progress took place also in the 70s, with the invention and application of renormalization group techniques in the seminal works of Wilson, Fisher, Kadanoff, Ma, Brezin, Jinn-Justin and many others, the description of which can be found in excellent monographs, e.g., [498]. Of particular relevance to the themes of this chapter are the works of Mermin and Wagner [499], Hohenberg [500], Fisher [501], Brezin and Jinn-Justin [502], to name just a few. For QFT in CST understanding phase transitions in the early universe is in fact an important driving force toward detailed studies of the infrared behavior of quantum fields in curved spacetime, especially after the advent of inflationary cosmology.

Phase Transitions in the Early Universe

The very early universe can at certain stages become vacuum energy dominated and undergo inflationary expansions. The vacuum energy source can be from Grand Unification interaction (GUT scale $M_{GUT} \sim 10^{14}$ GeV), quantum gravitational (QG) interactions (Planck scale, $M_P \sim 10^{19}$ GeV) or from interactions in some intermediate energy scales (supersymmetry, Kaluza-Klein, etc.). Guth [118] first pointed out how an inflationary stage in the early history can help to resolve some outstanding problems (horizon, flatness, fluctuations) in cosmology. The model he used, now called 'old' inflation, to illustrate these ideas is by tunneling, which may fail the mission because of inhomogeneities generated by infrequent bubble collisions. The model proposed by Albrecht and Steinhardt [119] and Linde [120], now called 'new' inflation, uses a flat Coleman–Weinberg potential [343] associated with a massless field, whereby inflation arises as 'slow-roll'. As it avoids the many bubble collision challenge – our universe is anchored in only one bubble – it offers a nice 'graceful exit' scheme, except for the fact that the self-interaction coupling constant of the field is too weak to be realistically implementable. Many inflationary models have been proposed in the last three decades since then, aiming at specific desirable features to better match with observations. Suffice it to say here that the study of symmetry breaking in curved spacetime and phase transitions in the early universe – old inflation being an example of first order, and new inflation, of second order – are essential for the understanding of early universe cosmology.

From a physical point of view one can also appreciate why we need to gain a better understanding of IR physics in de Sitter space. Take the simple case of eternal inflation, the scale factor $a(t)$ in the flat Friedmann–Lemaître–Robertson–Walker (FLRW) coordinatization of de Sitter space goes like $a(t) = e^{Ht}$ with t the cosmic time and $H = \dot{a}/a$ the Hubble expansion rate, also defining the de

Sitter horizon. What happens to the field under this inflationary expansion is that the tower of higher (frequency) modes collapses rapidly to the lowest mode. The heaping-on of these long wavelength modes generates IR divergence. This is both a mathematical problem and a physical issue: do we have the needed field-theoretical techniques to properly handle this IR divergence? As we shall soon see, the dominant IR contribution from these modes at superhorizon scale is a nonperturbative effect which defies loop expansion calculations. Physically, the late time behavior of quantum fields under inflation directly affects the physical processes in the post-inflationary eras and determines the cosmological parameters which enter into the later stages where observational data are taken. For example, are there secular effects of quantum fluctuations which persist till late times, and will they backreact on the cosmological constant and attenuate its value? Will the IR behavior of quantum fields render the de Sitter universe unstable? These are important questions to ask, but only a valid calculation based on sound techniques can provide trustworthy physical answers. We can see that this is no longer an academic issue; it begs for actual verifiable solutions.

What then are the relevant issues involved in, and the best methods useful for, the study of symmetry breaking in curved spacetimes, as applied to phase transitions in the early universe, and in particular, the infrared problem in dS space for inflationary cosmology?

Effective Potential for Symmetry Behavior in Curved Spacetimes

In the discussion of symmetry behavior it is important to know where and when minimum free energy states (local and global minimum) exist and how the system chooses between them – from energy and entropy considerations – and evolves from one to another – via nucleation for first-order, or spinodal decomposition for second-order, transitions. For these purposes the approach based on the effective action proves very useful. The effective action $\Gamma(\bar{\phi})$ gives the free energy density of the system as a functional of the order-parameter field $\bar{\phi}$. From $\Gamma(\bar{\phi})$ one can compute field-theoretical and thermodynamic quantities of interest in the system. Since contributions from the quantum and thermal fluctuations are built in, one does not need to solve separately the equations of motion for the background field and the fluctuation field as in the effective-mass approach often used in stability analysis, which may not satisfy the self-consistency condition in the iterations.

To focus on the effect of field coupling, curvature and topology on symmetry breaking, we can restrict our attention to static spacetimes like the Einstein universe or Euclidean spacetimes where the order parameter field is a constant. For constant background metrics and background fields, one can work with the effective potential $V(\bar{\phi}) = -(Vol)^{-1}\Gamma(\phi) \mid_{\phi=\bar{\phi}}$, where the spacetime four-volume Vol is factored out from the effective action. These effects were studied methodically by a few forerunners in the 80s, examining the symmetry behavior

of interacting quantum fields, specifically, in the Einstein universe, the Taub universe and the Euclidean de Sitter universe. See the Ph.D. thesis work of O'Connor and of Shen [461, 503]. For a taste of what this entails, recall in Chapter 4, after we had derived the effective potential by the zeta function method we gave a brief discussion of how to obtain the IR behavior of a ϕ^4 theory in the Einstein universe. In contrast to identifying the UV divergences in the small curvature limit, for IR behavior one should consider the high curvature regime (corresponding to small conformally related effective mass $M_{\rm I}^2 a^2 \ll 1$, with $M_{\rm I}^2$ defined in (2.128)).

The effect of the dynamics of spacetime on the symmetry behavior of a quantum field is more difficult since the background (order parameter) field $\bar{\phi}$ depends on time. One needs to use the effective action, not the effective potential, and include dynamical quantum processes such as particle production, discussed in Chapters 2 and 3, which can be significant if the phase transition occurs near the Planck scale. We will see some aspects of this in our discussion of QFT in Lorentzian dS.

Phase transitions in the de Sitter (dS) universe have been studied by many authors. The effective potential for $\lambda\Phi^4$, scalar quantum electrodynamics (SQED) and SU(5) gauge fields were worked out earlier in the 80s and 90s, but the understanding of the infrared behavior of these theories was incomplete. Here we will focus on the IR problem exclusively.

Three Veins of Developments and Four Parts in This Chapter

This chapter consists of four main parts. The first part explains in general terms how to identify the dominant contributions to the infrared (IR) behavior of interacting quantum fields in a compact space. If a band gap exists in the spectrum of the wave equation fluctuations operator one can identify the zero-mode or the lowest band as giving the dominant contributions to the IR behavior. Hu and O'Connor (H&O) [504] first pointed this out for quantum fields in curved spacetimes and derived the $\sqrt{\lambda}$ behavior of the (dynamically generated) effective mass (6.44) from the curvature of the effective potential at a minimum energy state for an $O(N)\lambda\Phi^4$ self-interacting scalar field in Euclidean de Sitter space in the leading-order large N approximation. This is a nonpertubative result for the leading IR behavior which cannot be obtained from perturbation theories via loop expansion. From the IR behavior of scalar quantum electrodynamics (SQED) they also drew implications for Planck scale inflations [462]. With this, H&O discussed the ensuing dimensional reduction in the infrared domain and introduced the notion of effective IR dimension (EIRD). They also posited that the concept of finite size effect in the description of critical phenomena of condensed matter systems captures the essence of the problem of symmetry breaking in curved spacetimes. We shall follow their exposition here, with examples in some familiar curved spacetimes. The second part of this chapter introduces some quantum field theory techniques for treating IR problems. This includes

the closed time path (CTP) or 'in-in' method developed in Chapters 3 and the two-particle-irreducible (2PI) effective action under large N expansion. We shall use the $O(N)\lambda\Phi^4$ field theory as an example to introduce these techniques. A more formal and systematic presentation of these advanced techniques will be presented in the next chapter.

Having shown a way to treat the IR behavior in Euclidean dS (EdS) – a 4-sphere S^4 in $5D$ Euclidean space E^5 – we turn in the third part to the Lorentzian field theories in dS (LdS) – a spatially flat ($k = 0$) FLRW coordinatization of dS, sometimes called the 'Poincaré patch' – with a dynamical description which is more often used in the discussion of physical processes in inflationary cosmology. But nonperturbative methods in Lorentzian quantum field theories are not so easily accessible (we will mention one method at the end, the nonperturbative renormalization group method). One has to rely on resummation techniques order by order, often resulting in rather complicated expressions where the physics is not so easily intrepretable (we will give an example of how this can be done with some clarity). Here, in the Lorentzian framework, the stochastic approach of Starobinsky and Yokayama (S&Y) [505] comes as a relief. Based on the stochastic inflationary model (SdS) of Starobinsky [506], the long wavelength sector is treated not as quantum fields but as classical stochastic variables – they become classical after decoherence and they become stochastic when driven by a noise source representing the effects of the short wavelength modes. The dominant nonperturbative IR behavior manifests in the correlation functions of stochastic field variables which can be evaluated by distributional averages rather than expectation values as for quantum field operators.

It is of interest to see how the symmetry behavior calculated in a Lorentzian (FLRW coordinate) de Sitter space compares with that calculated in the Euclidean coordinates. The time scale associated with exponential expansion $1/H$ maps to the length scale of the longest wavelength, namely, the radius of S^4 in the Euclidean coordinatization of de Sitter universe. In this Lorenzian QFT rendition the IR physics arises from what H&O called the *dynamical finite size effect* [507]. The 'finite size' in the Lorentzian picture particular to eternal inflation is dynamically generated: under continuous exponential expansion an observer would see physical processes as if they were static ('frozen' in time). The zero-mode in Euclidean dS corresponds to the longest wavelength mode in the Lorentzian dS. The zero-mode dominance of IR behavior in EdS has this correspondence in LdS: It is the late time behavior of these superhorizon modes which encode the IR behavior of the system. In this light we can see the Euclidean picture drawn by H&O and the Lorenzian stochastic picture of S&Y are equivalent.

These two key ideas in treating IR behavior of quantum fields in curved spacetimes, the Euclidean zero-mode proposed in the 80s and the stochastic approach proposed in the 90s lay dormant until much later, in 2005, when Tsamis and Woodard [508] discovered the potency of S&Y stochastic approach

for addressing IR issues, and in 2010, when Rajaraman [463] rediscovered the zero-mode dominance of IR in Euclidean dS, as known and used earlier in H&O's work. From 2008 onward we see an upsurge of activities on IR in dS issues. (For a review of this topic up to 2010, see [509].) Much of the developments in the IR in dS problem since around 2005 can be sorted out following these three main veins – the Euclidean, the Lorentzian QFT and the stochastic approach. In the second subsection we will provide a guide to the important recent work along these three main lines of development and the connections built around them, where further proofs of the equivalence of the IR results from the Euclidean and Lorentzian QFT, and stochastic approaches are provided.

In the last part of this chapter we highlight three topics, two of which are from very recent work (in 2018): (1) a resummation method [510] for taking into account the higher modes' influence in Lorentzian dS QFT over and beyond the zero mode dominance in Euclidean dS; (2) the nonperturbative renormalization group method [511, 512] which demonstrates curvature-induced effects, effective IR dimensional reduction, several of the salient features covered in this chapter, and through it, a proof of the stability of dS against infrared quantum fluctuations [513]; (3) IR behavior of gravitons and the gauge issue, graviton loop contributions and nonlinear effects.

We begin with a short calculation showing how the IR divergences appear in a free massless minimally coupled scalar field in the de Sitter spacetime.

6.1.1 Infrared Divergences in Massless Minimally Coupled Free Scalar Field

To see where the IR divergence appears we follow the exposition of Ford and Parker [346]. Consider a massless minimally coupled scalar field $\Phi(\mathbf{x}, t)$ obeying the Klein–Gordon equation

$$\Box\Phi(\mathbf{x}, t) = 0 \tag{6.1}$$

in a spatially-flat FLRW spacetime with metric

$$ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2. \tag{6.2}$$

Decomposing $\Phi(\mathbf{x}, t)$ into the spatial and temporal parts,

$$\Phi(\vec{x}, \tau) = \int \frac{d^3k}{(2\pi)^{3/2}} \left[A_{\mathbf{k}}\, e^{i\,\mathbf{k}\cdot\mathbf{x}} \phi_k(\tau) + \text{H.C.}\right] \tag{6.3}$$

(Here $k = |\mathbf{k}|$ and H.C. denotes the Hermitian conjugate of the preceding term.) In time $\tau \equiv \int^t dt'/a^3(t')$ chosen to simplify the form of the wave equation, the mode amplitude function $\phi_k(\tau)$ obeys

$$\frac{d^2\phi_k(\tau)}{d\tau^2} + k^2 a^4 \phi_k(\tau) = 0. \tag{6.4}$$

Ford and Parker treated two classes of evolution: (i) power law $a(t) = \sigma t^c$ and (ii) de Sitter in the $k = 0$ RW coordinate $a(t) = \sigma e^{Ht}$.

(i) The power law family is a solution of the Einstein equation with equation of state $p = \gamma\rho$ with $\gamma = (2 - 3c)/3c$. The general solution of (6.4) is given by the Hankel functions

$$\phi_k(\tau) = c_1 \tau^{\frac{1}{2}} H_\nu^{(1)}(k|b|^{-1} a_0^2 \tau^b) + c_2 \tau^{\frac{1}{2}} H_\nu^{(2)}(k|b|^{-1} a_0^2 \tau^b), \tag{6.5}$$

where $c_1(k)$, $c_2(k)$ are complex numbers and $b = (1 - c)/(1 - 3c)$, $\nu = (2|b|)^{-1}$.

Now consider a typical class of correlation functions

$$\frac{\langle 0|\Phi(x)\Phi(x')|0\rangle}{\langle 0|0\rangle}, \tag{6.6}$$

where $|0\rangle$ is the state defined by $A_{\mathbf{k}}|0\rangle = 0$ for all $\mathbf{k}$. From (6.3) one gets

$$\frac{\langle 0|\Phi(x)\Phi(x')|0\rangle}{\langle 0|0\rangle} = \int \frac{d^3k}{(2\pi)^3}\, e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \phi_k(\tau)\phi_k^*(\tau'). \tag{6.7}$$

From the general solution for $\phi_k(\tau)$ in (6.5), using the asymptotic expansion of the Hankel functions for $z \to 0$

$$H_\nu^{(1)}(z) \approx -H_\nu^{(2)}(z) \approx -\frac{i}{\pi}\Gamma(\nu)\left(\frac{z}{2}\right)^{-\nu}, \tag{6.8}$$

the integrand of (6.7) is seen to carry the factor $|c_1 - c_2|^2 k^{-2\nu}$. Thus if $|c_1 - c_2|$ is non-zero as $k \to 0$ the two-point function will diverge if $\nu \geq 3/2$, corresponding to the range of $c \geq 2/3$ for power law expansion. For c in this range, the two-point function will diverge if $c_1(k)$ and $c_2(k)$ are regular near $k = 0$, otherwise the Wronskian condition which ensures the unitarity of the field evolution will not be satisfied. Recall that each possible choice of $c_1(k)$ and $c_2(k)$ defines a choice of vacuum state.

However, as noted by Ford and Parker, it is possible to find choices of $c_1(k)$ and $c_2(k)$ which are singular at $k = 0$, and for which $|c_1 - c_2| \to 0$ as $k \to 0$. Thus, in contrast to the situation in flat space, here the well-behaved states correspond to singular coefficients, whereas any regular pair of coefficients yields a state with infrared divergence.

In examining the energy density they found that the infrared behavior in states with c_1, c_2 regular in a metric with power law expansion ($\frac{2}{3} \leq c < 1$, $1 < c \leq 2$) is such as to prevent these metrics from being self-consistent solutions of the Einstein equation, and thus should be excluded.

(ii) For exponential expansion the mode amplitude functions have the form (6.5) with $\nu = 3/2$ and $b = 1/3$. Again it is seen that choices of $c_1(k)$ and $c_2(k)$ which are regular lead to divergences in the two point functions.

The power law and the exponential cases can be considered together. Using the Hubble constant H and the deceleration parameter ϵ (the 'slow-roll' parameter in inflationary cosmology terminology)

$$H(t) \equiv \frac{\dot{a}}{a} \ , \ \ \epsilon(t) \equiv -\frac{\dot{H}}{H^2}, \tag{6.9}$$

the FLRW power law $a(t) = \sigma t^c$ and the eternal inflation $a(t) = \sigma e^{Ht}$ solutions correspond to $\epsilon = 1/c$ and 0 respectively. The mode amplitude function for D spacetime dimension in cosmic time t is given by [514]

$$\phi_k(t) = a^{-(\frac{D-1}{2})} \sqrt{\frac{\pi}{4(1-\epsilon)H}} \, H^{(1)}_{\nu}\left(\frac{-k}{(1-\epsilon)Ha}\right) \qquad \text{with} \qquad \nu = \frac{D-1-\epsilon}{2(1-\epsilon)} \ . \tag{6.10}$$

We note that for eternal inflation $\epsilon = 0$ the mode function for *massive* scalar field in open de Sitter coordinates has $\nu = \sqrt{(\frac{D-1}{2})^2 - \frac{M^2}{H^2}}$.[1]

From the small k form of the mode functions (6.10), one can see that this two-point function

$$\phi_k(t)\phi^*_k(t') \to \frac{4^{|\nu|}(1-\epsilon)^{|2\nu|}\Gamma^2(|\nu|)}{4\pi(1-\epsilon)\sqrt{Ha^{D-1}H'a'^{D-1}}} \times \left(\frac{HaH'a'}{k^2}\right)^{|\nu|} \Big[1 + O(k^2)\Big] \tag{6.11}$$

suffers from infrared divergences throughout the range from $\epsilon = 0$ (de Sitter) to $\epsilon = \frac{3}{2}, \nu = \frac{2}{3}$ (pressure-free dust).

The IR problem in inflationary cosmology is more aggravated owing to the rapid red-shifting of the spectrum towards the IR. Tsamis and Woodard [517] showed that infrared logarithm appears in pure quantum gravity, in gravity + fermions, in full scalar-driven inflation, in the scalar sector of scalar-driven inflation, in ϕ^4 theory, in scalar QED and in Yukawa theory. No matter how small the coupling constant in the theory is, the continued growth of the scale factor $a(t)$ over a prolonged period of inflation must eventually result in this 1-loop correction becoming order unity. Because higher-loop corrections also become order unity at about the same time, perturbation theory breaks down and that one must employ a nonperturbative technique to evolve further. The eventual breakdown of perturbation theory is a general feature of quantum field theories that exhibit infrared logarithms.

Thus, the question is, what are we to make of this IR divergence in de Sitter universe? Is it a nuisance to be rid of by, say, introducing a low frequency cut-off (which will artificially introduce a zero mode even if there isn't one), as was customarily done in earlier times, or does it contain some real physical meaning? Unlike in flat spacetime the IR divergences in QFT in curved spacetime are more complicated and they reflect real physical conditions. The IR behavior of

[1] The closed dS coordinate mode functions [515] can be derived by analytically continuing from Euclidean de Sitter [516]

quantum fields is determined by and also reflects the large-scale structure of spacetime. This is the main theme of this chapter.

6.1.2 Issues, Approaches and Methodology

Key Issues

From this simple example already two of the many factors involved in the infrared problem are apparent: (1) choice of the vacuum, (2) the dynamics. We may add a third factor, (3) coordinate representations. The commonly studied ones in de Sitter space are (a) the global (closed) de Sitter, in the closed ($k = 1$) FLRW coordinate, which covers the whole space, (b) the $a(t) = e^{Ht}$ in the flat ($k = 0$) FLRW coordinate (Poincaré patch) which covers half of the space in the conformal diagram, and (c) S^4 in the Euclidean coordinates of E^5. We may at times refer to case (b) as Lorentzian dS (LdS) and case (c) Euclidean dS (EdS). The Euclidean vacuum has been shown to be equivalent to the Bunch–Davies vacuum which is de Sitter invariant. (For more general cases, see [518, 519, 520].) The LdS (case b) has attracted greater attention after the advent of inflationary cosmology in the 80s. We shall discuss IR physics in both cases (b) and (c) below.

In view of the rapid surge of activities in recent years on this topic a short guide to the literature may be helpful to readers who prefer to read the original papers but are bewildered by the volume of publications. As a background, Ford and Parker [346] spelled out the infrared divergence issue for scalar fields in dS. Vilenkin and Ford [521] and Vilenkin [522] used the superdaisy diagrams to capture the leading IR contributions, a method detailed earlier by Dolan and Jackiw [429] for treating the IR problem in finite temperature QFT. Historically and still relevant, three groups of papers are noteworthy, here divided according to the approaches taken: (A1) Hu and O'Connor [462, 504] (H&O), treating the self-interacting massless minimally coupled scalar field in Euclidean dS, first pointed out that the IR behavior is dominated by the zero-mode. They also suggested the appropriate (B1) quantum field theoretical techniques for treating infrared behavior of QFT in CST, namely, the two-particle-irreducible (2PI) effective action and the large N expansion.

In a different approach (C1) in the context of stochastic inflation proposed by Starobinsky [506] where a Langevin equation with the short wavelength modes acting as a white noise drives the long wavelength modes treated classically, Starobinsky and Yokoyama (S&Y) [505] calculated the correlation function in flat ($k = 0$ FLRW)-dS arising from the longest wavelength modes at late times, and produced correctly the leading infrared logarithms of scalar potential models at arbitrary loop order. Tsamis and Woodard [508] who had long been studying two-loop perturbative quantum gravity effects in curved space exclaimed that the S&Y results 'miraculously' capture the leading-order graviton loop contributions

in the IR. This discovery opened the dialogue between the S&Y stochastic approach and its relation to the QFT methods.

Approaches and Methodologies

A. From zero-mode IR dominance in Euclidean de Sitter to nonperturbative renormalization group (NPRG): $A1 \to A2 \to A3 \to A4 \to A5$
B. Quantum field theory techniques in Lorentian de Sitter: 2PI effective action, Schwinger–Dyson equation: $B1 \to B2 \to B3 \to B4 \to B5$
C. Stochastic approach and relation to field theory methods: $C1 \to C2 \to C3 \to C4 \to C5$

For the Euclidean approach, in 2010 (A2) Rajaraman [463] called the attention to the dominance of the zero mode in determining the IR behavior in the massless minimally coupled ϕ^4 theory, an observation made earlier in Hu and O'Connor [504] who used this fact to define the effective infrared dimension (EIRD) of field theories in curved spacetimes. (A3) Beneke and Moch 2013 [523] showed that solutions based on truncations of the loop expansion can at best be approximate. With the proper exact treatment of the zero mode, the reorganized perturbative expansion is free from infrared divergences. They demonstrated that for the Schwinger–Dyson equations derived from a two-particle-irreducible effective action, the solutions for the two-point functions to leading IR order take the form of free propagators with a dynamical mass. They also showed that the functional integrals over just the zero mode coincide with the integrals over the probability distribution functions in stochastic inflation. (A4) Beyond the leading IR approximation, two-point functions for massive scalar fields on Euclidean de Sitter space to all orders in perturbation theory was investigated by Marolf and Morrison [524, 525], where it is found that these are well defined and that, in particular, the field correlations exhibit an exponentially decaying behavior for large separations. Hollands [526] derived the correlation functions of massless $\lambda\phi^{2n}$ fields in a de Sitter-invariant state which holds to arbitrary orders in perturbation theory by an analytic continuation from the corresponding objects on the sphere. He established that generic correlation functions cannot grow more than polynomially in proper time for large time-like separations of the points. Also noteworthy are the proof of the equivalence between the Euclidean and the 'in-in' formalisms in de Sitter QFT [527] and the so-called quantum no-hair theorem [528]. While addressing the residual gauge issue Tanaka and Urakawa [529] showed that when the Euclidean vacuum is chosen as the initial state, IR regularity and the absence of secular growth are ensured. Note, however, the warnings issued by Miao, Mora, Tamis and Woodard [514] in translating Euclidean results back to the Lorentian spacetime via analytic continuation. (A5) Guillieux and Serreau [530] investigated scalar field theories in de Sitter space by means of nonperturbative renormalization group techniques [531, 532]. They computed the functional flow equation for the effective potential of $O(N)$ theories

in the local potential approximation and studied the onset of curvature-induced effects as quantum fluctuations are progressively integrated out from subhorizon to superhorizon scales. This results in a dimensional reduction of the original action to an effective zero-dimensional Euclidean theory. They showed that the latter is equivalent both to the late-time equilibrium state of the stochastic approach of Starobinsky and Yokoyama [505] and to the effective theory for the zero mode on Euclidean de Sitter space. Their investigation of dimensional reduction, symmetry restoration and dynamical mass generation form a closure to these issues raised and explored by Hu and O'Connor [462, 504] and seals in its relation with the stochastic approach of Starobinsky and Yokoyama [506, 505]. We will describe this in a later section

In terms of field theory methods and IR issues, we mentioned the extensive work of Tsamis and Woodard and co-workers. (B2) Serreau [533] studied the quantum theory of an $O(N)$ scalar field on de Sitter geometry at leading order in a nonperturbative $1/N$ expansion which resums the infinite series of the superdaisy loop diagrams. He obtained the de Sitter symmetric solutions of the corresponding properly renormalized dynamical field equations and computed the complete effective potential. The self-interacting field acquires a strictly positive square mass which screens potential infrared divergences. Serreau and Parentani [534] computed the four-point vertex function in the deep infrared regime for the $O(N)$ scalar field theory with quartic self-coupling in de Sitter space in the large-N limit. They found that resummation of an infinite series of perturbative (bubble) diagrams leads to a modified power law which is analogous to the generation of an anomalous dimension in critical phenomena. The high momentum (subhorizon) modes influence the dynamics of infrared (superhorizon) modes only through a constant renormalization factor. Their calculation provides an explicit example of effective decoupling between high and low energy physics in an expanding space-time. (B3) Gautier and Serreau [535] showed that the results from the stochastic method are also obtainable by solving the Schwinger–Dyson equations. While it was known earlier that the two methods yield the same results for the leading IR-enhanced corrections from the local self-energy (seagull) diagram, from a leading-order large N expansion [533], these authors extended the agreement to nonlocal self-energy diagram. Gautier and Serreau [511] computed the self-energy of an $O(N)$ symmetric theory at next-to-leading order in a $1/N$ expansion in the regime of superhorizon momenta, and obtained an exact analytical solution of the corresponding Dyson–Schwinger equations for the two-point correlator. This amounts to resumming the infinite series of nonlocal self-energy insertions, which typically generate spurious infrared and/or secular divergences. The potentially large de Sitter logarithms resum into well-behaved power laws from which one can extract the field strength and mass renormalization.

(B4) Stability of de Sitter spacetime against infrared quantum scalar field fluctuations was shown recently by Moreau and Serreau [513, 536]. They studied

the backreaction of superhorizon fluctuations of a light quantum scalar field on a classical de Sitter geometry by means of the Wilsonian renormalization group. (B5) Finally, relating Euclidean with Lorentzian dS QFT, López Nacir, Mazzitelli and Trombetta [537] considered an $O(N)\lambda\phi^4$ scalar field in D-dimensional Euclidean de Sitter space and computed the two-point functions including corrections from the interaction of the zero mode with the higher modes. In [510] these authors computed the long wavelength limit of the two-point functions at next-to-leading order in $1/N$ and at leading order in λ which involves a further resummation of Feynman diagrams needed when the two-point functions are analytically continued. They prove that, after this extra resummation, up to $O(\sqrt{\lambda}, 1/N)$, including the contribution of the bubble chains diagrams changes the behavior of the two-point functions, which now tend to zero (instead of a constant) as $r \to \infty$.

For the connection between the stochastic approach and QFT approach (C2) Riotto and Sloth [538] investigated the infrared behavior of a $O(N)\phi^4$ theory in flat ($k = 0$ RW)$-$dS by solving the equation for the correlation functions $C^{++}(x, x)$ (where the superscripts are the CTP indices: $+$ for forward time and $-$ for backward time branches) and the gap equation derived by Ramsey and Hu [244] from a closed-time-path (CTP) 2PI effective action under a large N approximation. They first evaluated the correlation functions from Starobinsky and Yokoyama's Fokker–Planck equation, supposedly a good approximation for the late time behavior, insert this into the gap equation and solve for $C^{++}(x, x')$, obtaining a simple analytic form for it. (C3) Tsamis and Woodard [517] generalized Starobinsky's techniques to scalar models which involve fields that do not produce infrared logarithms such as fermions and photons and the derivative couplings of quantum gravity. (C4) The authors of [539, 540] provided an explicit linkage between these two approaches. In particular, from a QFT calculation truncated at leading IR order, [540] fully reproduces, at all loop order, the stochastic correlation functions found by Starobinsky and Yokohama, thus proving the proposed conjecture. Finally, (C5) Venin and Starobinsky [541] derived non-perturbative analytical expressions for all correlation functions of scalar perturbations in single-field, slow-roll inflation. They, and earlier the authors of [542, 543], alerted enthusiasts of the stochastic approach to take note that different choices of time variable in the Langevin equation account for different stochastic processes and the number of e-foldings $\alpha = \ln a$ (they call it $\mathcal{N}$) is the correct time to use for stochastic processes which matched with results from QFT.

6.2 Euclidean Zero-Mode, EIRD, 2PI Effective Action

In Chapter 4 we introduced the N-component self-interacting scalar field Φ^a ($a = 1, \ldots, N$) on an Euclidean manifold of dimension D and analyzed the spectrum of the fluctuation field wave operator. Here we show that in cases where the spectrum has an identifiable zero mode, it gives the dominant contribution to

the infrared behavior of the system. We then introduce the notion of effective infrared dimension (EIRD) to show dimensional reduction in the infrared limit and explain the physics as corresponding to finite size effect in condensed matter physics. Following this we provide a more rigorous analysis in terms of the two-particle-irreducible (2PI) effective action under a large N expansion of interacting $O(N)$ fields in curved spacetime. Our presentation here follows [504].

6.2.1 Euclidean Zero Mode: Effective IR Dimension

To identify the dominant source of infrared contributions, let us consider cases where the eigenvalues of the fluctuation operator take on a band structure. By band structure we mean that the eigenvalues occur in continua with each continuum having a higher lowest eigenvalue than the previous one. This is true for fields on spacetimes with compact sections or for fields with discrete spectrum. The procedure is to expand the fields in terms of the band eigenfunctions and convert the functional integral over the fields to an integral over the amplitudes of the individual modes. When the lowest mode is massless it will give the dominant contribution to the effective action. The low energy behavior corresponds to a lower-dimensional system.

Decoupling of the Higher Modes (or Bands)

On a manifold with topology $R^c \times B^b$ where B is compact, consider quantum fields where the fluctuation operator A in (4.62) has the general form of a direct sum of operators D and B

$$A^{ab}(x,y) = D^{ab}(x) + B^{ab}(y) \tag{6.12}$$

with coordinates x on R^c and y on B^b. Assume that the eigenvalues ω_n associated with the eigenfunctions $\psi_n(y)$ of B^{ab} are discrete:

$$B^{ab}\psi_n(y) = \omega_n^{ab}\psi_n(y). \tag{6.13}$$

Decomposing the field $\varphi^a(x,y)$ in terms of $\psi_n(y)$

$$\varphi^a(x,y) = \sum_n \varphi_n^a(x)\psi_n(y), \tag{6.14}$$

one obtains for the quadratic part of the action Eq. (4.59) for the $O(N)$ $\lambda\phi^4$ theory,

$$\frac{1}{2}\int d^cx d^by\ \varphi^a A^{ab}\varphi^b = \frac{1}{2}\int d^cx\ \left[\varphi_n^a f_{nm} D^{ab}\varphi_m^b + \omega_n^{ab} f_{nm}\varphi_n^a\varphi_m^b\right], \tag{6.15}$$

where $f_{nm} = \int d^by\ \psi_n(y)\psi_m(y)$. When φ_n are properly normalized $f_{nm} = \delta_{nm}$ (we will make such a choice here) the resulting theory in terms of the new fields φ_n^a will involve massive fields with masses determined by the eigenvalue matrix λ_{in}^{ab} of the operators $-\Box_{\rm E} + M_i$ even if the fields in terms of the old variables

appeared massless. We will take the smallest eigenvalue to be given by $n = 0$ and assume that its only degeneracy is labeled by the indices a and b. Assume also that the operator D^{ab} is simply minus the Laplacian $\Box_c$ on R^c times δ^{ab}, the $n = 0$ mode is then governed by the action whose quadratic term is

$$\frac{1}{2}\int d^c x\,\left[\varphi_0^a(-\Box_c)\delta^{ab}\varphi_0^b + \omega_0^{ab} f_{00}\varphi_0^a\varphi_0^b\right]. \tag{6.16}$$

Thus this appears like an c-dimensional field with an apparent mass matrix ω_0^{ab}. For the case of an N component $\lambda\phi^4$ theory the action after this decomposition takes the form

$$\begin{aligned} S_{\rm E}[\bar{\phi}+\varphi] = \int d^c x\,\Big[&\frac{1}{2}\varphi_n^a(-\Box_c\delta^{ab} + \omega_n^{ab})\varphi_n^b \\ &+ \frac{\lambda}{6} g^a_{n\ell m}\varphi_\ell^b\varphi_m^b + \frac{\lambda}{4!} f_{kn\ell m}\varphi_k^a\varphi_n^a\varphi_\ell^b\varphi_m^b\Big], \end{aligned} \tag{6.17}$$

where $g^a_{n\ell m} = \int d^b y\;\bar{\phi}^a\psi_n\psi_\ell\psi_m$ and $f_{kn\ell m} = \int d^b y\;\psi_k\psi_n\psi_\ell\psi_m$. The effective action is now given by the functional integral

$$e^{-\Gamma_{\rm E}[\bar{\varphi}]} = \int [d\varphi_n^a]\; e^{-S_{\rm E}[\bar{\phi}+\varphi]}. \tag{6.18}$$

The interesting case occurs when the lowest eigenvalue approaches zero. At low energy the Appelquist–Carazzone [497] decoupling theorem assures us that with higher modes decoupled from the dynamics, the infrared behavior is governed by the lowest band. We are then left with an effective lower dimensional theory.

Correlation Length and Effective Infrared Dimension

The above result of dimensional reduction from a formal derivation of mode decoupling can be understood in a more physical way by using the concept of effective infrared dimensions (EIRD). By effective IR dimension we mean the dimension of space or spacetime wherein the system at low energy behaves effectively. One well-known example is the Kaluza–Klein theory of unification and the cosmology based on it. For example, start with an 11-dimensional spacetime with full diffeomorphism symmetry at above the Planck energy scale. After spontaneous compactification it reduces at energy below the Planck energy to the physical four-dimensional space with $GL(4, R)$ covariance and a seven-dimensional internal space with symmetry group containing the standard $SU_3 \times SU_2 \times U_1$ subgroups of strong and electroweak interactions. For observers today at very low energy the effective IR dimension of spacetime is four, even though the complete theory is eleven dimensional.

For curved-space symmetry breaking considerations, the EIRD which the system "feels" is governed by a parameter η which is the ratio of the correlation length Ξ and the scale length L of the background space $\eta \equiv \Xi/L$. For compact spaces like S^4, L is simply 2π times the radius of S^4. For product spaces $R^c \times B^b$

with some compact space B, there are two scale lengths: L_b is finite in the compact dimensions and $L_c = \infty$ in the non-compact dimensions. The symmetry behavior of the system (described here by a $\lambda\phi^4$ scalar field as example) is determined by the *correlation length* Ξ defined as the inverse of the *effective mass* $M_{\rm eff}$ related to the effective potential $V_{\rm eff}$ by (we use subscript eff to denote quantities including higher loop corrections)

$$\Xi^{-2} = \frac{\partial^2 V_{\rm eff}}{\partial \bar{\phi}^2} \mid_{\bar{\phi}_{min}} \equiv M_{\rm eff}^2 = \begin{pmatrix} curvature-induced\ mass\ M_{1,2}^2 \\ +\ radiative\ corrections \end{pmatrix} \tag{6.19}$$

where $M_{1,2}$ were defined in (2.128) and (2.126). It measures the curvature of the effective potential at a minimum energy state ($\bar{\phi} = 0$ for the symmetric state, or the false vacuum, $\bar{\phi} = \bar{\phi}_{min}$ for the broken-symmetry state or the true vacuum.) The effective mass is defined to include radiative corrections to the same order corresponding to the effective potential. (This quantity is called the generalized susceptibility function in condensed matter physics.) The critical point of a system is reached when $\Xi \to \infty$ or $M_{\rm eff} \to 0$. In flat or open spaces or for bulk systems, the critical point can be reached without restriction from the geometry (note that in dynamical situations, exponential expansion can effectively introduce a finite size effect equivalent to event horizons, see [507].) However, in spaces with compact dimensions, the correlation length of fluctuations can only extend to infinity in the remaining non-compact dimensions, and thus the critical behavior becomes effectively equivalent to a lower c-dimensional system.

One can also think of Ξ as the Compton wavelength $\Lambda = 2\pi/M_{\rm eff}$ of a system of quasi-particles with effective mass $M_{\rm eff}$. Any fine structure of the background spacetime with scale L is relevant only if $\Lambda \leq L$. Thus when Λ is small or $\eta \ll 1$, (far away from critical point, at higher energy, with higher mode contributions) it sees the details of a spacetime of full dimensionality. At this wavelength, the apparent size of the universe is large in both compact and non-compact dimensions. When $\Lambda \to \infty$ or $\eta \gg 1$ (near critical point, IR limit, lowest mode dominant) structures of finite sizes or the compact dimensions will not be so important. The apparent size of the universe will be dominated by the non-compact directions and the EIRD is measured by the number of non-compact dimensions. The value of η getting very large is an indication of when dimensional reduction can take place. Notice that in curved space the coupling parameters also run with curvature or the scale length of the space. This makes the concept of EIRD even more interesting, as there is now an interplay between Ξ and L, and η can either decrease or increase with curvature.

For example, for $\lambda\phi^4$ fields in the Einstein universe near the symmetric state $\bar{\phi} = 0$, the EIRD is equal to 1 but near the global minimum of broken symmetry state $\bar{\phi} = \bar{\phi}_{min}$ is equal to 4. Near the symmetric state, $\eta \gg 1$ signifies reduction of EIRD to one. This is consistent with the theorem of Hohenberg [500], Mermin and Wagner [499] (for statistical mechanics on a lattice) and Coleman (for continuum field theory) [492] which states that in dimensions less than or equal

to two, the infrared divergence of the scalar field is so severe that there could be no possibility of spontaneous symmetry breaking: the only vacuum expectation value for $\bar{\phi}$ allowed is zero. Away from the region $\bar{\phi} \simeq 0$ the one-dimensional behavior no longer prevails. Indeed a global minimum of the effective potential exists at $\bar{\phi}_{min}$. Near $\bar{\phi}_{min}$, $\eta \ll 1$ and decreases with curvature. Thus the apparent size of the universe near the global minimum actually increases with increasing curvature. There is therefore no dimensional reduction and the system has a full four-dimensional IR behavior. A transition to the asymmetric ground state is not precluded as symmetry breaking via tunneling is in principle possible. The complete picture extending from $\bar{\phi} = 0$ to $\bar{\phi} = \bar{\phi}_{min}$ is a combination of one-dimensional and four-dimensional infrared behavior. Similar arguments can be applied to other spacetimes or field theories. Using this notion one can understand, for example, why it is often said that at high temperatures (small radius limit of S^1) the finite temperature theory becomes an effective three-dimensional theory.

6.2.2 2PI Effective Action and Infrared Behavior

In this section we discuss the nonperturbative regime of an $O(N)$ quantum field theory via the $1/N$ expansion [544]. Many quantum mechanical, statistical and field-theoretic models representing physical systems are known to possess internal symmetries. Several of these theories admit generalizations in which the number of internal degrees of freedom N parametrizing the symmetry group of the problem may be treated as a free variable parameter. The large N expansion looks at the limit when this parameter becomes very large, but does nothing to restrict the range of the coupling constants in the theory. Thus it is regarded as nonperturbative in the coupling constants.

The $1/N$ expansion scheme has proven its efficacy in dealing with a diverse class of theories ranging from single-particle potential problems in quantum mechanics to phase transitions and critical phenomena to quantum chromodynamics. (For a collection of representative papers see in, e.g., [544].) Take a flat space quantum field theory example: the spontaneous symmetry breaking of a finite temperature system studied via the one-loop effective potential. When expanded in powers of β^2/M^2 (here M is an effective mass for the $\lambda\phi^4$ theory and β is the inverse temperature) there are terms nonanalytic in the mass parameter near $M^2 = 0$. These terms arise from the infrared sector of the system (the $n = 0$ mode in the discrete sum and the $\mathbf{k} = 0$ region in the integration) and lead to a complex value for the critical temperature which is unphysical. To investigate the nonperturbative regime of the theory Dolan and Jackiw [429] considered an $O(N)$ $\lambda\phi^4$ theory in the $1/N$ approximation and found that the dominant graphs in the $1/N$ and high temperature expansion are the so-called daisy or cactus diagrams because for zero mass they are the most infrared singular. In this way they were able to eliminate the terms that give an imaginary contribution

to the critical temperature, and for weak coupling they found an acceptable expression for T_c. The divergence at high temperature of a finite temperature theory exemplifies the infrared problem for quantum fields which happens in many contexts. Here we shall develop the composite operator technique for treating such divergences in a general setting.

To study the nonperturbative properties of a field theory in the most general setting with nontrivial infrared behavior, we will use the effective action for composite operators introduced by Cornwall, Jackiw and Tomboulis [494]. We will see that using this method the sum of all daisy type graphs performed by Dolan and Jackiw can be reduced to the evaluation of only one graph. The main idea is to construct a generalized effective action from which one can derive by a variational calculation not only the full background field but also the dressed two-point function. The relevant object in this case in place of the one particle irreducible effective action $\Gamma_{\rm E}[\bar{\phi}]$ is the two-particle-irreducible (2PI) effective action $\Gamma_{\rm E}[\bar{\phi}, G]$ which can be deduced from the following generating functional

$$e^{-W_{\rm E}[J,K]} = \int [d\Phi]\mu[\Phi]\; e^{-S_{\rm E}[\Phi]-\Phi^a J_a-\Phi^a K_{ab}\Phi^b}, \tag{6.20}$$

where $\mu[\Phi]$ is the functional integration measure for the field Φ and we have used DeWitt's condensed index notation. Here K_{ab} is an external current, and functional derivatives with respect to K_{ab} generate vacuum expectation values of an even number of products of Φ^a. The generating functional in Φ for two-particle-irreducible Green functions expressed in terms of the propagator G is given by performing the Legendre transform of W

$$\Gamma_{\rm E}[\bar{\phi}, G] = W_{\rm E}[J, K] - \bar{\phi}^a J_a - (\bar{\phi}^a\bar{\phi}^b + G^{ab})K_{ab}, \tag{6.21}$$

where

$$\begin{aligned}\frac{\delta W_{\rm E}}{\delta J_a} &= \langle\Phi^a\rangle_{J,K} \equiv \bar{\phi}^a \\ \frac{\delta W_{\rm E}}{\delta K_{ab}} &= \langle\Phi^a\Phi^b\rangle_{J,K} \equiv G^{ab} + \bar{\phi}^a\bar{\phi}^b.\end{aligned} \tag{6.22}$$

Solving these equations with J_a and K_{ab} set to zero gives the true ground state and the connected two-point function on G^{ab} for the field. Substituting for these sources from the equation of motion one obtains the generating functional

$$\begin{aligned}e^{-\Gamma_{\rm E}[\bar{\phi},G]} = \int D\Phi\,\mu[\Phi]\,\exp\Bigg\{&-S_{\rm E}[\Phi] + \left(\Phi^a - \bar{\phi}^a\right)\left(\frac{\delta\Gamma_{\rm E}}{\delta\bar{\phi}^a} - \bar{\phi}^b\frac{\delta\Gamma_{\rm E}}{\delta G^{ab}}\right) \\ &+ \left(\Phi^a\Phi^b - \bar{\phi}^a\bar{\phi}^b - G^{ab}\right)\frac{\delta\Gamma_{\rm E}}{\delta G^{ab}}\Bigg\},\end{aligned} \tag{6.23}$$

which is an implicit equation for $\Gamma_{\rm E}[\bar{\phi}, G]$ that can be solved iteratively. Perform a background field decomposition $\Phi^a = \bar{\phi}^a + \varphi^a$, where φ^a is the fluctuation field, this expression simplifies to

$$e^{-\Gamma_{\rm E}[\bar\phi,G]} = \int D\varphi\,\mu[\bar\phi+\varphi]\,\exp\left\{-S_{\rm E}[\bar\phi+\varphi]+\varphi^a\frac{\delta\Gamma_{\rm E}}{\delta\bar\phi^a}+\left(\varphi^a\varphi^b-G^{ab}\right)\frac{\delta\Gamma_{\rm E}}{\delta G^{ab}}\right\}. \tag{6.24}$$

From this expression it is apparent that G^{ab} is the analog for the two-point function of what $\bar\phi^a$ is for the one-point function, the net effect being that we replace all internal propagators by G^{ab}. For the loop expansion we find that

$$\Gamma_{\rm E}[\bar\phi,G] = S_{\rm E}[\bar\phi]+\frac{1}{2}{\rm Tr}\ln[G^{-1}]+\frac{1}{2}A_{ab}[\bar\phi]G^{ab}-\frac{1}{2}{\rm Tr}(\mathbf{1})+\tilde\Gamma_{\rm E}[\bar\phi,G]. \tag{6.25}$$

Here $\tilde\Gamma_{\rm E}[\bar\phi,G]$ represents the sum of all two-particle-irreducible graphs that are present in the ordinary effective action but evaluated with the propagator G, and $A_{ab}[\bar\phi]$ is the small fluctuation operator in the background field $\bar\phi$.

We now apply the above formalism to the $O(N)$ self-interacting scalar field theory. The one-loop effective action is given by (4.65). The two loop contribution to the effective action is given by

$$\begin{aligned}\Gamma^{(2)}_{\rm E}[\bar\phi] = {}& \frac{\lambda}{4!}\int d^Dx\sqrt{g}\,\left\{\Big[{\rm Tr}\,G(x,x)\Big]^2+2\,{\rm Tr}\Big[G^2(x,x)\Big]\right\}\\ & -\frac{\lambda^2}{36}\int d^Dx\sqrt{g}\int d^Dy\sqrt{g}\;\;\bar\phi^a(x)\Big\{G^{ab}(x,y)\,{\rm Tr}\Big[G^2(x,y)\Big]\\ & +2G^{ac}(x,y)G^{cd}(y,x)G^{db}(x,y)\Big\}\bar\phi^b(y),\end{aligned} \tag{6.26}$$

where the trace is over the internal space indices. In the above expression the first term is represented by the graph where two loops are joined together in a 'figure eight' and the second term the 'setting sun' graph. Using the projection operators defined in (4.62) and used in (4.64) this expression takes the simplified form

$$\begin{aligned}\Gamma^{(2)}_{\rm E}[\bar\phi] = {}& \frac{\lambda}{4!}\int d^Dx\sqrt{g}\,\Big[3G_1^2(x,x)+2(N-1)G_1(x,x)G_2(x,x)+(N^2-1)G_2^2(x,x)\Big]\\ & -\frac{\lambda^2}{36}\int d^Dx\sqrt{g}\int d^Dy\sqrt{g}\,\bar\phi(x)\Big[3G_1^3(x,y)+(N-1)G_2^3(x,y)\Big]\bar\phi(y).\end{aligned} \tag{6.27}$$

It is evident from this expression that the dominant contribution in the large N limit comes from the 'figure eight' graph and from the G_2 propagator. The G_2 propagator also gives the dominant contribution to the one-loop effective action in the large N limit, as can be seen from (4.64).

We now evaluate the two-particle-irreducible effective action at the two-loop level in the large N limit. Recalling the definition of $\tilde\Gamma_{\rm E}[\bar\phi,G]$ and noticing that the 'figure eight' graph is two-particle-irreducible we have (we can now rename $G_2\to G$ which is now a dressed propagator to be solved for variationally along with the background field)

$$\Gamma_{\text{E}}[\bar{\phi}, G] = S_{\text{E}}[\bar{\phi}] + \frac{N}{2}\text{Tr}\ln G^{-1} + \frac{N}{2}\int d^D x\sqrt{g}\, A_2 G(x,x) - \frac{N}{2}\text{Tr}(\mathbf{1}) + \frac{\lambda}{4!}\int d^D x\sqrt{g}\, G^2(x,x), \tag{6.28}$$

where we have kept only the dominant $O(N^2)$–term from the two-loop contribution (notice that this term is also proportional to λ).

Let us pause here and ask how the usual effective action $\Gamma_{\text{E}}[\bar{\phi}]$ is related to this newly introduced $\Gamma_{\text{E}}[\bar{\phi}, G]$. The answer is, when the current K vanishes. Equivalently, it is $\Gamma^{(2)}_{\text{E}}[\bar{\phi}, G]$ at that value of $G(x,y)$ for which we have

$$\frac{\delta\Gamma_{\text{E}}[\bar{\phi}, G_0]}{\delta G(x,y)} = 0 \qquad \Rightarrow \qquad \Gamma_{\text{E}}[\bar{\phi}] = \Gamma_{\text{E}}[\bar{\phi}, G_0]. \tag{6.29}$$

The difference is, $\Gamma_{\text{E}}[\bar{\phi}, G]$ contains only two-particle-irreducible graphs, that is, graphs that do not split into disconnected nontrivial graphs upon opening two lines, while $\Gamma_{\text{E}}[\bar{\phi}]$ contains one particle irreducible graphs. Since $\Gamma_{\text{E}}[\bar{\phi}, G]$ does not contain all the graphs that are included in $\Gamma_{\text{E}}[\bar{\phi}]$, the remaining ones must be accounted by the fact that the lines in a graph belonging to $\Gamma_{\text{E}}[\bar{\phi}, G]$ represent the dressed propagator G and not the undressed one. From this line of reasoning, all the daisy graphs (they are two particle reducible, and dominate the large N contribution) must be contained in the dominant two-loop graph of $\Gamma_{\text{E}}[\bar{\phi}, G]$ since any daisy graph can be built from the same graph, with the lines representing the undressed propagator, by partially dressing the propagator lines.

From the variational equation (6.29) using our approximate effective action (6.28) we find (dropping the subscript from G_0)

$$G^{-1}(x,y) = -A_2(x,y) - \frac{\lambda N}{6} G(x,x)\delta(x,y). \tag{6.30}$$

Since

$$A_2(x,y) = \left(-\Box_{\text{E}} + {M_2}^2\right)\delta(x,y) \tag{6.31}$$

[M_2 was defined in (2.126)] if we write

$$G^{-1}(x,y) = \left(\Box_{\text{E}} - \chi\right)\delta(x,y) \tag{6.32}$$

then we have the consistency equation for the single component (no internal indices) scalar field χ:

$$\chi(x) = M_2^2 + \frac{\lambda N}{6} G(x,x). \tag{6.33}$$

We can now use (6.30) to eliminate A_2 from (6.28) to obtain

$$\Gamma_{\text{E}}[\bar{\phi}, G] = S_{\text{E}}[\bar{\phi}] + \frac{N}{2}\text{Tr}\ln G^{-1} - \frac{\lambda N^2}{4!}\int d^D x\sqrt{g}\, G^2(x,x). \tag{6.34}$$

Using (6.32) and (6.33) with χ constant, this becomes

$$\Gamma_{\rm E}[\bar{\phi},\chi] = S_{\rm E}[\bar{\phi}] + \frac{N}{2}\mathrm{Tr}\ln\Big[(-\Box_{\rm E}+\chi)\mu^{-2}\Big] - \frac{3}{2\lambda}\int d^D x\sqrt{g}\,\left(\chi - M_2^2\right)^2 . \tag{6.35}$$

The position-dependent effective mass χ satisfies the iterative equation

$$\chi = {M_2}^2 + \frac{\lambda N}{6(Vol)}\mathrm{Tr}(-\Box_{\rm E}+\chi)^{-1}, \tag{6.36}$$

where Vol is the spacetime volume. This can in principle be solved for $\chi = \chi(\bar{\phi})$ and when substituted back into (6.35) gives the one-particle-irreducible effective action $\Gamma_{\rm E}[\bar{\phi}]$ with the contribution of all daisy type graphs included. When $\bar{\phi}$ and χ are constants, (6.35) defines a two-particle-irreducible effective potential

$$V_{\rm eff}^{2PI}(\bar{\phi},\chi) = V(\bar{\phi}) - \frac{3}{2\lambda}(\chi - {M_2}^2)^2 + \frac{N}{2(\mathrm{Vol})}\mathrm{Tr}\ln\Big[(-\Box_{\rm E}+\chi)\mu^{-2}\Big]. \tag{6.37}$$

Having derived the equations for the effective mass $M_{\rm eff} = \chi$ (6.36) and the effective potential $V_{\rm eff}$ (6.37) we now study the symmetry behavior by seeking solutions to them for different geometries of interest. This will be followed by some remarks on the nature of finite-size effect in curved space related to its counterpart in condensed-matter physics, which characterizes the nature of infrared behavior in de Sitter spacetime.

6.2.3 Symmetry Behavior in Product Spaces

For spacetimes with topology $R^c \times B^b$ (where B is a b-dimensional compact space) χ is simply

$$\chi = M_2^2 + \frac{\lambda N}{6\Omega(B)}\int\frac{d^c k}{(2\pi)^c}\sum_n \frac{1}{k^2+\kappa_n+\chi}\ , \tag{6.38}$$

where $\Omega(B)$ is the volume of the subspace B and κ_n are the eigenvalues of $\Box_{\rm E}$ restricted to B. Considering only the dominant lowest mode contribution ($\kappa_0 = 0$), we obtain from integrating this lowest band the general expression for the effective mass:

$$\chi = M_2^2 + \frac{\lambda N}{6\Omega(B)\chi}\left(\frac{\chi}{4\pi}\right)^{c/2}\Gamma(1-c/2) \tag{6.39}$$

which, expressed explicitly for each individual dimension, reads:

$$\chi = M_2^2 + \frac{\lambda N}{6\Omega(B)\chi} \qquad \text{for} \qquad c = 0, \tag{6.40}$$

$$\chi = M_2^2 + \frac{\lambda N}{12\Omega(B)}\chi^{-1/2} \qquad \text{for} \qquad c = 1, \tag{6.41}$$

$$\chi = M_2^2 - \frac{\lambda N}{24\pi\Omega(B)}\left[\ln\left(\frac{\chi\mu^{-2}}{4\pi}\right) + \gamma_E\right] \qquad \text{for} \qquad c = 2, \tag{6.42}$$

$$\chi = M_2^2 - \frac{\lambda N}{24\pi\Omega(B)}\chi^{1/2} \qquad \text{for} \qquad c = 3. \tag{6.43}$$

From these formulas we see that the solution $\chi = 0$ for $M_2^2 = 0$ is not permissible for $c \leq 2$. This can be interpreted to mean that there cannot be a second-order phase transition for an $O(N)$ model in two or less dimensions (or, as Coleman put it: "There are no Goldstone bosons in two or less dimensions."), contained in the so-called Coleman–Hohenberg–Mermin–Wagner theorem.

Let us consider each case individually.

For $c = 0$ we are dealing with a compact space with finite volume, e.g., S^4, the Euclidean de Sitter, the case of special interest in this chapter. The effective mass or inverse correlation length is obtained by solving (6.40) when $M_2^2 = 0$:

$$M^2_{\text{eff,dyn}} = \chi = \left(\frac{\lambda N}{6\Omega}\right)^{1/2} \tag{6.44}$$

for N-component scalar fields to leading order in large N. For S^4 the Euclidean 'spacetime' volume $\Omega(S^4) = \text{Vol} = 8\pi^2 a^4/3$. This expression was first obtained by Hu and O'Connor [504] – the effective mass is considered as dynamically generated, thus also referred to as dynamical mass M_{dyn}. It depends on the size of the system and vanishes as $\Omega \to \infty$. A critical point which should exist in the infinite-volume system will disappear in a finite-volume theory: finite-size effect thus precludes a second-order phase transition from occurring in finite systems. This can be seen also from the form of the effective potential. For massless, minimally coupled scalar fields in S^4 near $\bar{\phi} = 0$, the effective potential obtained by inserting the solution of (6.40) for χ with $M_2^2 = (\lambda/6)\bar{\phi}^2$ into (6.37) is given by

$$V(\bar{\phi}) = \frac{8\pi^2}{3\Omega}\alpha(\lambda)_{IR} + \frac{1}{2}\left(\frac{\hbar\lambda N}{6\Omega}\right)^{1/2}\bar{\phi}^2 + \frac{\lambda}{48}\bar{\phi}^4 + \dots \ , \tag{6.45}$$

where

$$\alpha_{IR}(\lambda) = -\frac{3\hbar N}{32\pi^2}\Big[1 + \ln\frac{6\Omega a^4}{\hbar\lambda N}\Big].$$

Note that the $\ln\lambda$ dependence in (6.45) of α_{IR} renders the effective potential $V_{\bar{\phi}}$ singular in the limit $\lambda \to 0$. This means that there does not exist a ground state for the free ($\lambda = 0$) theory, but such a state does exist for the self-interacting field. Thus the free massless minimally-coupled field is not an appropriate starting

point for perturbation theory. One can, in fact, make the more general statement that the same effect occurs for any field theory defined in finite 4-volume for which the lowest eigenvalue of the fluctuation operator is zero. In the interacting case the effective potential has to have positive curvature near $\bar{\phi} = 0$ and the minimum value of this effective mass (inverse correlation length) is (6.44). The power 1/2 dependence is characteristic of a finite 4-volume situation as is 2/3 for a configuration with one-infinite dimension ($R^1 \times S^3$).

The above analysis shows that there is always a local minimum of the effective potential at $\bar{\phi} = 0$ for the massless minimally coupled field. The implication of this in critical phenomena is that there cannot be a second order phase transition in a finite volume.

As the volume becomes large in comparison with the volume sustained by the correlation length, the effect of the finite size of the system decreases and the higher mode contributions to the effective potential can no longer be neglected. To extract this large volume behavior a different approximation which treats the system as a small deviation from the infinite-volume limit is necessary. In general the approach to the infinite volume limit depends on the higher modes. For example, in the case of S^4, in the large-a limit, the deviations due to the finite size of the 4-sphere S^4 drop off as inverse powers of a; by contrast, for the 4-torus T^4 the finite-size effect drops off exponentially. The effective potential for T^4 in the large volume limit is given by

$$V_{\text{eff}}(\bar{\phi}) = V_{\text{eff}}^{\infty}(\bar{\phi}) - \frac{N\chi^2}{2(2\pi^3)^{1/2}} \Big[(\chi L_1^2)^{-5/4} e^{-(\chi L_1)^{1/2}} + \dots + (\chi L_4^2)^{-5/4} e^{-(\chi L_4)^{1/2}} \Big] + \dots \quad . \tag{6.46}$$

The exponential fall-off seems to be characteristic of periodic boundary conditions (i.e., of the torus) and not a generic property of the approach to the bulk system; it depends on the boundary conditions in the finite-volume setting [502]

Considering also the other three categories $c = 1, 2, 3$ (e.g., $c = 1, b = 3$ describes the Einstein universe, $c = 3, b = 1$ the imaginary-time finite temperature quantum field theory) Hu and O'Connor derived the forms of the 2PI effective potentials and the effective masses. See [504] for the details. For further studies of the critical behavior of a ϕ^4 theory in spherical and hyperbolic spaces, see, e.g., [545].

The next question that naturally arises is what happens when the higher modes are taken into account? Can the theory develop a minimum away from $\bar{\phi} = 0$ and thus allow the possibility of a first-order transition occurring in place of the second-order one? Physically, we know from condensed matter physics that phase transitions which were second order in the bulk can become first order in the finite volume setting. Technically, to explore this possibility we need an expression for the potential that interpolates between the large volume limit and the result for small $\bar{\phi}$. The leading behaviour of the effective potential is given

by the effective potential of the lowest mode plus the loop contributions to the effective potential of the higher (inhomogeneous) modes; the latter contributions are treated in the usual loop expansion fashion. When one goes beyond one loop corrections there are, of course, interactions between the lowest mode and the higher modes. It is considered in [504] that the leading contributions from these interactions are taken into account by substituting the solution for $\chi(\bar{\phi})$ back into the higher mode contribution to the 2PI effective potential.

How would these features in Euclidean dS, namely, the IR dominance of the zero mode, the contributions of the higher modes, manifest in the Lorentzian description? We will show some recent results of López Nacir et al [510] in a later section for the $O(N)\lambda\phi^4$ theory in Lorentzian dS by means of a double expansion in $\sqrt{\lambda}$ and $1/N$ using functional methods.

As for other fields, scalar electrodynamics (SQED) in de Sitter space is believed to be a more realistic class of theories for the description of phase transitions in inflationary cosmology. O'Connor [461] derived the one-loop-effective potential of scalar QED in de Sitter universe via the zeta function method. This was shown partially in Chapter 4. His results were used to explore the infrared behavior of SQED and draw implications about Planck scale phase transitions in [462]. More details of this theory in dS space can be found in the work of Prokopec, Tsamis and Woodard [464]. Functional renormalization group methods have been applied to SQED by González and Prokopec to obtain quantum loop effects [546]. For Yukawa theory, Miao and Woodard [547] studied the theory of a massless, minimally coupled scalar interacting with a massless fermion in a locally de Sitter geometry. Because Yukawa theory is quadratic in the fermion field it can be integrated out by using the classic solution of Candelas and Raine. They then follow Starobinsky's stochastic technique to get the leading log solution. For nonlinear sigma model one needs to extend the resummation formula to models with derivative interaction, as done in [548].

6.3 Lorentzian de Sitter: Late Time IR and Stochastic Approach

For de Sitter space in the S^4 coordinatization, we learn in the previous two sections how to analyze the infrared behavior of interacting quantum fields by way of the 2PI effective potential under the large N approximation, and the physical meaning of such behavior in terms of finite size effect and infrared dimensional reduction. Now we ask the question: If one views de Sitter spacetime in a dynamical setting, such as in the $k = 0$ or 1 RW coordinatizations, with topology $S^3 \times R^1$ or $R^3 \times R^1$, the compactness of S^4 is gone, would there still be a finite size effect? Would we expect different physics? Or, rather, since physics should be the same despite the differences in coordinate representations, how would the physics manifest in a dynamical setting? If we call the S^4 treatment as Euclidean, we will refer to the latter two cases as Lorentzian, since time evolution is explicit.

6.3.1 Lorentzian-de Sitter: Dynamical Finite Size Effect

The resolution of this puzzle brings out an interesting point related to the effect of spacetime dynamics on the symmetry behavior of a quantum field. Specifically, for the special case of exponential expansion, in the spectrum of the 4-dim (spacetime) wave operator, there is a gap which separates the zero mode from the rest. This is what gives the effective infrared dimension $d_{\rm eff} \simeq 0$ for the deep IR behavior of the scalar field in these other coordinate descriptions. Physically this arises from the fact that, as a result of exponential expansion the whole spectrum undergoes rapid red-shifting, and given long enough time (with only logarithmic difference) almost all the higher modes will heap upon the zero mode. (The S^3 or R^3 spatial sector also becomes immaterial.) That is why at late times the longest wavelength mode gives the dominant contribution to the system's IR behavior. The appearance of a scale (the event horizon H^{-1}) is a unique feature of this exponential class of expansion. It gives rise to effects similar to those originating from the existence of some finite size in static spacetimes.

Inflation as Scaling: Eternal Inflation as Static Critical Phenomenon

One can see the linkage between the Euclidean and the FLRW de Sitter formulations from the observation [241] that the exponential expansion of the scale factor can be viewed as a system undergoing a (Kadanoff–Migdal) scale transformation [549]. In a spatially flat FLRW metric with a constant scale factor – this is just the Minkowski spacetime – let us consider an ordered sequence of such static hyperspaces (foliation) with scales a_0, a_1, a_2, etc parameterized by $t_n = t_0 + n\Delta t, n = 0, 1, 2, \ldots$ These spacetimes have the same geometry and topology but differ only in the physical scale in space. One can always redefine the physical scale length $x'_{(n)} = a_n x$ to render them equivalent. If each copy has scale length magnified by a fixed factor s over the previous one in the sequence, i.e. $a_{n+1}/a_n \equiv s = e^{H\Delta t}$, we get exactly the physical picture as in an eternal inflation. After n-iterations $a_n/a_0 = e^{n(H\Delta t)}$, or, with a continuous parameter $a(t) = a_0 e^{Ht}$. It is important to recognize that t can be any real parameter not necessarily related to time. In other words, time in eternal inflation plays the role of a scaling parameter. Dynamics in this case is replaced by scaling, and every time step of eternal inflation can be viewed as a scaling transformation in a static setting.

In this light we see that in the 'dynamic' FLRW description of de Sitter, the constant time scale associated with exponential expansion, $1/H$, corresponds to the length scale in an equivalent static universe, the radius of S^4 in the Euclidean de Sitter space. In the Lorentzian QFT the infrared behavior is dominated by the late time behavior of the longest wavelength mode, and that is equivalent to the infrared behavior of the zero-mode in S^4 in a 'static' description. The IR physics in Lorentzian dS is called *dynamical finite size effect* by Hu and O'Connor [507], as the dynamical correspondence of finite size effect in a compact space.

As a reminder of the relation between the Euclidean de Sitter (EdS) and the Lorentzian de Sitter (LdS) metrics and to establish notations used in the sections below, we start with the global coordinate which covers the full d-dim LdS spacetime,

$$ds^2 = -dt^2 + \frac{1}{H^2}\cosh^2(Ht)d\Omega_{d-1}^2. \tag{6.47}$$

Analytically continue t to imaginary time $t \to i\tau$ with the periodicity condition $\tau \to \tau + 2\pi H^{-1}$ imposed to avoid a multivalued metric. The resultant EdS is a d-sphere of radius H^{-1}

$$ds^2 = H^{-2}\left[d\theta^2 + \sin(\theta)^2 d\Omega^2\right], \quad \theta = H\tau. \tag{6.48}$$

We now turn to the stochastic approach of Starobinsky and Yokoyama [505]. We shall see at the end that the infrared behavior induced by the dynamical effects of exponential expansion at late times is equivalent to the finite size effect associated with the zero mode in the Euclidean formulation described by Hu and O'Connor [462, 504].

6.3.2 Stochastic Approach for Long-Wavelength Late-Time Behavior

Working within the structural framework of stochastic inflation model[2] proposed by Starobinsky [506] in 1986 Starobinsky and Yokoyama [505] in 1994 calculated the two-point function of the long-wavelength field at late times. This result captured the attention of quantum field theorists working on inflationary cosmology because it is a nonperturbative result derived from a very different method. As we explained earlier, this can be understood from the dominance of the zero mode contribution to the infrared behavior of quantum fields in Euclidean dS. Beneke and Moch [523] gave a clear derivation of the late-time two-point functions in

[2] Some cautionary notes on the structural foundation of stochastic inflation may be helpful here. As pointed out in [550], two issues of fundamental concern are: (1) the decoherence of the long wavelength modes and (2) the justification of a white noise for the short wavelength sector. For issue (1) the justification for treating the long wavelength sector as classical need be provided. Usually one invokes decoherence, namely, that the long wavelength modes are readily decohered by the short wavelength or high frequency (>) sector, treated as noise via some environment-induced decoherence mechanism. The decoherence scheme of Polarsky and Starobinsky [551] asserting that the insignficance of the decaying mode renders the growing mode effectively decohered is too simplistic, because quantum wave functions even at small attenuated amplitude has a chance to recohere. A later version with Kiefer [552] comes closer to the open-system treatment as in [553]. A first example of cosmological decoherence was given in [554] using an interacting field theory based on the insight gained from the study of nonMarkovian processes in quantum Brownian motion models [555]. Lombardo and Mazzitelli [556] presented an open system treatment of decoherence in interacting quantum fields based on the use of the coarse-grained effective action introduced in [241] and the influence functional formalism [187]. For its relation to effective field theory, see [557].

the stochastic scheme and showed the relation between these two approaches. We summarize their derivations below.

Divide the scalar field into two sectors $\phi = \bar{\phi} + \phi_<$, where $\bar{\phi}$ contains all long-wavelength modes with wave number $k < \epsilon aH$ (a is the scale factor and $\epsilon \ll 1$ is a parameter which separates the long wavelength from the short wavelength modes), assumed to be superhorizon which behaves classically, and a short wavelength subhorizon part $\phi_<$ which retains the quantum field character but whose effect can be represented by a classical noise, assumed to be white.[3]

Now suppose that ϕ is acted on by a scalar field potential $V(\phi)$. From the field equation for ϕ, namely, $-\Box\phi + V'(\phi) = 0$ we obtain an equation governing $\bar{\phi}$ in the form of a Langevin equation

$$\dot{\bar{\phi}}(t, \mathbf{x}) = -\frac{1}{(d-1)H}\, V'(\bar{\phi}) + \xi(t, \mathbf{x}), \tag{6.49}$$

where d is the spacetime dimension and $\xi(t, \mathbf{x})$ denotes the stochastic force generated by the short-wavelength modes

$$\xi(t, \mathbf{x}) = \dot{\phi}_< = \epsilon a H^2 \int \frac{d^{d-1}k}{(2\pi)^{d-1}}\, \delta(k - \epsilon aH)\, \left(a_k \phi_k(t) e^{i\mathbf{k}\cdot\mathbf{x}} + \text{c.c.}\right), \tag{6.50}$$

where $\phi_k(t)$ are the mode functions. At leading order, we can neglect the self-interaction of the short-wavelength modes whence the fluctuations satisfy

$$\langle \xi(t_1, \mathbf{x}) \xi(t_2, \mathbf{x}) \rangle = \gamma\, \delta(t_1 - t_2) \tag{6.51}$$

with

$$\gamma = \frac{2\pi^{\frac{d-1}{2}}}{(2\pi)^{d-1}\Gamma(\frac{d-1}{2})} \times \frac{2^{d-3}\,[\Gamma(\frac{d-1}{2})]^2}{\pi} \times H^{d-1}. \tag{6.52}$$

The first factor arises from the volume of the $d-2$ dimensional momentum shell $k = \epsilon aH$, the second from the long-wavelength limit (since $k = \epsilon aH \ll aH$) of the Bunch–Davies mode functions $\phi_k(t)$. The Fokker–Planck equation associated with (6.49) for the one-particle probability density $\mathcal{P}[\varphi]$ has the form[4]

$$\frac{\partial \mathcal{P}}{\partial t} = \frac{1}{(d-1)H}\, \frac{\partial}{\partial \varphi}\, (V'(\varphi)\mathcal{P}) + \frac{\gamma}{2}\, \frac{\partial^2}{\partial \varphi^2}\, \mathcal{P}. \tag{6.53}$$

[3] For the noise issue (2), a sharp partitioning of the field into two sectors does not constitute an interaction between the two sectors, whereas an interaction is necessary for an environment (short wavelength sector)-induced decoherence (of the long wavelength sector) to occur. Winitzki and Vilenkin [558] examined the window function and expressed doubt that the sharp division generates a white noise spectrum. This is problematic from a quantum field theory viewpoint because in an expanding background the Hilbert space of each sector is changing in time, and a proper treatment of their interaction is rather involved. The open quantum systems treatment for noise and environment-induced decoherence provided the basis for later more vigorous investigations carried out by e.g., Lombardo and López Nacir [559]. (For further description of this topic, see, e.g., Chap 15 of [326].)

[4] In this context φ is a stochastic variable in contrast to it denoting the fluctuations of a quantum field variable used in earlier sections.

At late times it admits a stationary ('equilibrium') solution

$$\mathcal{P}[\varphi] = \mathcal{N} \exp\left(-\frac{2}{(d-1)\gamma H}\, V(\varphi)\right), \tag{6.54}$$

where the normalization factor $\mathcal{N}$ is determined by the condition $\int_{-\infty}^{\infty} d\varphi\, \mathcal{P}[\varphi] = 1$. In terms of this the two-point function of the long-wavelength field at late times is given by

$$\langle \bar{\phi}\bar{\phi} \rangle = \mathcal{N} \int_{-\infty}^{\infty} d\varphi\, \varphi^2 \exp\left(-\frac{2}{(d-1)\gamma H}\, V(\varphi)\right). \tag{6.55}$$

As shown by Beneke and Moch [523], this expression for the long-wavelength two-point functions (with $V(\phi) = \frac{\lambda}{4!}\phi^4$), and the 'dynamical masses' derived from it, agrees with the exact Euclidean result at leading order in the expansion in $\sqrt{\lambda}$ provided the dissipation and fluctuation coefficients in the Fokker-Planck equation are related to the volume of d-dimensional Euclidean de Sitter space with radius $1/H$ by $(d-1)H \times \frac{\gamma}{2} = \frac{1}{V_d}$, which can be easily verified.

Furthering this, worthy of mentioning, Garbrecht, Rigopoulos, and Zhu (GRZ) [539] found a perturbative diagrammatic representation of the stochastic function in the Starobinsky–Yokoyama approach. It goes as follows:

In the stochastic approach, the long wavelength modes $\bar{\phi}$ of the scalar inflaton field ϕ obey a Langevin-type equation

$$\dot{\bar{\phi}} + \frac{V'(\bar{\phi})}{3H} = \xi(t), \tag{6.56}$$

where ξ is a Gaussian random force with

$$\langle \xi(t)\xi(t') \rangle = \frac{H^3}{4\pi^2}\delta(t-t'). \tag{6.57}$$

At late times the stochastic process (6.56) is described by the probability distribution function [506, 505]

$$\mathcal{P}[\bar{\phi}] = \mathcal{N}\, e^{-\frac{8\pi^2}{3H^4}V(\bar{\phi})}. \tag{6.58}$$

Following Starobinsky's original approach, expectation values (at equal times) are obtained by weighing the variable by the probability distribution function, e.g.,

$$\langle \bar{\phi}^n \rangle = \int_{-\infty}^{\infty} d\bar{\phi}\, \bar{\phi}^n \mathcal{P}[\bar{\phi}]. \tag{6.59}$$

For a potential of the form

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4 \tag{6.60}$$

one can carry out the expansion

$$\mathrm{e}^{-\frac{8\pi^2}{3H^4}V(\phi)} = \mathrm{e}^{-\frac{4\pi^2}{3H^4}m^2\phi^2}\left(1 - \frac{\pi^2\lambda\phi^4}{9H^4} + \frac{\pi^4\lambda^2\phi^8}{162H^8} + \cdots\right). \tag{6.61}$$

Substituting this into Eq. (6.59), performing the integrals for the fluctuation and the normalization leads to the result

$$\langle\phi^2\rangle = \frac{3H^4}{8\pi^2m^2} - \frac{9\lambda H^8}{128\pi^4m^6} + \frac{9\lambda^2H^{12}}{256\pi^6m^{10}} + \cdots. \tag{6.62}$$

To bring the stochastic approach in a form closer to what can be checked against QFT methods, GRZ [539] introduced a functional representation of Eq (6.56) and (6.57) from which the expectation values of an operator can be calculated by means of distribution averages. In this framework, the expectation value of an operator $\mathcal{O}[\phi]$ is given by[5]

$$\langle\mathcal{O}[\phi]\rangle = \int D[\xi]\ e^{-\frac{1}{2}\int dt\,\xi^2\frac{4\pi^2}{H^3}}\int D[\phi]\ \mathcal{O}[\phi]\,\delta\Big(\dot{\phi} + \partial_\phi V/3H - \xi\Big). \tag{6.63}$$

By expressing the delta functional as a functional "Fourier transform" with the aid of an auxiliary field ψ, setting a functional determinant that appears in the integration to unity and performing the Gaussian ξ integral GRZ obtain

$$\langle\mathcal{O}[\phi]\rangle = \int D[\phi,\psi]\ \mathcal{O}[\phi]\,\mathrm{e}^{-\int dt\left[\mathrm{i}\psi\left(\dot{\phi}+\frac{\partial_\phi V}{3H}\right)+\frac{H^3}{8\pi^2}\psi^2\right]}. \tag{6.64}$$

To obtain a diagrammatic expansion one can rewrite the action by bringing the quadratic part in a more symmetric form. In fact, the analogue of Eq. (6.64) containing second time derivatives can be obtained directly from the more fundamental closed-time-path (CTP) integral after short wavelength modes are integrated out [540].

To construct the two point function of ϕ, we begin with the retarded and advanced Green functions $G^{(R,A)}(t,t')$ for the operator $\partial_t + \frac{m^2}{3H}$,

$$G^R(t,t') = G^A(t',t) = \mathrm{e}^{-\frac{m^2}{3H}(t-t')}\Theta(t-t'). \tag{6.65}$$

The 2-point function $F(t,t')$ of ϕ is then given by

$$\begin{aligned} F(t,t') \equiv \langle\phi(t)\phi(t')\rangle &= \frac{H^3}{4\pi^2}\int_0^{+\infty} d\tau\ G^R(t,\tau)G^A(\tau,t') \\ &= \frac{3H^4}{8\pi^2m^2}\left(\mathrm{e}^{-\frac{m^2}{3H}|t-t'|} - \mathrm{e}^{-\frac{m^2}{3H}(t+t')}\right). \end{aligned} \tag{6.66}$$

[5] The functional integration measures that appear in Eq. (6.63) is such that $<\mathbf{1}> = 1$. This corresponds to the retarded Ito regularization of the stochastic equation (6.56) [560].

If t and t' are taken to be sufficiently large, or, equivalently, the stochastic process is taken to have begun early enough, the correlator reduces to

$$F(t,t') \simeq \frac{3H^4}{8\pi^2 m^2} \mathrm{e}^{-\frac{m^2}{3H}|t-t'|}. \tag{6.67}$$

Note that in the massless limit, $G^R(t,t') \to \Theta(t-t')$, and the variance grows linearly with time

$$\langle \phi^2(t) \rangle_{m=0} \simeq \frac{H^3}{4\pi^2} t, \tag{6.68}$$

a fact known earlier from [522].

A diagrammatic representation of the stochastic processes can be found in Fig. 3 of [539]. At late times the equal time two-point function $\lim_{t\to\infty} \langle \phi(t)^2 \rangle$ has the same form as (6.62) obtained via the probability functional in the stochastic method. (Indeed the three terms in (6.62) can be obtained from Feynman diagrams by the following rules: For each vertex, assign a factor $-\lambda 8\pi^2/(3H^4)$ and for each propagator, a factor $3H^4/(8\pi^2 m^2)$, then divide by the appropriate symmetry factor. Compare this with the field-theoretical diagrams in Fig 1 of [539].)

GRZ showed that the result obtained from perturbative calculations for small massive scalar field agrees with the result from the QFT Schwinger–Dyson equations. Each individual contribution from the topologically distinct diagrams equals the QFT contribution from diagrams of corresponding topology. GRZ have also considered a massive theory with the mass of the scalar field ϕ chosen to be small enough, such that the amplification of superhorizon momentum modes leads to a significant enhancement of infrared correlations, typical of the massless case, but large enough such that perturbation theory remains valid. Using the closed-time-path approach, they calculated the infrared corrections to the two-point function of ϕ to 2-loop order. To this approximation, they find agreement with the correlation found using the stochastic method.

Continuing the investigations by GRZ for a light massive scalar field with $\lambda\phi^4$ self-interaction, Garbrecht, Gautier, Rigopoulos, and Zhu (GGRZ) [540] carried out a diagrammatic expansion that describes the field as driven by stochastic noise. This is compared with the Feynman diagrams in the Schwinger–Keldysh basis of the closed-time-path field-theoretical formalism. For all orders in the expansion, they find that the diagrams agree when evaluated in the leading infrared approximation, i.e. to leading order in m^2/H^2, where m is the mass of the scalar field and H is the Hubble rate. As a consequence, the correlation functions computed in both approaches also agree to leading infrared order. This perturbative correspondence shows that the stochastic theory is exactly equivalent to the field theory in the infrared. The former can then offer a nonperturbative resummation of the field theoretical Feynman diagram expansion, including fields with $0 \le m^2 \ll \sqrt{\lambda} H^2$ for which the perturbation expansion fails at late times.

Connecting the Stochastic Approach to Effective Action Method

The 2PI effective action we introduced in Sec. 2 is an Euclidean formulation. For QFT in de Sitter space an Euclidean formulation can be shown to be equivalent to the 'in-in' formulation [527]. We shall present in the next chapter the CTP ('in-in') 2PI effective action in Lorentzian QFT together with loop and $1/N$ expansion methods. For the $O(N)\lambda\phi^4$ quantum field theory in a general curved spacetime this has been derived by Ramsey and Hu [244] at two loops [(7.68), (7.75)], from which they obtained a set of local, covariant, nonperturbative equations for the mean-field $\bar{\phi}$ and the two-point function C^{++} [(7.77), (7.76)], which, at leading order $1/N$ expansion, take the form (7.87) and (7.85). Riotto and Sloth [538] used these prior results to investigate the infrared behavior of this field theory in the flat-FLRW representation of the de Sitter universe. As they were interested only in the IR properties in these equation they omitted the appropriate counterterms needed to cancel the UV divergences. The way they solved the gap equation for long wavelengths is to first evaluate $C^{++}(x,x)$ from S&Y's Fokker–Planck equation, which is a good approximation for the late time behavior. They then inserted this into the gap equation and solved for $C^{++}(x,x')$, obtaining a simple analytic form for it. We will give a summary description of it in the next chapter as a sample application of these advanced techniques.

While Riotto and Sloth borrowed some results from the stochastic approach using the Fokker–Planck equation of S&Y one can also adhere to the field theory route to calculate the higher order correlation function via a diagrammatic expansion of the 'in-in' effective action. Petri [561] obtained a resummation of the infrared divergences reproducing the results in [538]. Treatment of the infrared problem via diagrammatic expansion of the partition function in the stochastic approach was further pursued in [539, 540]. Finally we mention the work of Moss and Rigopoulos [562] who, using the Schwinger–Keldysh path integral formalism, derived an effective potential which at leading order and for time scales $\Delta t \gg H^{-1}$ produces Starobinsky's stochastic evolution, but which also allows for the computation of quantum UV corrections. The long wavelength stochastic dynamical equations are now second order in time, incorporating temporal scales $\Delta t \sim H^{-1}$ and resulting in a Kramers equation for the probability distribution, the Wigner function, in contrast to the usual Fokker–Planck equation. This feature allows for a non-perturbative evaluation, within the stochastic formalism, not only of the expectation values of field correlators, but also the stress-energy tensor of the inflaton field ϕ.

6.3.3 Resummation of Secular Terms and Long Distance Behavior of $O(N)$ Correlators in LdS

The three main approaches to the IR problem of QFT in dS space are, as we stated from the beginning, the Euclidean, the Lorentzian and the stochastic

formulations. We have seen the development in Euclidean formulation in the second section and the stochastic formulation in this section, where the relation between the two are discussed. Now we return to QFT and explore the relation between the Euclidean and the Lorentzian formulations in the depiction of IR in dS. Linking these two approaches involves taking the zero mode in the Euclidean dS as providing the dominant contribution, but also including the nonzero modes in the Lorentzian formulation. López Nacir, Mazzitelli and Trombetta (LMT) [510] pursue this with functional methods, performing a resummation in a double expansion in $\sqrt{\lambda}$ and $1/N$. We give a summary of their recent work below.

In Euclidean dS (EdS), recognizing that the dominant IR behavior comes from the zero mode in the fluctuation operator of a quantum field in S^4 is an important first step [462, 463, 523]. The effective or dynamically induced mass given by (6.44) provides to leading order in $\sqrt{\lambda}$ the correlator of the IR part of the field,

$$\langle \phi_{IR}(x)\phi_{IR}(x')\rangle \simeq \frac{3H^3}{8\pi^2 m^2_{\rm dyn}}. \tag{6.69}$$

The above nonperturbative result cures the divergence appearing in the free massless correlator. However, corrections beyond this constant IR contribution are necessary to understand the long distance limit of the correlators in the Lorentzian dS (LdS) spacetime. Consider again the $O(N)$-model with quartic self-interaction in an Euclidean dS space represented by a d dimensional sphere of radius H^{-1} around the unbroken symmetric vacuum

$$S_{\rm E} = -i\int d^dx\sqrt{g}\left[\frac{1}{2}\phi_a\left(-\Box_{\rm E}+m^2\right)\phi_a+\frac{\lambda}{8N}(\phi_a\phi_a)^2\right], \tag{6.70}$$

where ϕ_a are the components of an element of the adjoint representation of the $O(N)$ group, with $a=1,\ldots,N$. The sum over repeated indices is implied.[6] In [537] a systematic nonperturbative resummation scheme has been developed for a double expansion in $1/N$ and $\sqrt{\lambda}$ for sufficiently large number of fields N. Formulated in EdS it consists of a reorganization of the perturbative expansion followed by an analytic continuation to the LdS. The main advantages are that the improvement of the IR behavior of correlation functions in LdS is systematic, and the renormalization process is well understood. We summarize the simpler derivation in [510] below and discuss the results for the long wavelength limit of the correlators up to next-to-leading order (NTLO) in $1/N$.

Reorganizing the Perturbative Expansion

LMT begin by separating the field into two sectors $\phi_a(x) = \phi_{0a} + \tilde{\phi}_a(x)$, the homogeneous zero mode ϕ_{0a} and the inhomogeneous modes $\tilde{\phi}_a(x)$. The former is

[6] Beware of differences in the definitions of N, λ, such as in the factor multiplying the $(\phi_a\phi_a)^2$ term in (6.68).

treated nonperturbatively, following the method used in [504, 463, 523] for QFT in EdS. The interaction part of the action Eq. (6.70) is separated accordingly:

$$S_{int} = \frac{\lambda V_d}{8N} |\vec{\phi}_0|^4 + S^{(2)}_{int}[\phi_{0a}, \tilde{\phi}_a], \tag{6.71}$$

where

$$V_d = \int d^d x \sqrt{g} = \frac{2\pi^{\frac{d+1}{2}}}{\Gamma\left(\frac{d+1}{2}\right) H^d} \tag{6.72}$$

is the total volume of EdS space equal to the surface area of the d-sphere, and $S^{(2)}_{int}$ contains at least two powers of $\tilde{\phi}_a$ (note that the term linear in $\tilde{\phi}_a$ vanishes identically by orthogonality). The path integral over the constant zero modes is an ordinary integral which can be performed exactly (i.e., nonperturbatively in the coupling constant λ). The generating functional becomes

$$\begin{aligned} Z[J_0, \tilde{J}] &= \mathcal{N} \int d^N \phi_0 \int \mathcal{D}\tilde{\phi}\, e^{-S_{\rm E} - \int_x (\vec{J}_0 \cdot \vec{\phi}_0 + \tilde{J}_a \tilde{\phi}_a)} \\ &= \exp\left\{ -S^{(2)}_{int}\left[\frac{\delta}{\delta J_0}, \frac{\delta}{\delta \tilde{J}} \right] \right\} Z_0[J_0] \tilde{Z}_f[\tilde{J}], \end{aligned} \tag{6.73}$$

where J_{0a} and $\tilde{J}_a$ are external sources and the shorthand notation $\int_x = \int d^d x \sqrt{g}$ is introduced. The zero part $Z_0[J_0]$ is defined as the exact generating functional of the theory with the zero modes alone [463],

$$Z_0[J_0] = \frac{\displaystyle\int d^N \phi_0 \; e^{-V_d \left[\frac{\lambda}{8N} |\vec{\phi}_0|^4 + \frac{m^2}{2} |\vec{\phi}_0|^2 + \vec{J}_0 \cdot \vec{\phi}_0 \right]}}{\displaystyle\int d^N \phi_0 \; e^{-V_d \left[\frac{\lambda}{8N} |\vec{\phi}_0|^4 + \frac{m^2}{2} |\vec{\phi}_0|^2 \right]}}. \tag{6.74}$$

For a massless field $m = 0$, the variance of the zero modes is

$$\langle \phi_{0a} \phi_{0b} \rangle_0 = \frac{\displaystyle\int d^N \phi_0 \; \phi_{0a} \phi_{0b} \, e^{-\frac{V_d \lambda}{8N} |\vec{\phi}_0|^4}}{\displaystyle\int d^N \phi_0 \; e^{-\frac{V_d \lambda}{8N} |\vec{\phi}_0|^4}} = \delta_{ab} \sqrt{\frac{2}{V_d \lambda}} \frac{2}{\sqrt{N}} \frac{\Gamma\left(\frac{N+2}{4}\right)}{\Gamma\left(\frac{N}{4}\right)} \equiv \frac{\delta_{ab}}{V_d m^2_{\rm dyn}}, \tag{6.75}$$

which allows the identification of a dynamically-generated effective mass $m^2_{\rm dyn}$ (6.44) [504].

This result is valid at LO in $\sqrt{\lambda}$ and for all N. Corrections coming from the inhomogeneous modes can then be computed by treating $S^{(2)}_{int}$ in Eq. (6.73) perturbatively. The free correlators of the inhomogeneous modes can be written as $\tilde{G}^{(m)}_{ab}(r) = \delta_{ab} \tilde{G}^{(m)}$, with

$$\tilde{G}^{(m)}(r) = G^{(m)}(r) - \frac{1}{V_d m^2} = H^d \sum_{\vec{L}, |\vec{L}| > 0} \frac{Y_{\vec{L}}(x) Y^*_{\vec{L}}(x')}{H^2 L(L+d-1) + m^2}, \tag{6.76}$$

where $G^{(m)}(r)$ is the standard propagator for a free field with mass m and $Y_{\vec{L}}(x)$ are the characteristic functions of the Euclidean space, e.g., the hyperspherical harmonics with $L = (n, l, m)$ in the case of S^4. Although the correlator $\tilde{G}^{(m)}$ is finite for $m \to 0$, the perturbation theory built with the massless $\tilde{G}^{(0)}$ is still ill-defined when analytically continued to LdS and at long distances/late times, due to the divergent behavior given by

$$G^{(m)}(r) \simeq \frac{1}{V_d m^2}\, r^{-\frac{m^2}{d-1}}. \tag{6.77}$$

Solving this issue requires further resummations of contributions that also involve the inhomogeneous modes. A subclass of such contributions come from the terms in $S^{(2)}_{int}$ that are quadratic in both $\tilde{\phi}_a$ and ϕ_{0a}, which dress $\tilde{G}$ with a nonperturbative mass.

Resummation of Bi-Quadratic Terms

In the spirit of the separation of the interaction part of the action in Eq. (6.71), we further isolate the bi-quadratic terms,

$$S^{(2)}_{int} = \frac{\lambda}{8N} \int d^d x \sqrt{g} \left[2|\vec{\phi}_0|^2 |\vec{\tilde{\phi}}|^2 + 4(\vec{\phi}_0 \cdot \vec{\tilde{\phi}})^2 \right] + S^{(3)}_{int}, \tag{6.78}$$

where now $S^{(3)}_{int}$ contains terms with at least three powers of $\tilde{\phi}_a$. The main idea is to include the bi-quadratic terms in the definition of the propagator $\tilde{G}$ by defining $\vec{\phi}_0$-dependent inverse propagators

$$\tilde{G}^{-1}_{ab}(\vec{\phi}_0) = -\delta_{ab}\Box + m^2_{ab}(\vec{\phi}_0), \tag{6.79}$$

with the following mass matrix

$$m^2_{ab}(\vec{\phi}_0) = m^2 \delta_{ab} + \frac{\lambda}{2N} \left(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc} \right) \phi_{0c}\phi_{0d} = m_1^2\, P_{ab} + m_2^2\, \epsilon_a \epsilon_b, \tag{6.80}$$

where in the second line the matrix is split into the parallel and transverse components with respect to the $\epsilon_a \equiv \phi_{0a}/|\vec{\phi}_0|$ direction, by means of the projector $P_{ab} = \delta_{ab} - \epsilon_a \epsilon_b$. Here $m_1^2 = m^2 + \frac{\lambda}{2N}|\vec{\phi}_0|^2$ and $m_2^2 = m^2 + \frac{3\lambda}{2N}|\vec{\phi}_0|^2$. Diagonalizing the mass matrix m^2_{ab}, the generating functional becomes

$$Z[\vec{J}_0, \tilde{J}^{(1)}_i, \tilde{J}^{(2)}] = \exp\left(-S^{(3)}_{int}\left[\frac{\delta}{\delta J_0}, \frac{\delta}{\delta \tilde{J}} \right] \right) \mathcal{Z}[\vec{J}_0, \tilde{J}^{(1)}_i, \tilde{J}^{(2)}], \tag{6.81}$$

where $\mathcal{Z}$ is a new "free" generating functional,

$$\begin{aligned} \mathcal{Z}[\vec{J}_0, \tilde{J}^{(1)}_i, \tilde{J}^{(2)}] &= \mathcal{N} \left\langle \sqrt{\det \tilde{G}_1}^{N-1} \sqrt{\det \tilde{G}_2}\, \tilde{Z}_1[\tilde{J}^{(1)}_i]\, \tilde{Z}_2[\tilde{J}^{(2)}] \right\rangle_0^{\vec{J}_0} \\ &\equiv \left\langle \tilde{Z}_1[\tilde{J}^{(1)}_i]\, \tilde{Z}_2[\tilde{J}^{(2)}] \right\rangle_{\tilde{0}}^{\vec{J}_0}. \end{aligned} \tag{6.82}$$

Here we use the shorthand $\tilde{G}_\alpha \equiv \tilde{G}^{(m_\alpha)}, \tilde{Z}_\alpha \equiv \tilde{Z}_f\left[\tilde{J}^{(\alpha)}, m_\alpha^2\right]$ (which is the free generating functional of a single inhomogeneous field of mass m_α, normalized to $\tilde{Z}_\alpha[0] = 1$), with $\alpha = 1, 2$, and

$$\langle \dots \rangle_{\bar{0}}^{\vec{J}_0} \equiv \frac{\left\langle \sqrt{\det \tilde{G}_1}^{N-1} \sqrt{\det \tilde{G}_2} \dots \right\rangle_0^{\vec{J}_0}}{\left\langle \sqrt{\det \tilde{G}_1}^{N-1} \sqrt{\det \tilde{G}_2} \right\rangle_0}. \tag{6.83}$$

The normalization $\mathcal{N}$ factor is chosen to render $\mathcal{Z}[0] = 1$. The superindex $\vec{J}_0$ indicates that the $\bar{0}$-expectation value is taken over the zero modes in the presence of an external source $\vec{J}_0$.

If the interaction terms in $S_{int}^{(3)}$ are treated perturbatively, a new type of perturbative corrections introduced by [510] can be computed using new Feynman rules. The important differences from traditional Feynman rules are: As a consequence of the definition of the new "free" generating functional in Eq. (6.82) as a weighted average over the zero modes, there is no direct cancellation of disconnected graphs when computing perturbative corrections. Indeed, consider a correction $\Delta\tilde{Z}$ to the generating functional of the inhomogeneous modes, which at LO is just $\tilde{Z}_1\tilde{Z}_2$, prior to the zero mode average. The corrected complete (including both zero and inhomogeneous modes) generating functional $\mathcal{Z}'$ reads

$$\mathcal{Z}'[\vec{J}_0, \tilde{J}_i^{(1)}, \tilde{J}^{(2)}] = \left\langle \tilde{Z}_1[\tilde{J}_i^{(1)}]\, \tilde{Z}_2[\tilde{J}^{(2)}] + \Delta\tilde{Z}[\tilde{J}_i^{(1)}, \tilde{J}^{(2)}] \right\rangle_{\bar{0}}^{\vec{J}_0}, \tag{6.84}$$

which corrects Eq. (6.82).

Consider now a corrected n-point function of inhomogeneous fields computed from the previous expression, treating $\Delta\tilde{Z}$ perturbatively,

$$\begin{aligned} \frac{1}{\mathcal{Z}'} \frac{\delta^n \mathcal{Z}'}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0} &= \left\langle \frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{\tilde{J}=0} \right\rangle_{\bar{0}} \\ + \left\langle \frac{\delta^n\Delta\tilde{Z}}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{\tilde{J}=0} \right\rangle_{\bar{0}} &- \left\langle \Delta\tilde{Z} \right\rangle_{\bar{0}} \left\langle \frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0} \right\rangle_{\bar{0}}, \end{aligned} \tag{6.85}$$

where we used the normalization $\tilde{Z}_\alpha[0] = 1$. The first term on the right-hand side is the leading contribution obtained from Eq. (6.82), while the second and third terms are the corrections. In the usual case, the second term contains both connected and disconnected contributions, the latter of which is canceled by the third term. However, in the current situation this does not occur due to the weighting over the zero modes. Indeed,

$$\left\langle \Delta\tilde{Z} \frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0} \right\rangle_{\bar{0}} \neq \left\langle \Delta\tilde{Z} \right\rangle_{\bar{0}} \left\langle \frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0} \right\rangle_{\bar{0}}. \tag{6.86}$$

By adding and subtracting the left-hand side of the above equation to Eq. (6.86), we can identify two contributions to the correction of the n-point function as follows: a connected part

$$\Delta\langle\tilde{\phi}_{a_1}(x_1)\dots\tilde{\phi}_{a_n}(x_n)\rangle_C = \left\langle \frac{\delta^n\Delta\tilde{Z}}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{\tilde{J}=0} - \Delta\tilde{Z}\frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0}\right\rangle_{\bar{0}}, \tag{6.87}$$

which is built in the standard way with the connected Feynman diagrams using the new rules; and a 0-connected part

$$\Delta\langle\tilde{\phi}_{a_1}(x_1)\dots\tilde{\phi}_{a_n}(x_n)\rangle_{0C} = \left\langle \Delta\tilde{Z}\frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0}\right\rangle_{\bar{0}} - \left\langle\Delta\tilde{Z}\right\rangle_{\bar{0}}\left\langle \frac{\delta^n(\tilde{Z}_1\tilde{Z}_2)}{\delta\tilde{J}_{a_1}(x_1)\dots\delta\tilde{J}_{a_n}(x_n)}\bigg|_{J=0}\right\rangle_{\bar{0}}, \tag{6.88}$$

which accounts for contributions that are not connected by $\tilde{G}_\alpha$, but when written in terms of the original perturbation theory they are actually connected by correlations of the zero modes. A systematic evaluation of the integrals over the zero modes, $\langle\dots\rangle_{\bar{0}}$ can be implemented by an expansion in $1/N$ using the saddle-point approximation (Laplace method) [510].

Long Wavelength Behavior of the Two-Point Functions for Massless Fields

Once analytically continued back to LdS we can obtain the limiting value of the two-point functions for large distances/late times, $r \to \infty$. After the resummation of the bi-quadratic terms, the two-point function (without any corrections from $S^{(3)}_{int}$) is

$$\langle\phi_a(x)\phi_b(x')\rangle = \frac{\delta_{ab}}{N}\left\langle|\vec{\phi}_0|^2 + (N-1)\tilde{G}_1(r) + \tilde{G}_2(r)\right\rangle_{\bar{0}}. \tag{6.89}$$

Combining the constant contributions, we get

$$\langle\phi_a(x)\phi_b(x')\rangle^{LO} \to \frac{\delta_{ab}}{N}\left\langle\frac{2N}{\lambda}u - \left(N-1+\frac{1}{3}\right)\frac{1}{V_d u}\right\rangle_{\bar{0}} \simeq -\sqrt{\frac{2}{V_d\lambda}}\frac{4\delta_{ab}}{3N} + \mathcal{O}(N^{-2},\lambda^0), \tag{6.90}$$

where the expansion is at NLO in $1/N$. The reason why starting at this order we encounter a nonvanishing limiting value is that at NLO in $1/N$ there are diagrams contributing at LO in $\sqrt{\lambda}$ which are not of the type included in the

resummation of the bi-quadratic terms.[7] This stems from the fact that, although in EdS there is a hierarchy between interactions with $\vec{\phi}_0$ and $\vec{\tilde{\phi}}$ in terms of powers of $\sqrt{\lambda}$, upon analytic continuation to LdS, when $r \to \infty$ the correlators of inhomogeneous modes receive an enhancement that makes them as relevant as the zero mode correlators,

$$\langle |\vec{\phi}_0|^2 \rangle_{\bar{0}} \sim \frac{1}{\sqrt{\lambda}}, \qquad \langle \tilde{\phi}_a(x) \tilde{\phi}_b(x') \rangle_{\bar{0}} \to\sim \frac{\delta_{ab}}{\sqrt{\lambda}}. \tag{6.91}$$

Therefore, a consistent calculation of the two-point function at large distances in LdS, at the lowest order (LO) in $\sqrt{\lambda}$ and next-to-leading order (NLO) in $1/N$, requires further non-perturbative resummation. In [510] an additional resummation, consistent only at LO in $\sqrt{\lambda}$, but up to NLO in $1/N$ has been performed explicitly, with results showing that the two-point functions vanish as $r \to +\infty$ also at NLO in $1/N$. The strategy implemented to perform such calculation exploits the fact that the building blocks of the new Feynman rules involve the propagators $\tilde{G}_1$ and $\tilde{G}_2$ which are massive (with $|\vec{\phi}_0|$-dependent masses). This allows one to use general theorems proved in Refs. [524, 525] for massive propagators.

To understand the asymptotic IR behavior of the two-point functions, the next step is to study the decay of the full two-point function at large distances. In principle the same strategy can be applied systematically for a given order in $1/N$. This would involve the evaluation of the most relevant r-dependent parts of all the diagrams that contribute at that order. Such a calculation has not been done so far. What López Nacir, Mazzitelli and Trombetta presented in [510] are the results for the asymptotic behavior obtained after resumming only the bi-quadratic terms (with the exact treatment of the zero modes). The asymptotic scalings with r are summarized in Table 6.1 for different values of N, where also the standard perturbative results (prior to the resummation of bi-quadratic interactions) are shown for comparison.

In conclusion, the results obtained by [510] summarized in Table 6.1 show that when the bi-quadratic interactions between the zero and higher modes are treated nonperturbatively, the resummed two-point functions become convergent or less divergent in the large-distance limit, depending on the value of N. The improved behavior comes from the fact that they can be written as weighted averages of free, massive two-point functions. This was anticipated in [526] for $N = 1$. It is unclear whether additional corrections coming from the remaining diagrams that contribute at each order in $1/N$, when treated nonperturbatively, might change this behavior. This can be particularly important for small N, for which the decay given by the current resummation is milder, or absent altogether. For

[7] The family of diagrams which contribute at large-N is that of the daisy and superdaisy type, which add a local part to the self-energy, and whose leading contribution in $\sqrt{\lambda}$ is already completely taken into account by the resummation.

Table 6.1. *From [510] Asymptotic behavior of the resummed two-point function in the massless case ($m = 0$) for different values of the number of fields N as the de Sitter invariant distance $r \to +\infty$. After resummation of the bi-quadratic interaction terms, the usual logarithmic divergences present in the perturbative calculation (at all N) are softened when $N = 1, 2$ and cured for $N > 2$. Here $m^2_{\rm dyn} = \sqrt{\lambda/(2V_d)}$.*

	Free	L-loop	Resummed
$N = 1$			$\sqrt{\log(r)}$
$N = 2$			$\log(\log(r))$
$N > 2$	$\log(r)$	$\log(r)^L$	$\log(r)^{-(N-2)/2}$
$N \to \infty$			$r^{-m^2_{\rm dyn}/(d-1)}$

instance, if the higher modes were to have an additional dynamical mass coming from the interactions in $S^{(3)}_{int}$, this mechanism could dominate the IR behavior.

6.4 Nonperturbative RG. Graviton and Gauge Issues

In this last section we mention three topics on this subject: (i) the work of Guillieux and Serreau [530] using nonperturbative renormalization group techniques; (ii) the work of Moreau and Serreau [513] on the backreaction of superhorizon fluctuations of a light quantum scalar field on a classical de Sitter geometry by means of the Wilsonian renormalization group; (iii) the IR behavior of gravitons, where the gauge issue makes it more challenging. This last topic continues into the last chapters of the book.

6.4.1 Nonperturbative Renormalization Group

Nonperturbative renormalization group (NPRG) methods originally formulated in [531, 532] are useful for infrared physics from critical phenomena to long distance dynamics of non-Abelian gauge fields. They have been adopted to de Sitter spacetime [563, 564] for the study of renormalization group (RG) flow of $O(N)$ scalar field theories at superhorizon scales. Gravitationally enhanced infrared fluctuations renders the RG flow to be effectively dimensionally reduced to that of a zero-dimensional Euclidean field theory. This has various consequences, such as the radiative restoration of spontaneously broken symmetries in any spacetime dimension. The phenomenon of effective dimensional reduction allows one to establish a direct relation between the NPRG approach and the stochastic effective theory of Starobinsky and Yokoyama [505]. Guillieux and Serreau [530] show that the effective zero-dimensional field theory which results from integrating out the superhorizon degrees of freedom is equivalent to the

late-time equilibrium state in the stochastic description. They also showed that the dimensionally reduced theory in (Lorentzian) de Sitter spacetime at superhorizon scales is equivalent to the effective theory for the zero mode on the compact Euclidean de Sitter space [504, 463, 523].

For an $O(N)$ scalar field ϕ with a potential $V(\phi)$ being a function of the invariant $\phi_a\phi^a$ in an expanding flat-RW de Sitter space these authors derived a flow equation for the beta function of the effective potential $\beta(V(\phi)'', \kappa)$ as a function of the curvature of the potential (a prime denotes $d/d\phi$) or, alternatively, the order of the Bessel mode function $\nu = \sqrt{((d-1)/2)^2 - V(\phi)''}$ where d is the spacetime dimension. Both $\nu, V(\phi)''$ run with κ, an energy scale, with $\kappa > 1$ moving towards the ultraviolet and $\kappa = 0$ where all quantum fluctuations are integrated out. ($\ln\kappa$ is often called the RG 'time' in the flow equation.) Comparing it with the beta function of the same theory in Minkowski space they showed the de Sitter beta function coincides with the Minkowski one for all values of $V(\phi)''$ in the regime of subhorizon scales $\kappa \gg 1$ and for all values of κ when $V(\phi)'' \gg 1$. Spacetime curvature effects become sizable on superhorizon scales $\kappa \gg 1$ for $V(\phi)''_\kappa \approx (d-1)^2/4$. In a plot of the beta function $\beta(V(\phi)''; \kappa)$ as a function of $\ln\kappa$ for different values of the potential curvature $V(\phi)''$ [see their Fig. 2] its slope is dramatically reduced and even turns to zero for $V(\phi)'' \ll \kappa^2 \ll 1$ as a result of the gravitationally induced amplification of infrared fluctuations. This is an RG manifestation of the phenomenon of effective infrared dimensional reduction we discussed in the first part of this chapter. The authors also derived the RG flow in a d-dimensional sphere, the Euclidean de Sitter space. While the analysis of [564] was restricted to the deep infrared regime, where the flow is already dimensionally reduced, the analysis of [530] considered the complete flow from sub-horizon to super-horizon scales. This makes it possible to study how a possible broken phase in the Minkowski regime gets restored once gravitational effects become important in the infrared regime.

Recovery of Stochastic Results from NPRG Flow

It is instructive to see how the Euclidean zero mode and infrared dimensional reduction, the themes of the first part of this chapter, can be connected to the stochastic approach via the nonperturbative RG method. Infrared dimensional reduction suggests that the solution of the flow equation governed by the beta function can be written as an effective zero-dimensional field theory. Consider the following ordinary integral [530]

$$e^{-\Omega_{d+1}\mathcal{W}_\kappa(J)} = \int d^N\phi\, e^{-\Omega_{d+1}\left[V_{\rm eff}(\phi) + J_a\phi_a + \frac{\kappa^2}{2}\phi_a\phi_a\right]}, \tag{6.92}$$

where $\Omega_{d+1} = 4\pi^{(d+1)/2}/[(d-1)\Gamma((d-1)/2)]$ and $V_{\rm eff}(\phi)$ is a function to be specified below. Introduce the Legendre transform

$$V_\kappa(\phi) = \mathcal{W}_\kappa(J) - J_a\phi_a - \frac{\kappa^2}{2}\phi_a\phi_a, \tag{6.93}$$

with $\partial \mathcal{W}_\kappa(J)/\partial J_a = \phi_a$. It satisfies the flow equation for the case of small potential curvature $|V(\phi)''| \ll 1$. One can adjust the function $V_{\rm eff}(\phi)$ so as to produce the appropriate initial conditions[8] for the infrared flow at a scale $\kappa_0 \sim 1$. All solutions of the flow equation in the deep de Sitter regime can thus be written as (6.90). Notice that in this regime the original d-dimensional Lorentzian theory, with complex weight $\exp(iS)$ eventually flows to a zero-dimensional Euclidean-like integral, with real weight $\exp(-\Omega_{d+1}V_{\rm eff})$. Its physical meaning is ingrained in the dynamical finite size effect we discussed earlier.

Let us now examine the *stochastic approach* of Starobinsky and Yokoyama [505] in this light. Infrared dimensional reduction also holds the key to the utility of the effective theory for fields of long wavelength modes on superhorizon scales. Since these superhorizon modes, called $\bar\phi$ before, are almost frozen in time – recall an exponential expansion amounts to a scaling transformation in the effectively static picture – they can essentially be described by a single degree of freedom, a stochastic variable $\varphi_a(t)$ in each direction in field space, with t the cosmological time. These variables become stochastic because we have added on them the influence of the subhorizon modes of shorter wavelengths, represented by white noise in Starobinsky's model [506]. They obey the by-now familair Langevin equation

$$\partial_t \varphi_a(t) + \frac{1}{d-1}\frac{\partial V_{\rm soft}(\varphi)}{\partial \varphi_a(t)} = \xi_a(t), \tag{6.94}$$

where $V_{\rm soft}(\varphi)$ is the potential seen by the long wavelength modes. Treating the short wavelength modes as noninteracting fields in the Bunch–Davies vacuum, one has, generalizing the calculation of [505, 523] to arbitrary N,

$$\langle \xi_a(t)\xi_b(t')\rangle = \frac{\Gamma((d-1)/2)}{2\pi^{(d+1)/2}}\delta_{ab}\delta(t-t'). \tag{6.95}$$

Using standard manipulations, (6.94) can be turned into the following Fokker–Planck equation for the probability distribution $\mathcal{P}(\varphi, t)$ of the stochastic process

$$\partial_t \mathcal{P} = \frac{1}{d-1}\frac{\partial}{\partial \varphi_a}\left\{\frac{\partial V_{\rm soft}}{\partial \varphi_a}\mathcal{P} + \frac{1}{\Omega_{d+1}}\frac{\partial \mathcal{P}}{\partial \varphi_a}\right\}. \tag{6.96}$$

The latter admits an $O(N)$-symmetric stationary attractor solution at late times (i.e., in the deep infrared), given by

$$\mathcal{P}(\varphi) \propto \exp\big\{-\Omega_{d+1}V_{\rm soft}(\varphi)\big\}. \tag{6.97}$$

Equal-time correlation functions on superhorizon scales can then be computed as moments of this distribution. This is the point of contact one can make between

[8] In the case $N = 1$, one can show that $V_{\rm eff}(\phi) \approx V_{\kappa_0}(\phi)$ if $V''_{\rm eff}(\phi) \ll \kappa_0^2$. For arbitrary N, the inequality should be satisfied by the largest eigenvalue of the curvature matrix $\partial^2 V_{\rm eff}(\phi)/\partial\phi_a\partial\phi_b$.

the stochastic approach and the RG analysis described above, namely, (6.92) in the limit κ_0 provided one identifies $V_{\rm soft}(\varphi) = V_{\rm eff}(\varphi) \approx V_{\kappa_0}(\varphi)$. For instance, one has

$$\langle \varphi_a \varphi_b \rangle = \frac{\int d^N\varphi \; \varphi_a \varphi_b \, \mathcal{P}(\varphi)}{\int d^N\varphi \; \mathcal{P}(\varphi)} = \frac{1}{\Omega_{d+1}} \frac{\partial^2 \mathcal{W}_{\kappa=0}(J)}{\partial J_a \partial J_b}\bigg|_{J=0} . \tag{6.98}$$

Note that the relevant potential used in the stochastic approach is not the one at the UV scale Λ but the one evolved down to the horizon scale $\kappa \to 0$, a clear manifestation that this theory is designed for the infrared regime.

Going beyond the Local Effective Potential by Derivative Expansion

As was mentioned at the beginning of this chapter, to study the symmetry behavior of quantum fields in a dynamical spacetime one needs to take into consideration the time-varying background field. The effective potential which assumes a constant background field $\bar{\phi}$ is only the zeroth order approximation in a derivative expansion of the background field, the first order capturing slow variations. This was discussed in Chapter 2 under the topic of quasi-local effective Lagrangian following [375]. Now for the IR problem in de Sitter space one would expect that the late time behavior of the inflaton may not be so sensitive to this correction because the modes have virtually all been red-shifted exponentially fast, and the differences arising from the time variation of the background field may show up as a small logarithmic correction. This aspect has been addressed by Guillieu and Serreau [512] who introduced a so-called 'local potential approximation prime' which includes a running (but field-independent) field renormalization to the full effective potential. They explicitly computed the associated anomalous dimension for $O(N)$ theories and find that it can take on large values along the flow, leading to sizable differences as compared to the local potential approximation. However, it does not prevent the phenomenon of gravitationally-induced dimensional reduction. As a consequence, the effective potential at the end of the flow is unchanged as compared to the local potential approximation, the main effect of the running anomalous dimension being merely to slow down the flow.

Stability of dS against Backreaction of Superhorizon Quantum Fluctuations

We close this topic by mentioning the recent work of Moreau and Serreau [513] who have investigated the backreaction of a light quantum scalar field on a de Sitter geometry by means of NPRG techniques. They found a nontrivial renormalization of the spacetime curvature as superhorizon fluctutations are progressively integrated out. Perturbative loop corrections grow unbounded as a result of the gravitational amplification of such fluctuations. This signals the

breakdown of perturbation theory rather than an instability. Nonperturbative effects come into play with, in particular, the dynamical generation of a mass, which screens the growth of superhorizon fluctuations and freezes the RG flow of the effective spacetime curvature. Overall, the infrared renormalisation of the latter is controlled by the gravitational coupling which is small by assumption in their semiclassical treatment. This work adds weight to the belief that de Sitter spacetime is stable against infrared quantum fluctuations.

6.4.2 IR Divergence in Graviton and the Gauge Issue

On this challenging subject we highlight some key issues involved without belaboring, but point to the sources where one can find more details of the topics covered.

Graviton's IR Behavior

Quantized linear gravitational perturbations, otherwise known as gravitons, have two polarizations, each obeying an equation of motion identical to that of a massless minimally coupled scalar field [345, 346]. For this reason one may consider their infrared behavior to be identical. However, there are significant differences between a scalar field and the gravitational perturbations [491]. For one, the (partial) derivative character of the gravitational interaction improves the IR behavior over that of massless minimally coupled matter fields. A more distinct difference is the need to consider appropriate gauge-invariant observables for gravitons, which is a rather nontrivial aspect even in perturbative quantum gravity. Moreover, one should restrict one's attention to "sufficiently local" observables [565] that properly characterize the geometrical properties within a region of finite physical size. This is a crucial factor for the construction of IR-safe observables in situations which would otherwise lead to divergences in the absence of an IR cut-off [566, 567].

The gauge issue. It is essential to distinguish whether these IR divergences are restricted to the gauge sector of linearized gravity, or they appear also in the physical sector. Allen [568] and Higuchi [516] have shown that in the traceless-transverse-synchronous gauge the IR-divergent part of the graviton two-point function can be expressed in a nonlocal pure-gauge form. Higuchi, Marolf and Morrison [569] observed that a local gauge transformation on the graviton modes is sufficient to eliminate the IR divergences plaguing the graviton two-point function in that gauge. Other gauges and coordinate systems in which the graviton two-point function is IR finite have been worked out. See, e.g., references cited in [570].

Another way to examine the gauge nature of these IR divergences is through the linearized Weyl tensor, which is a local and gauge-invariant observable in the linearized theory. It was shown that the two-point function of the linearized

Weyl tensor is IR finite even if computed using a de Sitter noninvariant graviton two-point function with an IR cutoff [571, 572].

Residual Gauge conditions. In cosmological perturbation theory, one usually chooses gauge conditions on the entire constant time slice. But from an observational perspective one can only see within a causally connected region which changes in time. To regularize the IR contributions for the curvature perturbation, it is necessary to take into account this subtle issue. (See the series of papers by Tanaka and Yurakawa cited in [529].)

Since only physical quantities in the observable region are of relevance, one should compute observable quantities unaffected by what is outside of the observable region. The choice of gauge conditions should respect the compatibility of the 'outside' degrees of freedom with what is observed. These so-called 'residual' gauge conditions can be linked to the degrees of freedom in the boundary conditions of our observable local universe. It was shown that requesting the invariance under the change of such residual coordinate degrees of freedom in the local universe can ensure the IR regularity and the absence of the secular growth. The otherwise singular IR contributions are subsumed in the residual coordinate degrees of freedom [573]. On the issue whether the residual coordinate degrees of freedom can also affect the IR behavior of the graviton Tanaka and Urakawa [529] showed that when invariance under the residual coordinate transformations are required, the IR regularity and the absence of secular growth are also guaranteed for the graviton loops.[9]

Nonlinear effects. Measurement of primordial gravitational waves can provide information which cannot be obtained by the measurement of scalar perturbations. It could also provide finer discrimination between different models of inflation. Unlike the amplitude of the adiabatic curvature perturbation, the amplitude of the gravitational waves is sensitive to the detailed dynamics of the inflationary universe, and thus provide the energy scale of inflation. Measurements in the primordial gravitational waves for non-linear perturbations [576, 577] or interactions [578, 579] can for example reveal the impact of parity violation in the gravity sector on the bi-spectrum of the primordial gravitational waves.

[9] This is contested by Tsamis and Woodard and co-workers who have done extensive perturbative calculations on two-loop graviton contributions. They maintain that the secular behavior is not a gauge effect but is physical and can have important consequences such as causing the decay of the cosmological constant. See, e.g., the exchanges between Tsamis and Woodard [574] on the one side versus Garriga and Tanaka on the other [575].

Graviton Loop Corrections

We continue with the narrative in [529] on this issue. Since a massless scalar field yields a scale-invariant spectrum in the IR limit as $P(k) \propto 1/k^3$, a naive loop integral yields a factor $\int d^3k/k^3 \approx \int dk/k$, which diverges logarithmically. Free gravitons are expected to have the same IR behavior, as each of the two polarizations behaves like a massless minimally coupled scalar field. As the graviton loop corrections show divergences, some regularization scheme need be applied. A simple-minded way is to introduce an IR cutoff, say, at a comoving scale $k_{\rm IR}$. However, this will not provide a satisfactory solution, because the loop integral of the super-Hubble modes gives $\int_{k_{\rm IR}}^{aH} dk/k \approx \ln(aH/k_{\rm IR})$, which logarithmically increases in time. (Here a is the scale factor and H is the Hubble parameter of the background spacetime.) Compared with the tree-level contribution, the loop corrections are typically suppressed by the Planck scale as $(H/M_P)^2$ with $M_P^2 \equiv (8\pi G_{\rm N})^{-1}$.

Fröb, Roura and Verdaguer [491] calculated the one-loop correction to the tensorial metric perturbations, the key ingredient to obtain the 'sufficiently local' observables [565]. (The scalar and vectorial metric perturbations can be directly obtained from the stress-energy tensor correlation function.) By employing a large N expansion for N matter fields interacting with the gravitational field [190] they managed to avoid the consideration of graviton internal loops (the lowest-order contributions to the connected two-point function of the metric perturbations are of order $1/N$, whereas any contribution including graviton loops is suppressed by higher powers of $1/N$).

7

Advanced Field Theory Topics

In prior chapters various techniques of quantum field theory in curved spacetimes have been presented to tackle different types of problems in semiclassical gravity: The 'in-out' effective action for a self-consistent treatment of quantum processes such as particle creation and their backreaction on the dynamics of the background spacetime; the 'in-in' effective action for obtaining real and causal equations of motion showing dissipative dynamics; the large N expansion and the two-particle-irreducible effective action for eliciting the infrared behavior of interacting quantum fields, and the effects of spacetime curvature and topology on symmetry breaking. In this chapter we formalize these techniques and methodologies and collect them into a toolbox which can be used to tackle a range of problems, going beyond those discussed so far.

Again, in keeping with the philosophy stated at the beginning, we try not to present ideas, techniques or methods in abstract, but through model studies and the actual working out of concrete problems. Here we construct the CTP 2PI effective action to two-loop order for two types of quantum fields: (a) $O(N)$ self-interacting $\lambda\Phi^4$ fields, where the method of $1/N$ expansion in field theory is used, and (b) Yukawa coupling between scalar and spinor fields in curved spacetime, as an example of treating fermions. We have seen how the 2PI effective action formalism is suited for the treatment of infrared behavior of interacting quantum fields in the last chapter, and we will see how the results obtained here are put to use for treating the post-inflation, preheating problem in the next chapter.[1]

[1] The contents of this chapter base on [244] are more or less self-contained for ease of referencing. The more advanced readers who have a good background knowledge of quantum field theory in curved spacetime (from monographs such as [47, 53]) can start from here, pick up specific topics as needed from prior chapters and delve into semiclassical and stochastic gravity. For example, with the materials in this chapter plus the second halves of Chapters 4 and 5 for the fluctuations of the quantum fields from the correlations of the stress-energy tensor, they can begin to tackle the *backreaction and fluctuations* problems of quantum fields in curved spacetime, developed in Parts III and IV of this book.

7.1 2PI CTP Effective Action in Curved Spacetime

7.1.1 Recap of CTP formalism

In the closed-time-path formulation the classical action $S^F[\phi]$ for a quantum scalar field ϕ in a manifold M is generalized to

$$\mathcal{S}^F[\phi_+, \phi_-] = S^F[\phi_+] - S^F[\phi_-], \tag{7.1}$$

where ϕ_+ and ϕ_- denote the ϕ field on the $+$ and $-$ time branches respectively, joint at the hypersurface $x^0 = x^0_\star$ assumed to be far to the future of any dynamics we are interested in

$$\phi_+(x)\Big|_{x^0_\star} = \phi_-(x)\Big|_{x^0_\star}. \tag{7.2}$$

Thus the name 'closed' time-path.

The generating functional of vacuum n-point functions (i.e., expectation values in the $|0, \text{in}\rangle$ vacuum) for this theory is defined by

$$Z[J_+, J_-] = \int_{\text{CTP}} D\phi_+ D\phi_- \ \exp\left\{\frac{i}{\hbar}\left[\mathcal{S}^F[\phi_+, \phi_-] + \int_M d^4x (J_+\phi_+ - J_-\phi_-)\right]\right\}, \tag{7.3}$$

where J_+ and J_- are c-number sources on the $+$ and $-$ branches of the closed-time path. The designation "CTP" indicates that the functional integrals in Eq. (7.3) are over all field configurations (ϕ_+, ϕ_-) such that (i) $\phi_+ = \phi_-$ at the $x^0 = x^0_\star$ hypersurface and (ii) ϕ_+ (ϕ_-) consists of pure negative (positive) frequency modes at $x^0 = -\infty$. It is not necessary for the normal derivatives of ϕ_+ and ϕ_- to be equal at $x^0 = x^0_\star$. Because the theory is free in the asymptotic past, a positive frequency mode [2] is a solution to the spatial-Fourier transformed Euler–Lagrange equation for ϕ whose asymptotic behavior at $x^0 = -\infty$ is $\exp(-i\omega x^0)$, for $\omega > 0$.

The generating functional for connected diagrams is then defined by

$$W[J_+, J_-] = -i\hbar \ln Z[J_+, J_-]. \tag{7.4}$$

Mean fields on both $+$ and $-$ branches are then defined as

$$\bar{\phi}_a^{(J_\pm)}(x) = c^{ab} \frac{\delta W[J_+, J_-]}{\delta J_b(x)}, \tag{7.5}$$

where a, b are time branch indices with index set $\{+, -\}$. The matrix c^{ab} is defined by $c^{++} = 1, c^{--} = -1$, and $c^{+-} = c^{-+} = 0$. The functional differentiation in Eq. (7.5) is carried out with variations in δJ_+ and δJ_- which satisfy the constraint

[2] Here, the choice of vacuum boundary conditions corresponds to adding a small imaginary part $i\epsilon(\phi_+^2 - \phi_-^2)$ to the classical action $\mathcal{S}^{\text{F}}$. Alternatively, the boundary conditions correspond to the usual prescription $m^2 \to m^2 - i\epsilon$ in $S^{\text{F}}[\phi]$, but with S_{ctp} now redefined as $\mathcal{S}^{\text{F}}[\phi_+, \phi_-] = S^{\text{F}}[\phi_+] - S^{\text{F}}[\phi_-]^\star$, where superscripted $\star$ denotes complex conjugation [182].

that $\delta J_+ = \delta J_-$ on the $x^0 = x^0_\star$ hypersurface. The $J_\pm$ superscript in Eq. (7.5) indicates the functional dependence on $J_\pm$, which has been shown to be causal [181, 180]. In the limit $J_+ = J_- \equiv J$, the mean fields on the + and − time branches become equal,

$$\bar{\phi}^{(J_\pm)}_+(x)\Big|_{J_+=J_-\equiv J} = \bar{\phi}^{(J_\pm)}_-(x)\Big|_{J_+=J_-\equiv J} \equiv \bar{\phi}^{(J)}(x) = {}_J\langle 0,\mathrm{in}|\Phi(x)|0,\mathrm{in}\rangle_J, \quad (7.6)$$

where $|0,\mathrm{in}\rangle$ is the state which has evolved from the vacuum at t_0 under the interaction $\Phi\, J$, and becomes the expectation value $\langle\Phi\rangle$ in the limit $J = 0$. The effective action is defined via the usual Legendre transform, with c^{ab} now acting as a "metric" on the internal 2×2 CTP field space,

$$\Gamma[\bar{\phi}_+, \bar{\phi}_-] = W[J_+, J_-] - c^{ab} \int_M d^4x\; J_a(x)\bar{\phi}_b(x), \quad (7.7)$$

where the J superscripts on $\bar{\phi}_\pm$ are suppressed and the functional dependence of $J_\pm$ on $\bar{\phi}$ via inversion of Eq. (7.5) is understood. By direct computation, the inverse of Eq. (7.5) is found to be

$$J^{(\bar{\phi}_\pm)}_a(x) = -c_{ab}\, \frac{\delta\Gamma[\bar{\phi}_+, \bar{\phi}_-]}{\delta\bar{\phi}_b(x)}, \quad (7.8)$$

where we have indicated the explicit functional dependence of $J_\pm$ on $\bar{\phi}_\pm$ with a superscript, and c_{ab} is the inverse of the matrix c^{ab} defined above. In the limit $\bar{\phi}_+ = \bar{\phi}_- \equiv \bar{\phi}$, this yields the evolution equation for the expectation value ${}_J\langle\Phi\rangle_J \equiv \bar{\phi}_J$ in the state which has evolved from $|0,\mathrm{in}\rangle$ under the source interaction $\Phi\, J$. The evolution equation for $\bar{\phi}$, the vacuum expectation value $\langle 0,\mathrm{in}|\Phi|0,\mathrm{in}\rangle$, is therefore

$$\frac{\delta\Gamma[\bar{\phi}_+, \bar{\phi}_-]}{\delta\bar{\phi}_+}\Big|_{\bar{\phi}_+=\bar{\phi}_-\equiv\bar{\phi}} = -\frac{\delta\Gamma[\bar{\phi}_+, \bar{\phi}_-]}{\delta\bar{\phi}_-}\Big|_{\bar{\phi}_+=\bar{\phi}_-\equiv\bar{\phi}} = 0. \quad (7.9)$$

Using Eqs. (7.8) and (7.7), an integro-differential equation for Γ can be derived [181], in which the functional differentiations of Γ with respect to $\bar{\phi}_\pm$ are carried out with the constraint that the variations of $\bar{\phi}_\pm$ satisfy $\delta\bar{\phi}_+ = \delta\bar{\phi}_-$ when $x^0 = x^0_\star$. The difference $\phi_a - \bar{\phi}_a$ is naturally interpreted as the fluctuations of a particular history ϕ_a about the mean field configuration $\bar{\phi}_a$. Let us, therefore, define the *fluctuation field* $\varphi_a \equiv \phi_a - \bar{\phi}_a$ or, in terms of field operators,

$$\varphi \equiv \Phi - \langle\Phi\rangle = \Phi - \bar{\phi}, \quad (7.10)$$

where angular brackets around the field operator Φ denote an expectation value of Φ in the (time-independent) quantum state of the system. Performing the change of variables $\phi_a \to \varphi_a$ in the functional integral, where

$$\varphi_a \equiv \Phi_a - \bar{\phi}_a, \quad (7.11)$$

we obtain

$$\Gamma[\bar{\phi}_+,\bar{\phi}_-] = -i\hbar \ln\left\{\int_{\rm CTP} D\varphi_+ D\varphi_- \right.$$
$$\left. \times \exp\left[\frac{i}{\hbar}\left(\mathcal{S}^F[\bar{\phi}_+ + \varphi_+, \bar{\phi}_- + \varphi_-] - \frac{\delta\Gamma[\bar{\phi}_+,\bar{\phi}_-]}{\delta\bar{\phi}_a}\varphi_a\right)\right]\right\}. \quad (7.12)$$

This functional integro-differential equation has a formal solution [495]

$$\Gamma[\bar{\phi}_+,\bar{\phi}_-] = \mathcal{S}^F[\bar{\phi}_+,\bar{\phi}_-] - \frac{i\hbar}{2}\ln\,\det(\mathcal{A}^{-1}_{ab}) + \Gamma_1[\bar{\phi}_+,\bar{\phi}_-], \quad (7.13)$$

where $\mathcal{A}^{ab}(x,x')$, the second functional derivative of the classical action with respect to the field $\phi_\pm$, is

$$i\mathcal{A}^{ab}(x,x') = \frac{\delta^2\mathcal{S}^F[\bar{\phi}_+,\bar{\phi}_-]}{\delta\phi_a(x)\phi_b(x')}. \quad (7.14)$$

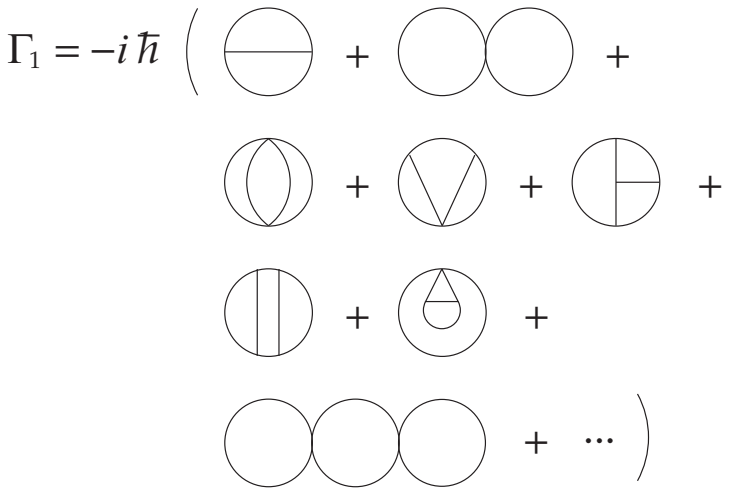
The inverse of $\mathcal{A}^{ab}$ is the one-loop correlator for the fluctuation field φ. The functional Γ_1 in Eq. (7.12) is defined as $-i\hbar$ times the sum of all one-particle-irreducible (1PI) vacuum-to-vacuum graphs with correlator given by $\mathcal{A}^{-1}_{ab}(x,x')$ and vertices given by a "shifted action" $\mathcal{S}^F_{\rm int}$, defined by

$$\mathcal{S}^F_{\rm int}[\varphi_+,\varphi_-] = \mathcal{S}^F[\varphi_+ + \bar{\phi}_+, \varphi_- + \bar{\phi}_-] - \mathcal{S}^F[\bar{\phi}_+,\bar{\phi}_-] - \int_M d^4x \left(\frac{\delta\mathcal{S}^F[\bar{\phi}_\pm]}{\delta\phi_a}\right)\varphi_a$$
$$- \frac{1}{2}\int_M d^4x \int_M d^4x' \left(\frac{\delta^2\mathcal{S}^F[\bar{\phi}_\pm]}{\delta\phi_a(x)\delta\phi_b(x')}\right)\varphi_a(x)\varphi_b(x'). \quad (7.15)$$

For simplicity, we do not explicitly indicate the functional dependence of $\mathcal{S}^F_{\rm int}$ on $\bar{\phi}_\pm$.

For the $O(N)$ self-interacting $\lambda\Phi^4$ field under study Fig. 7.1 shows the diagrammatic expansion for Γ_1. Each vertex carries a spacetime label in M and a time branch label in $\{+,-\}$. The lowest-order contribution is order $\hbar^2$, i.e., at two loops. The correlator $\mathcal{A}^{-1}$ does not depend on $\hbar$. The $\ln\det\mathcal{A}$ term in Eq. (7.13) is the one-loop (order $\hbar$) term in the CTP effective action. The CTP effective action, as a functional of $\bar{\phi}_\pm$, can be computed to any desired order in the loop expansion using Eq. (7.13). In general, this action contains divergences at each order in the loop expansion, which need to be renormalized.

Functionally differentiating $\Gamma[\bar{\phi}_+,\bar{\phi}_-]$ with respect to either $\bar{\phi}_+$ or $\bar{\phi}_-$ and making the identification $\bar{\phi}_+ = \bar{\phi}_- = \bar{\phi}$ [as shown in Eq. (7.9)] yields a dynamical, real, and causal evolution equation for the mean field $\bar{\phi}$. Thus the 1PI effective action $\Gamma[\bar{\phi}_\pm]$ yields *mean-field* dynamics for the theory, which is a lowest-order truncation of the correlation hierarchy [580, 581]. However, to follow the non-perturbative growth of quantum fluctuations linked to the mean-field dynamics


$\Gamma_1 = -i\hbar$
+
+
+

mean field $\bar{\phi}$, and setting $\delta\Gamma/\delta C = 0$ yields an evolution equation for C, the "gap" equation. The variance $\langle\varphi^2\rangle$ is the coincidence limit of the two-point function $\hbar\, C$, as seen from (7.16). As we have learned before, the closed-time-path method should be used in conjunction with the 2PI formalism in order to obtain real and causal dynamics for $\bar{\phi}$ and C [439, 580]. The 2PI effective action is placed at the second order in the correlation hierarchy generated by the master effective action [581, 582]. We give a formal treatment of the 2PI method below. For the original contribution see [494], for application to nonequilibrium quantum field theory see [439, 445].

7.1.3 2PI In-In Effective Action for $\lambda\Phi^4$ Theory in Minkowski Space

We develop the 2PI formalism for a scalar $\lambda\Phi^4$ theory with vacuum initial conditions first in Minkowski space, then generalize it to curved spacetime in the next subsection.

At the 2PI level, the field is coupled to two sources of c-number functions: a local source $J_a(x)$ via $\hbar\, c^{ab} J_a \Phi_a$ and a nonlocal source $K_{ab}(x,x')$ via $\hbar\, c^{ab} c^{cd} K_{ac}(x,x')\Phi_b(x)\Phi_d(x')$ interactions. Following (7.3), the CTP generating functional is defined as a vacuum persistence amplitude in the presence of the sources J and K, which has the path integral representation

$$Z[J,K] = \int_{\rm CTP} D\Phi_+ D\Phi_-\ \exp\left\{\frac{i}{\hbar}\left[\mathcal{S}^F[\phi_+,\phi_-] + \int_M d^4x\ c^{ab} J_a \Phi_b \right.\right.$$
$$\left.\left. + \frac{1}{2}\int_M d^4x \int_M d^4x'\ c^{ab} c^{cd} K_{ac}(x,x')\Phi_b(x)\Phi_d(x')\right]\right\}. \tag{7.17}$$

Here, $\mathcal{S}^F$ is as defined in (7.1), and we are using $Z[J,K]$ as a shorthand for $Z[J_+,J_-;K_{++},K_{--},K_{+-},K_{-+}]$. The generating functional for normalized n-point functions (connected diagrams) is defined by

$$W[J,K] = -i\hbar \ln Z[J,K]. \tag{7.18}$$

The mean field $\bar{\phi}_a^{(JK)}(x)$ and two-point correlation function $C_{ab}^{(JK)}(x,x')$ are then given by

$$\bar{\phi}_a^{(JK)}(x) = c_{ab}\frac{\delta W[J,K]}{\delta J_b(x)}, \tag{7.19}$$

$$\hbar\, C_{ab}^{(JK)}(x,x') = 2c_{ac}c_{bd}\frac{\delta W[J,K]}{\delta K_{cd}(x,x')} - \bar{\phi}_a^{(JK)}(x)\bar{\phi}_b^{(JK)}(x'), \tag{7.20}$$

where we use the superscript 'JK' to indicate that $\bar{\phi}_a$ and C_{ab} are functionals of the sources J and K.

In the limit $K = J = 0$, the mean field $\bar{\phi}_a$ satisfies

$$\bar{\phi}_+\Big|_{J=K=0} = \bar{\phi}_-\Big|_{J=K=0} = \langle\phi|\Phi|\phi\rangle \equiv \bar{\phi}, \tag{7.21}$$

i.e., it becomes the expectation value of the field operator Φ in the quantum state $|\phi\rangle$. In the same limit, the two-point function C_{ab} is the CTP correlator for the fluctuation field defined by (7.11). The four components of the CTP correlator are, for $J = K = 0$,

$$\hbar\, C_{++}(x,x')\Big|_{J=K=0} = \langle\phi|\,\mathcal{T}\,\{\varphi(x)\varphi(x')\}\,|\phi\rangle, \tag{7.22}$$

$$\hbar\, C_{--}(x,x')\Big|_{J=K=0} = \langle\phi|\tilde{\mathcal{T}}\,\{\varphi(x)\varphi(x')\}\,|\phi\rangle, \tag{7.23}$$

$$\hbar\, C_{+-}(x,x')\Big|_{J=K=0} = \langle\phi|\varphi(x')\varphi(x)|\phi\rangle, \tag{7.24}$$

$$\hbar\, C_{-+}(x,x')\Big|_{J=K=0} = \langle\phi|\varphi(x)\varphi(x')|\phi\rangle. \tag{7.25}$$

In the coincidence limit $x' = x$, all four components above are equivalent to the variance $\langle\varphi^2\rangle$ defined in (7.16). Provided we can invert (7.19) and (7.20) to obtain J and K in terms of $\bar{\phi}$ and C, the 2PI effective action can be defined as the double Legendre transform (in both J and K) of $W[J,K]$,

$$\begin{aligned}\Gamma[\bar{\phi},C] = W[J,K] &- \int_M d^4x\; c^{ab}J_a(x)\bar{\phi}_b(x) \\ &- \frac{1}{2}\int_M d^4x \int_M d^4x'\; c^{ab}c^{cd}K_{ac}(x,x')\Big[\hbar\, C_{bd}(x,x') + \bar{\phi}_b(x)\bar{\phi}_d(x')\Big].\end{aligned} \tag{7.26}$$

where $\Gamma[\bar{\phi},C]$ is a shorthand for $\Gamma[\bar{\phi}_+,\bar{\phi}_-;\,C_{++},C_{--},C_{+-},C_{-+}]$. The JK superscripting of $\bar{\phi}$ and C is suppressed, but the functional dependence of $\bar{\phi}$ and C on J and K through inversion of (7.19) and (7.20) is understood. By direct functional differentiation of (7.26), the inverses of equations (7.19) and (7.20) are found to be

$$\frac{\delta\Gamma[\bar{\phi},C]}{\delta\bar{\phi}_a(x)} = -c^{ab}J_b^{(\bar{\phi}C)}(x) - \frac{1}{2}c^{ab}c^{cd}\int_M d^4x'\;\Big[K_{bd}^{(\bar{\phi}C)}(x,x') + K_{db}^{(\bar{\phi}C)}(x',x)\Big]\bar{\phi}_c(x'), \tag{7.27}$$

$$\frac{\delta\Gamma[\bar{\phi},C]}{\delta C_{ab}(x,\xi')} = -\frac{\hbar}{2}c^{ac}c^{bd}K_{cd}^{(\bar{\phi}C)}(x,x'), \tag{7.28}$$

where the superscript '$\bar{\phi}C$' indicates that K and J are functionals of $\bar{\phi}$ and C. Once $\Gamma[\bar{\phi},C]$ has been calculated, the evolution equations for $\bar{\phi}$ and C are given by:

$$\frac{\delta\Gamma[\bar{\phi},C]}{\delta\bar{\phi}_a(x)}\bigg|_{\bar{\phi}_+=\bar{\phi}_-\equiv\bar{\phi}} = 0, \tag{7.29}$$

$$\frac{\delta\Gamma[\bar{\phi},C]}{\delta C_{ab}(x,y)}\bigg|_{\bar{\phi}_+=\bar{\phi}_-\equiv\bar{\phi}} = 0. \tag{7.30}$$

Of course, the two equations contained in (7.29) are not independent, just as in (7.9). In addition, only two of equations (7.30) are independent, one on the diagonal and one off-diagonal in the "internal" CTP space. Using both (7.26) and (7.19), an equation for $\Gamma[\bar{\phi}, C]$ in terms of the sources K and J can be derived,

$$\begin{aligned}\Gamma[\bar{\phi}, C] = -i\hbar \ln\Bigg[\int_{\text{CTP}} D\Phi_+ D\Phi_- \ \exp\Bigg\{\frac{i}{\hbar}\Big(\mathcal{S}^F[\Phi_+, \Phi_-] \\ + c^{ab}\int_M d^4x\ J_a(x)\big[\Phi_b(x) - \bar{\phi}_b(x)\big] \\ + \frac{1}{2}\,c^{ac}c^{bd}\int_M d^4x \int_M d^4x'\ K_{ab}(x,x') \\ \times \big[\Phi_c(x)\Phi_d(x') - \bar{\phi}_c(x)\bar{\phi}_d(x') - \hbar\, C_{cd}(x,x')\big]\Bigg\}\Bigg]. \end{aligned} \tag{7.31}$$

The sources K and J on the right hand side of (7.31) are functionals of $\bar{\phi}$, through (7.28). Expressing this functional dependence, we obtain a functional integro-differential equation for Γ,

$$\begin{aligned}\Gamma[\bar{\phi}, C] = \int_M d^4x \int_M d^4x'\ \frac{\delta\Gamma[\bar{\phi}, C]}{\delta C_{ba}(x', x)} C_{ab}(x, x') \\ - i\hbar \ln\Bigg[\int_{\text{CTP}} D\phi_+ D\phi_-\ \exp\Bigg\{\frac{i}{\hbar}\Big(\mathcal{S}^F[\Phi_+, \Phi_-] - \int_M d^4x\ \frac{\delta\Gamma[\bar{\phi}, C]}{\delta\bar{\phi}_a}(\Phi_a - \bar{\phi}_a) \\ - \frac{1}{\hbar}\int_M d^4x \int_M d^4x'\ \frac{\delta\Gamma[\bar{\phi}, C]}{\delta C_{ba}(x', x)}\big[\Phi_a(x) - \bar{\phi}_a(x)\big]\big[\Phi_b(x') - \bar{\phi}_b(x')\big]\Big)\Bigg\}\Bigg]. \end{aligned} \tag{7.32}$$

We have dropped the JK subscripting because the functional derivatives in the equation are only with respect to $\bar{\phi}$ and C. A change of variables $D\Phi_\pm \to D\varphi_\pm$ is carried out in the functional integral. The resulting equation,

$$\begin{aligned}\Gamma[\bar{\phi}, C] = \int_M d^4x \int_M d^4x'\ \frac{\delta\Gamma[\bar{\phi}, C]}{\delta C_{ba}(x', x)}\, C_{ab}(x, x') \\ - i\hbar \ln\Bigg[\int_{\text{CTP}} D\varphi_+ D\varphi_-\ \exp\Bigg\{\frac{i}{\hbar}\Big[\mathcal{S}^F[\varphi_+ + \bar{\phi}_+, \varphi_- + \bar{\phi}_-] \\ - \int_M d^4x \frac{\delta\Gamma[\bar{\phi}, C]}{\delta\bar{\phi}_a}\,\varphi_a - \frac{1}{\hbar}\int_M d^4x \int_M d^4x'\ \frac{\delta\Gamma[\bar{\phi}, C]}{\delta C_{ba}(x', x)}\,\varphi_a(x)\varphi_b(x')\Big]\Bigg\}\Bigg], \end{aligned} \tag{7.33}$$

has the formal solution [494]

$$\Gamma[\bar{\phi}, C] = \mathcal{S}^F[\bar{\phi}] - \frac{i\hbar}{2} \ln \det(C_{ab}) + \frac{i\hbar}{2} \int_M d^4x \int_M d^4x' \; \mathcal{A}^{ab}(x', x) C_{ab}(x, x') + \Gamma_2[\bar{\phi}, C], \tag{7.34}$$

where $\mathcal{A}^{ab}$ is the second functional derivative of the classical action $\mathcal{S}^F$, evaluated at $\bar{\phi}_a$

$$i\,\mathcal{A}^{ab}(x, x') = \frac{\delta^2 \mathcal{S}^F[\bar{\phi}]}{\delta \bar{\phi}_a(x) \bar{\phi}_b(x')} \tag{7.35}$$

The functional Γ_2 is $-i\hbar$ times the sum of all two-particle-irreducible vacuum-to-vacuum diagrams with lines given by C_{ab} and vertices given by a shifted action $\mathcal{S}^F_{int}$,

$$\mathcal{S}^F_{int}[\varphi] = \mathcal{S}^F[\varphi + \bar{\phi}] - \mathcal{S}^F[\bar{\phi}] - \int_M d^4x \; \left(\frac{\delta \mathcal{S}^F}{\delta \phi_a}[\bar{\phi}] \right) \varphi_a - \frac{i}{2} \int_M d^4x \int_M d^4x' \; \mathcal{A}^{ab}(x, x') \varphi_a(x) \varphi_b(x'). \tag{7.36}$$

The shifted action for the $\lambda\phi^4$ scalar field theory is

$$\mathcal{S}^F_{int}[\varphi] = S^F_{int}[\varphi_+] - S^F_{int}[\varphi_-], \tag{7.37}$$

in terms of

$$S^F_{int}[\varphi] = -\lambda \int_M d^4x \; \left(\frac{1}{4!} \varphi^4 + \frac{1}{3!} \bar{\phi} \varphi^3 \right), \tag{7.38}$$

where the functional dependence of $\mathcal{S}^F_{int}$ on $\bar{\phi}_\pm$ is not shown explicitly. Two types of vertices appear in (7.38); a vertex which terminates four lines, and a vertex terminating three lines which is proportional to the mean field $\bar{\phi}$. The expansion for Γ_2 in terms of G and $\bar{\phi}$ is depicted graphically up to three-loop order in Fig. 7.2. Each vertex carries a spacetime label in M and time branch label in $\{+, -\}$. In general, the 2PI effective action contains divergences at each order in the loop expansion. It has been shown formally that if the field theory is renormalizable in the "in-out" formulation, then the "in-in" equations of motion are renormalizable [439]. In the closed-time-path formalism it is easier to carry out explicit renormalization in the equations of motion, i.e., the mean-field and gap equations, which we will do in Section 8.3.2. Various approximations to the full quantum dynamics can be obtained by truncating the diagrammatic expansion for Γ_2. Throwing away Γ_2 in its entirety would yield the one-loop approximation. In Fig. 7.2, there are two two-loop diagrams, the "double bubble" and the "setting sun." Retaining just the double bubble diagram yields the time-dependent Hartree–Fock approximation [494]. Retaining both diagrams gives a

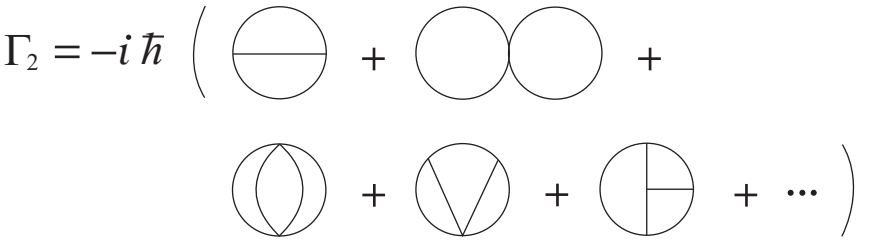


Figure 7.2 Diagrammatic expansion for Γ_2. Lines represent the correlator C, and vertices are given by $\mathcal{S}^F$. The vertex terminating three lines are proportional to $\bar{\phi}$.

two-loop approximation to the theory. This approximation will yield a non-time-reversal-invariant mean-field equation above threshold, due to the setting sun diagram [581]. We will discuss a different approximation, the $1/N$ expansion, in Section 7.2 for the study of the nonequilibrium dynamics of the $O(N)$ field theory. For further formal developments of nPI formalism see Berges [327], and as aided by functional renormalization group methods [531], see Carrington and co-workers [328, 583, 584].

7.1.4 2PI In-In Effective Action for $\lambda\Phi^4$ Theory in Curved Spacetime

We now show the construction of the CTP-2PI effective action in curved spacetime with the familiar $\lambda\Phi^4$ field as example. The scalar field action is given by

$$S^{\rm F}[\phi, g^{\mu\nu}] = -\frac{1}{2}\int d^4x\sqrt{-g}\,\left[\phi\Big(-\Box + m^2 + \xi R\Big)\phi + \frac{\lambda}{12}\phi^4\right], \tag{7.39}$$

The only difference is the definition of ξ: (unlike in Chapters 1 and 2 where we wished to emphasize the special role of conformal symmetry, thus setting the ξ there equal to zero for conformal coupling, in this chapter we use the more commonplace convention) $\xi = 0$ for minimal coupling and $\xi = 1/6$ for conformal coupling in $4D$. The classical stress-energy tensor $T_{\mu\nu}$ of the $\lambda\Phi^4$ field is given by

$$\begin{aligned} T_{\mu\nu} = {} & (1-2\xi)\phi_{;\mu}\phi_{;\nu} + \left(2\xi - \frac{1}{2}\right) g_{\mu\nu} g^{\rho\sigma}\phi_{;\rho}\phi_{;\sigma} - 2\xi\phi_{;\mu\nu}\phi \\ & + 2\xi g_{\mu\nu}\phi\Box\phi - \xi G_{\mu\nu}\phi^2 + \frac{1}{2}g_{\mu\nu}\left(m^2 + \frac{\lambda}{12}\phi^2\right)\phi^2. \end{aligned} \tag{7.40}$$

2PI 'In-In' Effective Action

In a curved spacetime, instead of a time x^0_* in the future where the forward and backward time paths are joint together ('closed'), they join in a Cauchy

hypersurface $\Sigma_\star$ chosen so that its past domain of dependence [585] contains all of the dynamics we wish to study. Namely, $g_+^{\mu\nu} = g_-^{\mu\nu}$ and $\phi_+ = \phi_-$ on $\Sigma_\star$. The CTP action for the scalar field is then

$$\mathcal{S}^F[\phi_\pm, g_\pm^{\mu\nu}] = S^F[\phi_+, g_+^{\mu\nu}] - S^F[\phi_-, g_-^{\mu\nu}], \tag{7.41}$$

where $g_\pm^{\mu\nu}$ is the metric tensor on the $+$ and $-$ time branches. Using (2.67) we can similarly define the gravity action $\mathcal{S}^G$ on $\mathcal{M}$,

$$\mathcal{S}^G[g_+^{\mu\nu}, g_-^{\mu\nu}] = S^G[g_+^{\mu\nu}] - S^G[g_-^{\mu\nu}], \tag{7.42}$$

where it is understood that only configurations of $g_\pm^{\mu\nu}$ satisfying the constraint $g_+^{\mu\nu} = g_-^{\mu\nu}$ on $\Sigma_\star$ are permitted.

The 2PI, CTP generating functional $Z[J, K, g^{\mu\nu}]$ is given by

$$Z[J, K, g^{\mu\nu}] = \int_{\text{CTP}} D\Phi_+ D\Phi_- \ \exp\left\{ \frac{i}{\hbar} \left[\mathcal{S}^F[\Phi_\pm, g_\pm^{\mu\nu}] + \int_M d^4x \sqrt{-g_c}\, c^{abc} J_a \Phi_b \right.\right.$$
$$\left.\left. + \frac{1}{2} \int_M d^4x \sqrt{-g_{a'}} \int_M d^4x' \sqrt{-g'_{c'}}\, c^{aba'} c^{cdc'} K_{ac}(x, x') \Phi_b(x) \Phi_d(x') \right] \right\}, \tag{7.43}$$

where we have written $Z[J, K, g^{\mu\nu}]$ as a shorthand for $Z[J_\pm, K_{\pm\pm}, g_\pm^{\mu\nu}]$. The three-index symbol c^{abc} is defined by

$$c^{abc} = \begin{cases} \pm 1, & \text{if } a = b = c = \pm, \\ 0, \text{otherwise.} \end{cases} \tag{7.44}$$

The boundary conditions on the functional integral define the initial quantum state (assumed here to be pure). If we are interested in the case of a quantum state corresponding to a nonzero mean field $\bar{\phi}$, with vacuum fluctuations around the mean field, as in the study of preheating dynamics of inflationary cosmology (to be discussed in Chapter 8), this requires a definition of the vacuum state for the fluctuation field, as in (7.10). In curved spacetime in general, there does not exist a unique Poincaré-invariant vacuum state for a quantum field [46, 48]. For an asymptotically free field theory, a choice of "in" vacuum state corresponds to a choice of a particular orthonormal basis of solutions of the covariant Klein–Gordon equation with which to canonically quantize the field.

From (7.43), we can derive the two-particle-irreducible (2PI) effective action $\Gamma[\bar{\phi}, C, g^{\mu\nu}]$ following the method of Section 7.1.3, with the understanding that Γ now depends functionally on the metric $g_\pm^{\mu\nu}$ on the $+$ and $-$ time branches. The functional differentiations should be carried out using a covariant generalization of the Dirac delta function to the manifold M (see, e.g.,[47]). The functional integro-differential equation (7.34) for the CTP-2PI effective action can then be generalized to the curved spacetime M in a straightforward fashion, modulo the curved-spacetime ambiguities in the boundary conditions of the functional integral (7.43).

The (bare) semiclassical field equations for the metric, mean field, and field variance can then be expressed in terms of variations of $\mathcal{S}^G[g^{\mu\nu}] + \Gamma[\bar{\phi}, C, g^{\mu\nu}]$ with respect to $g^{\mu\nu}_{\pm}$, $\phi_\pm$ and $C_{\pm\pm}$ respectively, followed by metric and mean-field identifications between the + and − time branches (CTP indices are suppressed inside functional arguments):

$$\left.\frac{\delta(\mathcal{S}^G[g^{\mu\nu}] + \Gamma[\bar{\phi}, C, g^{\mu\nu}])}{\delta g^{\mu\nu}_a}\right|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi};\quad g^{\mu\nu}_+=g^{\mu\nu}_-=g^{\mu\nu}} = 0, \tag{7.45}$$

$$\left.\frac{\delta\Gamma[\bar{\phi}, C, g^{\mu\nu}]}{\delta\bar{\phi}_a}\right|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi};\quad g^{\mu\nu}_+=g^{\mu\nu}_-=g^{\mu\nu}} = 0, \tag{7.46}$$

$$\left.\frac{\delta\Gamma[\bar{\phi}, C, g^{\mu\nu}]}{\delta C_{ab}}\right|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi};\quad g^{\mu\nu}_+=g^{\mu\nu}_-=g^{\mu\nu}} = 0. \tag{7.47}$$

Equation (7.45) should be understood as following after time branch indices have been reinstated on the metric tensor in the CTP-2PI-CGEA. Add to these three equations the following which defines the (unrenormalized) stress-energy tensor $\langle T_{\mu\nu}\rangle$

$$\langle T_{\mu\nu}\rangle = -\frac{2}{\sqrt{-g}}\left(\frac{\delta\Gamma[\bar{\phi}, C, g^{\mu\nu}]}{\delta g^{\mu\nu}_+}\right)_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi};\quad g^{\mu\nu}_+=g^{\mu\nu}_-=g^{\mu\nu}}, \tag{7.48}$$

together they constitute the fundamental equations for semiclassical gravity for the system described by the classical action $S^G[g^{\mu\nu}] + S^F[\phi, g^{\mu\nu}]$. The dynamics of $\bar{\phi}$ and C are given by the mean-field (7.46) and gap equations (7.47) while the spacetime dynamics given by the semiclassical Einstein equation (with bare parameters) for $g^{\mu\nu}$ obtained from (7.45) has the generic form given before.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + a^{(1)}H_{\mu\nu} + b^{(2)}H_{\mu\nu} = 8\pi G_{\rm N}\langle\hat{T}_{\mu\nu}\rangle, \tag{7.49}$$

where the stress-energy tensor operator $\hat{T}_{\mu\nu}$ has the form of (7.40) but with the field operator Φ substituted for ϕ in the classical theory, and the anglular brackets denote taking the expectation value with respect to a quantum state $|\phi\rangle$ defined by the boundary conditions on the functional integral in (7.43). In four spacetime dimensions (unrenormalized) $\langle T_{\mu\nu}\rangle$ has divergences which can be absorbed by the renormalization of $G_{\rm N}, \Lambda, a$, and b.

The stress-energy tensor as defined in (7.40) is obtained by a variation of the 2PI effective action Γ, which is a functional of the metric $g^{\mu\nu}_\pm$ on both the + and − time branches. From (7.43), it is possible to derive $\Gamma[\bar{\phi}, C, g^{\mu\nu}]$ as an arbitrary functional of $g^{\mu\nu}_+$ and $g^{\mu\nu}_-$. However, in practice it is often easier to work in the simplified case where the metric is fixed to be the same on both the + and − time branches, i.e., $g^{\mu\nu}_+ = g^{\mu\nu}_- \equiv g^{\mu\nu}$.

This is the result we set forth to attain: the set of coupled evolution equations for the mean field, variance and metric tensor in curved spacetime.

7.2 The $O(N)$ Field Theory in Curved Spacetime

The $O(N)$ model has been usefully applied to a great variety of problems in field theory and statistical mechanics. The $O(N)$ field theory has the advantage that it affords use of the $1/N$ expansion [494, 586, 587, 588], which yields nonperturbative evolution equations in the regime of strong mean field and a covariantly conserved stress-energy tensor [194]. More relevant to quantum field theory in curved spacetime, to leading order in large N, the quantum effective action for the matter fields can be interpreted as a leading-order term in the expansion of the full (matter plus gravity) quantum effective action [194], which produces equations of motion for semiclassical gravity; and, at the next-to-leading order in large N, stochastic gravity, as we shall show in later chapters.

In this section we derive coupled nonperturbative dynamical equations for the mean field $\bar{\phi}$ and variance $\langle\varphi^2\rangle$ for the minimally coupled $O(N)$ scalar field theory with quartic self-interaction and unbroken symmetry. The background spacetime dynamics is given by the semiclassical Einstein equation. These equations take into consideration the backreaction of quantum particle production on the mean field, and quantum fields on the dynamical spacetime, self-consistently.

The $O(N)$ field theory has the nice property that a systematic expansion in powers of $1/N$ yields a nonperturbative reorganization of the diagrammatic hierarchy which preserves the Ward identities order by order [493]. Unlike perturbation theory in the coupling constant, which is an expansion of the theory around the vacuum configuration, the $1/N$ expansion entails an enhancement of the mean field by $\sqrt{N}$; this corresponds to the limit of strong mean field. As discussed earlier the nonequilibrium initial conditions for the mean field as well as the nonperturbative aspect of the dynamics require use of both closed-time-path and two-particle-irreducible methods. The $1/N$ expansion can be achieved as a further approximation from the two-loop, two-particle-irreducible truncation of the Schwinger–Dyson equations [582] or the second order in the correlation hierarchy generated by the master effective action [581].

Although in this study we assume a pure state, the 2PI formalism is also useful for an open system calculation, in which the mean field is defined as the trace of the product of the reduced density matrix ρ and the field operator $\Phi, \mathrm{Tr}(\rho\,\Phi)$. When the position-basis matrix element $\langle\phi_1|\,\rho(\eta_0)|\phi_2\rangle$ can be expressed as a Gaussian functional of ϕ_1 and ϕ_2, the nonlocal source K can encompass the initial conditions coming from $\rho(t_0)$ in a natural way [439]. In order to incorporate a density matrix whose initial condition is beyond Gaussian order in the position basis, one can work with a higher-order truncation of the master effective action [581]. The leading-order $1/N$ approximation is equivalent to assuming a Gaussian initial density matrix, which suits a 2PI effective action treatment.

7.2.1 From Classical Action to Quantum Generating Functional

The $O(N)$ field theory consists of N spinless fields $\vec{\Phi} = \{\Phi^i\}, i = 1, \ldots, N$, with an action which is invariant under the N-dimensional real orthogonal group.[3] The generally covariant classical action for the $O(N)$ theory (with quartic self-interaction) plus gravity is given by

$$S[\phi^i, g^{\mu\nu}] = S^G[g^{\mu\nu}] + S^F[\Phi^i, g^{\mu\nu}], \tag{7.50}$$

where $S^G[g^{\mu\nu}]$ is defined in (2.67) for the spacetime manifold M with metric $g_{\mu\nu}$, and the matter field action $S^F[\Phi^i, g^{\mu\nu}]$ is given by

$$S^F[\Phi^i, g_{\mu\nu}] = -\frac{1}{2}\int_M d^4x\sqrt{-g}\ \left[\vec{\Phi}\cdot(-\Box + m^2 + \xi R)\vec{\Phi} + \frac{\lambda}{4N}(\vec{\Phi}\cdot\vec{\Phi})^2\right]. \tag{7.51}$$

The $O(N)$ inner product is defined by

$$\vec{\Phi}\cdot\vec{\Phi} = \Phi^i\Phi^j\delta_{ij}. \tag{7.52}$$

In (7.51), λ is a (bare) coupling constant with dimensions of $1/\hbar$, and ξ is the (bare) dimensionless coupling to gravity. The classical equations of motion are obtained by variation of the action S separately with respect to the metric tensor $g_{\mu\nu}$ and the matter fields Φ^i. In the classical action (7.51), the $O(N)$ symmetry is unbroken. However, the $O(N)$ symmetry can be spontaneously broken, when m^2 is shifted to $-m^2$ in S^F. In the symmetry-breaking case with tachyonic mass, the stable equilibrium configuration is found to be

$$\vec{\Phi}\cdot\vec{\Phi} = \frac{2Nm^2}{\lambda} \equiv v^2, \tag{7.53}$$

which is a constant. If we wish to study the dynamics of the system around the symmetry-broken equilibrium configuration, the $O(N)$ invariance of (7.51) implies that we can choose the minimum to be in any direction; hence we choose it to be in the 1st, i.e., $(\Phi^1)^2 = v^2$. In terms of the shifted field $\sigma = \Phi^1 - v$ and the unshifted fields $\pi^i = \Phi^i, i = 1, \ldots, N-1$, one can show that the effective mass of each of the 'pions' $\vec{\pi}$ defined as the second derivative of the potential is zero, due to Goldstone's theorem. The theorem holds for the quantum-corrected effective potential as well [589].

Here we continue with the unbroken symmetry case. The $O(N)$ model was used for the study of symmetry breaking in curved spacetime in Chapter 6, where infrared divergences become a central issue.

[3] Note the difference in the $O(N)$ theory index notation from the usual: since the beginning Latin letters a, b, c, d, e, f are used to designate CTP indices (with index set $\{+, -\}$) we use the middle Latin letters i, j, k, l, m, n to designate $O(N)$ indices (with index set $\{1, \ldots, N\}$). Beware that a, b, c, d are often used for covariant geometric objects such as the Riemann tensor (we use Greek indices for geometric objects here).

We now proceed with the construction of the 2PI CTP effective action, derive the mean-field and gap equations at two-loop order, then perform a large N expansion. The 2PI generating functional for the $O(N)$ theory in curved spacetime is defined using the closed-time-path method in terms of c-number sources J_a^i and nonlocal c-number sources K_{ab}^{ij}

$$Z[J_a^i, K_{ab}^{ij}, g_{\mu\nu}] = \prod_{i,a} \int_{\rm CTP} D\Phi_a^i \exp\left\{ \frac{i}{\hbar}\left[\mathcal{S}^F\left[\Phi, g^{\mu\nu}\right] + \int_M d^4x\sqrt{-g}\; c^{ab}\vec{J}_a\cdot\vec{\Phi}_b \right.\right.$$
$$\left.\left. + \frac{1}{2}\int_M d^4x\sqrt{-g}\int_M d^4x'\,\sqrt{-g'}\; c^{ab}c^{cd}K_{ac}^{ij}(x,x')\Phi_b^k(x)\Phi_d^l(x')\delta_{ik}\delta_{jl}\right]\right\}, \tag{7.54}$$

where the CTP classical action is defined as in (7.41), with $\Phi_\pm^i$ replacing $\Phi_\pm$, and the time branch indices on $g^{\mu\nu}$ suppressed. The sources J_a^i are coupled to the field by the $O(N)$ vector inner product,

$$\vec{J}_a\cdot\vec{\Phi}_b = J_a^i\Phi_b^j\delta_{ij}. \tag{7.55}$$

The time branch labels on the metric tensor are suppressed for simplicity of notation. (We will reinstate the time branch indices after the 2PI effective action has been explicitly computed, e.g., in the large-N approximation.) The designation 'ctp' on the functional integral indicates that we sum only over field configurations for Φ_a^i which satisfy the condition $\Phi_+^i = \Phi_-^i$ on $\Sigma_\star$, the hypersurface in the distant future where the two time branches meet. In addition, the boundary conditions on the asymptotic-past field configurations for $\Phi_\pm^i$ in the functional integral correspond to a choice of "in" quantum state $|\Phi\rangle$ for the system. The generating functional for normalized expectation values is given by (7.18), with the additional functional dependence of both W and Z on $g^{\mu\nu}$ understood. As above, we henceforth omit all indices in functional arguments. In terms of this functional, we can define the mean field $\bar{\phi}$ and two-point function C by functional differentiation,

$$\bar{\phi}_a^i(x) = \frac{c_{ab}}{\sqrt{-g}}\frac{\delta W}{\delta J_b^j(x)}\,\delta^{ij} \tag{7.56}$$

$$\bar{\phi}_a^i(x)\bar{\phi}_b^j(x') + \hbar\, C_{ab}^{ij}(x,x') = 2\frac{c_{ac}}{\sqrt{-g}}\frac{c_{bd}}{\sqrt{-g'}}\frac{\delta W}{\delta K_{cd}^{lm}(x,x')}\,\delta^{ik}\delta^{jl}. \tag{7.57}$$

In the zero-sources limit $K_{ab}^{ij} = J_a^i = 0$, the classical field $\bar{\phi}_a^i$ satisfies

$$\bar{\phi}_+^i\Big|_{J=K=0} = \bar{\phi}_-^i\Big|_{J=K=0} = \langle\phi|\Phi^i|\phi\rangle \equiv \bar{\phi}^i \tag{7.58}$$

as an expectation value of the field operator Φ^i in the quantum state $|\phi\rangle$. The fluctuation field is defined as the difference between the field operator Φ and the mean field $\bar{\phi}$ (times the identity operator),

$$\varphi^i = \Phi^i - \bar{\phi}^i. \tag{7.59}$$

In the same limit $J = K = 0$, the two-point correlation function C^{ij}_{ab} becomes the CTP correlator for the fluctuation field. The four components of the CTP correlator are, (for $J^i_a = K^{ij}_{ab} = 0$),

$$\hbar\, C^{ij}_{++}(x,x')\Big|_{J=K=0} = \langle\phi|\mathcal{T}\{\varphi^i(x)\varphi^j(x')\}|\phi\rangle \tag{7.60}$$

$$\hbar\, C^{ij}_{--}(x,x')\Big|_{J=K=0} = \langle\phi|\tilde{\mathcal{T}}\{\varphi^i(x)\varphi^j(x')\}|\phi\rangle \tag{7.61}$$

$$\hbar\, C^{ij}_{+-}(x,x')\Big|_{J=K=0} = \langle\phi|\varphi^j(x')\varphi^i(x)|\phi\rangle \tag{7.62}$$

$$\hbar\, C^{ij}_{-+}(x,x')\Big|_{J=K=0} = \langle\phi|\varphi^i(x)\varphi^j(x')|\phi\rangle. \tag{7.63}$$

In the coincidence limit $x' = x$, all four components above are equivalent to the mean-squared fluctuations (variance) about the mean field $\bar{\phi}^i$

$$\hbar\, C^{ii}_{++}(x,x)\Big|_{J=K=0} = \langle\phi|(\varphi^i)^2|\phi\rangle = \langle(\varphi^i)^2\rangle. \tag{7.64}$$

7.2.2 2PI CTP Effective Action

Provided that the above equations can be inverted to give J^i_a and K^{ij}_{ab} in terms of $\bar{\phi}^i_a$ and C^{ij}_{ab}, we can define the 2PI effective action as a double Legendre transform of W,

$$\Gamma[\bar{\phi}, C, g^{\mu\nu}] = W[J,K,g^{\mu\nu}] - \int_M d^4x\sqrt{-g}\; c^{ab} J^i_a \bar{\phi}^j_b \delta_{ij} - \frac{1}{2}\int_M d^4x\sqrt{-g} \times \int_M d^4x'\sqrt{-g'}\; c^{ab}c^{cd}K^{ij}_{ac}(x,x')\Big[\hbar\, C^{kl}_{bd}(x,x') + \bar{\phi}^k_b(x)\bar{\phi}^l_d(x')\Big]\delta_{ik}\delta_{jl}, \tag{7.65}$$

where J^i_a and K^{ij}_{ab} above denote the inverses of (7.56), (7.57). From this equation, it is clear that the inverses of (7.56), (7.57) can be obtained by straightforward functional differentiation of Γ,

$$\frac{1}{\sqrt{-g}}\frac{\delta\Gamma}{\delta\bar{\phi}^i_a(x)} = c^{ab}\delta_{ij}\left\{-J^j_b(x) - \frac{1}{2}c^{cd}\int_M d^4x'\sqrt{-g'}\left[K^{jk}_{bc}(x,x') + K^{jk}_{cb}(x',x)\right]\bar{\phi}^l_d\delta_{kl}\right\} \tag{7.66}$$

$$\frac{1}{\sqrt{-g}}\frac{\delta\Gamma}{\delta C^{ij}_{ab}(x,x')}\frac{1}{\sqrt{-g'}} = -\frac{\hbar}{2}\,c^{ac}c^{bd}K^{jk}_{cd}(x',x). \tag{7.67}$$

Performing the usual field-shifting involved in the background field approach [495], it can be shown that the 2PI effective action which satisfies (7.65) and (7.66),(7.67) can be written

$$\Gamma[\bar{\phi}, C, g^{\mu\nu}] = \mathcal{S}^F[\bar{\phi}, g^{\mu\nu}] + \frac{i\hbar}{2}\ \mathrm{Tr}\ \ln\big[\big(G^{ij}_{ab}\big)^{-1}\big] + \frac{i\hbar}{2}\int_M d^4x\sqrt{-g}\int_M d^4x'\sqrt{-g'}\ \mathcal{A}^{ab}_{ij}(x',x)C^{ij}_{ab}(x,x') + \Gamma_2[\bar{\phi}, C, g^{\mu\nu}], \tag{7.68}$$

where the kernel $\mathcal{A}$ is the second functional derivative of the classical action with respect to the field $\bar{\phi}$,

$$i\,\mathcal{A}^{ab}_{ij}(x,x') = \frac{1}{\sqrt{-g}}\left(\frac{\delta^2\mathcal{S}^F[\bar{\phi}, g^{\mu\nu}]}{\delta\bar{\phi}^i_a(x)\bar{\phi}^j_b(x')}\right), \tag{7.69}$$

and Γ_2 is a functional to be defined below. Evaluating $\mathcal{A}^{ab}_{ij}$ by differentiation of (7.51), we find

$$i\,\mathcal{A}^{ab}_{ij}(x,x') = -\Big\{\delta_{ij}c^{ab}\Big[-\Box_x + m^2 + \xi\,R(x)\Big] + \frac{\lambda}{2N}\,c^{abcd}\big[\bar{\phi}^k_c(x)\bar{\phi}^l_d(x)\delta_{ij}\delta_{kl} + 2\bar{\phi}^k_c(x)\bar{\phi}^l_d(x)\delta_{ik}\delta_{jl}\big]\Big\}\delta^4(x-x')\frac{1}{\sqrt{-g'}}, \tag{7.70}$$

where the four-index symbol c^{abcd} is defined in exact analogy with (7.44):

$$c^{abcd} = \begin{cases} +1 & \text{if } a=b=c=d=+, \\ -1 & \text{if } a=b=c=d=-, \\ 0 & \text{otherwise.} \end{cases} \tag{7.71}$$

In (7.68), Γ_2 is $-i\hbar$ times the sum of all two-particle-irreducible vacuum-to-vacuum graphs with correlator C and vertices given by the shifted action $\mathcal{S}^F_{int}$, defined by

$$\mathcal{S}^F_{int}[\varphi, g^{\mu\nu}] = \mathcal{S}^F[\varphi+\bar{\phi}, g^{\mu\nu}] - \mathcal{S}^F[\bar{\phi}, g^{\mu\nu}] - \int_M d^4x\left(\frac{\delta\mathcal{S}^F[\bar{\phi}, g^{\mu\nu}]}{\delta\phi^i_a}\right)\varphi^i_a - \frac{1}{2}\int_M d^4x\int_M d^4x'\left(\frac{\delta^2\mathcal{S}^F[\bar{\phi}, g^{\mu\nu}]}{\delta\phi^i_a(x)\phi^j_b(x')}\right)\varphi^i_a(x)\varphi^j_b(x'). \tag{7.72}$$

The expansion of Γ_2 in terms of C and $\bar{\phi}$ is depicted graphically in Fig. 7.2. From (7.72, 7.51), $\mathcal{S}^F_{int}$ is easily evaluated, and we find

$$\mathcal{S}^F_{int}[\varphi, g^{\mu\nu}] = S^F_{int}[\varphi_+, g^{\mu\nu}] - S^F_{int}[\varphi_-, g^{\mu\nu}], \tag{7.73}$$

in terms of an action S^F_{int} on M defined by

$$S^F_{int}[\varphi, g^{\mu\nu}] = -\frac{\lambda}{2N}\int_M d^4x\sqrt{-g}\,\left[\frac{1}{4}(\vec{\varphi}\cdot\vec{\varphi})^2 + (\vec{\bar{\phi}}\cdot\vec{\varphi})(\vec{\varphi}\cdot\vec{\varphi})\right]. \tag{7.74}$$

The two types of vertices in Fig. 7.2 are readily apparent in (7.74). The first term corresponds to the vertex which terminates four lines; the second term corresponds to the vertex which terminates three lines and is proportional to $\bar{\phi}$.

The action Γ including the full diagrammatic series for Γ_2 gives the full dynamics for $\bar{\phi}$ and C in the $O(N)$ theory. It is of course not feasible to obtain an exact, closed form expression for Γ_2 in this model. Various approximations to the full 2PI effective action can be obtained by truncating the diagrammatic expansion for Γ_2. Which approximation is most appropriate depends on the physical problem under consideration.

1. Retaining both the "setting sun" and the "double-bubble" diagrams of Fig. 7.2 corresponds to the two-loop, two-particle-irreducible approximation [581]. This approximation contains two-particle scattering through the setting sun diagram.
2. A truncation of Γ_2 retaining only the "double-bubble" diagram of Fig. 7.2 yields equations for $\bar{\phi}$ and C which correspond to the time-dependent Hartree–Fock approximation to the full quantum dynamics [494, 586]. This approximation does not preserve Goldstone's theorem, but is energy-conserving (in Minkowski space) [586].
3. Retaining only the $(\mathrm{Tr}\, C^{ij}_{ab})^2$ piece of the double-bubble diagram corresponds to taking the leading order $1/N$ approximation, shown below in Section 7.2.4.
4. Discarding Γ_2 altogether of course gives the one-loop approximation, whose severe limitations are precisely the reason we decided to go beyond.

7.2.3 2-Loop Approximation

Let us first evaluate the 2PI effective action at two loops [504, 581]. This is the most general of the various approximations described above. Here both two-loop diagrams in Fig. 7.2 are retained. The 2PI effective action is given by (7.68), and in this approximation, Γ_2 is given by:

$$\begin{aligned}\Gamma_2[\bar{\phi}, C, g^{\mu\nu}] = \frac{\lambda\hbar^2}{4N}\Bigg\{ &-\frac{1}{2}\,c^{abcd}\int_M d^4x\sqrt{-g}\,\Big[C^{ij}_{ab}(x,x)C^{kl}_{cd}(x,x) \\ &+ 2C^{ik}_{ab}(x,x)C^{jl}_{cd}(x,x)\Big]\delta_{ij}\delta_{kl} + \frac{i\lambda}{N}\,c^{abcd}c^{a'b'c'd'} \\ &\times\int_M d^4x\sqrt{-g}\int_M d^4x'\sqrt{-g'}\;\bar{\phi}^i_a(x)\bar{\phi}^{i'}_{a'}(x')\end{aligned}$$

$$\times \Big[C^{ii'}_{bb'}(x,x') C^{jj'}_{cc'}(x,x') C^{kk'}_{dd'}(x,x') + 2C^{ij'}_{bd'}(x,x') C^{jk'}_{cc'}(x,x') C^{ki'}_{db'}(x,x') \Big] \delta_{jk}\delta_{j'k'} \Bigg\}. \tag{7.75}$$

Functional differentiation of Γ with respect to $\bar{\phi}$ and C leads to the mean-field and gap equations, respectively. The gap equation obtained at two loops is given by

$$\begin{aligned}(C^{-1})^{ab}_{ij}(x,x') = {} & \mathcal{A}^{ab}_{ij}(x,x') + \frac{i\lambda\hbar}{2N}\, c^{abcd}\delta^{(4)}(x-x') \left[\delta^{ij}\delta_{kl} C^{kl}_{cd}(x,x) + 2C^{ij}_{cd}(x,x)\right] \\ & + \frac{\hbar\lambda^2}{2N^2}\, c^{acde} c^{bc'd'e'} \delta_{kk'}\delta_{ll'} \Big[\bar{\phi}^i_c(x)\bar{\phi}^j_{c'}(x')\, C^{kl}_{dd'}(x,x') C^{k'l'}_{ee'}(x,x') \\ & + 2\bar{\phi}^k_c(x)\bar{\phi}^l_{c'}(x')\, C^{k'l'}_{dd'}(x,x') C^{ij}_{ee'}(x,x') \\ & + 2\bar{\phi}^i_c(x)\bar{\phi}^k_{c'}(x')\, C^{lj}_{dd'}(x,x') C^{l'k'}_{ee'}(x,x') \\ & + 2\bar{\phi}^k_c(x)\bar{\phi}^l_{c'}(x')\, C^{k'j}_{dd'}(x,x') C^{il'}_{ee'}(x,x') \\ & + 2\bar{\phi}^k_c(x)\bar{\phi}^j_{c'}(x')\, C^{k'l}_{dd'}(x,x') C^{jl'}_{ee'}(x,x') \Big]. \end{aligned} \tag{7.76}$$

The mean-field equation is found to be:

$$\begin{aligned} & \Big[c^{cb}\Big(-\Box + m^2 + \xi R\Big) + \frac{\lambda}{2N}\, c^{abcd}\bar{\phi}^i_a\bar{\phi}^j_d\delta_{ij} \Big]\bar{\phi}^m_b - \frac{i\hbar^2\lambda^2}{4N^2}\int_M d^4x' \sqrt{-g'}\; \Sigma^{cm}(x,x') \\ & \quad + \frac{\hbar\lambda c^{abcd}}{2N} \Big\{ \delta_{ij}\bar{\phi}^m_d C^{ij}_{ab}(x,x) + \delta_{jl}\delta^m_i \bar{\phi}^l_d \Big[C^{ij}_{ab}(x,x) + C^{ji}_{ab}(x,x) \Big] \Big\} = 0, \end{aligned} \tag{7.77}$$

in terms of a nonlocal function $\Sigma^{cm}(x,x')$ defined by

$$\begin{aligned} \Sigma^{em}(x,x') = {} & c^{ebcd} c^{a'b'c'd'} \bar{\phi}^i_{a'}(x') \Big[C^{mi'}_{bb'}(x,x') C^{jj'}_{cc'}(x,x') C^{kk'}_{dd'}(x,x') \\ & + 2C^{mj'}_{bd'}(x,x') C^{jk'}_{cc'}(x,x') C^{ki'}_{b'd}(x,x') + C^{i'm}_{b'b}(x',x) C^{jj'}_{c'c}(x',x) C^{kk'}_{d'd}(x',x) \\ & + 2C^{i'j'}_{b'd}(x',x) C^{kk'}_{c'c}(x',x) C^{jm}_{d'b}(x',x) \Big] \delta_{jk}\delta_{j'k'}\delta_{ii'}. \end{aligned} \tag{7.78}$$

Taking the limit $\bar{\phi}^i_+ = \bar{\phi}^i_- = \bar{\phi}^i$ in (7.76) and (7.77) yields coupled equations for the mean field $\bar{\phi}^i$ and the CTP correlators C^{ij}_{ab}, on the fixed background spacetime $g^{\mu\nu}$. The equations, as well as the semiclassical Einstein equation obtained from (7.45), are real and causal, and correspond to expectation values in the $\bar{\phi}_+ = \bar{\phi}_- = \bar{\phi}$ limit. The $\mathcal{O}(\lambda^2)$ parts of the above equations are nonlocal and dissipative. Replacing the gravity sector by a gauge field and just focusing on the quantum matter sector this is the take off point of many investigations of dissipative processes in relativistic many-body systems such as heavy ion collision and quark-gluon plasma physics. For a detailed systematic investigation along these directions see Berges [445]. For a short introduction to this subject see Chapter 14 of [326].

7.2.4 Large-N Approximation

We now carry out the $1/N$ expansion to obtain local, covariant, nonperturbative mean-field and gap equations for the $O(N)$ field theory in a general curved spacetime. The $1/N$ expansion is a controlled nonperturbative approximation scheme which can be used to study nonequilibrium quantum field dynamics in the regime of strong quasi-classical field amplitude [586, 588, 587]. In the large-N approach, the large-amplitude quasi-classical field is modeled by N fields, and the quantum field-theoretic generating functional is expanded in powers of $1/N$. In this sense the method is a controlled expansion in a small parameter. Unlike perturbation theory in the coupling constant, which corresponds to an expansion of the theory around the vacuum, the large-N approximation corresponds to an expansion of the field theory about a strong quasi-classical field configuration [586]. At a particular order in the $1/N$ expansion, the approximation yields truncated Schwinger–Dyson equations which are gauge- and renormalization-group invariant, unitary, and (in Minkowski space) energy-conserving [586]. In contrast, the Hartree Fock approximation cannot be systematically improved beyond leading order,[4] and (in the case of spontaneous symmetry-breaking) violates Goldstone's theorem [588].

Let us implement the leading order large-N approximation in the two-loop, 2PI mean-field and gap equations (7.77) and (7.76) derived above. This amounts to computing the leading-order part of Γ in the limit of large N, which is $\mathcal{O}(N)$. In the unbroken symmetry case, this is easily carried out by scaling $\bar{\phi}$ by $\sqrt{N}$ and leaving C unscaled [494],

$$\bar{\phi}^i_a(x) \to \sqrt{N}\bar{\phi}_a(x), \qquad C^{ij}_{ab}(x,x') \to C_{ab}(x,x')\delta^{ij}, \tag{7.79}$$

$$\mathcal{A}^{ab}_{ij}(x,x') \to \mathcal{A}^{ab}(x,x')\delta_{ij}, \qquad \varphi^i_a(x) \to \varphi_a(x), \tag{7.80}$$

for all i, j. The field operator φ^i scales like φ^i_a in (7.80). In the above equations, the connection between the large-N limit and the strong mean-field limit is clear.

The truncation of the $1/N$ expansion should be carried out in the 2PI effective action, where it can be shown that the three-loop and higher-order diagrams do not contribute (at leading order in the $1/N$ expansion). Let us now also allow the metric $g_{\mu\nu}$ to be specified independently on the $+$ and $-$ time branches. We find, for the classical action,

$$\mathcal{S}^F[\phi, g^{\mu\nu}] = S^F[\phi_+, g^{\mu\nu}_+] - S^F[\phi_-, g^{\mu\nu}_-], \tag{7.81}$$

where

$$S^F[\phi, g^{\mu\nu}] = -\frac{N}{2}\int_M d^4x\sqrt{-g}\,\left[\phi(-\Box + m^2 + \xi R)\phi + \frac{\lambda}{2}\phi^4\right]. \tag{7.82}$$

[4] For an example of how one can use the 2PI CTP effective action method to go beyond the limitations of the Hartree–Fock–Bogoliubov approximation in cold atom many-body physics, see e.g.,[590]

The inverse of the one-loop correlator is[5]

$$i\,\mathcal{A}^{ab}(x,x') = -\Big\{c^{abc}\Big[-\Box_c^x + m^2 + \xi R_c(x)\Big] + \frac{\lambda}{2}c^{abcd}\bar{\phi}_c(x)\bar{\phi}_d(x)\Big\} \times \delta^{(4)}(x-x')\,\frac{1}{\sqrt{-g'}_b}. \tag{7.83}$$

Finally, for the CTP-2PI effective action at leading order in the $1/N$ expansion, we obtain

$$\Gamma[\bar{\phi},C,g^{\mu\nu}] = \mathcal{S}^F[\bar{\phi},g^{\mu\nu}] + \frac{i\hbar N}{2}\,\mathrm{Tr}\ln\left[(C_{ab}^{-1})\right] + \frac{i\hbar N}{2}\int_M d^4x\sqrt{-g}_a\int_M d^4x'\sqrt{-g'}_b\,\mathcal{A}^{ab}(x',x)C_{ab}(x,x') \frac{\lambda\hbar^2 N}{8}\,c^{abcde}\int_M d^4x\sqrt{-g}_e\,C_{ab}(x,x)C_{cd}(x,x) + \mathcal{O}(N^0). \tag{7.84}$$

Applying (7.46) and taking the limits $\bar{\phi}_+ = \bar{\phi}_- = \bar{\phi}$ and $g_+^{\mu\nu} = g_-^{\mu\nu} = g^{\mu\nu}$, we obtain the gap equation for C_{ab} at leading order in the $1/N$ expansion,

$$(C^{-1})^{ab}(x,x') = \bar{\mathcal{A}}^{ab}(x,x') + \frac{i\hbar\lambda}{2}\,c^{abcd}C_{cd}(x,x)\delta^{(4)}(x-x')\,\frac{1}{\sqrt{-g'}} + \mathcal{O}(\frac{1}{N}), \tag{7.85}$$

where

$$i\,\bar{\mathcal{A}}^{ab}(x,x') \equiv -\left\{c^{ab}\Big[-\Box + m^2 + \xi R(x)\Big] + \frac{\lambda}{2}\,c^{abcd}\bar{\phi}_c(x)\bar{\phi}_d(x)\right\}\delta^{(4)}(x-x')\,\frac{1}{\sqrt{-g'}}. \tag{7.86}$$

Similarly, we obtain the mean-field equation for $\bar{\phi}$ at leading order in the $1/N$ expansion,

$$\left(-\Box + m^2 + \xi R + \frac{\lambda}{2}\,\bar{\phi}^2 + \frac{\hbar\lambda}{2}\,C_{++}(x,x)\right)\bar{\phi}(x) + \mathcal{O}(\frac{1}{N}) = 0, \tag{7.87}$$

where we note that $C_{++}(x,x) = C_{ab}(x,x)$ for all a,b, which can be seen from (7.85); therefore, to get a consistent set of dynamical equations, we need only consider the $++$ component of (7.85). It should also be noted that $C_{ab}(x,x)$ is formally divergent. Regularization of the coincidence limit of the two-point function and the stress-energy tensor is necessary. Multiplying (7.85) by C and integrating over spacetime, we obtain a differential equation for the $++$ CTP Green function

[5] Note that the index b is not to be summed in the right hand side of (7.83), and the c subscript on $\Box$ and R is a CTP index.

$$\left[-\Box_x + m^2 + \xi R(x) + \frac{\lambda}{2}\,\bar{\phi}^2(x) + \frac{\hbar\lambda}{2}\,C_{++}(x,x)\right] C_{++}(x,x') + \mathcal{O}(\frac{1}{N})$$
$$= \delta^{(4)}(x-x')\,\frac{-i}{\sqrt{-g'}}, \tag{7.88}$$

where boundary conditions must be specified on C_{++}.

Equations (7.87) and (7.88) are the covariant evolution equations for the mean field $\bar{\phi}$ and the two-point function C_{++} at leading order in the $1/N$ expansion. Following (7.64), we denote the coincidence limit of $\hbar\,C_{++}(x,x)$ by $\langle\varphi^2\rangle$. With the inclusion of the semiclassical gravity field equation (7.49), these equations form a consistent, closed set of dynamical equations for the mean field $\bar{\phi}$, the time-ordered fluctuation-field correlation function C_{++}, and the metric $g_{\mu\nu}$.

The one-loop equations for $\bar{\phi}$ and G can be obtained from the leading-order equations by setting $\hbar = 0$ in (7.88), while leaving the mean-field equation (7.87) unchanged. In the Hartree approximation, the gap equation is unchanged from (7.85), and the mean-field equation is obtained from (7.87) by changing $\hbar \to 3\hbar$ [591, 592]. The principal limitation of the leading-order large-N approximation is that it neglects the setting sun diagram which is the lowest-order contribution to collisional thermalization of the system [439]. The system therefore does not thermalize at leading order in $1/N$, and the approximation breaks down on a time scale τ_2 which is on the order of the mean free time for binary scattering [588]. Thermalization of interacting quantum field is a long standing and complex issue. See, e.g., [582, 593], Chapter 12 of [326] and references therein. A new view [594] on decoherence and thermalization for a pure quantum state in quantum field theory evokes the truncation – more precisely, the 'slaving' of the lower-order correlations – in the correlation hierarchy described by the master effective action in the sense of [581].

7.2.5 Semiclassical Einstein Equation

We now use (7.45) to derive the bare semiclassical Einstein equation for the $O(N)$ theory at leading order in $1/N$. This equation contains two parts, $\delta\mathcal{S}^G/\delta g_+^{\mu\nu}$ and $\delta\Gamma/\delta g_+^{\mu\nu}$. The latter part is related to the bare stress-energy tensor $\langle T_{\mu\nu}\rangle$ by (7.40). At leading order in $1/N$, $\langle T_{\mu\nu}\rangle$ is given by a sum of classical and quantum parts

$$\langle T_{\mu\nu}\rangle = T^C_{\mu\nu} + T^Q_{\mu\nu} + \frac{\lambda N}{8}\langle\varphi^2_H\rangle^2 g_{\mu\nu}, \tag{7.89}$$

where we define the classical part of $\langle T_{\mu\nu}\rangle$ by

$$T^C_{\mu\nu} = N\Bigg[(1-2\xi)\bar{\phi}_{;\mu}\bar{\phi}_{;\nu} + \left(2\xi - \frac{1}{2}\right) g_{\mu\nu} g^{\rho\sigma}\bar{\phi}_{;\rho}\bar{\phi}_{;\sigma} - 2\xi\bar{\phi}_{;\mu\nu}\bar{\phi}$$
$$+ 2\xi g_{\mu\nu}\bar{\phi}\Box\bar{\phi} + \xi G_{\mu\nu}\bar{\phi}^2 - \frac{1}{2}g_{\mu\nu}\left(m^2 + \frac{\lambda}{4}\bar{\phi}^2\right)\bar{\phi}^2\Bigg], \tag{7.90}$$

and the quantum part of $\langle T_{\mu\nu}\rangle$ by

$$T^{Q}_{\mu\nu} = N\hbar \lim_{x'\to x}\Bigg\{\Bigg[(1-2\xi)\nabla_\mu\nabla'_\nu + \left(2\xi-\frac{1}{2}\right) g_{\mu\nu}g^{\rho\sigma}\nabla_\rho\nabla'_\sigma \\ - 2\xi\nabla_\mu\nabla_\nu + 2\xi g_{\mu\nu}g^{\rho\sigma}\nabla_\rho\nabla_\sigma + \xi G_{\mu\nu} \\ - \frac{1}{2}g_{\mu\nu}\left(m^2 + \frac{\lambda}{2}\bar{\phi}^2 + \frac{\hbar\lambda}{4}C_{++}(x,x')\right)\Bigg]C_{++}(x,x')\Bigg\} + \mathcal{O}(1). \tag{7.91}$$

The above expression for $T^{Q}_{\mu\nu}$ is divergent in four spacetime dimensions, and needs to be regularized or renormalized. The stress-energy tensor in the one-loop approximation is obtained by neglecting the $\mathcal{O}(\hbar^2)$ terms in (7.91). It can be shown using (7.88) that the stress-energy tensor at leading order in the $1/N$ expansion is covariantly conserved, up to terms of order $\mathcal{O}(1)$ (next-to-leading order). The bare semiclassical Einstein equation is then given (in terms of $\langle T_{\mu\nu}\rangle$ shown above) by (7.49).

At this point we formally set $N = 1$ since we are not including next-to-leading order diagrams in the $1/N$ expansion. This can be envisioned as a simple rescaling of the Planck mass by $\sqrt{N}$, since the matter field effective action Γ is entirely $\mathcal{O}(N)$. We now turn to the issue of regularization and renormalization.

7.2.6 Regularization

To renormalize the leading-order large-N CTP effective action in a general curved spacetime, one can use dimensional regularization, which requires formulating the effective action in n spacetime dimensions. This necessitates the introduction of a length parameter μ^{-1} into the classical action, $\lambda \to \lambda\mu^{4-n}$, in order for the classical action to have consistent units. As above, we maintain the restriction $g^{\mu\nu}_{+} = g^{\mu\nu}_{-} = g^{\mu\nu}$, and we suppress indices inside functional arguments.

Making a substitution of the gap equation into the leading-order large-N 2PI effective action, we obtain

$$\Gamma[\bar{\phi}, g^{\mu\nu}] = \mathcal{S}^{F}[\bar{\phi}, g^{\mu\nu}] + \frac{i\hbar N}{2}\,\mathrm{Tr}\,\ln\big[\big(C^{-1}\big)^{ab}\big] \\ + \frac{\hbar^2 N\lambda\mu^{4-n}}{8}\int_M d^n x\sqrt{-g}\; c^{abcd}\big[C_{ab}(x,x)C_{cd}(x,x)\big], \tag{7.92}$$

in terms of the CTP correlator $C_{ab}(x,x')$ which satisfies the gap equation

$$\big(C^{-1}\big)^{ab} = i\Big[-\Box_x c^{ab} + \chi^{ab}(x)\Big]\delta(x-x')\frac{1}{\sqrt{-g'}}, \tag{7.93}$$

in terms of a four-component "effective mass"

$$\chi^{ab}(x) = (m^2 + \xi R)\,c^{ab} + \frac{\lambda\mu^{4-n}}{2}c^{abcd}\Big[\bar{\phi}_c\bar{\phi}_d + \hbar C_{cd}(x,x)\Big]. \tag{7.94}$$

The divergences in the effective action can be made explicit with the use of the heat kernel $K^a{}_b(x,y;s)$ which was introduced in Chapter 4 under a Schwinger proper time (s) representation for the implementation of zeta-function regularization: $K^a{}_b(x,y,s)$ obeys the equation which satisfies

$$\frac{\partial K^a{}_b(x,y;s)}{\partial s} + \int_M d^n z \sqrt{-g(z)}\; c_{cd}\big(C^{-1}\big)^{ac}(x,z) K^d{}_b(z,y;s) = 0, \tag{7.95}$$

with boundary conditions

$$K^a{}_b(x,y;0) = \delta^a{}_b \delta(x-y) \frac{1}{\sqrt{-g(y)}} \tag{7.96}$$

at $s = 0$ [368]. From (7.95, 7.92) it follows that $K^+{}_- = K^-{}_+ = 0$ for all x, y, and s, and that $K^+{}_+$ ($K^-{}_-$) is a functional of $\bar\phi_+$ ($\bar\phi_-$) only. The CTP effective action can then be expressed as (the subscript IO on a quantity indicates it is of the 'in-out' type)

$$\Gamma[\bar\phi, g^{\mu\nu}] = \Gamma^+_{\rm IO}[\bar\phi_+, g^{\mu\nu}] - \Gamma^-_{\rm IO}[\bar\phi_-, g^{\mu\nu}], \tag{7.97}$$

in terms of a functional $\Gamma_{\rm IO}$ on M defined by

$$\begin{aligned}\Gamma^+_{\rm IO}[\bar\phi_+, g^{\mu\nu}] = S^F[\bar\phi_+, g^{\mu\nu}] &- \frac{i\hbar N}{2}\int_M d^n x\sqrt{-g}\int_0^\infty \frac{ds}{s}\; K^+{}_+(x,x;s) \\ &+ \frac{\hbar^2 N\lambda\mu^{4-n}}{8}\int_M d^n x\sqrt{-g}\left[\int_0^\infty ds\; K^+{}_+(x,x;s)\right]^2,\end{aligned} \tag{7.98}$$

and similarly for $\Gamma^-_{\rm IO}$. It follows from (7.95) that $K^+{}_+(x,x;s)[\bar\phi_+]$ is exactly the same functional of $\bar\phi_+$ as $K^-{}_-(x,x;s)[\bar\phi_-]$ is of $\bar\phi_-$; we denote it by $K(x,x;s)[\bar\phi]$, where $\bar\phi$ is a function on M. Note this amounts to the statement we made earlier, that the divergences of a quantum field in a CTP formulation are contained in the theory in either time branch. Thus it is sufficient to consider this issue in the 'in-out' (IO) formulation, as we demonstrated for the $\lambda\Phi^4$ theory in Chapter 2.

The divergences in the effective action arise in the small-s part of the integrations, so that in the equation

$$\int_0^\infty \frac{ds}{s}\; K(x,x;s) = \int_0^{s_0} \frac{ds}{s}\; K(x,x;s) + \int_{s_0}^\infty \frac{ds}{s}\; K(x,x;s), \tag{7.99}$$

only the first term on the right-hand side is divergent. Using the $s \to 0^+$ asymptotic expansion for $K(x,x;s)$ [368], one has (for a scalar field, such as in the unbroken-symmetry large-N limit of the $O(N)$ model),

$$K(x,x;s) \sim (4\pi s)^{-\frac{n}{2}} \sum_{m=0}^\infty s^m a_m(x), \tag{7.100}$$

where the $a_n(x)$ are the "HaMiDeW coefficients" we introduced in Chapter 2, made up of scalar invariants of the spacetime curvature. The divergences show up in dimensional regularization as poles in $1/(n-4)$ after the s integrations are performed. They have been evaluated for the $\lambda\Phi^4$ theory in a general spacetime and in the large-N limit of the $O(N)$ model (see, e.g. [368, 375] and [595]). At leading order in the $1/N$ expansion, the renormalization of $\lambda, \xi, m, G, \Lambda, a$, and b are required, but no field amplitude renormalization is required.

In Chapter 8 we shall apply the results obtained here, the CTP-2PI effective action for an $O(N)$ $\lambda\Phi^4$ theory in a leading order large N expansion, to the problem of preheating after inflation as an example of the backreaction problem in semiclassical gravity theory.

7.3 Remark: Consistent Renormalization of 2PI Effective Action

Before moving on to the next topic, we make a remark here, bringing the renormalization of the $(N = 1)$ $\lambda\Phi^4$ theory presented in Chapter 2 within the context of the 2PI effective action for the $O(N)$ theory discussed in this chapter. For the $O(N)$ model in curved spacetime [244] a nonperturbative regularization/renormalization procedure can be implemented systematically using $1/N$ as a small parameter. However, as Berges et al. [596] pointed out caution is needed when introducing non-systematic truncations to the 2PI effective action like the $N = 1$ Gaussian/Hartree approximation, or any finite value of N, for the renormalization program.

The single component field theory is equivalent to the Hartree or Gaussian approximation , in which one basically assumes that the n-point function can be written in terms of the two-point function as if the state were a Gaussian state. The Gaussian approximation is non-systematic: there is no small parameter to organize the next-to-leading-order (NTLO) correction. Mazzitelli and Paz [595] found that it is possible to renormalize the theory in the usual way, even in curved space-time. One can compute the same n-point function from the 2PI EA by taking derivatives with respect to the field or the two-point function. In an exact calculation, both approaches should give the same answer. When implementing a *systematic* approximation, both approaches also give the same results, order by order in the small parameter. However, if the effective action is truncated *non-systematically*, the results could differ.

Berges et al. [596] developed a *"consistent" renormalization procedure* to avert this pitfall. López Nacir, Mazzitelli and Trombetta implemented this procedure for $\lambda\Phi^4$ theory in curved spacetime. In [597] they analyzed the mean field and gap equations, computed the effective potential, and discussed spontaneous symmetry breaking in de Sitter space. They then showed that the semiclassical Einstein equations can be renormalized following this procedure [598], from which they also found self-consistent de Sitter solutions.

7.4 Solving the Gap Equation for the Infrared Behavior of $O(N)$ Field in dS

In Chapter 6 we discussed two methods proposed in the 80s–90s for exhibiting the infrared behavior of interacting quantum fields in de Sitter space which were precursors of the blossoming current activities on this topic: (1) the 2PI effective action calculated for interacting $O(N)$ fields in Euclidean de Sitter by Hu and O'Connor [462, 504] and (2) the stochastic field theory method of Starobinsky and Yokoyama [505] based on the stochastic inflationary model of Starobinsky [506]. Each has its own merit: in Euclidean dS the dominant infrared behavior is carried by the zero mode, and the 2PI EA provides a systematic method for deriving the mean field and the correlation functions, while the stochastic approach can also capture the leading infrared behavior in a nonperturbative way. We now show a pathway (in addition to those discussed in Chapter 6) in how these two methods can be connected in practice, employing the results we have just shown in the prior two sections obtained by Ramsey and Hu [244]. The following is a summary of the work of Riotto and Sloth [538].

Long Wavelength, Late Time Behavior

Consider the universe in the de Sitter stage expanding with a constant Hubble rate H. At late times the d'Alembertian in the wave equation for the scalar field is dominated by the $3H\partial_t$ term. Assuming that initially $\bar{\phi} = 0$, and that it will stay small due to the $\phi \to -\phi$ symmetry of the Lagrangian, in the massless case the gap equation will at late times simplify to

$$-3H\partial_t C_{++}(x,x) = \lambda\Big[C_{++}(x,x)\Big]^2 + 2i\,\frac{\delta(0)}{\sqrt{-g}}, \tag{7.101}$$

where the extra factor of two is a symmetry factor, accounting for the difference in taking the derivative of $C(x,x')$ before or after taking the coincidence limit. To solve this equation, we need to provide a boundary condition. In the Hartree approximation of Starobinsky and Yokoyama (S&Y) [505], it was obtained by requiring that one recovers the correct linear growth of the variance in the limit $\lambda \to 0$.

Fokker–Planck Equation in the Stochastic Approach

In S&Y's stochastic approach the expectation value of any function $F[\phi]$ of the coarse-grained field ϕ is determined by

$$\langle F[\phi]\rangle = \int d\phi\; F(\phi)\mathcal{P}(\phi,t), \tag{7.102}$$

where $\mathcal{P}$ is the probability density functional of the scalar field values at a given spatial point. Its dynamics is given by a Fokker–Planck equation of the form

$$\frac{\partial}{\partial t}\mathcal{P}(\phi,t) = \frac{\lambda}{6H}\frac{\partial}{\partial\phi}\left[\phi^3\mathcal{P}(\phi,t)\right] + \frac{H^3}{8\pi^2}\frac{\partial^2}{\partial\phi^2}\left[\mathcal{P}(\phi,t)\right]. \tag{7.103}$$

Consider the expectation value of the $2n$-th power of the field. Following [508], one can differentiate the expectation value of ϕ^{2n} and use the Fokker–Planck equation to obtain

$$\frac{\partial}{\partial t}\langle\phi^{2n}\rangle = \frac{n(2n-1)H^3}{4\pi^2}\langle\phi^{2n-2}\rangle - \frac{n\lambda}{3H}\langle\phi^{2n+2}\rangle, \tag{7.104}$$

if one assumes that the field vanishes at the boundaries of the field integration. With a redefinition

$$\bar{\alpha} \equiv \frac{1}{4\pi^2}\ln a, \qquad \bar{\lambda} \equiv 4\pi^2\lambda/3, \qquad \Xi = \frac{\phi}{H}, \tag{7.105}$$

one obtains the differential recursion relation

$$\frac{\partial}{\partial\bar{\alpha}}\langle\,\Xi^{2n}\,\rangle = n(2n-1)\langle\,\Xi^{2n-2}\,\rangle - n\bar{\lambda}\langle\,\Xi^{2n+2}\,\rangle \tag{7.106}$$

which can be solved iteratively to yield the perturbative solution of [508]

$$\langle\phi^{2n}\rangle = (2n-1)!!\left(\frac{H^2}{4\pi^2}\ln a\right)^n\left[1 - \frac{n}{2}(n+1)\frac{\lambda}{12\pi^2}\ln^2 a + \ldots\right]. \tag{7.107}$$

However, the equation can be solved exactly in the large N limit. Since $\langle\phi^4\rangle = \langle\phi^2\rangle^2$ (only the trace squared part contributes in large N), we obtain for $n = 1$

$$\frac{\partial}{\partial\bar{\alpha}}\langle\,\Xi^2\,\rangle = 1 - \bar{\lambda}\langle\,\Xi^2\,\rangle^2, \tag{7.108}$$

which has the solution

$$\langle\,\Xi^2\,\rangle = \frac{\tanh(\sqrt{\bar{\lambda}}\,\bar{\alpha})}{\sqrt{\bar{\lambda}}}. \tag{7.109}$$

This agrees with the asymptotic solution of S&Y for $\ln a \to \infty$ [505], and exemplifies how the logarithmic IR divergences can be resummed to yield a finite result.

Analytic Solution to the Gap Equation

Inserting $C_{++}(x,x)$ given by Eq. (7.109) into the gap equation Eq. (7.87) and solve for $C_{++}(x,x')$ in the long wavelength limit, with $x \neq x'$ we obtain an equation of the form

$$\Big[\partial_t + \frac{\lambda}{2}\frac{\tanh(\vartheta t)}{\upsilon}\Big]C(t) = 0, \tag{7.110}$$

where $\vartheta = \sqrt{\bar{\lambda}}H/4\pi^2$ and $\upsilon = 3\sqrt{\bar{\lambda}}/H$. The solution is

$$C(t) = A\big[\cosh(\vartheta t)\big]^{-\lambda/2\vartheta\upsilon}, \tag{7.111}$$

which scales like $\exp[-(\sqrt{3\lambda}/12\pi)Ht]$ at late times. Since at late times the variance $\langle\phi^2\rangle \to H^2/\sqrt{\lambda}$, the gap equation in Fourier space yields

$$\Big[\partial_t^2 + 3H\partial_t + \frac{k^2}{a^2} + \frac{\sqrt{3\lambda}}{4\pi}H^2\Big]C_{++}(t,t_0;k) = 0, \tag{7.112}$$

where we neglected only the gradient. Now the picture is clear: The nonperturbative infrared effect comes from the regulating mass given by $m_{\rm np}^2 = \frac{\sqrt{3\lambda}}{4\pi}H^2$. This is the same as the $m_{\rm eff}$ or the dynamically generated mass $m_{\rm dyn}$ in Euclidean de Sitter derived in the last chapter, albeit here in Lorentzian de Sitter. We can solve the equation by going to conformal time and rewriting the equation as

$$\Big[\partial_\eta^2 + k^2 - \frac{1}{\eta^2}\Big(\nu^2 - \frac{1}{4}\Big)\Big]\tilde{C}_{++}(\eta,\eta_0;k) = 0, \tag{7.113}$$

where

$$\nu^2 = \Big(\frac{9}{4} - \frac{m_{\rm np}^2}{H^2}\Big). \tag{7.114}$$

and $C = \tilde{C}/a$ is the conformally-related Green function. As a function of η the two independent solutions scale like $\sqrt{-\eta}\,H_\nu^{(1)}(-k\eta)$ and $\sqrt{-\eta}\,H_\nu^{(2)}(-k\eta)$. Thus the asymptotic late time behavior scales as $(-k\eta)^{1/2-\nu}$. Instead of going to a constant, $C_{++}(\eta,\eta_0;k)$ decays at late times as $(-k\eta)^{m_{\rm np}^2/H^2}$ in the limit $\eta \to 0$. This is the same late time behavior as given by the solution in Eq. (7.111). The exponential decay in physical time of the retarded Green function implies that the system has a finite correlation time which is proportional to $1/(\sqrt{\lambda}H)$.

After Tsamis and Woodard in 2005 [508] discovered the 'magic' in the stochastic approach, this result obtained by Riotto and Sloth in 2008 [538] was amongst the early successful attempts in bridging the Euclidean zero-mode approach proposed by Hu and O'Connor [504], advanced by a real-time (CTP) approach using the gap equation derived from the 2PI effective action by Ramsey and Hu in [244], and the stochastic approach of Starobinsky and Yokoyama [505] to the infrared problem of interacting quantum fields in de Sitter space.

We now turn to another example for the computation of CTP-2PI effective action of interacting quantum fields in curved spacetime, that of a scalar field Yukawa-coupled to a spinor based on [606], so we can see an example of how systems involving spinors operate in curved spacetime.[6]

7.5 Yukawa Coupled Scalar and Spinor Fields in Curved Spacetime

In this section, we present a first-principles derivation of the nonequilibrium, nonperturbative effective dynamics of a scalar field Φ coupled to a spinor field Ψ

[6] Some useful background for spinors in curved spacetime can be found in, e.g., [599, 600, 601, 47, 333, 53].

via a Yukawa interaction, in a general, curved classical background spacetime. The evolution equations for the scalar mean field and variance are derived from the two-loop, closed-time-path (CTP), two-particle-irreducible (2PI), coarse-grained effective action. As the name suggests, there are two approximations, one of a statistical mechanical, and the other, quantum field-theoretical nature. 'Coarse-grained' refers to some judicious way to represent the averaged or overall effects of the environment, a central theme of open quantum systems which is nicely implemented by the Feynman–Vernon influence functional formalism [187]. Coarse-grained effective action was first constructed in the 'in-out' formulation in [241] and in the 'in-in' formulation in [229]. The CTP CGEA is intimately related to the influence action. Here the scalar field is taken to be the system of interest and the fermion field acts as its environment, while both live in a curved background spacetime whose dynamics is in turn determined by these matter fields, all in a self-consistent manner. The other approximation, the two-particle-irreducible effective action, is a truncation of the correlation hierarchy for the quantum matter field [581]. As we have seen from the treatment of the $O(N)$ model, this formulation retains both the mean field and the variance as independent variables governed by a set of coupled dynamical equations. Thus, extending what we have learned so far, the two new elements in this problem are the spinor field and the coarse-grained effective action. Backreaction of the scalar and spinor field dynamics on the spacetime is incorporated into the semiclassical Einstein equation, which follows from functional differentiation of the 2PI CGEA with respect to the metric. The CTP formulation ensures that these dynamical equations are both real and causal.

The model we now study consists of a scalar field Φ of mass m (it will be the inflaton field when we apply it to preheating problem in inflationary cosmology) which is Yukawa-coupled to a spinor field Ψ of mass μ, in a curved, dynamical, classical background spacetime with metric $g_{\mu\nu}$. The total action

$$S[\Phi, \bar{\Psi}, \psi, g^{\mu\nu}] = S^G[g^{\mu\nu}] + S^F[\Phi, \bar{\Psi}, \Psi, g^{\mu\nu}], \tag{7.115}$$

consists of a part depicting gravity, $S^G[g^{\mu\nu}]$ given by Eq. (2.67), and a part for the matter fields,

$$S^F[\Phi, \bar{\Psi}, \psi, g^{\mu\nu}] = S^\Phi[\Phi, g^{\mu\nu}] + S^\Psi[\bar{\Psi}, \Psi, g^{\mu\nu}] + S^\Upsilon[\Phi, \bar{\Psi}, \Psi, g^{\mu\nu}], \tag{7.116}$$

whose scalar (inflaton) part is the $\lambda\Phi^4$ field, treated earlier in detail, while the spinor (fermion) and Yukawa-interaction parts are given by[7]

$$S^\Phi[\Phi, g^{\mu\nu}] = -\frac{1}{2}\int d^4x\sqrt{-g}\,\left[\Phi(-\Box + m^2 + \xi R)\Phi + \frac{\lambda}{12}\Phi^4\right], \tag{7.117}$$

[7] For bookkeeping, the scalar and spinor field masses m, μ have dimension of inverse length, ξ is dimensionless, λ, the self-coupling of the inflaton field, has dimension of $1/\sqrt{\hbar}$; and f, the Yukawa coupling constant, has dimension of $1/\sqrt{\hbar}$.

$$S^{\Psi}[\bar{\Psi},\Psi,g^{\mu\nu}] = \int d^4x\sqrt{-g}\,\left\{\frac{i}{2}\Big[\bar{\Psi}\gamma^{\mu}\nabla_{\mu}\Psi - (\nabla_{\mu}\bar{\Psi})\gamma^{\mu}\Psi\Big] - \mu\bar{\Psi}\Psi\right\}, \quad (7.118)$$

$$S^{\mathrm{Y}}[\Phi,\bar{\Psi},\Psi,g^{\mu\nu}] = -f\int d^4x\sqrt{-g}\;\Phi\bar{\Psi}\Psi. \quad (7.119)$$

where ∇_{μ} is the covariant derivative compatible with the metric; $\sqrt{-g}$ is the square root of the absolute value of the determinant of the metric. The curved spacetime Dirac matrices γ^{μ} satisfy the anticommutation relation[8]

$$\{\gamma^{\mu},\gamma^{\nu}\}_{+} = -2g^{\mu\nu}1_{\mathrm{sp}}, \quad (7.120)$$

in terms of the contravariant metric tensor $g^{\mu\nu}$. The symbol 1_{sp} denotes the identity element in the Dirac algebra. Motivated by a physical problem like preheating in inflationary cosmology, we may assume that the fermion field mass μ is much lighter than the inflaton field mass m, i.e., the renormalized parameters m and μ satisfy $m \gg \mu$.

7.5.1 CTP, Coarse-Grained (CG) Effective Action

We denote the quantum field operators of the scalar field and the spinor field by Φ and Ψ, respectively, and the quantum state[9] by $|\Omega\rangle$. For consistency with the truncation of the correlation hierarchy at second order, we assume Φ to have a Gaussian moment expansion in the position basis [595], in which case the relevant observables are the scalar mean field

$$\bar{\Phi}(x) \equiv \langle\Omega|\Phi(x)|\Omega\rangle, \quad (7.121)$$

and the mean-squared fluctuations, or variance, of the scalar field

$$\langle\Omega|\Phi^2(x)|\Omega\rangle - \langle\Omega|\Phi(x)|\Omega\rangle^2 \equiv \langle\Omega|\varphi^2(x)|\Omega\rangle, \quad (7.122)$$

where the last equality follows from the definition of the scalar fluctuation field

$$\varphi(x) \equiv \Phi(x) - \bar{\phi}(x). \quad (7.123)$$

The initial quantum state $|\Omega\rangle$ is assumed to be an appropriately defined vacuum state for the spinor field.

[8] Most textbooks in quantum field theory (e.g., [589]) use $(+,-,-,-)$ metric signature except notably Weinberg's monographs [602, 603] which use the same $(-,+,+,+)$ metric signature (MTW convention [604]) as here. Since we want to keep the definition of the Dirac gamma matrices with upper index the same as in the standard textbooks while working in the $(-,+,+,+)$ metric convention we opt to add a minus sign to this relation.

[9] We assume a well-defined quantum state of the matter fields exist for some specified background spacetime. Although in this case the particular initial conditions constitute a pure state, this formalism can encompass general mixed-state initial conditions [439].

The CTP matter field action is given by

$$\mathcal{S}^F[\Phi_-,\bar{\Psi}_-,\Psi_-,g_-^{\mu\nu};\Phi_+,\bar{\Psi}_+,\Psi_+,g_+^{\mu\nu}] \equiv S^F[\Phi_+,\bar{\Psi}_+,\Psi_+,g_+^{\mu\nu}] - S^F[\Phi_-,\bar{\Psi}_-,\Psi_-,g_-^{\mu\nu}], \tag{7.124}$$

In the spirit of the 'closed' time path, the restriction of Φ to the $+$ and $-$ time branches are subject to the boundary condition

$$\Phi_+\Big|_{\Sigma_\star} = \Phi_-\Big|_{\Sigma_\star} \tag{7.125}$$

at the hypersurface $\Sigma_\star$. Having established a reasonable familiarity with the CTP formulations, we will simplify notation by suppressing time branch indices in the argument of functionals, i.e.,

$$\mathcal{S}^F[\Phi,\bar{\Psi},\Psi,g^{\mu\nu}] \equiv \mathcal{S}^F[\Phi_-,\bar{\Psi}_-,\Psi_-,g_-^{\mu\nu};\Phi_+,\bar{\Psi}_+,\Psi_+,g_+^{\mu\nu}]. \tag{7.126}$$

Likewise we define the functional $\mathcal{S}^{\rm Y}$ for Yukawa interaction

$$\mathcal{S}^{\rm Y}[\Phi,\bar{\Psi},\Psi,g^{\mu\nu}] \equiv S^{\rm Y}[\Phi_+,\bar{\Psi}_+,\Psi_+,g_+^{\mu\nu}] - S^{\rm Y}[\Phi_-,\bar{\Psi}_-,\Psi_-,g_-^{\mu\nu}], \tag{7.127}$$

and $\mathcal{S}^G$ for gravity

$$\mathcal{S}^{\rm G}[g_+^{\mu\nu},g_-^{\mu\nu}] = S^{\rm G}[g_+^{\mu\nu}] - S^{\rm G}[g_-^{\mu\nu}]. \tag{7.128}$$

To formulate the CTP-2PI-CGEA, our first step is to define a generating functional for n-point functions of the scalar field, in terms of the initial quantum state $|\Omega\rangle$ which evolves under the influence of a local source J, and a nonlocal source K coupled to the scalar field (in the interaction picture with the external sources being treated as the "interaction"). This generating functional depends on both J and K, as well as the classical background metric $g^{\mu\nu}$. In the path integral representation, the generating functional $Z[J,K,g^{\mu\nu}]$ takes the form of a sum over scalar field configurations ϕ and complex Grassmann-valued configurations Ψ

$$\begin{aligned} Z[J,K,g^{\mu\nu}] \equiv \int_{\rm CTP} D\Phi_- D\bar{\Psi}_- D\Psi_- D\Phi_+ D\bar{\Psi}_+ D\Psi_+ \\ \times \exp\Bigg\{\frac{i}{\hbar}\Bigg[\mathcal{S}^F[\Phi,\bar{\Psi},\Psi,g^{\mu\nu}] + \int_M d^4x\sqrt{-g}\; c^{ab}\, J_a\Phi_b \\ + \frac{1}{2}\int_M d^4x\sqrt{-g}\int_M d^4x'\,\sqrt{-g'}\; c^{ab}c^{cd}\, K_{ac}(x,x')\Phi_b(x)\Phi_d(x')\Bigg]\Bigg\}, \end{aligned} \tag{7.129}$$

where $J_a(x)$ is a local c-number source and $K_{ab}(x,x')$ is a nonlocal c-number source. The subscript CTP on the functional integral denotes a summation over field configurations $\Phi_\pm,\bar{\Psi}_\pm$, and $\Psi_\pm$ which satisfy the boundary condition (7.125). The Latin indices $a,b,c,\ldots$ are time branch indices with index

set $\{+,-\}$ [182, 439]. The matrix c^{ab} is defined by $c^{++} = 1, c^{--} = -1$, and $c^{+-} = c^{-+} = 0$. The boundary conditions on the functional integral of Eq. (7.129) at the initial data surface determine the quantum state $|\Omega\rangle$. (The CTP indices have been dropped from $g^{\mu\nu}$ for ease of notation.)

7.5.2 2PI CTP Effective Action

The 2PI effective action is given by

$$\begin{aligned}\Gamma[\bar{\phi}, C, g^{\mu\nu}] = &\int_M d^4x \int_M d^4x' \, \frac{\delta\Gamma[\bar{\phi}, C]}{\delta C_{ba}(x', x)} \, C_{ab}(x, x') \\ &- i\hbar \ln\Bigg\{ \int_{\text{CTP}} D\varphi_+ D\bar{\Psi}_+ D\Psi_+ D\varphi_- D\bar{\Psi}_- D\Psi_- \\ &\times \exp\Bigg[\frac{i}{\hbar} \Big\{ \mathcal{S}^F[\varphi + \bar{\phi}, \bar{\Psi}, \Psi, g^{\mu\nu}] \\ &- \int_M d^4x \, \frac{\delta\Gamma[\bar{\phi}, C, g^{\mu\nu}]}{\delta\bar{\phi}_a} \varphi_a \\ &- \frac{1}{\hbar} \int_M d^4x \int_M d^4x' \, \frac{\delta\Gamma[\bar{\phi}, C, g^{\mu\nu}]}{\delta C_{ba}(x', x)} \, \varphi_a(x)\varphi_b(x') \Big\} \Bigg] \Bigg\}, \end{aligned} \tag{7.130}$$

which has the formal solution

$$\begin{aligned}\Gamma[\bar{\phi}, C, g^{\mu\nu}] = S^\Phi[\bar{\phi}] - \frac{i\hbar}{2} \ln \det C_{ab} - i\hbar \ln \det F_{ab} + \Gamma_2[\bar{\phi}, C] \\ + \frac{i\hbar}{2} \int_M d^4x \sqrt{-g} \int_M d^4x' \sqrt{-g'} \, \mathcal{A}^{ab}(x', x) C_{ab}(x, x'), \end{aligned} \tag{7.131}$$

where $\mathcal{A}^{ab}$ is the second functional derivative of the scalar part of the classical action S^Φ, evaluated at $\bar{\phi}$,

$$\begin{aligned} i\mathcal{A}^{ab}(x, x') &= \frac{1}{\sqrt{-g}} \left(\frac{\delta^2 S^\Phi[\bar{\phi}]}{\delta\bar{\phi}_a(x)\delta\bar{\phi}_b(x')} \right) \frac{1}{\sqrt{-g'}} \\ &= -\left[c^{ab} \Big\{ -\Box_x + m^2 + \xi R(x) \Big\} + c^{abcd} \, \frac{\lambda}{2} \, \bar{\phi}_c(x)\bar{\phi}_d(x) \right] \frac{\delta(x - x')}{\sqrt{-g'}}. \end{aligned} \tag{7.132}$$

The curved-spacetime Dirac δ function has been defined earlier, as in [47]. Notice that the tr $\ln F_{ab}$ in Eq. (7.131) differs from the usual one-loop term by a factor of 2, owing to the difference (in the exponent) between the Gaussian integral formulas for real and complex fields [494].

The symbol F_{ab} denotes the one-loop CTP spinor correlator, which is defined by

$$F_{ab}(x, x') \equiv \mathcal{B}^{-1}_{ab}(x, x'), \tag{7.133}$$

where we are suppressing spinor indices, and the inverse spinor correlator $\mathcal{B}^{ab}$ is defined by

$$
\begin{aligned}
i\mathcal{B}^{ab}(x,x') &= \frac{1}{\sqrt{-g}}\left[\frac{\delta^2(\mathcal{S}^{\Psi}[\bar{\Psi},\Psi]+\mathcal{S}^{\Upsilon}[\bar{\Psi},\Psi;\bar{\phi}])}{\delta\Psi_a(x)\delta\bar{\Psi}_b(x')}\right]\frac{1}{\sqrt{-g'}} \\
&= \left[c^{ab}\big(i\gamma^{\mu}\nabla'_{\mu}-\mu\big)-c^{abc}f\bar{\phi}_c(x')\right]\delta(x'-x)\frac{1}{\sqrt{-g}}1_{\rm sp}. \qquad (7.134)
\end{aligned}
$$

It is clear from Eq. (7.134) that the use of the one-loop spinor correlators in the construction of the CTP-2PI-CGEA represents a nonperturbative resummation in the Yukawa coupling constant, which goes beyond the standard time-dependent perturbation theory. The boundary conditions which define the inverses of Eqs. (7.132) and (7.134) are the boundary conditions at the initial data surface in the functional integral in Eq. (7.129), which in turn, define the initial quantum state $|\Omega\rangle$. The one-loop spinor correlator F_{ab} is related to the expectation values of the spinor field operators for a spinor field in the presence of the c-number background field $\bar{\phi}$,

$$
\hbar F_{++}(x,x')\Big|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi}} = \langle\Omega|T\big\{\Psi(x)\bar{\Psi}(x')\big\}|\Omega\rangle, \qquad (7.135)
$$

$$
\hbar F_{--}(x,x')\Big|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi}} = \langle\Omega|\tilde{T}\big\{\Psi(x)\bar{\Psi}(x')\big\}|\Omega\rangle, \qquad (7.136)
$$

$$
\hbar F_{+-}(x,x')\Big|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi}} = -\langle\Omega|\bar{\Psi}(x')\Psi(x)|\Omega\rangle, \qquad (7.137)
$$

$$
\hbar F_{-+}(x,x')\Big|_{\bar{\phi}_+=\bar{\phi}_-=\bar{\phi}} = \langle\Omega|\Psi(x)\bar{\Psi}(x')|\Omega\rangle, \qquad (7.138)
$$

where the spinor field operators obey the equations

$$
\begin{aligned}
\Big(i\gamma^{\mu}\nabla_{\mu}-\mu-f\bar{\phi}\Big)\Psi &= 0, \\
\Big(-i\gamma^{\mu}\nabla_{\mu}-\mu-f\bar{\phi}\Big)\bar{\Psi} &= 0. \qquad (7.139)
\end{aligned}
$$

The CTP spinor correlator components satisfy the relations (valid only when $\bar{\phi}_+=\bar{\phi}_-=\bar{\phi}$)

$$
\begin{aligned}
F_{++}(x,x')^{\dagger} &= F_{--}(x',x), \\
F_{--}(x,x')^{\dagger} &= F_{++}(x',x), \\
F_{-+}(x,x')^{\dagger} &= F_{-+}(x',x), \\
F_{+-}(x,x')^{\dagger} &= F_{+-}(x',x), \qquad (7.140)
\end{aligned}
$$

The functional $\Gamma_2[\bar{\phi},C]$ is defined as $-i\hbar$ times the sum of all vacuum diagrams drawn according to the following rules:

1. Vertices carry spacetime (x,x') and time branch $(a,b=\{+,-\})$ labels.
2. Scalar field lines denote $\hbar\,C_{ab}(x,x')$.

3. Spinor lines denote the one-loop CTP spinor correlator $\hbar F_{ab}(x,x')$ (spinor indices are suppressed), defined in Eq. (7.133).
4. There are three interaction vertices, given by $i\mathcal{S}^{\rm I}/\hbar$, which is defined by (following the notation of Eq. (7.127))

$$\mathcal{S}^{\rm I}[\bar\phi,\varphi,\bar\Psi,\Psi] = S^{\rm I}[\bar\phi_+,\varphi_+,\bar\Psi_+,\Psi_+] - S^{\rm I}[\bar\phi_-,\varphi_-,\bar\Psi_-,\Psi_-], \tag{7.141}$$

with

$$S^{\rm I}[\bar\phi,\varphi,\bar\Psi,\Psi] = -\int d^4x\sqrt{-g}\,\left[f\varphi\bar\Psi\Psi + \frac{\lambda}{24}\varphi^4 + \frac{\lambda}{6}\bar\phi\varphi^3\right]. \tag{7.142}$$

5. Only diagrams which are two-particle-irreducible with respect to cuts of *scalar* lines contribute to Γ_2.

The distinction between the CTP-2PI *coarse-grained* effective action which arises here, and the fully two-particle-irreducible effective action (2PI with respect to scalar *and* spinor cuts), is due to the fact that we only Legendre-transformed sources coupled to ϕ; i.e., the spinor field is treated as the environment.

7.5.3 Two Loop Diagrams

The functional $\Gamma_2[\bar\phi,C,g^{\mu\nu}]$ can be evaluated in a loop expansion, which corresponds to an expansion in powers of $\hbar$,

$$\Gamma_2[\bar\phi,C,g^{\mu\nu}] = \sum_{l=2}^{\infty}\hbar^l\Gamma^{(l)}[\bar\phi,C,g^{\mu\nu}], \tag{7.143}$$

starting with the two-loop term, $\Gamma^{(2)}$, which has a diagrammatic expansion shown in Fig. 7.3.

The $\lambda\phi^4$ self-interaction leads to two terms in the two-loop part of the effective action, the second and third graphs of Fig. 7.3. They are the "setting sun"

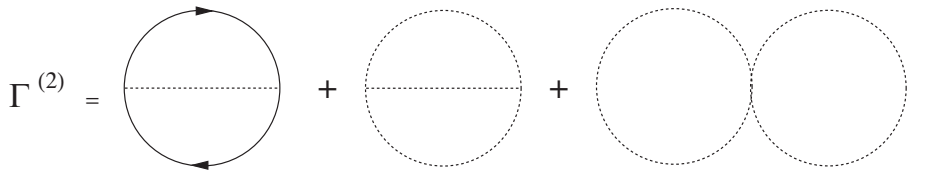


Figure 7.3 Diagrammatic expansion for $\Gamma^{(2)}$, the two-loop part of the CTP-2PI-CGEA. Solid lines represent the spinor correlator F defined in Eqs. (7.135)–(7.138) and dotted lines represent the scalar correlator C. The vertices terminating three ϕ lines are proportional to the scalar mean field $\bar\phi$. Each vertex carries spacetime (x) and CTP $(+,-)$ labels.

diagram, which is $O(\lambda^2)$, and the "double bubble," which is $O(\lambda)$, respectively. The Yukawa interaction leads to only one diagram in $\Gamma^{(2)}$, the first diagram of Fig. 7.3,

$$\frac{if^2}{2}c^{aa'a''}c^{bb'b''}\int d^4x\sqrt{-g}\int d^4x'\sqrt{-g'}\;C_{ab}(x,x')\,\mathrm{Tr}_{\mathrm{sp}}\left[F_{a'b'}(x,x')F_{b''a''}(x',x)\right], \tag{7.144}$$

where the trace is understood to be over the spinor indices which are not shown, and the three-index symbol c^{abc} was defined in (7.44). If the coupling λ is sufficiently small that the $O(\lambda^2)$ diagram is unimportant on the time scales of significant fermion production in the background scalar field (inflaton) dynamics one can then treat the λ self-interaction using the time-dependent Hartree–Fock (HF) approximation [494], which is equivalent to retaining only the $O(\lambda)$ (double-bubble) graph. As we forwarned in the last section where the mean-field and gap equations including both the $O(\lambda)$ and the $O(\lambda^2)$ diagrams were derived for the $O(N)$ model in a general curved spacetime, dropping the $O(\lambda^2)$ (setting sun) graph in the HF approximation ignores the dissipative effects from the backreaction of particle creation of the scalar field. in the last section. However, when there are other fields present, such as the spinor field in the Yukawa model under study, even under the HF approximation for the scalar field dynamics, there *will* be dissipation arising from the back reaction from fermionic particle production induced by the time-dependence of the inflaton variance, captured in the nonlocal kernel $\mathcal{K}(x,x'')$ in Eq. (7.155) of the next subsection.

7.5.4 Evolution Equations for $\bar{\phi}$ and C in Curved Spacetime

Just like in the $O(N)$ theory earlier, here for the Yukawa theory the (bare) semiclassical field equations for the two-point function, mean field, and metric can be obtained from the CTP-2PI-CGEA of the theory under consideration by functional differentiation with respect C_{ab}, $\bar{\phi}_a$, and $g^{\mu\nu}$, followed by identifications of $\bar{\phi}$ and $g^{\mu\nu}$ on the two time branches [605]. See discussions following Eq. (7.45)–(7.47) and Eq. (7.48).

Making the two-loop approximation to the CTP-2PI-CGEA, where we take $\Gamma_2 \simeq \hbar^2\Gamma^{(2)}$, and dropping the $O(\lambda^2)$ diagram from Γ_2, the mean-field equation (7.46) becomes

$$\left(-\Box + m^2 + \xi R(x) + \frac{\lambda}{6}\bar{\phi}^2(x) + \frac{\lambda\hbar}{2}C(x,x)\right)\bar{\phi} + \hbar f\,\mathrm{Tr}_{\mathrm{sp}}[F_{ab}(x,x)] - \hbar^2 g^3\Sigma(x) = 0, \tag{7.145}$$

where $C(x,x)$ is the coincidence limit of $C_{ab}(x,x')$, and in terms of a function $\Sigma(y)$ defined by

$$\Sigma(y) \equiv \int d^4x\sqrt{-g}\int d^4x'\sqrt{-g'}\Big\{C_{++}(x,x')\,\mathrm{Tr}_{\rm sp}\Big[F_{++}(x,y)F_{++}(y,x')F_{++}(x',x)\Big]$$
$$- C_{-+}(x,x')\,\mathrm{Tr}_{\rm sp}\Big[F_{-+}(x,y)F_{++}(y,x')F_{+-}(x',x)\Big]$$
$$- C_{+-}(x,x')\,\mathrm{Tr}_{\rm sp}\Big[F_{++}(x,y)F_{+-}(y,x')F_{-+}(x',x)\Big]$$
$$+ C_{--}(x,x')\,\mathrm{Tr}_{\rm sp}\Big[F_{-+}(x,y)F_{+-}(y,x')F_{--}(x',x)\Big]\Big\}. \tag{7.146}$$

Making use of the curved spacetime definitions of the scalar and spinor field Hadamard kernels [47]

$$G^{(1)}(x,x') = \langle\Omega|\{\varphi(x),\varphi(x')\}|\Omega\rangle, \tag{7.147}$$
$$F^{(1)}(x,x') = \langle\Omega|[\Psi(x),\bar{\Psi}(x')]|\Omega\rangle, \tag{7.148}$$

and retarded propagators

$$G_R(x,x') = i\,\theta(x,x')\langle\Omega|[\varphi(x),\varphi(x')]|\Omega\rangle, \tag{7.149}$$
$$F_R(x,x') = i\,\theta(x,x')\langle\Omega|\{\Psi(x),\bar{\Psi}(x')\}|\Omega\rangle, \tag{7.150}$$

the function $\Sigma(y)$ can be recast in a manifestly real and causal form,

$$\Sigma(y) = -2\int d^4x\sqrt{-g}\int d^4x'\sqrt{-g'}\ \mathrm{Re}\ \mathrm{Tr}_{\rm sp}\Big\{\Big[\theta(x,x')G^{(1)}(x',x)F^{(1)}(x,x')$$
$$- G_R(x,x')^\star F_R(x,x')\Big]F_R(y,x')^\star F_R(y,x)\Big\}, \tag{7.151}$$

from which it is clear that the integrand vanishes whenever x or x' is to the future of y. The "gap" equation for C_{ab} is obtained from Eq. (7.47),

$$(C^{-1})^{ba}(x,x') = \mathcal{A}^{ba}(x,x') + \frac{i\lambda\hbar}{2}c^{ba}C(x,x)\delta(x-x')\frac{1}{\sqrt{-g'}} \tag{7.152}$$
$$+ \hbar f^2 c^{aa'a''}c^{bb'b''}\,\mathrm{Tr}_{\rm sp}\left[F_{a'b'}(x,x')F_{b''a''}(x',x)\right].$$

Multiplying Eq. (7.152) through by C_{ab}, performing a spacetime integration, and taking the $++$ component, we obtain

$$\Big[-\Box + m^2 + \xi R + \frac{\lambda}{2}\bar{\phi}^2 + \frac{\lambda\hbar}{2}C(x,x)\Big]C_{++}(x,x')$$
$$+ \hbar f^2\int dx''\sqrt{-g''}\,\mathcal{K}(x,x'')C_{++}(x'',x') = -i\delta(x-x')\frac{1}{\sqrt{-g'}}, \tag{7.153}$$

in terms of a kernel $\mathcal{K}(x,x'')$ defined by

$$\mathcal{K}(x,x') = -i\,\mathrm{Tr}_{\rm sp}\left[F_{++}(x,x')F_{++}(x',x) - F_{+-}(x,x')F_{-+}(x',x)\right]. \tag{7.154}$$

Making use of Eqs. (7.148) and (7.150), this kernel takes the form

$$\mathcal{K}(x,x') = \mathrm{Re}\,\mathrm{Tr}_{\rm sp}\left[F_R(x,x')F^{(1)}(x',x)\right], \tag{7.155}$$

which shows that the gap equation (7.153) is manifestly real and causal. The set of evolution equations (7.145) for $\bar{\phi}$ and (7.153) for C, is formally complete to two loops.

As shown in [606] in a perturbative treatment, the kernel $\mathcal{K}(x, x')$ is dissipative. It reflects the backreaction from fermionic particle production induced by the time-dependence of the inflaton variance. The gap equation (7.153) is therefore damped for modes above threshold.[10] Note that this damping is not accounted for in the 1PI treatment of inflaton dynamics where only the inflaton mean field is dynamical [587, 586]. In contrast to assuming a local equation of motion for the inflaton propagator as in a 1PI treatment, the two-loop gap equation obtained from the CTP-2PI-CGEA includes a *nonlocal* kernel, which, as we have seen in many situations, *is a generic feature of backreaction from particle production.* The dissipative dynamics of the inflaton two-point function can be important when the inflaton variance is on the order of the square of the inflaton mean-field amplitude; such conditions may exist at the end of preheating. We will pick up this topic as an illustration of backreaction in semiclassical cosmology in Chapter 8.

[10] See [607] for a similar discussion in the context of spinodal decomposition in quantum field theory

8

Backreaction of Early Universe Quantum Processes

By backreaction we refer to the effects of quantum matter fields *on* the background spacetime, obtained by seeking self-consistent solutions of the semiclassical Einstein (SCE) equations, namely, the Einstein equation for the background metric driven by a quantum matter source, which, at the level of semiclassical gravity, is represented by the vacuum expectation value of the stress-energy tensor, together with the field equations describing quantum matter. For this reason they are also called semiclassical backreaction equations by some [608]. There are at least two ways to carry out the backreaction calculations, one at the level of the equations of motion and the other at the level of effective actions. Backreaction calculations performed at the equation of motion level are usually done iteratively in successive orders of some perturbative parameter (e.g., small anisotropy, inhomogeneity, weak coupling), because one has to first solve the equation of motion for the matter field ϕ assuming to begin with a background spacetime (at the zeroth order) $g^{(0)}$. After obtaining a (first-order) solution for the field, one uses this solution to construct the stress-energy tensor which is quadratic in ϕ, taking its expectation value with respect to some vacuum which acts as the source for the Einstein equation, which is now second order. One sees that the two parties enter in the coupled equations of motion at orders differing by one. Thus self-consistency in the equations of motion is kept at staggered orders.

The effective action treatment of the backreaction problem is somewhat different in this regard. The equation of motion for the matter field *and* the geometry are both obtained from taking the variational derivative of the same effective action at a specific loop order (orders of $\hbar$). Thus self-consistency is ingrained in the construction of the effective action. Backreaction of the quantum matter field processes such as particle creation and vacuum polarization are incorporated in the effective equations of motion for the background geometry.

In practical terms, derivations of the effective action usually require making some approximations because one can manage only Gaussian functional integrals. Thus, explicitly, backreaction problems may appear to be more approachable at the equations of motion level. That is how we will begin with each backreaction problem in this chapter, illustrating how to derive the SCE equation in some sample calculations and finding its solutions in this vein. Afterward, we shall adopt the 'in-in' effective action method with some suitable approximations to derive the SCE eqns.

Four groups of backreaction problems at the Planck and grand unified theory (GUT) scales are presented here: (1) Effect of *trace anomaly* (a) in facilitating possible avoidance of singularity [109], (b) in engendering inflation, as in the so-called Starobinsky inflation [359], (2) Effect of *particle creation*, in affecting the equation of state of matter [105, 107], in the damping of anisotropy [108, 110, 182] or inhomogeneity [423, 186]. For the effects of quantum processes on cosmological singularity and particle horizons, see e.g., [113, 114, 609]. Some extreme cases can also be of great cosmological significance: *topology change*, such as in the 'trousers' geometry studied by A. Anderson and DeWitt [610] where a bifurcation of space happens, presumably at energies higher than the Planck energy, or in the *'Birth' of the universe from 'Nothing' scenario* [611] –tunneling transition from a closed-FLRW to a de Sitter universe. They can bring forth catastrophic particle creation [612], the backreaction of which can be so strong as to prevent these processes from happening.

Turning to inflationary cosmology, we shall study (3) *post-inflationary reheating* by the dissipative effects of particle creation and interaction from the nonequilibrium inflaton dynamics, using an $O(N)\lambda\Phi^4$ theory as example [605]. Space limitation permits us to only mention (4a) *stochastic inflation* [506] where the short wavelength modes acting as noise backreact on the long wavelength modes, thereby decohering the latter into classical background modes [554, 550], and (4b) effect of inhomogeneous modes on the homogeneous mode in a $\lambda\Phi^4$ theory applied to the validity of minisuperspace approximation in *quantum cosmology* [229].

8.1 Vacuum Energy-Driven Cosmology

8.1.1 Casimir Energy and Quantum Vacuum Cosmology

To get acquainted with backreaction calculations let us begin with a simple example: a minimalist quantum cosmology where the only source present comes from the vacuum fluctuations of quantum fields. Consider the Casimir energy of a massless spin-j particle in the Einstein universe which is a static closed-FLRW universe with S^3 topology [613, 614]

$$E_{\rm C}^{(j)} = (-1)^{2j}\,\frac{h(j)}{240a}\,(30j^4 - 20j^2 + 1)\,, \tag{8.1}$$

where a is the radius of the universe and $h(j) = 1$ if $j = 0$ and $h(j) = 2$ if $j > 0$. Specifically, for scalar particles $E_{\rm C}^{(0)} = \frac{1}{240a}$ [59, 60, 61]; for spin 1/2 particles, $E_{\rm C}^{(1/2)} = \frac{17}{960a}$ and for vector particles $E_{\rm C}^{(1)} = \frac{11}{120a}$.

Assume that there is no classical matter and the vacuum energy is the only quantum source in a FLRW universe. The G_{00} component of the Einstein equation is

$$3\left[\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2}\right] = 8\pi G_{\rm N}\rho_0(a). \tag{8.2}$$

For the closed ($k = 1$) universe with radius a the vacuum energy density $\rho_0(a) \equiv E_{\rm C}^{(j)}/V$, with $V = 2\pi^2 a^3$ the volume of a 3-sphere. Now consider only a scalar field present in the Einstein universe. It being static, we can leave out the first (kinetic) term in (8.2):

$$\frac{3}{a^2} = \frac{8\pi G_{\rm N}}{480\pi^2 a^4}, \tag{8.3}$$

and obtain the solution

$$a = \frac{\kappa}{(2880\pi^2)^{\frac{1}{2}}}, \tag{8.4}$$

where $\kappa \equiv \sqrt{16\pi G_{\rm N}} = \sqrt{16\pi}\ell_p$ in geometric units. The factor $\varpi \equiv (2880\pi^2)^{-1}$ appears often in quantities related to vacuum fluctuations or polarizations, such as the trace anomaly. Self-consistent solutions can also exist in de Sitter spacetime, see, e.g., [96].

Now, restoring dynamics to the closed-FLRW quantum universe, and solving the full equation (8.2) we see that this quantum vacuum universe will expand to a maximum radius a_{max} (when $\dot{a}/a \to 0$), given by (8.4). The only scale length is the Planck length because there is no matter source present to introduce another scale. Simply by dimensional arguments we can see that with a quantum vacuum source the universe can only expand to the Planck size before it recontracts. Misner has an endearing name for it: a 'puff' universe [615].

From this simple example let us see the physical differences between quantum cosmology and classical cosmology. In the former, the vacuum energy density replaces the energy density for classical matter or radiation as source in the Einstein equation (8.3). The classical (Einstein) action $S_g^{(0)}$ of the background spacetime (g) is given by

$$S_g^{(0)}[g] = \frac{1}{16\pi\ell_p^2}\int_{\mathcal{D}} d^4x(-g)^{\frac{1}{2}}\ R + \text{surfaceterms}, \tag{8.5}$$

where $\mathcal{D}$ is the spacetime enclosed by the initial and final hypersurfaces. For the quantum matter field the action is given by

$$S_f[\Phi] = \int_{\mathcal{D}} d^4x(-g)^{1/2}\ L, \tag{8.6}$$

where L is the matter field Lagrangian density. For a massive scalar field conformally-coupled to a background spacetime with scalar curvature R

$$L = \frac{1}{2}\Big[-g^{ab}\partial_a\Phi\,\partial_b\Phi - \frac{R}{6}\Phi^2\Big]. \tag{8.7}$$

For classical radiation the energy density is $\rho_r = \tilde{\rho}_r\, a^{-4}$, where $\tilde{\rho}_r$ is a constant which measures the total number of photons (entropy content) in the universe. Its action is given by (2.78)

$$S_r = \int d^4x\ (-\tilde{\rho_r}). \tag{8.8}$$

During an adiabatic expansion the temperature T of the radiation varies as

$$T = \Theta/a \tag{8.9}$$

where Θ is a constant related to $\tilde{\rho}_r$. Using this we get $\rho_r = (\tilde{\rho_r}/\Theta^4)T^4$. Comparing this to the Stefan-Boltzmann law

$$\rho_r = \sigma T^4 \tag{8.10}$$

we get a relation between $\tilde{\rho_r}/\Theta^4$ and the Stefan–Boltzmann constant $\sigma = g\tilde{\rho_r}/\Theta^4$. Recall that σ contains the spin states g_{eff} of the particles where $g_{eff} \equiv \Sigma g_b + \frac{7}{8}\Sigma g_f$ has contributions from bosons (b) and fermions (f). Every polarization state contributes $\pi^2/30$ thus from (8.2), we see that the magnitude of $\tilde{\rho_r} = (\pi^2 g_{eff}/30)\Theta^4$ determines the maximum radius a_{max}. The 'natural' scale in the system with $\tilde{\rho}_r = 1$ would, from (8.4), give rise to a quantum universe with radius of the Planck size, the 'puff' universe. By this measure the number of photons $n_\gamma \approx 10^{80}$ in our present universe is *unnaturally* large, but that is precisely what makes our unusually large universe possible. This 'strangely' large (compare to unity) photon number or entropy content gives rise to the so-called 'oldness', 'flatness' or 'frigidity' problem special to our universe. (See., e.g. [616, 118]).

Eqs. (8.3) and (8.10) are the zero and high temperature limits of a more general expression for finite temperature energy density ρ_β, where β is the inverse temperature T. One way to calculate ρ_β when the spectrum of the invariant operators of the field is known is by a direct mode sum

$$\rho_\beta = \frac{1}{V}\sum_{k=0}^{\infty}\epsilon_k d_k \langle n_k\rangle, \tag{8.11}$$

where V is the volume of the 3-sphere, ϵ_k is the energy, d_k the degeneracy, and $\langle n_k\rangle$ the occupation number of the kth mode, $\langle n_k\rangle = (\exp\beta\epsilon_k \pm 1)^{-1}$, with the minus sign for bosons and the plus sign for fermions. For an Einstein Universe with radius a (see Chapter 4)

$$V = 2\pi^2 a^3, \qquad \epsilon_k = (k+1)/a, \qquad d_k = (n+1)^2. \tag{8.12}$$

Using the Poisson summation formula one can separate from ρ_β a Stefan-Boltzmann term describing black-body radiation, i.e., $\rho_r = (\pi^2/30)T^4$. The remainder is a correction term due to the compactness of the space

$$\rho_{cor} = \frac{12T^4}{\pi^2} \sum_{k=1}^{\infty} Re\zeta_R(4, 1 - 2i\,\pi k\Theta), \tag{8.13}$$

where the Riemann zeta function $\zeta_R(s, p)$ here is defined by

$$\zeta_R(s, p) = \sum_{m=0}^{\infty} (m + p)^{-s}. \tag{8.14}$$

For more than one polarization the energy density should be multiplied by a factor g_{eff} [given above in (8.10)]. The correction term can be evaluated by summing the series or by making use of the integral representations of generalized zeta functions [617, 458]. At large Θ, corresponding to high temperature T or small radius a,

$$\Theta \to \infty, \qquad \rho_{cor} \to -\rho_0, \tag{8.15}$$

whence the total energy density $\rho = \rho_0 + \rho_\beta \to \rho_{\rm r}$. At small Θ corresponding to low temperature T or large radii a,

$$\Theta \to 0, \qquad \rho_{cor} \to -\rho_r, \tag{8.16}$$

whence $\rho \to \rho_0$. These limits correspond respectively to the classical radiation-dominated regime and the quantum vacuum-dominated regime of the universe. These regimes are determined by the dimensionless parameter Θ, where Θ^4 gives a measure of the radiation content of the universe. For further studies of vacuum energy effects on the Einstein universe see, e.g., [618, 619].

Kaluza–Klein (KK) theories extend Einstein's general relativity to higher dimensions, with the metric components of the compactified extra dimensions acting as gauge potentials, thus providing a geometric basis for unifying gravity and particle fields. KK theory has the compactified fifth dimension embodying the electromagnetic interaction. Weinberg [620] has shown how the quantization of charge comes from the compactification. By extension to a deformed internal space, the Taub universe, Hu and Shen [621] show how the Weinberg angle of electro-weak interaction can be obtained. Likewise for the parameters in GUT theories from higher extra dimensions [622]. This is all very nice, but why would the extra dimensions curl up at the Planck scale and stay small while only four spacetime dimensions expand to the disparately large scale of our universe today? Vacuum energy is seen to play an important role in KK theories and its backreaction has been proposed as a possible mechanism for spontaneous compactification of the extra dimensions in Kaluza–Klein cosmologies [623]. For this purpose one needs to find self-consistent solutions to the higher dimensional Einstein equations with the vacuum energies as source [624, 625, 626, 627, 628].

These are good examples of vacuum quantum effects in cosmological backreaction problems. However, note that the vacuum cosmological models illustrated above, from the static Einstein universe to the KK cosmology, can exist only in a parametric or kinematical sense, but not in a dynamical context – parametric in the sense of a one parameter family of solutions for each of the static scale factors which are not dynamically determined. Appelquist, Chodos and Meyers [629] studied the question of whether quantum effects can cause certain classically allowed patterns of dimensional reduction to become unstable. In particular, in the case of general relativity with d toroidally compact dimensions in D-dimensional space-time, they show by means of a one-loop effective potential computation that the only stable possibility is for one of the d dimensions to contract and the other $(d-1)$ dimensions to expand. However, as we shall see below, dynamically induced quantum processes such as vacuum particle creation tend to act to isotropize the expansion rates in all directions, which subverts the goal of KK cosmology in providing a natural explanation why four dimensions open out while the remaining dimensions curl in at very disparate scales. This remains an open question.

8.1.2 Semiclassical Gravity: Effect of Trace Anomaly

Casmir energy is just one form of vacuum energy which could affect the state and the dynamics of the early universe near the Planck time. Trace anomaly is another such quantum source. The stress-energy tensor for classical conformal fields (massless, conformally coupled to a background spacetime) has zero trace, but not necessarily so for quantum conformal fields. As we may recall from discussions in Chapters 2–5 the appearance of an anomalous term for conformal fields in the trace $T \equiv T^{\mu}_{\mu}$, though unexpected, is legitimate and contains important physics. (See, e.g. [489] for a description of its discovery and development.) The trace anomaly T_A arises from the regularization of the stress-energy tensor of quantum conformal fields. We will see how the trace anomaly can backreact on spacetime and bring about significant effects such as engendering Planck scale inflation (e.g. [359]) and affecting the cosmological singularity (e.g., [109]).

Semiclassical Gravity

In general terms we learned from quantum field theories in curved spacetime that when radiative corrections of quantum fields are taken into account, we are naturally led to a modified Einstein theory with quadratic curvature terms $R^2, R_{ab}R^{ab}, C_{abcd}C^{abcd}$ added to the Einstein–Hilbert term R in the gravitational Lagrangian. Modification of the classical Einstein equation is a consequence of the natural demand for a physical renormalized or regularized stress-energy tensor of the quantum field. *Semiclassical gravity* (SCG) is the theory which

incorporates self-consistently the backreaction of quantum matter fields on a classical gravitational background spacetime in a dynamical way. SCG is expected to be operational in a regime where the spacetime curvature approaches but remains lower than the Planck scale ($\ell_p = \sqrt{G_{\rm N}\hbar/c^3} \approx 1.6 \times 10^{-33}\,{\rm cm}$). This includes the very early universe from the GUT epoch $\approx 10^{14}\,{\rm GeV}$ to the Planck era (characterized by the Planck energy $E_p \approx 10^{19}\,{\rm GeV}$) and evaporating black holes, when their masses are still much larger than the Planck mass ($M_p = \sqrt{\hbar c/G_{\rm N}} \approx 2.2 \times 10^{-5}\,{\rm g}$). In semiclassical gravity the gravitational field is treated classically, but its dynamics is governed by the semiclassical Einstein (SCE) equation with source given by the expectation values of the stress-energy tensor of the quantum matter field present. The wave equation governing the matter field is solved simultaneously with the SCE eqn in a consistent manner. We shall see that emphasis on self-consistency is the ultimate challenge to make SCG complete. Same requirement applies to stochastic gravity when the fluctuations of the quantum matter field are included as source in the Einstein–Langevin equation. In contrast, quantum field theory in curved spacetime studies the dynamics of a quantum field in a given, specified, but not dynamically determined, background spacetime. As such, it can be viewed as the test field limit of SCG. In addition to it being the proper theory for the exploration of quantum processes in the early universe and in black holes, semiclassical gravity is of theoretical significance because it dovetails to the low energy limit of superstring theory, a leading candidate theory of quantum gravity.

Semiclassical Einstein Equations for FLRW Spacetimes

Consider the class of FLRW spacetimes with line element

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1-kr^2} + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2 \right), \tag{8.17}$$

where $k = 1, 0, -1$ correspond to the (spatially) closed, flat and open cases respectively. In this universe in addition to the quantum fields of interest we assume the presence of classical radiation and allow for a nonzero cosmological constant. The semiclassical Einstein (SCE) equation has the generic form (2.76). In this conformally flat class of metrics with zero Weyl curvature, we can choose to use the ${}^{(1)}H_{ab}, {}^{(3)}H_{ab}$ combination for the H_{ab} term therein, as in (2.75). The SCE eqn reads:

$$R_{ab} - \frac{1}{2} g_{ab} R + \Lambda g_{ab} = 8\pi G_{\rm N} \left[T^m_{ab} + \hbar \langle T_{ab} \rangle + \hbar\alpha\, {}^{(1)}H_{ab} + \hbar\beta\, {}^{(3)}H_{ab} \right] \tag{8.18}$$

where Λ is the cosmological constant, T^m_{ab} is the stress-energy tensor for classical matter and $\langle T_{ab} \rangle$ denotes the vacuum expectation value of quantum matter field to the lowest order in $\hbar$. As shown in Chapter 2 the state-independent local quantum corrections of order $\hbar$ are included in the H terms which can be moved to the left-hand side of the SCE eqn.

In a FLRW universe with metric (8.17) filled with classical radiation the (t,t) component of the semiclassical Einstein equation reads

$$3\left(\frac{\dot{a}^2}{a^2}+\frac{k}{a^2}\right)-\Lambda=8\pi G_{\rm N}\left\{\rho_0\frac{a_0^4}{a^4}+12\hbar\beta\left(\frac{\dot{a}^2}{a^2}+\frac{k}{a^2}\right)^2 + 6\hbar\alpha\left[2\frac{\dddot{a}\dot{a}}{a^2}-\frac{\ddot{a}^2}{a^2}+2\frac{\ddot{a}\dot{a}^2}{a^3}-3\left(\frac{\dot{a}}{a}\right)^4-2k\frac{\dot{a}^2}{a^4}+\frac{k^2}{a^4}\right]\right\} \tag{8.19}$$

Here, ρ_0 is the radiation density at the time when the scale factor is a_0. The a^{-4} dependence of the ρ_0 term follows directly from the conservation of the stress-energy tensor and the fact that the expectation value of the trace, $T_a{}^a$, vanishes to lowest order, meaning, that the classical stress-energy tensor for conformal fields is trace-free. The order $\hbar$ quantum corrections to this trace is non-zero – thus the name 'anomaly' – carried by the α,β terms in the expression for the trace anomaly (2.62). The values of the α's depend on which massless fields make up the radiation. In the spatially closed universe ($k=1$), as a result of the non-local Casimer vacuum energy, the value of ρ_0 will have added to it a small constant value proportional to $\hbar$, as shown in (8.1). There is no additional non-local contribution to the stress-energy because we are dealing with conformally-invariant free radiation fields in Robertson-Walker metrics, which are all conformally flat.

Effect of Trace Anomaly on Cosmological Singularity

Consider a massless conformal field in a radiation-filled, spatially flat ($k=0$) FLRW universe. As before, the energy density of classical radiation is $\rho_r = \tilde{\rho}_r\, a^{-4}$, where $\tilde{\rho}_r$ is a constant which measures the total number of photons (entropy content) in the universe. The only quantum source present is the trace anomaly (2.62). Owing to the conformal-flatness of the geometry, there is no Weyl tensor-squared term ($\gamma=0$) in (2.68). Therefore the trace anomaly T_A reads

$$T_A=\frac{1}{a^4}\left\{6\alpha\left[-\frac{a''''}{a}+4\frac{a'''a'}{a^2}+3\left(\frac{a''}{a}\right)^2-6\frac{a''(a')^2}{a^3}\right]+12\beta\left[\left(\frac{a'}{a}\right)^4-\frac{a''(a')^2}{a^3}\right]\right\}, \tag{8.20}$$

where a prime denotes taking the derivative with respect to conformal time $\eta=\int dt/a$. Combining the action for classical radiation (2.78) and the action (2.66) for the trace anomaly [109]

$$\Gamma^{(1)}_{TA}[a]=V\int_0^\infty d\eta\left[-3\alpha\left(\frac{a''}{a}\right)^2+\beta\left(\frac{a'}{a}\right)^4\right], \tag{8.21}$$

we have the full effective action of the form

$$\Gamma_{0T}[a] = \Gamma^{(0)}[a] + \Gamma^{(1)}_{TA}[a], \tag{8.22}$$

where the superscripts $(0), (1)$ on Γ indicate they are of the zeroth and first order in $\hbar$, corresponding to classical matter and quantum field contributions, which in the present case consists only of the trace anomaly. Variation of this action with respect to the scale factor $\delta\Gamma_{0T}/\delta a = 0$ gives the semiclassical Einstein equation describing the backreaction of the trace anomaly

$$G^{00} = 8\pi G_{\rm N}(\rho_r + \rho_{TA}) \tag{8.23}$$

For a spatially flat, radiation-filled spacetime, it is more convenient to work with the rescaled variables defined by $(b, \tilde{\eta})$

$$a = \sqrt{16\pi G_{\rm N}}\ \tilde{\rho}_r^{1/4} b, \qquad\qquad \eta = 6^{1/2}\tilde{\rho}_r^{-1/4}\tilde{\eta}. \tag{8.24}$$

They are invariant under the scale transformations $a \to \Lambda a, x^\alpha \to \Lambda^{-1}x^\alpha, \tilde{\rho}_r \to \Lambda^4\tilde{\rho}_r$. Thus

$$\Gamma[b] = 6^{1/2}\tilde{\rho}_r^{3/4} V \int_0^\infty d\tilde{\eta}\ L(b'', b', b),$$

where

$$L = -(b')^2 - \frac{\alpha}{12}\left(\frac{b''}{b}\right)^2 + \frac{\beta}{36}\left(\frac{b'}{b}\right)^4 - 1, \qquad b' \equiv db/d\tilde{\eta}. \tag{8.25}$$

This is a fourth-order ordinary differential equation for b. But since L is independent of $\tilde{\eta}$, a first-order integral exists in the form [109, 113]

$$E = -b'\left(\frac{\partial L}{\partial b''}\right)' + b''\frac{\partial L}{\partial b''} + b'\frac{\partial L}{\partial b'} - L$$

or,

$$E - 1 = -b'^2 + \frac{\alpha}{3}\left[\frac{1}{2}\frac{b''''b'}{b^2} - \frac{b''(b')^2}{b^3} - \frac{1}{4}\left(\frac{b''}{b}\right)^2\right] + \frac{\beta}{12}\left(\frac{b'}{b}\right)^4. \tag{8.26}$$

Cosmological solutions from semiclassical Einstein equations incorporating vacuum fluctuation effects (and thus involving higher derivatives) was considered first by Ruzmaikina and Ruzmaikin [358] at a time when trace anomaly was not yet known. Their source which is similar in nature to the ${}^{(1)}H_{\mu\nu}$ and ${}^{(2)}H_{\mu\nu}$ of [85] is based on the vacuum polarization calculation of [70]. For these third-order nonlinear differential equations many families of solutions exist. To seek those physically meaningful solutions one needs to impose appropriate physical boundary conditions. Ref. [358] studied solutions which avoid the cosmological singularity ('bounce' solution), but does not approach the Friedmann solution at late times. Parker and Fulling [630] studied the behavior of a FLRW universe driven by a single mode of a massive scalar field and found a bounce solution at

the Compton wavelength of the quantum field. How much of the big picture can be determined out of one mode's behavior is not clear.

By contrast, Fischetti, Hartle and Hu (FHH) [109] imposed the requirement that all solutions approach the radiation-dominated Friedmann behavior at late time $a = \eta$ or $b \to \tilde{\eta}$ as $\tilde{\eta} \to \infty$ and looked for the behavior of a at $\tilde{\eta} = -\infty$ (or $t = 0$). As a consequence of the asymptotically Friedmann condition, $E = 0$. Writing $b' = f^{2/3}, x = b^3$, one can find a second-order equation for f

$$\frac{d^2 f}{dx^2} = -\frac{\beta}{12\alpha}\frac{f}{x^2} + \frac{1}{\alpha x^{2/3}}\left(\frac{1}{f^{1/3}} - \frac{1}{f^{5/3}}\right), \tag{8.27}$$

with the condition $f \to 1$ as $x \to \infty$. For different ranges in the value of β/α FHH found several classes of solutions with different behavior near the singularity. Their conclusion is that quantum effects due to the trace anomaly only serve to soften the approach to singularity but not avoiding it. Anderson [113, 114, 609] carried out more detailed analysis on this problem for a wider range of β/α with results showing more variety. He also considered the effect of trace anomaly on particle horizons, an issue brought to renewed interest with the advent of inflationary cosmology. Indeed, one version of inflationary cosmology proposed by Starobinsky [359] is based on the effect of trace anomaly or the like. Effect of finite temperature quantum fields including the trace anomaly was discussed in [631]. For theories with only the R^2 term in the Lagrangian their behavior related to the cosmological singularity, the horizon, and inflation have been studied by Mijic, Morris and Suen [632].

8.1.3 Trace Anomaly-Driven Inflationary Cosmologies

$f(R)$ Lagrangian Theory of Gravity

The 2015 analysis by the Planck Collaboration has provided significant constraints on many suggested inflationary models [633]. One scenario that is not excluded by the data is inflation driven by an R^2 term in the gravitational Lagrangian, often known as Starobinsky inflation [359, 634]. A surge of interest in these so-called $f(R)$ gravity theories followed. For reviews of $f(R)$ gravity, see [635, 636].

Note there are two Starobinsky inflationary models, the original 1980 model built on trace anomaly has higher-order derivatives in the equation which governs the scale factor. The runaway solutions taken as a model for inflationary cosmology were later shown by Simon to be spurious and unphysical. The modified 1983 model uses parameters which can in practice mitigate these pathologies, and can be regarded as an inflationary model with a certain 'scalaron' potential, bringing it more in line with the other inflationary models proposed around that time: old, new, chaotic, etc. Kofman, Linde and Starobinky [361] refer to these two

classes of models as Type I, based on vacuum polarizations, of the nature of the trace anomaly, and as Type II, based on an inflaton potentials. The Starobinsky inflation which survived observational scrutiny thus far is not the 1980 Type I model, but the 1983 Type II model.

Stability of de Sitter Solution in $f(R)$ Lagrangian Theories

A minimal necessary condition for a solution to be physically acceptable is its stability. With the introduction of higher-order curvature theories one needs to show that the de Sitter spacetime is a stable vacuum solution before it can be qualified for the depiction of the inflationary phase of the very early Universe. For this purpose Barrow and Ottewill [637] examined the existence and stability of homogeneous and isotropic cosmological solutions to general relativity (GR: theory with only an R term in the Einstein–Hilbert Lagrangian) with respect – not to perturbations within GR – but to perturbations of GR into a large class of metric gravity theories. (See, e.g., Chapter 39 of [604] for a description of metric theories of gravity versus non-metric theories.) They proved an existence theorem for the de Sitter cosmological solution of GR to exist as a solution of the general $f(R)$ Lagrangian theory of gravity and then examined the stability properties of any de Sitter solution that does exist. They also examined the existence and stability of the Friedmann solutions of GR in $f(R)$ theories and showed that the familiar de Sitter and Friedmann solutions almost always exist and, under fairly general circumstances, are stable.

At the level of semiclassical gravity, for a massless nonconformally coupled scalar field in a spatially flat FLRW universe driven by a cosmological constant Pérez-Nadal, Roura and Verdaguer [638] have proven the stability of de Sitter spacetime under isotropic perturbations, a result independent of the free renormalization parameters. They showed the existence of a self-consistent solution, associated with the Bunch–Davies vacuum for the quantum fields, with an effective cosmological constant slightly shifted from its classical value due to the vacuum polarization effects. They also emphasized that a complete analysis of the backreaction problem in de Sitter spacetime and its stability should take into account the effect of the quantum metric fluctuations as well, a topic to be expounded further in the following chapters. Later, with Papadopoulos [639], these authors solved nonperturbatively the semiclassical Einstein equation governing the dynamics of linear metric perturbations around de Sitter spacetime when the quantum backreaction of conformal scalar fields on the mean geometry is included. Their exact solutions establish the stability of de Sitter with respect to general linear metric perturbations (of scalar, vector and tensor type) and extend some of the existing no-hair results for de Sitter in classical general relativity to the case in which the effects of the quantum vacuum polarization of conformal fields on the semiclassical geometry are included. More discussions of de Sitter space quantum fields with fluctuations in the context of semiclassical stochastic gravity are contained in the later chapters.

Starobinsky Inflation

Consider[1] a massless conformal scalar field in a spatially-closed ($k = 1$) FLRW universe without a cosmological constant $\Lambda = 0$. Because of the conformal flatness of the FLRW metric the coefficient γ multiplying the Weyl curvature tensor squared in (2.68) takes on zero value. The SCE eqn (8.19) for the FLRW universe has a simple solution when $\beta < 0$

$$a(t) = \sqrt{|\beta|}\cosh\left(\frac{t}{\sqrt{|\beta|}}\right). \tag{8.28}$$

This is a de Sitter solution with an effective cosmological constant $\Lambda = 3/|\beta|$ and scalar curvature $R = 12/|\beta|$. Starobinsky showed that after a period of exponential expansion the universe evolves to a low curvature matter-like dominated stage driven by high-frequency oscillations of R (scalarons) which later decay to pairs of particles and antiparticles. This is the original 1980 Starobinsky inflation, a scenario endowed with a 'graceful exit' mechanism and fuel for reheating.

FHH Solutions vs Starobinsky Inflation

A remark is in place on how Starobinsky inflation relates to the FHH solutions [109] described in the previous section, since it is from the same SCE eqn driven by the trace anomaly of a conformal massless scalar field. As we recall, the time-time component of the SCE equation has up to three time derivatives of the scale factor, while the corresponding classical Einstein equation after being reduced to an equation for the constant of motion has only one. This means that there are many more solutions to the equations, and that one must fix the initial values not only of the scale factor a but also of its first two time derivatives. An overwhelming number of solutions diverge in time. FHH did not trust these runaway solutions to be physical. Instead, they demand that at late times $t \to \infty$ the universe settles into a radiation-dominated stage, i.e., $a(\eta) \sim \eta$ and examine the nature of the solutions at $t = 0$. They also asked if there exist physical solutions to this SCE equation beginning with a radiation-dominated phase in the *previous cosmological cycle*, assuming a cyclic universe scenario, passing a singularity or a bounce (with a minimum scale length), then evolving in the present cycle to a radiation-dominated phase. Thus FHH solved a boundary value problem for this SCE eqn while Starobinsky solved an initial value problem. The danger of regarding spurious solutions as physical which FHH noted and tried to avoid will be discussed further in a later subsection.

One conclusion relating the existence of bounce solutions to inflationary solutions which the FHH analysis can provide is, according to Carlson et al [608]

[1] If the reader finds the derivation in Starobinsky's 1980 paper too terse to follow try Vilenkin [360] especially Sec. II. Here we follow the narrative of Parker and Simon [357], feeding into FHH [109].

that, if the universe underwent Starobinsky inflation ($\alpha > 0$) there will be no bounce solutions. If a singularity is present classically, it will remain so in this semiclassical cosmology with a quantum source. We can see this as follows: if $\alpha > 0$, then all the terms on the right side of (8.19) either vanish or are positive if $H \equiv \dot{a}/a = 0$, and therefore this equation cannot have $H = 0$ at any time. This means that the scale factor will monotonically increase until either a singularity is reached or it becomes infinite.

Future Singularities?

An interesting question to ask, similar in nature to that posed by FHH on the possibility of a bounce solution in the past, and by Starobinsky on the existence of a de Sitter solution evolving into a matter-dominated universe, is, if Starobinsky inflation existed in the past, whether a cosmological singularity exists in the future, and if so of what type. This is the question Carlson et al. [608] raised. To address this issue one needs some way to incorporate the dark energy as it comprises a good fraction (0.7) of the total mass energy of the present universe. The authors of [608] assumed the dark energy can be described by classical matter with some equation of state. They then solved the SCE eqn with conformally invariant quantum fields plus this classical matter and were able to classify the different types of future singularities depending on the equation of state used for the dark energy.

8.1.4 The Attractions and Perils of R^2 Cosmology

The presence of an R^2 term in the gravitational Lagrangian and the presence of quantum fields results in the appearance of higher derivative terms in the SCE (semiclassical Einstein) or backreaction equations. These terms lead to a much larger number of solutions than those for classical general relativity. In many cases they also lead to solutions which may follow a solution to the classical Einstein equation for some time but are unstable and eventually deviate substantially from it, usually by going into a period of extremely rapid expansion or contraction. It has been shown [640, 357] that in cosmology such solutions can be eliminated if one follows a procedure called order reduction, in which the semiclassical backreaction equations are reduced to second-order equations. An important consequence observed by Simon [640] is that, if order reduction is used, then the 1980 Type I Starobinsky inflation does not occur.

No Starobinsky Inflation after Order Reduction

Let us see why starting from the perspective of a perturbative expansion can only serve as an approximation to the full quantum theory, that semiclassical gravity, like semiclassical electrodynamics, is an approximation at first order in $\hbar$ to the full, albeit as yet unknown, theory of quantum gravity. In this light any

appearance of nonperturbative solutions need be scrutinized closely within the validity perimeters of a perturbative theory and not simply be admitted as a *bona fide* solution which can stand on its own strength.

Nonperturbative spurious solutions in perturbative theories. The warning Simon issued was, despite the fact that both the effective action and the field equations governing the quantum corrections are perturbative expansions in $\hbar$, most of the solutions are not perturbatively expandable in $\hbar$ (i.e., not analytic functions of $\hbar$ as $\hbar \to 0$). The correct way to find a genuine solution is to take the perturbative expansion seriously and exclude all solutions not perturbatively expandable in $\hbar$ as fictitious. The guiding principle is self-consistency: the effective action is a formal perturbative expansion, the field equations are formal perturbative expansions, and so should the solutions be. The action and the field equations lose their interpretation as a perturbative expansion if evaluated at nonperturbative extrema. For an action or field equation derived as a perturbative expansion in powers of $\hbar$, all nonperturbative solutions of the theory has already been discarded by the perturbative approximation of the action. Nonperturbative solutions to the perturbative field equations are not expected to be related to nonperturbative solutions to the nonperturbative field equations. These pseudo-solutions must be discarded for the sake of self-consistency. Only solutions that are also perturbative expansions in powers of $\hbar$ can be expected to approximate the full theory. Simon [238, 287] provided several simple and familiar examples to ascertain this important point.

Higher derivatives, nonlocality, order reduction. The nonlocality in the equation of motion of one subsystem is a reflection of the difference in time scales (the cause of non-Markovianity) between the dynamics of two coupled subsystems. When each is considered alone the influence of the other is subsumed (integrated over). This point can be made clearer in the conceptual framework of open quantum systems, where the 'system' of interest, here, the classical gravity sector, is viewed as a subsystem of the whole (consisting of quantum gravity and quantum matter in full) while its 'environment' is the quantum matter field sector. When the backreaction of the coarse-grained environment represented by one or many colored noise is included the resultant open system dissipative dynamics has nonlocal kernels. See, e.g., [326]. Consider the dynamics of two interacting subsystems each obeying an ordinary differential equation with an interaction term signifying the presence of the other, as Zwanzig explained [134]: One can solve the coupled equations to get the dynamics of both. Or, if the dynamics of only one of the subsystems is of interest, one can recast the two coupled equations of motion into an integro-differential equation with no loss of information. The nonlocal kernels in this equation are the source of

non-Markovian (memory) effects, the dynamics of the other subsystem being subsumed but kept in the dynamics of the system of interest. Upon coarse-graining the environment and choosing a causal initial condition for the system, nonlocal dissipation generically appears in the open system dynamics. To find solutions to this integro-differential equation one can turn it into an ordinary differential equation, but higher derivative terms will appear, and with them, the ensuing problems of how initial conditions should be physically specified, and of possible acausality and instability issues in its dynamics. A familiar example is radiation reaction, where the system is the moving charge and the environment is the electromagnetic field. The Abraham–Lorentz–Dirac equation (see discussions in Chapter 1, Sec. 1.2.4) has higher derivatives arising from the integrating over of the field degrees of freedom.

Nonlocality is also a feature innate in effective theories, i.e., a theory rendered 'simpler' by integrating over – a form of coarse-graining – a subset of its degrees of freedom. When one inspects closely how an effective theory (of lower energy) emerges from a more basic theory, one can see how a noise term representing the coarse-grained (high energy) sector [641] can be defined which measures the degree of effectiveness of the effective theory. Those low energy effective theories which do not contain any dissipation or noise terms and thus whose dynamics is unitary usually requires the introduction of an averaged potential (e.g., the Vlasov equation [642, 125]) or some approximation (e.g., Hartree approximation [643]). See e.g. [244, 590] for examples of how different approximations representing different levels of coarse-graining lead to different dynamics, from unitary to mixing to dissipative (see, e.g. Chapter 1 of [326]).

Now, equipped with these perspectives, how should one proceed to find the solutions of the SCE eqn which is a fourth-order differential equation? The way to deal with equations with higher derivative terms, as Parker and Simon [357] show, is to perform an order reduction while keeping the correct perturbative orders. Note that a semiclassical theory, constrained to only solutions perturbatively expandable in $\hbar$, has the same dynamical degrees of freedom as the classical gravitational field, despite the presence of fourth-order derivatives in the field equations. The amount of initial data required to specify a physical solution is, however, the same as for classical gravity, which is given by solutions to a second-order differential equation. The reduction is performed iteratively, using lowest-(perturbative) order results to simplify the higher-order semiclassical corrections. It has the distinct advantage that unphysical solutions are bypassed. Specifically, for the SCE eqn derived earlier (8.19), reductions for corrections containing H_{ab} terms in (8.18) have been calculated for several cases by Bel and Sirousse-Zia [644], while Parker and Simon [357] carried out a full display of cases, with and without the cosmological constant, for spatially flat and closed FLRW universes. These reduced equations are expected to contain fewer numerical instabilities than the original fourth-order equations, and yield only physically relevant solutions.

In conclusion Simon's thesis asserts that, there are no de Sitter or de Sitter-like self-consistent solutions except in the presence of a cosmological constant, and inflation arise from spurious solutions in the 1980 Starobinsky model. Furthermore, linearized gravitational perturbations in a de Sitter background (with a cosmological constant) show no signs of instability from quantum effects.

Cautionary note for singularity analysis. Carlson et al. [608] study numerically the behaviors of the order-reduced equations in special cases and compare them to numerical solutions to the full semiclassical backreaction equations in those cases. They find that, once quantum effects become important, a solution to the order-reduced equation generally deviates significantly from the corresponding solution to the full SCE eqn and that if the order-reduced solutions are continued into the regions near the singularity, in many cases they have a qualitatively different effect on that singularity than the solutions to the SCE eqn. They claim that solutions to the order-reduced equations are often not very useful for studying quantum effects *near final singularities*. Note first that the FHH solutions to the full SCE equation pertaining to the past singularity are obtained upon imposing regular physical boundary conditions (radiation-filled FLRW universe) at late times. Second, it is desirable to try to obtain the full nonperturbative solutions near a singularity by both numerical and analytical methods, just so that we know the behavior attributed to the singularity is physical, and not from using approximations not applicable to that regime. Third, in terms of global analytical methods, one can investigate whether critical manifold exists in the solutions to the SCE eqn similar to that associated with the ALD equation, mentioned in Chapter 1, such as those found by Kunze and Spohn [645, 646].

R^2 *Lagrangian as 'Type II' Inflationary Models*

Our story does not end here. There is a Plan B to the rescue. A modified model was proposed by Starobinsky in 1983 [634] and the difference between these two classes of models are explained well in a later paper by Kofman, Linde and Starobinsky [361]. We follow their presentation here.

'Type II' inflation models. As mentioned earlier, there is an abundance of inflationary cosmological models, starting with Guth's 'old' inflation through a first-order phase transition (also Sato's) by nucleation (bubble formation and collision), the Albrecht–Steinhardt–Linde 'new inflation' through a second-order phase transition by spinodal decomposition, which inspired the now popular 'slow-roll' models. Add to this Linde's chaotic inflation operative at the Planck scale and Starobinsky's stochastic inflation, to name just a few important ones proposed in the 80s. In the intervening three decades, numerous

models proliferate, but most of them involve the postulate of a potential $V(\phi)$ describing how the inflaton field ϕ configuration changes: Guth's 'old' inflation has an uneven double well potential, 'new' inflation has a largely flat Coleman–Weinberg type of potential and Linde's chaotic inflation is via a $m^2\phi^2$ potential. They usually come with specified parameters aimed at getting $V(\phi)$ to perform certain desirable functions matching observational constraints.

One important point perhaps not stressed strongly enough is, the $f(R)$ type of models which enjoy the enthusiasm of today's cosmological community, although often called the Starobinsky inflation, should be understood as belonging to this 1983 Type II class, not the original 1980 Type I class of models based on vacuum polarizations, whose equation of motion has higher derivatives and the solutions are subjected to the criticism of Simon described above.

Modified Starobinsky inflationary model. Let us take another look at the expression for the conformal anomaly (2.62) noting that our (α, β, γ) correspond to the $\varpi \times (k_3, k_2, k_1)$ of [361] respectively, with $\varpi \equiv (2880\pi^2)^{-1}$. To get a long enough period of inflation one needs to assume $k_2 > 0, k_3 < 0, |k_3| \gg k_2$, which is not satisfied in typical cases. Kofman, Linde and Starobinsky referred to this class of trace anomaly-driven inflation as Type I model. Starobinsky in 1983 [634] proposed to add a local term $m_p R^2/(96\pi M^2)$ to the Lagrangian, with $M \ll m_p$, and show that it can sustain a prolonged quasi-de Sitter phase of expansion. Adding such a term amounts to a renormalization of the constant k_3, but with this extra freedom it is relieved from the duty of representing only the light quantum fields. More generally, a local term $f(R)$ satisfying the condition $\lim_{R\to 0}[f(R)/R] = 0$ can be added to the gravitational Lagrangian. The important additional condition $M \approx (10^{-4} - 10^{-5})m_p$ detaches this type of inflation away from Planck scale (quantum gravity) inflation, such as those caused by the trace anomaly in Starobinsky's 1980 model, and functionally closer to the Type II inflation based on a potential for the inflaton field with catered features. It was also shown that this type of inflation can be realized over a wide range of initial conditions for the scalar curvature R.

In this light, Kofman, Linde and Starobinsky entertained the possibility that both types of inflation can exist simultaneously with a scalar field ϕ under a potential $V(\phi)$ and a vacuum polarization term $m_p R^2/(96\pi M^2)$ in the Lagrangian, with $M \ll m_p$. Details of solutions to this extended model can be found in [361].

Screening Semiclassical Gravity Cosmological Models

Constraints from observation are important for screening out models which do not comply. For example, based on the Planck observatory 2013 data Huang

[647] considered a polynomial $f(R)$ inflation model of the form $f(R) = R + \frac{R^2}{6M^2} + \frac{\lambda_n}{2n}\frac{R^n}{(3M^2)^{n-1}}$ where λ_n is a dimensionless coupling and $n > 2$. The extra coefficient of the R^2 term is normalized to one by redefining the energy scale M. It reduces to Starobinsky's 1983 model when $\lambda_n \to 0$. Huang focused on the limit $\lambda_n \ll 1$ and the case when the R^n term can be taken as a small correction to Starobinsky's model. He calculated the spectral index and tensor-to-scalar ratio in this $f(R)$ inflation model, compare with the Planck data and concluded that the R^n term should be exponentially suppressed, i.e. $|\lambda_n| \leq 10^{-2n+2.6}$ For theories with R^2 Lagrangian this translates to an extraordinarily large value for the constant multiplying to the R^2 term in the Lagrangian, $\alpha = 10^9$ in (8.19), with rather odd yet perhaps interesting consequences.

By its defining nature that gravity is treated classically, solutions of semiclassical gravity are not expected to be reliable when spacetime curvature reaches the Planck scale. However, if Starobinsky inflation occurred, with the dimensionless parameter α measuring the magnitude of the R^2 term in the gravitational Lagrangian in the order of 10^9 [647], then it implies that quantum effects can be important when the spacetime curvature is much weaker, well below the Planck energy scale. This large value, along with the large number of effectively massless quantum fields present, implies that below the Planck energy quantum effects due to the *gravitational* field can be ignored to leading order. If so, this would ensure the semiclassical Einstein equation and its solutions are self-consistent and validate the use of semiclassical gravity for the depiction of such quantum processes in the very early universe. However, there are fundamental issues in whether the inflation solutions are physical, as we will discuss after a few remarks.

Even though a large value of $|\alpha|$ results in solutions for which quantum effects are important on scales well below the Planck energy, potential problems remain. One is that the semiclassical approximation may become invalid at scales much shorter than the Planck scale. This has been discussed [128] in the context of a large N expansion, with N the number of identical quantum fields. The main reason is that in the effective field theory approach [192] there is an infinite series of higher order terms in the gravitational Lagrangian, and this expansion is generally thought to break down when these terms become comparable to each other. Of course it is possible that the coupling constants for the other terms are very small compared with $|\alpha|$. So there might be a region where that term is large and the others are still small. We will give a fuller discussion of the validity issue of SCG in a later chapter, both in the framework of $1/N$ expansion where the leading order gives rise to SCG, and in the context of the Einstein–Langevin equation, where the effect of stress-energy fluctuations are also taken into considerations.

Quantum Gravity Weighs In

To explore what could have preceded Starobinsky inflation, Vilenkin is of the opinion that Starobinsky's model is consistent only if the universe is

spontaneously created [648, 360]. Creation of the universe as a quantum phenomenon was proposed by Brout, Englert and Gunzig [649], before inflation. Barvinsky and Kamenshchik [650] considered the quantum creation of the universe described by the density matrix defined by an Euclidean path integral. This yields an ensemble of universes what these authors consider as a cosmological landscape, in a mixed quasi-thermal state. They show that such a state is dynamically more preferable than the pure quantum state of the Hartle–Hawking type [222]. The latter is suppressed by the infinitely large positive action of its instanton, generated by the conformal anomaly of quantum matter. Thus, according to these authors, infrared effects of quantum gravity dynamically eliminates the Hartle–Hawking vacuum instantons. And, to the extent the Hartle–Hawking instantons can be regarded as posing initial conditions for the anomaly-driven de Sitter expansion, these solutions will also be ruled out, meaning, there will be no Starobinsky inflation. For topological effects noteworthy is the instanton-driven inflation of Hawking, Hertog and Reall [651].

8.2 Backreaction of Cosmological Particle Creation

In this section we consider the backreaction of particle creation on the dynamics of the early universe and the state of matter in the universe, using canonical quantization and adiabatic regularization methods. For this purpose we insert a summary discussion of adiabatic vacuum for quantum field theory in dynamical spacetimes and adiabatic regularization for backreaction problems in cosmology.

8.2.1 Adiabatic Vacuum and Regularization

Recall from Chapters 2 and 3 the case of a free quantum field in a time-dependent background field or a dynamical background spacetime. When a mode decomposition is possible the (c-number) amplitude function $f_{\mathbf{k}}$ of the $\mathbf{k}$ mode obeys a wave equation (2.6) or (2.27) where the natural frequency acquires additional terms coming from curvature and field coupling effects,

$$\frac{d^2 f_{\mathbf{k}}}{dt^2} + \omega_{\mathbf{k}}^2(t) f_{\mathbf{k}}(t) = 0. \tag{8.29}$$

All the same, the wave equation is in the form of a harmonic oscillator with a time-dependent natural frequency $\omega_{\mathbf{k}}$. This is where QFT in a dynamical background differs from QFT with constant fields, in Minkowski or in static spacetimes. When $\omega_{\mathbf{k}}$ is constant one can use the same Fock space representation of the field theory since it remains the same as originally defined at t_0. Staticity or constancy in these systems is represented by the existence of a Killing vector in time ∂_t, which enforces the positive and negative frequency components to remain separated. This means, in second quantized language, that the particles and antiparticles are separately well-defined and their numbers are kept constant.

Therefore, the possibility of defining a positive frequency component in a field theory is the precondition for a vacuum state to exist. Maintaining such a condition in a dynamical setting is not always possible.

However, if the external field or background spacetime changes sufficiently gradually one can construct an (n-th order) *adiabatic vacuum* or number state to match with how fast the background field or spacetime changes. Recall from classical wave scattering theory that a WKB solution can give a reasonable approximation to the wave equation with a time-dependent natural frequency when the system changes gradually enough. Successively higher order WKB (or adiabatic) solutions can encompass more rapid changes in the background field or dynamical spacetime. The sequence of successively higher order WKB solutions to this wave equation has been explored quite extensively by researchers working on wave propagation in inhomogeneous media. There, the reflection of waves due to successively higher-order derivatives in the dielectric media can be treated with successively higher-order WKB solutions. The condition for the reflection of waves is the same condition for particle creation, in a second quantization representation. Vacuum means no particles are present or created. Thus we demand that up to a certain adiabatic order (say, ℓ-th order), there should only be positive frequency component and no negative frequency component present– that is our adiabatic vacuum. Translating the variation in spatial inhomogeneity to how rapidly the background field or spacetime changes in time is an intuitively easier way to understand the physical meaning of the adiabatic vacuum.

Now, how is this useful for identifying ultraviolet divergences in the stress-energy tensor? The task for adiabatic regularization is to find a way to identify these divergences, and arrange them in adiabatic order so the subtraction of these divergences is carried out in an orderly manner. There are three types of divergences: quartic, quadratic and logarithmic (in wave number $\mathbf{k}$) which correspond to the a_0, a_1, a_2 'HaMideW' coefficients. In this light we should set $\ell = 6$, meaning, at the sixth adiabatic order the stress-energy tensor becomes finite and the vacuum is well defined to the fourth adiabatic order. Using the vacuum of this adiabatic order we can then identify the origin of the divergences, e.g., the quadratic and logarithmic divergences arise from the second and fourth adiabatic frequency corrections (or deviations).

This iterative-time WKB approximation proposed by Hu [71], inspired by wave scattering in inhomogeneous media, is similar in spirit to the 'n-wave regularization' of Zel'dovich and Starobinsky [69]. They produce the same results as the adiabatic regularization of Parker and Fulling [84, 85, 86]. Noteworthy recent developments include: Junker and Schrohe [652] have extended the definition of adiabatic vacua to general spacetime manifolds which is also applicable to interacting field theories; Landete et al. [653, 654] have extended adiabatic regularization to spin 1/2 fields, and Chu and Koyama [655] to gauge fields in curved spacetimes. Below we introduce adiabatic vacuum and regularization using the iterative-time WKB approach of Hu [71].

Adiabatic Vacuum

Consider the wave equation (2.10) in t time for the amplitude function of the $\mathbf{k}$ mode. (We shall omit the $\mathbf{k}$ subscript, as only one mode is being considered.) The idea is to use a transformation of both time t and dependent variable f to reduce this equation to one we can solve.

Define a new time variable $t_1 = t_1(t)$, and write Eq. (2.10) as

$$\left(\frac{dt_1}{dt}\right)^2 \frac{d^2 f}{dt_1^2} + \left(\frac{d^2 t_1}{dt^2}\right) \frac{df}{dt_1} + \omega^2 f = 0. \tag{8.30}$$

The equation is simplified by choosing

$$\frac{dt_1}{dt} = \omega(t), \tag{8.31}$$

whereby

$$\frac{d^2 f}{dt_1^2} + \frac{1}{\omega}\left(\frac{d\omega}{dt_1}\right) \frac{df}{dt_1} + f = 0. \tag{8.32}$$

The first-order term is eliminated by writing

$$f = \omega^{-1/2} f_1, \tag{8.33}$$

resulting in

$$\frac{d^2}{dt_1^2} f_1 + w_1^2 f_1 = 0, \tag{8.34}$$

where

$$w_1^2 = 1 + \epsilon_2, \quad \epsilon_2 = -\frac{1}{\omega^{1/2}} \frac{d^2}{dt_1^2}(\omega^{1/2}). \tag{8.35}$$

Observe that Eq. (8.34) has the same structure as the original Eq. (2.10). If ω varies sufficiently slowly, we can neglect ϵ_2, and it becomes trivial.

Higher order WKB approximations to the wave equation are obtained by iterating this procedure. Define

$$dt_r \equiv w_{r-1} dt_{r-1} \equiv W_r dt \quad (w_0 \equiv \omega, \quad t_0 \equiv t) \tag{8.36}$$

$$f_r \equiv w_{r-1}^{1/2} f_{r-1} = W_r^{1/2} f \tag{8.37}$$

$$W_r \equiv w_0 w_1 \cdots w_{r-1} \tag{8.38}$$

$$\Theta_r \equiv \int W_r dt. \tag{8.39}$$

The $n(=2r)$-th order WKB equation is given by $(r = 1, 2, \ldots)$

$$\frac{d^2}{dt_r^2} f_r + w_r^2 f_r = 0, \tag{8.40}$$

where, for $r = 1, 2, 3, \ldots$ on,

$$w_r^2 = 1 + \epsilon_{2r}, \qquad \epsilon_{2r} = -\frac{1}{w_{r-1}^{1/2}} \frac{d^2}{dt_r^2}(w_{r-1}^{1/2}). \tag{8.41}$$

The quantities ϵ_{2r} are called the *adiabatic frequency corrections* [86]. If $|\epsilon_{2r}| \ll 1$, the solution of the wave equation correct up to the $n(= 2r)$-th order of derivatives of the natural frequency $w^2(t)$ with respect to t_r is given by

$$f_{(n)}(t) = \frac{1}{(2W_r)^{1/2}} \left[A e^{-i\int W_r dt} + B e^{i\int W_r dt} \right], \tag{8.42}$$

where A, B are complex functions. The subscript (n) on f indicates that a solution to the full wave equation is sought *including up to* the n-th adiabatic order. In contradistinction, we define a $(n = 2r)$-th order adiabatic solution as the solution with ϵ_{2r} set equal to zero.

The *nth order adiabatic vacuum* is defined such that there is no negative frequency component in the nth order WKB solution. What this means is that, at the nth adiabatic order approximation, the nth order adiabatic number state is obtained by assuming that the wave function $f(t)$ is given only by the positive frequency nth order WKB solution

$$f(t) \simeq f_{(n)}^{+}(t) = \frac{A e^{-i\int^t W_{n/2} dt}}{\sqrt{2W_{n/2}}}. \tag{8.43}$$

So intrinsically this is a quasilocal (in time) expansion. It can be translated to orders in momentum k, but, as we pointed out in a footnote in Chapter 2, they are not one-to-one, because adiabatic order is measured by the power of time derivatives, not powers in k.

Adiabatic Regularization

To apply this method to the regularization of the stress energy tensor in an external field or a dynamical spacetime, we need to carry out a 4th order adiabatic expansion. Let us study a slightly more general wave equation (2.31) for $\chi_{\mathbf{k}}(\eta_{\mathbf{k}})$ with natural frequency $\sqrt{\omega_{\mathbf{k}}^2(\eta) + Q}$ where Q is a term of second adiabatic order. In the cosmological context Q stands for either Q_ξ for a nonconformally coupled scalar field in FLRW universe or for Q_β for a conformally coupled scalar field in an anisotropic Bianchi I universe.

As explained earlier, how high an adiabatic order one needs to expand to depends on what one aims to achieve. To identify up to and including the quartic ultraviolet divergences, we need the $(n = 4)$-th order adiabatic frequency corrections $\epsilon_{(4)}$ in (8.41). Eq. (8.39) tells us we need to get w_2, and hence up to $W_3 = w_0 w_1 w_2$. This means we need to have a well-defined adiabatic vacuum at the 6th adiabatic order, which in turn means the wave function at the $(n = 6)$-th order in (8.43) should have only a positive-frequency component,

which corresponds to the condition of no particle creation and a finite stress-energy tensor. Thus,

$$\chi_{(6)} = \frac{e^{-i\int W_3 dt}}{\sqrt{2W_3}} \tag{8.44}$$

where

$$W_3 = \omega(1+\epsilon_2+\epsilon_4)^{1/2} \tag{8.45}$$

(We will suppress the mode index $\mathbf{k}$ in $\chi, W, \omega, \epsilon$.) Assuming that the solution χ is well-approximated by $\chi_{(6)}$ we have

$$|\chi|^2 = (2W_3)^{-1}, \quad |\chi'|^2 = (2W_3)^{-1}\left[W_3^2 + \frac{1}{4}\left(\frac{d}{d\eta}\ln W_3\right)^2\right]. \tag{8.46}$$

The adiabatic frequency corrections are given by

$$\epsilon_{2(2)} = \frac{Q}{\omega^2} - \frac{\bar\omega^2}{4} - \frac{\bar\omega'}{2\omega}, \quad \epsilon'_{2(2)} = \frac{Q'}{\omega^2} - 2Q\frac{\bar\omega}{\omega} - \frac{\bar\omega''}{2\omega}, \tag{8.47}$$

where the subscripts in parentheses denote the adiabatic order and we have defined the nonadiabaticity parameter (here, for frequency ω in conformal time η) as $\bar\omega_{\mathbf{k}} \equiv \omega'_{\mathbf{k}}/\omega^2_{\mathbf{k}}$. Substituting these into (8.42)

$$N_{\mathbf{k}}(t) = \frac{1}{2\hbar\omega_{\mathbf{k}}}(\pi^2_{\mathbf{k}} + \omega^2_{\mathbf{k}} q^2_{\mathbf{k}}) - \frac{1}{2} = s_{\mathbf{k}}, \tag{8.48}$$

and keeping terms of the same adiabatic order (as measured by the time-derivatives) we get

$$\begin{aligned} s_{k(2)} &= \frac{1}{16}\bar\omega^2 \\ s_{k(4)} &= \frac{1}{16}\left(-\frac{1}{2}\frac{\bar\omega\bar\omega''}{\omega^2} + \frac{1}{4}\frac{\bar\omega'^2}{\omega^2} + \frac{1}{2}\frac{\bar\omega'\bar\omega^2}{\omega} + \frac{3}{16}\bar\omega^4 \right. \\ &\quad \left. + \frac{Q^2}{\omega^4} + Q'\frac{\bar\omega}{\omega^3} - Q\frac{\bar\omega'}{\omega^3} - 3Q\frac{\bar\omega^2}{\omega^2}\right). \end{aligned} \tag{8.49}$$

The adiabatic expansion for particle production in the high-frequency range at the zeroth, second and fourth adiabatic order above matches the quartic, quadratic and logarithmic divergences in the vacuum energy density respectively. Substituting these expressions for $s_{k(div)} = 1 + s_{k(2)} + s_{k(4)}$ for each $\mathbf{k}$ mode into the vacuum energy density from the conformal stress-energy tensor (2.59) we can identify the divergent vacuum energy density contributions as

$$\rho_{0(div)} = \frac{1}{a^4}\int \frac{d^3\mathbf{k}}{(2\pi)^3}\left[s_{\mathbf{k}(div)} + \frac{1}{2}\right]\hbar\omega_{\mathbf{k}}. \tag{8.50}$$

Subtracting these we get the regularized vacuum energy density given by $\rho_{0(reg)} = \rho_0 - \rho_{0(div)}$. These results were obtained by [69, 71, 86]. We shall use adiabatic

regularization [630, 107, 108] in the treatment of two examples of cosmological particle creation with backreaction in the following.

8.2.2 Graviton Creation and Equation of State

We begin with a simple example on the effect of gravitons created near the Planck time t_p in a FLRW universe containing some classical matter described by an equation of state $p_m = \bar{\gamma}\rho_m, (0 \leq \bar{\gamma} \leq 1)$. The result of this calculation suggests some interesting relation between graviton production and the equations of state of matter in the early universe. This was first noticed by Grishchuk [105]. Here we follow the treatment of Hu and Parker [107] which serves also to illustrate the physics behind the adopted approximation methods. For graviton creation treated at the level of quantum cosmology rather than quantum fields in curved spacetimes, see Berger [72, 73].

Consider linear gravitational perturbations in a spatially flat FLRW universe with line element (2.26)

$$ds^2 = -dt^2 + a^2(t)(dx_1^2 + dx_2^2 + dx_3^2). \tag{8.51}$$

In this spacetime (but not for spacetimes with lesser symmetry such as the mixmaster universe) one can impose both the synchronous condition $h_{0\mu} = 0$ and the transverse, traceless (TT) conditions, after which $h_i^j(x,t)$ obeys the Lifshitz equation [338, 656]:

$$\ddot{h}_i^j(x,t) + 3\,\frac{\dot{a}}{a}\,\dot{h}_i^j(x,t) + \frac{1}{a^2}\,\Delta h_i^j(x,t) = 0 \quad (i,j = 1,2,3), \tag{8.52}$$

where $\cdot \equiv d/dt$ and Δ is the Laplace–Beltrami operator on a constant t hypersurface. We perform a normal mode decomposition of $h_i^j(x,t) = \int d^3k\; h_k(t) G^{(k)}{}_i^j(x)$ in terms of the eigenfunctions $G^{(k)}{}_i^j(x)$ with eigenvalues k. The amplitude function $h_k(t)$ of the kth mode obeys the wave equation

$$h_k''(\tau) + \Omega_k^2(\tau)\, h_k(\tau) = 0. \tag{8.53}$$

Here a prime denotes $d/d\tau$ where τ time is defined by $\tau = \int dt\; V$, where $V = a^3$ and $\Omega_k = V\omega_k$ with $\omega_k = k/a$. After imposing the TT gauge we see that each of the two polarizations of the gravitational wave as tensor perturbations in a Friedmann universe obeys a wave equation which has the same form as that obeyed by a massless minimally-coupled scalar wave [345, 346]. The solution to the wave equation (8.53) can be written in the form [69]

$$h_k(\tau) = \frac{\alpha_k}{\sqrt{2\Omega_k}}\, e_+ + \frac{\beta_k}{\sqrt{2\Omega_k}}\, e_-, \qquad e_\pm = \exp\!\left(\pm i \int^\tau d\tau'\; \Omega_k\right), \tag{8.54}$$

where α_k and β_k are complex functions of τ. The Wronskian condition guarantees that $|\alpha_k|^2 - |\beta_k|^2 = 1$. The quantity

$$s_k = \beta_k \beta_k^* = \frac{1}{2}(\Omega_k |h_k|^2 + \Omega_k^{-1} |h_k'|^2) \tag{8.55}$$

gives a measure of the parametric amplification of the wave, corresponding to quantum mechanical particle production [68, 71]. Near yet below the Planck energy, for small t or low ω, $|\int^\tau d\tau\ \Omega| \ll 1$, one can adopt a sudden approximation to the wave equation (8.53), yielding

$$s_k = |c_1|^2 \Omega_k + \frac{|c_2|^2}{\Omega_k} - \frac{1}{2}, \tag{8.56}$$

where c_1 and c_2 are complex constants satisfying $c_1^* c_2 + c_1 c_2^* = \frac{1}{2}$.

Let us pause for a moment to explain the physical meaning of this sudden or short time approximation. Sudden is nonadiabatic, the opposite of gradual or adiabatic. The nonadiabaticity parameter $\omega_{\mathbf{k}}'/\omega_{\mathbf{k}}^2 \geq 1$ provides a measure of which modes will be most strongly amplified parametrically, engendering most particle creation, at a particular moment of time. Recall in finding the criterion for a proper definition of adiabatic vacuum we look for the condition where as little particles are produced as possible, having to reach out to W_3 in (8.39) to safely define a vacuum, the adiabatic vacuum. Also, to capture the logarithmic UV divergence we need to calculate the adiabatic frequency corrections up to $\epsilon_{(4)}$. These activities are all carried out in the high frequency domain and a WKB approximation is invoked as we look for gradual changes of the background field or spacetime, registered in the natural frequencies of the normal modes. The adiabatic expressions for the amount of particle creation given in the context of adiabatic vacuum and regularization are only in the high frequency modes when $\omega_{\mathbf{k}}'/\omega_{\mathbf{k}}^2 \leq 1$. Now the opposite, nonadiabatic, regime is what one should aim at for the most particle production, namely, under sudden changes corresponding to $\omega_{\mathbf{k}}'/\omega_{\mathbf{k}}^2 \gg 1$. A simple estimate on dimensional grounds shows nonadiabaticity corresponds to the condition $\omega_{\mathbf{k}} t < 1$, in the short time and low frequency ranges. Let us call the frequency range where particle creation is most pronounced the quantum domain $\mathcal{Q}(t)$.

The effective stress-energy tensor $T_{\mu\nu}$ of linear gravitational perturbations under the synchronous and TT gauge has the same form as that for two minimally coupled scalar fields. The energy density ρ and pressure p of the gravitons are given as its expectation values $\rho = -\langle T_0^0 \rangle, p\,\delta_i^j = \langle T_i^j \rangle$ with respect to some state. The state vector is chosen such that no quantum corresponding to waves of the form (8.54) are present. (A complete specification of the state vector still requires a choice of initial conditions on α_k and β_k.) We have

$$\rho = -\langle T_0^0 \rangle = \frac{1}{16\pi(2\pi)^3} \int d^3k\ \rho_0(k), \tag{8.57}$$

where

$$\rho_0(k) = |\dot{h}_k(t)|^2 + \omega_k^2(t)|h_k(t)|^2 = V^{-2}\Omega_k(2s_k + 1). \tag{8.58}$$

Adiabatic regularization of this quantity yields the finite energy density

$$\rho_{\rm reg} = \frac{1}{128\pi^4}\int d^3k\,\Big[\rho_0(k) - \rho_{\rm div}(k)\Big], \tag{8.59}$$

where

$$\rho_{\rm div}(k) = a^{-4}\Big[k + \frac{\dot{a}^2}{2k} + \frac{A}{(2k)^3}\Big] \tag{8.60}$$

and

$$A = a^4\left[\left(\frac{\ddot{a}}{a}\right)^2 - 2\left(\frac{\ddot{a}}{a}\right)\left(\frac{\dot{a}}{a}\right)^2 - 2\left(\frac{\dddot{a}}{a}\right)\left(\frac{\dot{a}}{a}\right) + 3\left(\frac{\dot{a}}{a}\right)^4\right]. \tag{8.61}$$

At early times the low frequency behavior dominates whereby s_k can be approximated by (8.56). We also assume that there is no particle creation before a time t_0. This can be true only if the background is static, or for radiation-dominated FLRW spacetimes [345]. From (8.57) we see that this corresponds to choosing c_1 and c_2 such that the quantity $\rho_0(k) - V^{-2}\Omega_k$ vanishes at time $t \le t_0$. We take $|c_1|^2 = (4\Omega_0)^{-1}, |c_2|^2 = \Omega_0/4$, where $\Omega_0 = a_0^2 k$, which corresponds to setting $\beta_k = 0$ at t_0. Thus at every moment $\mathcal{Q}_k(t)$ consists of all modes $\vec{k}$ where there is significant particle creation, marked by the condition $\omega_k \le t^{-1}$.

We can think of the graviton energy density ρ_g at any given time t as consisting of three parts $\rho_g = \rho_q + \rho_a + \rho_r$, where the q and r parts belong to the $\mathcal{Q}_k(t)$ and $\mathcal{R}_k(t)$ domains respectively. The first part ρ_q of quantum origin contains the contributions to (8.59) of all nonadiabatic modes $\vec{k}$ with $\omega \le \omega_q = 1/t$ in which quantum effects such as graviton creation and vacuum polarization are significant. The second part ρ_a contains the contributions of the adiabatic modes (higher frequency modes captured by the WKB approximation) at that moment of time t and the third part ρ_r contains the red-shifted remnants of the gravitons created at earlier times. Since ρ_a is the energy density of particles created in the high frequency modes which behave like a relativistic fluid with equation of state $p_{cg} = \rho_{cg}/3$ (under the Brill–Hartle–Isaacson average [604]), where p_{cg}, ρ_{cg} are the classical graviton pressure and energy density. These adiabatic modes which are exponentially insignificant will be red-shifted as the universe expands and join the low frequency graviton remnants created earlier at a later time. The ultra-high frequency modes which cause ultraviolet divergence will be subtracted as stipulated in the adiabatic regularization scheme.

Since we are interested in gravitons created from the vacuum let us focus on the $\mathcal{Q}_k(t)$ and $\mathcal{R}_k(t)$ sectors here. The quantum contribution ρ_q is found by

integrating (8.59) over all k with $k = |\vec{k}| < k_q = a/t$, using the approximation of (8.56). One finds[2]

$$\rho_q = \frac{1}{128\pi^3}\left[\frac{k_q^4}{2a^4}\left(\frac{a}{a_0}-\frac{a_0}{a}\right)^2 - \frac{k_q^2}{a^2}\left(\frac{\dot{a}}{a}\right)^2\right]$$
$$= \frac{1}{128\pi^3}\left[\frac{2\sinh^2 x}{t^4} - \frac{1}{t^2}\left(\frac{\dot{a}}{a}\right)^2\right], \tag{8.62}$$

where $x = \ln(a/a_0)$.

The energy density ρ_r at time t of the remnant of all previously created gravitons is given by the sum of all previous increments $\delta\rho_q(k_q(t'), t')$ at earlier times $t' < t$, arising from the change of $k_q(t')$ with time, being red-shifted by a factor $[a(t')/a(t)]^4$. Thus,

$$\rho_r(t) = \int_{t_0}^{t}\left[\frac{a(t')}{a(t)}\right]^4\left[-\frac{\partial\rho_q(k_q(t'),t')}{\partial k_q}\right]_a \frac{\partial k_q(t')}{\partial t'}dt', \tag{8.63}$$

where the subscript a on $\partial\rho_q/\partial k_q$ indicates that $a(t')$ is kept fixed during differentiation. One can write $\rho_r(t)$ in the form

$$\rho_r(t) = \frac{1}{16\pi^3}\left[\frac{a_0}{a(t)}\right]^4 F(t), \tag{8.64}$$

where $F(t)$ can be found by integration of (8.63) if $a(t')$ is given. For a FLRW universe containing a fluid with equation of state $p_m = \bar{\gamma}\rho_m, (0 \le \bar{\gamma} \le 1)$, which is not interacting with the created gravitons, the conservation law $T^{\mu\nu}{}_{,\nu} = 0$ yields

$$\rho_m = \rho_{m_0}\left(\frac{V}{V_0}\right)^{-(1+\gamma)}. \tag{8.65}$$

The effect of graviton creation is studied by solving the semiclassical Einstein equation

$$3\left(\frac{\dot{a}}{a}\right)^2 = 8\pi G_{\rm N}(\rho_m + \rho_{cg} + \rho_{qg}), \tag{8.66}$$

where ρ_{cg} is the classical graviton energy density and $\rho_{qg} = \rho_q + \rho_r$ is given by (8.62) and (8.63). This nonlocal property is characteristic of backreaction effects due to particle creation and other dynamical processes. With $\rho_r(t)$ given as a functional of $a(t)$, the above equation becomes an integro-differential equation. However, since the integrand of (8.63) becomes small after a short time interval,

[2] The expression for ρ_q does not vanish at t_0 because of the higher-order subtraction involving $(\dot{a}/a)^2$. That term is only significant for an interval of less than $0.1t_p$ after t_0, and has negligible influence on the time for graviton creation to alter the expansion to $a \sim t^{1/2}$.

one only needs $a(t)$ at very early times ($t \leq t_p$) to find $\rho_r(t)$. Making such simplifications enables one to solve the equation of $a(t)$ and find $\rho_{qg}(t)$. The relative importance of ρ_{qg} and ρ_m from such calculations enables one to make statements about the equation of state of matter affected by graviton creation in the early universe. See [107] for further details.

This example brings out the central characteristic of backreaction effect due to particle creation, i.e., that it is a history-dependent effect (nonlocal in time). This is because the energy density at any one time depends not only on the currently produced particles, but also on all previously created particles which were created under different circumstances with different background dynamics, and have undergone different evolutions. We have seen this effect expressed in terms of the nonlocal damping kernel derived from the effective action. We will see this again when we revisit the backreaction of particle creation in an anisotropically expanding spacetime below, presented at the equation of motion level.

8.2.3 Damping of Anisotropy and Inhomogenieites

Damping of Anisotropy

We have discussed the backreaction of particle creation in Bianchi Type I universe via the effective action method in Chapter 3. Here, we illustrate another method, via canonical quantization and by solving the coupled scalar wave and Einstein equations. The aim is to understand the physical processes and the meaning of approximations used.

Let us consider a massless conformally coupled scalar field in a Bianchi Type I (B1) universe with metric (3.56)

$$ds^2 = -dt^2 + \sum_{i=1}^{3} a_i^2(t)(dx^i)^2, \qquad (i = 1, 2, 3) \tag{8.67}$$

Recall from Chapter 3 that B1 is a class of spatially flat universes with three different scale factors $a_i \equiv e^{\alpha_i} (i = 1, 2, 3)$ where the expansion rates $\dot{\alpha}_i(t)$ are different in the three directions. In the wave equation for the mode amplitude $\chi_{\mathbf{k}}$ in conformal time, (2.31),

$$\chi''_{\mathbf{k}} + (\Omega^2_{\mathbf{k}} + Q)\chi_{\mathbf{k}} = 0 \tag{8.68}$$

the natural frequency $\Omega_{\mathbf{k}}$ is given by

$$\Omega_{\mathbf{k}} = a \left(\sum_{i}^{3} k_i^2/a_i^2 \right)^{\frac{1}{2}}, \tag{8.69}$$

where $a \equiv (a_1 a_2 a_3)^{1/3}$. The Q is given by the shear anisotropy Q_β defined as

$$Q_\beta \equiv -\frac{1}{18}[(\alpha'_1 - \alpha'_2)^2 + (\alpha'_2 - \alpha'_3)^2 + (\alpha'_3 - \alpha_1)^2]. \tag{8.70}$$

It is easily seen that Q_β is of second adiabatic order. We summarize the main features from the findings in [108]:

i) In a FLRW universe $Q_\beta = 0$ there is no production of massless conformal particles [657]. Thus particle production rate in B1 universe is proportional to the shear anisotropy. Zel'dovich and Starobinsky [69] first pointed out that particle production in anisotropic universes are significant, and its backreaction can isotropize the universe very effectively. Hu and Parker [108] solved the semiclassical Einstein equation with the adiabatic-regularized energy momentum tensor as source, showing explicitly the anisotropy damping process in real time. The approximations made are the same as those in the graviton production problem, i.e. a low-frequency or early time approximation

$$\exp\left(2i\int_{\eta_0}^{\eta} d\eta'\,\Omega_{\mathbf{k}}\right) \approx 1, \qquad \text{where } \eta = \int \frac{dt}{a(t)}. \tag{8.71}$$

ii) The nonadiabaticity condition defines at every moment of time in the momentum space selects out a quantum domain $\mathcal{Q}_{\mathbf{k}}(t)$ where particle creation is significant, consisting of all modes $\mathbf{k}$ satisfying the condition $\omega_{\mathbf{k}} \leq t^{-1}$. The adiabatic classical high frequency modes $\omega_{\mathbf{k}} > t^{-1}$ of a massless conformal field behave like a relativistic fluid. The created particles will be red-shifted and become the remnants from the quantum domain.

iii) In the models considered in [108], by about 5–10 Planck time t_p already the dominant contribution to the energy density comes from the remnant of created particles ρ_r so that the error introduced at later times by using the low-frequency approximation for $\omega_q \sim t^{-1}$ in ρ_q should have little influence on the dynamics, while at earlier times this short time or suddern approximation is expected to be valid.

iv) For modes $\mathbf{k}$ belonging to the quantum domain, the specification of the amplitude functions $\chi_{\mathbf{k}}$ in (2.31) amounts to a specification of the vacuum $|0_A\rangle$ state annihilated by all $A_{\mathbf{k}}$. Assume that at times before η_0 the universe was expanding isotropically so there was no particle present and there existed a well-defined vacuum state effected by $\alpha_{\mathbf{k}}(\eta) = 1, \beta_{\mathbf{k}}(\eta) = 0$ at $\eta \leq \eta_0$. Further assume that the initial vacuum state for an anisotropic expansion is the same as that for an isotropic expansion at η_0. This corresponds to choosing $\alpha_{\mathbf{k}}(\eta_0) = 1$ and $\beta_{\mathbf{k}}(\eta_0) = 0$ which in the early-time approximation determines $c_1(\mathbf{k})$ and $c_2(\mathbf{k})$ in Eq. (8.56).

v) At time $\eta + \Delta\eta$ the energy density of created particles ρ_q consists of ρ_r of the remnants from all particles created at earlier times up to $\eta + \Delta\eta$ and an increment $\delta\rho_q$ from the quantum particles just created at time η, both redshifted by a factor $[a(\eta)/a(\eta + \Delta\eta)]^4$:

$$\delta\rho_q(\eta) = \rho_q(k_q(\eta), \eta) - \rho_q(k_q(\eta + \Delta\eta), \eta) = -\left(\frac{\partial\rho_q}{\partial k_{iq}}\frac{dk_{iq}}{d\eta}\right)\,\Delta\eta \tag{8.72}$$

where $k_{iq} = (k_{1q}, k_{2q}, k_{3q})$ is the ellipsoid of maximum $\vec{k}$ in k-space, and k_q is the wave vector corresponding to $\varnothing_q$, the upper frequency in the nonadiabaticity range.

Using these approximations, one can derive an equation for the total energy density due to quantum particle creation, the remnants of created particles and the high frequency classical adiabatic modes. To simplify the computation Hu and Parker worked with a symmetric Bianchi Type I (B1) universe i.e., $a_1 = a_2 \neq a_3$. Upon numerically integrating the semiclassical Einstein equations with a source energy density ρ comprised of these parts, and the related pressure p_1, p_3 deduced from ρ and T, the trace of $\langle T_{\mu\nu} \rangle$, they obtained the solutions for different initial times from $t_0 = 0.5$ to $3t_p$ and for a wide range of initial expansion rates consistent with the constraints. They found that the expansion rates are equalized towards the value for a radiation-filled Friedmann universe in a time interval less than 10^3 Planck time (shorter if the particle creation begins at an earlier time). Anisotropy damping is strongly effective near the Planck time and is largely insensitive to the specification of the initial conditions.

In summary, backreaction due to particle creation represents an important class of quantum dissipative processes which can provide a working mechanism for the implementation of the chaotic cosmology philosophy of Misner [658, 419, 659]. Generalization of this calculation to higher dimensions, e.g., by viewing a_1 and a_3 as the scale factors of the external and internal spaces of Kaluza Klein (KK) cosmology (e.g. the five-dimensional Kasner solution of [625]) implies that particle creation near the Planck time tends to equalize the expansion rate of the external space (4D spacetime) to that of the internal spaces (six or seven extra dimensions). Particle creation leading to an isotropization of all dimensions will obliterate the division between expanding external dimensions and stable internal dimension. This could be the Achilles heel of Kaluza–Klein cosmology. Detailed calculation on this problem was carried out by Maeda [660] following the B1 example of Hu and Parker [108]. The moral seems to be: A more robust mechanism is needed to keep the internal dimensions spontaneously compactified at a very small scale, and remain small, while permitting the four spacetime dimensions to open up and expand to the vast space we live in.

Damping of Inhomogeneity in a Perturbed FLRW Universe

Instead of particle creation from a weakly anisotropic but spatially-homogeneous universe, one can also consider such processes from a weakly inhomogeneous but isotropic universe. Similar to what Hartle and Hu [110] and Calzetta and Hu [182] found for the production of massless conformal particles in the former case, production of such particles from an isotropic universe with weak inhomogeneities calculated by Campos and Verdaguer [423] shows similar behavior, namely, that the amount is proportional to the space-time integrated value of the Weyl tensor-squared. This is expected, since the Weyl tensor measures the

deviation from conformal invariance, and as we remarked in Chapter 2, particle creation originates from the departure from conformal symmetry, as when the mass is non-zero, the coupling is nonconformal, or in the present case, the background spacetime deviates from conformal flatness (FLRW). We shall show in Part IV, after stochastic gravity theory is introduced, how to calculate the effects of both the mean value and the correlations of the stress-energy tensor of quantum fields from the solutions of the semiclassical Einstein and the Einstein–Langevin equations.

8.3 Preheating from Inflaton Particle Creation and Interaction

Before we delve into the technical details it is useful to have an overview of the salient features of inflationary cosmology. Generally speaking there are three epochs characterizing its dynamics, namely, entry to a vacuum energy-dominated state, the inflating epoch and the reheating epoch. Inflation is engineered by a scalar field Φ called the inflaton. Its behavior is dictated by a potential $V(\Phi)$, usually depicted by several parameters, to be determined by observational constraints. For example, the number of photons or entropy content in the present universe constrains that inflation, if it occurred in the GUT epoch, should last 60-something e-folding time. To make this happen, the potential should be flat enough or the inflaton (background field ϕ) should 'roll down' the potential slowly enough. After this the universe will end inflation and enter the radiation-dominated phase which prevails in the later epochs. A natural way to convert the vacuum energy carried by the inflaton to radiation is to assume that the potential after a gentle downward slope (to enable 'slow-roll') dips into a deep well steeply. The rapid oscillations of the inflaton in the deep well parametrically amplifies the fluctuations $\varphi \equiv \Phi - \phi$ of the inflaton field into real particle pairs and the universe begins to reheat after it exits from the inflationary stage.

The beginning stage of reheating is known as 'preheating', when the inflaton field produces massive bosons due to parametric resonance. The second stage is the decay of previously produced particles. The third stage in the reheating epoch is thermalization. For more descriptions of the reheating epoch, see [661, 662, 663, 664, 665, 666, 592, 591, 667] and references therein. We will focus on the preheating stage here as it already contains a full display of the interlocking dynamics of quantum fluctuations amplified by (engendering particle creation) and backreacting on the background field (inflaton) dynamics governed by the background geometry. Note there are three players in this reheating game: the inflaton background field ϕ, the inflaton fluctuation field φ, and the background spacetime metric $g_{\mu\nu}$. It is the dynamical interplay amongst them which we want to capture; this means we need to derive the three coupled equations governing these three dynamical variables.

As a simple model of inflation, let us consider a scalar $\lambda\Phi^4$ field in semiclassical gravity, where the matter field is quantized on a classical, dynamical background

spacetime. The matter field and gravitational actions take on the forms (7.39) and (2.67) respectively.[3]

The inflaton field ϕ is quantized on the classical background spacetime; we denote the Heisenberg field operator by Φ, and the quantum state by $|\phi\rangle$. Of particular interest in a study of inflaton dynamics are the mean field

$$\bar{\phi}(x) \equiv \langle\phi|\Phi(x)|\phi\rangle, \tag{8.73}$$

the fluctuation field

$$\varphi(x) \equiv \Phi(x) - \bar{\phi}(x), \tag{8.74}$$

and the mean-squared fluctuations, or variance,

$$\langle\phi|\varphi^2(x)|\phi\rangle = \langle\phi|\Phi^2(x)|\phi\rangle - \langle\phi|\Phi(x)|\phi\rangle^2. \tag{8.75}$$

In Chapter 7, following [244], a systematic procedure was presented for deriving real and causal evolution equations for the mean field $\bar{\phi}$, the two-point function C, and the metric tensor $g^{\mu\nu}$ from functional differentiation (and subsequent field identifications) of the closed-time-path (CTP) two-particle-irreducible (2PI) effective action, $\Gamma[\bar{\phi}_\pm, C_{\pm\pm}, g^{\mu\nu}_\pm]$. We shall apply it now to the analysis of preheating in inflationary cosmology. Since this problem invokes many advanced techniques in field theory developed in earlier chapters, for pedagogical purpose, we shall try not to skip too many steps in this illustrative example of the solution of a class of important cosmological backreaction problems. We begin with an explanation of the set up of the problem, the adoption of the best available techniques, followed by a description of the renormalization of the stress-energy tensor of the inflaton field via adiabatic regularization. We finish with the choice of initial conditions and proceed to finding the self-consistent solutions of the coupled equations for the dynamical geometry, inflaton background field and the fluctuation field variables. Our presentation follows Ramsey and Hu [605].

8.3.1 $\lambda\phi^4$ Inflaton Dynamics in FLRW Spacetime

In restricting the spacetime to be a spatially–flat FLRW, we are reducing the number of degrees of freedom in the metric $g_{\mu\nu} \to a^2(\eta)\eta_{\mu\nu}$. This reduction should not be carried out in the 2PI generating functional $\Gamma[\bar{\phi}, C, g^{\mu\nu}]$, but only in the equations of motion. This is because functional differentiation of $\Gamma[\bar{\phi}, C, a^{-2}\eta^{\mu\nu}]$ with respect to the scale factor a gives only the *trace* of the stress-energy tensor, $a^{-2}\eta^{\mu\nu}\langle T_{\mu\nu}\rangle$, as we showed earlier, and not the additional constraint equation which the initial data must satisfy.

[3] In these formulas for $n = 4$ we adopt a more commonly used notation: $\xi_{here} = 0$ for minimal coupling and $\xi_{here} = 1/6$ for conformal coupling. When comparing them with corresponding formulas in earlier chapters where $\xi =$ signifies conformal coupling, replace $(1 - \xi)\xi_n$ there by ξ_{here}.

Owing to the spatial homogeneity and isotropy of FLRW spacetime there are only two algebraically independent components of the stress-energy tensor, $\langle T_{00}\rangle$ and $\langle T_{ii}\rangle$, both being functions of η only, while all other components are zero. For the purpose of numerically solving the semiclassical Einstein equation, it is convenient to work with the trace

$$T = g^{\mu\nu}\langle T_{\mu\nu}\rangle = a^{-2}\eta^{\mu\nu}\langle T_{\mu\nu}\rangle, \tag{8.76}$$

instead of $\langle T_{ii}\rangle$. The trace T enters into the dynamical equation for $a(\eta)$, and $\langle T_{00}\rangle$ enters into the constraint equation.

The spatial symmetries of FLRW spacetime also restrict the generality with which we may specify initial data for dynamical evolution. Let us choose to specify initial data on a Cauchy hypersurface Σ_{η_0} of constant η_0, the conformal time when the preheating stage begins. In the Heisenberg picture,[4] for consistency with spatial homogeneity, the quantum state $|\phi\rangle$ must satisfy

$$\langle\phi|\Phi(\eta_0,\mathbf{x})|\phi\rangle = \bar{\phi}(\eta_0), \tag{8.77}$$

$$\langle\phi|\Phi'(\eta_0,\mathbf{x})|\phi\rangle = \bar{\phi}'(\eta_0), \tag{8.78}$$

for all $\mathbf{x}\in\mathbb{R}^3$, where Φ is the Heisenberg field operator for the scalar field. The values of $\bar{\phi}(\eta_0)$ and $\bar{\phi}'(\eta_0)$ provide initial data for the mean field. In addition, the quantum state must satisfy

$$\langle\phi|\varphi(\eta_0,\mathbf{x})\varphi(\eta_0,\mathbf{x}')|\phi\rangle = F(\eta_0,|\mathbf{x}-\mathbf{x}'|), \tag{8.79}$$

$$\frac{\partial}{\partial\eta}_{|\eta_0}\langle\phi|\varphi(\eta,\vec{x})\varphi(\eta,\vec{x}')|\phi\rangle = F'(\eta_0,|\mathbf{x}-\mathbf{x}'|), \tag{8.80}$$

in terms of an equal-time correlation function $F(\eta_0,|\mathbf{x}-\mathbf{x}'|)$ which is invariant under simultaneous translations and rotations of $\mathbf{x}$ and $\mathbf{x}'$. As defined in Eq. (8.74), φ denotes the Heisenberg field operator for the fluctuation field. The spatial Fourier transform of F is related to the power spectrum of quantum fluctuations at η_0 for the quantum state $|\phi\rangle$. Alternatively, we may say that $F(\eta_0,r)$ and $F'(\eta_0,r)$ give initial data for the evolution of the two-point function C_{++} via the gap equation (7.47). The symmetry conditions (8.77), (8.78), (8.79), (8.80), along with the spatial symmetries of the classical action in FLRW spacetime, guarantee that the mean field and two-point function satisfy spatial homogeneity and isotropy for *all time,* i.e.,

$$\langle\Phi(x)\rangle = \bar{\phi}(\eta), \tag{8.81}$$

$$C_{++}(x,x') = C_{++}(\eta,\eta',|\mathbf{x}-\mathbf{x}'|). \tag{8.82}$$

[4] As discussed in Sec. 8.3.1, for our purposes it is sufficient to consider only the case of a pure state. The analysis can, however, be easily extended to encompass a mixed state with density matrix ρ.

The conditions (8.81), (8.82) permit a formal solution of the gap equation (7.47) for C_{++} in terms of homogeneous mode functions, via a Fourier transform in comoving momentum $\mathbf{k}$. By rotational invariance, the Fourier transform depends only on the magnitude $k \equiv \sqrt{\mathbf{k}\cdot\mathbf{k}}$. Of course, the quantum state $|\phi\rangle$ is not uniquely defined by the spatial symmetries; a unique choice of the initial conditions for $\bar{\phi}$ and C_{ab} at Σ_{η_0} is (in the Gaussian wave-functional approximation) equivalent to choosing $|\phi\rangle$. The choice of quantum state depends on the physics of the problem we wish to study.

As a consequence of covariant conservation of the stress-energy tensor

$$\nabla^{\mu}\langle T_{\mu\nu}\rangle = 0, \tag{8.83}$$

the functions $\langle T_{00}(\eta)\rangle$ and $\langle T_{ii}(\eta)\rangle$ satisfy

$$\frac{d}{d\eta}\Big(a\langle T_{00}\rangle\Big) = -\frac{\langle T_{ii}\rangle}{a^2}\frac{d}{d\eta}\left(a^3\right), \tag{8.84}$$

which comes from taking the $\nu = 0$ component of Eq. (8.83). In analogy with the continuity relation for a classical perfect fluid in FLRW spacetime,

$$\frac{d}{d\eta}\left(a^3\rho\right) = -p\frac{d}{d\eta}\left(a^3\right), \tag{8.85}$$

we may define the energy density ρ and pressure p, by

$$\rho(\eta) = \frac{1}{a^2}\langle T_{00}(\eta)\rangle, \qquad\qquad p(\eta) = \frac{1}{a^2}\langle T_{ii}(\eta)\rangle. \tag{8.86}$$

However, the quantity p should not be interpreted as the true hydrodynamic pressure until a perfect-fluid condition is shown to exist; otherwise, bulk viscosity corrections can enter into Eq. (8.86) [43]. The effective equation of state is defined as a time average (an overbar denotes time average over the time scale τ_1 for the matter field dynamics) of the ratio p/ρ, namely $\bar{\gamma} \equiv \frac{\bar{p}}{\bar{\rho}}$. The important cosmological stages corresponding to matter field effective equation of state include, with η being the conformal time:

(i) $\bar{\gamma} = 0$ for nonrelativistic matter giving rise to $a \propto \eta^2$.
(ii) $\bar{\gamma} = 1/3$ for relativistic matter giving rise to $a \propto \eta$.
(iii) $\bar{\gamma} = -1$ vacuum energy, giving rise to a 'vacuum-dominated' solution. This equation of state is disallowed in classical general relativity.

When expressed in the $k = 0$ FLRW coordinate (the Poincaré patch), $a(\eta) = -1/(H\eta)$, for $-\infty < \eta < 0$, where $H = \sqrt{8\pi G_{\rm N}\rho/3}$ and ρ is a constant (vacuum) energy density. This coordinate covers one-half of the de Sitter manifold [47] in the conformal diagram – note it is different from the 'static' coordinate of de Sitter, which maps into the Schwarzschild metric [322], but covers only a quarter of the de Sitter manifold.

Single Field Slow-Roll Inflation

Consider the case of inflation driven by a single self-interacting scalar field ϕ (with unbroken symmetry) in a spatially flat FLRW spacetime. The above arguments imply that one can model post-inflationary physics with a quantum state $|\phi\rangle$ which at η_0 corresponds to a coherent state for the field operator Φ [meaning, $\langle\phi|\Phi(x)|\phi\rangle = \bar{\phi}(\eta)$], and the fluctuation field $\varphi \equiv \Phi - \bar{\phi}$ is very nearly in the vacuum state. (The same method can be used to treat a quantum field in a mixed state, for example, a system initially in thermal equilibrium with a heat bath.) Then for $\eta < \eta_0$, $\langle T_{00}\rangle$ is dominated by the classical energy density of the mean field $\bar{\phi}$. The 00 component of the Einstein equation then yields

$$\frac{a'}{a^2} = \sqrt{\frac{8\pi G_{\rm N}\rho_c}{3}}, \tag{8.87}$$

where ρ_c is the classical energy density of the mean field, defined by

$$\rho_c = \frac{1}{2a^2}(\bar{\phi}')^2 + V(\bar{\phi}). \tag{8.88}$$

The mean field $\bar{\phi}$ satisfies the classical equation

$$\bar{\phi}'' + \frac{2a'}{a}\bar{\phi}' + a^2 V'(\bar{\phi}) = 0, \tag{8.89}$$

where $V(\bar{\phi})$ denotes the classical potential. For the $\lambda\Phi^4$ theory, the potential is

$$V(\bar{\phi}) = \frac{1}{2}m^2\bar{\phi}^2 + \frac{\lambda}{24}\bar{\phi}^4. \tag{8.90}$$

The assumption that the Universe is inflating (i.e., $\bar{\gamma} \simeq -1$) for $\eta < \eta_0$ requires that the energy density ρ_c be potential dominated,

$$V(\bar{\phi}) \gg \frac{1}{2a^2}(\bar{\phi}')^2, \tag{8.91}$$

and that the mean field satisfies the slow-roll condition,

$$\bar{\phi}'' \ll \frac{2a'}{a}\bar{\phi}'. \tag{8.92}$$

Given Eqs. (8.91) and (8.92), an approximate "0-th adiabatic order" solution to the Einstein equation can be obtained [normalized to $a(\eta_0) = 1$],

$$a(\eta) \simeq \frac{1}{1 + H(\eta)(\eta - \eta_0)}, \tag{8.93}$$

where H is a slowly varying function of η, given by

$$H(\eta) = \sqrt{\frac{8\pi G_{\rm N}\rho_c(\eta)}{3}}. \tag{8.94}$$

From (8.94), we can evaluate the expansion rate nonadiabaticity parameter [71]

$$\bar{\Omega}_H \equiv \frac{H'}{H^2} \tag{8.95}$$

for $\eta < \eta_0$ using (8.94). During slow-roll it follows from conditions (8.91) and (8.92) that

$$\bar{\Omega}_H = \frac{V'(\bar{\phi})\,\bar{\phi}'}{\sqrt{\dfrac{32\pi G}{3}\,V(\bar{\phi})^3}} \ll 1. \tag{8.96}$$

The solution (8.93) for $a(\eta)$ is exact in the limit of constant H (corresponding to a constant $\bar{\phi}$ at the tree level). For simplicity, let us assume that $\bar{\phi}$ goes to a constant value $\geq m_p$ in the asymptotic past, $\eta \to -\infty$. The spacetime is then asymptotically de Sitter, with the scale factor having an asymptotic cosmic-time dependence $a(t) \simeq \exp(Ht)$. Because the enormous cosmic expansion during the slow-roll period redshifts away all nonvacuum energy in the Universe, it is reasonable to assume that the quantum state $|\phi\rangle$ would register no particles for a comoving detector coupled to the fluctuation field φ at conformal past-infinity; i.e., that the fluctuation field φ is in the vacuum state at $\eta \to -\infty$. This would mean that $a \simeq -1/(H\eta)$ at $\eta \to -\infty$. This spacetime is *not* asymptotically static in the past, but it is conformally static with a conformal factor whose nonadiabaticity parameter vanishes at conformal past-infinity. Therefore, the best approximation to a "no-particle" state for a comoving detector in the asymptotic past is given by the *adiabatic vacuum* [84] constructed via matching at $\eta \to -\infty$. All higher-order adiabatic vacua will in this case agree at conformal past infinity.

To construct the n-th order *adiabatic vacuum* matched at an equal-time hypersurface Σ_{η_m} at η_m, one first exactly solves the conformal mode-function equation for the quantum field [see (8.102) below]. Since the mode-function equation is second order, each k mode has two independent solutions, which can be represented as u_k and u_k^*. A particular solution consists of a linear combination of u_k and u_k^*. The adiabatic vacuum is constructed by choosing (for each k) a linear combination which smoothly matches the nth-order positive frequency WKB mode function at Σ_{η_m}. The resulting orthonormal basis of mode functions is then used to expand the Heisenberg field operator $\Phi(x)$ in terms of a_k and $a_k^\dagger$. The vacuum state is defined by $a_k|vac\rangle = 0$ for all k, which can be shown to correspond (in the $\eta_m \to -\infty$ limit) to the de Sitter-invariant, adiabatic (Bunch–Davies) vacuum.

The $O(N)$ Model in the $1/N$ Expansion in FLRW Spacetime

The classical action for the unbroken symmetry O(N) model in a general curved spacetime is (7.51)

$$S^{\rm F}[\phi^i, g_{\mu\nu}] = -\frac{1}{2}\int_M d^4x\sqrt{-g}\,\left[\vec{\phi}\cdot\left(-\Box + m^2 + \xi R\right)\vec{\phi} + \frac{\lambda}{4N}(\vec{\phi}\cdot\vec{\phi})^2\right], \tag{8.97}$$

where the O(N) inner product is defined by[5]

$$\vec{\phi} \cdot \vec{\phi} = \phi^i \phi^j \delta_{ij}. \tag{8.98}$$

In Chapter 7, following [244], the covariant mean-field equation, gap equation, and geometrodynamical field equation were derived for this model at leading order in the $1/N$ expansion. The evolution equations follow from Eqs. (7.45)–(7.47), with the 2PI, CTP effective action truncated at leading order in the $1/N$ expansion. At leading order in the $1/N$ expansion, we need only keep track of one of the CTP two-point functions $C_{ab}(x, x')$; we choose $C_{++}(x, x')$, which is the Green function with Feynman boundary conditions. The covariant gap equation for C_{++} derived earlier in (7.88) at leading order in the $1/N$ expansion is

$$\left[-\Box_x + m^2 + \xi R(x) + \frac{\lambda}{2}\bar{\phi}^2(x) + \frac{\hbar\lambda}{2} C_{++}(x, x)\right] C_{++}(x, x') = \delta^4(x - x')\frac{-i}{\sqrt{-g'}}, \tag{8.99}$$

plus terms of $O(1/N)$. The covariant δ function is defined in Ref. [47]. The mean-field equation derived in (7.87) is, at leading order in $1/N$,

$$\left[-\Box + m^2 + \xi R + \frac{\lambda}{2}\bar{\phi}^2 + \frac{\hbar\lambda}{2} C_{++}(x, x)\right] \bar{\phi}(x) = 0, \tag{8.100}$$

The coincidence limit $C_{++}(x, x)$ is divergent in four spacetime dimensions, and the regularization method is described in Sec. 8.3.2 below.

We now consider initial conditions appropriate to post-inflation dynamics of the inflaton field in a spatially flat FLRW universe, with the scale factor to be determined consistently with the dynamics of the inflaton field and its fluctuation field. As discussed in Sec. 8.3.1, initial Cauchy data for $\bar{\phi}, C_{++}$, and a are specified on a spacelike hypersurface Σ_{η_0} (at conformal time η_0). The spatial symmetries of $\bar{\phi}$ and C_{++} for a quantum state $|\phi\rangle$ consistent with a spatially homogeneous and isotropic cosmology are given in (8.81) and (8.82). As a consequence of these symmetries, both the mean field $\bar{\phi}$ and variance $\langle\varphi^2\rangle$ are spatially homogeneous, i.e., functions of conformal time only.

Eq. (7.88) for C_{++} in spatially flat FLRW spacetimes has the formal solution

$$C_{++}(x, x') = a(\eta)^{-1} a(\eta')^{-1} \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \left[\Theta(\eta' - \eta)\tilde{u}_k(\eta)^\star \tilde{u}_k(\eta') + \Theta(\eta - \eta')\tilde{u}_k(\eta')\tilde{u}_k(\eta)^\star\right], \tag{8.101}$$

in terms of conformal mode functions $\tilde{u}_k$ which satisfy a harmonic oscillator equation with conformal-time dependent effective frequency,

$$\left[\frac{d^2}{d\eta^2} + \Omega_k^2(\eta)\right] \tilde{u}_k = 0. \tag{8.102}$$

[5] In our index notation, the latin letters i, j, k, l, m, n are used to designate O(N) indices (with index set $\{1, \ldots, N\}$), while the latin letters a, b, c, d, e, f are used below to designate CTP indices (with index set $\{+, -\}$).

The fact that $\tilde{u}_k(\eta)$ depends only on η and k (where k is comoving momentum) implies that C_{++} is invariant under simultaneous spatial translations and rotations of $\mathbf{x}$ and $\mathbf{x}'$. The effective frequency $\Omega_k(\eta)$ appearing in Eq. (8.102) is defined by

$$\Omega_k^2(\eta) = k^2 + a^2 M_\xi^2(\eta), \tag{8.103}$$

$$M_\xi^2(\eta) = M^2(\eta) + \left(\xi - \frac{1}{6}\right) R(\eta), \tag{8.104}$$

$$M^2(\eta) = m^2 + \frac{\lambda}{2}\bar{\phi}^2(\eta) + \frac{\lambda}{2}\langle\varphi^2(\eta)\rangle. \tag{8.105}$$

Initial conditions for the positive frequency conformal mode functions $\tilde{u}_k(\eta)$ must be specified (for all k) at η_0. A choice of initial conditions corresponds to a choice of quantum state $|\phi\rangle$ for the fluctuation field φ. (Initial conditions are discussed in Sec. 8.3.3 below.) The (bare) variance $\langle\varphi^2\rangle$ has a simple representation in terms of the conformal-mode functions:

$$\langle\phi|\varphi^2(x)|\phi\rangle = \hbar C_{++}(x,x) = \frac{\hbar}{a^2}\int \frac{d^3k}{(2\pi)^3}|\tilde{u}_k(\eta)|^2. \tag{8.106}$$

This expression is divergent, in consequence of our having computed the variance in terms of the bare (unrenormalized) constants of the theory. In terms of a physical upper momentum cutoff k_{max}, $C_{++}(x,x)$ diverges like k_{max}^2; there is also a logarithmic dependence on k_{max}. In addition, the mode functions $\tilde{u}_k$ depend on $\langle\varphi^2\rangle$ through Eq. (8.105). The leading-order, large–N, mean-field equation in spatially flat FLRW spacetime becomes

$$\bar{\phi}'' + \frac{2a'}{a}\bar{\phi}' + a^2 M^2(\eta)\bar{\phi} = 0, \tag{8.107}$$

where the time-dependent bare effective mass $M(\eta)$ is given by Eq. (8.105). For simplicity of notation, we will henceforth write M instead of $M(\eta)$, and similarly for $M_\xi(\eta)$.

Finally, we can express the bare stress-energy tensor in terms of the conformal mode functions $\tilde{u}_k(\eta)$. As mentioned, it is more convenient to work with the 00 component and the trace of the stress-energy tensor. The components of the classical part of the stress-energy tensor are spatially homogeneous, and given by

$$T^{\rm c}_{00}(\eta) = \frac{1}{2}(\bar{\phi}')^2 - \frac{3\xi}{2}\left(\bar{\phi}'' + \frac{2a'}{a}\bar{\phi}'\right)\bar{\phi} + \frac{1}{2}a^2\left(m^2 + \frac{\lambda}{4}\bar{\phi}^2 + \frac{3\xi(a')^2}{2a^4}\right)\bar{\phi}^2, \tag{8.108}$$

$$T^{\rm c}(\eta) = \frac{1}{a^2}\left[(6\xi - 1)(\bar{\phi}')^2 + 6\xi\left(\bar{\phi}'' + \frac{2a'}{a}\bar{\phi}'\right)\bar{\phi}\right] + 2\left(m^2 + \frac{\lambda}{4}\bar{\phi}^2 + \frac{\xi}{2}R\right)\bar{\phi}^2. \tag{8.109}$$

The quantum stress-energy tensor components are also spatially homogeneous. We find for the 00 component,

$$T^{\rm Q}_{00}(\eta) = \frac{\hbar}{2a^2}\int\frac{d^3k}{(2\pi)^3}\Big\{|\tilde{u}_k'|^2 + \Big(k^2 + a^2M^2 + (1-6\xi)\frac{(a')^2}{a^2}\Big)|\tilde{u}_k|^2 + (6\xi-1)\frac{a'}{a}\Big[(\tilde{u}_k')^\star\tilde{u}_k + \tilde{u}_k'\tilde{u}_k^\star\Big]\Big\}, \tag{8.110}$$

and for the trace,

$$T^{\rm Q}(\eta) = \frac{\hbar}{a^4}\int\frac{d^3k}{(2\pi)^3}\Big\{(6\xi-1)\Big[|\tilde{u}_k'|^2 - (k^2+a^2M^2)|\tilde{u}_k|^2 - \frac{a'}{a}\Big[(\tilde{u}_k')^\star\tilde{u}_k + \tilde{u}_k'\tilde{u}_k^\star\Big] + \Big(\frac{(a')^2}{a^2} - \xi a^2R\Big)|\tilde{u}_k|^2\Big] + a^2M^2|\tilde{u}_k|^2\Big\}. \tag{8.111}$$

It can be shown by asymptotic analysis that, in terms of a physical upper momentum cutoff k_{max}, the bare $T^{\rm Q}_{00}$ is quartically divergent, i.e., $O(k^4_{max})$, and that (for minimal coupling) $T^{\rm Q}$ is quadratically divergent. In addition, the components of the bare stress-energy tensor contain the effective mass M^2, which contains the divergent variance $\langle\varphi^2\rangle$. The energy density $\rho_{\rm Q}$ of quantum modes of the φ field is defined in terms of $T^{\rm Q}_{00}$ by

$$\rho_{\rm Q} = \frac{1}{a^2}T^{\rm Q}_{00} - \frac{\lambda}{8}\langle\varphi^2\rangle^2. \tag{8.112}$$

We shall also refer to $\rho_{\rm Q}$ as the energy density of the "fluctuation field."

8.3.2 Adiabatic Regularization

The variance $\langle\varphi^2\rangle$ and quantum components $T^{\rm Q}_{00}$ and $T^{\rm Q}$ are divergent in four spacetime dimensions, and must be regularized within the context of a systematic, covariant renormalization procedure. In the "in-out" formulation of quantum field theory, renormalization may be carried out via addition of counterterms to the effective action, which amounts to renormalization of the constants in the classical action. We have seen three such methods in Chapters 2, 4, and 5. The closed-time-path formulation of the effective dynamics is renormalizable provided the theory is renormalizable in the "in-out" formulation [182], as is the case with the O(N) field theory in curved spacetime [244, 368, 595]. For our purposes it is convenient (in this model) to carry out renormalization in the leading-order, large-N, evolution equations, rather than in the CTP effective action [181].

We illustrate this with the adiabatic regularization method of Parker, Fulling, and Hu [84, 71, 86]. The idea is to define an adiabatic approximation to the conformal mode function, and then to construct a regulator for the integrands of the bare stress-energy tensor and variance from the adiabatic mode functions [48]. Renormalization occurs when we define the renormalized variance and stress-energy tensor to be the difference between the bare expressions and the

regulators and simultaneously replace the bare quantities by their renormalized counterparts in the field m, λ, ξ in (7.51) and in the geometry G, a, b, Λ, as in (2.76) with (2.74). The equivalence of results produced in this procedure to other manifestly covariant methods (such as dimensional, zeta-function and point separation) has been shown for quantum fields in several commonly encountered curved spacetimes. (e.g., [668]).

We first perform adiabatic regularization of the variance and renormalize ξ, m, and λ. We then adiabatically regularize the stress-energy tensor and renormalize the semiclassical Einstein equation.

We define the adiabatic order of a conformal mode function as follows: let $\Omega_k(\eta) \to \Omega_k(\eta/\tau)$, where τ is introduced as a time scale which is formally taken to be unity at the end of the calculation. Then the adiabatic order of an expression involving derivatives of Ω_k is simply the inverse power of τ, of the leading-order term in an asymptotic expansion about $\tau \to \infty$. However, in order for the adiabatically regularized stress-energy tensor for an interacting scalar field theory to agree with the renormalized stress-energy tensor obtained by manifestly covariant methods (e.g., covariant point splitting [319]), it is necessary to define the adiabatic order of expressions involving λ and derivatives with respect to η, such as $\lambda(\bar{\phi}^2)''$, as the sum of the exponent of $\bar{\phi}$ and the number of conformal time differentiations [401]. Therefore, $\lambda\langle\varphi^2\rangle''$ is considered fourth adiabatic order, as is $\lambda(\bar{\phi}^2)''$.

From the discussion of adiabatic vacuum we see that the WKB ansatz

$$\tilde{u}_k(\eta) = \frac{1}{\sqrt{2W(\eta)}} \exp\left(i \int^{\eta} d\eta' \, W(\eta') \right) \tag{8.113}$$

turns the parametric oscillator equation (with time-dependent frequency Ω_k) into a nonlinear differential equation for W,

$$W(\eta)^2 = \Omega_k^2(\eta) + \frac{3[W'(\eta)]^2}{4W^2(\eta)} - \frac{W''(\eta)}{2W(\eta)}. \tag{8.114}$$

Starting with the lowest-order ansatz $W^{(0)}(\eta) = \Omega_k(\eta)$, one can iterate this equation; the nth-order iteration yields the nth-order WKB approximation for $\tilde{u}_k$. For the free field theory, the nth-order WKB approximation gives an expression for $\tilde{u}_k$ which is of adiabatic order $2n$. In the interacting case, the above definition of adiabatic order calls for removing terms such as $\lambda(\bar{\phi}^2)''''$ at 4th adiabatic order. Thus we have a method of deriving expressions for $T^{\rm Q}_{00}, T^{\rm Q}$, and $\langle\varphi^2\rangle$ at fourth, fourth, and second adiabatic orders, respectively. One then sets $\tau = 1$ in the truncated expression. We can thus obtain a fourth-order adiabatic approximation to the quantum $(T^{\rm Q}_{\mu\nu})_{\rm ad4}$, and a second-order adiabatic approximation to the variance $\langle\varphi\rangle_{\rm ad2}$. By subtracting $(T^{\rm Q}_{\mu\nu})_{\rm ad4}$ from the divergent $T^{\rm Q}_{\mu\nu}$ and $\langle\varphi^2\rangle_{\rm ad2}$ from the divergent $\langle\varphi^2\rangle$, finite expressions for the renormalized stress-energy tensor and variance are obtained.

Regularization of the Variance of the Field

First we regularize the variance $\langle\varphi^2\rangle$, and carry out a renormalization of λ, m, and ξ. In the leading-order, large N approximation, no terms appear in the mode-function equation (8.102) which would permit addition of counterterms; therefore, Ω_k must be finite. The effective frequency Ω_k which appears in Eq. (8.103) is the "bare" effective frequency, which we denote by $(\Omega_k)_{\rm B}$. In the process of regularization, upon adding counter terms, each bare quantity is replaced by a renormalized quantity (carrying a subscript R). E.g., in the expression

$$M_{\rm R}^2 = m_{\rm R}^2 + \frac{\lambda_{\rm R}}{2}\bar{\phi}^2 + \frac{\lambda_{\rm R}}{2}\langle\varphi^2\rangle_{\rm R} \tag{8.115}$$

each term therein replaces the corresponding bare quantity $\lambda_{\rm B}, m_{\rm B}$, and $\xi_{\rm B}$ in the classical action (7.51). The renormalized quantities in Eq. (8.103) are defined below. The bare $\langle\varphi^2\rangle_{\rm B}$ is a conformal-time-dependent function defined by Eq. (8.106),

$$\langle\varphi^2(\eta)\rangle_{\rm B} = \frac{\hbar}{a^2}\int\frac{d^3k}{(2\pi)^3}|\tilde{u}_k(\eta)|^2, \tag{8.116}$$

where the conformal-mode functions $\tilde{u}_k(\eta)$ obey Eq. (8.102). Now we demand that the renormalized theory be minimally coupled, i.e., we set $\xi_{\rm R} = 0$. We can also formally use $(\Omega_k^2)_{\rm R}$ in computing the adiabatic regulator for the variance $\langle\varphi^2\rangle_{\rm B}$. Computing the asymptotic series (in $1/\tau$) of the quantity $|\tilde{u}_k(\eta)|^2$ to $O(1/\tau^2)$, where $\Omega_k^2(\eta/\tau)$ is the effective frequency, we obtain (after setting $\tau = 1$)

$$\langle\varphi\rangle_{\rm ad2} = \frac{\hbar}{2C}\int\frac{d^3k}{(2\pi)^3}\left[\frac{1}{\tilde{\omega}_k} - \frac{(C')^2 - 2CC''}{8C^2\tilde{\omega}_k^3} + \frac{M_{\rm R}^2C''}{8\tilde{\omega}_k^5} - \frac{5M_{\rm R}^4(C')^2}{32\tilde{\omega}_k^7}\right], \tag{8.117}$$

in terms of auxiliary functions $C(\eta) = a^2(\eta)$ and $D(\eta) = \frac{C'(\eta)}{C(\eta)}$. In Eq. (8.117) the symbol $\tilde{\omega}_k$ is defined as follows

$$\tilde{\omega}_k^2 = k^2 + a^2M_{\rm R}^2. \tag{8.118}$$

In the adiabatic prescription, the renormalized variance $\langle\varphi^2\rangle_{\rm R}$ appearing in Eq. (8.115) is defined by

$$\langle\varphi^2\rangle_{\rm R} = \langle\varphi^2\rangle_{\rm B} - \langle\varphi^2\rangle_{\rm ad2}, \tag{8.119}$$

where the first term on the right-hand side is given by Eq. (8.116), and the second term is given by Eq. (8.117). Both can be expressed in terms of renormalized quantities, so this procedure is well defined. Written out explicitly, the renormalized variance satisfies the equation

$$\langle\varphi^2\rangle_{\rm R} = \frac{\hbar}{C}\int\frac{d^3k}{(2\pi)^3}\left(|\tilde{u}_k|^2 - \frac{1}{2\tilde{\omega}_k} - \frac{(C')^2 - 2CC''}{16C^2\tilde{\omega}_k^3} + \frac{M_{\rm R}^2C''}{16\tilde{\omega}_k^5} - \frac{5M_{\rm R}^4(C')^2}{64\tilde{\omega}_k^7}\right). \tag{8.120}$$

One can use the WKB approximation for $\tilde{u}_k(\eta)$ to compute the asymptotic series for the integrand in Eq. (8.120) in the limit $k \to \infty$, and show that the integral is convergent. Since $M_{\rm R}^2$ is contained in the integrand above, Eq. (8.120) leads to an integral equation for the renormalized effective mass $M_{\rm R}$,

$$M_{\rm R}^2 = m_{\rm R}^2 + \frac{\lambda_{\rm R}}{2}\bar{\phi}^2 + \frac{\hbar\lambda_{\rm R}}{2C}\int \frac{d^3k}{(2\pi)^3}\left[|\tilde{u}_k|^2 - \frac{1}{2\tilde{\omega}_k} - \frac{(C')^2 - 2CC''}{16C^2\tilde{\omega}_k^3} + \frac{M_{\rm R}^2 C''}{16\tilde{\omega}_k^5} - \frac{5M_{\rm R}^4(C')^2}{64\tilde{\omega}_k^7}\right]. \tag{8.121}$$

All physical quantities should now be expressed in terms of the renormalized parameters $m_{\rm R}$ and $\lambda_{\rm R}$ of the theory. The renormalized mean-field equation now reads

$$\bar{\phi}'' + \frac{2a'}{a}\bar{\phi}' + a^2 M_{\rm R}^2\bar{\phi} = 0, \tag{8.122}$$

where $M_{\rm R}^2$ is given by Eq. (8.121), and the mode functions in Eq. (8.121) obey the homogeneous equation,

$$\left(\frac{d^2}{d\eta^2} + k^2 + a^2 M_{\rm R}^2\right)\tilde{u}_k = 0. \tag{8.123}$$

Regularization of the Stress Energy Tensor

To be specific we add a subscript B to the bare stress-energy tensor components (8.110) and (8.111) calling them $(T^{\rm Q}_{00})_{\rm B}$ and $(T^{\rm Q})_{\rm B}$, respectively. In these equations we should use the renormalized $M_{\rm R}$ and $\xi_{\rm R}$ for the components of the quantum stress-energy tensor, with $\xi_{\rm R} = 0$ corresponding to minimal coupling. Renormalized constants a, b, G, and Λ should also be used for the semiclassical Einstein equation (7.49) Applying the method described above to construct the adiabatic regulator, the fourth adiabatic order expressions for $T^{\rm Q}_{00}$ and $T^{\rm Q}$ can be obtained – see [605] for details. In the free-field limit ($\lambda_{\rm R} = 0$) agree with the minimal-coupling, spatially flat limit of the adiabatic regulators obtained in [668].

We now proceed with the renormalization for the semiclassical Einstein equation (7.49). According to the adiabatic prescription, we define the quantum stress-energy tensor by

$$(T^{\rm Q}_{\mu\nu})_{\rm R} = (T^{\rm Q}_{\mu\nu})_{\rm B} - (T^{\rm Q}_{\mu\nu})_{\rm ad4}. \tag{8.124}$$

It can be checked that the momentum-integral expressions for the two independent components of Eq. (8.124) are convergent. In terms of $(T^{\rm Q}_{\mu\nu})_{\rm R}$, the total (after renormalization) is

$$\langle T_{\mu\nu}\rangle_{\rm R} = (T^{\rm C}_{\mu\nu})_{\rm R} + (T^{\rm Q}_{\mu\nu})_{\rm R} - \frac{\lambda_{\rm R}}{8}\left(\langle\varphi^2\rangle_{\rm R}\right)^2 g_{\mu\nu}, \tag{8.125}$$

where $(T^{\rm C}_{\mu\nu})_{\rm R}$ stands for $T^{\rm C}_{\mu\nu}$, and renormalized quantities are substituted for bare quantities. The bare quantities $G_{\rm B}$, $\Lambda_{\rm B}, a_{\rm B}$, and $b_{\rm B}$ in (2.76) with (2.74) are now replaced by $G_{\rm R}, \Lambda_{\rm R}, a_{\rm R}$, and $b_{\rm R}$ in the renormalized semiclassical Einstein equation (2.77).

$$G_{\mu\nu} - \Lambda_{\rm R} g_{\mu\nu} + a_{\rm R}{}^{(1)}H_{\mu\nu} + b_{\rm R}{}^{(2)}H_{\mu\nu} = 8\pi G_{\rm R}\langle T_{\mu\nu}\rangle_{\rm R}. \tag{8.126}$$

Note that $(a_{\rm R}, b_{\rm R})$ here are coefficients contained in Ξ in (2.77), thus equal to $-8\pi G_{\rm N}(a,b)_{ren}$, with a, b being the quantities in (2.74). At the preheating epoch the Hubble parameter is much less than the Planck mass, $H \ll m_P$. In the effective field theory sense, with a cut-off at the Planck scale m_P^2, $a_{\rm R}$ and $b_{\rm R}$ should be of order $\hbar^2 m_P^{-2}$, in which case $R \gg a_{\rm R}R^2$, and $R \gg b_{\rm R}R^{\alpha\beta}R_{\alpha\beta}$, provided $R_{\alpha\beta} \neq 0$. For the investigation of preheating we can therefore safely assume $a_{\rm R} = 0$ and $b_{\rm R} = 0$.

8.3.3 Renormalized Semiclassical Einstein Equation

If in addition we consider only cases where $\Lambda_{\rm R} = 0$, then the renormalized semiclassical Einstein equation Eq. (8.126) becomes

$$G_{\mu\nu} = 8\pi G_{\rm R}\left[(T^{\rm C}_{\mu\nu})_{\rm R} + (T^{\rm Q}_{\mu\nu})_{\rm R} - \frac{\lambda_{\rm R}}{8}(\langle\varphi^2\rangle_{\rm R})^2\, g_{\mu\nu}\right]. \tag{8.127}$$

Taking the trace of Eq. (8.127) in spatially-flat FLRW spacetime, we find

$$\frac{6a''}{a^3} = 8\pi G_{\rm R}\left[(T^{\rm C})_{\rm R} + (T^{\rm Q})_{\rm R} - \frac{\lambda_{\rm R}}{2}\langle(\varphi^2\rangle_{\rm R})^2\right]. \tag{8.128}$$

With minimal coupling $\xi_{\rm R} = 0$, using Eq. (8.109), the classical part of the trace of the renormalized stress-energy tensor is given by

$$(T^{\rm C})_{\rm R} = \frac{1}{a^2}\left[-(\bar\phi')^2 + 2\left(m_{\rm R}^2 + \frac{\lambda_{\rm R}}{4}\bar\phi^2\right)\bar\phi^2\right], \tag{8.129}$$

and the quantum part of the trace of the renormalized stress-energy tensor is given by

$$\begin{aligned}(T^{\rm Q})_{\rm R} = -\frac{\hbar}{a^4}\int\frac{d^3k}{(2\pi)^3}\Big\{&|\tilde u_k'|^2 - (k^2 - 2a^2M_{\rm R}^2)|\tilde u_k|^2 \\ &- \frac{a'}{a}\left[(\tilde u_k')^\star\tilde u_k + \tilde u_k'\tilde u_k^\star\right] + \frac{(a')^2}{a^2}|\tilde u_k|^2\Big\} - (\mathcal{T}^{\rm Q})_{\rm ad4}.\end{aligned} \tag{8.130}$$

The 00 component of the semiclassical Einstein equation is a constraint, given by

$$\frac{3(a')^2}{a^2} = 8\pi G_{\rm R}\left[(T^{\rm C}_{00})_{\rm R} + (T^{\rm Q}_{00})_{\rm R} - \frac{\lambda_{\rm R}}{8}a^2(\langle\varphi^2\rangle_{\rm R})^2\right]. \tag{8.131}$$

From Eq. (8.108), the expression for the classical and quantum parts of the 00 component of the renormalized stress-energy tensor are given respectively by

$$(T^{\rm C}_{00})_{\rm R} = \frac{1}{2}(\bar\phi')^2 + \frac{1}{2}a^2\left(m^2_{\rm R} + \frac{\lambda_{\rm R}}{4}\bar\phi^2\right)\bar\phi^2, \tag{8.132}$$

$$(T^{\rm Q}_{00})_{\rm R} = \frac{\hbar}{2a^2}\int\frac{d^3k}{(2\pi)^3}\left[|\tilde u_k'|^2 + (k^2 + a^2M^2_{\rm R})\,|\tilde u_k|^2 - \frac{a'}{a}\left[(\tilde u_k')^\star\tilde u_k + \tilde u_k'\tilde u_k^\star\right]\right] - (T^{\rm Q}_{00})_{\rm ad4}, \tag{8.133}$$

Eqs. (8.128) and (8.122) are coupled differential equations for a and $\bar\phi$, involving complex homogeneous conformal-mode functions $\tilde u_k$ which satisfy Eq. (8.123). The conformal mode functions are related to the variance $\langle\varphi^2\rangle_{\rm R}$ by Eq. (8.120). This is a closed, time-reversal-invariant system of equations. The initial data at η_0 must satisfy the constraint equation (8.131). We now drop all "R" subscripts, because we will henceforth work only with renormalized quantities.

Derivative Orders Reduction

The three adiabatic regulators (shown only one in (8.117)) for the variance and stress-energy tensor contain derivatives of up to fourth order in a and up to second order in $\bar\phi^2$ and $\langle\varphi^2\rangle$. The presence of the former can be understood as resulting in part from the well-known trace anomaly for a quantum field in curved spacetime [87], which contains higher-derivative local geometric terms, e.g., $\Box R$. In addition, there are nonanomalous finite terms which result from the renormalization of the stress-energy tensor and the choice of minimal coupling.

The problem with higher derivatives in the semiclassical Einstein equation having many solutions which are unphysical has been discussed earlier. Following the procedure of Simon and Parker [357] to reduce the order of the equations for $\bar\phi, a$, and $\langle\varphi^2\rangle$ to second order, we replace all expressions involving a''' and a'''' with expressions $a'''_{\rm cl}$ and $a''''_{\rm cl}$ obtained from the *classical* Einstein equation, i.e., Eq. (8.128) with $\hbar = 0$. This procedure is physically justifiable in this model for the following reason: at early times, the dominant contribution to the stress-energy tensor is classical, $T^{\rm C}_{\mu\nu}$. Therefore, the deviations $a''' - a'''_{\rm cl}$ and $a'''' - a''''_{\rm cl}$, which are entirely quantum in origin and $\propto\hbar$, are at early times expected to be very small. In addition, at late times the Universe is expected to become asymptotically radiation-dominated, in which case $a''' = a'''' = 0$. The classical approximations to the late-time behavior of a''' and a'''' should also have this property, regardless of whether the mean-field oscillations are harmonic or elliptic. This procedure is, therefore, physically justifiable in the system studied here.

Analysis and Results

The coupled dynamical equation (8.122) for the mean field ϕ, (8.123) for the conformal-mode functions $\tilde u_k$ of the fluctuation field, and (8.128) for the

scale factor a of the background FLRW spacetime are now ready to be solved self-consistently.

Initial conditions for the inflaton fluctuation field. At the Cauchy hypersurface at η_0, the time when the preheating stage begins, initial conditions are specified on the conformal mode functions $\tilde{u}_k$ which correspond to a choice of quantum state for the fluctuation field φ. Assuming the slow-roll condition (8.92), the potential-dominated condition (8.91), and the condition the variance $\langle\varphi^2\rangle$ satisfies

$$\frac{\lambda}{2}\langle\varphi^2\rangle \ll m^2 + \frac{\lambda}{2}\bar{\phi}^2 \tag{8.134}$$

for $\eta < \eta_0$, it follows that the spacetime is asymptotically de Sitter at conformal past-infinity. Using the approximate solution (8.93) for the scale factor for $\eta < \eta_0$, we can solve the mode function equation (8.123) for $\eta < \eta_0$ at the same (0th) adiabatic order. The general solution is of the form

$$\tilde{u}_k(\eta) \simeq \frac{1}{2}\sqrt{\pi(\eta - H^{-1} - \eta_0)}\Big\{c_k^{(1)}\, H_\nu^{(1)}\big[k(\eta - H^{-1}(\eta) - \eta_0)\big] + c_k^{(2)}\, H_\nu^{(2)}\big[k(\eta - H^{-1}(\eta) - \eta_0)\big]\Big\}, \tag{8.135}$$

where $H^{(1)}$ and $H^{(2)}$ are the Hankel functions of first and second kind, respectively, and ν is defined by

$$\nu^2 = \frac{9}{4} - \frac{M^2}{H^2}, \tag{8.136}$$

where M is defined in (8.115). The Hubble expansion function $H(\eta)$ defined in Eq. (8.94), now is slowly varying owing to $\rho_{\rm C} = a^2 T_{00}^{\rm C}$. The Wronskian condition on the mode functions (arising from the canonical field commutation relations) requires that $|c_k^{(1)}|^2 + |c_k^{(2)}|^2 = 1$. By choosing $c_k^{(1)}$ and $c_k^{(2)}$, different vacua are obtained. As described earlier, the 0-th order adiabatic vacuum (matched at $\eta = -\infty$) is constructed by choosing $c_k^{(1)}$ and $c_k^{(2)}$ so that $\tilde{u}_k$ smoothly matches the positive-frequency 0th-order WKB mode function at $\eta = -\infty$. This corresponds to $c_k^{(2)} = 1$ and $c_k^{(1)} = 0$, for all k. Using the asymptotic properties of the Hankel function, $\tilde{u}_k$ at the adiabatic limit, i.e., both $k, |\eta| \to \infty$, can be derived, and shown to have the correct form,

$$\lim_{k,|\eta|\to\infty} \tilde{u}_k \simeq \frac{1}{\sqrt{2k}} e^{-ik\eta}. \tag{8.137}$$

In addition, the high-momentum, flat-space limit ($k, H^{-1} \to \infty$) gives the same result. The initial conditions for the $\tilde{u}_k$ at η_0 are then defined by demanding that the $\tilde{u}_k$ functions smoothly match the approximate adiabatic mode function

solutions (for $\eta < \eta_0$) at $\eta = \eta_0$. This leads to the following initial conditions for the conformal mode functions:

$$\tilde{u}_k(\eta_0) = \left(\frac{-\pi}{4H_0}\right)^{1/2} H_\nu^{(2)}(-kH_0^{-1}), \tag{8.138}$$

$$\tilde{u}_k'(\eta_0) = \frac{d}{d\eta}\left[\left(\frac{\pi\eta}{4}\right)^{1/2} H_\nu^{(2)}(k\eta)\right]_{\eta=-H_0^{-1}}, \tag{8.139}$$

where $H_0 = H(\eta_0)$. The above conditions are valid only at 0th order in the above-defined adiabatic approximation, where terms of order H'/H are discarded. It is straightforward to show that the expansion rate nonadiabaticity parameter [669] $\bar{\Omega}_H \equiv \frac{H'}{H^2} \ll 1$ is valid given the slow-roll (8.92) and inflation (8.91) assumptions. In addition to the initial conditions for $\tilde{u}_k$ at η_0, we may freely choose initial values for $\bar{\phi}(\eta_0)$ and $\bar{\phi}'(\eta_0)$, subject to the constraint that $\bar{\phi}'$ must be small enough that conditions (8.91) and (8.92) are valid. We are already assuming that $a(\eta_0) = 1$. The initial value of $a'(\eta_0)$ is fixed by the constraint equation (8.131).

Results and Implications. From the CTP-2PI effective action under the leading-order large-N approximation Ramsey and Hu [244] derived and solved numerically the coupled dynamical equations for the mean field, variance, and scale factor for a variety of values for a parameter which measures the rate of mean-field oscillations compared to the cosmic expansion rate. We defer the details to the original paper but excerpt these general conclusions to this problem:

(i) *Preheating by parametric amplification.* For an inflaton field in a $V(\phi) = \frac{1}{2}m^2\Phi^2 + \frac{\lambda}{4!}\Phi^4$ potential minimally coupled to the curvature of spacetime and with a large initial inflaton amplitude at the end of slow roll, parametric amplification of the inflaton's *own* quantum fluctuations is not efficient enough to reheat the supercooled post-inflationary universe, unless the self-coupling is significantly increased, as is allowed in the model of [550] (see also [670, 671]) where λ can be as high as $\sim 10^{-5}$. The phenomenon of parametric amplification of quantum fluctuations can play an important role in the "preheating" period of inflationary cosmology. However, it is not sufficient for reheating, which involves particle interactions, quantum kinetic and possible turbulence processes.

(ii) *Effect of spacetime dynamics.* The results of this calculation testifies to the effect that at least in the chaotic inflation model [121] with large inflaton amplitude at the onset of reheating a correct study of the reheating period *must* take into account *the effects of spacetime dynamics.* The full program should be implemented with solutions to the coupled semiclassical Einstein equation and matter-field equations *self-consistently*, so that no *ad hoc* assumptions need be made about the effective equation of state and/or the relevant time scales involved.

(iii) *Two-loop 2PI effectve action.* The leading-order $1/N$ approximation which depicts collisionless processes, and hence adequately depicts the parametric

amplification of quantum fluctuations, is a subcase of the two-loop, 2PI effective action. A full two-loop treatment of the unbroken symmetry mean-field dynamics of the $O(N)$ field theory (which involves solving the nonlocal, integro-differential equations, Eq. (7.76) and Eq. (7.77) in Chapter 7) would include multiparticle scattering processes which contribute to reheating. Particle interactions are of a qualitatively different nature than the parametric resonance energy-transfer mechanism discussed above. In the model studied here, multiparticle scattering occurs on a time scale significantly longer than the time scale for parametric amplification of quantum fluctuations. That is the reason why the leading-order, large N (collisionless) approximation is sufficient for a study of parametric amplification of quantum fluctuations. In more realistic models of inflation, couplings of the inflaton to other fields could provide additional mechanisms of energy transfer away from the inflaton mean field into these other fields' quantum modes. The final thermalization stage of reheating would require a quantum kinetic field theory and collision-dominated hydrodynamic treatment (see, e.g., [326]) while efficient mechanisms may rely on complex hydrodynamic processes, especially turbulence.

8.4 Other Examples: Stochastic Inflation, Minisuperspace Cosmology

The first example of backreaction of quantum fields on the background spacetime we presented is the $\lambda\Phi^4$ theory, in Chapter 2, by way of the *in-out* effective action, and in Chapter 7, by way of the 2PI CTP effective action. There we performed a background field decomposition $\Phi = \bar{\phi} + \varphi$ on the self-interacting scalar field Φ and calculated the effect of the fluctuation field φ on the background field $\bar{\phi}$ as well as on the background geometry. This is a rather general practice in open quantum systems. For many problems especially those encountered in statistical mechanics where one is interested in the detailed behavior of only a part of the overall system (call it the system) interacting with its surrounding (call it the environment), one can decompose the field describing the overall system Φ according to $\Phi = \phi_S + \phi_E$ where ϕ_S denotes the system field and ϕ_E the environment field. One can calculate the backreaction of the latter on the former with the so-called 'coarse-grained effective action' [241, 242], where the steepest descent is not with respect to the quantum fluctuations in powers of $\hbar$, as in the conventional definition of an effective action, but rather with respect to a small parameter which can generally be expressed as the ratio of two time scales, length scales, mass scales or energy scales, characterizing the discrepancy between the system and its environment – in fact, justifying the separation of one from the other.

Stochastic Inflation

In the stochastic inflation scheme of Starobinsky [506] one regards the system field as containing only the lower frequency modes and the environmental field as containing the higher frequency modes. The high-low mode division is effected by the event horizon in de Sitter spacetime. A critical examination of two

fundamental issues related to stochastic inflation, namely, the decoherence of the long wavelength sector and the source of the noise in the Langevin equation can be found in [550], and a discussion of stochastic inflation as a backreaction problem by means of the coarse-grained effective action is given in [241, 242]. Further details on the open system perspective and the influence functional formalism can be found in [326]. For more recent work on coarse-graining and stochastic inflation see, e.g., Levasseur and McDonough [672] and Hollowood and McDonald [673].

Minisuperspace Quantum Cosmology

In Hamiltonian cosmology [674, 420] one often invokes the superspace, which is the space of all three-geometries ${}^{(3)}g$. A trajectory in the superspace represents the world history of the universe. On quantization, the three-geometry and its conjugate momentum are promoted to the status of operators obeying the canonical commutation relations. The classical Hamiltonian and momentum constraints are turned into operator equations acting on the "wave function of the Universe" $\Psi({}^{(3)}g, \Phi)$, with Φ representing the matter field present. Ψ, a functional on superspace, is the principal object of interest in *quantum cosmology*. It obeys the Wheeler–DeWitt equation [312, 313], which is an infinite-dimensional partial differential equation. Finding solutions in the general case is a formidable task. To make the problem more tractable one turns to minisuperspace (mss) quantization [615]. Minisuperspace quantization refers to the technique of restricting the quantum theory of gravity to spacetimes possessing some symmetry. In the most common examples, the symmetry is spatial homogeneity, which has the advantage of reducing the infinite dimensional superspace to a finite dimensional minsuperspace. Finding solutions to a finite dimensional Wheeler–DeWitt equation becomes tractable. A famous example is the quantization of the diagonal Bianchi Type IX universe, Misner's mixmaster universe [419]. Halliwell and Hawking [224] have gone beyond: they treated the homogeneous and isotropic degrees of freedom of the Friedmann Universe exactly and the inhomogeneous and anisotropic degrees of freedom in superspace to second order in the perturbations. For the same model Kiefer [226] calculated explicitly the wave functions for all multipoles of the matter field and for the tensor modes of the metric. D'Eath and Halliwell [675] have considered the backreaction of the fermionic perturbations on the homogeneous modes. See also [225, 676, 228] for the backreaction of quantized matter fields and the semiclassical limits in quantum cosmology.

The simplifying advantage in a minisuperspace approximation also carries certain penalty. Nonlinear interactions of the truncated superspace modes with the minisuperspace modes are being ignored, and since the truncation involves an infinite number of modes (save a handful), results obtained under such an approximation may not be reliable. It is also well-known that this truncation violates the uncertainty principle, since it implies setting the amplitudes and

momenta of the inhomogeneous modes simultaneously to zero. It is therefore important to understand what kind of effects the truncated modes have on the minsuperspace degrees of freedom and if there exist conditions when the minisuperspace wave function reflects enough the behavior of the full superspace. A serious attempt to assess the validity of the minisuperspace approximation was made by Kuchar and Ryan [227]. The same question, posed as a backreaction problem, was investigated by Sinha and Hu [229]. They use a $\lambda\phi^4$ scalar field coupled to a closed FLRW background spacetime as a model to explore this issue. In the open system conceptual framework, their system consists of the scale factor a plus the lowest mode of the scalar field, while the environment consists of the inhomogeneous modes in the interacting field. Note that in this example of minisuperspace, the scalar field should not be thought of as providing a matter source for the background metric. Instead, the inhomogeneous scalar field modes mimick the gravitational degrees of freedom in superspace. One then examines how nonlinearity and mode coupling affect the backreaction of the higher modes on the lower modes. The quantum cosmology problem studied in [229] shows the versatility of the 'in-in' coarse-grained effective action [241], which, as we remarked in an earlier chapter, is closely related to the influence action of Feynman and Vernon [187], and will play a significant role in the themes developed in the later chapters.

Part III

Stochastic Gravity

9

Metric Correlations at One-Loop: In-In and Large N

An important tool in quantum field theory is the effective action functional. This is the generating functional of one-particle-irreducible correlation functions and leads to the equations of motion for the vacuum expectation value of the field operator, or the mean field. In semiclassical gravity effective action methods were first used by Hartle and collaborators [106, 109, 110] to study the backreaction of quantum fields in cosmology.

However the use of the familiar in-out effective action leads to field equations that are neither real nor causal. The reason is that this effective action is tailored to connect in and out vacuum states, defined at early and late times. In contrast, in the gravitational context, of which cosmology is a primary example, one is interested in true correlation functions which are expectation values of operators rather than transition matrix elements, and one needs to impose initial conditions at early times rather than boundary conditions at both early and late times. As we have explained in previous chapters the appropriate formalism to deal with this was introduced long ago by Schwinger [239] (see also [415, 416]) and Keldysh [240], and further developed by Chou et al. [436], in what is known as the Schwinger–Keldysh, the in-in, or the closed time path (CTP) formalism. These functional techniques were adapted to the gravitational context [180, 181, 182, 677, 437, 423] and applied to different problems in cosmology. Using the CTP effective action, semiclassical Einstein equations describing the backreaction from quantum fields were derived.

In this chapter we first briefly summarize the in-in functional formalism and evaluate the in-in effective action for a scalar field in Minkowski spacetime. We then consider the interaction of N quantum matter fields interacting with the gravitational field in the effective field theory approach to quantum gravity and consider the quantization of metric perturbations around a semiclassical background in the CTP formalism. We derive expressions for the two-point metric correlations which are conveniently written in terms of the CTP effective action

that results from integrating out the matter fields. These correlations include loop corrections from matter fields but no graviton loops. This is consistently done in a large N expansion, as is illustrated in a simplified toy model of matter-gravity interaction. Here we use units in which $\hbar = c = 1$.

9.1 The In-In Formalism in Flat Spacetime

9.1.1 From the In-Out to the In-In Formalism

Quantum corrections to a classical field theory can be studied with the help of the effective action. For simplicity, we consider the quantization of a scalar field $\phi(x)$. The usual in-out effective action is rooted in the generating functional $Z[J]$, which is related to the vacuum persistence amplitude in the presence of some classical source $J(x)$ by

$$Z[J] \equiv \langle 0_+|0_-\rangle_J, \tag{9.1}$$

where $|0_-\rangle$ and $|0_+\rangle$ represent the "in" and "out" vacuum states, respectively. By taking functional derivatives with respect to the source J one generates the correlation functions of the field operators. It is also convenient to introduce the functional $W[J]$ by

$$Z[J] \equiv e^{iW[J]}. \tag{9.2}$$

This functional carries all the quantum information of the connected graphs of the theory, which are obtained by functional derivation of $W[J]$ with respect to the source J. $W[J]$ is called the generating functional of connected correlation functions.

When one couples an external field $J(x)$ it is convenient to use the interaction picture in which the states $|\psi\rangle$ evolve in time according to the Schrödinger equation $H_I|\psi\rangle = i\partial_t|\psi\rangle$, where H_I is the interaction Hamiltonian operator $H_I = \int d^{n-1}x J(x)\phi(x), \phi(x)$ is now the field operator in the Heisenberg picture and n the number of spacetime dimensions. The solution of this equation may be formally written as,

$$|\psi\rangle_{t_2} = \mathcal{T}\exp\left(i\int_{t_1}^{t_2} dt H_I\right)|\psi\rangle_{t_1}, \tag{9.3}$$

where $\mathcal{T}$ is the usual time-ordering operator, and (9.1) can be written as

$$Z[J] = \langle 0_+|\mathcal{T}\exp\left(i\int_{-\infty}^{\infty} dt H_I\right)|0_-\rangle. \tag{9.4}$$

It is easy to see from the classical field equations for $\phi(x)$ in the presence of $J(x)$ that $Z[J]$ satisfies the integro-differential Schwinger–Dyson equation, and that one may give a path integral representation for its solution as,

$$Z[J] = \int \mathcal{D}\phi \, e^{i(S[\phi]+J\phi)}, \tag{9.5}$$

where $S[\phi]$ is the classical action of the field theory and the common shorthand notation $J\phi$ for the integral $\int d^n x J(x)\phi(x)$ has been used. The functional integral is taken with the following boundary conditions: $\phi \to e^{\mp i\omega t}$, where $\omega > 0$, when the time $t \to \pm\infty$, i.e., the scalar field has only negative and positive frequency modes in the *in* and *out* regions respectively; the interaction is assumed to be switched off at these asymptotic regions.

To be more precise, in order to implement the standard flat-space choice of vacuum in the path integral (9.5), one slightly tilts the time integration contour on the complex plane to include an imaginary part

$$t \to t(1 - \mathrm{i}\epsilon) \tag{9.6}$$

with $\epsilon > 0$ (see, for instance, Sec. 4.2 in the textbook by Peskin and Schroeder [589]). This selects the asymptotic vacuum as the state of lowest energy of the full interacting theory, which includes appropriate correlations between the different fields or even different modes of the same field. This procedure selects the ground state of the interacting theory only for a time-independent Hamiltonian. Under the appropriate conditions, however, in the time-dependent case, this prescription can still select an adiabatic vacuum of the theory at early times. In practice, one calculates the integral from some initial time t_0 to some final time T, and takes the limits $t_0 \to -\infty, T \to \infty$ in a slightly imaginary direction.

By differentiating with respect to the source one generates matrix elements from $Z[J]$, or $W[J]$,

$$\frac{1}{Z[J]}\frac{\delta Z[J]}{i\delta J(x)} \equiv \frac{\delta W[J]}{\delta J(x)} = \frac{\langle 0_+|\phi(x)|0_-\rangle_J}{\langle 0_+|0_-\rangle_J} \equiv \bar{\phi}[J]. \tag{9.7}$$

If we assume that the above expression can be reversed, the *effective action* is defined as the Legendre transformation of the generating functional $W[J]$,

$$\Gamma[\bar{\phi}] = W[J] - J\bar{\phi}. \tag{9.8}$$

This functional of $\bar{\phi}$ is the generator of the one-particle-irreducible graphs (graphs that remain connected when any internal line is cut) and contains all the quantum corrections to the classical action. From (9.8) one may derive the dynamical equation for the effective mean field $\bar{\phi}[0]$, i.e., the matrix element of the field ϕ in the absence of the source $J(x)$, as

$$\left.\frac{\delta\Gamma[\bar{\phi}]}{\delta\bar{\phi}}\right|_{\bar{\phi}=\bar{\phi}[0]} = 0, \tag{9.9}$$

which expresses the quantum corrections to the classical equation as a variational problem of the effective action.

In order to work with expectation values rather than matrix elements one can define a new generating functional, the in-in generating functional, whose dynamics is determined by two different external classical sources J_+ and J_-, by letting the *in* vacuum evolve independently under these sources,

$$Z[J_\pm] = \sum_\alpha \langle 0_-|\alpha, T\rangle_{J_-} \langle \alpha, T|0_-\rangle_{J_+}. \tag{9.10}$$

Here we have assumed that $\{|\alpha, T\rangle\}$ is a complete basis of eigenstates of the field operator $\phi(x)$ at some future time T, *i.e.*, $\phi(T, \mathbf{x})|\alpha, T\rangle = \alpha(\mathbf{x})|\alpha, T\rangle$. Then (9.10) may be written according to (9.3) as

$$\begin{aligned} Z[J_\pm] = \int d\alpha \langle 0_-|\tilde{\mathcal{T}} e^{-i\int_{-\infty}^{T} dt \int d^{n-1}x J_-(x)\phi(x)}|\alpha, T\rangle \\ \times \langle \alpha, T|\mathcal{T} e^{i\int_{-\infty}^{T} dt \int d^{n-1}x J_+(x)\phi(x)}|0_-\rangle, \end{aligned} \tag{9.11}$$

where $\mathcal{T}$ and $\tilde{\mathcal{T}}$ mean, respectively, time and anti-time ordered operators and $d\alpha$ means $d\alpha = \Pi_{\mathbf{X}} d\alpha(\mathbf{x})$ where $\mathbf{x}$ are the points of the hypersurface Σ defined by $t = T$.

The in-in generating functional has also a path integral representation,

$$Z[J_\pm] = \int d\alpha \int \mathcal{D}\phi_- \, e^{-i\left(S[\phi_-] + J_-\phi_-\right)} \int \mathcal{D}\phi_+ \, e^{i\left(S[\phi_+] + J_+\phi_+\right)}, \tag{9.12}$$

with the boundary conditions that $\phi_+ = \phi_- = \alpha$ on Σ and that the fields ϕ_+ and ϕ_- are pure negative and pure positive modes, respectively, in the *in* region, i.e., $\phi_\pm \to e^{\pm i\omega t}$ at $t \to -\infty$ (vacuum boundary conditions in the remote past). In a more compact form one may write,

$$Z[J_\pm] = \int \mathcal{D}\phi_+ \mathcal{D}\phi_- e^{i\{S[\phi_+] + J_+\phi_+ - S[\phi_-] - J_-\phi_-\}}, \tag{9.13}$$

where it is understood that the sum is over all fields ϕ_+, ϕ_- with negative and positive frequency modes, respectively, in the remote past but which coincide at time $t = T$. These boundary conditions can be made explicit by substituting m^2 by $m^2 - i\epsilon$, where m is the field mass, in $S[\phi_+]$ and by substituting m^2 by $m^2 + i\epsilon$ in $S[\phi_-]$; the latter is also some times indicated by writing $S^*[\phi_-]$ instead of $S[\phi_-]$.

This integral can be thought of as the path sum of two different fields evolving in two different time branches, one going forward in time in the presence of J_+ from the *in* vacuum to a time $t = T$, and the other backward in time in the presence of J_- from the time $t = T$ to the *in* vacuum, with the constraint $\phi_+ = \phi_-$ on Σ. Because of such a path integral representation, this formalism is also called closed time path (CTP) formalism and $Z[J_\pm]$ the CTP generating functional.

The iϵ prescription to select the vacuum can also be carried over to the CTP formalism. However, due to the complex conjugation of the integrand in the path integral for the ϕ_- fields, we need to take the complex conjugate prescription for them

$$t \to t(1 + \mathrm{i}\epsilon). \tag{9.14}$$

One therefore has to integrate along a contour going from t_0^+ to T, returning to t_0^- with $t_0^- = (t_0^+)^*$, and taking at the end of the calculation $t_0^+ \to -\infty(1 - \mathrm{i}\epsilon)$. The dependence on T disappears in the final result as long as it is larger than all times of interest, as required by causality. Note that this prescription is suitable for selecting the asymptotic initial state in spacetimes where the behavior of the modes for free fields is dominated at past infinity by the same kind of oscillatory behavior as in Minkowski space (factors with a power-law or weaker time dependence are allowed); it then selects the adiabatic vacuum for the interacting theory as $t \to -\infty$.

It is also convenient to introduce the functional $W[J_\pm]$ defined by

$$Z[J_\pm] \equiv e^{iW[J_\pm]}, \tag{9.15}$$

which generates expectation values of connected correlation functions rather than matrix elements. For instance, we have,

$$\left.\frac{\delta W[J_\pm]}{\delta J_+}\right|_{J_\pm = J} = \langle 0_-|\phi(x)|0_-\rangle_J \equiv \bar{\phi}[J], \tag{9.16}$$

instead of Eq. (9.7). Functionally differentiating the CTP generating functional $Z[J_\pm]$ with respect to $\mathrm{i}J^+$ and $-\mathrm{i}J^-$ one now generates expectation values of ordered polynomials in the fields, which in general are *path-ordered* (later denoted by $\mathcal{P}$) along the CTP contour rather than time-ordered. For the ϕ^+ fields, the path ordering is the standard time ordering, while for the ϕ^- fields it is anti-time ordering as discussed above; and ϕ^- fields are always to be ordered as if they occurred at a later time than any ϕ^+ field.

$$\left.\frac{\delta Z[J_\pm]}{i\delta J_+(x_1)\cdots(-i)\delta J_-(y_1)\cdots}\right|_{J_\pm = J = 0} = \langle 0_-|\tilde{\mathcal{T}}(\phi(y_1)\cdots)\mathcal{T}(\phi(x_1)\cdots)|0_-\rangle. \tag{9.17}$$

In analogy with the in-out formalism the in-in effective action is defined as the Legendre transform of the new generating functional as,

$$\Gamma[\bar{\phi}_\pm] = W[J_\pm] - J_+\bar{\phi}_+ + J_-\bar{\phi}_-, \tag{9.18}$$

where the external sources are functionals of the fields $\bar{\phi}_+$ and $\bar{\phi}_-$, through the definitions

$$\frac{\delta W[J_\pm]}{\delta J_\pm} \equiv \pm\bar{\phi}_\pm[J_\pm], \tag{9.19}$$

which we assume can be reversed.

From the definitions (9.18) and (9.19) we get the equation for the expectation values $\bar{\phi}_\pm[J_\pm]_J$, i.e.,

$$\frac{\delta\Gamma[\bar{\phi}_\pm]}{\delta\bar{\phi}_\pm} = \mp J_\pm, \tag{9.20}$$

and by taking $J_\pm = 0$ in (9.19) we recover the equation for the vacuum expectation value of the field $\bar{\phi}[0] \equiv \bar{\phi}_\pm[0,0] = \langle 0_-|\phi(x)|0_-\rangle$,

$$\left.\frac{\delta\Gamma[\bar{\phi}_\pm]}{\delta\bar{\phi}_+}\right|_{\bar{\phi}_\pm=\bar{\phi}_\pm[0,0]\equiv\bar{\phi}[0]} = 0. \tag{9.21}$$

This equation does not follow from a simple variational principle in terms of a single field $\bar{\phi}$: in the in-in action we have two fields $\bar{\phi}_+$ and $\bar{\phi}_-$ which are treated independently and only when the sources have been eliminated they become the vacuum expectation values. Note also that $\Gamma[\bar{\phi},\bar{\phi}] = 0$ as a consequence of (9.18) and the fact that $W[J,J] = 0$, which follows from (9.10) and the usual normalization for the states. Eq. (9.21) is a dynamical field equation which admits an initial value formulation: the solution $\bar{\phi}[0] = \bar{\phi}_\pm[0,0]$ is real and causal [181], i.e., the solution at one spacetime point depends only on data on the past of that point.

Effective Action in a Quadratic Case

When the classical action is quadratic it is easy to see that the effective action coincides with it. This is, of course, the case of a free theory but it also includes the case when the action under consideration has been obtained by integration of degrees of freedom involving other fields. Let us consider an action such as

$$S[\phi] = -\frac{1}{2}\int d^n x \phi P \phi, \tag{9.22}$$

where P is an arbitrary invertible differential operator. Then the generating functional, after the Gaussian path integration, becomes

$$e^{iW[J]} = \int \mathcal{D}\phi\, e^{i(S[\phi]+J\phi)} = e^{\frac{i}{2}\int d^n x J P^{-1} J}, \tag{9.23}$$

so that $W[J] = \frac{1}{2}\int d^n x J P^{-1} J$. Here we use a compact notation which is applicable to both the in-out and the in-in cases (to be explained in the next section). From (9.7) we obtain that $P\bar{\phi} = J$ and then from (9.8) we have

$$\Gamma[\bar{\phi}] = \frac{1}{2}\int d^n x J P^{-1} J - \int d^n x J\bar{\phi} = -\frac{1}{2}\int d^n x \bar{\phi} P \bar{\phi} = S[\bar{\phi}], \tag{9.24}$$

which shows that the effective action coincides with the quadratic action (9.22).

9.1.2 The Propagators

For a free field theory, i.e., a theory with a quadratic action, we can compute $W_0[J_\pm]$ from (9.13), which becomes now a Gaussian integration for the two independent fields ϕ_+ and ϕ_-. The corresponding propagators will be determined by the very particular boundary conditions of this problem. In fact, let us assume that the free action for ϕ_+ is

$$S_0[\phi_+] = -\frac{1}{2}\int d^n x \left(\partial_\mu \phi_+ \partial^\mu \phi_+ + (m^2 - i\epsilon)\phi_+^2\right), \tag{9.25}$$

and that we have an analogous action $S_0^*[\phi_-]$ for ϕ_-; then the classical field equations are,

$$(\Box - m^2 \pm i\epsilon)\phi_\pm^0(x) = -J_\pm(x). \tag{9.26}$$

At this stage we can introduce the compact notation,

$$\begin{gathered} \mathcal{S}[\phi_A] \equiv S[\phi_+] - S^*[\phi_-], \\ \phi_A(x) \equiv \begin{pmatrix} \phi_+ \\ \phi_- \end{pmatrix}, \quad J_A(x) \equiv \begin{pmatrix} J_+ \\ -J_- \end{pmatrix}, \end{gathered} \tag{9.27}$$

where capital latin indices $A, B, \ldots$ take the two values + and −, to simplify the mathematical expressions. (Note we used $a, b, \ldots$ in Chapter 7 to denote CTP indices.) The solutions of the classical field equations satisfying the boundary conditions $\phi_\pm^0 \to e^{\pm i\omega t}$, when $t \to -\infty$ with $\omega \geq 0$, and $\phi_+^0(T, \mathbf{x}) = \phi_-^0(T, \mathbf{x})$ in the hypersurface Σ, which we take here at $t = T \to \infty$, can be written as

$$\phi_A^0(x) = -\int d^n y G_{AB}^0(x, y) J_B(y), \tag{9.28}$$

where $G_{AB}^0(x, y)$ is the matrix

$$G_{AB}^0 = \begin{pmatrix} \Delta_F & \Delta^- \\ \Delta^+ & \Delta_D \end{pmatrix}, \tag{9.29}$$

defined with the Feynman, Δ_F, Dyson, Δ_D, and the positive, Δ^+, and negative, Δ^-, Wightman functions:

$$\begin{aligned} \Delta_F(x - y) &= -\int \frac{d^n p}{(2\pi)^n} \frac{e^{ip\cdot(x-y)}}{p^2 + m^2 - i\epsilon}, \\ \Delta_D(x - y) &= \int \frac{d^n p}{(2\pi)^n} \frac{e^{ip\cdot(x-y)}}{p^2 + m^2 + i\epsilon}, \\ \Delta^\pm(x - y) &= -2\pi i \int \frac{d^n p}{(2\pi)^n} e^{ip\cdot(x-y)} \delta(p^2 + m^2)\theta(\pm p^0). \end{aligned} \tag{9.30}$$

These propagators are solutions of the equations

$$A_{BC}^0 \, G_{CD}^0(x, y) = \delta^n(x - y)\delta_{BD}, \tag{9.31}$$

where the operator A^0_{BC} is the diagonal matrix defined by

$$A^0_{BC} = \text{diag}\left[(\Box - m^2 + i\epsilon), -(\Box - m^2 - i\epsilon)\right]. \tag{9.32}$$

Furthermore the Feynman and Dyson Green functions have a mode decomposition

$$\begin{aligned}\Delta_F(x-y) &= \theta(x^0-y^0)\Delta^+(x-y) + \theta(y^0-x^0)\Delta^-(x-y),\\ \Delta_D(x-y) &= \theta(x^0-y^0)\Delta^-(x-y) + \theta(y^0-x^0)\Delta^+(x-y),\end{aligned} \tag{9.33}$$

which reflect the boundary conditions imposed over each classical field solution $\phi^0_A(x)$, because the Green functions $\Delta^{\pm}(x-y) \sim e^{\mp i\omega(x^0-y^0)}$ correspond to positive and negative frequency modes respectively (here $\omega = \sqrt{\mathbf{p}^2+m^2}$). Note that $\partial_t\phi^0_+(T,\mathbf{x}) = \partial_t\phi^0_-(T,\mathbf{x})$ at Σ is also satisfied. With these propagators to guarantee the boundary conditions the Gaussian integration of (9.13) for a free field is,

$$W_0[J_A] = -\frac{1}{2}\int d^nx d^ny J_A(x) G_{AB}(x-y) J_B(y), \tag{9.34}$$

where a term independent of J_A has been discarded to satisfy $W_0[J,J] = 0$, and one can use now (9.17) to generate time ordered and anti-time ordered expectation values of field operators. In particular we have that

$$\begin{aligned}\langle 0_-|\mathcal{T}\phi(x)\phi(y)|0_-\rangle &= i\Delta_F(x-y),\\ \langle 0_-|\tilde{\mathcal{T}}\phi(x)\phi(y)|0_-\rangle &= i\Delta_D(x-y,\\ \langle 0_-|\phi(x)\phi(y)|0_-\rangle &= i\Delta^+(x-y) = i\Delta^-(y-x).\end{aligned} \tag{9.35}$$

For interacting fields one can proceed as usual by writing

$$Z[J_A] \equiv e^{iW[J_A]} = e^{i\int d^nx \mathcal{L}_{\text{int}}\left(\frac{\delta}{i\delta J_A}\right)} e^{iW_0[J_A]}, \tag{9.36}$$

where we have separated the Lagrangian into a free and an interacting part, $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}$. Then one may continue in the usual perturbative fashion; in this chapter we are not going to consider self-interacting theories, and the interactions of the scalar field will be with the gravitational field. In this last case we need to introduce the gravitational $\pm$ fields and the external sources J_A of the generating functional that couple to that field. An example of this is given in Section 9.3.

9.2 The In-In or CTP Effective Action

Following the steps of previous chapters let us now proceed to the evaluation of the effective action $\Gamma[\bar{\phi}_A]$ up to the one loop order, which corresponds to the first-order expansion of $W[J_A]$ in powers of $\hbar$. As usual, see e.g. [678]. If we assume that the action is bounded from above then we can go to Euclidean space and solve (9.13) by the steepest descent method; we keep, however, the Minkowskian

notation. Let us denote by $\phi_+^{(0)}(x)$ and $\phi_-^{(0)}(x)$ the solutions of the classical field equations which may, or may not, include self-interactions,

$$\frac{\delta \mathcal{S}[\phi_B^{(0)}]}{\delta \phi_A^{(0)}(x)} = -J_A(x), \tag{9.37}$$

and let us expand the exponent in (9.13) about these background fields:

$$\begin{aligned}\mathcal{S}[\phi_A] + \int d^n x J_A(x)\phi_A(x) = \mathcal{S}[\phi_A^{(0)}] + \int d^n x J_A(x)\phi_A^{(0)}(x) \\ + \frac{1}{2}\int d^n x d^n y \left[\phi_M(x) - \phi_M^{(0)}(x)\right] \\ A_{MN}(x,y)\left[\phi_N(y) - \phi_N^{(0)}(y)\right] + \cdots ,\end{aligned} \tag{9.38}$$

where

$$\begin{aligned}A_{++}(x,y) &\equiv \left(\frac{\delta^2 S[\phi_+]}{\delta\phi_+(x)\delta\phi_+(y)}\right)_{\phi_+=\phi_+^{(0)}}, \\ A_{--}(x,y) &\equiv -\left(\frac{\delta^2 S^*[\phi_-]}{\delta\phi_-(x)\delta\phi_-(y)}\right)_{\phi_-=\phi_-^{(0)}},\end{aligned} \tag{9.39}$$

and, of course, $A_{+-}(x,y) = A_{-+}(x,y) \equiv 0$. Substituting this into (9.13) the integration is now Gaussian and we can write to this one-loop order,

$$e^{iW[J_A]} \simeq e^{iW^{(0)}[J_A]} \left(\det A_{MN}(x,y)\right)^{-1/2}, \tag{9.40}$$

where $W^{(0)}[J_A] = \mathcal{S}[\phi_A^{(0)}] + \int d^n x J_A \phi_A^{(0)}$. In terms of the propagator G, which is a functional of the background fields $\phi_A^{(0)}(x)$ and takes a 2×2 matrix form, i.e., $G(x,y) = A^{-1}(x,y)$, we can write (9.40) as

$$W[J_A] \simeq W^{(0)}[J_A] - \frac{i}{2}\mathrm{Tr}(\ln G). \tag{9.41}$$

The effective action, which is a functional of $\bar{\phi}_A$, can now be explicitly found to the same order. Using (9.18), (9.19), and the fact that $\bar{\phi}_A$ differs from $\phi_A^{(0)}$ by a term of order $\hbar$ we can show that $W^{(0)}[J_A] \simeq \mathcal{S}[\bar{\phi}_A] + \int d^n x J_A \bar{\phi}_A$, so that finally we have

$$\Gamma[\bar{\phi}_A] \simeq \mathcal{S}[\bar{\phi}_A] - \frac{i}{2}\mathrm{Tr}(\ln G). \tag{9.42}$$

Now the equations for $\bar{\phi}_A$ can be deduced from (9.20) using the explicit functional dependence on the fields given by (9.42). However we should note from (9.21) that in order to get the field equations for the expectation value of $\phi(x)$ we only need the explicit dependence of the effective action on one of the fields, $\bar{\phi}_+$ or $\bar{\phi}_-$. We are interested, however, in the full expression. This expression is formally very similar to the one loop in-out effective action except that now we have dependence on the full matrix G rather than on the Feynman propagator only.

9.2.1 Curved Spacetime. Including Metric Perturbations

As we have seen in previous chapters this formalism can be extended to curved spacetimes without difficulties assuming that the spacetime is globally hyperbolic. This will automatically describe the interaction of the matter fields with the classical gravitational field. The hypersufaces of constant time are now Cauchy hypersurfaces and the *in* state is defined in the Cauchy hypersurface corresponding to the far past. Now the spacetime integrals must be performed with the volume element $d^n x\sqrt{-g}$ where g_{ab} is the spacetime metric. The above expressions (9.28), (9.31) and (9.42) are still valid except that now the Feynman, Dyson and Wightman functions have a different representation to that of (9.30). If the spacetime is asymptotically flat in the *in* region the previous boundary conditions for the fields $\phi^0_\pm(x)$ will also apply; if not, in order to be able to define a physically meaningful *in* vacuum we must assume that we are still able to define positive and negative frequency solutions in the asymptotic region. This is always possible, for instance, if the asymptotic region admit approximate timelike Killing fields. Jordan [181] has shown that for quantum scalar fields in a curved spacetime the field equations are real and causal up to the two-loop order and he has also checked the unitarity of the formalism restricted to vacuum states.

Let us now write (9.42) as a perturbative expansion, which will be useful later. In general the propagator G_{AB} cannot be found exactly and has to be evaluated perturbatively. For instance, we may include metric perturbations $h_{ab}(x)$ to a given background metric and only the exact propagator corresponding to the background may be known. Thus we write

$$A_{MN} = A^0_{MN} + V^{(1)}_{MN} + V^{(2)}_{MN} + \cdots \tag{9.43}$$

where A^0_{MN} is the unperturbed (diagonal) operator whose propagator, G^0_{AB}, is known,

$$A^0_{BC} G^0_{CD} = \delta_{BD}, \tag{9.44}$$

and the diagonal operators $V^{(1)}_{BC} + V^{(2)}_{BC} + \cdots$ contain the perturbative terms, linear and quadratic in the metric perturbations.

We can write

$$\begin{aligned} G_{AB} &= G^0_{AB} - G^0_{AC}(V^{(1)}_{CD} + V^{(2)}_{CD} \cdots) G_{DB} \\ &= G^0_{AB} - G^0_{AC} V^{(1)}_{CD} G^0_{DB} - G^0_{AC} V^{(2)}_{CD} G^0_{DB} \\ &\quad + G^0_{AC} V^{(1)}_{CD} G^0_{DE} V^{(1)}_{EF} G^0_{FB} + \cdots , \end{aligned} \tag{9.45}$$

where the products are operator products. Expanding the logarithmic term in (9.45) and using that $A^0_{++}G^0_{+-} = 0$, we finally get the in-in or CTP effective action,

$$\begin{aligned}\Gamma[\bar\phi_\pm] \simeq S[\bar\phi_+] - S[\bar\phi_-] - \frac{i}{2}\mathrm{Tr}(\ln G^0_{AB}) \\ + \frac{i}{2}\mathrm{Tr}\Big[\ V^{(1)}_+ G^0_{++} - V^{(1)}_- G^0_{--} + V^{(2)}_+ G^0_{++} - V^{(2)}_- G^0_{--} \\ - \frac{1}{2}V^{(1)}_+ G^0_{++} V^{(1)}_+ G^0_{++} - \frac{1}{2}V^{(1)}_- G^0_{--} V^{(1)}_- G^0_{--} \\ + V^{(1)}_+ G^0_{+-} V^{(1)}_- G^0_{-+}\ \Big],\end{aligned} \tag{9.46}$$

where we have defined $V^{(i)}_+ \equiv V^{(i)}_{++}$ and $V^{(i)}_- \equiv -V^{(i)}_{--}$ following (9.39). Note that if it were not for the last term which involves the propagator G^0_{+-} this expression for the in-in effective action would agree with the in-out effective action which involves only one field $\phi(x)$; see for instance [110]. Therefore the term containing the propagator G^0_{+-} is the only new term that contributes to the field equation for $\bar\phi_+(x)$. It can be seen [437] that the effect of the last term in (9.46) is to make the field equation for $\bar\phi(x)$ causal: if one takes the derivative of the in-in effective action with respect to $\bar\phi_+(x)$ and puts $\bar\phi_+ = \bar\phi_- = \bar\phi$, the resulting field equation is causal.

9.3 Gravity and Matter Interaction in the CTP Formalism

Here we consider the interaction of the gravitational field with some matter fields ϕ. We are interested in the in-in or CTP effective action for the gravitational field, obtained by integrating out the matter fields, and in the evaluation of the two-point metric correlations. We want to include loop correction from the matter fields but no graviton loops, which can be consistently done in the large N expansion for N matter fields. That is, taking a mean field approach to quantum gravity, we will quantize the metric perturbations around a semiclassical background including the effects of matter loops.

9.3.1 Generating Functional for Metric Correlations and Large N

Although perturbative quantum gravity is power-counting nonrenormalizable, it can be consistently studied as an effective field theory (EFT) describing quantum gravitational phenomena with characteristic length scales well above the Planck length $\ell_{\rm P} = \sqrt{G_{\rm N}}$, see [192, 679, 680, 193]; remember that we use units in which $\hbar = c = 1$, and $G_{\rm N}$ is Newton's constant. In effective field theories the effect of the ultraviolet sector on the dynamics of the low energy theory are encoded at the level of the action through an expansion of local terms with an increasing

number of derivatives. In the gravitational case, one needs to introduce local counterterms in the bare action for the metric and the matter fields with an arbitrary number of covariant derivatives and powers of the curvature, suppressed by the corresponding negative power of the Planck mass, or, positive power of the Planck length. The key point is that for phenomena with a characteristic length scale L only a finite number of counterterms needs to be considered to achieve a certain precision, roughly given by $(\ell_{\rm P}/L)^{2n}$ with n being the total number of pairs of derivatives and/or powers of the curvature.

More specifically, using the effective action for the metric perturbations we want to calculate the two-point function of these perturbations around a fixed background g^0_{ab}, so that the total metric is

$$g_{ab} = g^0_{ab} + \kappa h_{ab}, \tag{9.47}$$

coupled to N scalar fields, where $\kappa^2 = 16\pi G_{\rm N}$. Note that here to keep the perturbative analysis more transparent we explicitly write the gravitational coupling parameter κ in the metric perturbation, which has now dimensions of inverse length.

Therefore, we will consider the following CTP generating functional containing external sources J_{ab} for the metric perturbations h_{ab} (since we are not interested in the matter field correlations, we do not need sources for the matter fields ϕ):

$$\begin{aligned} Z[J^\pm] = \int & \delta\big[\phi^+(T) - \phi^-(T)\big]\delta\big[h^+(T) - h^-(T)\big] \\ & \times \delta\big[h^\pm(t_0) - h^\pm_0\big]\delta\big[\phi^\pm(t_0) - \phi^\pm_0\big] \times \langle \phi^+_0, h^+_0|\hat\rho|\phi^-_0, h^-_0\rangle \\ & \times {\rm e}^{{\rm i}(S[h^+,\phi^+]+J^+h^+) - {\rm i}(S[h^-,\phi^-]+J^-h^-)} \mathcal{D}\phi^\pm \mathcal{D}h^\pm \mathcal{D}\phi^\pm_0 \mathcal{D}h^\pm_0, \end{aligned} \tag{9.48}$$

where

$$S[h,\phi] = S_{\rm g}[h] + S_{\rm m}[h,\phi], \tag{9.49}$$

is the total action for metric perturbations and matter fields, $S_{\rm g}[h]$ is the classical gravitational action and $S_{\rm m}[h,\phi]$ is the classical action for the matter fields. The $\phi^\pm_0$ and $h^\pm_0$ correspond to the field configurations at the initial time t_0, and $\hat\rho$ is the density matrix specifying the initial state; for an initially pure state $\hat\rho$ reduces to $|{\rm in}\rangle\langle{\rm in}|$.

Notice that for simplicity we generally adopt in this section a shorthand notation where we omit tensor indices, as well as spacetime integrals and the spatial dependence of the fields. Our calculation will be performed in two steps: first, we explicitly integrate over the matter fields $\phi^\pm$, and next we perform the remaining path integrals for the metric perturbations $h^\pm$. It is easier to proceed if one assumes that the initial density matrix factorizes so that $\langle \phi^+_0, h^+_0|\hat\rho|\phi^-_0, h^-_0\rangle = \langle \phi^+_0|\hat\rho_\phi|\phi^-_0\rangle\langle h^+_0|\hat\rho_h|h^-_0\rangle$.

We are actually interested in states of the interacting theory which involve correlations between the matter fields and the metric perturbations, for which the density matrix does not factorize. However, if one employs the iϵ prescription

method of Section 9.1 to select the adiabatic vacuum of the interacting theory, one can consider a factorized initial state involving the product of the free vacua for the matter fields and the metric perturbations both treated as free fields, integrate out the matter fields more easily and only at the final stage deform the CTP integration contour for the vertices appearing in the Feynman diagrams associated with the perturbative calculation of the path integral of the metric perturbations. In doing so we will neglect contributions from graviton loops. As we have emphasized earlier this can be formally implemented in a natural way by considering a large N expansion for N matter fields interacting with the gravitational field; see e.g. [194, 190]. For instance, the lowest-order contributions to the connected two-point function of the metric perturbations are of order $1/N$, whereas any contribution including graviton loops is suppressed by higher powers of $1/N$. A detailed description of this expansion is given in Section 9.4 using a simplified model.

After integrating out the matter fields, which amounts to a one-loop calculation, one is left with the following expression for the CTP generating functional (dropping an infinite proportionality constant, which anyway will cancel in the calculation of the correlation function):

$$Z[J^{\pm}] = \int \delta\big[h^{+}(T) - h^{-}(T)\big]\,\delta\big[h^{\pm}(t_0) - h_0^{\pm}\big]\,\langle h_0^{+}|\hat{\rho}_h|h_0^{-}\rangle \times \mathrm{e}^{\mathrm{i}(S_{\mathrm{g}}[h^{+}]+J^{+}h^{+})-\mathrm{i}(S_{\mathrm{g}}[h^{-}]+J^{-}h^{-})+\mathrm{i}S_{\mathrm{IF}}[h^{+},h^{-}]}\mathcal{D}h^{\pm}\mathcal{D}h_0^{\pm}, \tag{9.50}$$

where $S_{\mathrm{g}}[h]$, given in (9.49), is the gravitational part of the original bare action $S[h,\phi]$, i.e., all the terms depending only on the metric, and S_{IF}, the *influence action*, is the result of the functional integration over the matter fields, truncated at second order in h.

In the course of "integrating out" the matter fields ϕ, the ultraviolet (UV) divergences associated with matter loops which appear in S_{IF} must be regularized and appropriate counterterms must be included in the bare gravitational action S_{g}. For each additional term in the $1/N$ expansion one would need to include counterterms with higher and higher powers of the curvature and number of covariant derivatives in $S_{\mathrm{g}}[h]$. Moreover, in that case one would also need to include in the full bare action $S[h,\phi]$ counterterms coupling h and ϕ, and with a higher and higher number of covariant derivatives.

To leading order in $1/N$, it is only necessary to introduce counterterms in $S_{\mathrm{g}}[h]$ at most quadratic in the curvature, so that the bare gravitational action is given by

$$\begin{aligned} S_{\mathrm{g}}[g] = \frac{N}{\bar{\kappa}^2}\Bigg[& \int (R - 2\Lambda)\,\sqrt{-g}d^n x \\ & + a_1\bar{\kappa}^2 \int \left(R^{abcd}R_{abcd} - R^{ab}R_{ab}\right)\sqrt{-g}d^n x \\ & + a_2\bar{\kappa}^2 \int R^2\sqrt{-g}d^n x\Bigg] \end{aligned} \tag{9.51}$$

where a_i are dimensionless bare parameters and we included terms quadratic in the Riemann tensor with all possible contractions. Although one might expect three independent bare parameters associated to the square of the Ricci scalar, the square of the Ricci tensor and the square of the Riemann tensor, the Gauss Bonnet theorem ensures that only two of these parameters are independent in four dimensions [47]; compare with Eq. (2.67) and the discussion that follows. We write this action in n dimensions in order to use dimensional regularization and we have introduced a rescaled gravitational coupling constant

$$\bar{\kappa}^2 = N\kappa^2, \tag{9.52}$$

which helps to organize conveniently the calculation when using the $1/N$ expansion. Thus we now have a fundamental length-scale $\bar{\kappa}$, characterizing the cut-off scale of the low-energy gravitational EFT, which is much larger than the effective Planck length $\ell_{\rm P} = \sqrt{G_{\rm N}}$, the effective low energy length-scale determined by experiment. Note that this can happen naturally in braneworld models with warped extra dimensions [681] or models with a large number of fields [682]. In fact, in some cases both possibilities can be understood as equivalent descriptions within the framework of the AdS-CFT correspondence [683].

9.3.2 One-loop Effective Action for Metric Perturbations

The expression of the generating functional after integrating out the matter fields given in (9.50) suggests to define the following action for the metric perturbations,

$$S_{\rm eff}[h^\pm] = S_{\rm g}[h^+] - S_{\rm g}[h^-] + S_{\rm IF}[h^\pm]. \tag{9.53}$$

The ultraviolet divergences which appear in $S_{\rm IF}$ are regulated by appropriate counterterms in $S_{\rm g}$ and that sum becomes finite after the regulator is taken to its physical value. Note that the influence action $S_{\rm IF}$ provides an effective propagator for the metric perturbation h_{ab} which includes the interaction with ϕ, and $S_{\rm eff}$ is the *CTP effective action*. The fact that $S_{\rm eff}$, which can be read directly from the generating functional (9.50), is the effective action for the metric follows directly from the fact that $S_{\rm eff}$ is quadratic in the metric perturbations; see the observation at the end of Section 9.1.1.

Thus, from now on, we only consider $S_{\rm eff}[h^\pm]$ through quadratic order in the metric perturbations, since higher-order terms do not contribute to the generating functional and the connected correlation functions to leading order in $1/N$, as mentioned above. To analyze more closely the different contributions to $S_{\rm eff}$, let us expand it in powers of the metric perturbation h,

$$\begin{aligned} S_{\rm eff}[h^\pm] = {} & S_{\rm g}^{(0)}[h^+] + S_{\rm g}^{(1)}[h^+] + S_{\rm g}^{(2)}[h^+] - S_{\rm g}^{(0)}[h^-] \\ & - S_{\rm g}^{(1)}[h^-] - S_{\rm g}^{(2)}[h^-] + S_{\rm IF}^{(1)}[h^\pm] + S_{\rm IF}^{(2)}[h^\pm], \end{aligned} \tag{9.54}$$

where the superscript denotes how many powers of h appear.

The zeroth-order terms in Eq. (9.54) can be factored out of the path integral (9.50) and give a factor independent of the external source J which does not contribute to the correlation functions, obtained by functionally differentiating with respect to J, so we may disregard them.

The sum of all the first-order terms corresponds to integrating h^+ and h^- with the functional derivative of the CTP effective action for the matter fields in a semiclassical background with the metric of the two branches equal to g^0_{ab} after differentiation. When equated to zero, such a functional derivative corresponds to the semiclassical Einstein equation which governs the quantum backreaction of the matter fields on the mean background geometry [49, 128],

$$G_{ab}\,[g^0] - \Lambda g^0_{ab} = \frac{\kappa^2}{2}\langle \hat{T}^{\mathrm{R}}_{ab}\,[g^0; a_i]\rangle, \tag{9.55}$$

where the renormalized stress-energy tensor expectation value comes from the functional derivative of S_{IF} with respect to h_{ab} and here we also assume that it includes any finite contributions from the counterterms in S_{g} other than the Einstein tensor or the cosmological constant, which is indicated by its dependence on the parameters a_i. We will always consider a background g^0_{ab} which is a solution of the semiclassical equation (9.55). Therefore, the sum of the first-order terms on the right-hand side of Eq. (9.54) will vanish.

We are, thus, left with a purely quadratic expression for S_{eff}. It is now useful to explicitly expand this action in powers of κ,

$$S_{\mathrm{eff}}[h^\pm] = S^{(2)}_{\mathrm{g}}[h^+] - S^{(2)}_{\mathrm{g}}[h^-] + S^{(2)}_{\mathrm{IF}}[h^\pm] = S_0[h^\pm] + \kappa^2 S_2[h^\pm], \tag{9.56}$$

where S_0 is the free part of the gravitational action (quadratic in $h^\pm$ and of order κ^0) and S_2 are all the remaining terms of order κ^2 (quadratic in $h^\pm$), including the nonlocal terms in $S_{\mathrm{IF}}[h^\pm]$. Note that S_2 also includes any counterterms proportional to the terms in the Einstein–Hilbert action which can appear for massive fields or when using Pauli–Villars regularization.

Given an exponent quadratic in the perturbations h^+ and h^-, one can easily perform the functional integration in Eq. (9.50). In order to do so, it is convenient to adopt a matrix formulation with $h_A = (h^+, h^-), J_A = (J^+, -J^-)$, as in (9.27), with

$$A_{MN} = \begin{pmatrix} A_{++} & A_{+-} \\ A_{-+} & A_{--} \end{pmatrix},$$

and sum over repeated capital indices regardless of their position. Notice that general covariance of the terms $J^A h_A$ in the action coupling the metric perturbation to the external source requires that $J^A(x)$ transform as a density under coordinate transformations.

Then the effective action can be written as

$$S_{\mathrm{eff}}[h^\pm] = \frac{1}{2} h_M A_{MN} h_N = \frac{1}{2} h_M (A^0_{MN} + \kappa^2 V_{MN}) h_N, \tag{9.57}$$

where

$$A^0_{MN} = \begin{pmatrix} \frac{\delta^2 S_0[h^+]}{\delta h^+ \delta h^+} & 0 \\ 0 & -\frac{\delta^2 S_0[h^-]}{\delta h^- \delta h^-} \end{pmatrix}. \tag{9.58}$$

is the free theory differential operator acting on h, and

$$V_{MN} = \begin{pmatrix} \frac{\delta^2 S_2[h^\pm]}{\delta h^+ \delta h^+} & \frac{\delta^2 S_2[h^\pm]}{\delta h^+ \delta h^-} \\ \frac{\delta^2 S_2[h^\pm]}{\delta h^- \delta h^+} & \frac{\delta^2 S_2[h^\pm]}{\delta h^- \delta h^-} \end{pmatrix} \tag{9.59}$$

is the part arising from the interaction with the matter fields ϕ. Notice that here we are still using the simplified notation of omitting tensor indices and spacetime integrals.

Thus the effective action (9.56) can also be written as

$$S_{\rm eff}[h^\pm] = S_0[h^+] - S_0[h^-] + S_{\rm int}[h^\pm], \tag{9.60}$$

where we have denoted $S_{\rm int}[h^\pm] = \kappa^2 S_2[h^\pm]$, and the generating functional (9.50) can be rewritten as

$$Z[J^\pm] = e^{iS_{\rm int}[\mp i\delta/\delta J^\pm]}\, Z_0[J^\pm], \tag{9.61}$$

where $Z_0[J^\pm]$ is the generating functional associated with the free part; see also (9.36). This is a useful way of organizing the calculation which allows (by expanding the exponential) a simple derivation of the Feynman rules for general interacting theories and is directly applicable to both the in-out and the in-in or CTP formalisms.

To summarize, to compute connected correlation functions to leading-order in $1/N$, we only need to consider terms in $S_{\rm eff}[h^\pm]$ which are linear and quadratic in h since terms with higher powers of h give rise to contributions of higher order in $1/N$ [190]; there is also an irrelevant term which does not depend on the perturbation h. If we demand that the background metric g^0_{ab} is a solution of the semiclassical Einstein equation then the linear terms cancel out and we are left with a purely quadratic expression.

9.3.3 Two-Point Metric Correlations at One-Loop

The two-point functions, which are true expectation values with respect to the *in* state, are obtained by functionally differentiating the generating functional $Z[J^\pm]$ with respect to the sources,

$$\langle {\rm in}|\mathcal{P} h_A(x) h_B(x')|{\rm in}\rangle = \left[Z^{-1}[J^\pm] \frac{\delta^2 Z[J^\pm]}{i\delta J_A(x)\, i\delta J_B(x')} \right]_{J=0} = iG_{AB}(x,x'), \tag{9.62}$$

where, as before, the indices A and B can take the values $\pm$, and $|{\rm in}\rangle$ is some initial state. The path ordering denoted by $\mathcal{P}$ is a generalization of the time

ordering $\mathcal{T}$ and the anti-time ordering $\tilde{\mathcal{T}}$. Particularizing the four possible values of the index pair AB, we have

$$\begin{aligned} G_{++}(x,x') &= -i\langle \mathrm{in}|\mathcal{T}h(x)h(x')|\mathrm{in}\rangle, \qquad G_{-+}(x,x') = -i\langle \mathrm{in}|h(x)h(x')|\mathrm{in}\rangle, \\ G_{--}(x,x') &= -i\langle \mathrm{in}|\tilde{\mathcal{T}}h(x)h(x')|\mathrm{in}\rangle, \qquad G_{+-}(x,x') = -i\langle \mathrm{in}|h(x')h(x)|\mathrm{in}\rangle. \end{aligned} \tag{9.63}$$

From Eq. (9.60) for the effective action we can obtain the correlation functions up to order κ^2 (and leading order in $1/N$) by expanding the exponential in equation (9.61) up to that order; which corresponds, for free fields, to one-loop order in the matter fields:

$$Z[J^{\pm}] = (1 + \kappa^2 S_2\,[\mp i\,\delta/\delta J^{\pm}] + \mathcal{O}(\kappa^4))\, Z_0[J^{\pm}]. \tag{9.64}$$

The generating functional for the free part is given by

$$Z_0[J^{\pm}] = \exp\left[-\frac{i}{2}\int\int J^A(x) G^0_{AB}(x,x') J^B(x') d^4x\, d^4x'\right], \tag{9.65}$$

where G^0_{AB} corresponds to the free part of the two-point correlation functions (9.62). Here and in the rest of this section the spacetime integrals are over the background metric, we recall that $J^A(x)$ transforms as a density under coordinate transformtions.

Substituting equation (9.65) into equation (9.64) and functionally differentiating with respect to the sources $J^{\pm}$, we obtain the following expression for the two-point functions (9.62) up to order κ^2:

$$\begin{aligned} G_{AB}(x,x') = {} & G^0_{AB}(x,x') \\ & - \kappa^2 \iint G^0_{AM}(x,y) V_{MN}(y,y') G^0_{NB}(y',x') d^4y' d^4y + \mathcal{O}(\kappa^4), \end{aligned} \tag{9.66}$$

where V_{MN} is the amputated one-loop contribution from S_2, given by (9.59).

The CTP propagator of the free theory, G^0_{AB} satisfies the equation

$$\int A^0_{BM}(x,y) G^0_{MC}(y,x') d^4y = \delta^4(x-x')\delta_{BC}, \tag{9.67}$$

with the tree-level kinetic operator A^0_{BM} defined in (9.58). Note, however, that equation (9.67) does not specify completely the free propagator G^0_{AB} and one needs to provide, in addition, appropriate boundary conditions which determine the operator ordering of the different components as well as the quantum state of the field.

Eq. (9.66) involves time integrals between the initial time t_0 and the final time T with a suitable iϵ prescription that selects the true asymptotic vacuum state

of the interacting theory. As mentioned above, this consists in sending the initial time t_0 to $-\infty$ along a slightly imaginary direction:

$$t_0^{\pm} \to -\infty(1 \mp i\epsilon) \qquad \text{with } \epsilon > 0. \tag{9.68}$$

Whether we have to take t_0^+ or t_0^- depends on the corresponding index (M or N) in Eq. (9.66), *e.g.*, if $M = +$ the integral over y runs from t_0^+ to T.

Since, as seen in Eq. (9.59), V_{MN} is the second variation of the $S_2[h^{\pm}]$ part of the effective action (9.56), the integrand in (9.66) is given effectively by $S_2[h^{\pm}]$ with fields h replaced by the zero-order propagators G^0. This double integral can be calculated exactly in the case of massless conformally coupled matter fields; this will be done in Section 16.3.

9.4 Large N Expansion: A Toy Model

The large N expansion has been successfully used in quantum chromodynamics to compute some non-perturbative results. This expansion re-sums and rearranges Feynman perturbative series including self-energies. For gravity interacting with N matter fields it shows that graviton loops are of higher order than matter loops. To illustrate the large N expansion let us, first, consider the following toy model of gravity, which we will simplify as a dimensionless scalar field h, which mimics metric perturbations on a flat background, interacting with a single scalar field ϕ described by the action

$$\begin{aligned} S = &-\frac{1}{\kappa^2}\int d^4x\,(\partial_a h\partial^a h + h\,\partial_a h\partial^a h + \dots) \\ &-\frac{1}{2}\int d^4x\,(\partial_a\phi\partial^a\phi + m^2\phi^2) + \int d^4x\,(h\,\partial_a\phi\partial^a\phi + \dots)\,, \end{aligned} \tag{9.69}$$

where κ is the gravitational coupling introduced previously and we have assumed that the interaction is linear in the scalar gravitational field h and quadratic in the matter field ϕ to simulate in a simplified way the coupling of the metric with the stress-energy tensor of the matter fields. We include a self coupling graviton term of $\mathcal{O}(h^3)$ which appears in perturbative gravity beyond the linear approximation.

We may now compute the graviton dressed propagator perturbatively as the following series of Feynman diagrams of Figure 9.1. The first diagram is just the free graviton propagator which is of $\mathcal{O}(\kappa^2)$, as one can see from the kinetic term for the graviton in equation (9.69). The next diagram is one loop of matter with two external legs which are the graviton propagators. This diagram has two vertices with one graviton propagator and two matter field propagators. Since the vertices and the matter propagators contribute with 1 and each graviton propagator contributes with a κ^2 this diagram is of order $\mathcal{O}(\kappa^4)$. The next diagram contains two loops of matter and three gravitons, and consequently it is of order $\mathcal{O}(\kappa^6)$. There will also be terms with one graviton loop and two

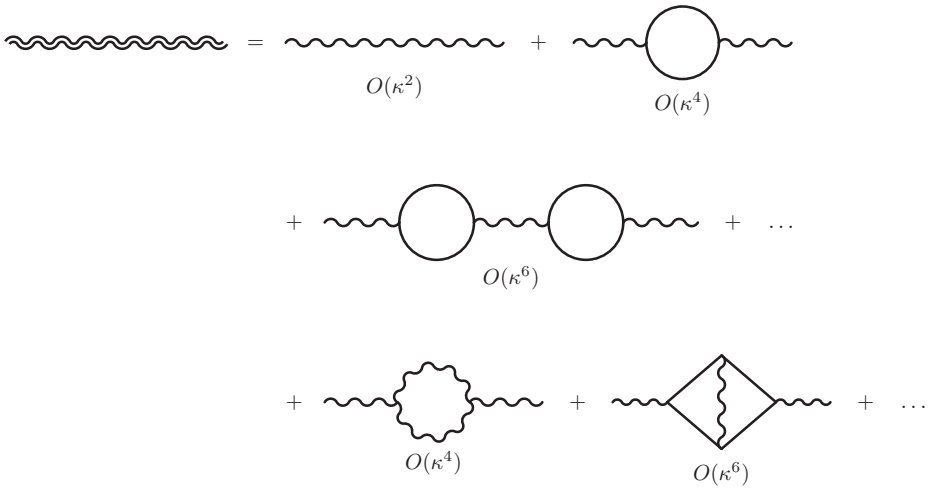


Figure 9.1 Series of Feynman diagrams expanded in powers of κ for the dressed propagator of the graviton interacting with a single matter field according to the toy model described by the action (9.69).

graviton propagators as external legs, with three graviton propagators at the two vertices due to the $\mathcal{O}(h^3)$ term in the action (9.69). Since there are four graviton propagators which carry a κ^8 but two vertices which have κ^{-4} this diagram is of order $\mathcal{O}(\kappa^4)$, like the term with one matter loop. Thus, in this perturbative expansion a graviton loop and a matter loop both contribute at the same order to the dressed graviton propagator.

Let us consider the large N expansion. We assume that the gravitational field is coupled with a large number of identical fields $\phi_j, j = 1, \ldots, N$ which couple only with h. We rescale the gravitational coupling in such a way that $\bar{\kappa}^2 = N\kappa^2$ is finite even when N goes to infinity. The action is:

$$S = -\frac{N}{\bar{\kappa}^2}\int d^4x\,(\partial_a h\partial^a h + h\,\partial_a h\partial^a h + \dots) - \frac{1}{2}\sum_j^N \int d^4x\,(\partial_a\phi_j\partial^a\phi_j + m^2\phi^2) + \sum_j^N \int d^4x\,(h\,\partial_a\phi_j\partial^a\phi_j + \dots)\,. \quad (9.70)$$

Now an expansion in powers of $1/N$ of the graviton dressed propagator is given by the following series of Feynman diagrams of Figure 9.2. The first diagram is the free graviton propagator which is now of order $\mathcal{O}(\bar{\kappa}^2/N)$ the following diagrams are N identical Feynman diagrams with one loop of matter and two graviton propagators as external legs, each diagram due to the two graviton propagators is of order $\mathcal{O}(\bar{\kappa}^4/N^2)$ but since there are N of them the sum can be represented by a single diagram with a loop of matter of weight N, therefore this diagram is of order $\mathcal{O}(\bar{\kappa}^4/N)$. Then there are the diagrams with two loops of matter and three graviton propagators; as before we can assign a weight of N to each

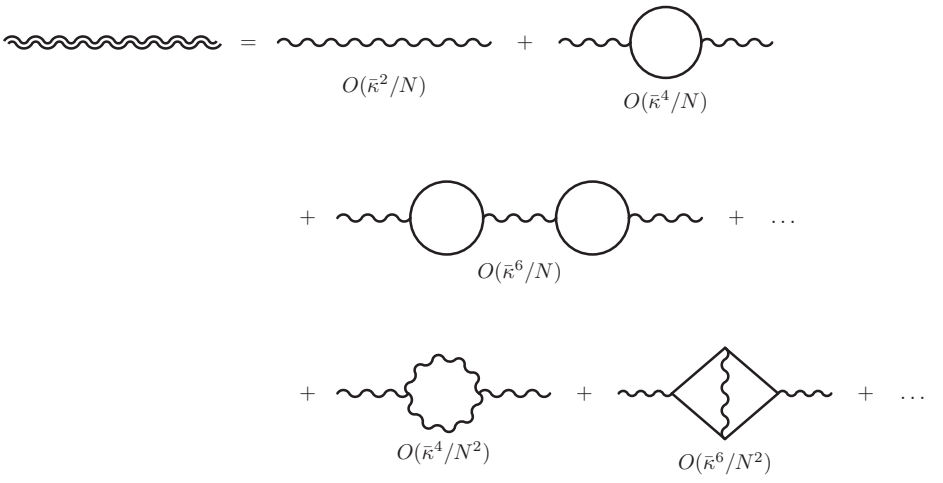


Figure 9.2 Series of Feynman diagrams expanded in powers of $1/N$ for the dressed propagator of the graviton interacting with N matter fields according to the toy model described by the action (9.70).

loop and taking into account the three graviton propagators this diagram is of order $\mathcal{O}(\bar{\kappa}^6/N)$, and so on. This means that to order $1/N$ the dressed graviton propagator contains all the perturbative series in powers of $\bar{\kappa}^2$ of the matter loops.

Next, there is a diagram with one graviton loop and two graviton legs: it contains four graviton propagators and two vertices, the propagators contribute as $(\bar{\kappa}^2/N)^4$ and the vertices as $(N/\bar{\kappa}^2)^2$, thus this diagram is of $\mathcal{O}(\bar{\kappa}^4/N^2)$. Therefore graviton loops contribute to higher order in the $1/N$ expansion than matter loops. Similarly there are N diagrams with one loop of matter with an internal graviton propagator and two external graviton legs. Thus we have three graviton propagators and since there are N of them, their sum is of order $\mathcal{O}(\bar{\kappa}^6/N^2)$.

To summarize, we have that when $N \to \infty$ there are no graviton propagators and gravity is classical yet the matter fields are quantized: this is semiclassical gravity as was first described by Hartle and Horowitz [194]. When we go to the next to leading order, $1/N$, the graviton propagator includes all matter loop contributions, but no contributions from graviton loops or internal graviton propagators in matter loops.

Notice that these conclusion do not change if we have self-interacting scalar fields of the type $\lambda \sum_j h\phi_j^4$; now there will be internal loops with four matter line vertices, but with weight N as the second diagram of Figure 9.2. On the other hand, if one considers self-interacting fields of the type $\lambda \sum_i \sum_j h(\phi_i)^2(\phi_j)^2$ then it is convenient to rescale the self-coupling parameter $\bar{\lambda} = N\lambda$ and the weight of the corresponding diagram with internal matter loops is also N.

10

The Einstein–Langevin Equation

Whereas semiclassical gravity is based on the semiclassical Einstein equation with sources given by the expectation value of the stress-energy tensor of quantum fields, stochastic semiclassical gravity is based on the Einstein–Langevin equation, which has in addition sources due to the noise kernel. The noise kernel is the vacuum expectation value of the stress-energy bitensor which describes the fluctuations of quantum matter fields in curved spacetimes. A new improved criterion for the validity of semiclassical gravity may also be formulated from the viewpoint of this theory. In this chapter we describe the fundamentals of this theory via two approaches: the axiomatic and the functional. The axiomatic approach is useful to see the structure of the theory from the framework of semiclassical gravity, showing the link from the mean value of the stress-energy tensor to the correlation functions. The functional approach uses the Feynman–Vernon influence functional and the Schwinger-Keldysh closed time path (CTP) effective action methods.

In Sections 10.1 and 10.2 we formulate *stochastic semiclassical gravity*, or *stochastic gravity* for short, in an axiomatic way. It is introduced as an extension of semiclassical gravity motivated by the search of self-consistent equations which describe the backreaction of the quantum stress-energy fluctuations on the gravitational field. We then discuss the equivalence between the stochastic correlation functions for the metric perturbations and the quantum correlation functions in the large N expansion, i.e. the quantum theory describing the interaction of the gravitational field with N arbitrary free fields expanded in powers of $1/N$, and illustrate the equivalence with a simple toy model. Based on the stochastic formulation a test is proposed for the validity of the semiclassical theory in Section 10.3. In the second part of the chapter, in Section 10.4, semiclassical gravity as well as stochastic gravity are formulated using a functional approach based on the open quantum system paradigm, discussed in the earlier chapters, in which the system of interest (the gravitational field) interacts with an environment (the matter fields). The functional approach brings

familiar quantum field theory techniques into play and provides powerful tools for explicit perturbative calculation. Finally, using this formalism an explicit form of the Einsein–Langevin equation is derived in Section 10.5.

In subsequent chapters we describe some applications of stochastic gravity. In Chapter 11 we consider metric perturbations in a Minkowski spacetime, compute the two-point correlation functions of these perturbations and prove that Minkowski spacetime is a stable solution of semiclassical gravity. In Chapter 12 we discuss the backreaction problem in cosmology when the gravitational field couples to a quantum conformal matter field, and derive the Einstein–Langevin equations describing the metric fluctuations on the cosmological background. In Chapter 13, we discuss structure formation, stochastic gravity is an alternative framework to study the generation of primordial inhomogeneities in inflationary models, which can easily incorporate effects that go beyond the linear perturbations of the inflaton field. In Chapter 14 using the Einstein–Langevin equation we discuss the backreaction of Hawking radiation and the behavior of metric fluctuations for both the quasi-equilibrium condition of a black hole in a box and the fully nonequilibrium condition of an evaporating black hole spacetime.

10.1 Semiclassical Gravity: Axiomatic Approach

Semiclassical gravity describes the interaction of a classical gravitational field with quantum matter fields. This theory can be formally derived as the leading $1/N$ approximation of quantum gravity interacting with N independent and identical free quantum fields which interact with gravity only [194, 684, 685, 686]. By keeping the value of $NG_{\rm N}$ finite, where $G_{\rm N}$ is Newton's gravitational constant, one arrives at a theory in which formally the gravitational field can be treated as a c-number field, i.e., quantized at tree level, while matter fields are fully quantized. The semiclassical theory may be summarized as follows.

Let $(\mathcal{M}, g_{ab})$ be a globally hyperbolic four-dimensional spacetime manifold $\mathcal{M}$ with metric g_{ab} and consider a real scalar quantum field ϕ of mass m propagating on that manifold; we just assume a scalar field for simplicity. The classical action $S_{\rm m}$ for this matter field is given by the functional

$$S_{\rm m}[g,\phi] = -\frac{1}{2}\int d^4x\sqrt{-g}\left[g^{ab}\nabla_a\phi\nabla_b\phi + (m^2+\xi R)\,\phi^2\right], \tag{10.1}$$

where ∇_a is the covariant derivative associated to the metric g_{ab}, ξ is a dimensionless coupling parameter between the field and the scalar curvature of the underlying spacetime R, and $g = \det g_{ab}$.

The field may be quantized in the manifold using the standard canonical quantization formalism [47, 48, 49]. The field operator in the Heisenberg picture $\hat{\phi}$ is an operator-valued distribution solution of the Klein–Gordon equation,

$$(\Box - m^2 - \xi R)\hat{\phi} = 0, \tag{10.2}$$

where $\Box = \nabla_a \nabla^a$. We may write the field operator as $\hat{\phi}[g;x)$ to indicate that it is a functional of the metric g_{ab} and a function of the spacetime point x. This notation will be used also for other operators and tensors.

The classical stress-energy tensor is obtained by functional derivation of this action in the usual way $T^{ab}(x) = (2/\sqrt{-g})\delta S_{\rm m}/\delta g_{ab}$, leading to

$$T^{ab}[g,\phi] = \nabla^a\phi\nabla^b\phi - \frac{1}{2}g^{ab}\left(\nabla^c\phi\nabla_c\phi + m^2\phi^2\right) + \xi\left(g^{ab}\Box - \nabla^a\nabla^b + G^{ab}\right)\phi^2, \tag{10.3}$$

where G_{ab} is the Einstein tensor. With the notation $T^{ab}[g,\phi]$ we explicitly indicate that the stress-energy tensor is a functional of the metric g_{ab} and the field ϕ. This is the same as the stress-energy tensor (2.59) where $(1-\xi)\xi_n$ has been replaced by ξ, so that now $\xi = 0$ and $\xi = \xi_n$ correspond to minimal and conformal couplings, respectively.

The next step is to define a stress-energy tensor operator $\hat{T}^{ab}[g;x)$. Naively one would replace the classical field $\phi[g;x)$ in the above functional by the quantum operator $\hat{\phi}[g;x)$, but this procedure involves taking the product of two distributions at the same spacetime point. This is ill-defined and a regularization procedure is needed. There are several regularization methods which one may use, one is the point-splitting or point-separation regularization method introduced by Christensen [100, 101] in which one takes a point y in a neighborhood of the point x and then uses as the regulator the vector tangent at the point x of the geodesic joining x and y; this method is discussed in a previous chapter.

Another well known method is dimensional regularization in which one works in n dimensions, where n is not necessarily an integer, and then uses as the regulator the parameter $\epsilon = n-4$; this method is implicitly used in this Section. The regularized stress-energy operator using the Weyl ordering prescription, i.e., symmetrical ordering, can be written as

$$\hat{T}^{ab}[g] = \frac{1}{2}\{\nabla^a\hat{\phi}[g]\,,\,\nabla^b\hat{\phi}[g]\} + \mathcal{D}^{ab}[g]\,\hat{\phi}^2[g], \tag{10.4}$$

where $\mathcal{D}^{ab}[g]$ is the differential operator:

$$\mathcal{D}^{ab} \equiv (\xi - 1/4)\,g^{ab}\Box + \xi\left(R^{ab} - \nabla^a\nabla^b\right). \tag{10.5}$$

Note that if dimensional regularization is used, the field operator $\hat{\phi}[g;x)$ propagates in a n-dimensional spacetime. Once the regularization prescription has been introduced a regularized and renormalized stress-energy operator $\hat{T}^{\rm R}_{ab}[g;x)$ may be defined as

$$\hat{T}^{\rm R}_{ab}[g;x) = \hat{T}_{ab}[g;x) + F^C_{ab}[g;x)\hat{I}, \tag{10.6}$$

which differs from the regularized $\hat{T}_{ab}[g;x)$ by the identity operator times some tensor counterterms $F^C_{ab}[g;x)$, which depend on the regulator and are local functionals of the metric. The field states can be chosen in such a way that for any

pair of physically acceptable states, i.e., Hadamard states in the sense of [49], $|\psi\rangle$ and $|\varphi\rangle$, the matrix element $\langle\psi|T^{\rm R}_{ab}|\varphi\rangle$, defined as the limit when the regulator takes the physical value, is finite and satisfies Wald's axioms [102, 48]. These counterterms can be extracted from the singular part of a Schwinger–DeWitt series [48, 100, 101, 91]. The choice of these counterterms is not unique but this ambiguity can be absorbed into the renormalized coupling constants which appear in the equations of motion for the gravitational field.

10.1.1 Semiclassical Einstein Equation

The *semiclassical Einstein equation* for the metric g_{ab} can then be written as

$$G_{ab}[g] + \Lambda g_{ab} - 2(\alpha_d A_{ab} + \beta_d B_{ab})[g] = 8\pi G_{\rm N}\langle\hat{T}^{\rm R}_{ab}[g]\rangle, \tag{10.7}$$

where $\langle\hat{T}^{\rm R}_{ab}[g]\rangle$ is the expectation value of the operator $\hat{T}^{\rm R}_{ab}[g,x)$ after the regulator takes the physical value in some physically acceptable state of the field on $(\mathcal{M}, g_{ab})$. Note that both the stress-energy tensor and the quantum state are functionals of the metric, hence the notation.

The parameters $G_{\rm N}$, Λ, α_d and β_d are the renormalized coupling constants, respectively, the gravitational constant, the cosmological constant and two parameters which are zero in the classical Einstein equation and have dimensions of square length (we use units in which $\hbar = c = 1$). In the effective field theory description of general relativity [192] these two parameters are naturally of the order of the Planck length square $\ell_{\rm P}^2 = G_{\rm N}$ and in later Chapters will be convenient to use instead dimensionless parameters a_1 and a_2 defined by $\alpha_d \equiv 8\pi\ell_{\rm P}^2 a_1$ and $\beta_d \equiv 8\pi\ell_{\rm P}^2 a_2$. The renormalized four constants of the semiclassical equation must be understood as the result of "dressing" the bare constants which appear in the classical action before renormalization. The values of these constants must be determined by experiment.

The left-hand side of Eq. (10.7) may be derived from the gravitational action

$$S_{\rm g}[g] = \frac{1}{8\pi G_{\rm N}}\int d^4x\sqrt{-g}\left[\frac{1}{2}R - \Lambda + \alpha_d C_{abcd}C^{abcd} + \beta_d R^2\right], \tag{10.8}$$

where C_{abcd} is the Weyl tensor. The parameters α_d and β_d are related to the parameters a, b, c introduced in the action (2.67). By writing the Ricci and the Riemann squares in terms of Ricci scalar and Weyl squares and the Euler density G, given in (2.61), and taking into account that by the Gauss–Bonnet theorem the Euler density does not contribute to the field equations one can see that these two actions are equivalent with $\alpha_d = b/4 + c$ and $\beta_d = a/2 + (b + c)/6$.

The tensors A_{ab} and B_{ab} come from the functional derivatives with respect to the metric of the terms quadratic in the curvature in Eq. (10.8), and are explicitly given by

$$A^{ab} = \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g_{ab}}\int d^4\sqrt{-g}C_{cdef}C^{cdef}$$
$$= \frac{1}{2}g^{ab}C_{cdef}C^{cdef} - 2R^{acde}R^{b}_{cde} + 4R^{ac}R^{b}_{c} - \frac{2}{3}RR^{ab}$$
$$- 2\Box R^{ab} + \frac{2}{3}\nabla^a\nabla^b R + \frac{1}{3}g^{ab}\Box R, \tag{10.9}$$
$$B^{ab} = \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g_{ab}}\int d^4\sqrt{-g}R^2$$
$$= \frac{1}{2}g^{ab}R^2 - 2RR^{ab} + 2\nabla^a\nabla^b R - 2g^{ab}\Box R, \tag{10.10}$$

where R_{abcd} and R_{ab} are the Riemann and Ricci tensors, respectively. These two tensors are, like the Einstein and metric tensors, symmetric and divergenceless, $\nabla^a A_{ab} = 0 = \nabla^a B_{ab}$, and they have been introduced previously in (2.71) and (2.69) with different names: $A_{ab} = {}^{(C)}H_{ab}$ and $B_{ab} = {}^{(1)}H_{ab}$.

A solution of semiclassical gravity consists of a spacetime $(\mathcal{M}, g_{ab})$, a quantum field operator $\hat{\phi}[g]$ which satisfies the evolution Eqs. (10.2), and a physically acceptable state $|\psi[g]\rangle$ for this field, such that Eq. (10.7) is satisfied when the expectation value of the renormalized stress-energy operator is evaluated in this state.

For a free quantum field this theory is robust in the sense that it is self-consistent and fairly well understood. As long as the gravitational field is assumed to be described by a classical metric, the above semiclassical Einstein equations seems to be the only plausible dynamical equation for this metric: the metric couples to matter fields via the stress-energy tensor and for a given quantum state the only physically observable c-number stress-energy tensor that one can construct is the above renormalized expectation value. However, lacking a full quantum theory of gravity the scope and limits of the theory are not so well understood. It is assumed that the semiclassical theory should break down at Planck scales, which is when simple order of magnitude estimates suggest that the quantum effects of gravity should not be ignored because the energy of a quantum fluctuation in a Planck size region, as determined by the Heisenberg uncertainty principle, is comparable to the gravitational energy of that fluctuation.

The theory is expected to break down when the fluctuations of the stress-energy operator are large [195]. A criterion based on the ratio of the fluctuations to the mean was proposed by Kuo and Ford [196] (see also work via zeta-function methods in [197, 205]). This proposal was questioned by Hu and Phillips [198, 199, 200] because it does not contain a scale at which the theory is probed or how accurately the theory can be resolved. They suggested the use of a smearing scale or point-separation distance, for integrating over the bitensor quantities, equivalent to a stipulation of the resolution level of measurements; see also the response by Ford and Wu [206, 207]. A different criterion was suggested by Anderson et al. [208, 209] based on linear response theory.

Later, Hu et al. [190, 214] proposed a criterion for the validity of semiclassical gravity which is based on the stability of the solutions of the semiclassical Einstein equations with respect to quantum metric fluctuations. The two-point correlations for the metric perturbations can be described in the framework of stochastic gravity, which is closely related to the quantum theory of gravity interacting with N matter fields, to the next to leading order in a $1/N$ expansion. We describe these developments in the following sections.

10.2 Stochastic Gravity: Axiomatic Approach

The purpose of stochastic gravity is to extend the semiclassical theory to account for the stress-energy tensor fluctuations in a self-consistent way. A physical observable that describes these fluctuations to lowest order is the *noise kernel* bitensor, which is defined through the two-point correlation of the stress-energy operator as

$$N_{abc'd'}[g;x,y) = \frac{1}{2}\langle\{\hat{t}_{ab}[g;x), \hat{t}_{c'd'}[g;y)\}\rangle, \tag{10.11}$$

where the curly brackets mean anticommutator,

$$\hat{t}_{ab}[g;x) \equiv \hat{T}_{ab}[g;x) - \langle\hat{T}_{ab}[g;x)\rangle, \tag{10.12}$$

and we use unprimed and primed indices to emphasize that this bitensor is a tensor with respect to the first two indices at the point x and a tensor with respect to the last two indices at the point y.

The noise kernel is defined in terms of the unrenormalized stress-energy tensor operator $\hat{T}_{ab}[g;x)$ on a given background metric g_{ab}, thus a regulator is implicitly assumed on the right-hand side of Eq. (10.11). However, for a linear quantum field the above kernel – the expectation function of a bitensor – is free of ultraviolet divergences because the regularized $T_{ab}[g;x)$ differs from the renormalized $T^{\rm R}_{ab}[g;x)$ by the identity operator times some tensor counterterms, see Eq. (10.6), so that in the subtraction (10.12) the counterterms cancel. Consequently the ultraviolet behavior of $\langle\hat{T}_{ab}(x)\hat{T}_{c'd'}(y)\rangle$ is the same as that of $\langle\hat{T}_{ab}(x)\rangle\langle\hat{T}_{c'd'}(y)\rangle$, and $\hat{T}_{ab}$ can be replaced by the renormalized operator $\hat{T}^{\rm R}_{ab}$ in Eq. (10.11). The noise kernel should be thought of as a distribution function; the limit of coincidence points has meaning only in the sense of distributions.

The bitensor $N_{abc'd'}[g;x,y)$, or $N_{abc'd'}(x,y)$ for short, is real and positive semidefinite, as a consequence of $\hat{T}^R_{ab}$ being self-adjoint. A simple proof of this property can be given as follows. Let $|\psi\rangle$ be a given quantum state and let $\hat{Q}$ be a self-adjoint operator, $\hat{Q}^\dagger = \hat{Q}$, then one can write $\langle\psi|\hat{Q}\hat{Q}|\psi\rangle = \langle\psi|\hat{Q}^\dagger\hat{Q}|\psi\rangle = |\hat{Q}|\psi\rangle|^2 \geq 0$. Now let $\hat{t}(x)$ be a self-adjoint operator (this is shorthand for $\hat{t}_{ab}$), then if we define $\hat{Q} = \int dx f(x)\hat{t}(x)$ for an arbitrary well defined function $f(x)$, the previous inequality can be written as follows $\int dxdy f(x)\langle\psi|\hat{t}(x)\hat{t}(y)|\psi\rangle f(y) \geq 0$,

which is the condition for the noise kernel to be positive semi-definite. Note that when considering the inverse kernel $N^{-1}(x,y)$ (shorthand notation), it is implicitly assumed that one is working in the subspace obtained from the eigenvectors which have strictly positive eigenvalues when the noise kernel is diagonalized.

Once the fluctuations of the stress-energy operator have been characterized we can perturbatively extend the semiclassical theory to account for such fluctuations. Thus we will assume that the background spacetime metric g_{ab} is a solution of the semiclassical Einstein equation (10.7) and we will write the new metric for the extended theory as $g_{ab}+h_{ab}$, where we will assume that h_{ab} is a perturbation to the background solution. The renormalized stress-energy operator and the state of the quantum field may now be denoted by $\hat{T}^{\rm R}_{ab}[g+h]$ and $|\psi[g+h]\rangle$, respectively, and $\langle\hat{T}^{\rm R}_{ab}[g+h]\rangle$ will be the corresponding expectation value.

Let us now introduce a Gaussian stochastic tensor field $\xi_{ab}[g;x)$ defined by the following correlators:

$$\langle\xi_{ab}[g;x)\rangle_s=0,\quad \langle\xi_{ab}[g;x)\xi_{c'd'}[g;y)\rangle_s=N_{abc'd'}[g;x,y), \tag{10.13}$$

where $\langle\dots\rangle_s$ means statistical average. The symmetry and positive semi-definite property of the noise kernel guarantees that the stochastic field tensor $\xi_{ab}[g,x)$, or $\xi_{ab}(x)$ for short, just introduced is well defined. Note that this stochastic tensor captures only partially the quantum nature of the fluctuations of the stress-energy operator since it assumes that cumulants of higher order are zero.

An important property of this stochastic tensor is that it is covariantly conserved in the background spacetime $\nabla^a\xi_{ab}[g;x)=0$. In fact, as a consequence of the conservation of $\hat{T}^{\rm R}_{ab}[g]$ one can see that $\nabla^a_x N_{abc'd'}(x,y)=0$. Taking the divergence in Eq. (10.13) one can then see that $\langle\nabla^a\xi_{ab}\rangle_s=0$ and $\langle\nabla^a_x\xi_{ab}(x)\xi_{c'd'}(y)\rangle_s$=0 so that $\nabla^a\xi_{ab}$ is deterministic and represents with certainty the zero vector field in $\mathcal{M}$.

For a conformal field, i.e., a field whose classical action is conformally invariant, ξ_{ab} is traceless: $g^{ab}\xi_{ab}[g;x)=0$; so that, for a conformal matter field the stochastic source gives no correction to the trace anomaly. In fact, from the trace anomaly result which states that $g^{ab}\hat{T}^{\rm R}_{ab}[g]$ is, in this case, a local c-number functional of g_{ab} times the identity operator, we have that $g^{ab}(x)N_{abc'd'}[g;x,y)=0$. It then follows from equation (10.13) that $\langle g^{ab}\xi_{ab}\rangle_s=0$ and $\langle g^{ab}(x)\xi_{ab}(x)\xi_{c'd'}(y)\rangle_s=0$; an alternative proof was given in the point separation chapter.

10.2.1 Einstein–Langevin equation

The previous properties of the stochastic field make it quite natural to incorporate into the Einstein equations the stress-energy fluctuations by using the stochastic tensor $\xi_{ab}[g;x)$ as the source of the metric perturbations. Thus we will write the following equation:

$$G^{(1)}_{ab}[g+h] + \Lambda(g_{ab}+h_{ab}) - 2(\alpha_d A^{(1)}_{ab} + \beta_d B^{(1)}_{ab})[g+h] = 8\pi G_{\rm N}\left(\langle \hat{T}^{(1)\rm R}_{ab}[g+h]\rangle + \xi_{ab}[g]\right), \tag{10.14}$$

where the background metric g_{ab} is a solution of the semiclassical Einstein equation (10.7). This is a dynamical equation for the metric perturbation h_{ab} to linear order, as indicated by the superscript (1), and is in the form of a *(semiclassical) Einstein–Langevin equation*. It describes the backreaction of the metric to the quantum fluctuations of the stress-energy tensor of matter fields, and gives a first order extension to semiclassical gravity as described by Eq. (10.7).

Note that we refer to the Einstein–Langevin equation as a first order extension to semiclassical Einstein equation of semiclassical gravity and the lowest level representation of stochastic gravity. However, stochastic gravity has a broader meaning; it refers to the range of theories based on second and higher-order correlation functions. Noise can be defined in effectively open systems (*e.g.*, correlation noise [582] in the Schwinger–Dyson equation hierarchy) to some degree but one should not expect the Langevin form to prevail. In this sense we say stochastic gravity is the intermediate theory between semiclassical gravity (a mean field theory based on the expectation values of the stress-energy tensor of quantum fields) and quantum gravity (the full hierarchy of correlation functions retaining complete quantum coherence [479, 482]).

The renormalization of the operator $\hat{T}_{ab}[g+h]$ is carried out exactly as for the operator $\hat{T}_{ab}[g]$, but now in the perturbed metric $g_{ab}+h_{ab}$. Note that the stochastic source $\xi_{ab}[g;x)$ is not dynamical, it is independent of h_{ab} since it describes the fluctuations of the stress-energy tensor on the semiclassical background g_{ab}.

An important property of the Einstein–Langevin equation is that it is gauge-invariant under the change of h_{ab} by $h'_{ab} = h_{ab} + \nabla_a\zeta_b + \nabla_b\zeta_a$, where ζ^a is a stochastic vector field on the background manifold $\mathcal{M}$. Note that a tensor such as $R_{ab}[g+h]$, transforms as $R_{ab}[g+h'] = R_{ab}[g+h] + \mathcal{L}_\zeta R_{ab}[g]$ to linear order in the perturbations, where $\mathcal{L}_\zeta$ is the Lie derivative with respect to ζ^a. Now, let us write the source tensors in Eqs. (10.14) and (10.7) to the left-hand sides of these equations. If we substitute h by h' in this new version of Eq. (10.14), we get the same expression, with h instead of h', plus the Lie derivative of the combination of tensors which appear on the left-hand side of the new Eq. (10.7). This last combination vanishes when Eq. (10.7) is satisfied, i.e., when the background metric g_{ab} is a solution of semiclassical gravity.

From the statistical average of Eq. (10.14) we have that $g_{ab} + \langle h_{ab}\rangle_s$ must be a solution of the semiclassical Einstein equation linearized around the background g_{ab}; this solution has been proposed as a test for the validity of the semiclassical approximation [208, 209], a point that will be further discussed in Section 10.3.

10.2.2 Stochastic vs. Quantum Correlations

The stochastic equation (10.14) predicts that the gravitational field has stochastic fluctuations over the background g_{ab}. This equation is linear in h_{ab}, thus its solutions can be written as follows,

$$h_{ab}(x) = h^0_{ab}(x) + 8\pi G_{\rm N} \int d^4x' \sqrt{-g(x')} G^{\rm ret}_{abc'd'}(x,x') \xi^{c'd'}(x'), \tag{10.15}$$

where $h^0_{ab}(x)$ is the solution of the homogeneous equation containing information on the initial conditions and $G^{\rm ret}_{abc'd'}(x,x')$ is the retarded propagator of Eq. (10.14) with vanishing initial conditions. Form this we obtain the two-point correlation functions for the metric perturbations:

$$\begin{aligned}\langle h_{ab}(x) h_{c'd'}(y)\rangle_s &= \langle h^0_{ab}(x) h^0_{c'd'}(y)\rangle_s + (8\pi G_{\rm N})^2 \int d^4x' d^4y' \sqrt{g(x')g(y')} \\ &\quad \times G^{\rm ret}_{ab\hat{e}\hat{f}}(x,x') N^{\hat{e}\hat{f}\bar{g}\bar{h}}(x',y') G^{\rm ret}_{c'd'\bar{g}\bar{h}}(y,y') \\ &\equiv \langle h_{ab}(x) h_{c'd'}(y)\rangle_{\rm int} + \langle h_{ab}(x) h_{c'd'}(y)\rangle_{\rm ind}. \end{aligned} \tag{10.16}$$

There are two different contributions to the two-point correlations, which we have distinguished in the second equality. The first one is connected to the fluctuations of the initial state of the metric perturbations, and we will refer to them as *intrinsic fluctuations*. The second contribution is proportional to the noise kernel and is thus connected with the fluctuations of the quantum fields; we will refer to them as *induced fluctuations*. To find these two-point stochastic correlation functions one needs to know the noise kernel $N_{abc'd'}(x,y)$. Explicit expressions of this kernel in terms of the two-point Wightman functions were given in [183], expressions based on point-splitting methods were given in [687, 200], and are reproduced at the end of this chapter. Note that, as remarked earlier, the noise kernel should be thought of as a distribution function, the limit of coincidence points has meaning only in the sense of distributions.

The two-point stochastic correlation functions for the metric perturbations of Eq. (10.16) satisfy an important property. They correspond exactly to the symmetrized two-point correlation functions for the quantum metric perturbations in the large N expansion, i.e. the quantum theory describing the interaction of the gravitational field with N arbitrary free fields and expanded in powers of $1/N$. To leading order for the graviton propagator one finds that

$$\langle \{\hat{h}_{ab}(x), \hat{h}_{c'd'}(y)\}\rangle = 2\,\langle h_{ab}(x) h_{c'd'}(y)\rangle_s, \tag{10.17}$$

where $\hat{h}_{ab}(x)$ mean the quantum operator corresponding to the metric perturbations and the statistical average in Eq. (10.16) for the homogeneous solutions is now taken with respect to the Wigner distribution that describes the initial quantum state of the metric perturbations. The Lorentz gauge condition $\nabla^a(h_{ab} - (1/2)\eta_{ab}h^c_c) = 0$ as well as some initial condition to completely fix the gauge of the initial state is implicitly understood. Moreover, since there are now N scalar fields the stochastic source has been rescaled so that the two-point correlation defined by Eq. (10.13) should be $1/N$ times the noise kernel of a single field.

This result was implicitly obtained in the Minkowski background in [189] where the two-point correlation in the stochastic context was computed for the

linearized metric perturbations. This stochastic correlation exactly agrees with the symmetrized part of the graviton propagator computed by Tomboulis [686] in the quantum context of gravity interacting with N Fermion fields, where the graviton propagator is of order $1/N$. This result can be extended to an arbitrary background in the context of the large N expansion, a sketch of the proof with explicit details in the Minkowski background is given in [190].

This connection between the stochastic correlations and the quantum correlations was noted and studied in detail in the context of simpler open quantum systems by Calzetta et al. [688]. Note that stochastic gravity goes beyond semiclassical gravity in the following sense. The semiclassical theory, which is based on the expectation value of the stress-energy tensor, carries information on the field two-point correlations only, since $\langle \hat{T}_{ab} \rangle$ is quadratic in the field operator $\hat{\phi}$. The stochastic theory on the other hand, is based on the noise kernel (10.11) which is quartic in the field operator. However, it does not carry information on the graviton-graviton interaction, which in the context of the large N expansion gives diagrams of order $1/N^2$. This is illustrated in Section 9.4.

Furthermore, the retarded propagator gives also information on the commutator. To leading order in $1/N$ the commutator is independent of the initial state of the metric perturbations and is given by

$$\langle [\hat{h}_{ab}(x), \hat{h}_{c'd'}(y)] \rangle = 16\pi i G_{\rm N} \left(G^{\rm ret}_{abc'd'}(y,x) - G^{\rm ret}_{abc'd'}(x,y) \right), \tag{10.18}$$

so that combining the commutator with the anticommutator the quantum two-point correlation functions are determined. Moreover, assuming a Gaussian initial state with vanishing expectation value for the metric perturbations any n-point quantum correlation function is determined by the two-point quantum correlations and thus by the stochastic approach. Consequently, one may regard the Einstein–Langevin equation as a useful tool to compute the correlation functions for the quantum metric perturbations in the large N expansion.

We should, however, emphasize also that Langevin like equations are obtained to describe the quantum to classical transition in open quantum systems, when quantum decoherence takes place by coarse graining of the environment as well as by suitable coarse graining of the system variables [147, 689, 149, 150, 151, 690]. In those cases the stochastic correlation functions describe actual classical correlations of the system variables. Examples can be found in the case of a moving charged particle in an electromagnetic field in quantum electrodynamics [246] and in several quantum Brownian models [691, 692, 688].

10.2.3 A Toy Model for Stochastic Gravity

To clarify the role of the noise kernel and illustrate the relationship between the semiclassical, stochastic and quantum descriptions it is useful to illustrate the theory with a simple toy model which minimizes the technical complications. Let us assume that the gravitational equations are described by a dimensionless

scalar linear gravitational field $h(x)$ whose source is a massless scalar field $\phi(x)$ which satisfies the Klein–Gordon equation in flat spacetime $\Box\phi(x) = 0$. The field stress-energy tensor is quadratic in the field, and independent of $h(x)$. The classical gravitational field equations will be given by

$$\Box h(x) = \kappa^2 T(x), \tag{10.19}$$

where as previously $\kappa^2 \equiv 16\pi G_{\rm N}$ and $T(x)$ is the (scalar) trace of the stress-energy tensor, $T(x) = \partial_a\phi(x)\partial^a\phi(x)$. Note that this is not a self-consistent theory since $\phi(x)$ does not react to the gravitational field $h(x)$. We should also emphasize that this model is not the standard linearized theory of gravity in which T is also linear in $h(x)$. It captures, however, some of the key features of linearized gravity.

In the Heisenberg picture the quantum field $\hat{h}(x)$ satisfies

$$\Box\hat{h}(x) = \kappa^2\hat{T}(x). \tag{10.20}$$

Since $\hat{T}(x)$ is quadratic in the field operator $\hat{\phi}(x)$ some regularization procedure has to be assumed in order for (10.20) to make sense. Since we work in flat spacetime we may simply use a normal ordering prescription to regularize the operator $\hat{T}(x)$. The solutions of this equation, i.e., the field operator at the point x, $\hat{h}(x)$, may be written in terms of the retarded propagator $G^{\rm ret}(x, y)$ as

$$\hat{h}(x) = \hat{h}^0(x) + \frac{1}{\kappa^2}\int dx' G^{\rm ret}(x, x')\hat{T}(x'), \tag{10.21}$$

where $\hat{h}^0(x)$ is the free field which carries information on the initial conditions and the state of the field. From this solution we may compute, for instance, the symmetric two point quantum correlation function (the anticommutator)

$$\begin{aligned}\frac{1}{2}\langle\{\hat{h}(x), \hat{h}(y)\}\rangle &= \frac{1}{2}\langle\{\hat{h}^0(x), \hat{h}^0(y)\}\rangle \\ &\quad + \frac{1}{2\kappa^4}\int\int dx' dy' G^{\rm ret}(x, x')G^{\rm ret}(y, y')\langle\{\hat{T}(x'), \hat{T}(y')\}\rangle,\end{aligned} \tag{10.22}$$

where the expectation value is taken with respect to the quantum state in which both fields $\phi(x)$ and $h(x)$ are quantized, and we assume for the free field, $\langle\hat{h}^0\rangle = 0$.

We can now consider the semiclassical theory for this problem. If we assume that $h(x)$ is classical and the matter field is quantum the semiclassical theory may just be described by substituting into the classical Eq. (10.19) the stress-energy trace by the expectation value of the stress-energy trace operator $\langle\hat{T}(x)\rangle$, in some quantum state of the field $\hat{\phi}(x)$. Since in our model $\hat{T}(x)$ is independent of $h(x)$ we may simply renormalize its expectation value using normal ordering,

then for the vacuum state of the field $\hat{\phi}(x)$, we would simply have $\langle\hat{T}(x)\rangle_0 = 0$. The semiclassical theory thus reduces to

$$\Box h(x) = \kappa^2\langle\hat{T}(x)\rangle. \tag{10.23}$$

The two-point function $h(x)h(y)$ that one may derive from this equation depends on the two-point function $\langle\hat{T}(x)\rangle\langle\hat{T}(y)\rangle$ and clearly cannot reproduce the quantum result (10.22) which depends on the expectation value of two point operator $\langle\{\hat{T}(x),\hat{T}(y)\}\rangle$. That is, the semiclassical theory entirely misses the fluctuations of the stress-energy operator $\hat{T}(x)$.

Let us now extend the semiclassical theory in order to account for such fluctuations. The first step is to characterize these fluctuations. For this, we introduce the noise kernel as the physical observable that measures the fluctuations of the stress-energy operator $\hat{T}$ as

$$N(x,y) = \frac{1}{2}\langle\{\hat{t}(x),\hat{t}(y)\}\rangle \tag{10.24}$$

where $\hat{t}(x) = \hat{T}(x) - \langle\hat{T}(x)\rangle$. As we have seen before the biscalar $N(x,y)$ is real and positive-semidefinite, as a consequence of $\hat{t}$ being self-adjoint. By this property it is possible to introduce a Gaussian stochastic field as follows:

$$\langle\xi(x)\rangle_s = 0, \quad \langle\xi(x)\xi(y)\rangle_s = N(x,y). \tag{10.25}$$

where the subscript s means a statistical average. These equations entirely define the stochastic process $\xi(x)$ since we have assumed that it is Gaussian.

The extension of the semiclassical equation is simply acomplished by adding to the right-hand side of the semiclassical Eq. (10.23) the stochastic source $\xi(x)$ which accounts for the fluctuations of $\hat{T}$ as follows,

$$\Box h(x) = \kappa^2\left(\langle\hat{T}(x)\rangle + \xi(x)\right). \tag{10.26}$$

This equation is in the form of a Langevin equation: the field $h(x)$ is classical but stochastic and the observables we may obtain from it are correlation functions for $h(x)$. In fact, the solution of this equation may be written in terms of the retarded propagator as,

$$h(x) = h^0(x) + \frac{1}{\kappa^2}\int dx' G^{\rm ret}(x,x')\left(\langle\hat{T}(x')\rangle + \xi(x')\right), \tag{10.27}$$

from where the two-point correlation function for the classical field $h(x)$, after using the definition of $\xi(x)$ and that $\langle h^0(x)\rangle_s = 0$, is given by

$$\langle h(x)h(y)\rangle_s = \langle h^0(x)h^0(y)\rangle_s + \frac{1}{2\kappa^4}\int\int dx'dy' G^{\rm ret}(x,x')G^{\rm ret}(y,y')\langle\{\hat{T}(x'),\hat{T}(y')\}\rangle. \tag{10.28}$$

Note that in writing $\langle \dots \rangle_s$ here we are assuming a double stochastic average, one is related to the stochastic field $\xi(x)$ and the other is related to the free field $h^0(x)$ which is assumed also to be stochastic with a distribution function to be specified.

Comparing (10.22) with (10.28) we see that the respective second terms on the right-hand side are identical provided the expectation values are computed in the same quantum state for the field $\hat{\phi}(x)$ (recall that we have assumed that $T(x)$ does not depend on $h(x)$). The fact that the field $h(x)$ is also quantized in (10.22) does not change the previous statement. The nature of the first term on the right-hand sides of Eqs. (10.22) and (10.28) is different: in the first case it is the two-point quantum expectation value of the free quantum field $\hat{h}^0$ whereas in the second case it is the stochastic average of the two-point classical homogeneous field h^0, which depends on the initial conditions. Now we can still make these terms equal to each other if we assume for the homogeneous field h^0 a Gaussian distribution of initial conditions such that

$$\langle h^0(x)h^0(y)\rangle_s = \frac{1}{2}\langle\{\hat{h}^0(x), \hat{h}^0(y)\}\rangle. \tag{10.29}$$

This Gaussian stochastic field $h^0(x)$ can always be defined due to the positivity of the anti-commutator. Thus, under this assumption on the initial conditions for the field $h(x)$ the two point correlation function of (10.28) equals the quantum expectation value of (10.22) exactly. An interesting feature of the stochastic description is that the quantum anticommutator of (10.22) can be written as the right-hand side of equation (10.28) where the first term contains all the information on initial conditions for the stochastic field $h(x)$ and the second term codifies all the information on the quantum correlations of the source. This separation is also seen in the description of some quantum Brownian motion models which are typically used as paradigms of open quantum systems; see, e.g., [691, 692, 688].

It is interesting to note that in the standard linearized theory of gravity $T(x)$ depends also on $h(x)$, both explicitly and also implicitly through the coupling of $\phi(x)$ with $h(x)$. The equations are not so simple but it is still true that the corresponding Langevin equation leads to the correct symmetrized two point quantum correlations for the metric perturbations [189]. Thus in a linear theory as in the model just described one may just use the statistical description given by (10.26) to compute the symmetric quantum two-point function of Eq. (10.21).

10.3 Validity of Semiclassical Gravity

As we have emphasized earlier the scope and limits of semiclassical gravity are not well understood because we still lack a quantum theory of gravity. From the semiclassical Einstein equations it seems also clear that the semiclassical theory should break down when the quantum fluctuations of the stress-energy tensor

are large. Ford [195] was among the first to have emphasized the importance of these quantum fluctuations. It is less clear, however, how to quantify the size of these fluctuations. Kuo and Ford [196] used the variance of the fluctuations of the stress-energy tensor operator compared to the mean value as a measure of the validity of semiclassical gravity. Hu and Phillips [198, 199] pointed out that such a criterion should be refined by considering the backreaction of those fluctuations on the metric. Ford and collaborators also noticed that the metric fluctuations associated to the matter fluctuations can be meaningfully classified as *active* [693, 210, 211] and *passive* [195, 196, 694, 207], which correspond to our *intrinsic* and *induced* fluctuations, respectively, and have studied their properties in different contexts; see e.g. [695, 696, 697]. However, these fluctuations are not treated in a unified way and their precise relation to the quantum correlation function for the metric perturbations is not discussed. Furthermore, the full averaged backreaction of the matter fields is not included self-consistently, and the contribution from the vacuum fluctuations in Minkowski space is discarded.

A different approach to the validity of semiclassical gravity was pioneered by Horowitz [213, 698] who studied the stability of a semiclassical solution with respect to linear metric perturbations. In the case of a free quantum matter field in its Minkowski vacuum state, flat spacetime is a solution of semiclassical gravity. The equations describing those metric perturbations involve higher-order derivatives, and Horowitz found unstable *runaway* solutions that grow exponentially with characteristic timescales comparable to the Planck time; see also the analysis by Jordan [677]. Later, Simon [238, 287] argued that those unstable solutions lie beyond the expected domain of validity of the theory and emphasized that only those solutions which resulted from truncating perturbative expansions in terms of the square of the Planck length are physically acceptable.

Further discussion was provided by Flanagan and Wald [128], who advocated the use of an *order reduction* prescription first introduced by Parker and Simon [357], which we briefly review in Section 10.3.1. Anderson et al. [208, 209] took up the issue of the validity of semiclassical gravity. Their starting point is the fact that the semiclassical Einstein equation will fail to provide a valid description of the dynamics of the mean spacetime geometry whenever the higher-order radiative corrections to the effective action, involving loops of gravitons or internal graviton propagators, become important. Next, they argue qualitatively that such higher-order radiative corrections cannot be neglected if the metric fluctuations grow without bound. Finally, they propose a criterion to characterize the growth of the metric fluctuations, and hence the validity of semiclassical gravity, based on the stability of the solutions of the linearized semiclassical equation. Following these approaches the Minkowski metric is shown to be a stable solution of semiclassical gravity with respect to small metric perturbations.

In a similar vein Fröb et al. [639] use the method of order reduction to solve nonperturbatively the semiclassical Einstein equation governing the dynamics of linear metric perturbations around de Sitter spacetime when the quantum

backreaction of conformal fields on the mean geometry is included. The analysis establishes the stability of de Sitter spacetime with respect to linear metric perturbations, and also the late-time attractor character of that spacetime. Thus extending the "no-hair" results for de Sitter in classical general relativity to the semiclassical context.

The above criteria may be understood as criteria based on semiclassical gravity itself. It is certainly true that stability is a necessary condition for the validity of a semiclassical solution, but one may also look for criteria within extensions of semiclassical gravity. In the absence of a quantum theory of gravity such criteria may be found in some more modest extensions. Thus, Ford [195] considered graviton production in linearized quantum gravity and compared the results with the production of gravitational waves in semiclassical gravity. Ashtekar [699] and Beetle [700] found large quantum gravity effects in three-dimensional quantum gravity models.

Finally Hu et al. [190], see also [214], advocated for a criterion within the stochastic gravity approach, and since stochastic gravity extends semiclassical gravity by incorporating the quantum stress-energy tensor fluctuations of the matter fields, this criterion is structurally the most complete to date. It turns out that this validity criterion is equivalent to the validity criterion that one might advocate within the large N expansion, that is the quantum theory describing the interaction of the gravitational field with N identical free matter fields. In the leading order, namely the limit in which N goes to infinity and the gravitational constant is appropriately rescaled, the theory reproduces semiclassical gravity. Thus, a natural extension of semiclassical gravity is provided by the next-to-leading order. As we have emphasized before the symmetrized two-point quantum correlation functions of the metric perturbations in the large N expansion are equivalent to the two-point stochastic metric correlation functions predicted by stochastic gravity.

Our validity criterion can then be formulated as follows: a solution of semiclassical gravity is valid when it is stable with respect to quantum metric perturbations. This criterion involves the consideration of quantum correlation functions of the metric perturbations, since the quantum field describing the metric perturbations $\hat{h}_{ab}(x)$ is characterized not only by its expectation value but also by its n-point correlation functions.

It is important to emphasize that the above validity criterion incorporates in a unified and self-consistent way the two main ingredients of the criteria exposed above, namely, the criteria based on the quantum stress-energy tensor fluctuations of the matter fields, and the criteria based on the stability of semiclassical solutions against classical metric perturbations. The former is incorporated through the induced metric fluctuations, and the latter through the intrinsic fluctuations introduced in Eq. (10.16). Whereas information on the stability of the intrinsic metric fluctuations can be obtained from an analysis of the solutions of the perturbed semiclassical Einstein equation, the homogeneous

part of Eq. (10.14), the effect of the induced metric fluctuations is accounted only in stochastic gravity, the full inhomogeneous equation (10.14). We will illustrate this criteria in Section 11.5 by studying the stability of Minkowski spacetime as a solution of semiclassical gravity.

10.3.1 Order Reduction

The semiclassical Einstein equation (10.7) contains terms with up to fourth-order derivatives of the metric, as seen from Eqs. (10.9) and (10.10). This also applies, of course, to the linear equation for the metric perturbation (10.14), with and without the stochastic source. Also the expectation value $\langle \hat{T}^R_{ab} \rangle$ involves higher derivative terms.

Such kind of higher-order time derivatives are common in backreaction problems. A well known example is the Abraham–Lorentz–Dirac equation, which describes the effect of radiation reaction on the motion of a point-like charge in classical electrodynamics [701, 702] (i.e., without considering the internal structure of the particle nor a finite size for the charge density distribution). In fact, they are a generic feature of *effective field theories* (EFT), where the effects of the ultraviolet (UV) sector on the dynamics of the low-energy degrees of freedom are encoded at the level of the action through an expansion of local terms with an increasing number of derivatives. The validity of the EFT expansion relies on the fact that for length-scales much larger than the inverse cut-off scale of the UV sector the higher-order terms in the expansion become increasingly smaller. In this regime their contribution amounts to a small correction to the equation of motion which results, when treated perturbatively, in locally small perturbations of the classical solutions.

In contrast, solving the corresponding higher-order equations exactly gives rise to additional solutions exhibiting exponential instabilities with characteristic time-scales comparable to the inverse cut-off scale of the EFT, or sometimes fast oscillations with the same kind of characteristic timescale, often referred to as "runaway" solutions. These are spurious solutions which should not be taken seriously since they involve characteristic scales for which the EFT expansion breaks down and the contributions from the higher-order terms to the equation of motion no longer correspond to small corrections but to dominant terms.

The simplest way of avoiding such spurious solutions is by solving the corrected equations of motion perturbatively. However, perturbative solutions may not be valid for long times. This happens when quantities like the total time appear to multiply the perturbative parameter so that the expansion contains so-called secular terms which grow with time and lead to a breakdown for sufficiently long times of the truncated perturbative expansion. Those limitations can be overcome with the *order reduction* method, which consists in taking the equation of motion with corrections up to a finite order and writing an alternative equation which is equivalent up to that order but contains no higher derivative terms.

This is achieved by taking successive derivatives of the original equation and substituting the higher-order derivatives in the correction terms to the appropriate order. The exact solutions of the equation obtained with this method agree locally with the perturbative solutions constructed around different times (each one with a finite domain of validity) and provides an interpolation between all of them valid for long times.

This is particularly important when considering situations where the effects of the corrections are locally small, but can build up over long times and give rise to substantial accumulated effects. Two examples of such situations are an electric charge following a quasi-circular trajectory in a uniform magnetic field and emitting electromagnetic radiation for a sufficiently long time so that the radius of its orbit decreases, say, to half of its initial value due to radiation reaction, or an evaporating black hole emitting Hawking radiation for such a long time that its mass, or horizon size, decreases to a small fraction of its initial value.

Related alternative methods which have been employed in the literature for discarding the spurious solutions mentioned above involve finding the exact solutions of the original backreaction equation and then selecting the appropriate subset either by demanding analyticity of the solutions with respect to the perturbative parameter or checking explicitly which solutions exhibit unphysical characteristic scales and disregarding them [703]. However, the latter method is less systematic and requires a case-by-case analysis, whereas the analyticity requirement may be too restrictive in some cases, as emphasized by Flanagan and Wald [128]. Furthermore, the order reduction method leads to equations of motion which are equivalent up to the order under consideration, but are often easier to solve.

The order reduction method has been applied to electromagnetic [702] and gravitational radiation reaction problems [704] as well as to higher derivative gravity [644]. It has also been employed in semiclassical gravity [357, 128, 639]. In this context it has been argued by Simon [640] that trace-anomaly-driven inflationary models (with no cosmological constant and driven entirely by the vacuum polarization of a large number of matter fields [359]) correspond to spurious solutions which lie beyond the EFT's domain of applicability and are automatically discarded when using order reduction.

It should be noted that order reduction cannot be always applied in a straightforward way. It may be ambiguous in integro-differential equations, or may lead to covariance breaking if the time derivatives and spatial derivatives are not simultaneously reduced; see [128] for a detailed discussion of these issues.

The order reduction method can be illustrated in a nutshell with the following simple example of a first order differential equation in time for a function $f(\eta)$ with a perturbative correction of order some parameter κ^2. Given

$$f' + bf = \kappa^2 P(f, f', f'', \ldots), \tag{10.30}$$

where b is a constant and P is an arbitrary function, order reduction uses that $f' = -bf + \mathcal{O}(\kappa^2)$, and by deriving one more time $f'' = -bf' + \mathcal{O}(\kappa^2) = b^2 f + \mathcal{O}(\kappa^2)$. Substituting those two equations into the right hand side, we get

$$f' + bf = \kappa^2 P(f, -bf, b^2 f, \ldots) + \mathcal{O}(\kappa^4), \tag{10.31}$$

an equation of first order which is valid to the same order in κ^2 as the original equation (10.30), but does not have unphysical solutions. Rather than considering a truncated perturbative expansion, this equation can now be solved exactly. It is clear how the method works for equations of more derivatives or partial differential equations: one takes the lowest order equation and substitutes it in the higher order terms, in κ^2, taking additional derivatives if necessary.

10.4 Functional Approach

Functional methods were used in semiclassical gravity to study the backreaction of quantum fields in cosmological models [106, 109, 110]. The primary advantage of the effective action approach is, in addition to the well-known fact that it is easy to introduce perturbation schemes like loop expansion and *nPI* formalisms, that it yields a *fully* self-consistent solution. For a general discussion in the semiclassical context of the two approaches, equation of motion versus effective action, contrast, *e.g.* [104, 105, 107, 108], with the above references and [111, 112, 113, 114].

As we have emphasized earlier, the well known in-out effective action method treated in textbooks, however, leads to equations of motion which are not real because they are tailored to compute transition elements of quantum operators rather than expectation values. The correct technique to use for the backreaction problem is the Schwinger–Keldysh, closed-time-path (CTP) or 'in-in' effective action; see e.g. [239, 705, 240, 436, 706, 424, 643]. These techniques were adapted to the gravitational context and applied to different problems in cosmology [180, 181, 182, 677, 437, 423]; see the chapter on backreaction of early universe quantum processes.

Furthermore, in this case the CTP functional formalism turns out to be related to the Feynman–Vernon *influence functional* formalism [187], since the full quantum system may be understood as consisting of a distinguished subsystem (the "system" of interest) interacting with the remaining degrees of freedom (the environment); see e.g. [706, 216, 186, 556, 707, 438, 708, 709, 710, 183, 223]. Integrating out the environment variables in a CTP path integral yields the influence functional, from which one can derive an effective action for the dynamics of the system [216, 185, 711, 707].

This approach to semiclassical gravity was motivated by the observation by Hu [184] that in some open quantum systems classicalization and decoherence [165, 166, 712, 215, 167, 169, 170, 175, 177] on the system may be brought about by interaction with an environment, the environment being in this case

the matter fields and some "high-momentum" gravitational modes [226, 713, 676, 714, 607, 715, 232, 690]. Unfortunately, since the form of a complete quantum theory of gravity interacting with matter is unknown, we do not know what these "high-momentum" gravitational modes are. Such a fundamental quantum theory might not even be a field theory, in which case the metric and scalar fields would not be fundamental objects [479]. Thus, in this case, we cannot attempt to evaluate the influence action of Feynman and Vernon starting from the fundamental quantum theory and performing the path integrations in the environment variables. Instead, we introduce the influence action for an effective quantum field theory of gravity and matter in which such "high-momentum" gravitational modes are assumed to have already been "integrated out" [716, 192, 679, 680, 229, 231].

10.4.1 Influence Action for Semiclassical Gravity

Let us formulate semiclassical gravity in the functional framework. As we have emphasized before we adopt the usual procedure of *effective field theories*; see e.g., [192, 679, 680, 193]. The effective action for the metric and the scalar field takes the most general local form compatible with general covariance:

$$S[g,\phi] \equiv S_{\rm g}[g] + S_{\rm m}[g,\phi] + \cdots , \tag{10.32}$$

where the gravitational action $S_{\rm g}[g]$ and the matter action $S_{\rm m}[g,\phi]$ are given by Eqs. (10.8) and (10.1), respectively, and the dots stand for terms of order higher than two in the curvature and in the number of derivatives of the scalar field. Here, we shall neglect the higher-order terms as well as self-interaction terms for the scalar field. The second-order terms are necessary to renormalize one-loop ultraviolet divergences of the scalar field stress-energy tensor, as we have already seen. Since $\mathcal{M}$ is a globally hyperbolic manifold, we can foliate it by a family of $t =$ constant Cauchy hypersurfaces Σ_t, and we will indicate the initial and final times by t_i and t_f, respectively.

The *influence functional* corresponding to the action (10.1) describing a scalar field in a spacetime (coupled to a metric field) may be introduced as a functional of two copies of the metric, denoted by g^{+}_{ab} and g^{-}_{ab}, which coincide at some final time $t = t_f$. Let us assume that, in the quantum effective theory, the state of the full system (the scalar and the metric fields) in the Schrödinger picture at the initial time $t = t_i$ can be described by a density operator which can be written as the tensor product of two operators on the Hilbert spaces of the metric and of the scalar field. Let $\rho_i(t_i) \equiv \langle \phi_+(t_i)|\hat{\rho}_\phi|\phi_-(t_i)\rangle$ be the matrix element of the density operator $\hat{\rho}_\phi$ describing the initial state of the scalar field. This is an assumption that must be handled with care to avoid some spurious results, which are discussed in detail in Section 16.4. The Feynman–Vernon influence functional is defined as the following path integral over the two copies of the scalar field:

$$\mathcal{F}_{\rm IF}[g^{\pm}] \equiv \int \mathcal{D}\phi_+ \mathcal{D}\phi_- \rho_i(t_i)\delta[\phi_+(t_f) - \phi_-(t_f)]\ e^{i\left(S_{\rm m}[g^+,\phi_+] - S_{\rm m}[g^-,\phi_-]\right)}. \quad (10.33)$$

Alternatively, the above double path integral can be rewritten as a closed time path (CTP) integral, namely, as a single path integral in a complex time contour with two different time branches, one going forward in time from t_i to t_f, and the other going backward in time from t_f to t_i (in practice one usually takes $t_i \to -\infty$). From this influence functional, the *influence action* $S_{\rm IF}[g^+, g^-]$, or $S_{\rm IF}[g^{\pm}]$ for short, defined by

$$\mathcal{F}_{\rm IF}[g^{\pm}] \equiv e^{iS_{\rm IF}[g^{\pm}]}, \quad (10.34)$$

carries all the information about the environment (the matter fields) relevant to the system (the gravitational field). Then we can define the CTP *effective action* for the gravitational field, $S_{\rm eff}[g^{\pm}]$, as

$$S_{\rm eff}[g^{\pm}] \equiv S_{\rm g}[g^+] - S_{\rm g}[g^-] + S_{\rm IF}[g^{\pm}]. \quad (10.35)$$

This is the effective action for the classical gravitational field in the CTP formalism. However, since the gravitational field is treated only at the tree level, this is also the effective classical action from which the classical equations of motion can be derived.

Expression (10.33) contains ultraviolet divergences that must be regularized. We shall assume that dimensional regularization can be applied, that is, it makes sense to dimensionally continue all the quantities that appear in Eq. (10.33). For this we need to work with the n-dimensional actions corresponding to $S_{\rm m}$ in (10.33) and $S_{\rm g}$ in (10.8). For example, the parameters $G_{\rm N}$, Λ, α_d and β_d of Eq. (10.8) (in n dimensions) are the bare parameters G_B, Λ_B, α_B and β_B, and in $S_{\rm g}[g]$, instead of the square of the Weyl tensor in Eq. (10.8), one must use $(2/3)(R_{abcd}R^{abcd} - R_{ab}R^{ab})$ which by the Gauss–Bonnet theorem leads to the same equations of motion as the action (10.8) when $n = 4$. The form of $S_{\rm g}$ in n dimensions is suggested by the Schwinger–DeWitt analysis of the ultraviolet divergences in the matter stress-energy tensor using dimensional regularization [47]. One can then write the effective action $S_{\rm eff}[g^{\pm}]$ in Eq. (10.35) in a form suitable for dimensional regularization. Since both $S_{\rm m}$ and $S_{\rm g}$ contain second-order derivatives of the metric, one should also add some boundary terms [585, 185]. The effect of these terms is to cancel out the boundary terms which appear when taking variations of $S_{\rm eff}[g^{\pm}]$ keeping the value of g^+_{ab} and g^-_{ab} fixed at Σ_{t_i} and Σ_{t_f}. Alternatively, in order to obtain the equations of motion for the metric in the semiclassical regime, we can work with the action without boundary terms and neglect all boundary terms when taking variations with respect to $g^{\pm}_{ab}$. From now on, all the functional derivatives with respect to the metric will be understood in this sense.

The semiclassical Einstein equation (10.7) can now be derived. Using the definition of the stress-energy tensor $T^{ab}(x) = (2/\sqrt{-g})\delta S_{\rm m}/\delta g_{ab}$ and the definition of the influence functional, equations (10.33) and (10.34), we see that

$$\langle \hat{T}^{ab}[g;x)\rangle = \frac{2}{\sqrt{-g(x)}} \frac{\delta S_{\rm IF}[g^{\pm}]}{\delta g^{+}_{ab}(x)}\bigg|_{g^{\pm}=g}, \tag{10.36}$$

where the expectation value is taken in the n-dimensional spacetime generalization of the state described by $\hat{\rho}_{\phi}$. Therefore, differentiating $S_{\rm eff}[g^{\pm}]$ in Eq. (10.35) with respect to g^{+}_{ab}, and then setting $g^{+}_{ab} = g^{-}_{ab} = g_{ab}$, we get the semiclassical Einstein equation in n dimensions. This equation is then renormalized by absorbing the divergences in the regularized $\langle \hat{T}^{ab}[g]\rangle$ into the bare parameters. Taking the limit $n \to 4$ we obtain the physical semiclassical Einstein equation (10.7).

10.4.2 Influence Action for Stochastic Gravity

In the spirit of the previous derivation of the Einstein–Langevin equation in Chapter 10, we now seek a dynamical equation for a linear perturbation h_{ab} to the semiclassical metric g_{ab}, solution of Eq. (10.7). Strictly speaking if we use dimensional regularization we must consider the n-dimensional version of that equation. From the results just described, if such an equation were simply a linearized semiclassical Einstein equation, it could be obtained from an expansion of the effective action $S_{\rm eff}[g+h^{\pm}]$. In particular, since, from Eq. (10.36), we have that

$$\langle \hat{T}^{ab}[g+h;x)\rangle = \frac{2}{\sqrt{-\det(g+h)(x)}} \frac{\delta S_{\rm IF}[g+h^{\pm}]}{\delta h^{+}_{ab}(x)}\bigg|_{h^{\pm}=h}, \tag{10.37}$$

the expansion of $\langle \hat{T}^{ab}[g+h]\rangle$ to linear order in h_{ab} can be obtained from an expansion of the influence action $S_{\rm IF}[g+h^{\pm}]$ up to second order in $h^{\pm}_{ab}$.

To perform the expansion of the influence action, we have to compute the first- and second-order functional derivatives of $S_{\rm IF}[g+h^{\pm}]$ and then set $h^{+}_{ab} = h^{-}_{ab} = h_{ab}$. If we do so using the path integral representation (10.33), we can interpret these derivatives as expectation values of operators. The relevant second-order derivatives are

$$\begin{aligned}
\frac{4}{\sqrt{-g(x)}\sqrt{-g(y)}} \frac{\delta^2 S_{\rm IF}[g+h^{\pm}]}{\delta h^{+}_{ab}(x)\delta h^{+}_{c'd'}(y)}\bigg|_{h^{\pm}=h} &= -H_{\rm S}^{abc'd'}[g;x,y) - K^{abc'd'}[g;x,y) \\
&\quad + iN^{abc'd'}[g;x,y), \\
\frac{4}{\sqrt{-g(x)}\sqrt{-g(y)}} \frac{\delta^2 S_{\rm IF}[g+h^{\pm}]}{\delta h^{+}_{ab}(x)\delta h^{-}_{c'd'}(y)}\bigg|_{h^{\pm}=h} &= -H_{\rm A}^{abc'd'}[g;x,y) - iN^{abc'd'}[g;x,y),
\end{aligned} \tag{10.38}$$

where

$$\begin{aligned}
N^{abc'd'}[g;x,y) &\equiv \frac{1}{2}\left\langle \{\hat{t}^{ab}[g;x), \hat{t}^{c'd'}[g;y)\}\right\rangle, \\
H_{\rm S}^{abc'd'}[g;x,y) &\equiv {\rm Im}\left\langle \mathcal{T}^{*}\left(\hat{T}^{ab}[g;x)\hat{T}^{c'd'}[g;y)\right)\right\rangle,
\end{aligned}$$

$$H_{\rm A}^{abc'd'}[g;x,y) \equiv -\frac{i}{2}\left\langle \left[\hat{T}^{ab}[g;x),\, \hat{T}^{c'd'}[g;y)\right]\right\rangle,$$

$$K^{abc'd'}[g;x,y) \equiv \frac{-4}{\sqrt{-g(x)}\sqrt{-g(y)}}\left\langle \frac{\delta^2 S_m[g+h,\phi]}{\delta h_{ab}(x)\delta h_{c'd'}(y)}\bigg|_{\phi=\hat{\phi}}\right\rangle,$$

with $\hat{t}^{ab}$ defined in Eq. (10.12), [,] denotes the commutator and { , } the anti-commutator. Here we use a Weyl ordering prescription for the operators. The symbol $\mathcal{T}^*$ denotes the following ordered operations: First, time order the field operators $\hat{\phi}$ and then apply the derivative operators which appear in each term of the product $T^{ab}(x)T^{c'd'}(y)$, where T^{ab} is the functional (10.3). This $\mathcal{T}^*$ "time ordering" arises because we have path integrals containing products of derivatives of the field, which can be expressed as derivatives of the path integrals which do not contain such derivatives. Notice, from their definitions, that all the kernels which appear in expressions (10.38) are real and also $H_{\rm A}^{abc'd'}$ is free of ultraviolet divergences in the limit $n \to 4$.

From Eqs. (10.36) and (10.38), since $S_{\rm IF}[g,g] = 0$ and $S_{\rm IF}[g^+,g^-] = -S_{\rm IF}^*[g^-,g^+]$, we can write the expansion for the influence action $S_{\rm IF}[g+h^\pm]$ around a background metric g_{ab} in terms of the previous kernels. Taking into account that these kernels satisfy the symmetry relations

$$\begin{aligned} &H_{\rm S}^{abc'd'}(x,y) = H_{\rm S}^{c'd'ab}(y,x), \quad H_{\rm A}^{abc'd'}(x,y) = -H_{\rm A}^{c'd'ab}(y,x), \\ &K^{abc'd'}(x,y) = K^{c'd'ab}(y,x), \end{aligned} \tag{10.39}$$

and introducing the new kernel

$$H^{abc'd'}(x,y) \equiv H_{\rm S}^{abc'd'}(x,y) + H_{\rm A}^{abc'd'}(x,y), \tag{10.40}$$

the expansion of $S_{\rm IF}$ can be finally written as

$$\begin{aligned} S_{\rm IF}[g+h^\pm] = {} & \frac{1}{2}\int d^4x\,\sqrt{-g(x)}\,\langle \hat{T}^{ab}[g;x)\rangle\,[h_{ab}(x)] \\ & - \frac{1}{8}\int d^4x\,d^4y\,\sqrt{-g(x)}\sqrt{-g(y)} \\ & \times [h_{ab}(x)]\left(H^{abc'd'}[g;x,y) + K^{abc'd'}[g;x,y)\right)\{h_{c'd'}(y)\} \\ & + \frac{i}{8}\int d^4x\,d^4y\,\sqrt{-g(x)}\sqrt{-g(y)} \\ & \times [h_{ab}(x)]\,N^{abc'd'}[g;x,y)\,[h_{c'd'}(y)] + \mathcal{O}(h^3), \end{aligned} \tag{10.41}$$

where we have used the notation

$$[h_{ab}] \equiv h_{ab}^+ - h_{ab}^-, \qquad \{h_{ab}\} \equiv h_{ab}^+ + h_{ab}^-. \tag{10.42}$$

From Eqs. (10.41) and (10.37) it is clear that the imaginary part of the influence action does not contribute to the perturbed semiclassical Einstein equation

(the expectation value of the stress-energy tensor is real); however, as it depends on the noise kernel, it contains information on the fluctuations of the operator $\hat{T}^{ab}[g]$.

We are now in a position to carry out the derivation of the semiclassical Einstein–Langevin equation. The procedure is well known; see e.g. [216, 185, 186, 440, 717, 718, 606]. It consists of deriving a new "stochastic" effective action from the observation that the effect of the imaginary part of the influence action (10.41) on the corresponding influence functional is equivalent to the averaged effect of a stochastic source ξ^{ab} coupled linearly to the perturbations $h^{\pm}_{ab}$. This observation follows from the identity first invoked by Feynman and Vernon [187] for such purpose. Thus we write

$$\exp\left(-\frac{1}{8}\int d^4x\, d^4y\, \sqrt{-g(x)}\sqrt{-g(y)}\; [h_{ab}(x)]\; N^{abc'd'}(x,y)\; [h_{c'd'}(y)]\right)$$
$$= \int \mathcal{D}\xi\, \mathcal{P}[\xi]\, \exp\left(\frac{i}{2}\int d^4x\, \sqrt{-g(x)}\, \xi^{ab}(x)\; [h_{ab}(x)]\right), \tag{10.43}$$

where $\mathcal{P}[\xi]$ is the probability distribution functional of a Gaussian stochastic tensor ξ^{ab} characterized by the correlators (10.13) with $N^{abc'd'}$ given by Eq. (10.11), and where the path integration measure is assumed to be a scalar under diffeomorphisms of $(\mathcal{M}, g_{ab})$.

The above identity follows from the identification of the right-hand side of (10.43) with the characteristic functional for the stochastic field ξ^{ab}. The probability distribution functional for ξ^{ab} is explicitly given by

$$\mathcal{P}[\xi] = \frac{1}{\det(2\pi N)^{1/2}} \exp\left[-\frac{1}{2}\int d^4x d^4y \sqrt{-g(x)}\sqrt{-g(y)}\xi^{ab}(x) N^{-1}_{abc'd'}(x,y)\xi^{c'd'}(y)\right]. \tag{10.44}$$

We may now introduce the *stochastic effective action* as

$$S^s_{\rm eff}[g+h^{\pm},\xi] \equiv S_{\rm g}[g+h^{+}] - S_{\rm g}[g+h^{-}] + S^s_{\rm IF}[g+h^{\pm},\xi], \tag{10.45}$$

where the "stochastic" influence action is defined as

$$S^s_{\rm IF}[g+h^{\pm},\xi] \equiv {\rm Re}\, S_{\rm IF}[g+h^{\pm}] + \frac{1}{2}\int d^4x\, \sqrt{-g(x)}\, \xi^{ab}(x)\, [h_{ab}(x)] + \mathcal{O}(h^3). \tag{10.46}$$

Note that, in fact, the influence functional can now be written as a statistical average over ξ^{ab}: $\mathcal{F}_{\rm IF}[g+h^{\pm}] = \langle \exp\left(iS^s_{\rm IF}[g+h^{\pm},\xi]\right)\rangle_s$. The stochastic equation of motion for h_{ab} reads

$$\left.\frac{\delta S^s_{\rm eff}[g+h^{\pm},\xi]}{\delta h^{+}_{ab}(x)}\right|_{h^{\pm}=h} = 0, \tag{10.47}$$

which is the Einstein–Langevin equation (10.14); notice that only the real part of $S_{\rm IF}$ contributes to the expectation value (10.37). To be precise we get first

the regularized n-dimensional equations with the bare parameters, and instead of the tensor A^{ab} we get $(2/3)D^{ab}$ where

$$\begin{aligned} D^{ab} &\equiv \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g_{ab}}\int d^n x\,\sqrt{-g}\left(R_{cdef}R^{cdef} - R_{cd}R^{cd}\right) \\ &= \frac{1}{2}\,g^{ab}\big(R_{cdef}R^{cdef} - R_{cd}R^{cd} + \Box R\big) - 2R^{acde}R^{b}_{cde} \\ &\quad - 2R^{acbd}R_{cd} + 4R^{ac}R_{c}{}^{b} - 3\Box R^{ab} + \nabla^a\nabla^b R. \end{aligned} \tag{10.48}$$

Of course, when $n = 4$ these tensors are related, $A^{ab} = (2/3)D^{ab}$. After that we renormalize and take the limit $n \to 4$ to obtain the Einstein–Langevin equations in the physical spacetime.

10.5 Explicit Form of the Einstein–Langevin Equation

We can write the Einstein–Langevin equation in a more explicit form by working out the expansion of $\langle\hat{T}^{ab}[g+h]\rangle$ up to linear order in the perturbation h_{ab}. From Eq. (10.37), we see that this expansion can be easily obtained from (10.41). The result is

$$\begin{aligned} \langle\hat{T}^{ab}_n[g+h;x)\rangle &= \langle\hat{T}^{ab}_n[g,x)\rangle + \langle\hat{T}^{(1)ab}_n[g,h;x)\rangle \\ &\quad - \frac{1}{2}\int d^n y\,\sqrt{-g(y)}H^{abc'd'}_n[g;x,y)h_{c'd'}(y) + \mathcal{O}(h^2). \end{aligned} \tag{10.49}$$

Here we use a subscript n on a given tensor to indicate that we are explicitly working in n-dimensions, as we use dimensional regularization, and we also use the superscript (1) to generally indicate that the tensor is the first order correction, linear in h_{ab}, in a perturbative expansion around the background g_{ab}.

Using the Klein–Gordon equation (10.2), and expressions (10.3) for the stress-energy tensor and the corresponding operator, we can write

$$\hat{T}^{(1)ab}_n[g,h] = \left(\frac{1}{2}\,g^{ab}h_{cd} - \delta^a_c h^b_d - \delta^b_c h^a_d\right)\hat{T}^{cd}_n[g] + \mathcal{F}^{ab}[g,h]\,\hat{\phi}^2_n[g], \tag{10.50}$$

where $\mathcal{F}^{ab}[g;h]$ is the differential operator

$$\begin{aligned} \mathcal{F}^{ab} \equiv \left(\xi - \frac{1}{4}\right)\left(h^{ab} - \frac{1}{2}\,g^{ab}h^c_c\right)\Box + \frac{\xi}{2}\big[&\nabla^c\nabla^a h^b_c + \nabla^c\nabla^b h^a_c \\ &-\Box h^{ab} - \nabla^a\nabla^b h^c_c - g^{ab}\nabla^c\nabla^d h_{cd} + g^{ab}\Box h^c_c + \big(\nabla^a h^b_c + \nabla^b h^a_c \\ &-\nabla_c h^{ab} - 2g^{ab}\nabla^d h_{cd} + g^{ab}\nabla_c h^d_d\big)\nabla^c - g^{ab}h_{cd}\nabla^c\nabla^d\big]. \end{aligned} \tag{10.51}$$

It is understood that indices are raised with the background inverse metric g^{ab} and that all the covariant derivatives are associated to the background metric g_{ab}.

Using (10.49) into the n-dimensional version of the Einstein–Langevin equation (10.24), taking into account that g_{ab} satisfies the semiclassical Einstein

equation (10.7), and substituting expression (10.50) we can write the Einstein–Langevin equation in dimensional regularization as

$$\begin{aligned}
&\frac{1}{8\pi G_B}\left[G^{(1)ab}-\frac{1}{2}\,g^{ab}G^{cd}h_{cd}+G^{ac}h^b_c+G^{bc}h^a_c+\Lambda_B\left(h^{ab}-\frac{1}{2}\,g^{ab}h^c_c\right)\right]\\
&\quad-\frac{4\alpha_B}{3}\left(D^{(1)ab}-\frac{1}{2}g^{ab}D^{cd}h_{cd}+D^{ac}h^b_c+D^{bc}h^a_c\right)\\
&\quad-2\beta_B\left(B^{(1)ab}-\frac{1}{2}g^{ab}B^{cd}h_{cd}+B^{ac}h^b_c+B^{bc}h^a_c\right)\\
&\quad-\mu^{-(n-4)}\,\mathcal{F}^{ab}_x\langle\hat{\phi}^2_n[g;x)\rangle+\frac{1}{2}\int d^ny\sqrt{-g(y)}\,\mu^{-(n-4)}H_n^{abc'd'}[g;x,y)h_{c'd'}(y)\\
&\quad=\mu^{-(n-4)}\xi^{ab}_n,
\end{aligned}\tag{10.52}$$

where the tensors G^{ab}, D^{ab} and B^{ab} are computed from the semiclassical metric g_{ab}, and where we have omitted the functional dependence on g_{ab} and h_{ab} in $G^{(1)ab}$, $D^{(1)ab}$, $B^{(1)ab}$ and $\mathcal{F}^{ab}$ to simplify the notation. The parameter μ is a mass scale which relates the dimensions of the physical field ϕ with the dimensions of the corresponding field in n-dimensions, $\phi_n=\mu^{(n-4)/2}\phi$.

Notice that, in equation (10.52), all the ultraviolet divergences in the limit $n\to 4$, which must be removed by renormalization of the coupling constants, are in $\langle\hat{\phi}^2_n(x)\rangle$ and the symmetric part $H^{abc'd'}_{\mathrm{S}n}(x,y)$ of the kernel $H_n^{abc'd'}(x,y)$, whereas the kernels $N_n^{abc'd'}(x,y)$ and $H^{abc'd'}_{\mathrm{A}n}(x,y)$ are free of ultraviolet divergences. If we introduce the bitensor $F_n^{abc'd'}[g;x,y)$ defined by

$$F_n^{abc'd'}[g;x,y)\equiv\left\langle\hat{t}^{ab}_n[g;x)\,\hat{t}^{c'd'}_n[g;y)\right\rangle\tag{10.53}$$

where $\hat{t}^{ab}$ is given by Eq. (10.12), then the kernels N and H_A can be written as

$$N_n^{abc'd'}[g;x,y)=\mathrm{Re}\,F_n^{abc'd'}[g;x,y),\quad H^{abc'd'}_{\mathrm{A}n}[g;x,y)=\mathrm{Im}\,F_n^{abc'd'}[g;x,y),\tag{10.54}$$

where we use that $2\langle\hat{t}^{ab}(x)\,\hat{t}^{c'd'}(y)\rangle=\langle\{\hat{t}^{ab}(x),\,\hat{t}^{c'd'}(y)\}\rangle+\langle[\hat{t}^{ab}(x),\,\hat{t}^{c'd'}(y)]\rangle$, and the fact that the first term on the right-hand side of this identity is real, whereas the second one is pure imaginary. Once we perform the renormalization procedure in Eq. (10.52), setting $n=4$ will yield the physical Einstein–Langevin equation. Due to the presence of the kernel $H_n^{abc'd'}(x,y)$, this equation will be usually nonlocal in the metric perturbation. In Chapter 11 we will carry out an explicit evaluation of the physical Einstein–Langevin equation which will illustrate the procedure.

10.5.1 The Kernels for the Vacuum State

When the expectation values in the Einstein–Langevin equation are taken in a vacuum state $|0\rangle$, such as, for instance, an "in" vacuum, we can be more explicit,

since we can write the expectation values in terms of the Wightman and Feynman functions, defined as

$$
\begin{aligned}
iG_n^+[g;x,y) &\equiv \langle 0|\hat{\phi}_n[g;x)\hat{\phi}_n[g;y)\,|0\rangle,\\
iG_{F_n}[g;x,y) &\equiv \langle 0|\mathcal{T}\left(\hat{\phi}_n[g;x)\hat{\phi}_n[g;y)\right)|0\rangle.
\end{aligned}
\tag{10.55}
$$

These expressions for the kernels in the Einstein–Langevin equation will be very useful for explicit calculations. To simplify the notation, we omit the functional dependence on the semiclassical metric g_{ab}, which will be understood in all the expressions below.

From Eqs. (10.54), we see that the kernels $N_n^{abc'd'}(x,y)$ and $H_{\mathrm{A}n}^{abc'd'}(x,y)$ are the real and imaginary parts, respectively, of the bitensor $F_n^{abc'd'}(x,y)$. From the expression (10.4) we see that the stress-energy operator $\hat{T}_n^{ab}$ can be written as a sum of terms of the form $\left\{\mathcal{A}_x\hat{\phi}_n(x),\,\mathcal{B}_x\hat{\phi}_n(x)\right\}$, where $\mathcal{A}_x$ and $\mathcal{B}_x$ are some differential operators.

It then follows that we can express the bitensor $F_n^{abc'd'}(x,y)$ in terms of the positive Wightman function as

$$
\begin{aligned}
F_n^{abc'd'}(x,y) = &-\nabla_x^a\nabla_y^{c'}G_n^+(x,y)\nabla_x^b\nabla_y^{d'}G_n^+(x,y) - \nabla_x^a\nabla_y^{d'}G_n^+(x,y)\nabla_x^b\nabla_y^{c'}G_n^+(x,y)\\
&- 2\mathcal{D}_x^{ab}\big(\nabla_y^{c'}G_n^+(x,y)\nabla_y^{d'}G_n^+(x,y)\big) - 2\mathcal{D}_y^{c'd'}\big(\nabla_x^aG_n^+(x,y)\nabla_x^bG_n^+(x,y)\big)\\
&- 2\mathcal{D}_x^{ab}\mathcal{D}_y^{c'd'}\big(G_n^{+2}(x,y)\big),
\end{aligned}
\tag{10.56}
$$

where $\mathcal{D}_x^{ab}$ is the differential operator (10.5). From this expression and the relations (10.54), we get expressions for the kernels N_n and $H_{\mathrm{A}n}$ in terms of the positive Wightman function $G_n^+(x,y)$.

Similarly the kernel $H_{\mathrm{S}n}^{abc'd'}(x,y)$, can be written in terms of the Feynman function as

$$
\begin{aligned}
H_{\mathrm{S}n}^{abc'd'}(x,y) = -\mathrm{Im}\Big[&\nabla_x^a\nabla_y^{c'}G_{F_n}(x,y)\nabla_x^b\nabla_y^{d'}G_{F_n}(x,y)\\
&+\nabla_x^a\nabla_y^{d'}G_{F_n}(x,y)\nabla_x^b\nabla_y^{c'}G_{F_n}(x,y)\\
&-g^{ab}(x)\nabla_x^e\nabla_y^{c'}G_{F_n}(x,y)\nabla_e^x\nabla_y^{d'}G_{F_n}(x,y)\\
&-g^{c'd'}(y)\nabla_x^a\nabla_y^{e'}G_{F_n}(x,y)\nabla_x^b\nabla_{e'}^yG_{F_n}(x,y)\\
&+\frac{1}{2}\,g^{ab}(x)g^{c'd'}(y)\nabla_x^e\nabla_y^{f'}G_{F_n}(x,y)\nabla_e^x\nabla_{f'}^yG_{F_n}(x,y)\\
&+\mathcal{K}_x^{ab}\Big(2\nabla_y^{c'}G_{F_n}(x,y)\nabla_y^{d'}G_{F_n}(x,y)\\
&-g^{c'd'}(y)\nabla_y^{e'}G_{F_n}(x,y)\nabla_{e'}^yG_{F_n}(x,y)\Big)\\
&+\mathcal{K}_y^{c'd'}\big(2\nabla_x^aG_{F_n}(x,y)\nabla_x^bG_{F_n}(x,y)
\end{aligned}
$$

$$-g^{ab}(x)\nabla_x^e G_{F_n}(x,y)\nabla_e^x G_{F_n}(x,y)\big)$$
$$+\,2\mathcal{K}_x^{ab}\mathcal{K}_y^{c'd'}\big(G^{\,2}_{F_n}(x,y)\big)\Big]. \tag{10.57}$$

Here $\mathcal{K}_x^{ab}$ is the differential operator

$$\mathcal{K}_x^{ab} \equiv \xi\left(g^{ab}(x)\Box_x - \nabla_x^a\nabla_x^b + G^{ab}(x)\right) - \frac{1}{2}\,m^2 g^{ab}(x). \tag{10.58}$$

Note that, in the vacuum state $|0\rangle$, the term $\langle\hat{\phi}_n^2(x)\rangle$ in Eq. (10.52) can also be written as $\langle\hat{\phi}_n^2(x)\rangle = iG_{F_n}(x,x) = iG_n^+(x,x)$.

Finally, the causality of the Einstein–Langevin equation (10.52) can be explicitly seen as follows. The non-local terms in that equation are due to the kernel $H(x,y)$ which is defined in equation (10.40) as the sum of $H_S(x,y)$ and $H_A(x,y)$. Now, when the points x and y are spacelike separated, $\hat{\phi}_n(x)$ and $\hat{\phi}_n(y)$ commute and, thus, $iG_n^+(x,y) = iG_{F_n}(x,y) = (1/2)\langle 0|\,\{\hat{\phi}_n(x),\hat{\phi}_n(y)\}\,|0\rangle$, which is real. Hence, from the above expressions, we have that $H_{\mathrm{A}_n}^{abc'd'}(x,y) = H_{\mathrm{S}_n}^{abc'd'}(x,y) = 0$, and thus $H_n^{abc'd'}(x,y) = 0$. This fact is expected since, from the causality of the expectation value of the stress-energy operator [102], we know that the non-local dependence on the metric perturbation in the Einstein–Langevin equation must be causal; see Eq. (10.49).

11

Metric Fluctuations in Minkowski Spacetime

Although the Minkowski vacuum is an eigenstate of the total four-momentum operator of a field in Minkowski spacetime, it is not an eigenstate of the stress-energy operator. Hence, even for those solutions of semiclassical gravity such as the Minkowski metric, for which the expectation value of the stress-energy operator can always be chosen to be zero, the fluctuations of this operator are non-vanishing. This fact leads to consider the stochastic metric perturbations induced by these fluctuations.

Here we derive the Einstein–Langevin equation for the metric perturbations in a Minkowski background. We solve this equation for the linearized Einstein tensor and compute the associated two-point correlation functions, as well as the two-point correlation functions for the metric perturbations. Even though, in this case, we expect to have negligibly small values for these correlation functions for points separated by lengths larger than the Planck length, there are several reasons why it is worth carrying out this calculation.

On the one hand, these are the first backreaction solutions of the full Einstein–Langevin equation. There are analogous solutions to a "reduced" version of this equation inspired in a "mini-superspace" model in [218, 219].

On the other hand, the results of this calculation, which confirm our expectations that gravitational fluctuations are negligible at length scales larger than the Planck length, but also predict that the fluctuations are strongly suppressed on small scales, can be considered a first test of stochastic semiclassical gravity. These results also reveal an important connection between stochastic gravity and the large N expansion of quantum gravity. In addition, they are used in Section 11.5 to study the stability of the Minkowski metric as a solution of semiclassical gravity, which constitutes an application of the validity criterion introduced in Section 10.3. This calculation requires also a discussion of the problems posed by the so called *runaway* solutions, which arise in the backreaction equations of

semiclassical and stochastic gravity, and some of the methods to deal with them. As a result we conclude that Minkowski spacetime is a stable and valid solution of semiclassical gravity.

11.1 Perturbations around Minkowski Spacetime

The Minkowski metric η_{ab}, in a manifold $\mathcal{M}$ which is topologically $\mathbb{R}^4$, and the usual Minkowski vacuum, denoted as $|0\rangle$, is the simplest solution to the semiclassical Einstein equation (10.7), the so-called trivial solution of semiclassical gravity [128]. It constitutes the ground state of semiclassical gravity. In fact, we can always choose a renormalization scheme in which the renormalized expectation value of the stress-energy tensor vanishes: $\langle 0|\,\hat{T}^{ab}_{\rm R}\,[\eta]|0\rangle = 0$. Thus, Minkowski spacetime $(\mathbb{R}^4, \eta_{ab})$ and the vacuum state $|0\rangle$ are a solution to the semiclassical Einstein equation with renormalized cosmological constant $\Lambda = 0$.

The fact that the vacuum expectation value of the renormalized stress-energy operator in Minkowski spacetime should vanish was originally proposed by Wald [102] and it may be understood as a renormalization convention [48, 51]. Note that other possible solutions of semiclassical gravity with zero vacuum expectation value of the stress-energy tensor are the exact gravitational plane waves, since these are known to be vacuum solutions of Einstein equations which induce neither particle creation nor vacuum polarization [719, 720, 721].

As we have already mentioned the vacuum $|0\rangle$ is an eigenstate of the total four-momentum operator in Minkowski spacetime, but not an eigenstate of $\hat{T}^{\rm R}_{ab}[\eta]$. Hence, even in the Minkowski background, there are quantum fluctuations in the stress-energy tensor and, as a result, the noise kernel does not vanish. This fact leads to consider the stochastic corrections to this class of trivial solutions of semiclassical gravity. Since, in this case, the Wightman and Feynman functions (10.55), their values in the two-point coincidence limit, and the products of derivatives of two of such functions appearing in expressions (10.56) and (10.57) are known in dimensional regularization, we can compute the Einstein–Langevin equation using the methods outlined in Chapter 10.

To perform explicit calculations it is convenient to work in a global inertial coordinate system $\{x^\mu\}$ and in the associated basis, in which the components of the flat metric are simply $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, \ldots, 1)$. In Minkowski spacetime, the components of the classical stress-energy tensor (10.3) reduce to

$$T^{\mu\nu}[\eta,\phi] = \partial^\mu\phi\partial^\nu\phi - \frac{1}{2}\,\eta^{\mu\nu}\partial^\rho\phi\partial_\rho\phi - \frac{1}{2}\,\eta^{\mu\nu}m^2\phi^2 + \xi\,(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)\,\phi^2, \quad (11.1)$$

where $\Box \equiv \partial_\mu\partial^\mu$, and the formal expression for the components of the corresponding "operator" in dimensional regularization, see Eq. (10.4), is

$$\hat{T}^{\mu\nu}_n[\eta] = \frac{1}{2}\{\partial^\mu\hat{\phi}_n, \partial^\nu\hat{\phi}_n\} + \mathcal{D}^{\mu\nu}\hat{\phi}^2_n, \quad (11.2)$$

where $\mathcal{D}^{\mu\nu}$ is the differential operator (10.5), with $g_{\mu\nu} = \eta_{\mu\nu}, R_{\mu\nu} = 0$, and $\nabla_\mu = \partial_\mu$. The field $\hat{\phi}_n(x)$ is the field operator in the Heisenberg picture in a n-dimensional Minkowski spacetime, which satisfies the Klein–Gordon equation (10.2). We use here a stress-energy tensor which differs from the canonical one, which corresponds to $\xi = 0$, both tensors, however, define the same total momentum.

The Wightman and Feynman functions (10.55) when $g_{\mu\nu} = \eta_{\mu\nu}$, are well known:

$$G_n^+(x,y) = \Delta_n^+(x-y), \quad G_{F_n}(x,y) = \Delta_{F_n}(x-y), \tag{11.3}$$

with

$$\Delta_n^+(x) = -2\pi i \int \frac{d^n k}{(2\pi)^n}\, e^{ik\cdot x}\, \delta(k^2+m^2)\, \theta(k^0),$$
$$\Delta_{F_n}(x) = -\int \frac{d^n k}{(2\pi)^n}\, \frac{e^{ik\cdot x}}{k^2+m^2-i\epsilon}, \epsilon \to 0^+, \tag{11.4}$$

where $k^2 \equiv \eta_{\mu\nu}k^\mu k^\nu$ and $k\cdot x \equiv \eta_{\mu\nu}k^\mu x^\nu$. Note that the derivatives of these functions satisfy $\partial_\mu^x \Delta_n^+(x-y) = \partial_\mu \Delta_n^+(x-y)$ and $\partial_\mu^y \Delta_n^+(x-y) = -\partial_\mu \Delta_n^+(x-y)$, and similarly for the Feynman propagator $\Delta_{F_n}(x-y)$.

To write down the semiclassical Einstein equation (10.7) in n-dimensions for this case, we need to compute the vacuum expectation value of the stress-energy operator components (11.2). Since, from (11.3), we have that $\langle 0|\hat{\phi}_n^2(x)|0\rangle = i\Delta_{F_n}(0) = i\Delta_n^+(0)$, which is a constant (independent of x), we have simply

$$\langle 0|\hat{T}_n^{\mu\nu}[\eta]|0\rangle = -i\int \frac{d^n k}{(2\pi)^n}\, \frac{k^\mu k^\nu}{k^2+m^2-i\epsilon} = \frac{\eta^{\mu\nu}}{2}\left(\frac{m^2}{4\pi}\right)^{n/2}\Gamma\left(-\frac{n}{2}\right), \tag{11.5}$$

where the integrals in dimensional regularization have been computed in the standard way and where $\Gamma(z)$ is the Euler's gamma function. The semiclassical Einstein equation (10.7) in n-dimensions before renormalization reduces now to

$$\frac{\Lambda_B}{8\pi G_B}\,\eta^{\mu\nu} = \mu^{-(n-4)}\langle 0|\hat{T}_n^{\mu\nu}[\eta]|0\rangle. \tag{11.6}$$

This equation, thus, simply sets the value of the bare coupling constant Λ_B/G_B. Note, from (11.5), that in order to have $\langle 0|\,\hat{T}_{\rm R}^{\mu\nu}\,|0\rangle[\eta] = 0$, the renormalized and regularized stress-energy tensor "operator" for a scalar field in Minkowski spacetime, see Eq. (10.6), has to be defined as

$$\hat{T}_{\rm R}^{\mu\nu}[\eta] = \mu^{-(n-4)}\,\hat{T}_n^{\mu\nu}[\eta] - \frac{\eta^{\mu\nu}}{2}\,\frac{m^4}{(4\pi)^2}\left(\frac{m^2}{4\pi\mu^2}\right)^{\frac{n-4}{2}}\Gamma\left(-\frac{n}{2}\right), \tag{11.7}$$

which corresponds to a renormalization of the cosmological constant

$$\frac{\Lambda_B}{G_B} = \frac{\Lambda}{G_{\rm N}} - \frac{2}{\pi}\,\frac{m^4}{n(n-2)}\,\kappa_n + \mathcal{O}(n-4), \tag{11.8}$$

where

$$\kappa_n \equiv \frac{1}{(n-4)}\left(\frac{e^\gamma m^2}{4\pi\mu^2}\right)^{\frac{n-4}{2}} = \frac{1}{n-4} + \frac{1}{2}\,\ln\left(\frac{e^\gamma m^2}{4\pi\mu^2}\right) + \mathcal{O}(n-4), \tag{11.9}$$

being γ the Euler's constant. In the case of a massless scalar field, $m^2 = 0$, one simply has $\Lambda_B/G_B = \Lambda/G_{\rm N}$. Introducing this renormalized coupling constant into Eq. (11.6), we can take the limit $n \to 4$. We find that, for $(\mathbb{R}^4, \eta_{ab}, |0\rangle)$ to satisfy the semiclassical Einstein equation, we must take $\Lambda = 0$.

We can now write down the Einstein–Langevin equations for the components $h_{\mu\nu}$ of the stochastic metric perturbation in dimensional regularization. In our case, using $\langle 0|\hat{\phi}_n^2(x)|0\rangle = i\Delta_{F_n}(0)$ and the explicit expression of Eq. (10.52) we obtain

$$\begin{aligned}
&\frac{1}{8\pi G_B}\left[G^{(1)\mu\nu} + \Lambda_B\left(h^{\mu\nu} - \frac{1}{2}\,\eta^{\mu\nu}h\right)\right](x) \\
&\quad - \frac{4}{3}\,\alpha_B D^{(1)\mu\nu}(x) - 2\beta_B B^{(1)\mu\nu}(x) - \xi G^{(1)\mu\nu}(x)\mu^{-(n-4)} i\Delta_{F_n}(0) \\
&\quad + \frac{1}{2}\int d^n y \mu^{-(n-4)} H_n^{\mu\nu\alpha'\beta'}(x,y) h_{\alpha'\beta'}(y) = \xi^{\mu\nu}(x).
\end{aligned} \tag{11.10}$$

The indices in $h_{\mu\nu}$ are raised with the Minkowski metric, $h \equiv h^\rho_\rho$, and here a superscript (1) denotes the components of a tensor linearized around the flat metric. Note that in n-dimensions the two-point correlation functions for the field $\xi^{\mu\nu}$ is written as

$$\langle \xi^{\mu\nu}(x)\xi^{\alpha'\beta'}(y)\rangle_s = \mu^{-2(n-4)} N_n^{\mu\nu\alpha'\beta'}(x,y), \tag{11.11}$$

Explicit expressions for $D^{(1)\mu\nu}$ and $B^{(1)\mu\nu}$ are given by

$$D^{(1)\mu\nu}(x) = \frac{1}{2}\,\mathcal{F}_x^{\mu\nu\alpha'\beta'}\,h_{\alpha'\beta'}(x), \qquad B^{(1)\mu\nu}(x) = 2\mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha\beta}h_{\alpha\beta}(x), \tag{11.12}$$

where we introduced the differential operators $\mathcal{F}_x^{\mu\nu} \equiv \eta^{\mu\nu}\Box_x - \partial_x^\mu\partial_x^\nu$ and $\mathcal{F}_x^{\mu\nu\alpha'\beta'} \equiv 3\mathcal{F}_x^{\mu(\alpha'}\mathcal{F}_x^{\beta')\nu} - \mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}$. Note that to distinguish tensor indices in x from tensor indices in y is unnecessary in the Minkowski background with standard coordinates. However, we will follow the convention of keeping the last two indices primed in bitensors in order to have an homogeneous notation with previous chapters.

11.2 The Kernels in the Minkowski Background

Since the two kernels (10.54) are free of ultraviolet divergences in the limit $n \to 4$, we can deal directly with the $F^{\mu\nu\alpha'\beta'}(x-y) \equiv \lim_{n\to 4}\mu^{-2(n-4)}\,F_n^{\mu\nu\alpha'\beta'}$ in Eq. (10.53). Also $N^{\mu\nu\alpha'\beta'}(x,y) = \mathrm{Re}\,F^{\mu\nu\alpha'\beta'}(x-y)$ and $H_{\rm A}^{\mu\nu\alpha'\beta'}(x,y) = \mathrm{Im}\,F^{\mu\nu\alpha'\beta'}(x-y)$ are actually the components of the "physical" noise and dissipation kernels that will appear in the Einstein–Langevin equations once the

renormalization procedure has been carried out. The bitensor $F^{\mu\nu\alpha'\beta'}$ can be expressed in terms of the positive Wightman function in four spacetime dimensions, according to (10.56). The different terms in this kernel can be easily computed using the integrals

$$I(p) \equiv \int \frac{d^4k}{(2\pi)^4}\, \delta(k^2+m^2)\, \theta(-k^0)\, \delta[(k-p)^2+m^2]\, \theta(k^0-p^0), \tag{11.13}$$

and $I^{\mu_1\cdots\mu_r}(p)$ which are defined as the previous one by inserting the momenta $k^{\mu_1}\ldots k^{\mu_r}$ with $r=1,2,3,4$ in the integrand. All these integrals can be expressed in terms of $I(p)$; see [189] for the explicit expressions.

It is convenient to separate $I(p)$ in its even and odd parts with respect to the variables p^μ as

$$I(p) = I_{\rm S}(p) + I_{\rm A}(p), \tag{11.14}$$

where $I_{\rm S}(-p) = I_{\rm S}(p)$ and $I_{\rm A}(-p) = -I_{\rm A}(p)$. These two functions are explicitly given by

$$\begin{aligned} I_{\rm S}(p) &= \frac{1}{8\,(2\pi)^3}\, \theta(-p^2-4m^2)\, \sqrt{1+4\,\frac{m^2}{p^2}}, \\ I_{\rm A}(p) &= \frac{-1}{8\,(2\pi)^3}\, \mathrm{sign}(p^0)\theta(-p^2-4m^2)\, \sqrt{1+4\,\frac{m^2}{p^2}}. \end{aligned} \tag{11.15}$$

After some manipulations, we find

$$\begin{aligned} F^{\mu\nu\alpha'\beta'}(x) = {}& \frac{\pi^2}{45}\, \mathcal{F}_x^{\mu\nu\alpha'\beta'} \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x} \left(1+4\frac{m^2}{p^2}\right)^2 I(p) \\ & + \frac{8\pi^2}{9} \mathcal{F}_x^{\mu\nu} \mathcal{F}_x^{\alpha'\beta'} \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x} \left(3\Delta\xi + \frac{m^2}{p^2}\right)^2 I(p), \end{aligned} \tag{11.16}$$

where $\Delta\xi \equiv \xi - 1/6$. The real and imaginary parts of the last expression, which yield the noise and dissipation kernels, are easily recognized as the terms containing $I_{\rm S}(p)$ and $I_{\rm A}(p)$, respectively.

To write them explicitly, it is useful to introduce the new kernels

$$\begin{aligned} N_{\rm A}(x;m^2) &\equiv \frac{1}{480\pi} \int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot x}\, \theta(-p^2-4m^2)\, \sqrt{1+4\,\frac{m^2}{p^2}} \left(1+4\,\frac{m^2}{p^2}\right)^2, \\ N_{\rm B}(x;m^2,\Delta\xi) &\equiv \frac{1}{72\pi} \int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot x}\, \theta(-p^2-4m^2) \\ &\quad \times \sqrt{1+4\,\frac{m^2}{p^2}} \left(3\Delta\xi + \frac{m^2}{p^2}\right)^2, \end{aligned}$$

$$D_{\rm A}(x;m^2) \equiv \frac{-i}{480\pi}\int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot x}\,\mathrm{sign}(p^0)\theta(-p^2-4m^2) \times\sqrt{1+4\,\frac{m^2}{p^2}}\left(1+4\,\frac{m^2}{p^2}\right)^2,$$

$$D_{\rm B}(x;m^2,\Delta\xi) \equiv \frac{-i}{72\pi}\int \frac{d^4p}{(2\pi)^4}\, e^{ip\cdot x}\,\mathrm{sign}(p^0)\theta(-p^2-4m^2) \times\sqrt{1+4\,\frac{m^2}{p^2}}\left(3\Delta\xi+\frac{m^2}{p^2}\right)^2, \tag{11.17}$$

and we finally obtain

$$N^{\mu\nu\alpha'\beta'}(x,y) = \frac{1}{6}\mathcal{F}_x^{\mu\nu\alpha'\beta'}N_{\rm A}(x-y;m^2) + \mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}N_{\rm B}(x-y;m^2,\Delta\xi),$$

$$H_{\rm A}^{\mu\nu\alpha'\beta'}(x,y) = \frac{1}{6}\mathcal{F}_x^{\mu\nu\alpha'\beta'}D_{\rm A}(x-y;m^2) + \mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}D_{\rm B}(x-y;m^2,\Delta\xi). \tag{11.18}$$

Notice that the noise and dissipation kernels defined in (11.17) are actually real because, for the noise kernels, only the $\cos(p\cdot x)$ terms of the exponentials $e^{ip\cdot x}$ contribute to the integrals, and, for the dissipation kernels, the only contribution of such exponentials comes from the $i\sin(p\cdot x)$ terms.

The evaluation of the kernel $H_{{\rm S}_n}^{\mu\nu\alpha'\beta'}(x,y)$ is a more involved task. Since this kernel contains divergences in the limit $n\to 4$, we use dimensional regularization. Using Eq. (10.57), this kernel can be written in terms of the Feynman propagator (11.4) as

$$\mu^{-(n-4)}H_{{\rm S}_n}^{\mu\nu\alpha'\beta'}(x,y) = \mathrm{Im}\,K^{\mu\nu\alpha'\beta'}(x-y), \tag{11.19}$$

where

$$\begin{aligned}K^{\mu\nu\alpha'\beta'}(x) \equiv -\mu^{-(n-4)}\Big\{ & 2\partial^\mu\partial^{(\alpha'}\Delta_{F_n}(x)\,\partial^{\beta')}\partial^\nu\Delta_{F_n}(x) \\ & + 2\mathcal{D}^{\mu\nu}\left(\partial^{\alpha'}\Delta_{F_n}(x)\partial^{\beta'}\Delta_{F_n}(x)\right) + 2\mathcal{D}^{\alpha'\beta'}\left(\partial^\mu\Delta_{F_n}(x)\,\partial^\nu\Delta_{F_n}(x)\right) \\ & + 2\mathcal{D}^{\mu\nu}\mathcal{D}^{\alpha'\beta'}\left(\Delta_{F_n}^2(x)\right) + \Big[\eta^{\mu\nu}\partial^{(\alpha'}\Delta_{F_n}(x)\partial^{\beta')} + \eta^{\alpha'\beta'}\partial^{(\mu}\Delta_{F_n}(x)\partial^{\nu)} \\ & + \Delta_{F_n}(0)\left(\eta^{\mu\nu}\mathcal{D}^{\alpha'\beta'}+\eta^{\alpha'\beta'}\mathcal{D}^{\mu\nu}\right) \\ & + \frac{1}{4}\eta^{\mu\nu}\eta^{\alpha'\beta'}\left(\Delta_{F_n}(x)\Box - m^2\Delta_{F_n}(0)\right)\Big]\delta^n(x)\Big\}.\end{aligned} \tag{11.20}$$

Let us define the integrals

$$J_n(p) \equiv \mu^{-(n-4)}\int \frac{d^nk}{(2\pi)^n}\,\frac{1}{(k^2+m^2-i\epsilon)\,[(k-p)^2+m^2-i\epsilon]}, \tag{11.21}$$

and $J_n^{\mu_1\cdots\mu_r}(p)$ obtained by inserting the momenta $k^{\mu_1}\dots k^{\mu_r}$ into the previous integral, together with

$$I_{0_n} \equiv \mu^{-(n-4)} \int \frac{d^n k}{(2\pi)^n} \frac{1}{(k^2+m^2-i\epsilon)}, \tag{11.22}$$

and $I_{0_n}^{\mu_1\cdots\mu_r}$ which are also obtained by inserting momenta in the integrand. Then, the different terms in Eq. (11.20) can be computed. These integrals are explicitly given in [189]. It is found that $I_{0_n}^{\mu} = 0$ and the remaining integrals can be written in terms of I_{0_n} and $J_n(p)$. It is useful to introduce the projector $P^{\mu\nu}$ orthogonal to p^μ and the tensor $P^{\mu\nu\alpha'\beta'}$ as

$$p^2 P^{\mu\nu} \equiv \eta^{\mu\nu}p^2 - p^\mu p^\nu, \qquad P^{\mu\nu\alpha'\beta'} \equiv 3P^{\mu(\alpha'}P^{\beta')\nu} - P^{\mu\nu}P^{\alpha'\beta'}, \tag{11.23}$$

then the action of the operator $\mathcal{F}_x^{\mu\nu}$ is simply written as $\mathcal{F}_x^{\mu\nu}\int d^n p\, e^{ip\cdot x} f(p) = -\int d^n p\, e^{ip\cdot x} f(p)\, p^2 P^{\mu\nu}$ where $f(p)$ is an arbitrary function of p^μ.

Finally after a rather long but straightforward calculation, and after expanding around $n = 4$, we get,

$$\begin{aligned}
K^{\mu\nu\alpha'\beta'}(x) = \frac{i}{(4\pi)^2}\Bigg\{ \kappa_n \Bigg[& \frac{1}{90}\mathcal{F}_x^{\mu\nu\alpha'\beta'}\delta^n(x) + 4\Delta\xi^2 \mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}\delta^n(x) \\
& + \frac{2}{3}\frac{m^2}{(n-2)}\left(\eta^{\mu\nu}\eta^{\alpha'\beta'}\Box_x - \eta^{\mu(\alpha'}\eta^{\beta')\nu}\Box_x + \eta^{\mu(\alpha'}\partial_x^{\beta')}\partial_x^\nu\right. \\
& \left. + \eta^{\nu(\alpha'}\partial_x^{\beta')}\partial_x^\mu - \eta^{\mu\nu}\partial_x^{\alpha'}\partial_x^{\beta'} - \eta^{\alpha'\beta'}\partial_x^\mu\partial_x^\nu\right)\delta^n(x) \\
& + \frac{4m^4}{n(n-2)}\left(2\eta^{\mu(\alpha'}\eta^{\beta')\nu} - \eta^{\mu\nu}\eta^{\alpha'\beta'}\right)\delta^n(x)\Bigg] \\
& + \frac{1}{180}\mathcal{F}_x^{\mu\nu\alpha'\beta'}\int\frac{d^n p}{(2\pi)^n}e^{ip\cdot x}\left(1+4\,\frac{m^2}{p^2}\right)^2\bar{\phi}(p^2) \\
& + \frac{2}{9}\,\mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}\int\frac{d^n p}{(2\pi)^n}\,e^{ip\cdot x}\left(3\Delta\xi + \frac{m^2}{p^2}\right)^2\bar{\phi}(p^2) \\
& - \left[\frac{4}{675}\,\mathcal{F}_x^{\mu\nu\alpha'\beta'} + \frac{1}{270}\,(60\xi-11)\,\mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}\right]\delta^n(x) \\
& - m^2\left[\frac{2}{135}\,\mathcal{F}_x^{\mu\nu\alpha'\beta'} + \frac{1}{27}\,\mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}\right]\Delta_n(x)\Bigg\} + \mathcal{O}(n-4),
\end{aligned} \tag{11.24}$$

where κ_n has been defined in (11.9), and $\bar{\phi}(p^2)$ and $\Delta_n(x)$ are given by

$$\begin{aligned}
\bar{\phi}(p^2) &\equiv \int_0^1 d\alpha\, \ln\left(1 + \frac{p^2}{m^2}\alpha(1-\alpha) - i\epsilon\right) \\
&= -i\pi\theta(-p^2-4m^2)\sqrt{1+4\frac{m^2}{p^2}} + \varphi(p^2),
\end{aligned} \tag{11.25}$$

$$\Delta_n(x) \equiv \int \frac{d^n p}{(2\pi)^n}\, e^{ip\cdot x}\, \frac{1}{p^2}, \tag{11.26}$$

where $\varphi(p^2) \equiv \int_0^1 d\alpha \ln|1+\frac{p^2}{m^2}\,\alpha(1-\alpha)|$. The imaginary part of (11.24) gives the kernel components $\mu^{-(n-4)} H_{\mathrm{S}_n}^{\mu\nu\alpha'\beta'}(x,y)$, according to (11.19). It can be easily obtained multiplying this expression by $-i$ and retaining only the real part, $\varphi(p^2)$, of the function $\bar{\phi}(p^2)$.

11.3 Einstein–Langevin Equation

With the previous results for the kernels we can now write the n-dimensional Einstein–Langevin equation (11.24), previous to the renormalization. Taking also into account Eqs. (11.5) and (11.6), we may finally write:

$$\begin{aligned}
&\frac{1}{8\pi G_B} G^{(1)\mu\nu}(x) - \frac{4}{3}\,\alpha_B D^{(1)\mu\nu}(x) - 2\beta_B B^{(1)\mu\nu}(x) \\
&\quad + \frac{\kappa_n}{(4\pi)^2}\left[-4\Delta\xi\,\frac{m^2}{(n-2)}\,G^{(1)\mu\nu} + \frac{1}{90}\,D^{(1)\mu\nu}\Delta\xi^2 B^{(1)\mu\nu}\right](x) \\
&\quad + \frac{1}{2880\pi^2}\left\{-\frac{16}{15}\,D^{(1)\mu\nu}(x) + \left(\frac{1}{6} - 10\Delta\xi\right) B^{(1)\mu\nu}(x)\right. \\
&\quad + \int d^n y \int \frac{d^n p}{(2\pi)^n}\, e^{ip\cdot(x-y)}\,\varphi(p^2)\left[\left(1+4\frac{m^2}{p^2}\right)^2 D^{(1)\mu\nu}(y)\right. \\
&\qquad\qquad \left.\left. +10\left(3\Delta\xi + \frac{m^2}{p^2}\right)^2 B^{(1)\mu\nu}(y)\right]\right. \\
&\quad \left. - \frac{m^2}{3}\int d^n y\,\Delta_n(x-y)\left(8D^{(1)\mu\nu} + 5B^{(1)\mu\nu}\right)(y)\right\} \\
&\quad + \frac{1}{2}\int d^n y\,\mu^{-(n-4)} H_{\mathrm{A}_n}^{\mu\nu\alpha'\beta'}(x,y)\, h_{\alpha'\beta'}(y) + \mathcal{O}(n-4) = \xi^{\mu\nu}(x).
\end{aligned} \tag{11.27}$$

Notice that the terms containing the bare cosmological constant have cancelled. These equations can now be renormalized, that is, we can now write the bare coupling constants as renormalized coupling constants plus some suitably chosen counterterms and take the limit $n \to 4$. In order to carry out such a procedure, it is convenient to distinguish between massive and massless scalar fields.

It is convenient to introduce the two new kernels

$$\begin{aligned}
H_{\mathrm{A}}(x;m^2) \equiv \frac{1}{480\pi^2}\int \frac{d^4 p}{(2\pi)^4}\, e^{ip\cdot x}\left\{\left(1+4\,\frac{m^2}{p^2}\right)^2\right. \\
\left.\times\left[-i\pi\,\mathrm{sign}(p^0)\,\theta(-p^2-4m^2)\sqrt{1+4\frac{m^2}{p^2}} + \varphi(p^2)\right] - \frac{8}{3}\,\frac{m^2}{p^2}\right\},
\end{aligned}$$

$$H_{\rm B}(x;m^2,\Delta\xi)\equiv\frac{1}{72\pi^2}\int\frac{d^4p}{(2\pi)^4}e^{ip\cdot x}\Bigg\{\left(3\Delta\xi+\frac{m^2}{p^2}\right)^2 \times\left[-i\pi{\rm sign}(p^0)\theta(-p^2-4m^2)\sqrt{1+4\frac{m^2}{p^2}}+\varphi(p^2)\right]-\frac{1}{6}\frac{m^2}{p^2}\Bigg\}, \tag{11.28}$$

where $\varphi(p^2)$ is given by the restriction to $n=4$ of expression (11.25). The renormalized coupling constants $1/G_{\rm N}$, α and β are easily computed as it was done in Eq. (11.8). Substituting their expressions into Eq. (11.27), we can take the limit $n\to 4$, using the fact that, for $n=4, D^{(1)\mu\nu}(x)=(3/2)\,A^{(1)\mu\nu}(x)$, we obtain the corresponding semiclassical Einstein–Langevin equation.

For the massless case one needs the limit $m\to 0$ of Eq. (11.27). In this case it is convenient to separate κ_n in (11.9) as $\kappa_n=\tilde{\kappa}_n+\frac{1}{2}\ln(m^2/\mu^2)+\mathcal{O}(n-4)$, where

$$\tilde{\kappa}_n\equiv\frac{1}{(n-4)}\left(\frac{e^\gamma}{4\pi}\right)^{\frac{n-4}{2}}=\frac{1}{n-4}+\frac{1}{2}\,\ln\left(\frac{e^\gamma}{4\pi}\right)+\mathcal{O}(n-4), \tag{11.29}$$

and use that, from Eq. (11.25), we have

$$\lim_{m^2\to 0}\left[\varphi(p^2)+\ln(m^2/\mu^2)\right]=-2+\ln\left|\frac{p^2}{\mu^2}\right|. \tag{11.30}$$

The coupling constants are then easily renormalized. We note that in the massless limit, the Newtonian gravitational constant is not renormalized and, in the conformal coupling case, $\Delta\xi=0$, we have that $\beta_B=\beta$. Note also that, by making $m=0$ in (11.17), the noise and dissipation kernels can be written as

$$\begin{aligned} &N_{\rm A}(x;m^2=0)=N(x), \qquad N_{\rm B}(x;m^2=0,\Delta\xi)=60\Delta\xi^2N(x),\\ &D_{\rm A}(x;m^2=0)=D(x), \qquad D_{\rm B}(x;m^2=0,\Delta\xi)=60\Delta\xi^2D(x), \end{aligned} \tag{11.31}$$

where

$$N(x)\equiv\frac{1}{480\pi}\int\frac{d^4p}{(2\pi)^4}e^{ip\cdot x}\theta(-p^2), \qquad D(x)\equiv\frac{-i}{480\pi}\int\frac{d^4p}{(2\pi)^4}e^{ip\cdot x}{\rm sign}\,p^0\theta(-p^2). \tag{11.32}$$

It is also convenient to introduce the new kernel

$$\begin{aligned} H(x;\mu^2)&\equiv\frac{1}{480\pi^2}\int\frac{d^4p}{(2\pi)^4}\,e^{ip\cdot x}\left[\ln\left|\frac{p^2}{\mu^2}\right|-i\pi\,{\rm sign}(p^0)\theta(-p^2)\right]\\ &=\frac{1}{480\pi^2}\lim_{\epsilon\to 0^+}\int\frac{d^4p}{(2\pi)^4}\,e^{ip\cdot x}\,\ln\left(\frac{-(p^0+i\epsilon)^2+p^ip_i}{\mu^2}\right). \end{aligned} \tag{11.33}$$

This kernel is real and can be written as the sum of an even part and an odd part in the variables x^μ, where the odd part is the dissipation kernel $D(x)$. The Fourier transforms (11.32) and (11.33) can actually be computed and, thus, in

this case we have explicit expressions for the kernels in position space; see for instance [722, 723, 213].

Finally, the Einstein–Langevin equation for the physical stochastic perturbations $h_{\mu\nu}$ can be written in both cases, for $m \neq 0$ and for $m = 0$, as

$$\frac{1}{8\pi G_{\rm N}} G^{(1)\mu\nu}(x) - 2\left(\bar{\alpha} A^{(1)\mu\nu}(x) + \bar{\beta} B^{(1)\mu\nu}(x)\right) + \frac{1}{4}\int d^4y \left[H_{\rm A}(x-y) A^{(1)\mu\nu}(y) + H_{\rm B}(x-y) B^{(1)\mu\nu}(y)\right] = \xi^{\mu\nu}(x), \quad (11.34)$$

where in terms of the renormalized constants α_d and β_d the new constants are $\bar{\alpha} = \alpha_d (8\pi G_{\rm N})^{-1} + (3600\pi^2)^{-1}$ and $\bar{\beta} = \beta_d (8\pi G_{\rm N})^{-1} - (1/12 - 5\Delta\xi)(2880\pi^2)^{-1}$. The kernels $H_{\rm A}(x)$ and $H_{\rm B}(x)$ are given by Eqs. (11.28) when $m \neq 0$, and $H_{\rm A}(x) = H(x;\mu^2)$, $H_{\rm B}(x) = 60\Delta\xi^2 H(x;\mu^2)$ when $m = 0$.

In the massless case, we can use the arbitrariness of the mass scale μ to eliminate one of the parameters $\bar{\alpha}$ or $\bar{\beta}$. The components of the Gaussian stochastic source $\xi^{\mu\nu}$ have zero mean value and their two-point correlation functions are given by $\langle \xi^{\mu\nu}(x) \xi^{\alpha'\beta'}(y)\rangle_s = N^{\mu\nu\alpha'\beta'}(x,y)$, where the noise kernel is given in (11.18), which in the massless case reduces to (11.31).

It is interesting to consider the massless conformally coupled scalar field, i.e., the case $\Delta\xi = 0$, of particular interest because of its similarities with the electromagnetic field, and also because of its interest in cosmology: matter fields in the standard model of particle physics become effectively conformally invariant in the very early universe. We have already mentioned that for a conformally coupled field, the stochastic source tensor must be traceless (up to first-order perturbations around semiclassical gravity), in the sense that the stochastic variable $\xi^\mu_\mu \equiv \eta_{\mu\nu}\xi^{\mu\nu}$ behaves deterministically as a vanishing scalar field. This can be directly checked by noticing, from Eqs. (11.18) and (11.31), that, when $\Delta\xi = 0$, one has $\langle \xi^\mu_\mu(x) \xi^{\alpha'\beta'}(y)\rangle_s = 0$, since $\mathcal{F}^\mu_\mu = 3\Box$ and $\mathcal{F}^{\mu\alpha}\mathcal{F}^\beta_\mu = \Box\mathcal{F}^{\alpha\beta}$. The Einstein–Langevin equations for this particular case (and generalized to a spatially flat FLRW background) were first obtained by Campos and Verdaguer [186]. We review this derivation in Chapter 12.

11.3.1 Stress-Energy Tensor

The expectation value of the renormalized stress-energy tensor for a scalar field can be obtained by comparing the previous equation (11.34) with the Einstein–Langevin equation (10.14).

For a massive scalar field, $m \neq 0$, we get

$$\langle \hat{T}^{(1)\mu\nu}_{\rm R}(x)\rangle = \frac{1}{2880\pi^2}\left[\frac{8}{5} A^{(1)\mu\nu}(x) - \left(\frac{1}{6} - 10\Delta\xi\right) B^{(1)\mu\nu}(x)\right] - \int d^4y \left[H_{\rm A}(x-y;m^2) A^{(1)\mu\nu}(y) + H_{\rm B}(x-y;m^2,\Delta\xi) B^{(1)\mu\nu}(y)\right] + \mathcal{O}(h^2). \quad (11.35)$$

For a massless scalar field, $m = 0$, we get

$$\langle \hat{T}_{\rm R}^{(1)\mu\nu}(x)\rangle = \frac{1}{2880\pi^2}\left[\frac{8}{5}\,A^{(1)\mu\nu}(x) - \left(\frac{1}{6} - 10\Delta\xi\right)B^{(1)\mu\nu}(x)\right] - \int d^4y\, H(x-y;\mu^2)\left[A^{(1)\mu\nu}(y) + 60\Delta\xi^2 B^{(1)\mu\nu}(y)\right] + \mathcal{O}(h^2), \tag{11.36}$$

Notice that in the massive case we have chosen, as usual, a renormalization scheme such that the expectation value of the renormalized stress-energy tensor does not have local terms proportional to the metric and the Einstein tensor.

The result (11.36) for a massless field agree with the general form found by Horowitz [213, 698] using an axiomatic approach and coincides with that given in [128]. The particular cases of conformal coupling, $\Delta\xi = 0$, and minimal coupling, $\Delta\xi = -1/6$, are also in agreement with the results for this cases given in [213, 698, 724, 423, 677], modulo local terms proportional to $A^{(1)\mu\nu}$ and $B^{(1)\mu\nu}$ due to different choices of the renormalization scheme. For the case of a massive minimally coupled scalar field, $\Delta\xi = -1/6$, the result (11.35) is equivalent to that found by Tichy and Flanagan [725].

11.4 Solutions of the Einstein–Langevin Equation

Here we solve the Einstein–Langevin equations (11.34) for the components $G^{(1)\mu\nu}$ of the linearized Einstein tensor. Then we use these solutions to compute the corresponding two-point correlation functions, which give a measure of the gravitational fluctuations predicted by the stochastic semiclassical theory of gravity in the present case. Since the linearized Einstein tensor is invariant under gauge transformations of the metric perturbations, these two-point correlation functions are also gauge-invariant. Once we have computed the two-point correlation functions for the linearized Einstein tensor, we find the solutions for the metric perturbations and compute the associated two-point correlation functions. The procedure used to solve the Einstein–Langevin equation is similar to the one used by Horowitz [213], see also [128], to analyze the stability of Minkowski spacetime in semiclassical gravity.

We first note that the tensors $A^{(1)\mu\nu}$ and $B^{(1)\mu\nu}$ can be written in terms of $G^{(1)\mu\nu}$ as

$$A^{(1)\mu\nu} = \frac{2}{3}\left(\mathcal{F}^{\mu\nu}G^{(1)\alpha}_{\alpha} - \mathcal{F}^{\alpha}_{\alpha}G^{(1)\mu\nu}\right), \qquad B^{(1)\mu\nu} = 2\mathcal{F}^{\mu\nu}G^{(1)\alpha}_{\ \ \alpha}, \tag{11.37}$$

where we have used that $3\Box = \mathcal{F}^{\alpha}_{\alpha}$. Therefore, the Einstein–Langevin equation (11.34) can be seen as a linear integro-differential stochastic equation for the components $G^{(1)\mu\nu}$. In order to find solutions to Eq. (11.34), it is convenient to Fourier transform this equation. With the convention $\tilde{f}(p) = \int d^4x e^{-ip\cdot x} f(x)$ for a given field $f(x)$, one finds, from (11.37),

$$\tilde{A}^{(1)\mu\nu}(p) = 2p^2\tilde{G}^{(1)\mu\nu}(p) - \frac{2}{3}p^2P^{\mu\nu}\tilde{G}^{(1)\alpha}{}_{\alpha}(p),$$
$$\tilde{B}^{(1)\mu\nu}(p) = -2p^2P^{\mu\nu}\tilde{G}^{(1)\alpha}{}_{\alpha}(p). \tag{11.38}$$

The Fourier transform of the Einstein–Langevin equation (11.34) now reads

$$F^{\mu\nu}_{\alpha'\beta'}(p)\,\tilde{G}^{(1)\alpha'\beta'}(p) = 8\pi G_{\rm N}\,\tilde{\xi}^{\mu\nu}(p), \tag{11.39}$$

where

$$F^{\mu\nu}_{\alpha'\beta'}(p) \equiv F_1(p)\,\delta^{\mu}_{(\alpha'}\delta^{\nu}_{\beta')} + F_2(p)\,p^2P^{\mu\nu}\eta_{\alpha'\beta'}, \tag{11.40}$$

with

$$F_1(p) \equiv 1 + 16\pi G_{\rm N}\,p^2\left[\frac{1}{4}\tilde{H}_{\rm A}(p) - 2\bar{\alpha}\right],$$
$$F_2(p) \equiv -\frac{16}{3}\,\pi G_{\rm N}\left[\frac{1}{4}\tilde{H}_{\rm A}(p) + \frac{3}{4}\tilde{H}_{\rm B}(p) - 2\bar{\alpha} - 6\bar{\beta}\right]. \tag{11.41}$$

In the Fourier transformed Einstein–Langevin equation (11.39), $\tilde{\xi}^{\mu\nu}(p)$, the Fourier transform of $\xi^{\mu\nu}(x)$, is a Gaussian stochastic source of zero average and

$$\langle\tilde{\xi}^{\mu\nu}(p)\tilde{\xi}^{\alpha'\beta'}(p')\rangle_s = (2\pi)^4\,\delta^4(p+p')\,\tilde{N}^{\mu\nu\alpha'\beta'}(p), \tag{11.42}$$

where we have introduced the Fourier transform of the noise kernel. The explicit expression for $\tilde{N}^{\mu\nu\alpha'\beta'}(p)$ is found from (11.17) and (11.18) to be

$$\tilde{N}^{\mu\nu\alpha'\beta'}(p) = \frac{1}{720\pi}\theta(-p^2-4m^2)\sqrt{1+4\,\frac{m^2}{p^2}}\left[\frac{1}{4}\left(1+4\,\frac{m^2}{p^2}\right)^2(p^2)^2P^{\mu\nu\alpha'\beta'}\right.$$
$$\left.+10\left(3\Delta\xi + \frac{m^2}{p^2}\right)^2(p^2)^2P^{\mu\nu}P^{\alpha'\beta'}\right], \tag{11.43}$$

which in the massless case reduces to

$$\lim_{m\to 0}\tilde{N}^{\mu\nu\alpha'\beta'}(p) = \frac{1}{480\pi}\theta(-p^2)\left[\frac{1}{6}\,(p^2)^2\,P^{\mu\nu\alpha'\beta'} + 60\Delta\xi^2(p^2)^2P^{\mu\nu}P^{\alpha'\beta'}\right]. \tag{11.44}$$

11.4.1 Linearized Einstein Tensor Correlator

In general, we can write $G^{(1)\mu\nu} = \langle G^{(1)\mu\nu}\rangle_s + G^{(1)\mu\nu}_{\rm f}$, where $G^{(1)\mu\nu}_{\rm f}$ is a solution to Eq. (11.34) with zero average, or (11.39) in the Fourier transformed version. The average $\langle G^{(1)\mu\nu}\rangle_s$ must be a solution of the linearized semiclassical Einstein equation obtained by averaging Eq. (11.34), or (11.39). Solutions to this semiclassical equation (specially in the massless case, $m = 0$) have been studied [213, 698, 726, 727, 728, 729, 730, 194, 287, 677, 128], particularly in connection

with the problem of the stability of the ground state of semiclassical gravity. The two-point correlation functions for the linearized Einstein tensor are defined by

$$\begin{aligned}\mathcal{G}^{\mu\nu\alpha'\beta'}(x,x') &\equiv \langle G^{(1)\mu\nu}(x)G^{(1)\alpha'\beta'}(x')\rangle_s - \langle G^{(1)\mu\nu}(x)\rangle_s\langle G^{(1)\alpha'\beta'}(x')\rangle_s \\ &= \langle G_{\rm f}^{(1)\mu\nu}(x)G_{\rm f}^{(1)\alpha'\beta'}(x')\rangle_s. \end{aligned} \tag{11.45}$$

Now we shall seek the family of solutions to the Einstein–Langevin equation which can be written as a linear functional of the stochastic source and whose Fourier transform, $\tilde{G}^{(1)\mu\nu}(p)$, depends locally on $\tilde{\xi}^{\alpha\beta}(p)$. Each of such solutions is a Gaussian stochastic field and, thus, it can be completely characterized by the averages $\langle G^{(1)\mu\nu}\rangle_s$ and the two-point correlation functions (11.45). For such a family of solutions, $\tilde{G}_{\rm f}^{(1)\mu\nu}(p)$ is the most general solution to Eq. (11.39) which is linear, homogeneous and local in $\tilde{\xi}^{\alpha\beta}(p)$. It can be written as

$$\tilde{G}_{\rm f}^{(1)\mu\nu}(p) = 8\pi G_{\rm N}\, D^{\mu\nu}_{\alpha'\beta'}(p)\,\tilde{\xi}^{\alpha'\beta'}(p), \tag{11.46}$$

where $D^{\mu\nu}_{\alpha'\beta'}(p)$ are the components of a Lorentz invariant tensor field distribution in Minkowski spacetime (by "Lorentz-invariant" we mean invariant under the transformations of the orthochronous Lorentz subgroup; see [213] for more details on the definition and properties of these tensor distributions). This tensor is symmetric under the interchanges of $\alpha \leftrightarrow \beta$ and $\mu \leftrightarrow \nu$, and is the most general solution of

$$F^{\mu\nu}_{\hat{\rho}\hat{\sigma}}(p)\, D^{\hat{\rho}\hat{\sigma}}_{\alpha'\beta'}(p) = \delta^{\mu}_{(\alpha'}\delta^{\nu}_{\beta')}. \tag{11.47}$$

In addition, we must impose the conservation condition: $p_\nu \tilde{G}_{\rm f}^{(1)\mu\nu}(p) = 0$, where this zero must be understood as a stochastic variable which behaves deterministically as a zero vector field. We can write $D^{\mu\nu}_{\alpha'\beta'}(p) = D^{\mu\nu}_{p\,\alpha'\beta'}(p) + D^{\mu\nu}_{h\,\alpha'\beta'}(p)$, where $D^{\mu\nu}_{p\,\alpha'\beta'}(p)$ is a particular solution to Eq. (11.47) and $D^{\mu\nu}_{h\,\alpha'\beta'}(p)$ is the most general solution to the homogeneous equation. Consequently, see Eq. (11.46), we can write $\tilde{G}_{\rm f}^{(1)\mu\nu}(p) = \tilde{G}_{p}^{(1)\mu\nu}(p) + \tilde{G}_{h}^{(1)\mu\nu}(p)$.

To find the particular solution, we try an ansatz of the form

$$D^{\mu\nu}_{p\,\alpha'\beta'}(p) = d_1(p)\,\delta^{\mu}_{(\alpha'}\delta^{\nu}_{\beta')} + d_2(p)\,p^2 P^{\mu\nu}\eta_{\alpha'\beta'}. \tag{11.48}$$

Substituting this ansatz into Eq. (11.47), it is easy to see that it solves this equation if we take

$$d_1(p) = \left[\frac{1}{F_1(p)}\right]_r, \qquad d_2(p) = -\left[\frac{F_2(p)}{F_1(p)F_3(p)}\right]_r, \tag{11.49}$$

with

$$F_3(p) \equiv F_1(p) + 3p^2F_2(p) = 1 - 48\pi G_{\rm N}\,p^2\left[\frac{1}{4}\tilde{H}_{\rm B}(p) - 2\bar{\beta}\right], \tag{11.50}$$

and where the notation $[\ \ ]_r$ means that the zeros of the denominators are regulated with appropriate prescriptions in such a way that $d_1(p)$ and $d_2(p)$

are well defined Lorentz-invariant scalar distributions. This yields a particular solution to the Einstein–Langevin equation:

$$\tilde{G}^{(1)\mu\nu}_{p}(p) = 8\pi G_{\rm N}\, D^{\mu\nu}_{p\,\alpha'\beta'}(p)\,\tilde{\xi}^{\alpha'\beta'}(p), \tag{11.51}$$

which, since the stochastic source is conserved, satisfies the conservation condition. Note that, in the case of a massless scalar field, $m = 0$, the above solution has a functional form analogous to that of the solutions of linearized semiclassical gravity found in the appendix of [128]. Notice also that, for a massless conformally coupled field, $m = 0$ and $\Delta\xi = 0$, the second term on the right-hand side of Eq. (11.48) will not contribute in the correlation functions (11.45), since in this case the stochastic source is traceless.

A detailed analysis given in [189] concludes that the homogeneous solution $\tilde{G}^{(1)\mu\nu}_{h}(p)$ gives no contribution to the correlation functions (11.45). Consequently $\mathcal{G}^{\mu\nu\alpha'\beta'}(x,x') = \langle G^{(1)\mu\nu}_{p}(x) G^{(1)\alpha'\beta'}_{p}(x')\rangle_s$, where $G^{(1)\mu\nu}_{p}(x)$ is the inverse Fourier transform of (11.51), and the correlation functions (11.45) are

$$\langle \tilde{G}^{(1)\mu\nu}_{p}(p) \tilde{G}^{(1)\alpha'\beta'}_{p}(p')\rangle_s = 64(2\pi)^6 G^2_{\rm N}\delta^4(p+p') D^{\mu\nu}_{p\hat{\rho}\hat{\sigma}}(p) D^{\alpha'\beta'}_{p\bar{\lambda}\bar{\gamma}}(-p) \tilde{N}^{\hat{\rho}\hat{\sigma}\bar{\lambda}\bar{\gamma}}(p). \tag{11.52}$$

It is easy to see from the above analysis that the prescriptions $[\ ]_r$ in the factors D_p are irrelevant in the last expression and, thus, they can be suppressed. Taking into account that $F_l(-p) = F^*_l(p)$, with $l = 1, 2, 3$, we get from Eqs. (11.48) and (11.49)

$$\begin{aligned}\langle \tilde{G}^{(1)\mu\nu}_{p}(p)\, \tilde{G}^{(1)\alpha'\beta'}_{p}(p')\rangle_s = 64(2\pi)^6\, G^2_{\rm N}\, \frac{\delta^4(p+p')}{|F_1(p)|^2} \\ \times \left[\tilde{N}^{\mu\nu\alpha'\beta'}(p) - \frac{F_2(p)}{F_3(p)}\, p^2 P^{\mu\nu} \tilde{N}^{\alpha'\beta'\rho}{}_{\rho}(p)\right. \\ - \frac{F^*_2(p)}{F^*_3(p)}\, p^2 P^{\alpha'\beta'} \tilde{N}^{\mu\nu\rho'}{}_{\rho'}(p) \\ \left. + \frac{|F_2(p)|^2}{|F_3(p)|^2} p^2 P^{\mu\nu} p^2 P^{\alpha'\beta'} \tilde{N}^{\rho\,\sigma'}_{\rho\,\sigma'}(p)\right].\end{aligned} \tag{11.53}$$

This last expression is well defined as a bi-distribution and can be easily evaluated using Eq. (11.43). The final explicit result for the Fourier transformed correlation function for the Einstein tensor is thus

$$\begin{aligned}\langle \tilde{G}^{(1)\mu\nu}_{p}(p)\, \tilde{G}^{(1)\alpha'\beta'}_{p}(p')\rangle_s = \frac{2G^2_{\rm N}}{45}(2\pi)^5 \frac{\delta^4(p+p')}{|F_1(p)|^2}\theta(-p^2-4m^2)\sqrt{1+4\frac{m^2}{p^2}} \\ \times \left[\frac{1}{4}\left(1+4\,\frac{m^2}{p^2}\right)^2 (p^2)^2\, P^{\mu\nu\alpha'\beta'}\right. \\ \left. +10\left(3\Delta\xi + \frac{m^2}{p^2}\right)^2 (p^2)^2 P^{\mu\nu} P^{\alpha'\beta'} \left|1 - 3p^2\frac{F_2(p)}{F_3(p)}\right|^2\right].\end{aligned} \tag{11.54}$$

To obtain the correlation functions in coordinate space, i.e., Eq. (11.45), we take the inverse Fourier transform. The final result is:

$$\mathcal{G}^{\mu\nu\alpha'\beta'}(x,x') = \frac{\pi G_{\rm N}^2}{45}\mathcal{F}_x^{\mu\nu\alpha'\beta'}\mathcal{G}_{\rm A}(x-x') + \frac{8\pi G_{\rm N}^2}{9}\mathcal{F}_x^{\mu\nu}\mathcal{F}_x^{\alpha'\beta'}\mathcal{G}_{\rm B}(x-x'), \quad (11.55)$$

with

$$\tilde{\mathcal{G}}_{\rm B}(p) \equiv \theta(-p^2-4m^2)\sqrt{1+4\frac{m^2}{p^2}}\left(3\Delta\xi+\frac{m^2}{p^2}\right)^2\frac{1}{|F_1(p)|^2}\left|1-3p^2\frac{F_2(p)}{F_3(p)}\right|^2,$$

$$\tilde{\mathcal{G}}_{\rm A}(p) \equiv \theta(-p^2-4m^2)\sqrt{1+4\,\frac{m^2}{p^2}}\left(1+4\frac{m^2}{p^2}\right)^2\frac{1}{|F_1(p)|^2}, \quad (11.56)$$

where $F_l(p), l=1,2,3$, are given in (11.41) and (11.50).

Notice that, for a massless field ($m=0$), we have

$$\begin{aligned} F_1(p) &= 1+4\pi G_{\rm N}p^2\tilde{H}(p;\bar{\mu}^2),\\ F_2(p) &= -\frac{16}{3}\pi G_{\rm N}\left[(1+180\Delta\xi^2)\frac{1}{4}\tilde{H}(p;\bar{\mu}^2)-6\Upsilon\right],\\ F_3(p) &= 1-48\pi G_{\rm N}p^2\left[15\Delta\xi^2\tilde{H}(p;\bar{\mu}^2)-2\Upsilon\right], \end{aligned} \quad (11.57)$$

with $\bar{\mu} \equiv \mu\exp(1920\pi^2\bar{\alpha})$ and $\Upsilon \equiv \bar{\beta}-60\Delta\xi^2\bar{\alpha}$, and where $\tilde{H}(p;\mu^2)$ is the Fourier transform of $H(x;\mu^2)$ given in (11.33).

11.4.2 Two-Point Correlations for Metric Perturbations

Starting from the solutions found for the linearized Einstein tensor, which are characterized by the two-point correlation functions (11.55) or, in terms of Fourier transforms, (11.54), we can now solve the equations for the metric perturbations. Working in the harmonic gauge, $\partial_\nu\bar{h}^{\mu\nu}=0$ (this zero must be understood in a statistical sense) where $\bar{h}_{\mu\nu} \equiv h_{\mu\nu}-(1/2)\eta_{\mu\nu}h^\alpha_\alpha$, the equations for the metric perturbations in terms of the Einstein tensor are

$$\Box\bar{h}^{\mu\nu}(x) = -2G^{(1)\mu\nu}(x), \quad (11.58)$$

or, in terms of Fourier transforms, $p^2\tilde{\bar{h}}^{\mu\nu}(p) = 2\tilde{G}^{(1)\mu\nu}(p)$. Similarly to the analysis of the equation for the Einstein tensor, we can write $\bar{h}^{\mu\nu}=\langle\bar{h}^{\mu\nu}\rangle_s+\bar{h}^{\mu\nu}_{\rm f}$, where $\bar{h}^{\mu\nu}_{\rm f}$ is a solution to these equations with zero average, and the two-point correlation functions are defined by

$$\begin{aligned} \mathcal{H}^{\mu\nu\alpha'\beta'}(x,x') &\equiv \langle\bar{h}^{\mu\nu}(x)\bar{h}^{\alpha'\beta'}(x')\rangle_s - \langle\bar{h}^{\mu\nu}(x)\rangle_s\langle\bar{h}^{\alpha'\beta'}(x')\rangle_s\\ &= \langle\bar{h}^{\mu\nu}_{\rm f}(x)\bar{h}^{\alpha'\beta'}_{\rm f}(x')\rangle_s. \end{aligned} \quad (11.59)$$

We can now seek solutions of the Fourier transform of Eq. (11.58) of the form $\tilde{\bar{h}}^{\mu\nu}_{\rm f}(p) = 2D(p)\tilde{G}^{(1)\mu\nu}_{\rm f}(p)$, where $D(p)$ is a Lorentz-invariant scalar distribution

in Minkowski spacetime, which is the most general solution of $p^2 D(p) = 1$. Note that, since the linearized Einstein tensor is conserved, solutions of this form automatically satisfy the harmonic gauge condition. As in Section 11.4.1, we can write $D(p) = [1/p^2]_r + D_h(p)$, where $D_h(p)$ is the most general solution to the associated homogeneous equation and, correspondingly, we have $\tilde{\bar{h}}^{\mu\nu}_{\rm f}(p) = \tilde{\bar{h}}^{\mu\nu}_{p}(p) + \tilde{\bar{h}}^{\mu\nu}_{h}(p)$. However, since $D_h(p)$ has support on the set of points for which $p^2 = 0$, it is easy to see from Eq. (11.54) (from the factor $\theta(-p^2 - 4m^2)$) that $\langle \tilde{\bar{h}}^{\mu\nu}_{h}(p) \tilde{G}^{(1)\alpha'\beta'}_{\rm f}(p') \rangle_s = 0$ and, thus, the two-point correlation functions (11.59) can be computed from $\langle \tilde{\bar{h}}^{\mu\nu}_{\rm f}(p) \tilde{\bar{h}}^{\alpha'\beta'}_{\rm f}(p') \rangle_s = \langle \tilde{\bar{h}}^{\mu\nu}_{p}(p) \tilde{\bar{h}}^{\alpha'\beta'}_{p}(p') \rangle_s$.

From Eq. (11.54) and due to the factor $\theta(-p^2 - 4m^2)$, it is also easy to see that the prescription $[\ ]_r$ is irrelevant in this correlation function and we obtain

$$\langle \tilde{\bar{h}}^{\mu\nu}_{p}(p) \tilde{\bar{h}}^{\alpha'\beta'}_{p}(p') \rangle_s = \frac{4}{(p^2)^2} \langle \tilde{G}^{(1)\mu\nu}_{p}(p)\, \tilde{G}^{(1)\alpha'\beta'}_{p}(p') \rangle_s, \tag{11.60}$$

where $\langle \tilde{G}^{(1)\mu\nu}_{p}(p)\, \tilde{G}^{(1)\alpha'\beta'}_{p}(p') \rangle_s$ is given by Eq. (11.54). The right-hand side of this equation is a well-defined bi-distribution, at least for $m \neq 0$ (the θ function provides the suitable cutoff). In the massless field case, since the noise kernel is obtained as the limit $m \to 0$ of the noise kernel for a massive field, it seems that the natural prescription to avoid the divergences on the lightcone $p^2 = 0$ is a Hadamard finite part, see [731, 732] for its definition. Taking this prescription, we also get a well-defined bi-distribution for the massless limit of the last expression.

The final result for the two-point correlation function for the field $\bar{h}^{\mu\nu}$ is:

$$\mathcal{H}^{\mu\nu\alpha'\beta'}(x,x') = \frac{4\pi G_{\rm N}^2}{45} \mathcal{F}^{\mu\nu\alpha'\beta'}_{x}\, \mathcal{H}_{\rm A}(x-x') + \frac{32\pi G_{\rm N}^2}{9} \mathcal{F}^{\mu\nu}_{x} \mathcal{F}^{\alpha'\beta'}_{x}\, \mathcal{H}_{\rm B}(x-x'), \tag{11.61}$$

where $\tilde{\mathcal{H}}_{\rm A}(p) \equiv [1/(p^2)^2]\, \tilde{\mathcal{G}}_{\rm A}(p)$ and $\tilde{\mathcal{H}}_{\rm B}(p) \equiv [1/(p^2)^2]\, \tilde{\mathcal{G}}_{\rm B}(p)$, with $\tilde{\mathcal{G}}_{\rm A}(p)$ and $\tilde{\mathcal{G}}_{\rm B}(p)$ given by (11.55). The two-point correlation functions for the metric perturbations can be easily obtained using $h_{\mu\nu} = \bar{h}_{\mu\nu} - (1/2)\eta_{\mu\nu}\bar{h}^{\alpha}_{\alpha}$.

11.4.3 Conformally Coupled Field

For a conformally coupled field, i.e., when $m = 0$ and $\Delta\xi = 0$, the previous correlation functions are greatly simplified and can be approximated explicitly in terms of analytic functions. The detailed results are given in [189], here we outline the main features.

When $m = 0$ and $\Delta\xi = 0$ we have $\mathcal{G}_{\rm B}(x) = 0$ and $\tilde{\mathcal{G}}_{\rm A}(p) = \theta(-p^2)\, |F_1(p)|^{-2}$. Thus the two-point correlations functions for the Einstein tensor is

$$\mathcal{G}^{\mu\nu\alpha'\beta'}(x,x') = \frac{\pi G_{\rm N}^2}{45} \mathcal{F}^{\mu\nu\alpha'\beta'}_{x} \int \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-x')}\, \theta(-p^2)}{|1 + 4\pi G_{\rm N} p^2 \tilde{H}(p;\bar{\mu}^2)|^2}, \tag{11.62}$$

where $\tilde{H}(p,\bar{\mu}^2) = (480\pi^2)^{-1} \ln[-((p^0 + i\epsilon)^2 + p^i p_i)/\bar{\mu}^2]$, see Eq. (11.33).

To estimate this integral let us consider spacelike separated points $(x-x')^\mu = (0, \mathbf{x}-\mathbf{x}')$, and define $\mathbf{y} \equiv \mathbf{x}-\mathbf{x}'$. We may now formally change the momentum variable p^μ by the dimensionless vector s^μ: $p^\mu = s^\mu/|\mathbf{y}|$, then the previous integral denominator is $|1+16\pi(\ell_{\rm P}/|\mathbf{y}|)^2 s^2 \tilde{H}(s)|^2$, where we have introduced the Planck length $\ell_{\rm P} = \sqrt{G_{\rm N}}$ (we use units in which $\hbar = c = 1$). It is clear that we can consider two regimes: (a) when $\ell_{\rm P} \ll |\mathbf{y}|$ which corresponds to a safe limit, and (b) when $|\mathbf{y}| \sim \ell_{\rm P}$ which corresponds to the limit of the semiclassical theory. In case (a) the correlation function, for the 0000 component, say, will be of the order

$$\mathcal{G}^{0000}(\mathbf{y}) \sim \frac{\ell_{\rm P}^4}{|\mathbf{y}|^8}.$$

In case (b) when the denominator has zeros a detailed calculation carried out by [189] shows that:

$$\mathcal{G}^{0000}(\mathbf{y}) \sim e^{-|\mathbf{y}|/\ell_{\rm P}} \left(\frac{\ell_{\rm P}}{|\mathbf{y}|^5} + \cdots + \frac{1}{\ell_{\rm P}^2 |\mathbf{y}|^2} \right)$$

which indicates an exponential decay at distances around the Planck scale. Thus it seems that short scale fluctuations are strongly suppressed by the effect of matter fields.

For the two-point metric correlation the results are similar. In this case we have

$$\mathcal{H}^{\mu\nu\alpha'\beta'}(x,x') = \frac{4\pi G_{\rm N}^2}{45} \mathcal{F}_x^{\mu\nu\alpha'\beta'} \int \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-x')}\theta(-p^2)}{(p^2)^2 |1+4\pi G_{\rm N} p^2 \tilde{H}(p;\bar{\mu}^2)|^2}. \quad (11.63)$$

The integrand has the same behavior of the correlation function of Eq. (11.62), thus matter fields tends to suppress the short scale metric perturbations. In this case we find, as for the correlation of the Einstein tensor, that for case (a) above we have,

$$\mathcal{H}^{0000}(\mathbf{y}) \sim \frac{\ell_{\rm P}^4}{|\mathbf{y}|^4},$$

and for case (b) we have

$$\mathcal{H}^{0000}(\mathbf{y}) \sim e^{-|\mathbf{y}|/\ell_{\rm P}} \left(\frac{\ell_{\rm P}}{|\mathbf{y}|} + \dots \right).$$

It is interesting to write expression (11.63) in an alternative way. If we use the dimensionless tensor $P^{\mu\nu\alpha'\beta'}$ introduced in Eq. (11.23), which accounts for the effect of the operator $\mathcal{F}_x^{\mu\nu\alpha'\beta'}$, we can write

$$\mathcal{H}^{\mu\nu\alpha'\beta'}(x,x') = \frac{4\pi G_{\rm N}^2}{45} \int \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-x')}\, P^{\mu\nu\alpha'\beta'}\, \theta(-p^2)}{|1+4\pi G_{\rm N} p^2 \tilde{H}(p;\bar{\mu}^2)|^2}. \quad (11.64)$$

This expression allows a direct comparison with the graviton propagator for linearized quantum gravity in the $1/N$ expansion found by Tomboulis [686].

One can see that the imaginary part of the graviton propagator leads to Eq. (11.64). Thus, the two-point correlation functions for the metric perturbations derived from the Einstein–Langevin equation are equivalent to the symmetrized quantum two-point correlation functions for the metric fluctuations in the large N expansion of quantum gravity interacting with N matter fields.

The main results of this section are the correlation functions (11.55) and (11.61). In the case of a conformal field, the correlation functions of the linearized Einstein tensor have been explicitly estimated. From the exponential factors $e^{-|\mathbf{y}|/\ell_{\rm P}}$ in these results for scales near the Planck length, we see that the correlation functions of the linearized Einstein tensor have the Planck length as the correlation length. A similar behavior is found for the correlation functions of the metric perturbations. Since these fluctuations are induced by the matter fluctuations we infer that the effect of the matter fields is to suppress the fluctuations of the metric at very small scales. On the other hand, at scales much larger than the Planck length the induced metric fluctuations are small compared with the free graviton propagator which goes like $\ell_{\rm P}^2/|\mathbf{y}|^2$, since the action for the free graviton goes like $S_h \sim \int d^4x\, \ell_{\rm P}^{-2} h \Box h$.

For background solutions of semiclassical gravity with other scales present apart from the Planck scales (for instance, for matter fields in a thermal state), stress-energy fluctuations may be important at larger scales. For such backgrounds, stochastic semiclassical gravity might predict correlation functions with characteristic correlation lengths larger than the Planck scales. It seems quite plausible, nevertheless, that these correlation functions would remain non-analytic in their characteristic correlation lengths. This would imply that these correlation functions could not be obtained from a calculation involving a perturbative expansion in the characteristic correlation lengths. Hence, stochastic semiclassical gravity might predict a behavior for gravitational correlation functions different from that of the analogous functions in perturbative quantum gravity. This is not necessarily inconsistent with having neglected action terms of higher order in $\ell_{\rm P}$ when considering semiclassical gravity as an effective field theory [128]. It is, in fact, consistent with the closed connection of stochastic gravity with the large N expansion of quantum gravity interacting with N matter fields.

11.5 Stability of Minkowski Spacetime

In this section we apply the validity criterion for semiclassical gravity introduced in Section 10.3 to flat spacetime. The Minkowski metric is a particularly simple and interesting solution of semiclassical gravity. In fact, as we have seen in Section 11.1, when the quantum fields are in the Minkowski vacuum state one may take the renormalized expectation value of the stress-energy tensor as $\langle \hat{T}^{\rm R}_{\mu\nu}[\eta] \rangle = 0$ (this is equivalent to assuming that the cosmological constant is zero), then the Minkowski metric $\eta_{\mu\nu}$ is a solution of the semiclassical Einstein equation (10.7).

Thus, we can look for the stability of Minkowski spacetime against quantum matter fields. According to the criteria we have established we have to look for the behavior of the two-point quantum correlations for the metric perturbations $h_{\mu\nu}(x)$ over the Minkowski background which are given by Eqs. (10.15) and (10.16). As we have emphasized before these metric fluctuations separate in two parts: the first term on the right hand side of Eq. (10.16) which corresponds to the *intrinsic* metric fluctuations, and the second term which corresponds to the *induced* metric fluctuations.

11.5.1 Intrinsic Metric Fluctuations

Let us first consider the intrinsic metric fluctuations,

$$\langle h_{\mu\nu}(x) h_{\alpha'\beta'}(y)\rangle_{\rm int} = \langle h^0_{\mu\nu}(x) h^0_{\alpha'\beta'}(y)\rangle_s, \tag{11.65}$$

where $h^0_{\mu\nu}$ are the homogeneous solutions of the Einstein–Langevin equation (10.14), or equivalently the linearly perturbed semiclassical equation, and where the statistical average is taken with respect to the Wigner distribution that describes the initial quantum state of the metric perturbations. Since these solutions are described by the linearized semiclassical equation around flat spacetime we can make use of the results derived in [213, 128, 209, 208]. The solutions for the case of a massless scalar field were first discussed by Horowitz [213] and an exhaustive description can be found in appendix A of [128].

It is convenient to decompose the perturbation around Minkowski spacetime into scalar, vectorial and tensorial parts, as

$$h_{\mu\nu} = \bar{\phi}\,\eta_{\mu\nu} + (\nabla_{(\mu}\nabla_{\nu)} - \eta_{\mu\nu}\nabla_\alpha\nabla^\alpha)\,\psi + 2\nabla_{(\mu}v_{\nu)} + h^{\rm TT}_{\mu\nu}, \tag{11.66}$$

where v^μ is a transverse vector and $h^{\rm TT}_{\mu\nu}$ is a transverse and traceless symmetric tensor, *i.e.*, $\nabla_\mu v^\mu = 0$, $\nabla^\mu h^{\rm TT}_{\mu\nu} = 0$ and $(h^{\rm TT})^\mu_\mu = 0$. A vector field ζ^μ characterizes the gauge freedom due to infinitesimal diffeomorphisms $h_{\mu\nu} \to h_{\mu\nu} + \nabla_\mu\zeta_\nu + \nabla_\nu\zeta_\mu$. We may use this freedom to choose a gauge, a convenient election is the Lorentz or harmonic gauge defined as

$$\nabla^\mu\left(h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h^\alpha_\alpha\right) = 0. \tag{11.67}$$

When this gauge is imposed we have the following conditions on the metric perturbations $\nabla_\mu\nabla^\mu v^\nu = 0$ and $\nabla_\nu\bar{\phi} = 0$, which implies $\bar{\phi} = {\rm const}$. A remaining gauge freedom compatible with the Lorentz gauge is still possible provided the vector field ζ^μ satisfies the condition $\nabla_\mu\nabla^\mu\zeta^\nu = 0$. One can easily see that the vectorial and scalar part $\bar{\phi}$ can be eliminated, as well as the contribution of the scalar part ψ which corresponds to Fourier modes $\tilde{\psi}(p)$ with $p^2 = 0$ [190]. Thus we will assume that we impose the Lorentz gauge with additional gauge transformations which leave only the tensorial component and the modes of the scalar component ψ with $p^2 \neq 0$ in Fourier space.

Using the metric decomposition (11.66) we may compute the linearized Einstein tensor $G^{(1)}_{\mu\nu}$. It is found that the vectorial part of the metric perturbation gives no contribution to this tensor, and the scalar and tensorial components give rise, respectively, to scalar and tensorial components: $G^{(1)\,(S)}_{\mu\nu}$ and $G^{(1)\,(T)}_{\mu\nu}$, respectively. Thus let us now write the Fourier transform of the homogeneous Einstein–Langevin equation (11.39), which is equivalent to the linearized semi-classical Einstein equation,

$$F^{\mu\nu}_{\alpha'\beta'}(p)\,\tilde{G}^{(1)\alpha'\beta'}(p) = 0. \tag{11.68}$$

With the previous decomposition of the Einstein tensor this equation can be re-written in terms of its scalar and tensorial parts as

$$[F_1(p) + 3p^2 F_2(p)]\,\tilde{G}^{(1)\,(\mathrm{S})}_{\mu\nu}(p) = 0, \tag{11.69}$$

$$F_1(p)\tilde{G}^{(1)\,(\mathrm{T})}_{\mu\nu}(p) = 0. \tag{11.70}$$

where $F_1(p)$ and $F_2(p)$ are given by Eqs. (11.41), and $\tilde{G}^{(1)\,(\mathrm{S})}_{\mu\nu}$ and $\tilde{G}^{(1)\,(\mathrm{T})}_{\mu\nu}$ denote, respectively, the Fourier transformed scalar and tensorial parts of the linearized Einstein tensor. To simplify the problem and to illustrate, in particular, how the runaway solutions arise we will consider the case of a massless and conformally coupled field; see [128] for the massless case with arbitrary coupling and [189, 209] for the general massive case. Thus substituting $m = 0$ and $\xi = 1/6$ into the functions $F_1(p)$ and $F_2(p)$, and using Eq. (11.33), the above equations become

$$\left(1 + 96\pi G_{\mathrm{N}}\bar{\beta}p^2\right)\tilde{G}^{(1)\,(\mathrm{S})}_{\mu\nu}(p) = 0, \tag{11.71}$$

$$\lim_{\epsilon\to 0^+}\left[1 + \frac{G_{\mathrm{N}}p^2}{120\pi}\ln\left(\frac{-(p^0+i\epsilon)^2 + \vec{p}^{\,2}}{\mu^2}\right)\right]\tilde{G}^{(1)\,(\mathrm{T})}_{\mu\nu}(p) = 0. \tag{11.72}$$

Let us consider these two equations separately.

For the scalar component when $\bar{\beta} = 0$ the only solution is $\tilde{G}^{(1)\,(\mathrm{S})}_{\mu\nu}(p) = 0$. When $\bar{\beta} > 0$ the solutions for the scalar component exhibit an oscillatory behavior in spacetime coordinates which corresponds to a massive scalar field with $m^2 = (96\pi G_{\mathrm{N}}|\bar{\beta}|)^{-1}$; for $\bar{\beta} < 0$ the solutions correspond to a tachyonic field with $m^2 = -(96\pi G_{\mathrm{N}}|\bar{\beta}|)^{-1}$: in spacetime coordinates they exhibit an exponential behavior in time, growing or decreasing, for wavelengths larger than $4\pi(24\pi G_{\mathrm{N}}|\bar{\beta}|)^{1/2}$, and an oscillatory behavior for wavelengths smaller than $4\pi(24\pi G_{\mathrm{N}}|\bar{\beta}|)^{1/2}$. On the other hand, the solution $\tilde{G}^{(1)\,(\mathrm{S})}_{\mu\nu}(p) = 0$ is completely trivial since any scalar metric perturbation $\tilde{h}_{\mu\nu}(p)$ giving rise to a vanishing linearized Einstein tensor can be eliminated by a gauge transformation.

For the tensorial component, when $\mu \leq \mu_{\mathrm{crit}} = \ell_{\mathrm{P}}^{-1}(120\pi)^{1/2}e^{\gamma}$, the first factor in Eq. (11.72) vanishes for four complex values of p^0 of the form $\pm\omega$ and $\pm\omega^*$, where ω is some complex value. This means that in the corresponding propagator, there are two poles on the upper half plane of the complex p^0 plane and two poles in the lower half plane. We will consider here the case in which $\mu < \mu_{\mathrm{crit}}$; a detailed description of the situation for $\mu \geq \mu_{\mathrm{crit}}$ can be found in appendix A

of [128]. The two zeros on the upper half of the complex plane correspond to solutions in spacetime coordinates which exponentially grow in time, whereas the two on the lower half correspond to solutions exponentially decreasing in time. Strictly speaking, these solutions only exist in spacetime coordinates, since their Fourier transform is not well defined. They are commonly referred to as *runaway* solutions and for $\mu \sim \ell_{\rm P}^{-1}$ they grow exponentially on time scales comparable to the Planck time.

Consequently, in addition to the solutions with $G^{(1)}_{\mu\nu}(x) = 0$, there are other solutions that in Fourier space take the form $\tilde{G}^{(1)}_{\mu\nu}(p) \propto \delta(p^2 - p_0^2)$ for some particular values of p_0, but all of them exhibit exponential instabilities with characteristic Planckian time scales. In order to deal with those unstable solutions, one possibility is to make use of the *order reduction* prescription, which is briefly summarized in Section 10.3.1. Note that the p^2 terms in Eqs. (11.71) and (11.72) come from two spacetime derivatives of the Einstein tensor, and the $p^2 \ln p^2$ term comes from the nonlocal term of the expectation value of the stress-energy tensor.

Order Reduction Prescription

The order reduction prescription amounts here to neglecting these higher derivative terms. Thus, neglecting the terms proportional to p^2 in Eqs. (11.71) and (11.72), we are left only with the solutions which satisfy $\tilde{G}^{(1)}_{\mu\nu}(p) = 0$. The result for the metric perturbation in the gauge introduced above can be obtained by solving for the Einstein tensor, which in the Lorentz gauge of Eq. (11.67) reads in terms of the metric perturbation:

$$\tilde{G}^{(1)}_{\mu\nu}(p) = \frac{1}{2}p^2 \left(\tilde{h}_{\mu\nu}(p) - \frac{1}{2}\eta_{\mu\nu}\tilde{h}^{\alpha}_{\alpha}(p) \right). \tag{11.73}$$

The solutions for $\tilde{h}_{\mu\nu}(p)$ of $\tilde{G}^{(1)}_{\mu\nu}(p) = 0$ simply correspond to free linear gravitational waves propagating in Minkowski spacetime expressed in the transverse and traceless (TT) gauge. When substituting back into Eq. (11.65) and averaging over the initial conditions we simply get the symmetrized quantum correlation function for free gravitons in the TT gauge for the state given by the Wigner distribution. As far as the intrinsic fluctuations are concerned, the order reduction prescription may seem too drastic, at least in the case of Minkowski spacetime, since no effects due to the interaction with the quantum matter fields are left.

Hawking, Hertog and Reall Prescription

A second possibility, proposed by Hawking et al. [651, 733], is to impose boundary conditions which discard the runaway solutions that grow unbounded in time. These boundary conditions correspond to a special prescription for the integration contour when Fourier transforming back to spacetime coordinates. As we will discuss in some more detail in Section 11.5.2, this prescription reduces here to

integrating along the real axis in the p^0 complex plane. Following that procedure we get, for example, that for a massless conformally coupled matter field with $\bar{\beta} > 0$ the intrinsic contribution to the symmetrized quantum correlation function coincides with that of free gravitons plus an extra contribution for the scalar part of the metric perturbations. This extra massive scalar renders Minkowski spacetime stable, but also plays a crucial role in providing a graceful exit in inflationary models driven by the vacuum polarization of a large number of conformal fields. Such a massive scalar field would not be in conflict with present observations because, for the range of parameters considered, the mass would be far too large to have observational consequences [651].

11.5.2 Induced Metric Fluctuations

The induced metric fluctuations are described by the second term in Eq. (10.16). They are dependent on the noise kernel that describes the stress-energy tensor fluctuations of the matter fields,

$$\langle h_{\mu\nu}(x) h_{\alpha'\beta'}(y)\rangle_{\rm ind} = \frac{64\pi^2 \bar{G}_{\rm N}^2}{N} \int d^4x' d^4y' \sqrt{g(x')g(y')} \times G^{\rm ret}_{\mu\nu\hat{\rho}\hat{\sigma}}(x,x') N^{\hat{\rho}\hat{\sigma}\bar{\tau}\bar{\epsilon}}(x',y') G^{\rm ret}_{\alpha'\beta'\bar{\tau}\bar{\epsilon}}(y,y'), \tag{11.74}$$

where here we have written the expression in the large N limit, so that $\bar{G}_{\rm N} \equiv N G_{\rm N}$ and N is the number of independent free scalar fields. As explained in Section 9.3.1, it is important to recall that when working in the large N expansion, the length-scale characterizing the cut-off scale of the low energy effective field theory is $\bar{\ell}_{\rm P} = \sqrt{N}\ell_{\rm P}$, which is much larger than the Planck scale.

The contribution corresponding to the induced quantum fluctuations is equivalent to the stochastic correlation function obtained by considering just the inhomogeneous part of the solution to the Einstein–Langevin equation. We can make use of the results for the metric correlations obtained in Sections 11.3 and 11.4 for solving the Einstein–Langevin equation. In fact, one should simply take $N = 1$ to transform our expressions here to those of Sections 11.3 and 11.4 or, more precisely, one should multiply the noise kernel in the expressions of Sections 11.3 and 11.4 by N in order to use those expressions here, as follows from the fact that now we have N independent matter fields.

As we have seen in Section 11.4 the Einstein–Langevin equation can be entirely written in terms of the linearized Einstein tensor. The equation involves second spacetime derivatives of that tensor and in terms of its Fourier components is given in Eq. (11.39) as

$$F^{\mu\nu}{}_{\alpha'\beta'}(p)\, \tilde{G}^{(1)\alpha'\beta'}(p) = 8\pi \bar{G}_{\rm N}\, \tilde{\xi}^{\mu\nu}(p), \tag{11.75}$$

where we have used now the rescaled coupling $\bar{G}_{\rm N}$. The solution for the linearized Einstein tensor is given in Eq. (11.51) in terms of the retarded propagator

$D_{\mu\nu\rho'\sigma'}(p)$ defined in Eq. (11.47). Now this propagator, which is written in Eq. (11.48), exhibits two poles in the upper half complex p^0 plane and two poles in the lower half plane, as we have seen analyzing the zeros in Eqs. (11.71) and (11.72) for the massless and conformally coupled case.

The retarded propagator in spacetime coordinates is obtained as usual by taking the appropriate integration contour in the p^0 plane. It is convenient in this case to deform the integration path along the real p^0 axis so as to leave the two poles of the upper half plane below that path. In this way when closing the contour by un upper half circle, in order to compute the anti-causal part of the propagator, there will be no contribution. The problem now is that when closing the contour on the lower half plane, in order to compute the causal part, the contribution of the upper half plane poles gives an unbounded solution, a runaway instability. If we adopt the criterion in [651, 733] of imposing final boundary conditions which discard solutions growing unboundedly in time this implies that we just need to take the integral along the real axis, as was done in Section 11.4.2. But now the propagator is no longer strictly retarded, there are causality violations in time scales of the order of $\sqrt{N}\ell_{\rm P}$, which should have no observable consequences. This propagator, however, has well-defined Fourier transform.

Following the steps after Eq. (11.51) the Fourier transform of the two-point correlation for the linearized Einstein tensor can be written in our case as,

$$\langle \tilde{G}^{(1)}_{\mu\nu}(p)\tilde{G}^{(1)}_{\alpha'\beta'}(p')\rangle_{\rm ind} = \frac{64\pi^2\bar{G}_{\rm N}^2}{N}(2\pi)^4\delta^4(p+p') \times D_{\mu\nu\hat{\rho}\hat{\sigma}}(p)D_{\alpha'\beta'\bar{\lambda}\bar{\gamma}}(-p)\tilde{N}^{\hat{\rho}\hat{\sigma}\bar{\lambda}\bar{\gamma}}(p), \tag{11.76}$$

where the noise kernel $\tilde{N}^{\rho\sigma\lambda'\gamma'}(p)$ is given by Eq. (11.43). Note that these correlation functions are invariant under gauge transformations of the metric perturbations because the linearized Einstein tensor is invariant under those transformations.

We may also take the *order reduction* prescription which amounts in this case to neglecting terms in the propagator which are proportional to p^2, corresponding to two spacetime derivatives of the Einstein tensor. The propagator then becomes a constant, and we have

$$\langle \tilde{G}^{(1)}_{\mu\nu}(p)\tilde{G}^{(1)}_{\alpha'\beta'}(p')\rangle_{\rm ind} = \frac{64\pi^2\bar{G}_{\rm N}^2}{N}(2\pi)^4\delta^4(p+p')\tilde{N}_{\mu\nu\alpha'\beta'}(p). \tag{11.77}$$

Finally we may derive the correlations for the metric perturbations from these Eqs. (11.76) or (11.77). In the Lorentz or harmonic gauge the linearized Einstein tensor takes the particularly simple form of Eq. (11.73) in terms of the metric perturbation. One may derive the correlation functions for $\tilde{h}_{\mu\nu}(p)$ as it was done in Section 11.4.2 to get

$$\langle \tilde{\bar{h}}_{\mu\nu}(p)\tilde{\bar{h}}_{\alpha'\beta'}(p')\rangle_{\rm ind} = \frac{4}{(p^2)^2}\,\langle \tilde{G}_{\mu\nu}(p)\tilde{G}_{\alpha'\beta'}(p')\rangle_{\rm ind}. \tag{11.78}$$

There will be one possible expression for the two-point metric correlation which corresponds to the Einstein tensor correlation of Eq. (11.76) and another expression corresponding to Eq. (11.77), when the order reduction prescription is used.

We should note that contrary to the correlation functions for the Einstein tensor, the two-point metric correlation is not gauge invariant (it is given in the Lorentz gauge). Moreover, when taking the Fourier transform to get the correlations in spacetime coordinates there is an apparent infrared divergence when $p^2 = 0$ in the massless case. This can be seen from the expression for the noise kernel $\tilde{N}_{\mu\nu\alpha'\beta'}(p)$ defined in Eq. (11.43). For the massive case no such divergence exists due to the factor $\theta(-p^2 - 4m^2)$, but as one takes the limit $m \to 0$ it will show up. This infrared divergence, however, is a gauge artifact that has been enforced by the use of the Lorentz gauge. A gauge different from the Lorentz gauge should be used in the massless case; see Hu et al. [190] for a more detailed discussion of this point.

Let us now write the two-point metric correlation function in spacetime coordinates for the massless and conformally coupled fields. In order to avoid runaway solutions we use the prescription that the propagator should have a well defined Fourier transform, by integrating along the real axis in the complex p^0 plane. This was, in fact, done in Section 11.4.3, see Eq. (11.64), which we may write now as

$$\langle \tilde{\bar{h}}_{\mu\nu}(x) \tilde{\bar{h}}_{\alpha'\beta'}(y) \rangle_{\rm ind} = \frac{4\pi \bar{G}_{\rm N}^2}{45N} \int \frac{d^4p}{(2\pi)^4} \frac{e^{ip\cdot(x-y)}\, P_{\mu\nu\alpha'\beta'}\, \theta(-p^2)}{|1 + 4\pi \bar{G}_{\rm N} p^2 \tilde{H}(p;\bar{\mu}^2)|^2}, \tag{11.79}$$

where the projector $P_{\mu\nu\alpha'\beta'}$ is defined in Eq. (11.23). As explained earlier this correlation function for the metric perturbations is in agreement with the real part of the graviton propagator obtained by Tomboulis [686] using a large N expansion with Fermion fields. Note that when the *order reduction* prescription is used the terms in the denominator of Eq. (11.79) which are proportional to p^2 are neglected. Thus, in contrast to the intrinsic metric fluctuations, there is still a nontrivial contribution to the induced metric fluctuations due to the quantum matter fields in this case.

To estimate the above integral let us follow Section 11.4.3, consider space-like separated points $x - y = (0, \mathbf{r})$ and introduce the Planck length $\ell_{\rm P}$. For space separations $|\mathbf{r}| \gg \ell_{\rm P}$ we have that the two-point correlation (11.79) goes like $\sim N\ell_{\rm P}^4/|\mathbf{r}|^4$, and for $|\mathbf{r}| \sim \sqrt{N}\ell_{\rm P}$ we have that they go like $\sim \exp(-|\mathbf{r}|/\sqrt{N}\ell_{\rm P})\ell_{\rm P}/|\mathbf{r}|$. Since these metric fluctuations are induced by the matter stress fluctuations we infer that the effect of the matter fields is to suppress metric fluctuations at small scales. On the other hand, at large scales the induced metric fluctuations are small compared to the free graviton propagator, which goes like $\ell_{\rm P}^2/|\mathbf{r}|^2$. We thus conclude that, once the instabilities giving rise to the unphysical runaway solutions have been discarded, the fluctuations of the metric perturbations around the Minkowski spacetime induced by the interaction with quantum scalar

fields are indeed stable (instabilities lead to divergent results when Fourier transforming back to spacetime coordinates).

Let us summarize our main results on the stability of Minkowski spacetime. An analysis of the stability of any solution of semiclassical gravity with respect to small quantum perturbations should include not only the evolution of the expectation value of the metric perturbations around that solution, but also their fluctuations, encoded in the quantum correlation functions. We have analyzed the symmetrized two-point quantum correlation function for the metric perturbations around the Minkowski spacetime interacting with N scalar fields initially in the Minkowski vacuum state. Once the instabilities that arise in semiclassical gravity which are commonly regarded as unphysical, have been properly dealt with by using the *order reduction* prescription or the procedure proposed by Hawking et al. [651, 733], both the intrinsic and the induced contributions to the quantum correlation function for the metric perturbations are found to be stable, and consequently Minkowski spacetime is stable [190]. Thus, we conclude that Minkowski spacetime is a valid solution of semiclassical gravity.

Part IV

Cosmological and Black Hole Backreaction with Fluctuations

12

Cosmological Backreaction with Fluctuations

As an example of cosmological backreaction including fluctuations we consider in this chapter a massless conformally coupled quantum scalar field on a weakly perturbed spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) spacetime and derive the semiclassical Einstein–Langevin equation for the metric perturbations off this spacetime. We use the in-in or CTP functional formalism described in previous chapters. The first step is the computation of the in-in effective action to one-loop order, which is then used to derive the Einstein–Langevin equation.

Since along this computation one obtains also all the terms needed for the in-out effective action, this action is also discussed, and it is used to derive the probability for pair creation; the results agree with those obtained by other methods. From the in-in effective action, or the related influence action, the stress-energy tensor for the quantum field is obtained and the semiclassical correction to Einstein's equations is written down.

Considering massless conformally coupled fields in a conformally flat background such as the FLRW cosmology makes the calculation of the in-in effective action reasonable straightforward, and the expressions obtained are analytic and compact. Moreover, although conformal matter may seem a too restrictive kind of matter, it can in fact be a reasonable approximation for matter fields in the very earlier universe in the high-curvature limit when their masses may be neglected, i.e. when $m^2 \ll R$ where R is the spacetime curvature scalar. Now, massless spinor fields are described by conformally invariant equations, and the same is true for massless vector fields in four dimensions. This means that matter fields in the standard model of particle physics could be expected to become effectively conformally invariant. Also, the fact that there are nearly one hundred fields makes the use of the large N approximation quite reasonable.

Hartle and Horowitz [194] pointed out that the one-loop order quantum corrections to the classical action of the gravitational field due to its interacting with N identical non self-interacting matter fields reduce in the leading-order $1/N$ approximation to the semiclassical theory. This is the viewpoint that we also adopt here together with that of general relativity as the low energy limit of an effective field theory of quantum gravity.

12.1 The Backreaction Problem in Cosmology

Our picture of the evolution of the early universe relies largely on the semiclassical theory of gravity which describes the interaction of quantum fields with the classical gravitational field. The semiclassical approach provides the framework for some realistic scenarios which explain some features of the present universe. One of these scenarios is inflation, which gives an explanation of the homogeneity and flatness problems of the standard big bang cosmology [118, 120, 359]. In the inflationary model the quantum fluctuations of the inflaton field may be the source of the small gravitational inhomogeneities which seed galaxies or gravitational waves. This can explain the universe large scale structure [734, 735, 736, 737, 738], and the presence of a hypothetical background of gravitational radiation [345, 739]. Another scenario is the possible formation of topological defects as the universe undergoes some phase transitions [740, 360]. Topological defects, in particular cosmic strings, can also seed structure and may be an alternative to inflation for the generation of structure in the universe [741, 742].

In both scenarios the picture that emerges is that of a conformally flat FLRW background in which small gravitational perturbations are present. There are two related but different problems to consider due to the existence of these gravitational perturbations. The first is the quantum effects that these perturbations may produce, which is an issue of *quantum field theory in curved spacetime*. The second is viewing these perturbations as the backreaction effect from the interaction with quantum fields, which is an issue properly related to *semiclassical gravity*.

Concerning the first problem, namely the study of the quantum effects produced by the presence of small perturbations in cosmological backgrounds the history goes back to the early stages of the formulation of quantum field theory in curved spacetimes. Quantum effects due to small anisotropies were first considered by Zel'dovich and Starobinsky [68, 69], Hu and Parker [108] and Hartle and Hu [110], who computed the creation of conformally coupled particles interacting with the anisotropies. Conformally coupled particles are not created in conformally flat backgrounds such as the FLRW spacetimes, see e.g. [743, 66], but the anisotropies break the conformal symmetry. Different techniques were used for such computations, and these range from a perturbative evaluation of the Bogoliubov transformations relating two vacua of the quantum field, to the evaluation of the in-out effective action of this field in the given gravitational

background. These results were extended to the presence of arbitrary perturbations, including inhomogeneities, in [75, 76, 744] by a technique based in the perturbative evaluation of the scattering matrix which had been used in flat backgrounds by Sexl and Urbantke [67] and Zel'dovich and Strobinsky [69].

In relation to the second problem, i.e. the quantum effects on the geometry, or *backreaction* effect, this is more difficult to evaluate because it requires, on the one hand, the computation of the renormalized stress-energy tensor of the quantum field in order to derive the semiclassical equations which modify the classical Einstein equations, and on the other hand, it requires the solution to these semiclassical equations. It was argued by Zel'dovich [68] that the backreaction would tend to dissipate the inhomogeneities as in a sort of gravitational Lenz's law effect. Although this is a mechanism to homogenize the universe, in the standard scenario of a decelerating universe, one cannot explain the present large-scale homogeneity by any causal mechanism after the Planck era; on the other hand the inflationary scenario, which assumes an era of accelerated expansion, seems to solve the homogeneity problem quite naturally, see e.g. [745]. Early work on the backreaction effect on the geometry due to anisotropies was done by Lukash, Starobinsky, Novikov and Zel'dovich [104, 746], who assumed very special conditions near the Planck time, and by Hu and Parker [108] who considered a Bianchi type I anisotropic model, evaluated the stress-energy tensor in the low frequency approximation and computed the resulting modified Einstein's equations numerically. The results of such work indicate that the dynamical mechanism of particle production achieves a rapid damping of the anisotropy if the calculations are extrapolated to the Planck era.

The backreaction problem took an important step forward when effective action methods borrowed from quantum field theory were introduced into the field, as we explained in previous chapters where most of the relevant literature is also given. The first steps were made by Hartle, Fischetti and Hu [106, 109, 110], who studied the effect of anisotropies. But in their formalism the basic element is the usual effective action which is related to the generating functional of the in-out vacuum persistence amplitude. As explained before this in-out formalism leads to matrix elements rather than expectation values for the quantum operators. This means that from the in-out effective action one does not get the vacuum expectation value of the stress-energy tensor of the quantum field, and thus one still needs to compute the Bogoliubov transformation between the *in* and *out* vacua. This method is, however, very useful for the computation of the particles created, since the probability amplitude for particle creation is directly related to the vacuum persistence amplitude.

One of the most powerful and efficient methods, which is very well adapted to a perturbative scheme, to derive the expectation value of the stress-energy tensor is based on the one-loop order computation of the in-in or CTP effective action for the gravitational field due to its interaction with quantum fields. As we have seen in previous chapters this is a technique adapted to compute expectation values of

quantum operators. It was first proposed by Schwinger [239] and Keldysh [240] and was then extended to curved spacetimes by Jordan [181], by Calzetta and Hu [182] and also by Campos and Verdaguer [423].

The next step forward on the backreaction consists in the inclusion of the effect of the stress-energy tensor fluctuations, and this can be done in the framework of stochastic gravity. Here we provide an explicit computation of the in-in effective action for gravitational perturbations, on a spatially flat FLRW background, interacting with a massless conformally coupled scalar field to one-loop order. This in-in effective action is used to derive the quantum stress-energy tensor and the corresponding semiclassical Einstein equations. The effect of the stress-energy tensor fluctuations is included through the Einstein–Langevin equation which can also be derived from the in-in effective action once the noise kernel has been identified.

12.2 Influence Action for Cosmological Perturbations

To derive the Einstein–Langevin equation we can compute the influence action $S_{\rm IF}$, defined in Eq. (10.41), which is equivalent to evaluating the different kernels introduced after Eq. (10.38). This can be done directly from the expressions for the kernels in terms of products of the Feynman and the Wightman propagators for the scalar field in the background metric which were derived in Chapters 10 and 11, or it can be done by an explicit evaluation of the path integrals which define the influence action in Eqs. (10.33) and (10.34). We follow this second, more direct, route in this chapter.

The metric of our spacetime is given by

$$\tilde{g}_{\mu\nu}(x) = a^2(\eta)\left(\eta_{\mu\nu} + h_{\mu\nu}(x)\right), \tag{12.1}$$

where $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, \ldots, +1)$, $a(\eta) \equiv e^{\omega(\eta)}$ is the scale factor, η is the conformal time which is related to the cosmological or physical time t by $dt = ad\eta$, $h_{\mu\nu}(x)$ is a dimensionless symmetric tensor which represents arbitrary small metric perturbations, and we work in an n-dimensional spacetime in preparation for dimensional regularization.

The classical action for a free massless conformally coupled real scalar field $\Phi(x)$ is given in n dimensions by

$$S_{\rm m}[\tilde{g}_{\mu\nu}, \Phi] = -\frac{1}{2}\int d^n x\sqrt{-\tilde{g}}\left[\tilde{g}^{\mu\nu}\partial_\mu\Phi\partial_\nu\Phi + \xi(n)\tilde{R}\Phi^2\right], \tag{12.2}$$

where $\xi(n) = (n-2)/[4(n-1)]$, and $\tilde{R}$ is the Ricci scalar for the metric $\tilde{g}_{\mu\nu}$. Because of the conformal coupling $\xi(n)$ we can define a new field $\phi(x)$ and a new metric $g_{\mu\nu}$ as

$$\phi(x) \equiv e^{(n/2-1)\omega(\eta)}\Phi(x), \quad g_{\mu\nu}(x) \equiv e^{-2\omega(\eta)}\tilde{g}_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}(x), \tag{12.3}$$

so that $g_{\mu\nu}$ is conformally related to $\tilde{g}_{\mu\nu}$. After one integration by parts the classical action (12.2) takes the form

$$S_{\rm m}[g_{\mu\nu},\phi] = -\frac{1}{2}\int d^n x\sqrt{-g}\,[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + \xi(n)R\phi^2], \tag{12.4}$$

which is the action for a free massless conformally coupled real scalar field $\phi(x)$ in a nearly flat spacetime with metric $g_{\mu\nu}$. Although the physical field is $\Phi(x)$ the fact that it is related to the field $\phi(x)$ by a power of the conformal factor implies that a positive frequency mode of the field $\phi(x)$ in flat spacetime will correspond to a positive frequency mode in the conformally related space; these modes define the conformal vacuum. Thus quantum effects such as particle creation will be due to the breaking of conformal flatness which, in this case, is produced by the perturbations $h_{\mu\nu}(x)$.

Expanding the above action in terms of these perturbations, after integrations by parts we have

$$S_{\rm m}[g_{\mu\nu},\phi] = \frac{1}{2}\int d^n x\,\phi[\Box + V^{(1)} + V^{(2)} + \dots]\phi, \tag{12.5}$$

where, the operators $V^{(1)}$ and $V^{(2)}$ are defined as

$$\begin{aligned} V^{(1)}(x) &= -\bar{h}^{\mu\nu}\partial_\mu\partial_\nu - \left(\partial_\mu\bar{h}^{\mu\nu}\right)\partial_\nu - \xi(n)R^{(1)},\\ V^{(2)}(x) &= \hat{h}^{\mu\nu}\partial_\mu\partial_\nu + \left(\partial_\mu\hat{h}^{\mu\nu}\right)\partial_\nu - \xi(n)\left(R^{(2)} + \frac{1}{2}hR^{(1)}\right), \end{aligned} \tag{12.6}$$

where

$$\begin{aligned} \bar{h}_{\mu\nu} &\equiv h_{\mu\nu} - \frac{1}{2}h\eta_{\mu\nu},\\ \hat{h}_{\mu\nu} &\equiv h_\mu{}^\alpha h_{\alpha\nu} - \frac{1}{2}hh_{\mu\nu} + \frac{1}{8}h^2\eta_{\mu\nu} - \frac{1}{4}h_{\alpha\beta}h^{\alpha\beta}\eta_{\mu\nu}, \end{aligned} \tag{12.7}$$

and $R^{(1)}$ and $R^{(2)}$ are the first- and second-order terms, respectively, in the metric perturbations of the scalar curvature; see Section 12.5.1. We may compare the expansion in (12.5) to the analogous expansion in (9.38).

To the classical action for the matter fields $S_{\rm m}$ we have to add the action of the physical metric $\tilde{g}_{\mu\nu}$: $S_{\rm g}[\tilde{g}_{\mu\nu}]$. As it was emphasized in Chapter 10 in order to renormalize the effective action we have to introduce appropriate counterterms quadratic in the curvature. In this case, using dimensional regularization the only terms needed for renormalization are, see e.g. Birrell and Davies book [47],

$$\begin{aligned} S_{\rm g}[\tilde{g}_{\mu\nu}] \equiv \int d^n x\sqrt{-\tilde{g}(x)}\Bigg\{ &\frac{1}{16\pi G_{\rm N}}\tilde{R}(x)\\ &+\frac{\mu^{n-4}}{2880\pi^2(n-4)}\left[\tilde{R}_{\mu\nu\alpha\beta}(x)\tilde{R}^{\mu\nu\alpha\beta}(x) - \tilde{R}_{\mu\nu}(x)\tilde{R}^{\mu\nu}(x)\right]\Bigg\}, \end{aligned} \tag{12.8}$$

where μ is a mass renormalization parameter. Note that in four dimensions

$$\tilde{R}_{\mu\nu\alpha\beta}\tilde{R}^{\mu\nu\alpha\beta} - \tilde{R}_{\mu\nu}\tilde{R}^{\mu\nu} = \frac{3}{2}\tilde{C}_{\mu\nu\alpha\beta}\tilde{C}^{\mu\nu\alpha\beta} - \frac{1}{2}\tilde{\mathcal{E}}_4, \tag{12.9}$$

where $\tilde{\mathcal{E}}_4 \equiv \tilde{R}^{\mu\nu\alpha\beta}\tilde{R}_{\mu\nu\alpha\beta} - 4\tilde{R}^{\mu\nu}\tilde{R}_{\mu\nu} + \tilde{R}^2$, is the Euler density which does not contribute to the dynamics. Thus the previous counterterm leads to the square of the Weyl tensor. We can add also a term proportional to $\tilde{R}^2$ in the gravitational action with an arbitrary coefficient, but since this term is not needed for renormalization we do not introduce it here, for simplicity.

To compute the influence action $S_{\rm IF}$ from Eqs. (10.30) and (10.31) we have to introduce two scalar fields $\phi_+(x)$ and $\phi_-(x)$ which coincide at some future time T, $\phi_+(T) = \phi_-(T)$, and which evolve in two different geometries given by $h^+_{\mu\nu}$ and $h^-_{\mu\nu}$ such that $h^+_{\mu\nu}(T) = h^-_{\mu\nu}(T)$. The standard Gaussian path integral computation, as was explained in Section 9.2, leads to

$$S_{\rm IF}[h^\pm_{\mu\nu}] = -\frac{i}{2}{\rm Tr}(\ln G), \tag{12.10}$$

where G stands for the 2×2 matrix propagator G_{AB} for the fields $\phi_+(x)$ and $\phi_-(x)$ that may be obtained from the action (12.5) for the + field minus the same action for the – field. This propagator cannot be found exactly but it can be obtained perturbatively in powers of the metric perturbations. The unperturbed matrix propagator, G^0_{AB}, is the inverse of the kinetic operator ${\rm diag}(\Box - i\epsilon, -(\Box + i\epsilon))$, where we introduced the usual prescriptions for the vacuum state. It has the following components: $G^0_{++} = \Delta_F$, $G^0_{--} = \Delta_D$, $G^0_{+-} = \Delta^-$ and $G^0_{-+} = \Delta^+$, where Δ_F and Δ_D are the Feynman and Dyson propagators, respectively, and $\Delta^\pm$ are the Wightman functions, which were introduced in (9.30).

Up to second order in the metric perturbations we get (see Eqs. (9.42) and (9.46))

$$\begin{aligned} S_{\rm IF}[h^\pm_{\mu\nu}] \simeq & -\frac{i}{2}{\rm Tr}(\ln G^0) \\ & + \frac{i}{2}{\rm Tr}\Big[\ V^{(1)}_+ G^0_{++} - V^{(1)}_- G^0_{--} + V^{(2)}_+ G^0_{++} - V^{(2)}_- G^0_{--} \\ & - \frac{1}{2}V^{(1)}_+ G^0_{++} V^{(1)}_+ G^0_{++} - \frac{1}{2}V^{(1)}_- G^0_{--} V^{(1)}_- G^0_{--} \\ & + V^{(1)}_+ G^0_{+-} V^{(1)}_- G^0_{-+}\ \Big], \end{aligned} \tag{12.11}$$

where we have defined $V^{(i)}_+ \equiv V^{(i)}_{++}$ and $V^{(i)}_- \equiv -V^{(i)}_{--}$ $(i = 1, 2)$, since the operators $V^{(i)}$ are obviously diagonal. The first trace term is independent of the metric perturbations, its divergences are cancelled by terms which lead to the conformal anomaly. The tadpole terms of type ${\rm Tr}(VG)$ involve n-dimensional integrals which are identically zero in dimensional regularization. Thus, all calculation reduces to the computation of the three terms of type ${\rm Tr}(VGVG)$. To streamline

this section the detailed evaluation of these terms is given in Section 12.4, completed by Section 12.5 which contains some useful mathematical expressions.

After dimensional regularization of the divergent terms and renormalization with the action of the gravitational action (12.8), we obtain the renormalized effective action, see (10.35), for the gravitational field. Up to second order it can be written as

$$S_{\rm eff}[\tilde{g}^{\pm}_{\mu\nu}] = S^{\rm R}_{\rm g}[\tilde{g}^{+}_{\mu\nu}] - S^{\rm R}_{\rm g}[\tilde{g}^{-}_{\mu\nu}] + S^{\rm R}_{\rm IF}[h^{\pm}_{\mu\nu}]. \tag{12.12}$$

The renormalized influence action is given by (see the details in Section 12.4)

$$\begin{aligned} S^{\rm R}_{\rm IF}[h^{\pm}_{\mu\nu}] = &-\frac{1}{8}\int d^4x d^4y [h_{\mu\nu}(x)] H^{\mu\nu\alpha'\beta'}(x,y;\mu)\{h_{\alpha'\beta'}(y)\} \\ &+ \frac{i}{8}\int d^4x d^4y [h_{\mu\nu}(x)] N^{\mu\nu\alpha'\beta'}(x,y)[h_{\alpha'\beta'}(y)]. \end{aligned} \tag{12.13}$$

The $S^{\rm R}_{\rm g}[\tilde{g}^{\pm}_{\mu\nu}]$ comes from the gravitational action (12.8) after renormalization and is given by (see the details in Section 12.4)

$$\begin{aligned} S^{\rm R}_{\rm g}[\tilde{g}_{\mu\nu}] = &\int d^4x\sqrt{-\tilde{g}(x)}\left[\frac{\tilde{R}(x)}{16\pi G_{\rm N}} - \frac{1}{12}\frac{1}{2880\pi^2}\tilde{R}^2(x)\right] \\ &+ \frac{1}{1440\pi^2}\int d^4x\sqrt{-g(x)}\left[G^{\mu\nu}(x)\omega_{;\mu}\omega_{;\nu} + \Box_g\omega(\omega_{;\nu}\omega^{;\nu}) + \frac{1}{2}(\omega_{;\mu}\omega^{;\mu})^2\right] \\ &+ \frac{1}{2880\pi^2}\int d^4x\sqrt{-g(x)}\left[R_{\mu\nu\alpha\beta}(x)R^{\mu\nu\alpha\beta}(x) - R_{\mu\nu}(x)R^{\mu\nu}(x)\right]\omega(x), \end{aligned} \tag{12.14}$$

where the terms with and without tilde refer to tensors obtained with metrics $\tilde{g}_{\mu\nu}$ and $g_{\mu\nu}$, respectively.

Here we have used the notation of Eq. (10.39) and the kernels $H^{\mu\nu\alpha'\beta'}$ and $N^{\mu\nu\alpha'\beta'}$ are given by

$$H^{\mu\nu\alpha'\beta'}(x,y;\mu) = \frac{2}{3}F^{\mu\nu\alpha'\beta'}_{x} H(x-y;\mu), \tag{12.15}$$

where $F^{\mu\nu\alpha'\beta'}_{x}$ is the differential operator

$$F^{\mu\nu\alpha'\beta'}_{x} \equiv 3F^{\mu(\alpha'}_{x}F^{\beta')\nu}_{x} - F^{\mu\nu}_{x}F^{\alpha'\beta'}_{x}, \quad F^{\mu\nu}_{x} \equiv \eta^{\mu\nu}\Box_x - \partial^{\mu}_{x}\partial^{\nu}_{x}, \tag{12.16}$$

and

$$H(x;\mu) = \frac{1}{1920\pi^2}\int \frac{d^4p}{(2\pi)^4}e^{ip\cdot x}\left[\ln\frac{|p^2|}{\mu^2} - i\pi\,{\rm sgn}(p^0)\theta(-p^2)\right]. \tag{12.17}$$

The noise kernel is given by

$$N^{\mu\nu\alpha'\beta'}(x,y) = \frac{2}{3}F^{\mu\nu\alpha'\beta'}_{x} N(x-y), \tag{12.18}$$

where

$$N(x) = \frac{1}{1920\pi} \int \frac{d^4p}{(2\pi)^4} e^{ip\cdot x} \theta(-p^2). \tag{12.19}$$

These kernels result from suitable combinations of the kernels derived in Section 12.4, to be explained later.

12.3 Einstein–Langevin Equation

We are now in a position to derive the Einstein–Langevin equation. Following Section 10.4 we can introduce the stochastic effective action

$$S^s_{\rm eff}[\tilde{g}^{\pm}_{\mu\nu}, \xi] = S^{\rm R}_{\rm g}[\tilde{g}^{+}_{\mu\nu}] - S^{\rm R}_{\rm g}[\tilde{g}^{-}_{\mu\nu}] + S^{{\rm R},s}_{\rm IF}[h^{\pm}_{\mu\nu}, \xi], \tag{12.20}$$

where the renormalized stochastic influence action is given by

$$S^{{\rm R},s}_{\rm IF}[h^{\pm}_{\mu\nu}, \xi] = {\rm Re}\, S^{\rm R}_{\rm IF}[h^{\pm}_{\mu\nu}] + \frac{1}{2} \int d^4x\, \xi^{\mu\nu}(x)[h_{\mu\nu}(x)], \tag{12.21}$$

with the Gaussian stochastic field $\xi^{\mu\nu}$ defined by $\langle \xi^{\mu\nu}(x) \rangle_s = 0$ and

$$\langle \xi^{\mu\nu}(x) \xi^{\alpha'\beta'}(y) \rangle_s = N^{\mu\nu\alpha'\beta'}(x, y). \tag{12.22}$$

The Einstein–Langevin equation can be obtained by functional derivation according to Eq. (10.47). Note that we have to take the derivative with respect to the physical metric $\tilde{g}_{\mu\nu}$, and we use that for an arbitrary functional $A[\tilde{g}_{\mu\nu}]$,

$$\frac{\delta A[\omega, g_{\mu\nu}]}{\sqrt{-g}\delta g_{\mu\nu}} = e^{6\omega} \frac{\delta A[\tilde{g}_{\mu\nu}]}{\sqrt{-\tilde{g}}\delta \tilde{g}_{\mu\nu}}. \tag{12.23}$$

The final result is:

$$\begin{aligned} e^{6\omega} \Bigg[& -\frac{1}{8\pi G_{\rm N}} \left(\tilde{G}^{\mu\nu}_{(0)} + \tilde{G}^{\mu\nu}_{(1)} \right) - \frac{1}{6} \frac{1}{2880\pi^2} \left(\tilde{B}^{\mu\nu}_{(0)} + \tilde{B}^{\mu\nu}_{(1)} \right) \\ & + \frac{1}{2880\pi^2} \left(\tilde{H}^{\mu\nu}_{(0)} + \tilde{H}^{\mu\nu}_{(1)} \right) - \frac{1}{1440\pi^2} \tilde{R}^{(0)}_{\alpha\beta} \tilde{C}^{\mu\alpha\nu\beta}_{(1)} \\ & - \frac{3}{720\pi^2} e^{-6\omega} (C^{\mu\alpha\nu\beta}_{(1)} \omega)_{,\alpha\beta} \Bigg] \\ & - \int d^4y A^{\mu\nu}_{(1)}(y) {\rm H}(x - y; \mu) + \xi^{\mu\nu} = \mathcal{O}(h^2_{\mu\nu}), \end{aligned} \tag{12.24}$$

where the (0) and (1) subscripts refer to the zero- and first-order terms, respectively, in the metric perturbation $h_{\mu\nu}$. The tensors $A^{\mu\nu}(x)$ and $B^{\mu\nu}(x)$ are given by Eqs. (10.9) and (10.10), respectively, and $H^{\mu\nu}(x)$ is given by

$$H^{\mu\nu} = -R^{\mu\alpha} R_{\alpha}{}^{\nu} + \frac{2}{3} R R^{\mu\nu} + \frac{1}{2} g^{\mu\nu} R_{\alpha\beta} R^{\alpha\beta} - \frac{1}{4} g^{\mu\nu} R^2, \tag{12.25}$$

which was called ${}^{(3)}H^{\mu\nu}$ in (2.73). When comparing the Einstein-Langevin equation (12.24) with Eq. (10.14) we recall that in our renormalization scheme [see Eq. (12.8)], we have implicitly fixed the arbitrary parameter β_d, whereas the parameter α_d appears related to the parameter μ. In fact, if we change μ by μ' in the kernel (12.17) we have

$$H(x-y;\mu) = H(x-y;\mu') + \frac{1}{1920\pi^2}\delta^{(4)}(x-y)\ln\frac{\mu^2}{\mu'^2}, \tag{12.26}$$

therefore the arbitrariness of α_d corresponds to that of the renormalization parameter μ. To be specific, one must assume that $\alpha_d(\mu)$ and that if we change μ by μ' also α_d changes appropriately so that the physical parameters in the Einstein–Langevin equation do not change. Note that $\tilde{A}^{\mu\nu}_{(0)} = 0$ because $\tilde{A}^{\mu\nu}$ is obtained by taking the functional derivative of an action density proportional to the square of the Weyl tensor, see (10.9), but the conformal symmetry is broken only by the metric perturbations $h_{\mu\nu}$, at zero order in the perturbation the physical metric is conformally flat and the Weyl tensor is zero. Since the parameters α_d and β_d as well as the gravitational constant can only be fixed by experiment we will introduce a (rescaled) parameter β in our final equations.

12.3.1 The Stress-Energy Tensor

Let us write the Einstein–Langevin equation (12.24) with the effective source term on the right-hand side as

$$\tilde{G}^{\mu\nu} = 8\pi G_{\rm N}\left(\langle T^{\mu\nu}\rangle + e^{-6\omega}\xi^{\mu\nu}\right), \tag{12.27}$$

where $\langle T^{\mu\nu}\rangle$ is the vacuum expectation value of the stress-energy tensor of the quantum field up to first order in $h_{\mu\nu}$; it includes the terms quadratic in the curvature which are needed for renormalization. It is given by

$$\begin{aligned}
\langle T^{\mu\nu}_{(0)}\rangle &= \frac{1}{2880\pi^2}\left[\tilde{H}^{\mu\nu}_{(0)} - \frac{1}{6}\tilde{B}^{\mu\nu}_{(0)}\right]\\
\langle T^{\mu\nu}_{(1)}\rangle &= \frac{1}{2880\pi^2}\left[\tilde{H}^{\mu\nu}_{(1)} - \frac{1}{6}\tilde{B}^{\mu\nu}_{(1)} - 2\tilde{R}^{(0)}_{\alpha\beta}\tilde{C}^{\mu\alpha\nu\beta}_{(1)} - 12e^{-6\omega}(C^{\mu\alpha\nu\beta}_{(1)}\omega)_{,\alpha\beta}\right.\\
&\qquad \left. +3e^{-6\omega}\int d^4y A^{\mu\nu}_{(1)}(y)\mathrm{H}(x-y;\mu)\right].
\end{aligned} \tag{12.28}$$

This tensor was first derived by Campos and Verdaguer [423] using the in-in or CTP formalism, although it had been derived previously by other means. It was obtained by Horowitz and Wald [684, 685], who used an axiomatic approach to derive it, and by Starobinsky [724] who used a modified Pauli–Villars regularization method, see e.g. [69]. One should note that the stress-energy tensor computed does not include the energy of the particles created which is a second-order correction to the computed terms; it includes only first-order vacuum

polarization effects. Although the energy of the particles created is small it might have a long-term cumulative effect.

To summarize, the stochastic equation (12.27) is the semiclassical Einstein–Langevin equation for weakly inhomogeneous perturbations on spatially flat FLRW spacetimes with a conformally coupled massless scalar field. Here we have also derived a stochastic correction to the vacuum expectation value of the quantum field stress-energy tensor (12.28), which accounts for the noise associated with the fluctuations of the stress-energy tensor on the homogeneous background spacetime.

12.3.2 The Trace Anomaly

We can recover the trace anomaly result, see e.g. [47]. In fact, to first order in $h_{\mu\nu}$ we have from Eq. (12.28) that

$$\begin{aligned}\langle T^{\mu}{}_{\mu}\rangle &= \langle T_{(0)\mu}^{\ \ \mu}\rangle + \langle T_{(1)\mu}^{\ \ \mu}\rangle + O(h_{\mu\nu}^2) \\ &= \frac{1}{2880\pi^2}\left[\Box_{\tilde{g}}\tilde{R} + \left(\tilde{R}^{\mu\nu}\tilde{R}_{\mu\nu} - \frac{1}{3}\tilde{R}^2\right)\right] + \mathcal{O}(h_{\mu\nu}^2),\end{aligned} \tag{12.29}$$

which agrees with (2.62) for a scalar field, to linear order in the metric perturbations. Note that the tensor $\tilde{A}^{\mu\nu}$ is traceless due to the conformal invariance of $\int d^4x\sqrt{-g}\tilde{C}_{\mu\nu\alpha\beta}\tilde{C}^{\mu\nu\alpha\beta}$.

As we have seen in Section 10.2 the stochastic correction to the stress-energy tensor is traceless for conformal fields and has vanishing divergence to first order in the metric perturbations. These properties can be easily checked in this case. That $\langle\xi_{\mu}^{\mu}\rangle_s = 0$ follows directly from Eq. (12.22) and (12.18) by noticing that $\langle N_{\mu}^{\mu\ \alpha'\beta'}(x,y)\rangle = 0$ as a consequence of $F^{\mu(\alpha}F_{\mu}^{\beta)} = \Box F^{\alpha\beta}$ and $F_{\mu}^{\mu} = 3\Box$. Furthermore, using $\tilde{g}_{\mu\nu} = e^{2\omega}g_{\mu\nu}$ and that $\xi^{\mu\nu}$ is symmetric and traceless, it is easy to see that $\tilde{\nabla}_{\nu}\left(e^{-6\omega}\xi^{\mu\nu}\right) = e^{-6\omega}\nabla_{\nu}\xi^{\mu\nu}$. Then, from equation (12.22) and the symmetries of ξ, we obtain that $\nabla_{\nu}\xi^{\mu\nu} = \mathcal{O}(h_{\mu\nu})$. It is thus consistent to write this term on the right-hand side of Einstein equations and consider it as a correction of order higher than $\langle T_{(0)}^{\mu\nu}\rangle$ (note that $\tilde{\nabla}_{\nu}\langle T_{(0)}^{\mu\nu}\rangle = \mathcal{O}(h_{\mu\nu}^2)$).

12.3.3 The Semiclassical Equation

Taking the mean value of the Einstein–Langevin equation with respect to the tensor field source $\xi^{\mu\nu}$ with Gaussian probability distributions (12.22), we obtain the semiclassical Einstein equation. This equation can be used to study the stability of the zero order semiclassical equation (12.27),

$$\tilde{G}_{(0)}^{\mu\nu} = 8\pi G_{\rm N}\langle T_{(0)}^{\mu\nu}\rangle, \tag{12.30}$$

where the stress-energy tensor is given by Eqs. (12.28). This stress-energy tensor $\langle T_{(0)}^{\mu\nu}\rangle$ for a scalar field in a conformally flat spacetime exactly agrees with

that found by other techniques; see e.g. [47]. Comparing with the semiclassical equation (10.7) we see that since the tensor $\tilde{A}^{\mu\nu}_{(0)} = 0$ there is no α_d parameter. On the other hand the parameter β_d is arbitrary and should be determined by experiment. Moreover, the coefficient of the tensor $\tilde{H}^{\mu\nu}_{(0)}$ is fixed, its value differs for different types and number of fields, it has been computed here for a single scalar field. Note that the tensor $\tilde{H}^{\mu\nu}$ is conserved only in conformally flat spacetimes and cannot be obtained by functional derivation of a geometrical term in the action, so that it plays a very different role than the tensors $\tilde{A}^{\mu\nu}$ and $\tilde{B}^{\mu\nu}$.

Eq. (12.30) can be solved to find the conformal factor $\omega(\eta)$. It was shown by Starobinsky [359], see also [360], that this equation describes the so-called trace anomaly driven inflation. Note that we could use the Einstein–Langevin equation (12.27) to compute the two-point correlations for the metric perturbations induced by the stress-energy tensor fluctuations; in Chapter 16 we explain how to compute these correlations to avoid spurious divergencies. These metric correlations are relevant for the generation of primordial inhomogeneities in this inflationary scenario [651].

12.3.4 Particle Creation

It is interesting to notice also that the imaginary term in the regularized influence action (12.13) can be written after an integration by parts in the following alternative form, see [186] for the details,

$$\mathrm{Im}\, S^{\mathrm{R}}_{\mathrm{IF}}[h^{\pm}_{\mu\nu}] = \frac{1}{2}\int d^4x d^4y [C_{\mu\nu\alpha\beta}(x)] N(x-y) [C^{\mu\nu\alpha\beta}(y)], \tag{12.31}$$

which shows that the noise couples to the Weyl conformal tensor. Here we have used the notation (10.42) for the square brackets.

The fact that the stochastic source couples to the conformal tensor is not a surprise. For a conformal quantum field nontrivial quantum effects are a consequence of breaking the conformal symmetry of the spacetime, which is characterized by the conformal tensor. For instance, it is known that the probability density of pair creation in this case [76], or in the presence of small anisotropy [185], is determined by the square of the Weyl tensor. Thus, as it was shown by Calzetta and Hu [216] and by Martín and Verdaguer [189] there is a direct relation between particle creation and noise.

12.4 Detailed Computation of the Trace Terms

As it was pointed out after Eq. (12.11) the computation of the influence action S_{IF} is reduced to the evaluation of the three trace terms of the type $\mathrm{Tr}(VGVG)$ in the expression (12.11), or equivalently (9.46). The first term appears also in

the evaluation of the in-out effective action and the second and third are typical of the in-in contribution to this order.

Let us evaluate the first term, i.e., $T^+ \equiv -\frac{i}{4}\mathrm{Tr}(V_+^{(1)}G^0_{++}V_+^{(1)}G^0_{++})$, and recall that $G^0_{++} = \Delta_F$ is given in (9.30) with $m = 0$,

$$\begin{aligned} T^+ &= -\frac{i}{4}\int d^n x d^n x' V_+^{(1)}(x)\Delta_F(x,x')V_+^{(1)}(x')\Delta_F(x',x) \\ &= -\frac{i}{4}\int d^n x d^n x' \int \frac{d^n p}{(2\pi)^n}\frac{d^n q}{(2\pi)^n} \\ &\quad\times\left[\left(\left(\partial_\mu \bar{h}_+^{\mu\nu}(x)\right)\partial_\nu + \bar{h}_+^{\mu\nu}(x)\partial_\mu\partial_\nu + \xi(n)R_+^{(1)}(x)\right)\frac{e^{iq\cdot(x-x')}}{q^2-i\epsilon}\right] \\ &\quad\times\left[\left(\left(\partial_{\alpha'}\bar{h}_+^{\alpha'\beta'}(x')\right)\partial_{\beta'} + \bar{h}_+^{\alpha'\beta'}(x')\partial_{\alpha'}\partial_{\beta'}\right.\right. \\ &\quad\left.\left.+\xi(n)R_+^{(1)}(x')\right)\frac{e^{ip\cdot(x'-x)}}{p^2-i\epsilon}\right]. \end{aligned} \tag{12.32}$$

We now introduce the projector $P^{\mu\nu} = \eta^{\mu\nu} - p^\mu p^\nu u/p^2$, the symbol $\eta^{\mu\nu\alpha\beta}$, $\bar{h}^{\mu\nu} \equiv \eta^{\mu\alpha\nu\beta}h_{\alpha\beta}$, change the p integration by $p' \equiv q - p$, rename p' as p again, and write T^+ as,

$$T^+ = -i\int d^n x d^n y h^+_{\mu\nu}(x)h^+_{\alpha'\beta'}(y)\int\frac{d^n p}{(2\pi)^n}e^{ip\cdot(x-y)}\hat{K}^{\mu\nu\alpha'\beta'}(p), \tag{12.33}$$

where

$$\begin{aligned} \hat{K}^{\mu\nu\alpha'\beta'}(p) &= \frac{1}{4}\int\frac{d^n q}{(2\pi)^n}\frac{1}{(q^2-i\epsilon)\left[(p-q)^2-i\epsilon\right]} \\ &\quad\times\left(\eta^{\rho\mu\tau\nu}(q-p)_\rho q_\tau - \xi(n)p^2P^{\mu\nu}\right) \\ &\quad\times\left(\eta^{\lambda\alpha'\sigma\beta'}(q-p)_\lambda q_\sigma - \xi(n)p^2P^{\alpha'\beta'}\right). \end{aligned}$$

The momentum integrals can be computed in the standard way, see Section 12.5.3, and expanding around $n = 4$ we get, after a rather long calculation,

$$\begin{aligned} \hat{K}^{\mu\nu\alpha'\beta'}(p) = \frac{p^4 I_1(p)}{1440}\Big[\ &(3P^{\mu\beta'}P^{\nu\alpha'} - P^{\mu\nu}P^{\alpha'\beta'}) \\ &+\frac{(n-4)}{15}\left(8(P^{\mu\nu}P^{\alpha'\beta'} - 3P^{\mu\beta'}P^{\nu\alpha'}) + 5P^{\mu\nu}P^{\alpha'\beta'}\right) \\ &+\mathcal{O}(n-4)^2\ \Big] \end{aligned} \tag{12.34}$$

where

$$I_1(p) = -\frac{i}{8\pi^2}\left(\frac{1}{n-4} + \frac{1}{2}\ln\left[\frac{p^2-i\epsilon}{\mu_0^2}\right] + \mathcal{O}(n-4)\right), \tag{12.35}$$

and the parameter μ_0 is a fixed parameter involving Euler's constant γ. Now using the expressions of the Riemann components in terms of the projectors $P_{\mu\nu}$ of Section 12.5.1 we can write,

$$
\begin{aligned}
T^{+} = -\frac{\alpha}{4}\Big\{ & \frac{1}{n-4}\int d^4x \left(3R^{+}_{\mu\nu\alpha\beta}(x)R^{+\mu\nu\alpha\beta}(x) - R^{+2}(x)\right) \\
& - \int d^4x d^4y \left[3R^{+}_{\mu\nu\alpha\beta}(x)R^{+\mu\nu\alpha\beta}(y) - R^{+}(x)R^{+}(y)\right] \mathrm{K}^{+}(x-y;\mu_0) \\
& + \frac{1}{3}\int d^4x R^{+2}(x) + \mathcal{O}(n-4) \ \Big\}, \qquad (12.36)
\end{aligned}
$$

where

$$
\mathrm{K}^{\pm}(x-y;\mu_0) \equiv -\frac{1}{2}\int \frac{d^4p}{(2\pi)^4} e^{ip\cdot(x-y)} \ln\left[\frac{(p^2 \mp i\epsilon)}{\mu_0^2}\right], \qquad (12.37)
$$

and where we have introduced the dimensionless constant

$$
\alpha \equiv \frac{N}{2880\pi^2}, \qquad (12.38)
$$

with N the number of fields, being $N = 1$ in the present case.

Note that we also write the kernel $\mathrm{K}^{-}(x-y;\mu_0)$ for the analogous evaluation of T^{-}, the second of the trace terms which involves the Dyson propagator. Note that the divergent terms with a pole at $n = 4$ are local and quadratic in the curvature. They may be compensated by counterterms in the gravitational part of the action $S^{(2)}_{\mathrm{g}}$ coming from (12.8). Recall that the curvature terms here depend on the metric $g_{\mu\nu}$ rather than the physical metric $\tilde{g}_{\mu\nu}$.

Let us compute now the third trace term in the influence action (12.11), i.e., $T \equiv \frac{i}{2}\mathrm{Tr}(V^{(1)}_{+}G^{0}_{+-}V^{(1)}_{-}G^{0}_{-+})$, which depends on the propagators $G^{0}_{+-} = \Delta^{-}$ and $G^{0}_{-+} = \Delta^{+}$, given in (9.30) with $m = 0$. We can write

$$
\begin{aligned}
T &= \frac{i}{2}\int d^nx d^nx' V^{(1)}_{+}(x)\Delta^{-}(x,x')V^{(1)}_{-}(x')\Delta^{+}(x',x) \\
&= (-2i\pi^2)\int d^nx d^nx' \int \frac{d^np}{(2\pi)^n}\frac{d^nq}{(2\pi)^n}\delta(q^2)\theta(-q^0)\delta(p^2)\theta(p^0) \\
&\quad \times \left[\left(\partial_\mu \bar{h}^{\mu\nu}_{+}(x)\right)\partial_\nu + \bar{h}^{\mu\nu}_{+}(x)\partial_\mu\partial_\nu + \xi(n)R^{(1)}_{+}(x)\right] e^{iq\cdot(x-x')} \\
&\quad \times \Big[\left(\partial_{\alpha'}\bar{h}^{\alpha'\beta'}_{-}(x')\right)\partial_{\beta'} + \bar{h}^{\alpha'\beta'}_{-}(x')\partial_{\alpha'}\partial_{\beta'} \\
&\quad + \xi(n)R^{(1)}_{-}(x')\Big] e^{ip\cdot(x'-x)}, \qquad (12.39)
\end{aligned}
$$

changing the integration variable from p to $q - p$ as in the previous case we can write,

$$
T = -8i\pi^2 \int d^nx d^ny h^{+}_{\mu\nu}(x)h^{-}_{\alpha'\beta'}(y)\int \frac{d^np}{(2\pi)^n} e^{ip\cdot(x-y)}\hat{L}^{\mu\nu\alpha'\beta'}(p), \qquad (12.40)
$$

where

$$\hat{L}^{\mu\nu\alpha'\beta'}(p) = \frac{1}{4}\int \frac{d^n q}{(2\pi)^n}\delta((p-q)^2)\theta(q^0-p^0)\delta(q^2)\theta(-q^0) \times (\eta^{\rho\mu\tau\nu}(q-p)_\rho q_\tau - \xi(n)p^2 P^{\mu\nu}) \times \left(\eta^{\lambda\alpha'\sigma\beta'}(q-p)_\lambda q_\sigma - \xi(n)p^2 P^{\alpha'\beta'}\right).$$

After performing the phase space integrals and expanding around $n = 4$, see Section 12.5.3, we obtain

$$\hat{L}^{\mu\nu\alpha'\beta'}(p) = \frac{p^4 I_2(p)}{1440}\left[(3P^{\mu\beta'}P^{\nu\alpha'} - P^{\mu\nu}P^{\alpha'\beta'}) + \mathcal{O}(n-4)\right], \tag{12.41}$$

where

$$I_2(p) = \frac{1}{8\pi^2}\left[\frac{1}{4\pi}\theta(-p^2)\theta(-p^0) + \mathcal{O}(n-4)\right], \tag{12.42}$$

which has no poles at $n = 4$, therefore the term T requires no counterterms in the action to be renormalized. Using the expressions of the Riemann components in terms of $P_{\mu\nu}$ (see Section 12.5.1) we get,

$$T = \frac{\alpha}{2}\int d^4x d^4y \left[3R^{+}_{\mu\nu\alpha\beta}(x)R^{-\mu\nu\alpha\beta}(y) - R^{+}(x)R^{-}(y)\right]\mathrm{K}(x-y) + \mathcal{O}(n-4), \tag{12.43}$$

where

$$\mathrm{K}(x-y) \equiv -\frac{1}{2}\int \frac{d^4p}{(2\pi)^4}e^{ip\cdot(x-y)}(2\pi i)\theta(-p^2)\theta(-p^0). \tag{12.44}$$

Here again the Riemann components refer to the metric $g_{\mu\nu}$.

We now have the influence action up to second order in the metric perturbations, $S^{(2)}_{\mathrm{IF}} = T^{+} + T^{-} + T$, which must be renormalized by adding the gravitational action with the counterterms up to the second order in $h_{\mu\nu}$. The explicit expansion of S_g, in (12.8), up to this order in terms of the curvature components of the metric $g_{\mu\nu}$ is:

$$\begin{aligned} S_{\mathrm{g}}[\tilde{g}_{\mu\nu}] &\equiv S^{(0)}_{\mathrm{g}} + S^{(1)}_{\mathrm{g}} + S^{(2)}_{\mathrm{g}} + \cdots \\ &= \frac{1}{16\pi G_{\mathrm{N}}}\int d^4x(-g(x))^{1/2}e^{2\omega}[R(x) + 6\omega_{;\mu}\omega^{;\mu}] \\ &\quad + \frac{\alpha}{4(n-4)}\int d^4x[3R_{\mu\nu\alpha\beta}(x)R^{\mu\nu\alpha\beta}(x) - R^2(x)] \\ &\quad + \alpha\int d^4x[R_{\mu\nu\alpha\beta}(x)R^{\mu\nu\alpha\beta}(x) - R_{\mu\nu}(x)R^{\mu\nu}(x)]\ln(\mu e^{\omega}) \\ &\quad + \alpha\int d^4x(-g(x))^{1/2}\,[2R^{\mu\nu}\omega_{;\mu}\omega_{;\nu} + R\Box_g\omega - 4(\Box_g\omega)\omega_{;\nu}\omega^{;\nu} \\ &\quad -3(\Box_g\omega)^2 - 2(\omega_{;\nu}\omega^{;\nu})^2] + \mathcal{O}(n-4), \end{aligned} \tag{12.45}$$

where we have dropped the + sign on the fields for simplicity and we have used the expansions collected in Section 12.5.2.

Finally adding these terms to the bare $S^{(2)}_{\rm IF}$ which is given by the calculated expressions (12.36) and (12.43) we get the renormalized influence action

$$\begin{aligned} S^{\rm R}_{\rm IF}[h^{\pm}_{\mu\nu}] = & \frac{3\alpha}{2}\int d^4x d^4y C^{+}_{\mu\nu\alpha\beta}(x) C^{+\mu\nu\alpha\beta}(y) {\rm K}^{+}(x-y;\bar\mu) \\ & - \frac{3\alpha}{2}\int d^4x d^4y C^{-}_{\mu\nu\alpha\beta}(x) C^{-\mu\nu\alpha\beta}(y) {\rm K}^{-}(x-y;\bar\mu) \\ & + 3\alpha \int d^4x d^4y C^{+}_{\mu\nu\alpha\beta}(x) C^{-\mu\nu\alpha\beta}(y) {\rm K}(x-y), \end{aligned} \tag{12.46}$$

together with the renormalized $S^{\rm R}_{\rm g}$ given by (12.14). Note that $\bar\mu \equiv \mu\mu_0$, and also that for the variational formulae needed in the computations of the field equations one may introduce the $\sqrt{-g}$ in all the integrals involving quadratic curvature terms of the metric $g_{\mu\nu}$.

Finally we can write the above influence action $S^{\rm R}_{\rm IF}$ in the form (12.13). This is easily done after integration by parts using the explicit expressions given in Section 12.5.1. In doing this one finds the new kernel ${\rm H}(x-y;\bar\mu)$ as the sum of the integrals (12.37) and (12.44),

$$\begin{aligned} {\rm H}(x-y;\bar\mu) \equiv & \, {\rm K}^{+}(x-y;\bar\mu) + {\rm K}(x-y) \\ = & -\frac{1}{2}\int \frac{d^4p}{(2\pi)^4} e^{ip\cdot(x-y)} \left\{ \ln\left[\frac{(p^2-i\epsilon)}{\bar\mu^2}\right] \right. \\ & \left. + (2\pi i)\theta(-p^2)\theta(-p^0) \right\}, \end{aligned} \tag{12.47}$$

which can be simplified to the form given by (re-scaled with α) (12.17), i.e.,

$${\rm H}(x-y;\bar\mu) = -\frac{1}{2}\int \frac{d^4p}{(2\pi)^4} e^{ip\cdot(x-y)} \left\{ \ln\left[\frac{|p^2|}{\bar\mu^2}\right] + i\pi\theta(-p^2){\rm sgn}(-p^0) \right\}, \tag{12.48}$$

after using $\ln(\pm i) = \pm i\frac{\pi}{2}$ and $\lim_{\epsilon\to 0}\ln(\epsilon + ix) = \ln|x| + {\rm sgn}(x) i\frac{\pi}{2}$. Note that this kernel is real, in spite of appearances, because the imaginary part of the integrand is an odd term with respect to the integration variable p^μ.

12.5 Mathematical Supplement

In this section we collect a series of expressions useful for the detailed calculations carried out in the previous section. The expressions are self explanatory: they include expansions of curvature quantities around flat spacetime up to second order in the metric perturbations, relations involving curvature tensors corresponding to conformally related metrics, momentum integrals in n dimensions for the one-loop corrections, and explicit functional differentiations of expressions needed in this chapter.

12.5.1 Expansions around Flat Space

Curvature Tensors

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad (-,+,\ldots,+). \tag{12.49}$$

$$g^{\mu\nu}(x) = \eta^{\mu\nu} - h^{\mu\nu}(x) + h^{\mu}{}_{\alpha}(x)h^{\alpha\nu}(x) + \mathcal{O}(h^3_{\mu\nu}). \tag{12.50}$$

$$(-g(x))^{1/2} = 1 + \frac{1}{2}h + \frac{1}{8}h^2 - \frac{1}{4}h_{\mu\nu}h^{\mu\nu} + \mathcal{O}(h^3_{\mu\nu}). \tag{12.51}$$

$$R^{\alpha}{}_{\beta\gamma\delta} = \partial_\gamma\Gamma^{\alpha}_{\beta\delta} - \partial_\delta\Gamma^{\alpha}_{\beta\gamma} + \Gamma^{\alpha}_{\mu\gamma}\Gamma^{\mu}_{\beta\delta} - \Gamma^{\alpha}_{\mu\delta}\Gamma^{\mu}_{\beta\gamma}. \tag{12.52}$$

$$\Gamma^{\alpha}_{\beta\delta} = \frac{1}{2}\eta^{\alpha\lambda}S_{\lambda\delta,\beta} - \frac{1}{2}h^{\alpha\lambda}S_{\lambda\delta,\beta} + \mathcal{O}(h^3_{\mu\nu}). \tag{12.53}$$

$$S_{\lambda\delta,\beta} \equiv h_{\lambda\delta,\beta} + h_{\beta\lambda,\delta} - h_{\beta\delta,\lambda}. \tag{12.54}$$

$$R_{\sigma\beta\gamma\delta} = S_{\sigma[\delta,\beta\gamma]} - h^{\lambda}{}_{\sigma,[\gamma}S_{\lambda\delta],\beta} + \frac{1}{2}\eta^{\mu\rho}S_{\sigma[\gamma,\mu}S_{\rho\delta],\beta} + \mathcal{O}(h^3_{\mu\nu}). \tag{12.55}$$

$$\begin{aligned} R_{\beta\delta} &= \frac{1}{2}\eta^{\mu\nu}(h_{\mu\delta,\beta\nu} - h_{\beta\delta,\mu\nu} - h_{\mu\nu,\beta\delta} + h_{\beta\nu,\mu\delta}) \\ &- \left(h^{\mu\nu}{}_{,\mu} - \frac{1}{2}h^{,\nu}\right)h_{\nu(\delta,\beta)} + \frac{1}{2}(h^{\mu\nu}h_{\beta\delta,\nu})_{,\mu} + \frac{1}{4}h^{\mu\nu}{}_{,\delta}h_{\mu\nu,\beta} - \frac{1}{4}h^{,\mu}h_{\beta\delta,\mu} \\ &+ h_{\mu\delta,\nu}h_{\beta}{}^{[\mu,\nu]} + \frac{1}{2}h^{\mu\nu}h_{\mu\nu,\beta\delta} - h^{\mu\nu}h_{\mu(\beta,\delta)\nu} + \mathcal{O}(h^3_{\mu\nu}) \end{aligned} \tag{12.56}$$

$$\begin{aligned} R &= h^{\alpha\beta}{}_{,\alpha\beta} - h_{,\alpha}{}^{\alpha} \\ &- h^{\mu\nu}{}_{,\mu}h_{\nu\alpha}{}^{,\alpha} + h^{\mu\nu}{}_{,\mu}h_{,\nu} + \frac{3}{4}h^{\mu\nu,\alpha}h_{\mu\nu,\alpha} - \frac{1}{4}h^{,\mu}h_{,\mu} - \frac{1}{2}h^{\mu\nu,\alpha}h_{\alpha\mu,\nu} \\ &- 2h^{\mu\nu}h_{\mu\alpha,\nu}{}^{\alpha} + h^{\mu\nu}h_{,\mu\nu} + h^{\mu\nu}h_{\mu\nu,\alpha}{}^{\alpha} + \mathcal{O}(h^3_{\mu\nu}). \end{aligned} \tag{12.57}$$

$$R_{\sigma\beta\gamma\delta}R^{\sigma\beta\gamma\delta} = h_{\sigma\delta,\beta\gamma}h^{\sigma\delta,\beta\gamma} - 2h_{\sigma\delta,\beta\gamma}h^{\beta\delta,\sigma\gamma} + h_{\sigma\delta,\beta\gamma}h^{\beta\gamma,\sigma\delta} + \mathcal{O}(h^3_{\mu\nu}). \tag{12.58}$$

$$\begin{aligned} R_{\beta\delta}R^{\beta\delta} = \frac{1}{4}\big[&2h^{\mu\nu}{}_{,\alpha\mu}h_{\beta\nu}{}^{,\alpha\beta} - 4h^{\mu\nu}{}_{,\alpha\mu}h^{\alpha}{}_{\nu,\beta}{}^{\beta} - 4h^{\mu\nu}{}_{,\alpha\mu}h^{,\alpha}{}_{\nu} + 2h^{\mu\nu}{}_{,\alpha\mu}h^{\alpha\beta}{}_{,\beta\nu} \\ &+ h^{\mu\nu}{}_{,\alpha}{}^{\alpha}h_{\mu\nu,\beta}{}^{\beta} + 2h^{\mu\nu}{}_{,\alpha}{}^{\alpha}h_{,\mu\nu} + h^{,\mu\nu}h_{,\mu\nu} + \mathcal{O}(h^3_{\mu\nu})\big]. \end{aligned} \tag{12.59}$$

$$R^2 = h^{\alpha\beta}{}_{,\alpha\beta}h^{\mu\nu}{}_{,\mu\nu} - 2h^{\alpha\beta}{}_{,\alpha\beta}h_{,\mu}{}^{\mu} + h_{,\alpha}{}^{\alpha}h_{,\mu}{}^{\mu} + \mathcal{O}(h^3_{\mu\nu}). \tag{12.60}$$

Curvature Tensors in Terms of the Projector

$$P_{\mu\nu} = \eta_{\mu\nu} - \frac{q^\mu q^\nu u}{q^2}. \tag{12.61}$$

$$\int d^n x \left(R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta} - 4R_{\mu\nu}R^{\mu\nu} + R^2\right) = \mathcal{O}(h^3_{\mu\nu}). \tag{12.62}$$

$$\int d^n x R^2(x) = \int d^n x d^n y h_{\mu\nu}(x)h_{\alpha\beta}(y)\int \frac{d^n q}{(2\pi)^n}e^{iq\cdot(x-y)}P^{\mu\nu}P^{\alpha\beta}q^4 + \mathcal{O}(h^3_{\mu\nu}). \tag{12.63}$$

$$\int d^n x R_{\mu\nu\alpha\beta}(x) R^{\mu\nu\alpha\beta}(x) = \int d^n x d^n y h_{\mu\nu}(x) h_{\alpha\beta}(y) \times \int \frac{d^n q}{(2\pi)^n} e^{iq\cdot(x-y)} P^{\mu\beta} P^{\nu\alpha} q^4 + \mathcal{O}(h^3_{\mu\nu}). \quad (12.64)$$

$$\int d^n x d^n y R(x) R(y) \mathrm{K}^+(x-y;\mu) = -\frac{1}{2} \int d^n x d^n y h_{\mu\nu}(x) h_{\alpha\beta}(y) \times \int \frac{d^n q}{(2\pi)^n} e^{iq\cdot(x-y)} \ln\left[\frac{q^2 - i\epsilon}{\mu^2}\right] P^{\mu\nu} P^{\alpha\beta} q^4 + \mathcal{O}(h^3_{\mu\nu}). \quad (12.65)$$

$$\int d^n x d^n y R_{\mu\nu\alpha\beta}(x) R^{\mu\nu\alpha\beta}(y) \mathrm{K}^+(x-y;\mu) = -\frac{1}{2} \int d^n x d^n y h_{\mu\nu}(x) h_{\alpha\beta}(y) \times \int \frac{d^n q}{(2\pi)^n} e^{iq\cdot(x-y)} \ln\left[\frac{q^2 - i\epsilon}{\mu^2}\right] \times P^{\mu\beta} P^{\nu\alpha} q^4 + \mathcal{O}(h^3_{\mu\nu}). \quad (12.66)$$

$$\int d^n x d^n y R(x) R(y) \mathrm{K}(x-y) = -\frac{1}{2} \int d^n x d^n y h_{\mu\nu}(x) h_{\alpha\beta}(y) \times \int \frac{d^n q}{(2\pi)^n} e^{iq\cdot(x-y)} (2\pi i) \theta(-q^2) \theta(-q^0) \times P^{\mu\nu} P^{\alpha\beta} q^4 + \mathcal{O}(h^3_{\mu\nu}). \quad (12.67)$$

$$\int d^n x d^n y R_{\mu\nu\alpha\beta}(x) R^{\mu\nu\alpha\beta}(y) \mathrm{K}(x-y) = -\frac{1}{2} \int d^n x d^n y h_{\mu\nu}(x) h_{\alpha\beta}(y) \times \int \frac{d^n q}{(2\pi)^n} e^{iq\cdot(x-y)} (2\pi i) \theta(-q^2) \theta(-q^0) \times P^{\mu\beta} P^{\nu\alpha} q^4 + \mathcal{O}(h^3_{\mu\nu}). \quad (12.68)$$

12.5.2 Curvature Tensors in Conformally Related Spaces

$$\tilde{g}_{\mu\nu}(x) = e^{2\omega} g_{\mu\nu}(x) \quad (12.69)$$

$$\tilde{R}_{\mu\nu\alpha\beta} = e^{2\omega} \left[R_{\mu\nu\alpha\beta} + 2 g_{\beta[\mu} \delta_{\nu]}{}^{\rho} (\omega_{;\alpha\rho} - \omega_{;\alpha} \omega_{;\rho}) - 2 g_{\alpha[\mu} \delta_{\nu]}{}^{\rho} (\omega_{;\beta\rho} - \omega_{;\beta} \omega_{;\rho}) - 2 g_{\alpha[\mu} g_{\beta\nu]} \omega_{;\rho} \omega^{;\rho} \right]. \quad (12.70)$$

$$\tilde{R}_{\mu\alpha} = R_{\mu\alpha} - (n-2)\omega_{;\mu\alpha} - g_{\mu\alpha} \Box_g \omega + (n-2)\omega_{;\mu}\omega_{;\alpha} - (n-2) g_{\mu\alpha} \omega_{;\rho} \omega^{;\rho}. \quad (12.71)$$

$$\tilde{R} = e^{-2\omega} \left[R - 2(n-1)\Box_g \omega - (n-1)(n-2)\omega_{;\rho}\omega^{;\rho} \right]. \quad (12.72)$$

$$\begin{aligned}\tilde{R}_{\mu\nu\alpha\beta}\tilde{R}^{\mu\nu\alpha\beta} &= e^{-4\omega}\,[\, R_{\mu\nu\alpha\beta}R^{\mu\nu\alpha\beta} - 8R^{\mu\nu}(\omega_{;\mu\nu} - \omega_{;\mu}\omega_{;\nu}) - 4R\omega_{;\mu}\omega^{;\mu} \\ &\quad + 4(n-2)\omega_{;\mu\nu}\omega^{;\mu\nu} - 8(n-2)\omega^{;\mu\nu}\omega_{;\mu}\omega_{;\nu} + 4(\Box_g\omega)^2 \\ &\quad + 8(n-2)\omega_{;\mu}\omega^{;\mu}\Box_g\omega + 2(n-1)(n-2)(\omega_{;\mu}\omega^{;\mu})^2\,]\,. \qquad (12.73)\\ \tilde{R}_{\mu\nu}\tilde{R}^{\mu\nu} &= e^{-4\omega}\,[\, R_{\mu\nu}R^{\mu\nu} - 2(n-2)R^{\mu\nu}(\omega_{;\mu\nu} - \omega_{;\mu}\omega_{;\nu}) - 2R\Box_g\omega \\ &\quad - 2(n-2)R\omega_{;\mu}\omega^{;\mu} + (n-2)^2\omega_{;\mu\nu}\omega^{;\mu\nu} + (3n-4)(\Box_g\omega)^2 \\ &\quad - 2(n-2)^2\omega^{;\mu\nu}\omega_{;\mu}\omega_{;\nu} + 2(n-2)(2n-3)\omega_{;\mu}\omega^{;\mu}\Box_g\omega \\ &\quad + (n-1)(n-2)^2(\omega_{;\mu}\omega^{;\mu})^2\,]\,. \qquad (12.74)\end{aligned}$$

$$\begin{aligned}\tilde{R}^2 = e^{-4\omega}\,[\, & R^2 - 4(n-1)R\Box_g\omega - 2(n-1)(n-2)R\omega_{;\mu}\omega^{;\mu} + 4(n-1)^2(\Box_g\omega)^2 \\ & + 4(n-1)^2(n-2)\omega_{;\mu}\omega^{;\mu}\Box_g\omega + (n-1)^2(n-2)^2(\omega_{;\mu}\omega^{;\mu})^2\,]\,. \qquad (12.75)\end{aligned}$$

$$\begin{aligned}\tilde{C}_{\mu\nu\alpha\beta}\tilde{C}^{\mu\nu\alpha\beta} &= \tilde{R}_{\mu\nu\alpha\beta}\tilde{R}^{\mu\nu\alpha\beta} - \frac{4}{(n-2)}\tilde{R}_{\mu\nu}\tilde{R}^{\mu\nu} + \frac{2}{(n-1)(n-2)}\tilde{R}^2 \\ &= e^{-4\omega}C_{\mu\nu\alpha\beta}C^{\mu\nu\alpha\beta} \qquad (12.76)\end{aligned}$$

12.5.3 Momentum Integrals and Dimensional Regularization

$$I(q) \equiv \int \frac{d^n p}{(2\pi)^n} f(p,q). \qquad (12.77)$$

$$I_\mu = \int \frac{d^n p}{(2\pi)^n} f(p,q) p_\mu = \frac{I(q)}{2} q_\mu. \qquad (12.78)$$

$$I_{\mu\nu} = \int \frac{d^n p}{(2\pi)^n} f(p,q) p_\mu p_\nu = \frac{I(q)}{4}\left[q_\mu q_\nu - \frac{q^2}{(n-1)}P_{\mu\nu}\right]. \qquad (12.79)$$

$$\begin{aligned}I_{\mu\nu\alpha} &= \int \frac{d^n p}{(2\pi)^n} f(p,q) p_\mu p_\nu p_\alpha \\ &= \frac{I(q)}{8}\left[q_\mu q_\nu q_\alpha - \frac{q^2}{(n-1)}(P_{\mu\nu}q_\alpha + P_{\mu\alpha}q_\nu + P_{\alpha\nu}q_\mu)\right]. \qquad (12.80)\\ I_{\mu\nu\alpha\beta} &= \int \frac{d^n p}{(2\pi)^n} f(p,q) p_\mu p_\nu p_\alpha p_\beta \\ &= \frac{I(q)}{16}\Big\{\; q_\mu q_\nu q_\alpha q_\beta - \frac{q^2}{(n-1)}\,[P_{\mu\nu}q_\alpha q_\beta + P_{\nu\alpha}q_\mu q_\beta + P_{\nu\beta}q_\mu q_\alpha \\ &\quad + P_{\mu\alpha}q_\nu q_\beta + P_{\mu\beta}q_\nu q_\alpha + P_{\alpha\beta}q_\mu q_\nu] \\ &\quad + \frac{q^4}{(n^2-1)}\,[P_{\mu\nu}P_{\alpha\beta} + P_{\mu\beta}P_{\nu\alpha} + P_{\nu\beta}P_{\mu\alpha}]\;\Big\}. \qquad (12.81)\end{aligned}$$

$$I_1(p) = \int \frac{d^n q}{(2\pi)^n} \frac{1}{(q^2 - i\epsilon)[(p-q)^2 - i\epsilon]}$$
$$= i(p^2 - i\epsilon)^{\frac{n}{2}-2} \frac{\Gamma\left(2 - \frac{n}{2}\right)\left[\Gamma\left(\frac{n}{2} - 1\right)\right]^2}{(4\pi)^{\frac{n}{2}}\Gamma(n-2)}. \tag{12.82}$$

$$I_1^{(4D)}(p) = \left(\frac{-i}{8\pi^2}\right)\left[\frac{1}{n-4} + \frac{1}{2}\ln(p^2 - i\epsilon) + \frac{1}{2}\left(\gamma - 2 - \ln 4\pi\right) + \mathcal{O}(n-4)\right]. \tag{12.83}$$

$$I_2(p) = \int \frac{d^n q}{(2\pi)^n} \delta(q^2)\theta(-q^0)\delta[(p-q)^2]\theta(q^0 - p^0)$$
$$= \frac{\theta(-p^0)\theta(-p^2)(p^2)^{n-3}}{(n-3)2^{n-1}(2\pi)^3|\vec{p}|}\left[\frac{1}{(p^0 + |\vec{p}|)^{n-3}} - \frac{1}{(p^0 - |\vec{p}|)^{n-3}}\right]. \tag{12.84}$$

$$I_2^{(4D)}(p) = \frac{\theta(-p^0)\theta(-p^2)}{4(2\pi)^3}. \tag{12.85}$$

12.5.4 Variational Expressions

$$g_{\mu\nu}(x) \longrightarrow g_{\mu\nu}(x) + \delta g_{\mu\nu}(x). \tag{12.86}$$

$$\delta g^{\alpha\lambda} = -g^{\alpha\mu} g^{\lambda\nu} \delta g_{\mu\nu}. \tag{12.87}$$

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g} g^{\mu\nu}\delta g_{\mu\nu}. \tag{12.88}$$

$$\delta\Gamma^{\alpha}_{\beta\delta} = \frac{1}{2} g^{\alpha\lambda}\left[\delta g_{\lambda\delta;\beta} + \delta g_{\beta\lambda;\delta} - \delta g_{\beta\delta;\lambda}\right]. \tag{12.89}$$

$$\delta R^{\alpha}{}_{\beta\gamma\delta} = \frac{1}{2} g^{\alpha\lambda}\left[\delta g_{\lambda\delta;\beta\gamma} + \delta g_{\beta\lambda;\delta\gamma} - \delta g_{\beta\delta;\lambda\gamma}\right.$$
$$\left. - \delta g_{\lambda\gamma;\beta\delta} - \delta g_{\beta\lambda;\gamma\delta} + \delta g_{\beta\gamma;\lambda\delta}\right]. \tag{12.90}$$

$$\delta R_{\beta\delta} = \frac{1}{2} g^{\alpha\lambda}\left[\delta g_{\lambda\delta;\beta\alpha} + \delta g_{\beta\lambda;\delta\alpha} - \delta g_{\beta\delta;\lambda\alpha} - \delta g_{\lambda\alpha;\beta\delta}\right]. \tag{12.91}$$

$$\delta R = -R^{\beta\delta}\delta g_{\beta\delta} + g^{\beta\delta} g^{\alpha\lambda}\left[\delta g_{\lambda\delta;\beta\alpha} - \delta g_{\beta\delta;\lambda\alpha}\right]. \tag{12.92}$$

$$\delta(\Box_g \omega) = -\omega^{;\mu\nu}\delta g_{\mu\nu} - \frac{1}{2}\omega^{;\lambda} g^{\mu\nu}\left[\delta g_{\lambda\nu;\mu} + \delta g_{\mu\lambda;\nu} - \delta g_{\mu\nu;\lambda}\right]. \tag{12.93}$$

13

Structure Formation in the Early Universe

Structure formation in the early universe is a key problem in modern cosmology. It is believed that galaxies, clusters or any large-scale structure observed today originated from random sources of primordial inhomogeneities in the early universe amplified during the universe expansion. Theories of structure formation based on general relativity theory have been in existence for over 70 years [338, 656, 116], see e.g. [747, 748, 749], long before the advent of inflationary cosmology [750, 751, 745]. But the inflation paradigm [118, 119, 120, 122] provided at least two major improvements in the modern theory of cosmological structure formation [734, 736, 737, 752, 753].

First: *the sources*. Instead of a classical white noise source arbitrarily specified, the seeds of structures of the new theory are from quantum fluctuations which obey equations derivable from the dynamics of the inflaton field, which is responsible for driving inflation.

Second: *the spectrum*. The almost scale-invariant spectrum (masses of galaxies as a function of their scales) has a more natural explanation from the almost exponential expansion of the inflationary universe than from the power law expansion of the Friedmann–Lemaître–Robertson–Walker (FLRW) universe in the traditional theory.

Stochastic gravity provides a sound and natural formalism for the derivation of the cosmological perturbations generated during inflation. It was shown by Roura and Verdaguer [203] that the correlation functions that follow from the Einstein–Langevin equation which emerges in the framework of stochastic gravity coincide with that obtained with the usual quantization procedures when both the metric perturbations and the inflaton fluctuations are linearized; see, e.g., Mukhanov et al. [753]. An explicit computation of the curvature perturbation correlations was performed by Urakawa and Maeda [754] in the stochastic gravity formulation. Stochastic gravity, however, can also deal very naturally with the fluctuations of the inflaton field beyond the linear approximation.

Here, we illustrate the stochastic approach with one of the simplest chaotic inflationary models in which the background spacetime is a quasi de Sitter universe, and prove the equivalence of the stochastic and quantum correlations to linear order.

13.1 The Model

In this chaotic inflationary model [751] the inflaton field ϕ of mass m is described by the following Lagrangian density,

$$\mathcal{L}(\phi) = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} m^2 \phi^2. \tag{13.1}$$

The conditions for the existence of an inflationary period, which is characterized by an accelerated cosmological expansion, is that the value of the field over a region with the typical size of the Hubble radius is higher than the Planck mass $m_{\rm P} = 1/\sqrt{G_{\rm N}}$ (we use $\hbar = c = 1$). This is because in order to solve the cosmological horizon and flatness problem more than 60 e-folds of expansion are needed; to achieve this the scalar field should begin with a value higher than $3\, m_{\rm P}$. The inflaton mass is small: as we will see, the large-scale anisotropies measured in the cosmic background radiation [755] restrict the inflaton mass to be of the order of $10^{-6} m_{\rm P}$. We will not discuss the naturalness of this inflationary model but simply assume that if one such region is found (inside a much larger universe) it will inflate to become our observable universe.

We want to study the metric perturbations produced by the stress-energy tensor fluctuations of the inflaton field on the homogeneous background of a flat FLRW model, described by the cosmological scale factor $a(\eta)$, where η is the conformal time, which is driven by the homogeneous inflaton field $\phi(\eta) = \langle \hat{\phi} \rangle$. Thus we write the inflaton field in the following form

$$\hat{\phi}(x) = \phi(\eta) + \hat{\varphi}(x), \tag{13.2}$$

where $\hat{\varphi}(x)$ corresponds to a free massive quantum scalar field with zero expectation value on the homogeneous background metric: $\langle \hat{\varphi} \rangle = 0$. We will restrict ourselves to scalar-type metric perturbations because these are the ones that couple to the inflaton fluctuations in the linear theory. We note that this is not so if we were to consider inflaton fluctuations beyond the linear approximation, where tensorial and vectorial metric perturbations would also arise. The perturbed metric $\tilde{g}_{\mu\nu} = g_{\mu\nu} + h_{\mu\nu}$ can be written in the longitudinal gauge as

$$ds^2 = a^2(\eta)[-(1 + 2\Phi(x))d\eta^2 + (1 - 2\Psi(x))\delta_{ij} dx^i dx^j], \tag{13.3}$$

where the scalar metric perturbations $\Phi(x)$ and $\Psi(x)$ correspond to Bardeen gauge invariant variables [748].

13.2 Einstein–Langevin Equation for Scalar Perturbations

The Einstein–Langevin equation as described in Chapter 10 is gauge-invariant, and thus we can work in a desired gauge and then extract the gauge invariant quantities. The Einstein–Langevin equation (10.14) now reads

$$G^{(0)}_{\mu\nu} - 8\pi G_{\rm N}\langle\hat{T}^{(0)}_{\mu\nu}\rangle + G^{(1)}_{\mu\nu}(h) - 8\pi G_{\rm N}\langle\hat{T}^{(1)}_{\mu\nu}(h)\rangle = 8\pi G_{\rm N}\xi_{\mu\nu}. \tag{13.4}$$

Note that the two first terms cancel, that is,

$$G^{(0)}_{\mu\nu} - 8\pi G_{\rm N}\langle\hat{T}^{(0)}_{\mu\nu}\rangle = 0, \tag{13.5}$$

if the background metric is a solution of the semiclassical Einstein equations. Here the superscript (0) refers to functions in the background metric $g_{\mu\nu}$, and the superscript (1) refers to functions which are linear in the metric perturbation $h_{\mu\nu}$. The stress-energy tensor operator $\hat{T}_{\mu\nu}$ for the minimally coupled inflaton field in the perturbed metric is.

$$\hat{T}_{\mu\nu} = \tilde{\nabla}_\mu\hat{\phi}\tilde{\nabla}_\nu\hat{\phi} - \frac{1}{2}\tilde{g}_{\mu\nu}(\tilde{\nabla}_\alpha\hat{\phi}\tilde{\nabla}^\alpha\hat{\phi} + m^2\hat{\phi}^2). \tag{13.6}$$

Using the decomposition of the scalar field into its homogeneous and inhomogeneous part, see Eq. (13.2), and the metric $\tilde{g}_{\mu\nu}$ into its homogeneous background $g_{\mu\nu}$ and its perturbation $h_{\mu\nu}$, the renormalized expectation value for the stress-energy tensor operator can be written as

$$\langle\hat{T}^{\rm R}_{\mu\nu}[\tilde{g}]\rangle = \langle\hat{T}_{\mu\nu}[\tilde{g}]\rangle_{\phi\phi} + \langle\hat{T}_{\mu\nu}[\tilde{g}]\rangle_{\phi\varphi} + \langle\hat{T}^{\rm R}_{\mu\nu}[\tilde{g}]\rangle_{\varphi\varphi}, \tag{13.7}$$

where the subindices indicate the degree of dependence on the homogeneous field ϕ and its perturbation φ.

The first term on the right-hand side of Eq. (13.7) will be denoted by $\mathcal{T}_{\mu\nu} \equiv \langle\hat{T}_{\mu\nu}[g]\rangle_{\phi\phi}$. Taking into account the form of the spatially flat FLRW metric, in the basis associated with the conformal time and comoving spatial coordinates these components can be rewritten as

$$\mathcal{T}_{00} = \frac{1}{2}\left((\phi')^2 + m^2a^2\phi^2\right), \tag{13.8}$$

$$\mathcal{T}_{ij} = \frac{1}{2}\left((\phi')^2 - m^2a^2\phi^2\right)\delta_{ij}, \tag{13.9}$$

where primes denote derivatives with respect to the conformal time η. In this coordinate system the time-time and space-space components of $a^{-2}(\eta)\mathcal{T}_{\mu\nu}$ can be respectively identified with the energy density $\rho(\eta)$ and the isotropic pressure $p(\eta)$ of a perfect fluid.

The components of the background equation (13.5) become then the usual Friedmann equations

$$4\pi G_{\rm N}\left((\phi')^2 + m^2a^2\phi^2\right) = 3\mathcal{H}^2, \tag{13.10}$$

$$4\pi G_{\rm N}\left((\phi')^2 - m^2a^2\phi^2\right) = -\left(2\mathcal{H}' + \mathcal{H}^2\right), \tag{13.11}$$

where $\mathcal{H} = a'/a$. Thus, the first term in this equation depends only on the homogeneous field and it is given by the classical expression. The second term in (13.7) is proportional to $\langle\hat{\varphi}[\tilde{g}]\rangle$ which is not zero because the field dynamics is considered on the perturbed spacetime, i.e., this term includes the coupling of the field with $h_{\mu\nu}$ and may be obtained from the expectation value of the linearized Klein–Gordon equation,

$$(\Box_{g+h} - m^2)\,\hat{\varphi} = 0. \tag{13.12}$$

The last term in Eq. (13.7) corresponds to the expectation value of the stress-energy tensor for a free scalar field on the spacetime of the perturbed metric. It has a similar structure to that of Eqs. (13.8) and (13.9) with diagonal non-vanishing components which can be regarded as corrections $\Delta\rho(\eta)$ and $\Delta p(\eta)$ to the energy density and pressure.

13.2.1 Einstein Equations for Linear Perturbations

It is now convenient to write the linearly perturbed classical Einstein equations in terms of the scalar metric perturbations and the inflaton perturbations. The components of the linear perturbation of the Einstein tensor are

$$G^{(1)\,0}_{\ \ 0} = 2a^{-2}\left(3\mathcal{H}(\mathcal{H}\Phi + \Psi') - \nabla^2\Psi\right), \tag{13.13}$$

$$G^{(1)\,i}_{\ \ 0} = 2a^{-2}\partial_i(\mathcal{H}\Phi + \Psi'), \tag{13.14}$$

$$G^{(1)\,j}_{\ \ i} = 2a^{-2}\left[\left(2\mathcal{H}' + \mathcal{H}^2\right)\Phi + \mathcal{H}\Phi' + \Psi'' + 2\mathcal{H}\Psi' + \frac{1}{2}\nabla^2 D\right]\delta^j_i - a^{-2}\delta^{jk}\partial_k\partial_i D, \tag{13.15}$$

where $D \equiv \Phi - \Psi$ and $\nabla^2 = \delta^{ij}\partial_i\partial_i$.

The contribution to the stress-energy tensor of the linear perturbations will be denoted by $\delta\mathcal{T}^{\nu}_{\mu}$. Hence, taking all this into account, the components of the Einstein equation for linear scalar perturbations of the metric become

$$G^{(1)\,\nu}_{\ \ \mu} = 8\pi G_{\rm N}\delta\mathcal{T}^{\nu}_{\mu}. \tag{13.16}$$

The components for the linear perturbations of the stress-energy tensor, $\delta\mathcal{T}^{\nu}_{\mu}$, in the basis associated with the conformal time and comoving spatial coordinates are then straightforwardly obtained:

$$\delta\mathcal{T}^0_0 = a^{-2}\left(\Phi(\phi')^2 - \phi'\varphi' - m^2a^2\phi\varphi\right), \tag{13.17}$$

$$\delta\mathcal{T}^i_0 = a^{-2}\phi'\partial_i\varphi, \tag{13.18}$$

$$\delta\mathcal{T}^j_i = a^{-2}\delta^j_i\left(-\Phi(\phi')^2 + \phi'\varphi' - m^2a^2\phi\varphi\right). \tag{13.19}$$

Taking into account the fact that $\delta\mathcal{T}^j_i$ is diagonal, one can use Eq. (13.16) (see also Eq. (13.15)) with $i \neq j$ to conclude that Φ and Ψ are equal [753], except

for a possible homogeneous component (independent of the spatial coordinates), which should be included in the background scale factor. Alternatively, the same conclusion can also be reached by considering the sum of all the diagonal elements of the ij-components of Eq. (13.16) together with Eqs. (13.18) and (13.19) to substitute φ in terms of Φ and Ψ, which yields $\nabla^2(\Phi - \Psi) = 0$ and hence $\Psi = \Phi$ provided that they vanish at infinity.

From the Friedmann Eqs. (13.10) and (13.11) for the background solution, we have

$$4\pi G_{\rm N}(\phi')^2 = \mathcal{H}^2 - \mathcal{H}', \tag{13.20}$$

and we can use this equation to reexpress the terms in Eqs. (13.17)–(13.19) which are linear in Φ. Substituting these terms into the perturbed Einstein equation (13.16), we finally get

$$\nabla^2\Phi - 3\mathcal{H}\Phi' - (\mathcal{H}' + 2\mathcal{H}^2)\,\Phi = 4\pi G_{\rm N}\,(\phi'\varphi' + m^2a^2\phi\varphi)\,, \tag{13.21}$$

$$\partial_i(\Phi' + \mathcal{H}\Phi) = 4\pi G_{\rm N}\,\phi'\partial_i\varphi, \tag{13.22}$$

$$(\mathcal{H}' + 2\mathcal{H}^2)\,\Phi + 3\mathcal{H}\Phi' + \Phi'' = 4\pi G_{\rm N}\,(\phi'\varphi' - m^2a^2\phi\varphi)\,, \tag{13.23}$$

which are the linearly perturbed classical Einstein equations.

13.2.2 Equations Linear in the Inflaton Fluctuations

We can now go back to the Einstein–Langevin equations (13.4). After using the decomposition of the expectation value of the stress-energy tensor given by (13.7), the noise kernel $N_{\mu\nu\alpha'\beta'}[g;x,y)$ defined in Eq. (10.11) can be written as

$$\begin{aligned}\langle\{\hat{t}_{\mu\nu}[g;x),\hat{t}_{\alpha'\beta'}[g;y)\}\rangle &= \langle\{\hat{t}_{\mu\nu}[g;x),\hat{t}_{\alpha'\beta'}[g;y)\}\rangle_{(\phi\varphi)^2}\\ &\quad + \langle\{\hat{t}_{\mu\nu}[g;x),\hat{t}_{\alpha'\beta'}[g;y)\}\rangle_{(\varphi\varphi)^2},\end{aligned} \tag{13.24}$$

where we have used the fact that $\langle\hat{\varphi}\rangle = 0 = \langle\hat{\varphi}\hat{\varphi}\hat{\varphi}\rangle$ for Gaussian states on the background geometry. We consider the vacuum state to be the Euclidean vacuum, which is Gaussian and is the preferred state in the de Sitter background. In the above equation the first term is quadratic in $\hat{\varphi}$ whereas the second one is quartic; both contributions to the noise kernel are separately conserved since both $\phi(\eta)$ and $\hat{\varphi}$ satisfy the Klein–Gordon field equations on the background spacetime. Consequently, the two terms can be considered separately. On the other hand, if one treats $\hat{\varphi}$ as a small perturbation the second term in (13.24) is of lower order than the first and may be consistently neglected. This corresponds to neglecting the last term of Eq. (13.7), which is quadratic in the inflaton perturbations. The stress-energy tensor fluctuations due to a term of that kind were considered by Roura and Verdaguer [756].

Thus, we can write the Einstein–Langevin equations (13.4) particularized to the case of scalar-type metric perturbations around a spatially flat FLRW background geometry to linear order in the inflaton perturbations:

$$4\pi G_{\mathrm{N}} a^2 \left(\langle \delta\hat{\mathcal{T}}_0^0 \rangle_\Phi + \xi_0^0 \right) = 3\mathcal{H}(\mathcal{H}\Phi + \Psi') - \nabla^2\Psi, \tag{13.25}$$

$$4\pi G_{\mathrm{N}} a^2 \left(\langle \delta\hat{\mathcal{T}}_0^i \rangle_\Phi + \xi_0^i \right) = \partial_i(\Psi' + \mathcal{H}\Phi), \tag{13.26}$$

$$4\pi G_{\mathrm{N}} a^2 \left(\langle \delta\hat{\mathcal{T}}_i^j \rangle_\Phi + \xi_i^j \right) = \left[(2\mathcal{H}' + \mathcal{H}^2)\,\Phi + \mathcal{H}\Phi' + \Psi'' + 2\mathcal{H}\Psi' + \frac{1}{2}\nabla^2 D \right] \delta_i^j - \frac{1}{2}\delta^{jk}\partial_k\partial_i D. \tag{13.27}$$

where we have used Eqs. (13.13)–(13.15) for the linearized Einstein tensor and we have considered the stress-energy tensor operator $\delta\hat{\mathcal{T}}_\mu^\nu$, which results from keeping terms linear in either the inflaton perturbations or the metric perturbations. The notation $\langle \delta\hat{\mathcal{T}}_\mu^\nu \rangle_\Phi$ is equivalent to $\langle \delta\hat{\mathcal{T}}_\mu^\nu[g+h] \rangle$ for the particular case of scalar metric perturbations that we are considering. Note, in addition, that all the contributions to $\langle \delta\hat{\mathcal{T}}_\mu^\nu \rangle_\Phi$ are, explicitly or implicitly, proportional to the metric perturbations since otherwise they would be proportional to $\langle \hat{\varphi}[g] \rangle$, which vanishes. In fact, this turns out to be important so that $\langle \delta\hat{\mathcal{T}}_\mu^\nu[g+h] \rangle$ coincides with $\langle \hat{T}^{(1)}{}_\mu^\nu[g+h] \rangle$, which is the object that appears in the Einstein–Langevin equation, when one keeps to linear order in the inflaton perturbations, i.e., when only the first two terms on the right-hand side of Eq. (13.7) are considered.

The three equations corresponding to the spatial components with equal indices of the Einstein–Langevin equation are equivalent owing to the symmetries of the FLRW metric and those of the Gaussian state of the inflaton perturbations being considered, which was chosen to be compatible with those symmetries. On the other hand, the equation for the spatial components with different indices can be used in a similar way to the previous perturbed classical Einstein equations in order to show that the gauge-invariant functions for the scalar metric perturbations $\Phi(x)$ and $\Psi(x)$ coincide. In this case it is also necessary that the spatial components of the stochastic source $\xi_{ij}(x)$ with indices $i \neq j$ vanish identically. Indeed, since $\xi_{\mu\nu}(x)$ is a Gaussian stochastic process with zero mean, $\xi_{ij}(x)$ will vanish provided that $\langle \xi_{ij}(x)\xi_{\alpha'\beta'}(y) \rangle_\xi = 0$, which can be argued as follows. The correlation function for $\xi_{\mu\nu}$ is defined by the noise kernel and, as we are keeping to linear order in the inflaton perturbations, only the first contribution in Eq. (13.24), $\langle \{ \hat{t}_{ij}[g], \hat{t}_{\alpha'\beta'}[g] \} \rangle_{\phi\varphi} = \langle \{ \delta\hat{t}_{ij}[g], \delta\hat{t}_{\alpha'\beta'}[g] \} \rangle$, where $\delta\hat{t}_{\mu\nu} \equiv \delta\hat{\mathcal{T}}_{\mu\nu} - \langle \delta\hat{\mathcal{T}}_{\mu\nu} \rangle$, should be considered. Finally, $\langle \{ \delta\hat{t}_{ij}[g], \delta\hat{t}_{\alpha'\beta'}[g] \} \rangle$ vanishes for $i \neq j$ since $\delta\hat{\mathcal{T}}_{ij} = 0$ in that case, as follows from Eq. (13.19) with the inflaton perturbation promoted to a quantum operator. Hence, from now on we will take $\Psi = \Phi$. Eq. (13.27) is then trivially satisfied for $i \neq j$ and for $i = j$ it reduces to (no summation should be understood over the repeated index i)

$$4\pi G_{\mathrm{N}} a^2 \left(\langle \delta\hat{\mathcal{T}}_i^i \rangle_\Phi + \xi_i^i \right) = (2\mathcal{H}' + \mathcal{H}^2)\,\Phi + 3\mathcal{H}\Phi' + \Phi''. \tag{13.28}$$

13.3 Correlation for Scalar Perturbations

We will now use the previously derived Einstein–Langevin equations to find correlation functions for the scalar metric perturbations Φ at large scales. The large scale assumption will considerably simplify our equations. The equation for Φ can be obtained from the $0i$-component of the Einstein–Langevin equation (13.26) which, neglecting a nonlocal term, reads in Fourier space as

$$ik_i(\mathcal{H}\Phi_k + \Phi_k') = 4\pi G_{\rm N}(\xi_{0i})_k, \tag{13.29}$$

where k_i is the comoving momentum component associated to the comoving coordinate x^i, and we have used the definition $\Phi_k(\eta) = \int d^3x \exp(-i\vec{k}\cdot\vec{x})\Phi(\eta,\vec{x})$. A nonlocal term of dissipative character on the left-hand side of Eq. (13.26) should also appear on Eq. (13.29), but we have neglected it because the large scale spectrum does not change in a substantial way. A calculation without that assumption was carried out by Roura and Verdaguer [203]. We must emphasize, however, that the proof of the equivalence of the stochastic approach to linear order in $\hat{\varphi}$ to the usual linear cosmological perturbations approach does not assume that simplification, as we will see in Section 13.4.

To solve Eq. (13.29), whose left-hand side comes from the linearized Einstein tensor for the perturbed metric, we need the retarded propagator for the gravitational potential Φ_k,

$$G_k(\eta,\eta') = -i\frac{4\pi}{k_i m_{\rm P}^2}\left(\theta(\eta-\eta')\frac{a(\eta')}{a(\eta)} + f(\eta,\eta')\right), \tag{13.30}$$

where f is a homogeneous solution of Eq. (13.29) related to the initial conditions chosen. For instance, if we take $f(\eta,\eta') = -\theta(\eta_0-\eta')a(\eta')/a(\eta)$ the solution would correspond to "turning on" the stochastic source at η_0. With the solution of the Einstein–Langevin equation (13.29) for the scalar metric perturbations we are in the position to compute the two-point correlation functions for these perturbations.

The two-point correlation function for the scalar metric perturbations induced by the inflaton fluctuations is thus given by

$$\begin{aligned}\langle\Phi_k(\eta)\Phi_{k'}(\eta')\rangle_s = (2\pi)^2\delta(\vec{k}+\vec{k}')\\ \times\int^{\eta} d\eta_1\int^{\eta'} d\eta_2 G_k(\eta,\eta_1)G_{k'}(\eta',\eta_2)\langle(\xi_{0i})_k(\eta_1)(\xi_{0'i'})_{k'}(\eta_2)\rangle_s.\end{aligned} \tag{13.31}$$

Here the two-point correlation function for the stochastic source, which is connected to the stress-energy tensor fluctuations through the noise kernel, is given by,

$$\begin{aligned}\langle(\xi_{0i})_k(\eta_1)(\xi_{0'i'})_{-k}(\eta_2)\rangle_s &= \frac{1}{2}\langle\{(\hat{t}_{0i})_k(\eta_1),(\hat{t}_{0'i'})_{-k}(\eta_2)\}\rangle_{\phi\varphi}\\ &= \frac{1}{2}k_i k_i\phi'(\eta_1)\phi'(\eta_2)G_k^{(1)}(\eta_1,\eta_2),\end{aligned} \tag{13.32}$$

where $G_k^{(1)}(\eta_1,\eta_2) = \langle\{\hat{\varphi}_k(\eta_1),\hat{\varphi}_{-k}(\eta_2)\}\rangle$ is the kth-mode Hadamard function for a free minimally coupled scalar field in the appropriate vacuum state on the FLRW background.

In practice, to make the explicit computation of the Hadamard function we will assume that the field state is in the Euclidean vacuum and the background spacetime is de Sitter. Furthermore we will compute the Hadamard function for a massless field, and will make a perturbative expansion in terms of the dimensionless parameter $m/m_{\rm P}$. Thus we consider

$$\bar{G}_k^{(1)}(\eta_1,\eta_2) = \langle 0|\{\hat{y}_k(\eta_1),\hat{y}_{-k}(\eta_2)\}|0\rangle = 2\mathcal{R}\left(u_k(\eta_1)u_k^*(\eta_2)\right), \tag{13.33}$$

with

$$\hat{y}_k(\eta) = a(\eta)\hat{\varphi}_k(\eta) = \hat{a}_k u_k(\eta) + \hat{a}^\dagger_{-k} u^*_{-k}(\eta), \tag{13.34}$$

where

$$u_k = \frac{1}{\sqrt{2k}}\, e^{ik\eta}\left(1 - \frac{i}{k\eta}\right), \tag{13.35}$$

are the positive frequency k-modes for a massless minimally coupled scalar field on a de Sitter background, which define the Euclidean or Bunch–Davies vacuum state [47].

The assumption of a massless field for the computation of the Hadamard function is made because massless modes in de Sitter are much simpler to deal with than massive modes. We can see that this is nonetheless a reasonable approximation as follows. For a given mode the $m = 0$ approximation is reasonable when its wavelength λ is shorter than the Compton wavelength, $\lambda_c = 1/m$. In our case we have a very small mass m and the horizon size H^{-1} satisfies that $H^{-1} < \lambda_c$ (here H is the Hubble constant $H = \dot{a}/a$ with $a(t)$ where t is the physical time $dt = ad\eta$). Thus, for modes inside the horizon $\lambda < \lambda_c$ and $m = 0$ is a good approximation. Outside the horizon massive modes decay in amplitude as $\sim \exp(-m^2 t/3H)$ whereas massless modes remain constant; thus when modes leave the horizon the approximation will eventually break down. However, we only need to ensure that the approximation is still valid after 60 e-folds, i.e., $H\Delta t \sim 60$ (Δt being the time between horizon exit and the end of inflation). But this is the case provided $3H^2/m^2 > 60$, since the decay factor $\sim \exp[-(m^2/3H^2)H\Delta t]$ will not be too different from unity for those modes that left the horizon during the last 60 e-folds of inflation. This condition is indeed satisfied given that $m \ll H$ in most slow-roll inflationary models [750, 749], and in particular for the model considered here where $m \sim 10^{-6} m_{\rm P}$.

We note that the background geometry is not exactly that of de Sitter spacetime, for which $a(\eta) = -(H\eta)^{-1}$ with $-\infty < \eta < 0$. One can expand in terms of the "slow-roll" parameters and assume that to first order $\dot{\phi}(t) \simeq m_{\rm P}^2(m/m_{\rm P})$. The correlation function for the metric perturbation (13.31) can then be easily

computed; see [203] for details. The final result, however, is very weakly dependent on the initial conditions as one may understand from the fact that the accelerated expansion of the quasi-de Sitter spacetime during inflation erases the information about the initial conditions. Thus one may take the initial time to be $\eta_0 = -\infty$ and obtain, to lowest order in $m/m_{\rm P}$, the expression

$$\langle \Phi_k(\eta)\Phi_{k'}(\eta')\rangle_s \simeq 8\pi^2 \left(\frac{m}{m_{\rm P}}\right)^2 k^{-3}(2\pi)^3\delta(\vec{k}+\vec{k}')\cos k(\eta-\eta'). \qquad (13.36)$$

From this result two main conclusions can be derived. First, the prediction of an almost scale-invariant Harrison–Zel'dovich spectrum for large scales, i.e., small values of k. Second, since the correlation function is of the order of $(m/m_{\rm P})^2$ a severe bound to the mass m is imposed by the gravitational fluctuations derived from the small values of the Cosmic Microwave Background (CMB) anisotropies first detected by COBE. This bound is of the order of $(m/m_{\rm P}) \sim 10^{-6}$ [755, 753].

We now comment on some differences with some related work [550, 757, 671, 670] which used a self-interacting scalar field or a scalar field interacting nonlinearly with other fields. In those works an important relaxation of the ratio $m/m_{\rm P}$ was found. The long wavelength modes of the inflaton field were regarded as an open system in an environment made out of the shorter wavelength modes. Then, Langevin-type equations were used to compute the correlations of the long wavelength modes driven by the fluctuations of the shorter wavelength modes. In order to get a significant relaxation on the above ratio, however, one had to assume that the correlations of the free long wavelength modes, which correspond to the dispersion of the system's initial state, had to be very small. Otherwise they dominate by several orders of magnitude those fluctuations that come from the noise of the environment. This would require a great amount of fine-tuning for the initial quantum state of each mode [203].

We should remark that in the linear model discussed here there is no environment for the inflaton fluctuations. When one linearizes with respect to both the scalar metric perturbations and the inflaton perturbations, the system cannot be regarded as a true open quantum system. The reason is that Fourier modes decouple and the dynamical constraints due to diffeomorphism invariance link the metric perturbations of scalar type with the perturbations of the inflaton field so that only one true dynamical degree of freedom is left for each Fourier mode. Nevertheless, the inflaton fluctuations are responsible for the noise that induces the metric perturbations.

13.4 Equivalence of the Stochastic–Quantum Correlations

In this section we provide a proof of the equivalence between stochastic and quantum correlation functions for linearized cosmological perturbations. The key step is to show that the Einstein–Langevin equation for linearized cosmological

perturbations implies Eq. (6.48) of the review by Mukhanov et al. [753], which is derived below.

Let us begin with the Einstein–Langevin equation for scalar metric perturbations when one also linearizes with respect to the inflaton field, whose different components are given by Eqs. (13.25)–(13.27). We will take $\Psi = \Phi$, which is justified by the discussion following these equations, and work in Fourier space for the spatial coordinates. Next, we add Eq. (13.25), the $i = j = 1$ component of Eq. (13.26) and Eq. (13.27) contracted with $F_i \equiv 2m^2a^2(\phi/\phi')(k_i/k^2)$, which leads to the following result:

$$\Phi_k'' + 2\left(\mathcal{H} - \frac{\phi''}{\phi'}\right)\Phi_k' + k^2\Phi_k + 2\left(\mathcal{H}' - \mathcal{H}\frac{\phi''}{\phi'}\right)\Phi_k = 4\pi G_{\rm N}a^2\left[\langle\delta\hat{\mathcal{T}}_0^0\rangle_\Phi + \langle\delta\hat{\mathcal{T}}_1^1\rangle_\Phi + F_i\langle\delta\hat{\mathcal{T}}_0^i\rangle_\Phi + \xi_0^0 + \xi_1^1 + F_i\xi_0^i\right]_k. \tag{13.37}$$

In deriving Eq. (13.37) we made use of the two next relations

$$6\mathcal{H} + 2m^2a^2\frac{\phi}{\phi'} = \frac{2}{\phi'}\left(3\mathcal{H}\phi' + m^2a^2\phi\right) = 2\mathcal{H} - 2\frac{\phi''}{\phi'}, \tag{13.38}$$

$$2\mathcal{H}' + 2\mathcal{H}\left(2\mathcal{H} + m^2a^2\frac{\phi}{\phi'}\right) = 2\mathcal{H}' - 2\mathcal{H}\frac{\phi''}{\phi'}, \tag{13.39}$$

which follow from the Klein–Gordon equation for the background field ϕ: $\phi'' + 2(a'/a)\phi' + m^2a^2\phi = 0$.

The final step is to prove that the right-hand side of Eq. (13.37) vanishes. In order to do this, it is convenient to consider first the Fourier-transformed version of Eqs. (13.17)–(13.19) for the linearized stress-energy tensor. It is then straightforward to show that

$$\left(\delta\mathcal{T}_0^0\right)_k + \left(\delta\mathcal{T}_1^1\right)_k + F_i\left(\delta\mathcal{T}_0^i\right)_k = 0. \tag{13.40}$$

The same conclusion applies when φ is promoted to a Heisenberg operator $\hat{\varphi}$, which implies that the first three terms on the right-hand side of Eq. (13.37) cancel out. On the other hand, since $(\xi_0^0 + \xi_1^1 + F_i\xi_0^i)_k$ is a Gaussian stochastic process with vanishing mean; in order to prove that it vanishes it is sufficient to see that $\left\langle(\xi_0^0 + \xi_1^1 + F_i\xi_0^i)_k(\eta)\left(\xi_{\mu'}^{\nu'}\right)_{k'}(\eta')\right\rangle_\xi$, which is proportional to $\left\langle\left\{\left(\delta\hat{t}_0^0 + \delta\hat{t}_1^1 + F_i\delta\hat{t}_0^i\right)_k(\eta), \left(\delta\hat{t}_{\mu'}^{\nu'}\right)_{k'}(\eta')\right\}\right\rangle$, vanishes. Indeed, taking $\Phi = 0$ in Eq. (13.40) and promoting φ to a Heisenberg operator, it follows that $\left(\delta\hat{t}_0^0 + \delta\hat{t}_1^1 + F_i\delta\hat{t}_0^i\right)_k = 0$. Thus, the right-hand side of Eq. (13.37) vanishes and one is left with

$$\Phi'' + 2\left(\mathcal{H} - \frac{\phi''}{\phi'}\right)\Phi' - \nabla^2\Phi + 2\left(\mathcal{H}' - \mathcal{H}\frac{\phi''}{\phi'}\right)\Phi = 0, \tag{13.41}$$

which coincides with Eq. (6.48) of [753].

Several remarks about Eq. (13.41) are in order. First, the non-local terms associated with $\langle \delta\hat{\mathcal{T}}_\mu^\nu \rangle_\Phi$ are not present so that, when working in Fourier space for the spatial coordinates, one is left with an ordinary differential equation rather than an integro-differential one. Second, the equation exhibits no dependence on the stochastic source. However, the solutions of the Einstein–Langevin equation should also satisfy the constraint equations at the initial time in addition to Eq. (13.41). This implies a dependence on the stochastic source for the initial conditions $\Phi_k(\eta_0)$ and $\Phi'_k(\eta_0)$, which will involve a linear combination of terms linearly proportional to the stochastic source. The solution of the linearized Einstein–Langevin equation can then be written as $\Phi_k(\eta) = u_1(\eta)\Phi_k(\xi_\mu^\nu(\eta_0);\eta_0) + u_2(\eta)\Phi'_k(\xi_\mu^\nu(\eta_0);\eta_0)$, where $u_1(\eta)$ and $u_2(\eta)$ are solutions of Eq. (13.41) with initial conditions $u_1(\eta_0) = 1$, $u_1'(\eta_0) = 0$ and $u_2(\eta_0) = 0$, $u_2'(\eta_0) = 1$. Such a dependence of the initial conditions on the stochastic source at the initial time is responsible for the non-trivial part of the stochastic correlation functions of Φ at later times.

Now in the standard approach one quantizes both the linearized metric perturbation and the inflaton field, so that Φ and φ are promoted to Heisenberg operators in the linear equations (13.21)–(13.23). One can easily conclude, proceeding analogously to the previous case, that the operator $\hat{\Phi}$ for the linearized metric perturbations also satisfies Eq. (13.41). Furthermore, the constraints at the initial time give the same results for $\hat{\Phi}_k(\eta_0)$ and $\hat{\Phi}'_k(\eta_0)$ as in the stochastic case but with $\delta\hat{t}_\mu^\nu$ in place of ξ_μ^ν, so that the solutions of the quantum version of Eqs. (13.21)–(13.23) can be written as $\hat{\Phi}_k(\eta) = u_1(\eta)\hat{\Phi}_k(\delta\hat{t}_\mu^\nu(\eta_0);\eta_0) + u_2(\eta)\hat{\Phi}'_k(\delta\hat{t}_\mu^\nu(\eta_0);\eta_0)$. Taking into account that $\left\langle \xi_\mu^\nu[g;x)\,\xi_{\rho'}^{\sigma'}[g;y) \right\rangle_\xi = (1/2)\left\langle \left\{ \delta\hat{t}_\mu^\nu[g;x)\,, \delta\hat{t}_{\rho'}^{\sigma'}[g;y) \right\} \right\rangle$, it is straightforward to see that the result for symmetrized two-point quantum correlation function $(1/2)\langle\{\hat{\Phi}_k(\eta_1), \hat{\Phi}_{k'}(\eta_2)\}\rangle$ is equivalent to that for the stochastic correlation function $\langle \Phi_k(\eta_1)\Phi_{k'}(\eta_2)\rangle_\xi$, i.e.,

$$\langle \Phi_k(\eta_1)\Phi_{k'}(\eta_2)\rangle_\xi = \frac{1}{2}\langle\{\hat{\Phi}_k(\eta_1), \hat{\Phi}_{k'}(\eta_2)\}\rangle. \tag{13.42}$$

This constitutes a proof of the equivalence between quantum and stochastic correlation functions for linearized cosmological perturbations.

13.4.1 Curvature Perturbation

We close this section by showing that there is no discrepancy between stochastic gravity and the standard treatment for superhorizon modes. More specifically, in [754] the Einstein–Langevin equation for linearized cosmological perturbations was solved using certain approximations and the correlation function for the gauge-invariant variable ζ was computed. This variable corresponds to the curvature perturbation in the uniform density gauge, which coincides with the comoving gauge for modes outside the horizon [758]. It was found that for modes outside the horizon and for a sufficiently large number of e-folds the correlation function was not constant in time, contrary to the standard result. This would

be in conflict with the equivalence for linear perturbations that we have shown to hold in general. However, one can provide an exact argument which shows that the linearized Einstein–Langevin equation actually implies that for modes outside the horizon the correlation function of ζ remains constant in time. This means that the result in [754] seems to imply a problem with some of the approximations used rather than a shortcoming of stochastic gravity in that regime.

The exact argument is the following. One starts with the expression for ζ in terms of Φ [745]:

$$\zeta = \frac{2}{3(1+w)}(\Phi + \mathcal{H}^{-1}\Phi') + \Phi, \tag{13.43}$$

with $w = p/\rho$, where the relation between the background density ρ and pressure p and the background field ϕ is given by Eqs. (13.8) and (13.9) and the text after them. Differentiating Eq. (13.43) with respect to the conformal time and multiplying by $(3/2)\mathcal{H}(1+w)$, one obtains

$$\frac{3(1+w)}{2}\mathcal{H}\zeta' = \Phi'' + 2\left(\mathcal{H} - \frac{\phi''}{\phi'}\right)\Phi' + 2\left(\mathcal{H}' - \mathcal{H}\frac{\phi''}{\phi'}\right)\Phi, \tag{13.44}$$

where we made use of the following relations, which can be derived from Eqs. (13.10) and (13.11),

$$1 + w = \frac{\rho + p}{\rho} = \frac{8\pi G(\phi')^2}{3\mathcal{H}^2} = \frac{2(\mathcal{H}^2 - \mathcal{H}')}{3\mathcal{H}^2}, \tag{13.45}$$

$$\frac{d}{d\eta}\ln(1+w) = 2\frac{\phi''}{\phi'} - 2\frac{\mathcal{H}'}{\mathcal{H}}. \tag{13.46}$$

Note that the right hand-side of Eq. (13.44) coincides with the left-hand side of Eq. (13.41) except for the $\nabla^2\Phi$ term. This means that for modes outside the horizon, for which $\nabla^2\Phi$ can be neglected, the right-hand side of Eq. (13.44) vanishes and, consequently, the superhorizon Fourier mode ζ_k of the curvature perturbation remains constant in time.

13.5 Including One-Loop Contributions

Stochastic gravity provides an alternative framework to study the generation of primordial inhomogeneities in inflationary models. Besides the interest of the problem in its own right, there are also other reasons that make this problem worth discussing from the point of view of stochastic gravity. The Einstein–Langevin equation is not restricted to linearized perturbations of the inflaton field. In practice this may not be very important for inflationary models which are driven by an inflaton field which takes a non-zero expectation value, because the linear perturbations will give the leading contribution. However, the importance of considering corrections due to one-loop contributions from the inflaton field

perturbations, beyond the tree level of the linear cosmological perturbation theory, was emphasized by Weinberg [427, 759] as a means to understand the consequences of perturbation theory even where in practice its predictions may not be observed.

The correlation functions for the metric perturbations obtained using the Einstein–Langevin equation are equivalent to the correlation functions that would follow from a purely quantum field theory calculation up to the leading-order contribution in the large N limit. This result was proved on a Minkowski background by Hu et al. [190, 214]. This has important implications on the use of the Einstein–Langevin equation to address situations in which the background configuration for the scalar field vanishes. This includes not only the case of a Minkowski background spacetime, but also the interesting case of the trace anomaly-driven inflation, that is, inflationary models driven by the vacuum polarization of a large number of conformal fields [359, 360, 651], where the usual approaches based on the linearization of both the metric perturbations and the scalar field perturbations and their subsequent quantization can no longer be applied.

In the trace anomaly-driven inflation model Hawking et al. [651] were able to compute the two-point quantum correlation function for scalar and tensorial metric perturbations in a spatially closed de Sitter universe, making use of the anti-de Sitter/conformal field theory correspondence. They find that short-scale metric perturbations are strongly suppressed by the conformal matter fields. This is similar to what we obtained in Chapter 11 for the induced metric fluctuations in Minkowski spacetime.

14

Black Hole Backreaction and Fluctuations

Backreaction[1] refers to the quantum effects of matter fields such as vacuum polarization, quantum fluctuations and particle creation on the spacetime structure and dynamics. We have studied cosmological particle creation in Chapters 2 and 3, cosmological backreaction effects in Chapter 8, at the level of semiclassical gravity, where the expectation value of the stress-energy tensor is the sole quantum matter source in the semiclassical Einstein equation, and in Chapters 13 and 12 at the level of stochastic gravity theory, where the expectation value of the fluctuations or correlations of the stress-energy tensor, known as the noise kernel, act additionally as source of the Einstein–Langevin equation.

14.1 Issues, Proposals and Scenarios

Schematically, the way the backreaction problem is set up and approached (e.g., via the 'in-in', Schwinger–Keldysh or closed-time-path effective action) in these previously studied cosmological models can also be applied to black holes.[2] Backreaction studies of quantum field processes in cosmological spacetimes have progressed further than the corresponding problems in black holes, especially in an evaporating black hole spacetime, largely because of the relative technical simplicity associated with the higher symmetries of the relevant cosmological background geometries, such as the de Sitter spacetime which is a maximally symmetric space in four dimensions, or the FLRW universes which have $SO(4)$ symmetry on the 3-sphere, or the spatially homogeneous Bianchi universes with groups of motion acting on the spatial hypersurfaces. In contrast, even the simplest static black hole model, the Schwarzschild spacetime, has symmetry only on

[1] We dedicate this chapter to the memory of Bill Hiscock.
[2] As preparation for the topics discussed in this chapter the reader may find useful background materials in Chapter 9 of [47] and Chapter 4 of [53]

the 2-sphere. Technically, since the equation governing the radial function admits no analytical solutions, many problems involving the radial function must be solved by numerical means, which makes understanding the qualitative behavior less accessible. This situation is already encountered in classical backreaction problems such as the gravitational radiation reaction and self-force problem touched on in Chapter 1.

We begin with a brief summary of its development. The first stage in a black hole backreaction investigation is the determination of the regularized stress-energy tensor of quantum matter fields in the relevant black hole background spacetimes. We can perhaps regard the works of DeWitt [46], Davies, Fulling and Unruh [760] and Christensen and Fulling [761] as opening this stage. The effort expanded from 1980 onward with the works of Candelas and Howard [762, 763, 764], Page [488], Frolov and Zel'nikov [765, 766], Brown and Ottewill [767] Fawcett and Whiting [768, 769] and others. The problem of interacting quantum fields around a black hole was explored by Hawking [770]. The first attempt toward finding solutions to the semiclassical backreaction equations for quasi-static black holes with thermal radiation was made by York [771, 772], which we shall describe in a later section. Whiting and York [773] derived an Euclidean action for the black hole entropy plus quantum corrections in the canonical ensemble. For evaporating black holes, owing to its dynamical character, the backreaction problem is more challenging, and more important, because it bears on the end-state and information-loss issues. The first papers on evaporating black hole dynamics were by Hájíček and Israel [774] and by Bardeen [775], who used the Vaidya metric as his model of a radiating black hole. The first paper on black hole fluctuations was by Bekenstein [776]. We shall discuss both Bardeen's and Bekenstein's works in later sections. Among the first numerical calculations for the dynamics of evaporating black holes was the work of Parentani and Piran [777]. They obtained solutions to the semiclassical equations which they claim can describe the entire spacetime geometry from the collapse to the end of the evaporation. Their solution does confirm that black holes evaporate thermally. By the mid-90s the state of the art for computing quantities needed for tackling the black hole backreaction problem can be found in the works of Anderson, Hiscock, Massar, Ottewill, York and their collaborators [777, 778, 779, 780, 781].

A welcoming recent development is that some leaders of gravitational self-force numerical computations in the 2000s have extended their know-how to tackle the backreaction of quantum fields on black hole spacetimes. For example, the group led by Ori [782, 784] has recently developed a mode-sum renormalization scheme [783, 784] based on covariant point-splitting. They numerically computed the renormalized expectation value $\langle \Phi^2 \rangle$ of a minimally coupled massless quantum scalar field in the interior of a four-dimensional Reissner–Nordström black hole, in both the Hartle–Hawking [82] and Unruh [321] states. (See also [785, 786].) The backreaction effects on the event horizon and the black hole interior have

gained renewed interest thanks in part to the firewall controversy, as we will depict in the next section.

Technical difficulties for solving the radial equation aside, there are some advantageous factors: the static Schwarzschild spacetime being asymptotically flat permits the establishment of a viable quantum field theory based on the well-defined asymptotic vacuum. On it one can also construct canonical quantum gravity [787] or S-matrix theories for quantum black holes [788]. A black hole with its emitted thermal radiation when placed in a box can maintain thermodynamic quasi-quilibrium, unlike one without such a boundary which will become dynamically and thermodynamically unstable. Likewise, when a radiating black hole is placed in an anti-de Sitter (AdS) space, the asymptotically AdS region allows for an unambiguous definition of the vacuum state and a quantum field theory in the Schwarzschild–AdS spaces.

The linkage of quantum field processes in black hole geometries to those in cosmological spacetimes also provides some technical advantage and adds a physical dimension to our understanding. Physically, since the interior of a black hole can be described by a cosmological model (e.g., the Kantowski–Sachs universe for a spherically symmetric black hole), some cosmological backreaction features may carry over. See, e.g., the work of Kennedy et al [1016] on quantum effects on the gravitational collapse of a dust cloud, and Hiscock et al. [789] on the backreaction in the interior of a black hole. From a broader perspective, amongst the many geometrical and topological structures the solutions to the Einstein equation admit, besides the well-studied event horizons, there are also naked singularities and 'tunnels' or 'bridges' to other universes. It is of fundamental interest to find out whether these structures permitted in classical general relativity (GR) may continue to exist or be altered when the backreaction from quantum field processes are taken into consideration.

An example of these 'exotic' situations is the inner horizon of a Reissner–Nordström black hole, where it is known in classical GR that the metric may be analytically continued into a concatenation of asymptotically flat spacetime regions ('other universes), accessible to an observer in the 'universe where the black hole originally formed. Quantitative analysis using the enlarged framework of semiclassical gravity is needed to weigh these possibilities. For example, Balbinot and Poission [1017] studied the quantum effects near the black-hole Cauchy horizon; Herman and Hiscock [790] considered the backreaction effects of Schwinger pair creation from the electromagnetic field within the black-hole interior in a simplified model and found no naked singularity, Cauchy horizon or tunnel in that region of spacetime. Anderson et al [1015] found that macroscopic uncharged zero temperature black hole solutions to the semiclassical backreaction equations (remnants) may exist if the coefficient of one of the higher derivative terms in the gravitational Lagrangian is large enough and of the right sign for Starobinsky inflation to have occurred in the early Universe.

14.1.1 Hawking Effect and Evaporating Black Holes

Going beyond classical general relativity the challenge begins when one adds in the consideration of quantum field effects. This is well exemplified by the celebrated discovery of Hawking [77, 78] that a black hole is not black, as classical gravity ordains, but emits thermal radiation with a temperature inversely proportional to its mass [79, 80, 81]. Several key points from Hawking's original paper are relevant to the themes of this chapter. In his words [78] [Beginning of quote]:

(i) Considering a collapsing object in the formation of a black hole it might seem that since all the time dependence of the metric is in the collapsing phase, all the particle creation must take place in the collapsing body just before the formation of the event horizon, and that an infinite number of created particles would hover just outside the event horizon, escaping to future infinity at a steady rate.[3]

(ii) Particle creation by the black hole is not localized in the collapse but a global process. An observer falling through the event horizon would not see an infinite number of particles coming out from the collapsing body.

(iii) The negative energy density needed to account for the decrease in the area of the horizon should be thought of as arising from the indeterminacy of order of M^{-4} of the local energy density at the horizon.

(iv) Equivalently, one can think of the area decrease as resulting from the fact that quantum fluctuations of the metric will cause the position and the very concept of the event horizon to be somewhat indeterminate. [End of quote]

We shall see that point (ii) has bearings on later proposed drastic events near the horizon, as in the firewall postulate [792]. Point (iv) is a statement which fits right into the context of stochastic gravity, the theory which deals with metric fluctuations from the backreaction of particle creation and other quantum processes. This topic will be discussed in greater detail in the last section of this chapter based on more recent calculations.

As the radiation carries energy away the black hole's mass decreases. For a black hole of mass large compared to the Planck mass 10^{-5} g, the black hole evolves very slowly (compared to the characteristic time for light to cross the Schwarzchild radius). This regime is aptly described by quantum field theory in curved spacetime. In this quasi-stationary stage, by using a normal-ordered energy momentum tensor Hawking showed the average energy flux across a surface of constant radius between two retarded times is directed outwards and

[3] [Authors' footnote] This impression may have been mistakenly formed in the early 70s in equating particle creation in spacetimes with event horizons to cosmological particle creation. The latter is based on a different mechanism, namely, the parametric excitation of vacuum fluctuations by a time-dependent source, as in the expanding or contracting universe.

is equal to the energy flux for the thermal emission from a hot body. If one extends this quasi-stationary stage forever and considers the idealized case of an eternal black hole, then the thermal radiation can be easily understood by way of the thermal properties of the Green function induced by the existence of an event horizon in the Schwarzschild spacetime. This is the condition for the Hartle-Hawking vacuum [82]. The state of an observer freely falling into a black hole is more aptly described by the Unruh vacuum [321].

Note, however, in the realistic situation of an evaporating black hole the static Schwarzschild geometry does not provide an accurate description. The mass decrease due to particle creation predicted by Hawking is one important qualitative feature of black hole backreaction effects. The spacetime for this dynamical process is determined by an interplay between the emitted radiation and the time-dependent geometry.

As the black hole mass decreases further the emitted radiation temperature increases. The conversion of the mass of a black hole into thermal radiation is often referred to as an 'evaporation' process. Close to the Plank mass this process gets faster and faster, ending in a final explosion. What is the end state of black hole evaporation? This is an important question because it is tied in with the intriguing black hole information loss issue, which we shall comment on later without going into much detail.

14.1.2 Backreaction and Fluctuations

The two topics covered in this chapter are, respectively,

(1) the backreaction of thermal radiation on a black hole under the quasi-stationary condition, and
(2) metric fluctuations of the black hole horizon in an evaporating black hole spacetime.

An exemplary work of (1) is that of Bardeen [775] (see also Hájíček and Israel [774] and Masser [781]) who used the Vaidya metric as a model of a black hole immersed in a null congruence of radiation. His finding shows that Hawking's result for the quasi-stationary stage is valid and semiclassical gravity is a reasonable theory for energy scales below the Planck energy. Most works on the backreaction of Hawking radiation in semiclassical gravity are for this quasi-stationary phase. They are relevant as it is the longest span of the black hole's lifetime, yet the backreaction effects are not expected to be drastic in this phase, as the work of York and others shows [771, 793, 772, 794, 795, 778]. The real challenge is in calculating the backreaction of Hawking radiation at the final explosion stage, because the end state of black hole collapse has great astrophysical as well as theoretical significance, theoretical in the sense that black holes have been compared to hydrogen atoms [796, 797] and elementary particles [798] and black hole collapse to the crisis of the classical radiating atom,

the resolution of which helped to usher in quantum mechanics. Furthermore, one may ask whether with a better understanding of quantum black hole physics the underlying principles of quantum mechanics can be sharpened [799] or how candidate theories of quantum gravity (such as strings or loops, see below) reveal their particularities. But to achieve these goals one needs to be working with dynamical spacetimes co-determined by the emitted radiation self-consistently, a much more demanding problem than working with a quasi-stationary background. Issue (2) requires probing into metric fluctuations near the event horizon, both of an intrinsic and induced nature by quantum matter fields.

To find out the end state of black holes without guessing one needs to seek exact solutions of quantum gravity or string theory. Luckily they exist for 2D dilaton black holes [800, 801, 802, 803]. The Callen–Giddings–Harvey–Strominger (CGHS) 'evanescent' black hole solution has been studied in greater detail than any model with interesting results. For example, Ashtekar, Pretorius and Ramazanoglu [804] analyzed CGHS black holes in the mean-field approximation with backreaction and found that if the black hole is initially macroscopic, the evaporation process exhibits remarkable universal properties which are distinct from the features observed in the simplified, exactly soluble models. Their results also support the full quantum gravity scenario of Ashtekar, Taveras and Varadarajan [805] who claimed that there is no information loss. As for 4D black hole dynamics with Hawking radiation there are no exact solutions to date. The only way is to carry out backreaction calculations as described above, with analytical and numerical methods.

Before we begin discussions of the black hole backreaction problem, in view of the fact that this is the only chapter in this book devoted to black holes while there are so many interesting topics related to black holes and quantum fields, we provide a list of important topics and supply some background references to help orient the interested reader toward appreciating the key issues without going into the details, which can be found in the reviews and research papers.

Issues of Quantum Fields in Black Holes

We mention three classes of issues and develop one of them further in the following.

(1) On *black hole mechanics* [806] and *black hole thermodynamics* [807, 808, 809, 810, 811] Wald's book [49] and reviews [812, 813] are canonical sources. Read, on the formal side, his discovery that black hole entropy is a Noether charge [814, 815] and on the lighter side, his short historical account of the backdrop when Bekenstein made this fundamental discovery. See also [816, 817, 818, 819, 820, 821, 822, 823, 824] and a recent survey by Wall [825] where one can find up-to-date references on this topic.

(2) *Black hole formation: Hawking–Page transition.* Gravitating systems are dynamically unstable, with a natural tendency toward coalescence and

collapse. They are also known to have negative specific heat [826, 827, 828], and are thus thermodynamically unstable, meaning their temperatures will increase when energy is released. Therefore strictly speaking only microcanonical ensembles are appropriate for the description of the thermodynamical properties of black holes. To achieve quasi-equilibrium condition of a black hole with its emitted radiation one needs to put the system in a box or in an asymptotically anti-de Sitter (AdS) space because the gravitational potential of AdS space acts as a box of finite volume. Black holes in asymptotically AdS space have positive specific heat and can be in stable equilibrium with thermal radiation at a fixed temperature, thus allowing the use of the canonical ensemble for its description. Gross, Perry and Yaffe [928] and Whiting and York [773] calculated the probability for the nucleation of a black hole from hot flat space Hawking and Page [829] calculated the energy ranges where the formation of black holes will be thermodynamically more favored than pure radiation and vice versa and described their scenarios. Such transitions in asymptotically AdS space are known as Hawking–Page phase transitions.

(3) *Black hole end-state and information loss.* This issue was first raised by Hawking after his discovery of black hole radiation [811, 339, 736, 830, 831]. He phrased it in terms of the breakdown of predictability, namely, part of the information about the state of a system going down a black hole is lost; a pure quantum state goes into a mixed state described by a density matrix. The reason why this issue has persisted for over four decades and will likely continue to attract the attention of theoretical physicists for a long time is because many of them feel its magnitude in the same way as expressed by Preskill [832], who, despite his holding the position – like most particle and string theorists – that there is no information loss in black hole evaporation – namely, there exists a unitary S-matrix which describes the evolution from infalling matter to outgoing Hawking-like radiation [833, 834, 835]) – maintains that "the information loss paradox may well presage a revolution in fundamental physics."

14.1.3 End-States and Information Loss

The continual excitement over this issue of whether there is information loss in quantum processes involving black holes comes not from a gradual convergence of opinions over time, but from greater disagreements in the face of more novel ideas proposed. Interestingly almost every decade some new thoughts emerge and waves of activities follow. Space limitation precludes any detailed description of these activities. Nonetheless we wish to highlight the key issues involved, such as nonlocality in QFT inspired by strings, the complementarity principle and holography and fire walls from quantum information considerations, as they all have stimulating intellectual value.

Based on predictable behaviors in quantum black holes Hawking pronounced in 1982 [340] that quantum gravity adds a new level of unpredictability into physics over and above that normally associated with the uncertainty principle, because the metric of spacetime can fluctuate from being globally hyperbolic. In other words, the evolution is not completely determined by Cauchy data at past or future infinity.

The information loss issue is naturally tied to the end state issue as each scenario may offer a new way to interpret where the information is lost or how it may be retrieved. (See Page [836] for a summary of the information paradox and literature up to 1995.) In the 80s the four main possibilities, as related by Hawking [340], are:

1. The evaporation produces a naked singularity of negative mass which persists.
2. The evaporation slows down and stops, leaving a remnant black hole of about the Planck mass.
3. The black hole disappears completely but all the information about the black hole states and any other locally conserved quantities escape to infinity.
4. The black hole disappears completely taking with it the information about the black hole states, and any other quantities which are not coupled to long range fields.

Fast forward 35 years to 2017: the four kinds of end states, depicted by Unruh and Wald [837], are now:

(a) The black hole never forms, fuzzballs instead (e.g., [838]),
(b) A firewall near the black hole event horizon [792],
(c) Remnants, baby universe [839],
(d) Final burst releasing all the stored information ending in a pure state.

The most important new ingredient is the proposal of the existence of a firewall near the event horizon. For a recent review of black hole information with a comprehensive coverage of arguments and counterarguments related to firewalls and other proposals, see Marolf [840]. The main stimulus for such a drastic proposal comes from quantum information considerations, black hole being fast scramblers, but its antithesis is the *black hole complementarity principle*. So we need to begin with the latter, formulated in 1993 by Susskind, Thorlacius and Uglum [835] in the framework of semiclassical gravity.

Black Hole Complementarity

According to the principle of black hole complementarity, paraphrasing [841], an observer who remains outside the horizon of a black hole can describe the black hole as a very hot membrane, the stretched horizon, which absorbs any matter, energy and information that fall onto it. The information which is absorbed by the stretched horizon is eventually re-emitted in the Hawking radiation, albeit in a very scrambled form. An observer (Alice) who falls freely into the black hole sees things very differently: no membrane, no high temperature, no irregularities

of any kind as the observer crosses the event horizon. The principle of black hole complementarity asserts the consistency of these apparently contradictory descriptions. According to this principle, the matter which has fallen past the event horizon and the Hawking radiation are not different objects. They are complementary descriptions of a single system, viewed from very different reference frames which are related by an enormous Lorentz boost.

The setting of this principle should be acceptable because (a) already in classical general relativity we are familiar with different observers reporting on different scenarios,[4] and (b) the membrane paradigm was introduced in general relativity [843] to describe the energetics of a black hole in a phenomenological way. The new quantum element is Hawking radiation, and this principle serves to connect the narratives of the two different observers.

Susskind [844] further showed that this complementarity is commensurate with string theory and AdS/CFT or the holography principle in the following sense: to a freely infalling observer a string falling through a black hole horizon appears to be a Planck size object. To an outside observer the string and all the information it carries begin to spread as the string approaches the horizon. In a time of order the "information retention time" it fills the entire area of the horizon while information is radiated away. For a recent review on black hole information with more emphasis on AdS/CFT and string theory, see Polchinsky [845].

String theory usually conjures up some type of nonlocality beyond regular QFT, as is used by [841] to rebuke the assumption of locality in the low energy 'nice-slice' theory that many proofs of information loss are based on (see [846, 847] on how it is related to the trans-Planckian mode issue). We shall refrain from dwelling on the nonlocality issue, a very important one notwithstanding, but refer readers to the amendments suggested by Giddings [848, 849] to the arguments in [841], and the description of nonlocality in [565]. For more thoughts on this issue in this period see, e.g., [850, 832, 851, 852, 853, 854, 855].

Fire Wall

The firewall scenario came about because its proposers perceived inconsistencies in the four postulates of the BHC principle. Let us reexamine them:

Postulate 1 (quantum mechanics). The process of formation and evaporation of a black hole, as viewed by a distant observer, can be described entirely within the context of standard quantum theory.

Postulate 2 (semiclassical gravity). Outside the stretched horizon of a massive black hole, physics can be described to good approximation by a set of semiclassical field equations.

[4] E.g., Alice, the free-fall observer versus the 'shell' observer in the terminology of [842] – the quantum state defined on a shell situated close to the event horizon is the Hartle–Hawking vacuum; that defined on a shell infinitely far away from the black hole is the Boulware [1012] vacuum, while the quantum state of Alice is the Unruh vacuum.

Postulate 3. To a distant observer, a black hole appears to be a quantum system with discrete energy levels. The dimension of the subspace of states describing a black hole of mass M is the exponential of the Bekenstein entropy $S(M)$.

The authors of [792] put what was stated in [835] as an additional postulate,

Postulate 4 ('no drama'). A freely falling observer experiences nothing out of the ordinary when crossing the horizon.

They argued that postulates 1, 2 and 4 are not consistent with one another for a sufficiently old black hole and suggested a ('most conservative') resolution: that the infalling observer burns up close to the horizon – the firewall.

Black hole complementarity (BHC) is a common and easily acceptable physics framework. The new ingredient which triggers this drastic firewall proposal comes from *quantum information* considerations (there were quantum information arguments proposed, see e.g., Smolin and Oppenheim [856] and Braunstein and Pati [857]), especially after the papers of Hayden and Preskill [858] and Sekino and Susskind [859], identifying black holes as 'fast scramblers', using the earlier results of Page [860] which states the following: for a black hole that begins in a pure state, the black hole entropy S_{BH} which is proportional to its surface area is reduced to half of its initial value in a time (known as the Page time) t_{Page}. For 4d asymptotically-flat Schwarzschild black holes of initial radius R_0 the Page time is $t_{Page} = R_0^3/\ell_{\rm P}^2$ where $\ell_{\rm P}$ is the Planck length. Thus for a large R_0, at t_{Page} where information has for all this time been continuously released through Hawking radiation the black hole remains far from the Planck scale as the area of the black hole has decreased only by a factor of 2. What this means is that black hole information is unrelated to the quantum structure of the black hole and largely independent of the late time quantum processes. The authors (AMPS) of [792] separate the black hole into an early and a late regime according to the Page time and use information-theoretical considerations to argue that postulates 1 and 2 in BHC force one to the conclusion that the infalling observer will encounter high-energy modes, the basis for their firewall proposal. There are arguments for and against the complementarity principles which one can find in the reviews by Polchinsky [845] and Marolf [840].

A lucid explanation of the quantum informational aspects of this issue is offered by Susskind [861] in terms of two types of highly entangled quantum states, the highly ordered ground states of quantum field theories including condensed matter systems, and the highly random scrambled states represented by black holes. These two almost diametrically opposite states come into play in black hole physics. The highly ordered ground state seen by a freely falling observer at the horizon is 'dual' to a highly scrambled state seen by an observer who stays outside the black hole; the entanglements of the ordered infalling state are dual to the entanglements of the random thermal state of the exterior description. Moreover, Susskind pointed out that AMPS's argument relies on a 'proximity postulate' which says that the interior of a black hole must be constructed from degrees of freedom that are physically near the black hole. The proximity postulate contradicts the complementarity principle which asserts

that interior information is redundant with information in Hawking radiation, which is very far from the black hole.

As for the firewall proposal, AMPS argue that a violation of the proximity postulate (such as adhering to the BHC principle) would lead to a contradiction in a thought experiment in which Alice distills the Hawking radiation and brings a bit back to the black hole. According to AMPS the only way to protect against the contradiction is for a firewall to form at the Page time. Thus one is faced with the choice between complementarity or firewall. Susskind's response is, the measurement that Alice must make is of such a fine-grained nature that carrying it out before the black hole evaporates may be impossible. He cites Harlow and Hayden [862] who show that the quantum computations required to do these experiments would take a time which is exponential in the entropy of the black hole, and for a wide variety of black holes this prevents the experiments from being done.

No Loss in Information Lost

Before we close this short summary of colorful thoughts on the black hole information problem it is perhaps appropriate to return to Hawking's original viewpoint, preserved by Unruh and Wald [837]. They stated three objections by others to information loss and presented arguments to counter these objections, two of which are given below.

(a) *Unitarity*: Unruh and Wald emphasized that one should distinguish between (i) conservation of probability and (ii) evolution from a pure to a mixed state. The second case is common, but bears no relation to the first.
(b) *Energy nonconservation in decoherence*: They show that there exists dynamics where a system can be decohered without dissipation [863, 864]. This serves as a counterexample to the assertion in [865] that any evolution law taking a pure state to a mixed state would necessarily result in unacceptable violations of energy conservation, which is usually used as a rationale against information loss.

Finally, no account of the recent impact of quantum information on the foundational issues of theoretical physics would be complete without mentioning the works of *entanglement entropy* advanced with the help of the Ryu and Takayanagi formula [866, 867] derived using AdS/CFT correspondence (see also [868, 869, 870, 871]). The lecture note by van Raamsdonk [872] is useful to see the connections between quantum gravity and quantum information theory in the AdS/CFT correspondence.

14.1.4 Related Broader Topics

To end this introduction to black hole quantum field-theoretic processes we mention additional topics related to the central ones discussed above. First, the elegant formulation of black hole thermodynamics is possible under the assumption of quasi-equilibrium conditions, but we also know that the challenging

black hole collapse and end-state issues should be considered in the context of an evaporating black hole in dynamical spacetimes. Thus it is desirable to extend the well-known black hole thermodynamics to *nonequilibrium conditions* [873] with the hope that nonequilibrium thermodynamics can offer a different perspective. However, as we stated earlier, gravitating systems are intrinsically out of equilibrium and thermodynamically unstable; thus, beware that this effort may not be as clean-cut and the results as readily interpretable as before.

Second, the logic behind Bekenstein's derivation of the black hole entropy, which uses Einstein's equations, can be reversed, as done by Jacobson [874] to derive the Einstein's equations from a thermodynamic relation. This has spurred new views of gravity as *spacetime thermodynamics* [873, 875]. Viewing general relativity as the hydrodynamics limit of quantum gravity – theories of the microscopic constituents of spacetime, or *geometro-hydrodynamics* [876, 480, 877, 878, 879, 880] – is in line with general relativity as an effective theory [192, 193], gravity as an *emergent phenomenon* [881, 882, 483, 883] and a new *quantum order* of strongly correlated systems in a condensed matter theoretical perspective of quantum gravity [884, 885, 886, 887].

Third, *analog gravity* has seen promising developments since the seminal paper of Unruh [888], with analog gravitational systems found in atom-optical, BEC, fluid, plasma, electro-mechanical, opto-mechanical and superconducting devices, to mention just a few [889, 890, 891, 892, 893, 894, 895, 896, 897, 898]. Analog gravity offers a practical and accessible way for laboratory experiments to probe physical regimes where calculations for real black hole or early universe quantum processes pose a challenge (e.g., for Planckian scale physics [899]). For example, in analog models where the dispersion relations are modified at very short wavelengths, there is no change in the prediction of Hawking radiation and its entanglement across the horizon at wavelengths long compared with the scale which characterizes this modification [846, 900]. Also, in relation to the backreaction problems we are about to discuss, it is useful to remember that, unlike in theoretical treatments where one separates a system (e.g., classical background spacetime) from its environment (e.g., quantum matter fields) – which is possible only if there exists some discrepancy parameter separating these two sectors, and then considers the backreaction of the environment on the system, experimental measurements of the whole setup always contain the backreaction effects. In fact they should be matched with the exact physical solutions! Naturally, analog gravity experiments are useful for checking the validity and consistency of field-theoretical backreaction effects.

14.2 General Issues on Backreaction

We have seen in the previous section many innovative proposals and colorful scenarios in dealing with the black hole end state and information problem. Despite the strong disagreement and diversity of thoughts there is one solid ground which almost all of the proposed theories agree to honor and keep, and

that is semiclassical gravity. The reason is simple: it is the natural union of two well-tested theories, general relativity for classical spacetime and quantum field theory for matter. We now return to this modest yet reliable theoretical framework to carry out our investigations into the backreaction and fluctuation effects of quantum fields in black hole spacetimes for the rest of this chapter.

14.2.1 Regularized Stress-Energy Tensor

The first step in this investigation is to determine the regularized stress-energy tensor of the quantum matter fields entering into the backreaction problem of interest, using reasonable techniques, since its expectation value acts as the source in the semiclassical Einstein equation. For this, much work started in the 80s was concerned with finding the right approximations for the regularized stress-energy tensor. Even in the simplest spherically symmetric spacetime, which includes the important Schwarzschild metric, this is technically quite involved. As a precursor of this effort it was quite remarkable that Christensen and Fulling [761] as early as 1977 used simple arguments for the form of the stress-energy tensor T_{mn} and obtained results important for the study of Hawking radiation. They started by seeking the spherically symmetric, static solution of the divergence-free condition of T_{mn} in the exterior Schwarzschild metric. Noting that it can be expressed in terms of two integration constants and two arbitrary functions, one of which is the trace of T_{mn}, they pieced up the form of T_{mn} from different observations and arguments. An important result they found was that in a two-dimensional Schwarzschild black hole, the magnitude of the Hawking black body effect at infinity is directly proportional to the magnitude of the anomalous trace. In four dimensions, they obtained a relation constraining the remaining undetermined function, thus laying down the groundwork for explicit calculations of $\langle T_{mn} \rangle$.

We mention some noteworthy works since then in this pursuit. Using a covariant point-separation scheme Candelas [762] calculated the renormalized value of $\langle \Phi^2 \rangle$ for the Boulware, Hartle–Hawking, and Unruh vacua. He also presented the form of the renormalized expectation value of the stress-energy tensor near the horizon and at infinity for each of these three states. In exploring thermal stress tensors in static Einstein spaces Page [488] used the Bekenstein–Parker [487] Gaussian path-integral approximation to evaluate the thermal propagators and found that it gives a very good approximation to the expectation value of the stress-energy tensor. We discussed the Page approximation in Chapter 5, with warnings about its limitation for the calculation of the noise kernel. Using both analytical and numerical techniques Fawcett and Whiting [768] derived the zeta-function regularized mean-square field in the vicinity of a black hole. Fawcett [769] calculated the energy-momentum tensor for a conformally invariant scalar field near a Schwarzschild black hole in thermal equilibrium with radiation. His results agree well with those of Candelas, but differ significantly from the

Gaussian approximation of Page. Numerical results by Elster [901] for a massless conformally coupled scalar field in the Schwarzschild spacetime provided a bridge between the analytical expressions derived by Candelas for the expectation values of the stress-energy tensor with respect to the Unruh and the Hartle–Hawking vacuua for all distances from the black hole. For conformally invariant fields Zannias [902] calculated the renormalized thermal stress-energy tensor for arbitrary static spacetimes. Howard and Candelas [763, 764] computed the vacuum stress-energy tensor for a conformal scalar field and Jensen and Ottewill [903] that of a massless vector field in the Schwarzschild geometry. Candelas and Jensen [904] presented the Feynman Green function inside a Schwarzschild black hole.

Effective action method has been used with the Schwinger–DeWitt approximation by Frolov and Zel'nikov [765, 766] to obtain explicit expressions for the vacuum polarization in the vacuum expectation values of the energy-momentum tensor for massive scalar, spinor and vector fields in Kerr spacetime. They later explored the Killing approximation for vacuum and thermal stress-energy tensor in static spacetimes [905]. Brown and Ottewill [767] also used the effective action method for conformally invariant fields in general curved spacetimes. They showed that the conformal transformation law for the effective action can be used to obtain almost all known renormalized stress-energy tensors. (Conformal transformation was also discussed in Chapter 5.) They derived as a special case Page's approximation for static Einstein spacetimes [906]. Ottewill and Winstanley [907] calculated the renormalized stress-energy tensor in Kerr spacetime.

Noteworthy albeit in a different vein is the work of Jensen, McLaughlin and Ottewill [780], who studied the one-loop perturbative quantum gravity in a Schwarzschild spacetime using the Newman–Penrose formalism. They calculated the expectation values of the modulus squared of the operators corresponding to the perturbed Weyl scalars $\dot{\Psi}_0, \dot{\Psi}_4$ in the Boulware, Unruh, and Hartle–Hawking quantum states.

Anderson, Hiscock and Samuel [908, 779] introduced a numerical method for computing the stress-energy tensor of quantized scalar fields coupled arbitrarily to a general static spherically symmetric spacetime at any temperature. This is aided by a new analytical approximation they devised which extends the approximation schemes of Page and Frolov and Zel'nikov to nonconformally coupled scalar field. As illustration they presented numerical results for massive and massless scalar fields in Schwarzschild and Reissner–Nordström spacetimes.

Quantum fields in the Reissner–Nordström geometry have attracted much attention for the reasons we gave earlier, related to the special spacetime properties of its inner horizon. Anderson, Hiscock and Loranz [909] showed the semiclassical stability of the extreme Reissner–Nordström black hole. Anderson, Mottola and Vaulin [910] calculated the stress-energy tensor from the trace

anomaly in Reissner–Nordström spacetimes. More recently, Lanir et al. [785] derived explicit expressions for the two-point function of a massless scalar field in the interior region of a Reissner–Nordström black hole, in both the Unruh and the Hartle–Hawking quantum states. They are the internal counterparts of the well-known results of Christensen and Fulling [761] in the external region. For a massless, minimally coupled quantum scalar field Sela [786] studied the leading asymptotic behavior of the expectation value of the renormalized stress-energy tensor $\langle \hat{T}_{mn} \rangle$ and $\langle \hat{\Phi}^2 \rangle_{ren}$ near the inner horizon of a Reissner–Nordström black hole background, again, in both the Unruh and the Hartle–Hawking quantum states.

Though arduous and demanding, the effort continues because of its importance in finding the backreaction effects of Hawking radiation on the evolution of black holes and the quantum structure of spacetime.

In our present discussion we wish to address the black hole backreaction problem with the help of some new insights and methods provided by stochastic gravity theory. It is not our intention here to seek better approximations for the regularized energy momentum tensor, but to point out new ingredients lacking in the existing semiclassical gravity framework. In particular, one needs to consider both the dissipation and the fluctuations aspects in the backreaction of particle creation or vacuum polarization.

In a short note Hu, Raval and Sinha [911] first adopted the stochastic gravity framework to address the backreaction of evaporating black holes. A more detailed analysis is given by Hu and Roura [912, 490]. For the class of quasi-static black holes, the formulation of the problem in this new light was sketched out in [341]. In what follows we expound the stochastic gravity theory approach to the black hole fluctuations and backreaction problems based on these two latter works.

14.2.2 Backreaction Represented by Fluctuation–Dissipation Relation

From the statistical field theory perspective provided by stochastic gravity one can understand that backreaction effect is the manifestation of a fluctuation–dissipation relation [913, 914, 915, 916, 917, 918]. This was first conjectured by Candelas and Sciama [919, 920, 921] for a dynamic Kerr black hole emitting Hawking radiation, and Mottola [922] for a static black hole (in a box) in quasi-equilibrium with its radiation via linear response theory [441, 923, 924, 925, 926]. This postulate was shown to hold for fully dynamical spacetimes. From a cosmological backreaction problem Hu and Sinha [185] derived a generalized fluctuation–dissipation relation relating dissipation (of anisotropy in Bianchi Type I universes) and fluctuations (measured by particle numbers created in neighboring histories). The same can be shown for a weakly inhomogeneous universe from the work of Campos and Verdaguer [186].

While the fluctuation–dissipation relation in a linear response theory captures the response of the system (e.g., dissipation of the black hole) to the environment (in these cases, the quantum matter field) linear response theory (in the way it is commonly presented in statistical thermodynamics) cannot provide a full description of self-consistent backreaction on at least two counts: first, because it is usually based on the assumption of a specified background spacetime (static in this case) and state (thermal) of the matter field(s) (see, e.g., [922]). The spacetime and the state of matter should be determined in a self-consistent manner by their dynamics and mutual influence. Second, the fluctuation part represented by the noise kernel is amiss (e.g. [208, 209]). This is also a problem in the fluctuation–dissipation relation proposed by Candelas and Sciama [919, 920, 921]. As demonstrated by many authors, backreaction is intrinsically a dynamical process [185, 186]. The Einstein–Langevin equation in stochastic gravity overcomes both of these deficiencies.

For Candelas and Sciama (CS) [919, 920, 921], the classical formula they give, which relates the dissipation in area linearly to the squared absolute value of the shear amplitude, is suggestive of a fluctuation–dissipation relation. When the gravitational perturbations are quantized (they choose the quantum state to be the Unruh vacuum) they argue that it approximates a flux of radiation from the hole at large radii. Thus the dissipation in area due to the Hawking flux of gravitational radiation is allegedly related to the quantum fluctuations of gravitons. In the opinion of the authors of [911], what CS proposed is not a fluctuation–dissipation relation in the truly statistical mechanical sense because it does not relate dissipation of a certain quantity (in this case, horizon area) to the fluctuations of *the same quantity*. To do so would require one to compute the two-point function of the area, which, being a four-point function of the graviton field, is related to a two-point function of the stress tensor. The stress-energy tensor is the true "generalized force" acting on the spacetime via the equations of motion, and the dissipation in the metric must eventually be related to the fluctuations of this generalized force for the relation to qualify as a fluctuation–dissipation relation.

14.2.3 Noise and Fluctuations; the Missing Ingredient

From this reasoning, we see that the vacuum expectation value of the stress energy bitensor, i.e., the noise kernel, is the necessary new ingredient in addition to the dissipation kernel, and stochastic gravity as an extension of semiclasical gravity is the appropriate framework for backreaction considerations. We have presented the calculations of the noise kernel in various spacetimes of physical relevance in previous chapters.

14.3 Backreaction under Quasi-Static Conditions

As an illustration of the application of stochastic gravity theory we outline the essential steps in a black hole backreaction calculation, focusing on a more manageable class of spacetimes, the quasi-static, which is slightly more restrictive than quasi-stationary (in the sense that Schwarzschild is static whereas Kerr is stationary). We adopt the Hartle–Hawking picture [82] where the black hole is bathed eternally – actually, in quasi-thermal equilibrium – in the Hawking radiance it emits. It is described here by a massless scalar quantum field at the Hawking temperature. As is well-known, this quasi-equilibrium condition is possible only if the black hole is enclosed in a box that is suitably larger than the event horizon.

We can divide our consideration into the far field case and the near horizon case. Campos and Hu [438, 708] have treated a relativistic thermal plasma in a weak gravitational field. Since the far field limit of a Schwarzschild metric is just the perturbed Minkowski spacetime, one can perform a perturbation expansion off hot flat space using the thermal Green functions [927]. Strictly speaking the location of the box holding the black hole in equilibrium with its thermal radiation is as far as one can go, thus the metric may not reach the perturbed Minkowski form. But one can also put the black hole and its radiation in an anti-de Sitter space, which contains such a region [829]. Hot flat space has been studied before for various purposes. See, e.g., [928, 929, 930, 931, 932]. Campos and Hu derived a stochastic CTP effective action and from it an equation of motion, the Einstein–Langevin equation, for the dynamical effect of a scalar quantum field on a background spacetime. To perform calculations leading to the Einstein–Langevin equation one needs to begin with a self-consistent solution of the semiclassical Einstein equation for the thermal field and the perturbed background spacetime. For a black hole background, a semiclassical gravity solution is provided by York [771, 793, 772]. The theoretical framework used is similar to that of finding self-consistent solutions to the semiclassical Einstein equations for finite temperature quantum fields in an Einstein universe by Dowker et al. [617, 458, 618, 619] and in a FLRW cosmology by Hu [631] described in Chapter 8.

Following the scheme of Campos and Hu [438], Sinha et al. [341] presented a strategy for treating the backreaction and fluctuations of quantum matter fields on a static black hole in the near horizon case. In both the far-field and the near-horizon cases two new terms appear which are absent in semiclassical gravity considerations: a nonlocal dissipation and a (generally colored) kernel. When the noise average is taken one recovers the semiclassical equations for radially perturbed quasi-static black holes [771, 793, 772]. For the near-horizon case one cannot obtain the full details yet, because the Green function for a scalar field in the Schwarzschild metric comes only in an approximate form, e.g., under the Page approximation [488], which, though reasonably accurate

for the stress-energy tensor, fails badly for the noise kernel [201]. In addition, a formula is derived in [341] expressing the in-in or CTP effective action in terms of the Bogoliubov coefficients. Since it measures not only the number of particles created, but also the difference of particle creation in alternative histories, this provides a useful avenue to explore the wider set of issues in black hole physics related to noise and fluctuations.

Since backreaction calculations in semiclassical gravity have been under study for a much longer time than in stochastic gravity we will concentrate on explaining how the new stochastic features arise from the framework of semiclassical gravity, i.e., noise and fluctuations and their consequences. The goal is to obtain an influence action for this model of a black hole coupled to a scalar field and to derive an Einstein–Langevin equation from it. As a byproduct, from the fluctuation–dissipation relation, one can derive the vacuum susceptibility function and the isothermal compressibility function for black holes, two quantities of fundamental interest in characterizing the nonequilibrium thermodynamic properties of black holes.

14.3.1 The Model

In this model the black hole spacetime is described by a spherically symmetric static metric with line element of the following general form written in advanced time Eddington–Finkelstein coordinates

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -e^{2\psi}\left(1-\frac{2m}{r}\right)dv^2 + 2e^{2\psi}dvdr + r^2\, d\Omega^2, \tag{14.1}$$

where $\psi = \psi(r)$ and $m = m(r)$, $v = t + r + 2Mln\left(\frac{r}{2M} - 1\right)$ and $d\Omega^2$ is the line element on the two sphere. Hawking radiation is described by a massless, conformally coupled quantum scalar field ϕ with the classical action

$$S_{\rm m}[\phi, g_{\mu\nu}] = -\frac{1}{2}\int d^n x\sqrt{-g}\,[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + \xi(n)R\phi^2]\,, \tag{14.2}$$

where $\xi(n) = \frac{(n-2)}{4(n-1)}$ (n is the dimension of spacetime) and R is the curvature scalar of the spacetime.

Let us consider linear dimensionless perturbations $h_{\mu\nu}$ off a background Schwarzschild metric $g^{(0)}_{\mu\nu}$

$$g_{\mu\nu} = g^{(0)}_{\mu\nu} + h_{\mu\nu}, \tag{14.3}$$

with standard line element

$$(ds^2)^{(0)} = \left(1-\frac{2M}{r}\right)dv^2 + 2dvdr + r^2 d\Omega^2. \tag{14.4}$$

We look for this class of perturbed metrics in the form given by (14.1), (thus restricting our consideration only to spherically symmetric perturbations):

$$e^{\psi} \simeq 1 + \epsilon\rho(r), \tag{14.5}$$

and

$$m \simeq M[1 + \epsilon\mu(r)], \tag{14.6}$$

where $\epsilon/(\lambda M^2) = \frac{1}{3}aT_H^4$, $a = \pi^2/30$, $\lambda = 90(8^4)\pi^2$ with T_H the Hawking temperature.

In this particular parametrization of the perturbation, following York [771, 793, 772], the only non-zero components of $h_{\mu\nu}$ are

$$h_{vv} = -\left[\left(1 - \frac{2M}{r}\right) 2\epsilon\rho(r) + \frac{2M\epsilon\mu(r)}{r}\right], \tag{14.7}$$

and

$$h_{vr} = \epsilon\rho(r). \tag{14.8}$$

So this represents a metric with small static and radial perturbations about a Schwarzschild black hole. The initial quantum state of the scalar field is taken to be the Hartle–Hawking vacuum, which is essentially a thermal state at the Hawking temperature and it represents a black hole in quasi-thermal equilibrium with its own Hawking radiation. In the far field limit, the gravitational field is described by a linear perturbation from Minkowski spacetime. In equilibrium the thermal bath can be characterized by a relativistic fluid with a four-velocity (time-like normalized vector field) u^μ, and temperature in its own rest frame β^{-1}.

To facilitate later comparisons we briefly recall the results from [771, 793, 772]. (See also [794] for a massless vector field, [795] for a massless spinor field, and [778] for a quantized massless scalar field with arbitrary coupling to spacetime curvature.) York considered the semiclassical Einstein equation (SCE)

$$G_{\mu\nu}(g_{\alpha\beta}) = 8\pi G_{\rm N}\langle T_{\mu\nu}\rangle, \tag{14.9}$$

with $G_{\mu\nu} \simeq G^{(0)}_{\mu\nu} + \delta G_{\mu\nu}$ where $G^{(0)}_{\mu\nu}$ is the Einstein tensor for the background spacetime. The zeroth order solution gives a background metric in empty space, i.e. the Schwarzschild metric, and $\delta G_{\mu\nu}$ is the linear correction to the Einstein tensor in the perturbed metric. The semiclassical Einstein equation in this approximation therefore reduces to

$$\delta G_{\mu\nu}(g^{(0)}, h) = 8\pi G_{\rm N}\langle T_{\mu\nu}\rangle. \tag{14.10}$$

York solved this equation to first order by using the expectation value of the stress-energy tensor for a conformally coupled scalar field in the Hartle–Hawking vacuum in the unperturbed (Schwarzschild) spacetime on the right-hand side and using (14.7) and (14.8) to calculate $\delta G_{\mu\nu}$ on the left-hand side. Since no exact analytical expression is available for the $\langle T_{\mu\nu}\rangle$ in a Schwarzschild metric with the quantum field in the Hartle–Hawking vacuum York adopts the approximate expression given by Page [488] which fortunately gives very good agreement with numerical results. Page's approximate expression for $\langle T_{\mu\nu}\rangle$ was constructed using

a thermal Feynman Green's function obtained by a conformal transformation of a WKB approximated Green's function for an optical Schwarzschild metric. York then solves the semiclassical Einstein equation (14.10) self consistently to obtain the corrections to the background metric induced by the backreaction encoded in the functions $\mu(r)$ and $\rho(r)$. There was no mention of fluctuations or its effects. As we shall see, in the language of the previous section, the procedure which York followed working at the equation of motion level is equivalent to looking at the noise-averaged backreaction effects in semiclassical gravity theory. Solutions to the semiclassical Einstein equation are useful as the background spacetimes for calculating the metric fluctuations from solutions to the Einstein–Langevin equation in stochastic gravity theory.

14.3.2 CTP Effective Action for the Black Hole

We first derive the in-in or CTP effective action for the model described in the previous section. Using the metric (14.4) (and neglecting the surface terms that appear in an integration by parts) we have the action for the scalar field written perturbatively as

$$S_{\rm m}[\phi, h_{\mu\nu}] = \frac{1}{2}\int d^n x\sqrt{-g^{(0)}}\;\phi\left[\Box^{(0)} + V^{(1)} + V^{(2)} + \cdots\right]\phi, \tag{14.11}$$

where the first- and second-order perturbative operators $V^{(1)}$ and $V^{(2)}$ are given by

$$\begin{aligned} V^{(1)} &\equiv -\frac{1}{\sqrt{-g^{(0)}}}\left\{\partial_\mu\left(\sqrt{-g^{(0)}}\bar{h}^{\mu\nu}\right)\partial_\nu + \bar{h}^{\mu\nu}\partial_\mu\partial_\nu + \xi(n)R^{(1)}\right\}, \\ V^{(2)} &\equiv -\frac{1}{\sqrt{-g^{(0)}}}\left\{\partial_\mu\left(\sqrt{-g^{(0)}}\hat{h}^{\mu\nu}\right)\partial_\nu + \hat{h}^{\mu\nu}\partial_\mu\partial_\nu \right. \\ &\qquad \left. -\xi(n)(R^{(2)} + \frac{1}{2}hR^{(1)})\right\}. \end{aligned} \tag{14.12}$$

In the above expressions, $R^{(k)}$ is the k-order term in the perturbation $h_{\mu\nu}(x)$ of the scalar curvature R and $\bar{h}_{\mu\nu}$ and $\hat{h}_{\mu\nu}$ denote a linear and a quadratic combination of the perturbation, respectively,

$$\begin{aligned} \bar{h}_{\mu\nu} &\equiv h_{\mu\nu} - \frac{1}{2}hg^{(0)}_{\mu\nu}, \\ \hat{h}_{\mu\nu} &\equiv h_\mu{}^\alpha h_{\alpha\nu} - \frac{1}{2}hh_{\mu\nu} + \frac{1}{8}h^2 g^{(0)}_{\mu\nu} - \frac{1}{4}h_{\alpha\beta}h^{\alpha\beta}g^{(0)}_{\mu\nu}. \end{aligned} \tag{14.13}$$

From quantum field theory in curved spacetime considerations discussed above we take the following action for the gravitational field:

$$S_{\rm g}[g_{\mu\nu}] = \frac{1}{(16\pi G_{\rm N})^{\frac{n-2}{2}}}\int d^n x\ \sqrt{-g(x)}R(x) + \frac{\alpha\bar{\mu}^{n-4}}{4(n-4)}\int d^n x\ \sqrt{-g(x)}$$
$$\times\left\{3R_{\mu\nu\alpha\beta}(x)R^{\mu\nu\alpha\beta}(x) - \left[1 - 360\left(\xi(n) - \frac{1}{6}\right)^2\right]R^2(x)\right\}. \quad (14.14)$$

The first term is the classical Einstein–Hilbert action and the second term is the counterterm in four dimensions used to renormalize the divergent effective action. In this action $\alpha = (2880\pi^2)^{-1}$ and $\bar{\mu}$ is an arbitrary mass scale.

We are interested in computing the CTP effective action (14.11) for the matter action when the field ϕ is initially in the Hartle–Hawking vacuum. This is equivalent to saying that the initial state of the field is described by a thermal density matrix at a finite temperature $T = T_H$. Following Section 9.2, see equation (9.42), the CTP effective action at finite temperature $T \equiv 1/\beta$ can be written for this model as

$$S^{\beta}_{\rm eff}[h^{\pm}_{\mu\nu}] = S_{\rm g}[h^{+}_{\mu\nu}] - S_{\rm g}[h^{-}_{\mu\nu}] - \frac{i}{2}{\rm Tr}\{\ln\bar{G}^{\beta}_{AB}[h^{\pm}_{\mu\nu}]\}, \quad (14.15)$$

where $\pm$ denote the forward and backward time path of the CTP formalism and $\bar{G}^{\beta}_{AB}[h^{\pm}_{\mu\nu}]$ is the complete 2×2 matrix propagator ($A, B, \ldots$ take $\pm$ values: G_{++}, G_{+-} and G_{--} correspond to the Feynman, Wightman and Dyson Greens functions respectively) with thermal boundary conditions for the differential operator $\sqrt{-g^{(0)}}(\Box + V^{(1)} + V^{(2)} + \cdots)$. The actual form of $\bar{G}^{\beta}_{AB}$ cannot be explicitly given. However, it is easy to obtain a perturbative expansion in terms of $V^{(k)}_{AB}$, the k-order matrix version of the complete differential operator defined by $V^{(k)}_{\pm\pm} \equiv \pm V^{(k)}_{\pm}$ and $V^{(k)}_{\pm\mp} \equiv 0$, and G^{β}_{AB}, the thermal matrix propagator for a massless scalar field in Schwarzschild spacetime. As seen in equation (9.45), to second order $\bar{G}^{\beta}_{AB}$ reads,

$$\bar{G}^{\beta}_{AB} = G^{\beta}_{AB} - G^{\beta}_{AC}V^{(1)}_{CD}G^{\beta}_{DB} - G^{\beta}_{AC}V^{(2)}_{CD}G^{\beta}_{DB}$$
$$+ G^{\beta}_{AC}V^{(1)}_{CD}G^{\beta}_{DE}V^{(1)}_{EF}G^{\beta}_{FB} + \cdots \quad (14.16)$$

Expanding the logarithm and dropping one term that is independent of the perturbation $h^{\pm}_{\mu\nu}(x)$, the CTP effective action may be perturbatively written as

$$S^{\beta}_{\rm eff}[h^{\pm}_{\mu\nu}] = S_{\rm g}[h^{+}_{\mu\nu}] - S_{\rm g}[h^{-}_{\mu\nu}]$$
$$+ \frac{i}{2}{\rm Tr}\left[V^{(1)}_{+}G^{\beta}_{++} - V^{(1)}_{-}G^{\beta}_{--} + V^{(2)}_{+}G^{\beta}_{++} - V^{(2)}_{-}G^{\beta}_{--}\right]$$
$$- \frac{i}{4}{\rm Tr}\left[V^{(1)}_{+}G^{\beta}_{++}V^{(1)}_{+}G^{\beta}_{++} + V^{(1)}_{-}G^{\beta}_{--}V^{(1)}_{-}G^{\beta}_{--}\right.$$
$$\left. -2V^{(1)}_{+}G^{\beta}_{+-}V^{(1)}_{-}G^{\beta}_{-+}\right]. \quad (14.17)$$

In computing the traces, some terms containing divergences are canceled using counterterms introduced in the classical gravitational action after dimensional regularization.

14.3.3 Far-Field, Nearly-Flat Case

At this point we divide our considerations into two cases. In the far field limit $h_{\mu\nu}$ represents perturbations about flat space, *i.e.*, $g^{(0)}_{\mu\nu} = \eta_{\mu\nu}$. The exact "unperturbed" thermal propagators for scalar fields are known, *i.e.*, the Euclidean propagator with periodicity β. Using the Fourier transformed form (those quantities are denoted with a tilde) of the thermal propagators $\tilde{G}^{\beta}_{AB}(k)$, the trace terms of the form $\mathrm{Tr}[V^{(1)}_A G^{\beta}_{MN} V^{(1)}_B G^{\beta}_{RS}]$ can be written as [438, 708],

$$\mathrm{Tr}[V^{(1)}_A G^{\beta}_{MN} V^{(1)}_B G^{\beta}_{RS}] = \int d^n x d^n x' \; h^A_{\mu\nu}(x) h^B_{\alpha'\beta'}(x') \times \int \frac{d^n k}{(2\pi)^n} \frac{d^n q}{(2\pi)^n} e^{ik\cdot(x-x')} \tilde{G}^{\beta}_{MN}(k+q) \tilde{G}^{\beta}_{RS}(q) \mathrm{T}^{\mu\nu\alpha'\beta'}(q,k), \tag{14.18}$$

where the tensor $\mathrm{T}^{\mu\nu\alpha'\beta'}(q,k)$ is defined by Campos and Hu [438, 708] after an expansion in terms of a basis of 14 tensors given by Rebhan [929, 930]. In particular, the last trace of (14.17) may be split in two different kernels $\mathrm{N}^{\mu\nu\alpha'\beta'}(x-x')$ and $\mathrm{D}^{\mu\nu\alpha'\beta'}(x-x')$,

$$\frac{i}{2} Tr[V^{(1)}_{+} G^{\beta}_{+-} V^{(1)}_{-} G^{\beta}_{-+}] = -\int d^4 x d^4 x' \; h^{+}_{\mu\nu}(x) h^{-}_{\alpha'\beta'}(x') \times [\mathrm{D}^{\mu\nu\alpha'\beta'}(x-x') + i\mathrm{N}^{\mu\nu\alpha'\beta'}(x-x')]. \tag{14.19}$$

One can express the Fourier transforms of these kernels, respectively, as

$$\tilde{\mathrm{N}}^{\mu\nu\alpha'\beta'}(k) = \pi^2 \int \frac{d^4 q}{(2\pi)^4} \; \{\theta(k^o+q^o)\theta(-q^o) + \theta(-k^o-q^o)\theta(q^o) + n_\beta(|q^o|) + n_\beta(|k^o+q^o|) + 2n_\beta(|q^o|)n_\beta(|k^o+q^o|)\} \times \delta(q^2)\delta[(k+q)^2]\mathrm{T}^{\mu\nu\alpha'\beta'}(q,k), \tag{14.20}$$

and

$$\tilde{\mathrm{D}}^{\mu\nu\alpha'\beta'}(k) = -i\pi^2 \int \frac{d^4 q}{(2\pi)^4} \; \{\theta(k^o+q^o)\theta(-q^o) - \theta(-k^o-q^o)\theta(q^o) + \mathrm{sgn}(k^o+q^o)n_\beta(|q^o|) - \mathrm{sgn}(q^o)n_\beta(|k^o+q^o|)\} \times \delta(q^2)\delta[(k+q)^2]\mathrm{T}^{\mu\nu\alpha'\beta'}(q,k). \tag{14.21}$$

Using the property $\mathrm{T}^{\mu\nu\alpha'\beta'}(q,k) = \mathrm{T}^{\mu\nu\alpha'\beta'}(-q,-k)$, it is easy to see that the kernel $\mathrm{N}^{\mu\nu\alpha'\beta'}(x-x')$ is symmetric and $\mathrm{D}^{\mu\nu\alpha'\beta'}(x-x')$ antisymmetric in their arguments; that is, $\mathrm{N}^{\mu\nu\alpha'\beta'}(x) = \mathrm{N}^{\mu\nu\alpha'\beta'}(-x)$ and $\mathrm{D}^{\mu\nu\alpha'\beta'}(x) = -\mathrm{D}^{\mu\nu\alpha'\beta'}(-x)$.

The physical meanings of these kernels can be extracted if we write the renormalized CTP effective action at finite temperature (14.17) in an influence functional form [168, 933, 171, 172]. N, the imaginary part of the CTP effective action can be identified with the noise kernel and D, the antisymmetric piece

of the real part, with the dissipation kernel. One can show [438, 708] that these kernels identified as such indeed satisfy a thermal fluctuation–dissipation relation.

If we denote the difference and the sum of the perturbations $h^{\pm}_{\mu\nu}$ defined along each branch $C_{\pm}$ of the complex time path of integration C by $[h_{\mu\nu}] \equiv h^{+}_{\mu\nu} - h^{-}_{\mu\nu}$ and $\{h_{\mu\nu}\} \equiv h^{+}_{\mu\nu} + h^{-}_{\mu\nu}$, respectively, the influence functional form of the thermal CTP effective action may be written to second order in $h_{\mu\nu}$ as

$$\begin{aligned} S^{\beta}_{\rm eff}[h^{\pm}_{\mu\nu}] \simeq \frac{1}{2(16\pi G_{\rm N})} \int d^4x\ d^4x'\ [h_{\mu\nu}](x) \mathrm{L}^{\mu\nu\alpha'\beta'}_{(o)}(x-x')\{h_{\alpha'\beta'}\}(x') \\ + \frac{1}{2}\int d^4x\ [h_{\mu\nu}](x) T^{\mu\nu}_{(\beta)} \\ + \frac{1}{2}\int d^4x\ d^4x'\ [h_{\mu\nu}](x) \mathrm{H}^{\mu\nu\alpha'\beta'}(x-x')\{h_{\alpha'\beta'}\}(x') \\ - \frac{1}{2}\int d^4x\ d^4x'\ [h_{\mu\nu}](x) \mathrm{D}^{\mu\nu\alpha'\beta'}(x-x')\{h_{\alpha'\beta'}\}(x') \\ + \frac{i}{2}\int d^4x\ d^4x'\ [h_{\mu\nu}](x) \mathrm{N}^{\mu\nu\alpha'\beta'}(x-x')[h_{\alpha'\beta'}](x'). \end{aligned} \tag{14.22}$$

The first line is the Einstein–Hilbert action to second order in the perturbation $h^{\pm}_{\mu\nu}(x)$. $\mathrm{L}^{\mu\nu\alpha'\beta'}_{(o)}(x)$ is a symmetric kernel (*i.e.* $\mathrm{L}^{\mu\nu\alpha'\beta'}_{(o)}(x) = \mathrm{L}^{\mu\nu\alpha'\beta'}_{(o)}(-x)$). In the near- flat case its Fourier transform is given by

$$\begin{aligned} \tilde{\mathrm{L}}^{\mu\nu\alpha'\beta'}_{(o)}(x) = \frac{1}{4}\Big[-k^2 \mathrm{T}^{\mu\nu\alpha'\beta'}_{1}(q,k) + 2k^2 \mathrm{T}^{\mu\nu\alpha'\beta'}_{4}(q,k) \\ + \mathrm{T}^{\mu\nu\alpha'\beta'}_{8}(q,k) - 2\mathrm{T}^{\mu\nu\alpha'\beta'}_{13}(q,k)\Big]. \end{aligned} \tag{14.23}$$

The 14 elements of the tensor basis $\mathrm{T}^{\mu\nu\alpha'\beta'}_{i}(q,k)$ $(i = 1, \ldots, 14)$ are defined by Rebhan [929, 930]. The second is a local term linear in $h^{\pm}_{\mu\nu}(x)$. Far away from the hole it takes the form of the stress-energy tensor of massless scalar particles at temperature β^{-1}, which has the form of a perfect fluid stress-energy tensor

$$T^{\mu\nu}_{(\beta)} = \frac{\pi^2}{30\beta^4}\left[u^\mu u^\nu + \frac{1}{3}(\eta^{\mu\nu} + u^\mu u^\nu)\right], \tag{14.24}$$

where u^μ is the four-velocity of the plasma and the factor $\frac{\pi^2}{30\beta^4}$ is the familiar thermal energy density for massless scalar particles at temperature β^{-1}. Thus, in the far field limit, taking into account the four-velocity u^μ of the fluid, a manifestly Lorentz-covariant approach to thermal field theory may be used [934].

However, in order to simplify the involved tensorial structure we work in the comoving coordinate system of the fluid where $u^\mu = (1,0,0,0)$. In the third line, the Fourier transform of the symmetric kernel $\mathrm{H}^{\mu\nu\alpha'\beta'}(x)$ can be expressed as

$$\tilde{\mathrm{H}}^{\mu\nu\alpha'\beta'}(k) = -\frac{\alpha k^4}{4}\left\{\frac{1}{2}\ln\frac{|k^2|}{\mu^2}\mathrm{Q}^{\mu\nu\alpha'\beta'}(k) + \frac{1}{3}\bar{\mathrm{Q}}^{\mu\nu\alpha'\beta'}(k)\right\}$$
$$+\frac{\pi^2}{180\beta^4}\left\{-\mathrm{T}_1^{\mu\nu\alpha'\beta'}(u,k) - 2\mathrm{T}_2^{\mu\nu\alpha'\beta'}(u,k)\right.$$
$$\left.+\mathrm{T}_4^{\mu\nu\alpha'\beta'}(u,k) + 2\mathrm{T}_5^{\mu\nu\alpha'\beta'}(u,k)\right\}$$
$$+\frac{\xi}{96\beta^2}\left\{k^2\mathrm{T}_1^{\mu\nu\alpha'\beta'}(u,k) - 2k^2\mathrm{T}_4^{\mu\nu\alpha'\beta'}(u,k)\right.$$
$$\left.-\mathrm{T}_8^{\mu\nu\alpha'\beta'}(u,k) + 2\mathrm{T}_{13}^{\mu\nu\alpha'\beta'}(u,k)\right\}$$
$$+\pi\int\frac{d^4q}{(2\pi)^4}\left\{\delta(q^2)n_\beta(|q^o|)\mathcal{P}\left[\frac{1}{(k+q)^2}\right]\right.$$
$$\left.+\delta[(k+q)^2]n_\beta(|k^o+q^o|)\mathcal{P}\left[\frac{1}{q^2}\right]\right\}\times\mathrm{T}^{\mu\nu\alpha'\beta'}(q,k), \qquad (14.25)$$

where μ is a simple redefinition of the renormalization parameter $\bar{\mu}$ given by $\mu \equiv \bar{\mu}\exp(\frac{23}{15} + \frac{1}{2}\ln 4\pi - \frac{1}{2}\gamma)$, and the tensors $\mathrm{Q}^{\mu\nu\alpha'\beta'}(k)$ and $\bar{\mathrm{Q}}^{\mu\nu\alpha'\beta'}(k)$ are defined, respectively, by

$$\mathrm{Q}^{\mu\nu\alpha'\beta'}(k) = \frac{3}{2}\left\{\mathrm{T}_1^{\mu\nu\alpha'\beta'}(q,k) - \frac{1}{k^2}\mathrm{T}_8^{\mu\nu\alpha'\beta'}(q,k) + \frac{2}{k^4}\mathrm{T}_{12}^{\mu\nu\alpha'\beta'}(q,k)\right\}$$
$$-\left[1 - 360\left(\xi - \frac{1}{6}\right)^2\right]\left\{\mathrm{T}_4^{\mu\nu\alpha'\beta'}(q,k)\right.$$
$$\left.+\frac{1}{k^4}\mathrm{T}_{12}^{\mu\nu\alpha'\beta'}(q,k) - \frac{1}{k^2}\mathrm{T}_{13}^{\mu\nu\alpha'\beta'}(q,k)\right\}, \qquad (14.26)$$

and

$$\bar{\mathrm{Q}}^{\mu\nu\alpha'\beta'}(k) = \left[1 + 576\left(\xi - \frac{1}{6}\right)^2 - 60\left(\xi - \frac{1}{6}\right)(1 - 36\xi')\right]$$
$$\times\left\{\mathrm{T}_4^{\mu\nu\alpha'\beta'}(q,k) + \frac{1}{k^4}\mathrm{T}_{12}^{\mu\nu\alpha'\beta'}(q,k) - \frac{1}{k^2}\mathrm{T}_{13}^{\mu\nu\alpha'\beta'}(q,k)\right\}. \qquad (14.27)$$

In the above and subsequent equations, we denote the coupling parameter in four dimensions $\xi(4)$ by ξ and $d\xi(n)/dn$ evaluated at $n = 4$ by ξ'. The upshot is, $\tilde{\mathrm{H}}^{\mu\nu\alpha'\beta'}(k)$ gives the complete contribution of a free massless quantum scalar field to the thermal graviton polarization tensor [929, 930, 931, 932] and it is responsible for the instabilities found in flat spacetime at finite temperature [928, 929, 930, 931, 932].

Note that the addition of the contribution of other kinds of matter fields to the effective action, even graviton contributions, does not change the tensor structure of these kernels and only the overall factors are different to leading order [929, 930]. Equation (14.25) reflects the fact that the kernel $\tilde{\mathrm{H}}^{\mu\nu\alpha'\beta'}(k)$ has

thermal as well as non-thermal contributions. Note that it reduces to the first term in the zero temperature limit ($\beta \to \infty$)

$$\tilde{\mathrm{H}}^{\mu\nu\alpha'\beta'}(k) \simeq -\frac{\alpha k^4}{4}\left\{\frac{1}{2}\ln\frac{|k^2|}{\mu^2}\mathrm{Q}^{\mu\nu\alpha'\beta'}(k) + \frac{1}{3}\bar{\mathrm{Q}}^{\mu\nu\alpha'\beta'}(k)\right\}, \tag{14.28}$$

and at high temperatures the leading term (β^{-4}) may be written as

$$\tilde{\mathrm{H}}^{\mu\nu\alpha'\beta'}(k) \simeq \frac{\pi^2}{30\beta^4}\sum_{i=1}^{14}\mathrm{H}_i(r)\mathrm{T}_i^{\mu\nu\alpha'\beta'}(u,K), \tag{14.29}$$

where we have introduced the dimensionless external momentum $K^\mu \equiv k^\mu/|\vec{k}| \equiv (r,\hat{k})$. The $\mathrm{H}_i(r)$ coefficients were first given by Rebhan [929, 930] and generalized to the next-to-leading order (β^{-2}) by de Almeida *et al.* [931, 932]. They are given with the MTW sign [604] convention by Campos and Hu [438, 708].

Finally, as defined above, $\mathrm{N}^{\mu\nu\alpha'\beta'}(x)$ is the dissipation kernel representing the random fluctuations of the thermal radiance and $\mathrm{D}^{\mu\nu\alpha'\beta'}(x)$ is the dissipation kernel, describing the dissipation of energy of the gravitational field.

14.3.4 Near-Horizon Case

In the near-horizon case, the perturbations are taken around the Schwarzschild spacetime. The simplicity of the Green functions for the near-flat metric in the case treated before is no longer available, and the exact expressions for the corresponding unperturbed propagators $G^\beta_{AB}[h^\pm_{\mu\nu}]$ are not known. Therefore, in addition to the approximations needed for computing the CTP effective action to a certain order in perturbation theory, an appropriate approximation scheme for the unperturbed Green functions is also required. This feature manifested itself in York's calculation of backreaction as well, where, in writing the $\langle T_{\mu\nu}\rangle$ on the right-hand side of the semiclassical Einstein equation in the unperturbed Schwarzschild metric, he had to use an approximate expression for $\langle T_{\mu\nu}\rangle$ in the Schwarzschild metric given by Page [488]. The additional complication here is, while to obtain $\langle T_{\mu\nu}\rangle$ as in York's calculation, the knowledge of only the thermal Feynman Green function is required, to calculate the CTP effective action one needs the knowledge of the full matrix propagator, which involves the Feynman, Dyson and Wightman functions. This said, we want to emphasize that it is indeed possible to construct the full thermal matrix propagator $G^\beta_{AB}[h^\pm_{\mu\nu}]$ based on Page's approximate Feynman Green function by using identities relating the Feynman Green function with the other Green functions with different boundary conditions. This done, one can then proceed to explicitly compute a CTP effective action and hence the influence functional based on this approximation. Another way is to utilize the near-horizon expressions of the noise kernels in the static de Sitter spacetime [477, 478] (described at the end of Chapter 5) which has the same causal structure as in the Schwarzschild metric. In what follows, we shall focus on the noise term which is the new ingredient in the backreaction.

As we recall, the noise term is related to the variance of fluctuations in $T_{\mu\nu}$, i.e., schematically, to $\langle T^2_{\mu\nu}\rangle$ which gives a stochastic contribution $\xi^{\mu\nu}$ to the Einstein–Langevin equation (10.14). A calculation of $\langle T^2_{\mu\nu}\rangle$ in the Hartle–Hawking state in a Schwarzschild background using the Page approximation was performed by Phillips and Hu [200, 201]. Unfortunately, though the approximation is excellent as far as $\langle T_{\mu\nu}\rangle$ is concerned, it gives unacceptably large errors for $\langle T^2_{\mu\nu}\rangle$ at the horizon. In fact, similar errors will be propagated in the non-local dissipation term as well, because both terms originate from the same source, that is, they come from the last trace term in (14.17) which contains terms quadratic in the Green function. However, the influence functional or CTP formalism itself does not depend on the nature of the approximation, so we will attempt to exhibit the general structure of the calculation without resorting to a specific form for the Green function. We shall outline the procedures schematically. A more accurate computation can be performed adhering to this formal structure once a better approximation becomes available.

The general structure of the CTP effective action arising from the calculation of the traces in equation (14.17) remains the same. But to write down explicit expressions for the non-local kernels one requires the input of the explicit form of $G^{\beta}_{AB}[h^{\pm}_{\mu\nu}]$ in the Schwarzschild metric, which is not available in closed form. We can make some general observations about the terms in there. The first line containing L does not have an explicit Fourier representation as given in the far field case, neither will $T^{\mu\nu}_{(\beta)}$ in the second line representing the zeroth order contribution to $\langle T_{\mu\nu}\rangle$ have a perfect fluid form. The third and fourth terms containing the remaining quadratic component of the real part of the effective action will not have any simple or even complicated analytic form. The symmetry properties of the kernels $H^{\mu\nu\alpha'\beta'}(x,x')$ and $D^{\mu\nu\alpha'\beta'}(x,x')$ remain intact, i.e., they are respectively even and odd in x, x'. The last term in the CTP effective action gives the imaginary part of the effective action and the kernel $N(x,x')$ is symmetric.

Continuing our general observations from this CTP effective action, using the connection between this thermal CTP effective action to the influence action [706, 216] via an equation in the schematic form (10.35), we see that the non-local imaginary term containing the kernel $N^{\mu\nu\alpha'\beta'}(x,x')$ is responsible for the generation of the stochastic noise term in the Einstein–Langevin equation and the real non-local term containing kernel $D^{\mu\nu\alpha'\beta'}(x,x')$ is responsible for the non-local dissipation term.

With the identification of the noise and dissipation kernels, one can write down a linear, non-local relation of the form,

$$N(t-t') = \int d(s-s')K(t-t', s-s')\gamma(s-s'), \tag{14.30}$$

where γ is related to the dissipation kernel by $D(t,t') = -\partial_{t'}\gamma(t,t')$. This is the general functional form of a fluctuation–dissipation relation and $K(t,s)$ is called

the fluctuation–dissipation kernel [168, 933, 171, 172]. In the present context this relation depicts the backreaction of thermal Hawking radiation on a black hole in quasi-equilibrium with it.

14.3.5 Einstein–Langevin Equation

In this section we show how a semiclassical Einstein–Langevin equation can be derived from the previous thermal CTP effective action. This equation depicts the stochastic evolution of the perturbations of the black hole under the influence of the fluctuations of the thermal scalar field.

The influence functional $\mathcal{F}_{\rm IF} \equiv \exp(iS_{\rm IF})$ previously introduced in equation (10.34) can be written in terms of the the CTP effective action $S^{\beta}_{\rm eff}[h^{\pm}_{\mu\nu}]$ derived in equation (14.22) using equation (10.35). The Einstein–Langevin equation follows from taking the functional derivative of the stochastic effective action (10.45) with respect to $[h_{\mu\nu}](x)$ and imposing $[h_{\mu\nu}](x) = 0$. This leads to

$$\frac{1}{16\pi G_{\rm N}} \int d^4x' \mathrm{L}^{\mu\nu\alpha'\beta'}_{(o)}(x-x') h_{\alpha'\beta'}(x') + \frac{1}{2}\, T^{\mu\nu}_{(\beta)} + \int d^4x' \Big(\mathrm{H}^{\mu\nu\alpha'\beta'}(x-x') - \mathrm{D}^{\mu\nu\alpha'\beta'}(x-x') \Big) h_{\alpha'\beta'}(x') + \xi^{\mu\nu}(x) = 0, \quad (14.31)$$

where

$$\langle \xi^{\mu\nu}(x) \xi^{\alpha'\beta'}(x') \rangle_s = \mathrm{N}^{\mu\nu\alpha'\beta'}(x-x'). \quad (14.32)$$

In the far field limit this equation should reduce to that obtained by Campos and Hu [438, 708]: For gravitational perturbations $h^{\mu\nu}$ defined in (14.13) under the harmonic gauge $\bar{h}^{\mu\nu}{}_{,\nu} = 0$, their Einstein–Langevin equation is given by

$$\Box \bar{h}^{\mu\nu}(x) + \frac{1}{16\pi G_{\rm N}^2} \Big\{ T^{\mu\nu}_{(\beta)} + 2P_{\hat{\rho}\hat{\sigma}\alpha'\beta'} \int d^4x' \Big(\mathrm{H}^{\mu\nu\alpha'\beta'}(x-x') - \mathrm{D}^{\mu\nu\alpha'\beta'}(x-x') \Big) \bar{h}^{\hat{\rho}\hat{\sigma}}(x') + 2\xi^{\mu\nu}(x) \Big\} = 0, \quad (14.33)$$

where the tensor $P_{\rho\sigma\alpha'\beta'}$ is given by

$$P_{\rho\sigma\alpha'\beta'} = \frac{1}{2} \left(\eta_{\rho\alpha'}\eta_{\sigma\beta'} + \eta_{\rho\beta'}\eta_{\sigma\alpha'} - \eta_{\rho\sigma}\eta_{\alpha'\beta'} \right). \quad (14.34)$$

The expression for $P_{\rho\sigma\alpha'\beta'}$ in the near horizon limit of course cannot be expressed in such a simple form. Note that this differential stochastic equation includes a non-local term responsible for the dissipation of the gravitational field and a noise source term which accounts for the fluctuations of the quantum field. Note also that this equation in combination with the correlation for the stochastic variable (14.32) determines the two-point correlation for the stochastic metric fluctuations $\langle \bar{h}_{\mu\nu}(x) \bar{h}_{\alpha'\beta'}(x') \rangle_s$ self-consistently.

As we have seen before and here, the Einstein–Langevin equation is a dynamical equation governing the dissipative evolution of the gravitational field under the influence of the fluctuations of the quantum field, which, in the case of Hawking radiation from a black hole has a thermal character. From its form we can see that even for the quasi-static case under study the backreaction of Hawking radiation on the black hole spacetime has an innate dynamical nature. This is Hawking's quasi-stationary phase in the evolution of an evaporating black hole.

For the far field case making use of the explicit forms available for the noise and dissipation kernels, Campos and Hu [438, 708] proved the existence of a fluctuation–dissipation relation at all temperatures between the quantum fluctuations of the thermal radiance and the dissipation of the gravitational field. They also showed the formal equivalence of this method with linear response theory for lowest order perturbations of a near-equilibrium system, and how the response functions such as the contribution of the quantum scalar field to the thermal graviton polarization tensor can be derived. An important quantity not usually obtained in linear response theory, but of equal importance, manifest in the in-in or CTP stochastic approach is the noise term arising from the quantum and statistical fluctuations in the thermal field. The example given in this section shows that the backreaction described by the Einstein–Langevin equation is intrinsically a dynamical stochastic process. By comparison, traditional response theories can show how the system responds to an external drive, but take no account of how the bath dynamically backreacts on the system, much less the reverse, because the bath is characterized by a fixed parameter, the temperature or the chemical potential in a canonical ensemble, and does not possess any dynamical degrees of freedom. Thus linear response is not a suitable theory for backreaction considerations.

14.3.6 Comments on the Approach Used

As remarked earlier, except for the near-flat case, an analytic form of the Green function is not available. Even the Page approximation [488] which gives unexpectedly good results for the stress energy tensor has been shown to fail in the fluctuations of the energy density [201]. Thus using such an approximation for the noise kernel will give unreliable results for the Einstein–Langevin equation. If we confine ourselves to Page's approximation and derive the equation of motion without the stochastic term, namely, retaining only the zeroth order contribution, i.e., the first two terms in the expression for the CTP effective action in Eq. (14.22) recover York's semiclassical Einstein's equation. Of course our purpose is to go beyond semiclassical gravity, which means using the full CTP effective action to get to the stochastic gravity results.

Another point worth noting is that a non-local dissipation term arises from the fourth term in Eq. (14.22) in the CTP effective action which is absent in

York's treatment. This difference exists primarily due to the difference in the way backreaction is treated: York works with the equation of motion using iterative approximations while here we work with the effective action in the influence functional approach. In York's treatment, the Einstein tensor is computed to first order in perturbation theory, while $\langle T_{\mu\nu} \rangle$ on the right-hand side of the semiclassical Einstein equation is replaced by the zeroth order term. In the influence functional treatment the full effective action is computed to second order in perturbation, and hence includes the higher-order non-local terms.

The other important conceptual point that comes to light from this more complete approach is that related to the fluctuation–dissipation relation. In the quantum Brownian motion analog (see e.g. [168, 933, 171, 172] and references therein), the dissipation of the energy of the Brownian particle as it approaches equilibrium and the fluctuations at equilibrium are connected by the fluctuation–dissipation relation. Here the backreaction of quantum fields on black holes also consists of two forms – dissipation and fluctuation or noise – corresponding to the real and imaginary parts of the influence functional as embodied in the dissipation and noise kernels. A fluctuation–dissipation relation has been shown to exist for the near flat case by Campos and Hu [438, 708] and is expected to exist between the noise and dissipation kernels for the general case, as it is a categorical relation [168, 933, 171, 172, 243]. Martín and Verdaguer [183] have also proved the existence of a fluctuation–dissipation relation when the semiclassical background is a stationary spacetime and the quantum field is in thermal equilibrium. Their result was then extended to a conformal field in a conformally stationary background [188].

As discussed earlier the existence of a fluctuation–dissipation relation for the black hole case has been suggested by some authors previously [919, 920, 921, 922]. This relation and the relevant physical quantities contained therein, such as the black hole susceptibility function which characterizes the statistical mechanical and dynamical responses of a black hole interacting with its quantum field environment, will allow us to study the *nonequilibrium* thermodynamic properties of the black hole, one of several broadening issues we mentioned at the end of the last section worthy of further investigations.

There are limitations of a technical nature in the quasi-static case studied above. First, as mentioned earlier, there is no reliable approximation to the Schwarzschild thermal Green function to explicitly compute the noise and dissipation kernels. Second, there is a technical limitation in the following sense: although we have allowed for backreaction effects to modify the initial state in the sense that the temperature of the Hartle–Hawking state gets affected by the backreaction, our analysis is essentially confined to a Hartle–Hawking thermal state of the field. It does not directly extend to a more general class of states, for example, to the case where the initial state of the field is in the Unruh vacuum. To study the dynamics of a radiating black hole and the fluctuations of its event horizon under the influence of a quantum field, i.e., the radiation it emits we need

to consider a different model and use a different approach. This is discussed in the next section.

14.4 Metric Fluctuations of an Evaporating Black Hole

The semiclassical gravity level of description of particle production and black hole evaporation is believed to be valid before the Planck scale is reached [775, 781]. However, semiclassical gravity is a mean field description which neglects the fluctuations of the spacetime geometry [47, 49, 128]. A number of studies since the 90s have suggested the existence of large fluctuations near the black hole horizon, with characteristic time-scales much shorter than the black hole evaporation time. For example, Casher et al. consider the fluctuations of the horizon induced by the "atmosphere" of high angular momentum particles near the horizon, while Sorkin and Sudarsky [935, 936, 937] calculate the fluctuations of the shape of the horizon induced by the quantum field fluctuations under a Newtonian approximation – a relativistic generalization was later provided by Marolf [938]. Both groups of authors came to the conclusion that horizon fluctuations become large at scales much longer than the Planck length (Mazur and Mottola [939] even claim instabilities exist near the event horizon.) On the other hand Ford and Svaiter [693] present results contrary to this claim. All sides considered, there is one thing in common: any inquiry into the fate of a radiating black hole needs to factor in the importance of metric fluctuations near the event horizon, one of the key points made by Hawking [78] when he discovered that black holes radiate.

A comment on these earlier works is in order: the authors who claim large fluctuations near the horizon either consider states which are singular on the horizon (such as the Boulware vacuum for Schwarzschild spacetime), or they compute fluctuations with respect to those states. What causes these huge fluctuations are not clear: are they physical and generic, or are they mere artifacts arising from the consideration of states that are singular on the horizon? By contrast, if one considers states regular on the horizon, as Wu and Ford [940] did, the fluctuations are found to be small even when integrated over a time of the order of the evaporation time. These contradictory claims and the fact that most claims on black hole horizon fluctuations were based on qualitative arguments and/or semi-quantitative estimates indicate that a more rigorous and quantitative description is needed. Stochastic gravity theory serves this purpose.

Such a program of research was pursued by Hu and Roura [941, 490]. In contrast to the claims made before, they find that even for states regular on the horizon the accumulated fluctuations become significant by the time the black hole mass has changed substantially, well before reaching the Planckian regime. This result is different from those obtained in prior studies,[5] but in

[5] The apparent difference from the conclusions drawn in the earlier work of Hu et al. [911], which was also based on stochastic gravity, will be explained later.

agreement with earlier work by Bekenstein [776]. We begin with a description of the evolution of the mean geometry.

14.4.1 Mean Geometry Evolution

Backreaction of the Hawking radiation emitted by the black hole on the dynamics of spacetime geometry has been studied in some detail for spherically symmetric black holes by Bardeen [775] and Massar [781]. For a general spherically symmetric metric there always exists a system of coordinates in which it takes the form

$$ds^2 = -e^{2\psi(v,r)}\left(1 - \frac{2m(v,r)}{r}\right)dv^2 + 2e^{\psi(v,r)}dvdr + r^2\left(d\theta^2 + \sin^2\theta d\varphi^2\right). \tag{14.35}$$

In general this metric exhibits an *apparent horizon*, where the expansion of the outgoing radial null geodesics vanish. It separates regions with positive and negative expansion for those geodesics, at those radii that correspond to (odd degree) zeroes of the vv metric component. We mark the location of the apparent horizon by $r_{AH}(v) = 2M(v)$, where $M(v)$ satisfies the equation $2m(2M(v), v) = 2M(v)$.

The non-zero components of the semiclassical Einstein equation associated with the metric in equation (14.35) become

$$\frac{\partial m}{\partial v} = 4\pi r^2 \langle T^r_v \rangle, \tag{14.36}$$

$$\frac{\partial m}{\partial r} = -4\pi r^2 \langle T^v_v \rangle, \tag{14.37}$$

$$\frac{\partial \psi}{\partial r} = 4\pi r \langle T_{rr} \rangle, \tag{14.38}$$

where in the above and henceforth we use $\langle T_{\mu\nu} \rangle$ to denote the renormalized or regularized vacuum expectation value of the stress-energy tensor $\langle \hat{T}^{\rm R}_{\mu\nu}[g] \rangle$ and employ Planckian units (with $G_{\rm N} = \hbar = c = 1$).

Solving equations (14.36)–(14.38) is not easy. However, one can introduce a useful adiabatic approximation in the regime where the mass of the black hole is much larger than the Planck mass, which is in any case a necessary condition for the semiclassical treatment to be valid. What this entails is that when $M \gg 1$ (remember we are using Planckian units) for each value of v one can simply substitute $\langle T_{\mu\nu} \rangle$ by its "parametric value" – by this we mean the expectation value of the stress-energy tensor of the quantum field in a Schwarzschild black hole with a mass corresponding to $M(v)$ evaluated at that value of v. This is in contrast to its dynamical value, which should be determined by solving self-consistently the semiclassical Einstein equation for the spacetime metric and the equations of motion for the quantum matter fields. This kind of approximation

introduces errors of higher order in $L_H \equiv B/M^2$ (B is a dimensionless parameter that depends on the number of massless fields and their spins and accounts for their corresponding grey-body factors; the errors have been estimated to be of order 10^{-4} by Page [83]), which is very small for black holes well above Planckian scales. These errors are due to the fact that $M(v)$ is not constant and that, even for a constant $M(v)$, the resulting static geometry is not exactly Schwarzschild because the vacuum polarization of the quantum fields gives rise to a non-vanishing $\langle \hat{T}^{\rm R}_{\mu\nu}[g] \rangle$ [793].

The expectation value of the stress-energy tensor for Schwarzschild spacetime has been found to correspond to a thermal flux of radiation (with $\langle T^r_v \rangle = L_H/(4\pi r^2)$) for large radii and is of order L_H near the horizon [762, 488, 763, 764, 779]. This shows the consistency of the adiabatic approximation for $L_H \ll 1$: the right-hand side of equations (14.36)–(14.38) contains terms of order L_H and higher, so that the derivatives of $m(v, r)$ and $\psi(v, r)$ are indeed small. We note that the natural quantum state for a black hole formed by gravitational collapse is the Unruh state, which corresponds to the absence of incoming radiation far from the horizon. The expectation value of the stress-energy tensor operator for that state is finite on the future horizon of the black hole,

One can use the v component of the stress-energy conservation equation

$$\frac{\partial \left(r^2 \langle T^r_v \rangle\right)}{\partial r} + r^2 \frac{\partial \langle T^v_v \rangle}{\partial v} = 0, \tag{14.39}$$

to relate the $\langle T^r_v \rangle$ components on the horizon and far from it. Integrating equation (14.39) radially, one gets

$$(r^2 \langle T^r_v \rangle)(r = 2M(v), v) = (r^2 \langle T^r_v \rangle)(r \approx 6M(v), v) + \mathcal{O}(L_H^2), \tag{14.40}$$

where we consider a radius sufficiently far from the horizon, but not arbitrarily far (i.e. $2M(v) \ll r \ll M(v)/L_H$). Hence, equation (14.40) relates the positive energy flux radiated away far from the horizon and the negative energy flux crossing the horizon. Taking into account this connection between energy fluxes and evaluating equation (14.36) on the apparent horizon, we arrive at the equation governing the evolution of its size:

$$\frac{dM}{dv} = -\frac{B}{M^2}. \tag{14.41}$$

Unless $M(v)$ is constant, the event horizon and the apparent horizon do not coincide. However, in the adiabatic regime their radii are related, differing by a quantity of higher order in L_H: $r_{EH}(v) = r_{AH}(v)\,(1 + \mathcal{O}(L_H))$.

14.4.2 Spherically Symmetric Induced Fluctuations

We now consider metric fluctuations around a background metric $g_{\mu\nu}$ that corresponds to a given solution of semiclassical gravity. Their dynamics is governed by the Einstein–Langevin equation (10.14)

$$G^{(1)}_{\mu\nu}\left[g+h\right] = 8\pi G_{\rm N}\left(\left\langle \hat{T}^{(1)\rm R}_{\mu\nu}[g+h]\right\rangle + \xi_{\mu\nu}\left[g\right]\right), \tag{14.42}$$

where the superscript (1) indicates that only the terms linear in the metric perturbations should be considered, and the terms quadratic in the curvature are here included in the expression of the renormalized stress-energy tensor. Here $\xi_{\mu\nu}$ is a Gaussian stochastic source with vanishing expectation value and correlation function $\langle\xi_{\mu\nu}(x)\xi_{\alpha'\beta'}(x')\rangle_s = (1/2)\langle\{\hat{t}_{\mu\nu}(x),\hat{t}_{\alpha'\beta'}(x')\}\rangle$ ($\hat{t}_{\mu\nu}\equiv\hat{T}_{\mu\nu}-\langle\hat{T}_{\mu\nu}\rangle$), *i.e.*, the noise kernel $N_{\mu\nu\alpha'\beta'}(x,x')$ defined in (10.11).

As explained in Chapter 10 the symmetrized two-point function consists of two contributions: *intrinsic* and *induced* fluctuations. The intrinsic fluctuations are a consequence of the quantum width of the initial state of the metric perturbations. They are obtained in stochastic gravity by averaging over the initial conditions for the solutions of the homogeneous part of equation (14.42) distributed according to the reduced Wigner function associated with the initial quantum state of the metric perturbations. On the other hand, the induced fluctuations are due to the quantum fluctuations of the matter fields interacting with the metric perturbations. They are obtained by solving the Einstein–Langevin equation using a retarded propagator with vanishing initial conditions.

In this section we study the spherically symmetric sector (i.e., the monopole contribution, which corresponds to $l=0$, in a multipole expansion in terms of spherical harmonics $Y_{lm}(\theta,\phi)$) of metric fluctuations for an evaporating black hole. Restricting one's attention to the spherically symmetric sector of metric fluctuations necessarily implies a partial description of the fluctuations because, contrary to the case for semiclassical gravity solutions, even if one starts with spherically symmetric initial conditions, the stress-energy tensor fluctuations will induce fluctuations involving higher multipoles. Thus, the multipole structure of the fluctuations is far richer than that of spherically symmetric semiclassical gravity solutions, but this also means that obtaining a complete solution (including all multipoles) for fluctuations rather than the mean value is much more difficult.

For spherically symmetric fluctuations only induced fluctuations are possible. The fact that intrinsic fluctuations cannot exist can be clearly seen if one neglects vacuum polarization effects, since Birkhoff's theorem forbids the existence of spherically symmetric free metric perturbations in the exterior vacuum region of a spherically symmetric black hole that keep the ADM mass constant.[6] Even

[6] This fact should sound an alarm in the approach of York and Schmekel [942] to the black hole fluctuation problem. The degrees of freedom corresponding to spherically symmetric

when vacuum polarization effects are included, spherically symmetric perturbations, characterized by $m(v,r)$ and $\psi(v,r)$, are not independent degrees of freedom. This follows from equations (14.36)–(14.38), which can be regarded as constraint equations.

Considering only spherically symmetric fluctuations is a simplification, but, even so, doing it in the stochastic gravity approach gives more accurate results than that of the two-dimensional dilaton gravity models resulting from simple dimensional reduction [943, 944, 945]. This is because the solutions of the Einstein–Langevin equation are projected only at the end, rather than considering only the contribution of the s-wave modes to the classical action for both the metric and the matter fields from the beginning. Hence, an infinite number of modes for the matter fields with $l \neq 0$ contribute to the $l = 0$ projection of the noise kernel. If dimensional reduction had been imposed from the start, as is done in [946, 947, 948] as well as in studies of two-dimensional dilaton gravity models, then only the s-wave modes for each matter field would be included in the noise kernel leading to greater discrepancies.

The Einstein–Langevin equation for the spherically symmetric sector of metric perturbations can be obtained by considering linear perturbations of $m(v,r)$ and $\psi(v,r)$, projecting the stochastic source that accounts for the stress-energy tensor fluctuations to the $l = 0$ sector, and adding it to the right-hand side of equations (14.36)–(14.38). We will focus our attention on the equation for the evolution of $\eta(v,r)$, the perturbation of $m(v,r)$,

$$\frac{\partial(m+\eta)}{\partial v} = -\frac{B}{(m+\eta)^2} + 4\pi r^2 \xi_v^r + \mathcal{O}\left(L_H^2\right), \tag{14.43}$$

which reduces, after neglecting terms of order L_H^2 or higher, to the following equation to linear order in η:

$$\frac{\partial \eta}{\partial v} = \frac{2B}{m^3}\eta + 4\pi r^2 \xi_v^r. \tag{14.44}$$

It is important to emphasize that in equation (14.43) we assumed that the change in time of $\eta(v,r)$ is sufficiently slow so that the adiabatic approximation used to obtain the mean evolution of $m(v,r)$ can also be applied to the perturbed quantity $m(v,r) + \eta(v,r)$. This is guaranteed as long as the term corresponding to the stochastic source is of order L_H or higher, a point which will be discussed below.

A more serious issue raised by Hu and Roura [490] is that in most previous investigations, such as those of Bekenstein [776] and Wu and Ford [940], of the problem of metric fluctuations driven by quantum matter field fluctuations of states regular on the horizon (as far as the expectation value of the stress tensor

perturbations are constrained by the Hamiltonian and momentum constraints at both the classical and quantum levels. Therefore, they will not exhibit quantum fluctuations unless they are coupled to some quantum matter field.

is concerned), the authors assumed the existence of correlations between the outgoing energy flux far from the horizon and a negative energy flux crossing the horizon. (See, however, Parentani [947, 948], where those correlators were shown to vanish in an effectively two-dimensional model.) In semiclassical gravity, using energy conservation arguments, *such correlations have been confirmed for the expectation value of the energy fluxes*, provided that the mass of the black hole is much larger than the Planck mass. A more careful analysis of Hu and Roura [490] shows that *no such simple connection exists for energy flux fluctuations.* It also reveals that the fluctuations on the horizon are in fact divergent. This requires that one modify the classical picture of the event horizon from a sharply defined three-dimensional hypersurface to that possessing a finite width, i.e., a fluctuating geometry. One needs to find an appropriate way to probe the metric fluctuations near the horizon and extract physically meaningful information. (E.g., the use of different types of smearing functions in evaluating fluctuations is described in an appendix of [490].) It also testifies to the necessity of a complete reexamination of all previous cases afresh. One can draw the conclusion that obtaining expressions for the noise kernel near the horizon (rather than just relying on the expectation value of the stress-energy tensor) is unavoidable for the consideration of fluctuations and backreaction in black hole spacetimes.

Having registered this cautionary note, Hu and Roura show that, contrary to the case of the *mean flux*, there is no relation between the *fluctuations* of the fluxes. They then state the essential elements required in understanding the operational meaning of metric fluctuations. Their argument goes as follows.

Thesis: assuming that there is a relation between fluctuations. Since the generation of Hawking radiation is especially sensitive to what happens near the horizon, we can concentrate on the metric perturbations near the horizon and consider $\eta(v) = \eta(v, 2M(v))$. This means that possible effects on the Hawking radiation due to the fluctuations of the potential barrier for the radial mode functions will be amiss. Assuming that the fluctuations of the energy flux crossing the horizon and those far from it are exactly correlated, from Eq. (14.44) we have

$$\frac{d\eta(v)}{dv} = \frac{2B}{M^3(v)}\eta(v) + \xi(v), \tag{14.45}$$

where $\xi(v) \equiv 4\pi(r^2\,\xi^r_v)(v, r \approx 6M(v))$.

The correlation function for the spherically symmetric fluctuation $\xi(v)$ is determined by the integral over the whole solid angle of the $N^{r\ r'}_{v\ v'}$ component of the noise kernel, which is given by $(1/2)\langle\{\hat{t}^r_v(x), \hat{t}^{r'}_{v'}(x')\}\rangle$. The $l = 0$ fluctuations of the energy flux of Hawking radiation far from a black hole formed by gravitational collapse, characterized also by $(1/2)\langle\{\hat{t}^r_v(x), \hat{t}^{r'}_{v'}(x')\}\rangle$ averaged over the whole solid angle, have been studied by Wu and Ford [940]. Its main features are a correlation time of order M and a characteristic fluctuation amplitude of order ϵ_0/M^4 (this is the result of smearing the stress-energy tensor two-point

function, which diverges in the coincidence limit, over a period of time of the order of the correlation time). The order of magnitude of ϵ_0 has been estimated to lie between $0.1B$ and B [776, 940]. For simplicity, we will consider quantities smeared over a time of order M. We can then introduce the Markovian approximation $(\epsilon_0/M^3(v))\delta(v-v')$, which coarse-grains the information on features corresponding to time scales shorter than the correlation time M. Under those conditions $r^2\xi^r_v$ is of order $1/M^2$ and thus the adiabatic approximation made when deriving equation (14.43) is justified.

The stochastic equation (14.45) for η can be solved in the usual way and the correlation function for $\eta(v)$ can then be computed. Alternatively, one can obtain an equation for $\langle\eta^2(v)\rangle_s$ by first multiplying Eq. (14.45) by $\eta(v)$ and then taking the expectation value. This brings out a term $\langle\eta(v)\xi(v)\rangle_s$ on the right-hand side. For delta-correlated noise (the Stratonovich prescription is the appropriate one here), it is equal to one half the time-dependent coefficient multiplying the delta function $\delta(v-v')$ in the correlator $\langle\xi(v)\xi(v')\rangle_s$, which is given by $\epsilon_0/M^3(v)$ in our case. Finally, changing from the v coordinate to the mass function $M(v)$ for the background solution, we obtain

$$\frac{d}{dM}\langle\eta^2(M)\rangle_s = -\frac{4}{M}\langle\eta^2(M)\rangle_s - \frac{(\epsilon_0/B)}{M}. \tag{14.46}$$

The solutions of this equation are given by

$$\langle\eta^2(M)\rangle_s = \langle\eta^2(M_0)\rangle_s \left(\frac{M_0}{M}\right)^4 + \frac{\epsilon_0}{4B}\left[\left(\frac{M_0}{M}\right)^4 - 1\right]. \tag{14.47}$$

Provided that the fluctuations at the initial time corresponding to $M = M_0$ are negligible (much smaller than $\sqrt{\epsilon_0/4B} \sim 1$), the fluctuations become comparable to the background solution when $M \sim M_0^{2/3}$. Note that fluctuations of the horizon radius of order one in Planckian units do not correspond to Planck scale physics because near the horizon $\Delta R = r - 2M$ corresponds to a physical distance $L \sim \sqrt{M\,\Delta R}$, as can be seen from the line element for Schwarzschild, $ds^2 = -(1-2M/r)dt^2 + (1-2M/r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2)$, by considering pairs of points at constant t. So $\Delta R \sim 1$ corresponds to $L \sim \sqrt{M}$, whereas a physical distance of order one is associated with $\Delta R \sim 1/M$, which corresponds to an area change of order one for spheres with those radii. One can, therefore, have initial fluctuations of the horizon radius of order one for physical distances well above the Planck length for a black hole with a mass much larger than the Planck mass. One expects that the fluctuations for states that are regular on the horizon correspond to physical distances not much larger than the Planck length, so that the horizon radius fluctuations would be much smaller than those for sufficiently large black hole masses. Nevertheless, that may not be the case when dealing with states which are singular on the horizon, with estimated fluctuations of order $M^{1/3}$ or even $\sqrt{M}$ [949, 938, 939].

The result by Hu and Roura [490] for the growth of the fluctuations of the size of the black hole horizon agrees with the result obtained by Bekenstein [776] and implies that, for a sufficiently massive black hole (with a few solar masses or a supermassive black hole), the fluctuations become important before the Planckian regime is reached.

This growth of the fluctuations found by Bekenstein and confirmed by the use of the Einstein–Langevin equation seems to be in conflict with the estimate given by Wu and Ford [940]. According to their estimate, the accumulated mass fluctuations over a period of the order of the black hole evaporation time ($\Delta t \sim M_0^3$) would be of the order of the Planck mass. The discrepancy is due to the fact that the first term on the right-hand side of equation (14.45), which corresponds to the perturbed expectation value $\langle \hat{T}_{\mu\nu}^{(1)\mathrm{R}}[g+h] \rangle$ in Eq. (14.42), was not taken into account by Wu and Ford [940]. The larger growth obtained here is a consequence of the secular effect of that term, which builds up in time (slowly at first, during most of the evaporation time, and becoming more significant at late times when the mass has changed substantially) and reflects the unstable nature of the background solution for an evaporating black hole.

As for the relation between the results reported here and earlier results of Hu, Raval and Sinha [911], since both adopted the stochastic gravity framework and performed their analysis based on the Einstein–Langevin equation there should not be any discrepancy. Indeed the claim in [911] was based on a qualitative argument that focused on the dynamics of the stochastic source alone. The stochastic component of course rides on the dynamics of the mean. If one adds in the consideration that the perturbations around the mean are unstable for an evaporating black hole, their results agree with Hu and Roura [490].

All this can be qualitatively understood in a simple way as follows. Consider an evaporating black hole with initial mass M_0 and suppose that the initial mass is perturbed by an amount $\delta M_0 = 1$. The mean evolution for the perturbed black hole (without taking into account any fluctuations) leads to a mass perturbation that grows like $\delta M = (M_0/M)^2\, \delta M_0 = (M_0/M)^2$, so that it becomes comparable to the unperturbed mass M when $M \sim M_0^{2/3}$, which coincides with the result obtained above. Such a coincidence has a simple explanation: the fluctuations of the Hawking flux slowly accumulated during most of the evaporating time, which are of the order of the Planck mass, as found by Wu and Ford, give a dispersion of that order for the mass distribution at the time when the instability of the small perturbations around the background solution start to become significant.

Antithesis: when no such relation exists and its consequences. For conformal fields in two-dimensional spacetimes, Hu and Roura [490] showed that the correlations between the energy flux crossing the horizon and the flux far from it vanish. The correlation function for the outgoing and incoming null energy fluxes in an effectively two-dimensional model was explicitly computed by Parentani [947, 948] and it was also found to vanish. On the other hand, in four dimensions

the correlation function does not vanish in general and correlations between outgoing and incoming fluxes do exist near the horizon (at least partially).

For black hole masses much larger than the Planck mass, one can use the adiabatic approximation for the background mean evolution. Therefore, to lowest order in L_H one can compute the fluctuations of the stress-energy tensor in Schwarzschild spacetime. In Schwarzschild, the amplitude of the fluctuations of $r^2\langle T^r_v\rangle$ far from the horizon is of order $1/M^2 (= M^2/M^4)$ when smearing over a correlation time of order M, which one can estimate for a hot thermal plasma in flat space [438, 708]. See also Wu and Ford [940] for a computation of the fluctuations of $r^2\langle T^r_v\rangle$ far from the horizon. The amplitude of the fluctuations of $r^2\langle T^r_v\rangle$ is thus of the same order as its expectation value. However, their derivatives with respect to v are rather different: since the characteristic variation times for the expectation value and the fluctuations are M^3 and M respectively, $\partial(r^2\langle T^r_v\rangle)/\partial v$ is of order $1/M^5$ whereas $\partial(r^2\xi^r_v)/\partial v$ is of order $1/M^3$. This implies an additional contribution of order L_H from the second term in Eq. (14.39) if one radially integrates the same equation applied to stress-energy tensor fluctuations (the stochastic source in the Einstein–Langevin equation). Hence, in contrast to the case of the mean value, the contribution from the second term in Eq. (14.39) cannot be neglected when radially integrating since it is of the same order as the contributions from the first term, and one can no longer obtain a simple relation between the outgoing energy flux far from the horizon and the energy flux crossing the horizon.

What then? Without the convenience of this relation, which almost all researchers before had taken for granted, to get a more precise description we need to compute the noise kernel near the horizon. However, as shown by Phillips and Hu [201] earlier when examining the coincidence limit of the noise kernel, and confirmed by Hu and Roura [490] using smearing functions, the noise kernel smeared over the horizon is divergent, and so are the induced metric fluctuations. Hence, one cannot study the fluctuations of the horizon as a three-dimensional hypersurface for each realization of the stochastic source because the amplitude of the fluctuations is infinite, even when restricting one's attention to the $l = 0$ sector. Instead, one should regard the horizon as possessing a finite effective width due to quantum fluctuations. In order to characterize its width one must find a sensible way of probing the metric fluctuations near the horizon and extracting physically meaningful information, such as their effect on the Hawking radiation emitted by the black hole. How to probe and quantify metric fluctuations is a fundamental issue at the root of all fluctuation phenomena.

14.4.3 Summary and Prospects

The work of Hu and Roura [490] based on the stochastic gravity program found that the spherically symmetric fluctuations of the event horizon of an evaporating black hole become important at late times, even comparable to its mean value

when $M \sim M_0^{2/3}$, where M_0 is the mass of the black hole at some initial time when the fluctuations of the horizon radius are much smaller than the Planck length (remember that for large black hole masses this can still correspond to physical distances much larger than the Planck length). This is consistent with the result previously obtained by Bekenstein [776].

It is important to realize that for a sufficiently massive black hole, the fluctuations become significant well before the Planckian regime is reached. More specifically, for a solar mass black hole they become comparable to the mean value when the black hole radius is of the order of 10 nm, whereas for a supermassive black hole with $M \sim 10^7 M_\odot$, that happens when the radius reaches a size of the order of 1 mm. One expects that in those circumstances the low-energy effective field theory approach of stochastic gravity should provide a reliable description.

Due to the nonlinear nature of the backreaction equations, such as Eq. (14.43), the fact that the fluctuations of the horizon size can grow and become comparable to the mean value implies non-negligible corrections to the dynamics of the mean value itself. This can be seen by expanding Eq. (14.43) (evaluated on the horizon) in powers of η and taking the expectation value. Through order η^2 we get

$$\begin{aligned}\frac{d(M(v)+\langle\eta(v)\rangle_s)}{dv} &= -\left\langle \frac{B}{(M(v)+\eta(v))^2}\right\rangle_s \\ &= -\frac{B}{M^2(v)}\left[1-\frac{2}{M(v)}\langle\eta(v)\rangle_s+\frac{3}{M^2(v)}\langle\eta^2(v)\rangle_s+\mathcal{O}\left(\frac{\eta^3}{M^3}\right)\right].\end{aligned} \tag{14.48}$$

When the fluctuations become comparable to the mass itself, the third term (and higher-order terms) on the right-hand side is no longer negligible and we get non-trivial corrections to Eq. (14.41) for the dynamics of the mean value. These corrections can be interpreted as higher-order radiative corrections to semiclassical gravity that include the effects of metric fluctuations on the evolution of the mean value.

Finally we remark on the relation of this finding with earlier results. Do significant deviations in the evolution of the mean mentioned above invalidate the commonly accepted results of Bardeen [775] and Massar [781] based on semiclassical gravity? First, those deviations start to become significant only after a period of the order of the evaporation time when the mass of the black hole has decreased substantially. Second, since fluctuations were not considered in those works, a direct comparison cannot be established. Nevertheless, we can compare the average of the fluctuating ensemble. Doing so exhibits an evolution that deviates significantly when the fluctuations become important. However, if one considers a single member of the ensemble at that time, its evolution will be accurately described by the corresponding semiclassical gravity solution until the fluctuations around that particular solution become important again, after

a period of the order of the evaporation time associated with the new initial value of the mass at that time. This problem of metric fluctuations in black hole spacetimes is probably the most telling example testifying to the necessity of including fluctuations beyond the mean in a correct description of gravitational systems.

14.5 Work on Metric Fluctuations without Backreaction

In closing, we mention some work on metric fluctuations where no backreaction is considered. Barrabès et al. [950, 951] when considering the propagation of null rays and massless fields in a black hole fluctuating geometry have shown that the stochastic nature of the metric leads to a modified dispersion relation which helps to confront the trans-Planckian frequency problem. However, in this case the stochastic noise is put in by hand. It does not arise from coarse-graining a quantum field as done in the stochastic gravity program. Neither was there any backreaction. It would be interesting to explore how a stochastic black hole metric, obtained from a solution to the Einstein–Langevin equation, hence fully incorporating backreaction and fluctuations, would fare in the trans-Planckian problem.

As mentioned earlier, Ford and his collaborators [693, 207, 940] have explored the issue of metric fluctuations in considerable detail. In particular they have studied the fluctuations of the black hole horizon induced by metric fluctuations. However, the fluctuations they considered are in the context of a fixed background, not involving any backreaction. Thompson and Ford [952] discuss the possible role of quantum horizon fluctuations on black hole radiance, especially whether they can invalidate Hawkings analysis based upon transplanckian modes. They are particularly concerned with enhanced fluctuations produced by gravitons or matter fields in squeezed vacuum states sent into the black hole after the collapse process, because this allows for the possibility of increasing the fluctuations well above the vacuum level. They find that these enhanced fluctuations could significantly alter stimulated emission, but have little effect upon the spontaneous emission. They concluded that at least in this regard the thermal character of the Hawking radiation is robust.

Another work on metric fluctuations with no backreaction is that of Hu and Shiokawa [953], who studied effects associated with electromagnetic wave propagation in a Robertson–Walker universe and the Schwarzschild spacetime with a small amount of given metric stochasticity . They find that time-independent randomness can decrease the total luminosity of Hawking radiation due to multiple scattering of waves outside the black hole and gives rise to event horizon fluctuations and fluctuations in the Hawking temperature. The stochasticity in a background metric in their work is assumed rather than derived (as induced by quantum field fluctuations). It would be of interest to compare their results with those incorporating backreaction, as in stochastic gravity, so one can get a

sense of the different sources of stochasticity and their weights (see, e.g., [479] for a list of possible sources of stochasticity.)

In a subsequent paper Shiokawa [481] showed that the scalar and spinor waves in a stochastic spacetime behave similarly to the electrons in a disordered system. Viewing this as a quantum transport problem, he expressed the conductance and its fluctuations in terms of a nonlinear sigma model in the closed-time-path formalism and showed that the conductance fluctuations are universal, independent of the volume of the stochastic region and the amount of stochasticity. This result may have significant importance in characterizing the mesoscopic behavior of spacetimes resting between the semiclassical and the quantum regimes.

Part V

Quantum Curvature Fluctuations in de Sitter Spacetime

15

Stress-Energy Tensor Fluctuations in de Sitter Space

The geometry of most inflationary models as well as the late time behavior of our expanding universe are well approximated by the de Sitter geometry. For this reason quantum metric perturbations around de Sitter and near-de Sitter spacetimes have been extensively studied in the context of inflationary models. The standard analysis is based on the linearized computation of the two-point function of the metric perturbations [954, 734, 753]. But the need for matter loop corrections was emphasized by Weinberg [427], who pointed out also the importance of testing perturbation theory in cosmology given the success of the tree-level approximation; see also Senatore and Zaldarriaga [567]. A series of works including matter loop corrections in different models have been performed using different approaches and models [955, 956, 957] including stochastic gravity [958, 203].

However, certain aspects such as the significance of possible large infrared effects, have not been completely settled yet. The standard theoretical framework also assumes that the background dynamics of de Sitter and near-de Sitter spacetimes is not affected by large quantum corrections. However, there is also a number of studies that claim that one-loop corrections could give a significant backreaction in inflationary models [959, 960, 961, 576, 577]. It has also been argued that higher-order radiative corrections from graviton two-loops could give rise to a secular screening of the cosmological constant for pure gravity in de Sitter spacetime [962, 963]. Such nonlinear effects are dominated by infrared modes which are copiously generated by the exponential expansion and should be fairly insensitive to the short-distance behavior of quantum gravity. However, whether these large effects really take place is still uncertain, mainly due to the gauge dependence of the results [964, 965, 966, 508, 575].

The existence of large quantum nonlinear effects like those mentioned above could lead to a breaking of de Sitter invariance, as suggested by Tsamis and Woodard [962, 963] or by Polyakov [967]. In order to gain insight into this

possibility, it seems convenient to study quantum metric perturbations around de Sitter spacetime keeping de Sitter invariance always manifest. This can be achieved with the formalism of the de Sitter-invariant bitensors introduced by Allen and Jacobson [476]. The tree-level graviton two-point function has been computed using this formalism by different authors such as Allen and Turyn [968], Antoniadis and Mottola [969] and Higuchi and Kouris [970]. A conclusion that can be drawn from their results is that the two-point function of the linearized Riemann tensor $R_{ab}{}^{cd}$, which is gauge-invariant around de Sitter spacetime, decays like the fourth power of the physical distance between the two points; see Kouris [971].

We wish to go beyond the linear level and study the effects of matter loops on the quantum metric fluctuations. However, a first step in this direction is to study the stress-energy tensor fluctuations of the matter fields described by the noise kernel. Calculating the noise kernel in de Sitter spacetime, in a de Sitter-invariant vacuum, and with the formalism of de Sitter-invariant bitensors, is the main goal of this chapter. In subsequent chapters quantum gravitational field correlations, including one-loop corrections from matter fields, in de Sitter spacetime are studied.

We start by reviewing in Section 15.1 the basic geometric properties of de Sitter spacetime and the invariant bitensors that will be used in the rest of the chapter and in later chapters. These tools are employed in Section 15.2 to write the noise kernel for spacelike separated points in de Sitter-invariant form, and explicit expressions for the case of a free minimally coupled scalar field are derived. Furthermore, closed results in terms of elementary functions can be given for the particular cases of small masses, vanishing mass, and large separations. A massless limit discontinuity is found and is discussed in some detail. In Section 15.3 an alternative analysis of the noise kernel based on the approximate solutions of the field modes is given, which also helps to clarify the above discontinuity. Finally we discuss the implications of our results for the quantum metric fluctuations around de Sitter spacetime in Section 15.4.

15.1 De Sitter Geometry and Invariant Bitensors

The D-dimensional de Sitter spacetime, dS_D, may be defined as a submanifold embedded in $(D+1)$-dimensional Minkowski spacetime, whose points satisfy the equation

$$\eta_{AB} X^A(x) X^B(x) = H^{-2}, \tag{15.1}$$

where H is a parameter with dimensions of mass or inverse length (we use $\hbar = c = 1$), η_{AB} the Minkowski metric, and $X^A(x)$ is the position vector in Minkowski spacetime of the point x. The hyperboloid is invariant under Lorentz transformations $X^A = \Lambda^A_B X^B$ and these define the isometries σ of dS_D: $X^A(\sigma(x)) = \Lambda^A_B X^B(x)$. Thus, the isometries of de Sitter form the Lorentz group

$O(D,1)$. Another isometry that will also play a role is the antipodal isometry $\alpha(x)$ defined by $X^A(\alpha(x)) = -X^A(x)$.

One may introduce spatially flat coordinates by labelling the points in dS_D with the coordinates $\{t, x^i\}$, with $i = 1, \ldots, D-1$, defined by

$$\begin{aligned} X^0 &= \frac{1}{H}\sinh(Ht) + \frac{1}{2}He^{Ht}\delta_{ij}x^i x^j, \\ X^i &= x^i e^{Ht}, \\ X^D &= \frac{1}{H}\cosh(Ht) - \frac{1}{2}He^{Ht}\delta_{ij}x^i x^j, \end{aligned} \tag{15.2}$$

then the line element induced on the hypersurface takes the form

$$ds^2 = \eta_{AB}dX^A(x)dX^B(x) = -dt^2 + e^{2Ht}\delta_{ij}dx^i dx^j, \tag{15.3}$$

so that the hypersurfaces Σ_t of constant t are flat. These coordinates do not cover the whole de Sitter spacetime (they cover the Poincaré patch), but it is not difficult to introduce coordinates that do and which, moreover, are useful to construct a conformal diagram for de Sitter spacetime. To see this we first rewrite the previous metric using the conformal time $ad\eta = dt$, where a is the scale factor $a = \exp(Ht)$, and spherical coordinates,

$$ds^2 = \frac{1}{H^2\eta^2}\left(-d\eta^2 + dr^2 + r^2\left(d\theta^2 + \sin^2\theta d\varphi^2\right)\right), \tag{15.4}$$

where $-\infty < \eta < 0$, $0 \le r < \infty$, $0 \le \theta < \pi$ and $0 \le \varphi < 2\pi$. We may then introduce new time and space coordinates $(\tilde{\eta}, \chi)$ which span a finite range as [745],

$$\eta = \frac{\sin\tilde{\eta}}{\cos\tilde{\eta} + \cos\chi}, \quad r = \frac{\sin\chi}{\cos\tilde{\eta} + \cos\chi}. \tag{15.5}$$

where $-\pi < \tilde{\eta} < 0$ and $0 \le \chi < \pi$, and the metric now reads

$$ds^2 = \frac{1}{H^2\sin^2\tilde{\eta}}\left(-d\tilde{\eta}^2 + d\chi^2 + \sin^2\chi\left(d\theta^2 + \sin^2\theta d\varphi^2\right)\right). \tag{15.6}$$

the section of $\tilde{\eta} =$ const are three spheres. These coordinates cover the entire de Sitter spacetime. They represent a closed universe which contracts for $-\pi < \tilde{\eta} < -\pi/2$ and expands for $-\pi/2 < \tilde{\eta} < 0$. The de Sitter conformal diagram, for the conformal metric $(H^2\sin^2\tilde{\eta})\,ds^2$, may be represented in terms of these coordinates $(\chi, \tilde{\eta})$, each point of the conformal diagram represents a S^2 sphere of radius $\sin\chi$.

The metric of de Sitter spacetime is a solution of Einstein equation in vacuum with a cosmological constant

$$G_{ab} + \Lambda g_{ab} = 0, \tag{15.7}$$

where the cosmological constant is $\Lambda = [(D-1)(D-2)/2]\,H^2$. The metric can be interpreted as a cosmological FLRW metric for a perfect fluid whose pressure

equals the constant energy density (vacuum energy). Because the energy density is constant any hypersurface is an hypersurface of constant energy density.

15.1.1 De Sitter Invariant Bitensors

To introduce the formalism of the de Sitter invariant bitensors it is convenient to define the basic biscalar

$$Z(x,x') = H^2 \eta_{AB} X^A(x) X^B(x'), \tag{15.8}$$

where $x, x' \in dS_D$. Note that Z is invariant under the de Sitter group; if σ is an isometry, then $Z(\sigma(x), \sigma(x')) = Z(x,x')$ and under the antipodal transformation $Z(\alpha(x), x') = -Z(x,x')$. This biscalar is related to the Minkowskian distance, defined by

$$d^2(x,x') = \eta_{AB}[X^A(x) - X^A(x')][X^B(x) - X^B(x')], \tag{15.9}$$

which is the length of the straight line in the embedding $(D+1)$-Minkowski spacetime that joins x and x'. It follows from the two previous equations that Z and d are related through

$$d^2(x,x') = 2H^{-2}(1 - Z(x,x')). \tag{15.10}$$

Therefore, $Z = 1$ when x' is on the light cone of x, $Z > 1$ when the points are timelike separated, and $Z < 1$ when the points are spacelike separated. It should also be noted that, as the squared Minkowskian distance between two points on the hyperboloid can take any value, $Z \in (-\infty, \infty)$.

Now, without loss of generality we can take the point x in the conformal diagram ($\chi = 0, \tilde{\eta} = -\pi/2$) and consider the spacelike straight curve with $\tilde{\eta} =$ const connecting x and its antipodal $\alpha(x)$ which in that diagram is given by ($\chi = \pi, \tilde{\eta} = -\pi/2$). This curve is a geodesic, because it is a straight line on a cylinder tangent to the hyperboloid. The Minkowskian distance between x and any other point x' in this geodesic satisfies $0 \leq d(x,x') \leq 2$. This implies

$$-1 \leq Z(x,x') \leq 1. \tag{15.11}$$

Any other spacelike geodesic passing through x can be obtained from this one by an isometry that leaves x invariant (it also leaves $\alpha(x)$ invariant). This is analogous to the case of the sphere, where any geodesic (meridian) passing through the north pole is obtained from a given one by a rotation that leaves the north pole (and south pole) invariant. This means that any point y' connected to x by a spacelike geodesic satisfies $y' = \sigma(x')$ for some x' in the above geodesic, and where σ is an isometry such that $\sigma(x) = x$. De Sitter invariance of Z then implies $Z(x,y') = Z(\sigma(x), \sigma(x')) = Z(x,x')$. Consequently, Eq. (15.11) holds for any point x' connected with x by a spacelike geodesic. In other words, spacelike separated points x, x' such that $Z(x,x') < -1$ cannot be joined by a geodesic.

There is no similar restriction for timelike geodesics, as there are theorems that guarantee the existence of geodesics between any two timelike separated points when the spacetime is globally hyperbolic; see Wald [585].

Any two spacelike separated points can be brought to the same flat hypersurface by an isometry. Now, for $x, x' \in \Sigma_t$ we can write the distance $d(x, x')$ defined by (15.9) from (15.3),

$$d^2(x, x') = e^{2Ht}\delta_{ij}[x^i(x) - x^i(x')][x^j(x) - x^j(x')]. \tag{15.12}$$

Hence, for $x, x' \in \Sigma_t$ the Minkowskian distance coincides with the length of the straight line of Σ_t that joins x and x'. This is the "physical distance" or "proper distance" in cosmology.

A bitensor is said to be maximally symmetric if it is invariant under all the isometries of the manifold. The following bitensors are maximally symmetric [476] in dS_D: the distance, $\mu(x, x')$, along the shortest geodesic joining x and x', also called the geodesic distance; the unit vectors, $n_a(x, x')$ and $n_{a'}(x, x')$, tangent to the geodesic at the points x and x' respectively, pointing outward from it; the parallel propagator, $g_{ab'}(x, x')$, which parallel-transports a vector $w_{a'}(x')$ from x' to x along the geodesic, *i.e.*, $w_a(x) = g_a^{b'}(x, x')w_{b'}(x')$; and the metric tensors, $g_{ab}(x, x')$ and $g_{a'b'}(x, x')$, at the points x and x' respectively. A theorem by Allen and Jacobson [476] states that any maximally symmetric bitensor is a linear combination of products of $n_a, n_{a'}, g_{ab}, g_{a'b'}$ and $g_{ab'}$, with coefficients that depend only on μ.

From now on in this and the next section, we assume $H = 1$ in order to simplify some expressions. The value of H is restored in the final equations. If $x(\lambda)$ is the shortest geodesic joining x and x', and its tangent vector is $v^a(\lambda)$, and we take $x(0) = x$ and $x(1) = x'$, the geodesic distance between x and x' is

$$\mu(x, x') = \int_0^1 d\lambda \, [g_{ab}(x(\lambda))v^a(\lambda)v^b(\lambda)]^{1/2}. \tag{15.13}$$

Note that $\mu(x, x')$ is a real number for spacelike separated points, and an imaginary number for timelike separated points. The sign prescription for the square root is chosen in such a way that $\mu(x, x') > 0$ for spacelike separated points, and $\mu(x, x') = i\xi(x, x')$, with $\xi(x, x') > 0$, for timelike separated points. The bitensors n_a and $n_{a'}$ are defined by the equations

$$n_a = \nabla_a \mu \qquad n_{a'} = \nabla_{a'} \mu. \tag{15.14}$$

The geometric interpretation of $n_a(x, x')$ and $n_{a'}(x, x')$ is obtained by differentiating Eq. (15.13). One finds that they are the unit vectors tangent to the geodesic at the points x and x', pointing outward from it, and multiplied by the imaginary unit in the case where x and x' are timelike separated. This implies that $n_a n^a = n_{a'} n^{a'} = 1$ for both spacelike and timelike separated points.

Some useful properties of $n_a, n_{a'}$ and $g_{ab'}$, which follow directly from the definition of these bitensors, are

$$\begin{aligned} g_{ab'}(x,x')n^{b'}(x,x') &= -n_a(x,x'), \\ g_a{}^{c'}(x,x')g_{c'b}(x',x) &= g_{ab}(x). \end{aligned} \tag{15.15}$$

From the last of these equations, and the fact that parallel transport leaves the scalar product invariant, one can prove the additional relation

$$g_{ab'}(x,x') = g_{b'a}(x',x). \tag{15.16}$$

Allen and Jacobson's theorem can be applied to the computation of the covariant derivatives of $n_a, n_{a'}$ and $g_{ab'}$, which are also maximally symmetric bitensors. The corresponding μ-dependent coefficients were also computed by Allen and Jacobson for all maximally symmetric spaces. In the particular case of dS_D, the result is

$$\begin{aligned} \nabla_a n_b &= \cot\mu\,(g_{ab} - n_a n_b), \\ \nabla_a n_{b'} &= -\csc\mu\,(g_{ab'} + n_a n_{b'}), \\ \nabla_a g_{bc'} &= (\csc\mu - \cot\mu)(g_{ab}n_{c'} + g_{ac'}n_b). \end{aligned} \tag{15.17}$$

For spacelike geodesics, $-1 \le Z < 1$, the geodesic distance μ of dS_D is given by the equation

$$\mu = \cos^{-1} Z, \tag{15.18}$$

where $\cos^{-1}$ is the principal value of the inverse cosine, which is single-valued. For example, $\cos^{-1} 1 = 0$, and $\cos^{-1}(-1) = \pi$. This is easily seen for the geodesic with $\tilde{\eta} = \text{const}$ joining $(\chi = 0, \tilde{\eta} = -\pi/2)$ and $(\chi = \pi, \tilde{\eta} = -\pi/2)$. Other spacelike geodesics are obtained from this one by an isometry, and therefore, by de Sitter invariance of μ and Z, Eq. (15.18) is true for all spacelike geodesics.

Using a similar argument, it can be shown that an analogous equation holds for timelike separated points, $Z > 1$:

$$\mu = \cos^{-1}(Z - i\epsilon), \tag{15.19}$$

where ϵ is a positive real infinitesimal. This $-i\epsilon$ term is added because the function $\cos^{-1} Z$ has a branch cut at $Z < -1$ and $Z > 1$ along the real axis, across which its imaginary part changes sign.

For spacelike separated points not joined by a geodesic one may still define μ by analytical continuation. The definition of n_a and $n_{a'}$ is also easily extended. The definition of the parallel propagator can also be extended using the second equation in (15.17), which gives $g_{ab'}$ in terms of μ, n_a and $n_{a'}$. Since these definitions have been obtained by analytic continuation, all the previous results concerning $\mu, n_a, n_{a'}$ and $g_{ab'}$ still hold. A geometric interpretation of this analytic extension for spacelike separated points not joined by geodesics was given by Pérez-Nadal et al. [972].

15.2 Noise Kernel in de Sitter Spacetime

Now let us consider a quantum field in dS_D, whose stress-energy tensor operator is T_{ab}, in a vacuum state $|0\rangle$ that we assume to be de Sitter invariant. The physical observable that measures the stress-energy tensor fluctuations is the *noise kernel* which was defined by Eqs. (10.11) and (10.12) as the four index bitensor describing the expectation value of the anticommutator of the stress-energy tensor operator T_{ab}. In the notation of this chapter reads

$$N_{abc'd'}(x,x') = \frac{1}{2}\langle\{T_{ab}(x), T_{c'd'}(x')\}\rangle - \langle T_{ab}(x)\rangle\langle T_{c'd'}(x')\rangle. \tag{15.20}$$

This expectation value is taken in the background metric g_{ab} which we assume to be a solution of the semiclassical equation.

Since $|0\rangle$ is a de Sitter-invariant state, the expectation value of any observable must be de Sitter-invariant. Therefore, $N_{abc'd'}$ is a de Sitter-invariant bitensor. According to Allen and Jacobson's theorem stated in the previous section, $N_{abc'd'}$ must be a linear combination of products of n_a, $n_{a'}$, g_{ab}, $g_{a'b'}$ and $g_{ab'}$, with coefficients that depend only on μ. On the other hand, it is obvious from its definition that $N_{abc'd'}(x,x') = N_{c'd'ab}(x',x)$, and, since the stress-energy tensor is symmetric, we also have that $N_{abc'd'} = N_{bac'd'} = N_{abd'c'}$.

The most general bitensor that is a linear combination of products of $n_a, n_{a'}, g_{ab}, g_{a'b'}$ and $g_{ab'}$, with coefficients that depend only on μ, and that satisfies the above constraints is

$$\begin{aligned} N_{abc'd'} = {} & P(\mu) n_a n_b n_{c'} n_{d'} + Q(\mu)(n_a n_b g_{c'd'} + n_{c'} n_{d'} g_{ab}) \\ & + R(\mu)(n_a n_{c'} g_{bd'} + n_b n_{d'} g_{ac'} + n_a n_{d'} g_{bc'} + n_b n_{c'} g_{ad'}) \\ & + S(\mu)(g_{ac'} g_{bd'} + g_{bc'} g_{ad'}) + T(\mu) g_{ab} g_{c'd'}. \end{aligned} \tag{15.21}$$

Moreover, the stress-energy tensor conservation law implies $\nabla^a N_{abc'd'} = 0$ which imposes some differential equations for the coefficients P, Q, R, S, T. Also there are further restrictions on these coefficients imposed by the condition that the noise kernel should be unambiguous when x and x' are antipodal; see [972].

Now let us assume a free massive minimally coupled ($\xi = 0$) scalar field $\phi(x)$, whose stress-energy tensor is given by the coincidence limit, see Eq. (10.3),

$$T_{ab}(x) = \lim_{x'' \to x} \mathcal{D}_{ab}(x, x'')\phi(x)\phi(x''), \tag{15.22}$$

with

$$\mathcal{D}_{ab} = g_b{}^{c''} \nabla_a \nabla_{c''} - \frac{1}{2} g_{ab} g^{cd''} \nabla_c \nabla_{d''} - \frac{1}{2} m^2 g_{ab}. \tag{15.23}$$

The stress-energy tensor fluctuations of the free scalar field is obtained by substituting (15.22) into (15.20). Since the field ϕ is free, and $|0\rangle$ is a vacuum

state, the four-point function appearing in this equation can be written in terms of the positive Wightman function, $iG^+(x,x') = \langle 0|\phi(x)\phi(x')|0\rangle$,

$$\begin{aligned}\langle 0|\phi(x)\phi(x'')\phi(x')\phi(x''')|0\rangle = &-G^+(x,x'')G^+(x',x''') - G^+(x,x')G^+(x'',x''') \\ &- G^+(x,x''')G^+(x'',x'). \end{aligned} \tag{15.24}$$

This equation is obtained by expanding the field ϕ in terms of creation and annihilation operators. Note that, in the particular case where the product of fields on the left-hand side is time-ordered, this equation also follows from Wick's theorem. Substituting (15.24) into (15.20) using (15.22), and taking the limit $x'' \to x$ and $x''' \to x'$, we find (see also (10.53) and (10.56))

$$\begin{aligned} N_{abc'd'} = &-\nabla_a\nabla_{c'}G^+\nabla_b\nabla_{d'}G^+ - \nabla_a\nabla_{d'}G^+\nabla_b\nabla_{c'}G^+ \\ &+ g_{ab}\left(\nabla_e\nabla_{c'}G^+\nabla^e\nabla_{d'}G^+ + m^2\nabla_{c'}G^+\nabla_{d'}G^+\right) \\ &+ g_{c'd'}\left(\nabla_a\nabla_{e'}G^+\nabla_b\nabla^{e'}G^+ + m^2\nabla_aG^+\nabla_bG^+\right) \\ &- \frac{1}{2}g_{ab}g_{c'd'}\left[\nabla_e\nabla_{e'}G^+\nabla^e\nabla^{e'}G^+ \right. \\ &\left. +m^2\left(\nabla_eG^+\nabla^eG^+ + \nabla_{e'}G^+\nabla^{e'}G^+\right) + m^4G^{+2}\right]. \end{aligned} \tag{15.25}$$

Note that, after substitution of (15.24) into (15.20), there has been a cancellation of terms that diverge in the limit $x'' \to x, x''' \to x'$, so that this result is finite as long as x is not on the light cone of x'.

At this point we impose that the vacuum $|0\rangle$ is de Sitter-invariant, and that the points x and x' are spacelike separated. For spacelike separated points $[\phi(x), \phi(x')] = 0$ and the Wightman function is the expectation value of a Hermitian operator, i.e., an observable (the product of two Hermitian operators that commute is Hermitian). Therefore the Wightman function for spacelike separated points must be a de Sitter-invariant biscalar, thus it must have the form $G^+ = G(\mu)$. On the other hand, the Wightman function satisfies the Klein–Gordon equation:

$$G''(\mu) + (D-1)\cot\mu G'(\mu) - m^2G(\mu) = 0. \tag{15.26}$$

Imposing as boundary conditions that $G(\mu)$ diverges only at the coincidence limit, $\mu \to 0$, and that it does so in the same way as the standard Wightman function in Minkowski spacetime, the only solution is

$$iG(\mu) = c_{m,D}F\left(h_+, h_-; \frac{D}{2}; \frac{1+Z}{2}\right), \tag{15.27}$$

where F denotes the hypergeometric function, $h_\pm$ and $c_{m,D}$ are constants,

$$h_\pm = \frac{1}{2}\left[(D-1) \pm \sqrt{(D-1)^2 - 4m^2}\right],$$
$$c_{m,D} = \frac{\Gamma(h_+)\Gamma(h_-)}{(4\pi)^{D/2}\Gamma(\frac{D}{2})}, \tag{15.28}$$

and we are using that $Z = \cos\mu$, as follows from (15.18). A vacuum whose Wightman function for spacelike separated points is (15.27), is the Bunch-Davies vacuum [47].

Substituting $G(\mu)$ into (15.25), and using (15.17), we find that the stress-energy tensor two-point function in the Bunch–Davies vacuum has the form (15.21), as corresponds to a de Sitter-invariant state. The associated μ-dependent functions are

$$\begin{aligned}
P &= -2\left(G'' - G'\csc\mu\right)^2, \\
Q &= \left(G''\right)^2 - \left(G'\right)^2\left(\csc^2\mu - m^2\right), \\
R &= G'\csc\mu\left(G'' - G'\csc\mu\right), \\
S &= -\left(G'\right)^2\csc^2\mu, \\
T &= -\frac{1}{2}\left(G''\right)^2 - \left(G'\right)^2\left(\frac{D-5}{2}\csc^2\mu - m^2\right) - \frac{1}{2}m^4G^2,
\end{aligned} \tag{15.29}$$

where, again, the prime stands for derivative with respect to μ. The functions (15.29) satisfy the stress-energy tensor conservation equations and the conditions for antipodal points. Thus, the noise kernel is given by expression (15.21) with the values of $P(\mu)$, $Q(\mu)$, $R(\mu)$, $S(\mu)$ and $T(\mu)$ obtained above, which are given in terms of the hypergeometric function (15.21). A detailed analysis of this noise kernel in different limits and for different values of the field mass is dicussed by Pérez-Nadal et al. [972]. Here we focus in some of the most relevant aspects.

15.2.1 Super/Sub-horizon scales and massless limit

It is interesting to give the two-point noise kernel at long distances $d \gg H^{-1}$, i.e., superhorizon scales, for spacelike separated points. From Eq. (15.10), this corresponds to large negative values of Z. Using the properties of the hypergeometric function one obtains, for $D = 4$,

$$\begin{aligned}
P, Q, R, S &\sim d^{-4} + O(m^2) \\
T &\sim \text{constant} + O(m^2).
\end{aligned} \tag{15.30}$$

Note that the leading term in T does not decay to zero, which means that the noise kernel in the limit $m \to 0$ does not vanish at long distances.

The origin of this behavior is the term proportional to m^4G^2 in Eq. (15.29). Indeed, the terms with derivatives of G are either $O(m^2)$, and thus subleading,

or vanishing in the limit $Z \to -\infty$, whereas the term proportional to $m^4 G^2$ is a constant of order $O(m^0)$, because the Wightman function $G(\mu)$ diverges in the limit $m \to 0$. This is the infrared divergence of the Bunch–Davies vacuum. Since the divergent term is independent of μ, the derivative of the Wightman function is finite in this limit.

It is worth comparing this result with the stress-energy tensor fluctuations in Minkowski spacetime. Note, first, that since Minkowski spacetime is also a maximally symmetric space the theorem stated above also applies and the noise kernel can be written in the form (15.21). The corresponding functions P, Q, R, S, T can be estimated, by dimensional analysis, to behave as

$$P, Q, R, S, T \propto d^{-8}, \tag{15.31}$$

for $D = 4$. Hence, the stress-energy tensor fluctuations in the limit $m \to 0$ do decay with the distance in Minkowski spacetime.

Finally, it is worth noting that, for small distances ($Z \to 1$), i.e., when $d \ll H^{-1}$, the functions (15.29) have the same form as in Minkowski spacetime, which is also (15.31), as expected of subhorizon scales. This implies that the massless limit discontinuity that we describe next is not manifested at short distances.

15.2.2 Massless Field

We now consider a strictly massless field, $m = 0$. As is well-known in this case there exists no de Sitter-invariant vacuum [518]. The vacuum states we choose for the massless field are the $O(D)$-invariant vacua, $|0\rangle_\alpha$, introduced by Allen and Folacci [519]. These states form a one-parameter family with $\alpha \in (0, \infty)$. We will be interested in the limit $\alpha \to 0$, because, although $|0\rangle_\alpha$ are not de Sitter-invariant, there is a class of observables whose expectation values are de Sitter-invariant in this limit.

The noise kernel of a free scalar field is given by Eq. (15.25), for any mass m and for any vacuum state. In the massless case we just need to take $m = 0$ in that equation. Note that, in this case, all terms involve derivatives of the Wightman functions; in particular they are products of $\nabla_a \nabla_{b'} G^+$. We want to compute the noise kernel in the vacuum $|0\rangle_\alpha$ in the limit $\alpha \to 0$. It can be shown [972] that

$$\lim_{\alpha \to 0} \nabla_a \nabla_{b'} G^+_\alpha = \lim_{m \to 0} \nabla_a \nabla_{b'} G^+_{BD}, \tag{15.32}$$

where G^+_α is the Wightman function of the massless field in the vacuum $|0\rangle_\alpha$, and G^+_{BD} is the Wightman function of the massive field in the Bunch–Davies vacuum. Therefore, the stress-energy tensor fluctuations of a massless field in the limit $\alpha \to 0$ can be computed as follows. First, write Eq. (15.25) without the mass terms; second, compute the resulting equation for a massive field in the Bunch–Davies vacuum; and then let $m \to 0$. It is important to notice that this is not exactly the same as computing the stress-energy tensor fluctuations of a

massive field in the limit $m \to 0$, which is what we have done in the previous subsection.

Owing to the absence of the constant piece, the behavior at long distances or superhorizon scales changes dramatically with respect to the case of small masses, and we have

$$P, Q, R, S, T \sim d^{-4}. \tag{15.33}$$

Therefore, the noise kernel of the massless field vanishes at long distances. There is a "discontinuity" at $m = 0$: taking the limit $m \to 0$ is not the same as setting $m = 0$.

One might think that this discontinuity is due to our choice of states for both the massive and the massless fields, and that by choosing appropriate states the discontinuity would disappear. However, this is not the case. In the paper by Kirsten and Garriga [520] the expectation value ${}_\alpha\langle 0|T_{ab}|0\rangle_\alpha$ was computed for all values of α. In the limit $\alpha \to 0$ the result is de Sitter-invariant and it disagrees with the limit $m \to 0$, similarly to what we have just seen for the noise kernel. Furthermore, at late times one has

$${}_\alpha\langle 0|T_{ab}|0\rangle_\alpha \to \lim_{\alpha\to 0} {}_\alpha\langle 0|T_{ab}|0\rangle_\alpha.$$

That is, $\lim_{\alpha\to 0} {}_\alpha\langle 0|T_{ab}|0\rangle_\alpha$ is an attractor at late times. As is well-known, the same behavior occurs for the massive field in the Bunch–Davies vacuum. This means that, in the case of the expectation value of T_{ab}, the discontinuity at $m = 0$ appears at late times regardless of the state of the quantum field.

It is important to note that we are comparing the massless, minimally coupled field with a very particular limit in parameter space. In general, in de Sitter (and in any curved spacetime) the Klein–Gordon equation includes the term $\xi R\phi$, where R is the Ricci scalar and ξ is the dimensionless parameter coupling to the curvature. Here we have considered only minimally coupled fields, i.e., $\xi = 0$, and we have compared the case $\xi = 0, m = 0$ with the case $\xi = 0, m \to 0$. In [520] the expectation value of the stress-energy tensor in the case $\xi = 0, m = 0$ was compared with the same expectation value in the more general limit $\xi \to 0$, $m \to 0$. It was found that such a limit is not unique. In particular, for $D = 4$ it was shown that if the limit is taken in such a way that $\xi R/m^2 \to -2$, the discontinuity disappears. It is reasonable to think that the noise kernel will exhibit a similar behavior.

15.3 Analysis Based on Field Modes

We have seen that the noise kernel in the limit of $m \to 0$ does not vanish at long distances, whereas it does go like d^{-4}, where d is the physical distance, when the mass vanishes exactly. Consequently there is a discontinuity in the massless limit.

These results have been obtained in a coordinate-independent way, using the properties of the de Sitter-invariant bitensors and the hypergeometric function.

Here we provide an alternative explanation, in a language that is more familiar in cosmology. The analysis, based on the approximate solutions of the field modes, will allow us to derive the exact dependence of the noise kernel at long distances on the field mass. Again, for simplicity we will restrict the discussion to $D = 4$ de Sitter spacetime dimensions and the dependence on H will be explicit throughout.

We will use the coordinate system $\{t, x^i\}$ defined by Eqs. (15.2), in which the line element takes the spatially flat form

$$ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j, \tag{15.34}$$

with scale factor $a(t) = e^{Ht}$. In these coordinates, the scalar field ϕ can be expanded in plane waves,

$$\phi(t, \vec{x}) = \int d^3k \left[a_{\vec{k}} u_k(t) e^{i\vec{k}\cdot\vec{x}} + a^\dagger_{\vec{k}} u^*_k(t) e^{-i\vec{k}\cdot\vec{x}}\right], \tag{15.35}$$

where $\vec{v}$ denotes a vector of components v^i, $\vec{v}\cdot\vec{w} = \delta_{ij}v^i w^j$, and $a^\dagger_{\vec{k}}$ and $a_{\vec{k}}$ are creation and annihilation operators. We assume that the modes $u_k(t)$ depend on $\vec{k}$ only through its norm $k = |\vec{k}| = (\vec{k}\cdot\vec{k})^{1/2}$, which corresponds to having a homogeneous and isotropic vacuum. Substituting the equation above in the Klein–Gordon equation,

$$(\nabla_a \nabla^a - m^2)\phi = 0, \tag{15.36}$$

one obtains the following equation for the modes:

$$\ddot{u}_k + 3H\dot{u}_k + (k^2/a^2 + m^2)\, u_k = 0. \tag{15.37}$$

It is worth noticing that the Bunch–Davies vacuum [319, 47] (a state defined by $a_{\vec{k}}|0\rangle = 0$) is associated with the following particular solution of this equation

$$u_k = H^{-1/2} a^{-3/2} H^{(1)}_\lambda\,(k/aH)\,, \tag{15.38}$$

where $H^{(1)}_\lambda$ is a Hankel function, and $\lambda = [9/4 - (m/H)^2]^{1/2}$.

The Wightman function $iG^+(x, x') = \langle 0|\phi(x)\phi(x')|0\rangle$ can also be written in terms of the modes. Assuming that the points x and x' are simultaneous in our coordinate system, $t = t'$, we have

$$iG^+(t, \vec{x}; t, \vec{x}') = \int_0^\infty dk\, k^2\, \frac{\sin kr}{kr}\, |u_k(t)|^2, \tag{15.39}$$

where $r = |\vec{x} - \vec{x}'|$. Global numerical coefficients are unimportant for our present purposes, and for this reason they are omitted from the equation above and all the equations that follow in this section. For pairs of points separated by a physical distance much larger than the horizon, $d = ar \gg H^{-1}$, the integral above is dominated by terms with physical wavelengths also much larger than

the horizon, i.e., $k \ll aH$. This condition is satisfied in the time interval $t \gg t_k$, where t_k is defined by $k = a(t_k)H$, and is usually referred to as the time of horizon crossing.

Since k/a decreases exponentially, in most of this time interval we have $k/a \ll m$. Hence, the term proportional to $(k/a)^2$ in Eq. (15.37) can be neglected, and that equation takes the form of the Klein–Gordon equation for a homogeneous scalar field in de Sitter,

$$\ddot{u}_k + 3H\dot{u}_k + m^2 u_k \simeq 0, \tag{15.40}$$

This equation has two linearly independent solutions, which are proportional to e^{-3Ht} and $e^{-m^2 t/3H}$ respectively. The second solution, which decays much more slowly than the first, eventually dominates. Therefore, we can concentrate on the second, slowly decaying solution. In order to find the dependence on k, we have to impose an initial condition. This is obtained by matching with the behavior when the mode is inside the horizon, i.e., when $t \ll t_k$. In this time interval, for the vacuum to be adiabatic we must have (see Birrell and Davies [47]),

$$u_k(t) \simeq \frac{1}{a(t)} \frac{1}{\sqrt{k}} e^{-ik\eta}, \tag{15.41}$$

where $\eta = -1/aH$ is the conformal time. Assuming that this equation holds also at $t \simeq t_k$, we obtain the initial condition $u_k(t_k) \simeq H/k^{3/2}$. Therefore we have

$$u_k(t) \simeq \frac{H}{k^{3/2}} e^{-\frac{m^2}{3H}(t-t_k)} = \frac{H}{k^{3/2}} \left(\frac{k}{aH}\right)^{\frac{m^2}{3H^2}} \tag{15.42}$$

for $t \gg t_k$ or, equivalently, for $k \ll aH$.

This result has been obtained using a few semi-qualitative arguments, but it can be derived more rigorously from the exact form (15.38) of the modes associated with the Bunch–Davies vacuum. Substituting the mode (15.42) into Eq. (15.39), and defining the dimensionless variable $s \equiv kr$, we obtain the following approximate result for the Wightman function:

$$iG^+(t, \vec{x}; t, \vec{x}') \simeq H^2 (Hd)^{-\frac{2m^2}{3H^2}} \int_0^\infty ds\, s^{-2+\frac{2m^2}{3H^2}} \sin s, \tag{15.43}$$

where we have used that the physical distance is $d = ar$. The integral above is just a constant, dimensionless factor that depends on H and m. It can nevertheless be computed, giving rise to the result

$$iG^+(t, \vec{x}; t, \vec{x}') \simeq \frac{H^4}{m^2} (Hd)^{-\frac{2m^2}{3H^2}}. \tag{15.44}$$

This result is valid for $d \gg H^{-1}$ and $m \ll H$, up to a numerical factor.

Thus, we see that the Wightman function can decay arbitrarily slowly with the physical distance, provided that the mass is sufficiently small. Similar results hold for the time derivatives of the Wightman function evaluated at simultaneous

points (which are given by expressions similar to (15.39), but involving time derivatives of the modes),

$$(i\partial_t G^+)(t,\vec{x};t,\vec{x}') \simeq H^3 (Hd)^{-\frac{2m^2}{3H^2}}$$
$$(i\partial_t \partial_{t'} G^+)(t,\vec{x};t,\vec{x}') \simeq H^2 m^2 (Hd)^{-\frac{2m^2}{3H^2}}. \tag{15.45}$$

In contrast, spatial derivatives of the Wightman function decay faster with the distance, because they are obtained by deriving Eq. (15.44).

The noise kernel is given by Eq. (15.25) in terms of the Wightman function and its derivatives. From what we have just seen, it follows that the terms that decay most slowly are those proportional to the squares of $G^+, \partial_t G^+$ and $\partial_t \partial_{t'} G^+$. It then follows that, owing to these terms, the noise kernel decays like $d^{-4m^2/(3H^2)}$. This is exactly the long-distance behavior, particularized to small masses and to $D = 4$. In the massless limit, $m \to 0$, the term of (15.25) proportional to G^{+2} tends to a finite constant. Contrarily, the terms proportional to time derivatives of G^+ vanish, due to the powers of m appearing in Eqs. (15.25) and (15.45). Hence, in the massless limit the noise kernel is a nonvanishing constant at long distances, and this is entirely due to the term proportional to G^{+2} in (15.25). In the strictly massless case, the term proportional to G^{+2} is absent, and thus the noise kernel vanishes at long distances. Therefore, taking the limit $m \to 0$ is not the same as setting $m = 0$, as was found in Section 15.2.2. To summarize, the discontinuity arises because in the massive case, there is a contribution from the mass term in the stress-energy tensor that does not vanish in the limit $m \to 0$ due to the infrared divergence of the Bunch–Davies vacuum.

15.4 Implications for Gravitational Fluctuations

The quantum correlations of the stress-energy tensor described by the noise kernel can be used to obtain information on the quantum fluctuations of the gravitational field in de Sitter spacetime. We may consider the quantization of a metric perturbation around a de Sitter background within the framework of the effective field theory approach to quantum gravity [192, 193], and neglect the contributions from graviton loops. As explained earlier in Section 9.3 this is naturally implemented in the context of a large N expansion for N matter fields coupled to the gravitational field, where one introduces a rescaled gravitational coupling constant $\bar{\kappa} = \sqrt{N}\kappa$, where $\kappa = \sqrt{16\pi G_{\rm N}} = \sqrt{16\pi}\ell_{\rm P}$ (in our units $\ell_{\rm P} = \sqrt{G_{\rm N}}$), and expands in powers of $1/N$ keeping $\bar{\kappa}$ fixed. Neglecting graviton loops amounts then to considering only the leading order in $1/N$. The leading contribution to the two-point function of the metric perturbations is of order $1/N$.

To characterize the gravitational fluctuations we may compute the two-point function of the linearized Riemann tensor. The Riemann tensor of the background de Sitter spacetime with suitably raised indices,

$$R_{ab}{}^{cd} = \frac{R}{6}\,\delta_{[a}{}^{[c}\delta_{b]}{}^{d]}, \tag{15.46}$$

has gauge-invariant linear perturbations. The Riemann tensor is entirely determined by the Ricci tensor (or equivalently the Einstein tensor) and the Weyl tensor. Both the linearized Ricci and the Weyl tensors over a de Sitter background are separately gauge-invariant, and the Weyl tensor also vanishes in the background.

The noise kernel can be directly used to get information on the quantum metric fluctuations in de Sitter spacetime. It turns out that the lowest-order contribution to the two-point function for the linearized Einstein tensor is quartic in $\bar{\kappa}$ and comes directly from the one-loop contributions of the matter fields, both from the stochastic gravity formalism and from the Einstein equation to linear order in the metric perturbations as an equation for operators in the Heisenberg picture, including the matter fields as well as the metric perturbations. The two quadratic curvature tensors which appear as counterterms in the renormalization of the stress-energy tensor both vanish in this case because they become zero for a de Sitter background.

Thus only the linearized Einstein tensor survives, and one obtains the following result for the (connected) symmetrized two-point function of the Einstein tensor at that order:

$$\begin{aligned}\langle G^{a\,(1)}_{\ b}(x)\, G^{c'\,(1)}_{\ d'}(x')\rangle_c &\equiv \frac{1}{2}\langle\{G^{a\,(1)}_{\ b}(x)\,, G^{c'\,(1)}_{\ d'}(x')\}\rangle - \langle G^{a\,(1)}_{\ b}(x)\rangle\langle G^{c'\,(1)}_{\ d'}(x')\rangle \\ &= \frac{\bar{\kappa}^{\,4}}{4N}\, N^{a\ c'}_{\ b\ d'}(x,x'),\end{aligned} \tag{15.47}$$

where $G^{a(1)}_{\ b}$ is the linearized Einstein tensor, and $N^{a\ c'}_{\ b\ d'}$ is the noise kernel that has been computed previously. This corresponds to part of the *induced* metric fluctuations described in Eq. (10.16). Therefore, the noise kernel in de Sitter gives the leading order contribution to the symmetrized two-point function of the linearized Einstein tensor. Thus, from the results for the noise kernel we have that for massless fields, $m = 0$, and long distances d, i.e., on superhorizon scales, we have

$$\langle G^{a\,(1)}_{\ b}(x)\, G^{c'\,(1)}_{\ d'}(x')\rangle_c \sim \frac{\bar{\kappa}^{\,4} H^4}{N\, d^4}\,, \tag{15.48}$$

and for low mass fields, $m \ll H$, we have

$$\langle G^{a\,(1)}_{\ b}(x)\, G^{c'\,(1)}_{\ d'}(x')\rangle_c \sim \frac{\bar{\kappa}^{\,4} H^8}{N}\,(Hd)^{-\frac{4m^2}{3H^2}}\,, \tag{15.49}$$

which implies very long range correlations.

On the other hand the two-point function of the Weyl tensor, which is also gauge-invariant, cannot be obtained from the noise kernel. A non-vanishing tree-level contribution of order $\bar{\kappa}^{\,2}$ even in the absence of matter loops, was calculated by Kouris [971]. For spacelike separated points and long distances he found

$$\langle C^{(1)}_{abcd}(x) C^{(1)}_{a'b'c'd'}(x') \rangle_c \sim \frac{\bar{\kappa}^2 H^2}{N\, d^4} + \mathrm{O}\left(\bar{\ell}_{\mathrm{P}}^4 H^4\right) . \tag{15.50}$$

The tree-level contribution to this two-point function corresponds to the metric *intrinsic* fluctuations of Eq. (10.16) but at the matter loop order there are contributions of both the *intrinsic* and the *induced* metric fluctuations. The calculation of the quantum correlations of the full linearized Riemann tensor is the subject of Chapter 17. That calculation, however requires knowing first the full quantum two-point metric perturbations, which are derived in Chapter 16.

16

Two-Point Metric Perturbations in de Sitter

The linearized Riemann two-point correlation function on a de Sitter background including the one-loop corrections from matter fields is not only gauge-invariant but it is also an infrared-safe observable. In Chapter 17 we derive this correlator for conformal matter fields, but a necessary step for this calculation is the derivation of the full two-point metric perturbations. This involves the computation of the two-point tensor metric perturbations, which are true dynamical degrees of freedom, as well as the scalar and vector metric perturbations, which are not dynamical. In this chapter we derive the full two-point metric perturbations.

We do the calculation using spatially flat coordinates in the Poincaré patch of de Sitter spacetime, and we follow the in-in or CTP formalism presented in Section 9.3 selecting an asymptotic initial state by an $i\epsilon$ prescription which corresponds to the vacuum of the interacting theory. Exact de Sitter invariance will only happen for a very particular choice of initial state corresponding to the de Sitter-invariant vacuum of the interacting theory (rather than the free one): in general results will not be exactly invariant for any other state. However, by analogy with the results for scalar theories found by Marolf and Morrison [525] and Rajaraman [463], if such a de Sitter invariant state exists and is a late-time attractor, as suggested by the intuition that regular initial excitations get red-shifted away by the exponential expansion, one expects that for sufficiently regular initial states the result for observables characterizing the local geometry in a finite region should become insensitive to the initial conditions and tend at sufficiently late times to the de Sitter invariant result.

As we have argued in Chapter 12 massless conformally coupled fields in cosmology may be a good approximation for matter fields in the standard model of particle physics in the high-curvature limit when their masses are negligible compared to the spacetime curvature. Moreover, the standard model of particle physics contains nearly one hundred fields so that the large N approximation is reasonable and justifies ignoring the effect of graviton loops.

The quantum two-point function at one-loop can be written in terms of the effective action that results from integrating out the massless conformally coupled fields. Such an effective action is presented in Section 16.1 for metric perturbations around a FLRW background. The decomposition of the metric perturbations, taking into account the symmetries of the background spatial sections, into scalar, vectorial and tensorial perturbations is reviewed in Section 16.2. We write the perturbations in terms of spatial Fourier modes and fix the gauge for quantization, and an expression for the effective action in terms of this scalar, vector and tensor decomposition is given. In Section 16.3 we follow closely Fröb et al. [491] and compute the two-point function of the tensor perturbations around a de Sitter background for an arbitrary pair of points including one-loop corrections from conformal fields; some technical details are given in Section 16.7. The relation with the intrinsic and induced fluctuations of stochastic gravity is discussed in Section 16.4. In Section 16.5 we compute the two-point metric perturbations for the scalar and vectorial modes, which are constrained degrees of freedom. The result for the full two-point metric perturbations is of course gauge-dependent and does not exhibit de Sitter invariance explicitly, but it is invariant under spatial rotations and translations as well as under a simultaneous rescaling of the spatial and conformal time coordinates, which are necessary conditions for de Sitter invariance. Furthermore, although we mainly focus on loop corrections due to free massless scalar fields with conformal coupling to the curvature, our results can be extended to general conformal field theories, even strongly interacting ones; the effective action for a general conformal field theory is derived in Section 16.6.

16.1 Effective Action for Cosmological Perturbations

Let us consider N massless and conformally coupled free real scalar fields interacting with the metric perturbations around a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) background driven by a cosmological constant. Since we are interested in calculating the correlation functions of the metric perturbations to leading order in $1/N$, in addition to the Einstein–Hilbert action for gravity with a cosmological constant we only need to include counterterms quadratic in the curvature for renormalization. Thus, with the gravitational part $S_{\rm g}[\tilde{g}]$ given in Eq. (9.51) and adding the matter field action for N conformal fields $\tilde{\phi}_k$, we have the total bare action,

$$\begin{aligned} S[\tilde{g},\tilde{\phi}] = {} & \frac{N}{\bar{\kappa}^2}\int \left(\tilde{R}-2\Lambda\right)\sqrt{-\tilde{g}}d^nx \\ & + Na_1\int \left(\tilde{R}^{abcd}\tilde{R}_{abcd}-\tilde{R}^{ab}\tilde{R}_{ab}\right)\sqrt{-\tilde{g}}d^nx + Na_2\int \tilde{R}^2\sqrt{-\tilde{g}}d^nx \\ & - \frac{1}{2}\sum_{k=1}^{N}\int \left[\left(\tilde{\nabla}^a\tilde{\phi}_k\right)\left(\tilde{\nabla}_a\tilde{\phi}_k\right)+\xi(n)\tilde{R}\tilde{\phi}_k^2\right]\sqrt{-\tilde{g}}d^nx. \end{aligned} \tag{16.1}$$

In order to use dimensional regularization we consider the action in n dimensions, and all parameters are bare parameters, except for $\xi(n) = (n-2)/[4(n-1)]$ which depends on n to make the scalar field action conformally invariant in all dimensions.

It is convenient to rewrite the first counterterm quadratic in the curvature in terms of the square of the Weyl tensor and the integrand of the four-dimensional Euler invariant, $\tilde{\mathcal{E}}_4 \equiv \tilde{R}^{abcd}\tilde{R}_{abcd} - 4\tilde{R}^{ab}\tilde{R}_{ab} + \tilde{R}^2$, as follows:

$$\int \left(\tilde{R}^{abcd}\tilde{R}_{abcd} - \tilde{R}^{ab}\tilde{R}_{ab}\right)\sqrt{-\tilde{g}}d^n x = -\left(\frac{1}{2} - \frac{3}{4}(n-4)\right)\int \tilde{\mathcal{E}}_4\sqrt{-\tilde{g}}d^n x + \frac{n-4}{12}\int \tilde{R}^2\sqrt{-\tilde{g}}d^n x + \left(\frac{3}{2} - \frac{3(n-4)}{4}\right) \times \int \tilde{C}^{abcd}\tilde{C}_{abcd}\sqrt{-\tilde{g}}d^n x + \mathcal{O}((n-4)^2). \tag{16.2}$$

It follows from the action (16.1) that in the large N limit and after the rescaling of the gravitational constant $\bar{\kappa}^2 = N\kappa^2$, graviton loops are suppressed by higher powers of $1/N$ with respect to matter loops, and one can neglect them at leading order. We assume that the background metric is conformal to the flat metric, $g^0_{ab} = a^2\eta_{ab}$, and write the physical metric $\tilde{g}_{ab}$ as

$$\tilde{g}_{ab} = a^2(\eta)g_{ab} = a^2(\eta)(\eta_{ab} + \kappa h_{ab}), \tag{16.3}$$

where g_{ab} is a nearly flat metric and η denotes the conformal time. Neglecting graviton loops then amounts to truncating the effective action $S_{\rm eff}$ to second order in the metric perturbation h_{ab}.

The functional integral over the conformal matter fields was calculated in Chapter 12, following [423, 186], for the case of a single scalar field ($N = 1$) and for $\Lambda = 0$. The basic idea was to use the conformal transformation formulae given in Section 12.5.2 together with the conformal invariance of the classical action for the conformal fields to rewrite Eq. (16.1) in terms of the metric g_{ab} for small perturbations around flat space. One could then perform the path integral for the scalar fields in flat space treating perturbatively the interaction with the metric perturbations. In doing so, ultraviolet divergences arise which are cancelled by an appropriate choice of the bare parameter

$$Na_1 = Na_1^{\rm R}(\mu) + \alpha\left[(n-4)^{-1} + \ln\mu + \mathcal{O}(n-4)\right], \tag{16.4}$$

where $a_1^{\rm R}$ is the renormalized parameter and we have used the dimensionless constant $\alpha = N(2880\pi^2)^{-1}$ defined in (12.38). In contrast, the second bare parameter, Na_2, does not get renormalized, which is a peculiarity of the conformal case [47]. Instead of Na_2 it is convenient to introduce a new dimensionless parameter $\beta = \alpha - 12Na_2$, which will be used from now on.

Taking all this into account, the final result for $S_{\rm eff}$ was obtained in Eqs. (12.14) and (12.46), which are valid to quadratic order in the metric perturbations. The effective action can be written as

$$\begin{aligned} S_{\rm eff}[\tilde{g}^{\pm}_{ab}] = {} & S^{\rm R}_{\rm g}[a, g^{+}_{ab}] - S^{\rm R}_{\rm g}[a, g^{-}_{ab}] \\ & + 3\alpha \iint C^{+}_{abmn}(x) C^{-abmn}(y) K(x-y) d^4x d^4y \\ & + \frac{3}{2}\alpha \iint C^{+}_{abmn}(x) C^{+abmn}(y) K^{+}(x-y;\mu) d^4x d^4y \\ & - \frac{3}{2}\alpha \iint C^{-}_{abmn}(x) C^{-abmn}(y) K^{-}(x-y;\mu) d^4x d^4y. \end{aligned} \tag{16.5}$$

The renormalized $S_{\rm g}$ terms are,

$$\begin{aligned} S^{\rm R}_{\rm g}[a, g_{ab}] = {} & \frac{1}{\kappa^2} \int \left(a^2 R - 6a\nabla^m \nabla_m a - 2\Lambda a^4\right) \sqrt{-g} d^4x \\ & + \alpha \int \left(R^{abcd} R_{abcd} - R^{ab} R_{ab}\right) \ln a \sqrt{-g} d^4x \\ & - \frac{\beta}{12} \int \left(R - 6a^{-1} \nabla^m \nabla_m a\right)^2 \sqrt{-g} d^4x \\ & + 2\alpha \int G^{mn} a^{-2} \left(\nabla_m a\right) \left(\nabla_n a\right) \sqrt{-g} d^4x \\ & + \alpha \int a^{-4} \left[2a\nabla^m \nabla_m a - \left(\nabla^m a\right) \left(\nabla_m a\right)\right] \\ & \times \left(\nabla^n a\right) \left(\nabla_n a\right) \sqrt{-g} d^4x, \end{aligned} \tag{16.6}$$

where here we use $a(\eta)$ instead of $\omega(\eta)$, which are related by $\omega = \ln(a)$, and we include the cosmological constant Λ.

The kernels K, see (12.37) and (12.44), are given by

$$\begin{aligned} K(x) &= -{\rm i}\pi \int \Theta(-p^2) \Theta(-p^0) {\rm e}^{{\rm i}p\cdot x} \frac{d^4p}{(2\pi)^4} \\ K^{\pm}(x;\mu) &= \frac{1}{2} \int \left[-\ln \left| \frac{p^2}{\mu^2} \right| \pm {\rm i}\pi \Theta(-p^2) \right] {\rm e}^{{\rm i}p\cdot x} \frac{d^4p}{(2\pi)^4}, \end{aligned} \tag{16.7}$$

where μ is the renormalization scale. It should be stressed that the dependence on μ of these kernels and of the renormalized parameters cancels out, so that $S_{\rm eff}$ is invariant under the renormalization group.

More specifically, we have

$$N a^{\rm R}_1(\mu) = N a^{\rm R}_1(\mu_0) - \alpha \ln \left(\frac{\mu}{\mu_0} \right), \tag{16.8}$$

for a fixed scale μ_0, and one can see that the logarithmic dependence on μ of the local terms quadratic in the Weyl tensor exactly matches that of the kernels

$K^{\pm}(x;\mu)$. Given any value of $a_1^{\rm R}(\mu_0)$, one can always choose a renormalization scale $\bar{\mu}$ such that $a_1^{\rm R}(\bar{\mu}) = 0$; this corresponds to taking a renormalization scale in which the dimensional parameter $\alpha_d = 0$ in Eq. (10.8). This is the choice that was made in Chapter 12, and we will also employ it here to get simpler expressions. However, once we obtain the final result in the next section, we will specify the small changes needed so that the result is valid for arbitrary μ. On the other hand, since the parameter a_2 does not get divergent contributions for conformal fields, it has no dependence on μ. In this case the functional integration of the scalar fields just gives a finite contribution, $-\alpha/12$, to the coefficient of the term with the square of the Ricci scalar. To take that into account, the new parameter β has been introduced in Eq. (16.5), see Eq. (16.6). In Chapter 12 the particular choice $\beta = \alpha$, corresponding to $a_2 = 0$, was made; this corresponds to choosing the dimensional parameter $\beta_d = 0$ in Eq. (10.8).

As we have emphasized earlier the effective action $S_{\rm eff}$ in Eq. (16.5) must be understood as truncated to second order in h_{ab}, the perturbations of the metric around flat space. The expansion of the effective action up to that order can be obtained using the expansions around flat space given in Section 12.5.1, noting that we now have the factor κ in the metric perturbation.

The linearized Weyl tensor in four dimensions can be written as

$$\begin{aligned} C_{abcd} &= \frac{\kappa}{3} T_{abcdmnpq} \partial^m \partial^n h^{pq} + \mathcal{O}(\kappa^2) \\ T_{abcdmnpq} &= 6\eta_{p[a}\eta_{b][c}\eta_{d][n}\eta_{q]m} - 6\eta_{m[a}\eta_{b][c}\eta_{d][n}\eta_{q]p} \\ &\quad - 6\eta_{m[a}\eta_{b]p}\eta_{n[c}\eta_{d]q} + 2\eta_{a[c}\eta_{d]b}\eta_{m[p}\eta_{n]q}. \end{aligned} \tag{16.9}$$

Taking into account that the scale factor a only depends on the conformal time η, we can write the effective action as

$$\begin{aligned} S_{\rm eff}[\tilde{g}^{\pm}_{ab}] &= S^{\rm R}_{\rm g}[\tilde{g}^{+}_{ab}] - S^{\rm R}_{\rm g}[\tilde{g}^{-}_{ab}] \\ &\quad + \frac{\alpha}{3}\kappa^2 T_{abcd}{}^{mnpq} T^{abcdrskl} \iint \left[\partial_m\partial_n h^{+}_{pq}(x)\right] \left[\partial_r\partial_s h^{-}_{kl}(x')\right] \\ &\quad \times K(x-x') d^4x d^4x' \\ &\quad + \frac{\alpha}{6}\kappa^2 T_{abcd}{}^{mnpq} T^{abcdrskl} \iint \left[\partial_m\partial_n h^{+}_{pq}(x)\right] \left[\partial_r\partial_s h^{+}_{kl}(x')\right] \\ &\quad \times K^{+}(x-x';\mu) d^4x d^4x' \\ &\quad - \frac{\alpha}{6}\kappa^2 T_{abcd}{}^{mnpq} T^{abcdrskl} \iint \left[\partial_m\partial_n h^{-}_{pq}(x)\right] \left[\partial_r\partial_s h^{-}_{kl}(x')\right] \\ &\quad \times K^{-}(x-x';\mu) d^4x d^4x'. \end{aligned} \tag{16.10}$$

The renormalized $S_{\rm g}$ terms can be separated by the contribution of the metric perturbations as

$$S^{\rm R}_{\rm g}[\tilde{g}_{ab}] = S^{(0)}_{\rm g}[a] + S^{(1)}_{\rm g}[h_{ab}] + S^{(2)}_{\rm g}[h_{ab}], \tag{16.11}$$

where, denoting the derivative with respect to η by a prime, we have

$$S_{\rm g}^{(0)}[a] = \frac{1}{\kappa^2}\int [6aa'' - 2\Lambda a^4 + \alpha\kappa^2 a^{-4}\left(2a(a')^2a'' - (a')^4\right) \\ -3\beta\kappa^2 a^{-2}(a'')^2]\, d^4x, \tag{16.12}$$

which is independent on the metric perturbation h_{ab}.

The second term,

$$S_{\rm g}^{(1)}[h_{ab}] = \frac{1}{\kappa}\int [2aa'' - (a')^2 - \Lambda a^4 \\ +\frac{1}{2}\alpha\kappa^2 a^{-4}\left(5(a')^4 - 4a(a')^2a''\right)\Big] h d^4x \\ -\frac{\beta\kappa}{2}\int a^{-3}\left(2a^2a^{(4)} - 10aa'a''' - 5a(a'')^2 + 16(a')^2a''\right) h d^4x \\ +\frac{2}{\kappa}\int [aa'' - 2(a')^2 - \alpha\kappa^2 a^{-4}\left(a(a')^2a'' - 2(a')^4\right)]\, h^{00} d^4x \\ -\beta\kappa\int a^{-3}\left(a^2a^{(4)} - 8aa'a''' - a(a'')^2 + 14(a')^2a''\right) h^{00} d^4x, \tag{16.13}$$

is linear in the metric perturbation. The third term $S_{\rm g}^{(2)}[h_{ab}]$ is quadratic in the metric perturbation and is given in Section 16.2.

16.1.1 Background Solution: de Sitter

As explained in Section 9.3, we will consider a background metric g^0_{ab} which satisfies the semiclassical Einstein equation (9.55). Assuming a FLRW background of the form $a^2(\eta)\eta_{ab}$, the dynamics for the scale factor $a(\eta)$ is entirely determined by the Friedmann equation, which corresponds to the 00 component of the semiclassical equation. It can be immediately obtained by functionally differentiating with respect to h^{00} the expression for $S_{\rm g}^{(1)}$ in Eq. (16.13). Taking into account that $\delta h(x)/\delta h^{00}(y) = -\delta^{(4)}(x-y)$, we get

$$6(a')^2 - 2\Lambda a^4 = 3\alpha\kappa^2 a^{-4}(a')^4 + 3\beta\kappa^2 a^{-3}\left[2aa'a''' - a(a'')^2 - 4(a')^2a''\right]. \tag{16.14}$$

The quantum corrections on the right-hand side of Eq. (16.14) involve higher derivative terms, which lead to spurious solutions that lie beyond the domain of validity of the effective field theory expansion, see Simon [640] and Flanagan and Wald [128]. As we have dicussed in Section 10.3 a suitable way of dealing with that problem is the order reduction method of Parker and Simon [357].

The basic idea, applied to this case, is to start with the equation $6(a')^2 - 2\Lambda a^4 = \mathcal{O}(\kappa^2)$ and write a' in terms of a up to terms of order κ^2, which is $a' = \sqrt{\Lambda/3}\, a^2 + \mathcal{O}(\kappa^2)$. Differentiating this expression and proceeding recursively, one can express higher-order derivatives such as a'' and a''' in terms of a up to order κ^2. Next, one substitutes these results into the right-hand side of Eq. (16.14) and obtains

an equation without higher-order derivative terms, but equivalent to Eq. (16.14) through order κ^2, which is anyway the order up to which that equation is valid within the effective field theory approach to perturbative quantum gravity. When this is done, the terms proportional to β cancel out. This was to be expected because they correspond to the trace of the functional derivative with respect to the metric of the term quadratic in the Ricci scalar, which is proportional to $\Box R$, whereas the equation that we are substituting implies that $R = 4\Lambda$, which is a constant.

Order reduction thus leads to

$$3(a')^2 - \Lambda\left(1 + \frac{1}{6}\alpha\kappa^2\Lambda\right)a^4 = \mathcal{O}(\kappa^4). \tag{16.15}$$

We can, therefore, define an effective cosmological constant

$$\Lambda_{\text{eff}} = \Lambda\left(1 + \frac{1}{6}\alpha\kappa^2\Lambda\right), \tag{16.16}$$

which includes a small constant shift due to the quantum backreaction of the matter fields on the background geometry. The scale factor is then given by

$$a(\eta) = -\frac{1}{H\eta} \tag{16.17}$$

with the Hubble parameter $H = \sqrt{\Lambda_{\text{eff}}/3}$. This corresponds to de Sitter spacetime, more precisely to the part which can be covered with a foliation involving flat spatial sections, i.e., the Poincaré patch.

We hereby rederive the well-known fact (see e.g., [96, 359, 973]) that one can have self-consistent solutions of Eq. (9.55) which correspond to de Sitter spacetime with a modified cosmological constant, since, owing to de Sitter invariance, the renormalized expectation value of the stress-energy tensor for the Bunch–Davies vacuum is a cosmological constant like term proportional to the metric.

16.2 Classification of Metric Perturbations

Here we classify the metric perturbations into different sectors according to their behavior under spatial transformations, as was first done by Lifshitz [338] and by Bardeen [748], and fix the gauge. In contrast to the Fadeev–Popov gauge-fixing procedure, which is implemented at the level of the generating functional and leads to the introduction of additional ghost fields to cancel residual unphysical modes, we fix the gauge completely proceeding in a way analogous to that advocated by Maldacena [974] and which is essentially equivalent to working in reduced phase-space. This ensures that only physical modes contribute.

In the Poincaré patch of de Sitter spacetime, the de Sitter metric with spatially flat sections can be written using Cartesian coordinates as $\hat{g}_{ab} = a^2(\eta)\eta_{ab}$, where

η_{ab} is the Minkowski metric, $-\infty < \eta \leq 0$ is the conformal time and $a(\eta) = (-H\eta)^{-1}$. To this metric we add perturbations h_{ab} and we have

$$\tilde{g}_{ab} = a^2(\eta) g_{ab} = a^2(\eta) \left(\eta_{ab} + \kappa h_{ab}\right) = \hat{g}_{ab} + a^2(\eta)\kappa h_{ab}, \tag{16.18}$$

the first two equations coincide with (16.3) but we now also use a hat to describe the background metric. The metric perturbation h_{ab} is called the "pseudo-graviton field" by Tsamis and Woodard [975]. In general we will use the tilde over a symbol to refer to terms defined with the perturbed physical metric $\tilde{g}_{ab}$, while hats over a symbol denote quantities defined in the de Sitter background $\hat{g}_{ab}$. Under an infinitesimal coordinate transformation of the full metric generated by a vector field $\hat{\xi}_a$, the physical perturbation $a^2 h_{ab}$ changes by the gauge transformation

$$a^2 h_{ab} \to a^2 h_{mab} + \hat{\nabla}_a \hat{\xi}_b + \hat{\nabla}_b \hat{\xi}_a, \tag{16.19}$$

which upon rescaling of the vector field $\hat{\xi}_a = a^2 \xi_a$ gives

$$h_{ab} \to h_{ab} + \partial_a \xi_b + \partial_b \xi_a + \frac{2}{\eta} \eta_{ab} \xi_0. \tag{16.20}$$

To fix the gauge, we first decompose the perturbation h_{ab} according to its transformation properties under spatial diffeomorphisms. The spatial part $h_{\alpha\beta}$ ($\alpha, \beta = 1, 2, 3$) can be decomposed as

$$h_{\alpha\beta} = h^{\mathrm{TT}}_{\alpha\beta} + 2\partial_{(\alpha} w^{\mathrm{T}}_{\beta)} + \partial_\alpha \partial_\beta \sigma + \tau \delta_{\alpha\beta}, \tag{16.21}$$

where $\delta^{\mu\alpha}\partial_\mu h^{\mathrm{TT}}_{\alpha\beta} = 0 = \delta^{\alpha\beta} h^{\mathrm{TT}}_{\alpha\beta}$ and $\delta^{\alpha\beta}\partial_\alpha w^{\mathrm{T}}_\beta = 0$. The temporal components similarly can be decomposed as

$$h_{0\alpha} = v^{\mathrm{T}}_\alpha + \partial_\alpha \psi, \qquad h_{00} = \varphi, \tag{16.22}$$

where $\delta^{\alpha\beta}\partial_\alpha v^{\mathrm{T}}_\beta = 0$. In total, we have four scalars φ, ψ, σ, τ, two transverse vectors w_α and v_α with two kinematical degrees of freedom each, and a transverse traceless tensor $h^{\mathrm{TT}}_{\alpha\beta}$ with also two kinematical degrees of freedom, which amount to 10 kinematical degrees of freedom as expected for a symmetric tensor h_{ab} in four dimensions.

Decomposing also the vector field ξ_a as

$$\xi_\alpha = \chi^{\mathrm{T}}_\alpha + \partial_\alpha \xi, \tag{16.23}$$

where $\delta^{\alpha\beta}\partial_\alpha \chi^{\mathrm{T}}_\beta = 0$, we can see that the different components change under a gauge transformation as

$$
\begin{aligned}
h^{\mathrm{TT}}_{\alpha\beta} &\to h^{\mathrm{TT}}_{\alpha\beta}, & w^{\mathrm{T}}_{\alpha} &\to w^{\mathrm{T}}_{\alpha} + \chi^{\mathrm{T}}_{\alpha},\\
\sigma &\to \sigma + 2\xi, & \tau &\to \tau + \frac{2}{\eta}\xi_0,\\
v^{\mathrm{T}}_{\alpha} &\to v^{\mathrm{T}}_{\alpha} + \chi^{\mathrm{T}\,\prime}_{\alpha}, & \psi &\to \psi + \xi' + \xi_0.\\
\varphi &\to \varphi + 2\xi_0' - \frac{2}{\eta}\xi_0.
\end{aligned}
\tag{16.24}
$$

Choosing $\chi^{\mathrm{T}}_{\alpha}$, ξ and ξ_0 appropriately, we can therefore set w_α, σ and τ to zero. Then, the spatial part – now in the transverse traceless gauge – describes the dynamical, physical perturbations, which are gravitational waves with two independent polarizations.

We take now the Fourier transform with respect to the spatial arguments

$$
h_{ab}(\eta, \vec{x}) = \int \bar{h}_{ab}(\eta, \vec{p}) e^{i\vec{p}\cdot\vec{x}} \frac{d^3p}{(2\pi)^3}. \tag{16.25}
$$

Here we also introduce the notation of using an overbar over a quantity to indicate its spatial Fourier transform, which will be used throughout the chapter. After fixing the gauge $w^{\mathrm{T}}_{\alpha} = \sigma = \tau = 0$, the metric perturbation (16.21)–(16.22) reads in terms of its Fourier modes as

$$
\bar{h}_{ab} = \delta^0_a \delta^0_b \bar{\varphi} + 2\delta^0_{(a} \left(\bar{v}^{\mathrm{T}}_{b)} + i p_{b)} \bar{\psi}\right) + \bar{h}^{\mathrm{TT}}_{ab}, \tag{16.26}
$$

where now

$$
\bar{v}_0 = p_0 = p^a \bar{v}^{\mathrm{T}}_a = p^a \bar{h}^{\mathrm{TT}}_{ab} = \bar{h}^{\mathrm{TT}}_{0a} = \eta^{ab} \bar{h}^{\mathrm{TT}}_{ab} = 0. \tag{16.27}
$$

In the next section we quantize these perturbations. As we will see, whereas the tensorial perturbations are true dynamical degrees of freedom of the gravitational field, the scalar and vectorial parts are constrained degrees of freedom since they are determined by the matter content of the theory, i.e., by the stress-energy tensor through the Einstein equations. Thus, at the operator level these perturbations could be eliminated in favor of the matter fields, as is often done in cosmology. However, in our case it is more convenient to deal with these constrained perturbations in a different way, which is nevertheless fully equivalent to their elimination at the operator level.

16.2.1 Separating Scalar, Vector and Tensor Parts

Following Fröb et al. [572] we separate the effective action into scalar, vectorial and tensorial parts. Let us first write the terms quadratic in the metric perturbation of the effective action following the notation of Eq. (9.56), which gives the effective action in powers of κ, namely

$$
S_{\mathrm{eff}}[h^{\pm}] = S^{(2)}_{\mathrm{g}}[h^{+}] - S^{(2)}_{\mathrm{g}}[h^{-}] + S^{(2)}_{\mathrm{IF}}[h^{\pm}] = S_0[h^{\pm}] + \kappa^2 S_2[h^{\pm}], \tag{16.28}
$$

where on the second equation we explicitly expand the action in powers of κ : S_0 is the free part of the gravitational action (quadratic in $h^{\pm}$ and of order κ^0) and S_2 are all the remaining terms of order κ^2 (quadratic in $h^{\pm}$), including the nonlocal terms in the influence action $S_{\rm IF}[h^{\pm}]$, which will be written as $S_{\rm IF}[h^{\pm}] \equiv \kappa^2 S_2^{\rm K}[h^{\pm}]$ to explicitly indicate the dependence on the kernels.

Inserting the decomposition of the metric perturbation (16.26) into expressions (16.5) and (16.6), we see that the tensorial, vectorial and scalar perturbations decouple. Specifically, when expanded in powers of κ as done in Eq. (9.56), the effective action (16.28), which only contains terms quadratic in $h^{\pm}$, can be written as

$$\begin{aligned} S_0[h] &= S_0^{\rm S}[\varphi,\psi] + S_0^{\rm V}[v_m^{\rm T}] + S_0^{\rm TT}[h_{mn}^{\rm TT}], \\ S_2[h^{\pm}] &= S_2^{\rm g}[h^{+}] - S_2^{\rm g}[h^{-}] + S_2^{\rm K}[h^{\pm}], \\ S_2^{\rm g}[h] &= S_2^{\rm g,S}[\varphi,\psi] + S_2^{\rm g,V}[v_m^{\rm T}] + S_2^{\rm g,TT}[h_{mn}^{\rm TT}], \end{aligned} \tag{16.29}$$

where the scalar (S), vectorial (V) and tensorial (TT) parts have been separated, except for the nonlocal term $S_2^{\rm K}$, which will be decomposed below.

For the scalar part we have

$$\begin{aligned} S_0^{\rm S}[\varphi,\psi] &= -\frac{3}{2}H^2\int a^4\varphi^2 d^4x + 2\int Ha^3\varphi\Delta\psi d^4x, \\ S_2^{\rm g,S}[\varphi,\psi] &= \frac{5}{2}\alpha H^4\int a^4\varphi^2 d^4x - 2\left(3\alpha-\beta\right)H^2\int Ha^3\varphi\Delta\psi d^4x \\ &\quad + \alpha\int\left[\frac{1}{2}\left(\Delta\varphi\right)^2 - 2\left(\Delta\varphi\right)\Delta\psi' + 2\left(\Delta\psi'\right)^2\right]\ln a d^4x \\ &\quad + \beta\int\Big[-\frac{7}{4}H^2a^2\varphi\Delta\varphi - \frac{3}{4}H^2a^2\left(\varphi'\right)^2 \\ &\quad - \frac{1}{12}\left(\Delta\varphi\right)^2 - Ha\varphi\Delta\psi'' + Ha\left(\Delta\varphi\right)\Delta\psi \\ &\quad + \frac{1}{3}\left(\Delta\varphi\right)\Delta\psi' - 2H^2a^2\left(\Delta\psi\right)^2 - \frac{1}{3}\left(\Delta\psi'\right)^2\Big]d^4x. \end{aligned} \tag{16.30}$$

where a prime denotes a derivative with respect to the conformal time η, $\Delta = \eta^{\mu\nu}\partial_\mu\partial_\nu$, $\Box = \eta^{mn}\partial_m\partial_n = -\partial_\eta^2 + \Delta$, and $H = \sqrt{\Lambda_{\rm eff}/3}$ where $\Lambda_{\rm eff}$ is defined in Eq. (16.16).

For the vectorial part we have

$$\begin{aligned} S_0^{\rm V}[v_m^{\rm T}] &= -\frac{1}{2}\int a^2 v_{\rm T}^m\Delta v_m^{\rm T} d^4x, \\ S_2^{\rm g,V}[v_m^{\rm T}] &= \frac{1}{2}\left(\alpha+2\beta\right)H^2\int a^2 v_{\rm T}^m\Delta v_m^{\rm T} d^4x \\ &\quad - \frac{3}{2}\alpha\int\left[\left(\Delta v_{\rm T}^m\right)\left(\Delta v_m^{\rm T}\right) + v_{\rm T}'^m\Delta v_m^{\rm T\prime}\right]\ln a d^4x. \end{aligned} \tag{16.31}$$

For the tensorial part we get

$$S_0^{\rm TT}[h_{mn}^{\rm TT}] = -\frac{1}{4}\int a^2(\partial_s h_{mn}^{\rm TT})(\partial^s h_{\rm TT}^{mn})d^4x,$$
$$S_2^{\rm g,TT}[h_{mn}^{\rm TT}] = -\frac{1}{4}(5\alpha-2\beta)H^2\int a^2(\partial_s h_{mn}^{\rm TT})(\partial^s h_{\rm TT}^{mn})d^4x$$
$$-\frac{3}{2}\alpha H^2\int a^2 h_{mn}^{\rm TT\prime}h_{\rm TT}^{\prime mn}d^4x+\frac{3}{4}\alpha\int(\Box h_{mn}^{\rm TT})(\Box h_{\rm TT}^{mn})\ln a d^4x. \tag{16.32}$$

The nonlocal part $S_2^{\rm K}$ can also be decomposed into scalar, vectorial and tensorial parts when we take into account that the product of the two linearized Weyl tensors, integrated with a kernel as in Eq. (16.5), can be decomposed in that way. In fact, for the scalar sector we have

$$C^{+abcd}(x)C^{-}_{abcd}(y)\big|_{\rm S} = \frac{\kappa^2}{3}\Big[\left(\Delta\varphi^+(x)\right)\left(\Delta\varphi^-(y)\right)+4\left(\Delta\psi^{+\prime}(x)\right)\left(\Delta\psi^{-\prime}(y)\right)$$
$$-2\left(\Delta\varphi^+(x)\right)\left(\Delta\psi^{-\prime}(y)\right)$$
$$-2\left(\Delta\psi^{+\prime}(x)\right)\left(\Delta\varphi^-(y)\right)\Big], \tag{16.33}$$

for the vectorial sector we find

$$C^{+abcd}(x)C^{-}_{abcd}(y)\big|_{\rm V} = -\frac{\kappa^2}{2}\eta^{ab}\Big[\left(\Box v_a^{+\rm T}(x)\right)\left(\Delta v_b^{-\rm T}(y)\right)$$
$$+\left(\Delta v_a^{+\rm T}(x)\right)\left(\Box v_b^{-\rm T}(y)\right)\Big], \tag{16.34}$$

and for the tensorial sector we have

$$C^{+abcd}(x)C^{-}_{abcd}(y)\big|_{\rm TT} = \frac{\kappa^2}{2}\left(\Box h_{ab}^{+\rm TT}(x)\right)\eta^{ac}\eta^{bd}\left(\Box h_{cd}^{-\rm TT}(y)\right). \tag{16.35}$$

Thus, up to the order at which we are working and which does not include nonlinear interactions for the metric perturbations, the effective action separates into three independent parts corresponding to the scalar, vectorial and tensorial perturbations of the metric. This in turn implies that the two-point function can also be written as the sum of scalar, vectorial and tensorial parts:

$$G^{-+}_{abc'd'}(x,x') = G^{-+\rm S}_{abc'd'}(x,x')+G^{-+\rm V}_{abc'd'}(x,x')+G^{-+\rm TT}_{abc'd'}(x,x'), \tag{16.36}$$

where we have now restored the tensorial indices explicitly. In Section 16.3 we calculate the one-loop contribution for the tensorial sector and in Section 16.5 the vectorial and scalar sectors.

The two-point function (16.36) is given in the exact gauge defined by Eq. (16.26), and is not de Sitter-invariant. However, this should not be taken to conclude that the corresponding "in" vacuum state is not de Sitter-invariant, since it results from having chosen a gauge which is not de Sitter invariant. Indeed, the result for the gauge-invariant Riemann tensor correlator obtained in Chapter 17 will instead be manifestly de Sitter invariant.

16.3 Two-Point Functions for Tensor Metric Perturbations

Let us consider now the tensorial metric perturbations around the mean background. From now on in this section we restrict to tensor perturbations, which fulfill the TT gauge conditions, $h = 0$, $h^{0n} = 0$, $\partial_m h^{mn} = 0$, and we drop the TT superscript to simplify the notation. Inserting the background scale factor solution (16.17) into the gravitational action (16.13), the linear term $S_{\rm g}^{(1)}$ vanishes as expected. From Eq. (16.32) the quadratic term of the gravitational action becomes

$$S_{\rm g}^{(2)}[h_{ab}] = -\frac{1}{4}\left(1+(5\alpha-2\beta)\kappa^2H^2\right)\int a^2(\partial_s h_{mn})(\partial^s h^{mn})d^4x \\ -\frac{3}{2}\alpha\kappa^2H^2\int a^2 h'_{mn}h'^{mn}d^4x+\frac{3}{4}\alpha\kappa^2\int(\Box h_{mn})(\Box h^{mn})\ln a d^4x. \tag{16.37}$$

Substituting (16.35) into Eq. (16.5) the quadratic part of the effective action can then be written as

$$S_{\rm eff}[\tilde{g}^{\pm}_{ab}] = S_{\rm g}^{(2)}[h^{+}_{ab}] - S_{\rm g}^{(2)}[h^{-}_{ab}] \\ +\frac{3}{2}\alpha\kappa^2\iint\left(\Box h^{+}_{pq}(x)\right)\left(\Box' h^{-pq}(x')\right)K(x-x')d^4xd^4x' \\ +\frac{3}{4}\alpha\kappa^2\iint\left(\Box h^{+}_{pq}(x)\right)\left(\Box' h^{+pq}(x')\right)K^{+}(x-x';\bar{\mu})d^4xd^4x' \\ -\frac{3}{4}\alpha\kappa^2\iint\left(\Box h^{-}_{pq}(x)\right)\left(\Box' h^{-pq}(x')\right)K^{-}(x-x';\bar{\mu})d^4xd^4x'. \tag{16.38}$$

Note that we can alternatively shift the last term in the gravitational action (16.37) into the kernels $K^{\pm}$, which amounts to replacing $\bar{\mu}$ with $\bar{\mu}a = -\bar{\mu}/(H\eta)$. Eq. (16.38) corresponds to the effective action expanded in powers of κ given by the expression (9.56) and is the starting point for the computation of the metric correlations, which is the next step.

16.3.1 The Tree Level

To calculate the two-point function to first order in V (which is proportional to κ^2), as given by expression (9.66), we need to compute the lowest order propagator. For this we note that the gravitational action (16.37) is given at lowest order by

$$S_{\rm g}^{(2)}(\kappa=0) = S_0 = -\frac{1}{4}\int a^2(\partial^s h^{mn})(\partial_s h_{mn})d^4x. \tag{16.39}$$

We follow Ford and Parker [346] and express this as the action of two scalar fields, corresponding to the two physical polarizations of the tensor perturbations.

We first Fourier transform the field with respect to the spatial coordinates as in Eq. (16.25) and note that the TT gauge-fixing conditions for tensor perturbations correspond to

$$\bar{h} = 0, \qquad \bar{h}^{0n} = 0, \qquad p_m \bar{h}^{mn} = 0. \tag{16.40}$$

Next, we choose a right-handed orthogonal set of three real vectors $\vec{e}_A(\vec{p})$ such that $\vec{p} = |\vec{p}|\vec{e}_3(\vec{p})$, and define the following two real scalars corresponding to the two polarizations:

$$\begin{aligned} \bar{h}_+(\eta, \vec{p}) &= \frac{i}{2} \bar{h}_{mn}(\eta, \vec{p}) \left[e_1^m(\vec{p}) e_1^n(\vec{p}) - e_2^m(\vec{p}) e_2^n(\vec{p}) \right] \\ \bar{h}_\times(\eta, \vec{p}) &= \bar{h}_{mn}(\eta, \vec{p}) e_1^m(\vec{p}) e_2^n(\vec{p}). \end{aligned} \tag{16.41}$$

As is well known, such a definition is not possible in a way that would be continous for every orientation of $\vec{p}$. However, with respect to the transformation $\vec{p} \to -\vec{p}$ one can adopt consistently the definition

$$\vec{e}_3(-\vec{p}) = -\vec{e}_3(\vec{p}), \qquad \vec{e}_1(-\vec{p}) = \vec{e}_2(\vec{p}), \qquad \vec{e}_2(-\vec{p}) = \vec{e}_1(\vec{p}). \tag{16.42}$$

Different definitions would be possible, but lead to the same result for the graviton propagator. The relation (16.41) can be inverted to give

$$\begin{aligned} \bar{h}^{mn}(\eta, \vec{p}) = &-i \bar{h}_+(\eta, \vec{p}) \left[e_1^m(\vec{p}) e_1^n(\vec{p}) - e_2^m(\vec{p}) e_2^n(\vec{p}) \right] \\ &+ \bar{h}_\times(\eta, \vec{p}) \left[e_1^m(\vec{p}) e_2^n(\vec{p}) + e_2^m(\vec{p}) e_1^n(\vec{p}) \right]. \end{aligned} \tag{16.43}$$

Using the reality of h_{mn}, i.e., $\bar{h}_{mn}(\eta, -\vec{p}) = \bar{h}^*_{mn}(\eta, \vec{p})$, it follows that

$$\begin{aligned} S_0 = \frac{1}{2} \iint a^2(\eta) \Big[&\bar{h}'_+(\eta, \vec{p}) \bar{h}^{*\prime}_+(\eta, \vec{p}) - \vec{p}^{\,2} \bar{h}_+(\eta, \vec{p}) \bar{h}^*_+(\eta, \vec{p}) \\ &+ \bar{h}'_\times(\eta, \vec{p}) \bar{h}^{*\prime}_\times(\eta, \vec{p}) - \vec{p}^{\,2} \bar{h}_\times(\eta, \vec{p}) \bar{h}^*_\times(\eta, \vec{p}) \Big] \frac{d^3p}{(2\pi)^3} d\eta, \end{aligned} \tag{16.44}$$

and after inverting the Fourier transform it reads

$$S_0 = -\frac{1}{2} \int a^2 \left[(\partial^s h_+) (\partial_s h_+) + (\partial^s h_\times) (\partial_s h_\times) \right] d^4x, \tag{16.45}$$

which is the action for a pair of minimally coupled real scalar fields in a spatially flat FLRW background, whose canonically conjugate momenta are given by

$$\pi_+ = a^2 h'_+, \qquad \pi_\times = a^2 h'_\times. \tag{16.46}$$

As usual we can quantize these fields by imposing that the field operators satisfy the canonical commutation relations at equal times,

$$[\hat{h}_+(\eta, \vec{x}), \hat{\pi}_+(\eta, \vec{y})] = [\hat{h}_\times(\eta, \vec{x}), \hat{\pi}_\times(\eta, \vec{y})] = i\delta^3(\vec{x} - \vec{y}), \tag{16.47}$$

with all other commutators vanishing. Fourier transforming the spatial coordinates, this is equivalent to

$$[\bar{\hat{h}}_+(\eta,\vec{p}),\bar{\hat{\pi}}_+(\eta,\vec{q})] = [\bar{\hat{h}}_\times(\eta,\vec{p}),\bar{\hat{\pi}}_\times(\eta,\vec{q})] = i(2\pi)^3\delta^3(\vec{p}+\vec{q}). \tag{16.48}$$

From the action it follows that the equations of motion for the operators in the Heisenberg picture are

$$\Big(\Box - 2Ha\partial_\eta\Big)\hat{h}_+ = \Big(\Box - 2Ha\partial_\eta\Big)\hat{h}_\times = 0. \tag{16.49}$$

To calculate the propagator, we need to find a complete set of solutions to those equations, which will determine the expansion of the perturbation operators $\hat{h}_+$ and $\hat{h}_\times$ and the associated vacuum state. In the following, we perform the analysis for $\hat{h}_+$ and trivially extend the results for $\hat{h}_\times$. Fourier transforming the first equation in (16.49) with respect to spatial coordinates, it becomes

$$\bar{\hat{h}}_+''(\eta,\vec{p}) - \frac{2}{\eta}\bar{\hat{h}}_+'(\eta,\vec{p}) + \vec{p}^{\,2}\bar{\hat{h}}_+(\eta,\vec{p}) = 0, \tag{16.50}$$

whose general solution is given by

$$\bar{\hat{h}}_+(\eta,\vec{p}) = C(\vec{p})\hat{a}_+(\vec{p})(|\vec{p}|\eta - i)e^{-i|\vec{p}|\eta} + C^*(-\vec{p})\hat{a}_+^\dagger(-\vec{p})(|\vec{p}|\eta + i)e^{i|\vec{p}|\eta} \tag{16.51}$$

with an arbitrary operator $\hat{a}_+$ and a normalization constant C, and where we have used the reality condition for $h_+(x)$, which corresponds to hermiticity for $\hat{h}_+(x)$. For the canonical momentum, we get

$$\bar{\hat{\pi}}_+(\eta,\vec{p}) = -i\frac{\vec{p}^{\,2}}{H^2\eta}\left[C(\vec{p})\hat{a}_+(\vec{p})e^{-i|\vec{p}|\eta} - C^*(-\vec{p})\hat{a}_+^\dagger(-\vec{p})e^{i|\vec{p}|\eta}\right]. \tag{16.52}$$

The canonical commutation relations (16.48), together with the vanishing of the equal-time commutators for $\hat{h}_+$ and $\hat{\pi}_+$ with themselves, are valid if and only if

$$\begin{aligned} C(\vec{p})C^*(\vec{p}) &= C(-\vec{p})C^*(-\vec{p}) \\ [\hat{a}_+(\vec{p}),\hat{a}_+(\vec{q})] &= [\hat{a}_+^\dagger(\vec{p}),\hat{a}_+^\dagger(\vec{q})] = 0 \\ [\hat{a}_+(\vec{p}),\hat{a}_+^\dagger(\vec{q})] &= \frac{(2\pi)^3\delta^3(\vec{p}-\vec{q})H^2}{2|\vec{p}|^3C(\vec{p})C^*(\vec{p})}. \end{aligned} \tag{16.53}$$

By choosing $C(\vec{p}) = H\,(2|\vec{p}|^3)^{-\frac{1}{2}}$, the operators $\hat{a}_+$ and $\hat{a}_+^\dagger$ fulfill the standard commutation relations for creation and annihilation operators. We then define the free field vacuum $|0\rangle$ by

$$\hat{a}_+(\vec{p})|0\rangle = \hat{a}_\times(\vec{p})|0\rangle = 0. \tag{16.54}$$

This vacuum which (for $|\vec{p}|\eta \gg 1$) resembles the Minkowski vacuum with respect to the choice of mode functions (16.51), is the Bunch–Davies vacuum for the gravitons [319], and we will use it in the following; a subtlety concerning infrared divergences is briefly discussed below.

The lowest order two-point function, $\langle 0|\hat{h}^{ab}(x)\hat{h}^{c'd'}(y)|0\rangle$, of the tensor perturbations can now be evaluated directly using the decomposition (16.43) into the + and × polarizations and the mode expansion (16.51). A straightforward calculation then leads to

$$\langle 0|\bar{\hat{h}}^{ab}(\eta,\vec{p})\bar{\hat{h}}^{c'd'}(\eta',\vec{q})|0\rangle = (2\pi)^3\delta^3(\vec{p}+\vec{q})f(\eta,\eta',|\vec{p}|)P^{abc'd'}(\vec{p}), \tag{16.55}$$

where

$$f(\eta,\eta',|\vec{p}|) = \frac{H^2}{2|\vec{p}|^3}(|\vec{p}|\eta - i)(|\vec{p}|\eta' + i)e^{-i|\vec{p}|(\eta-\eta')} = f^*(\eta',\eta,|\vec{p}|), \tag{16.56}$$

and

$$\begin{aligned} P^{abc'd'}(\vec{p}) &= \left(e_1^a(\vec{p})e_2^b(\vec{p}) + e_2^a(\vec{p})e_1^b(\vec{p})\right)\left(e_1^{c'}(\vec{p})e_2^{d'}(\vec{p}) + e_2^{c'}(\vec{p})e_1^{d'}(\vec{p})\right) \\ &\quad + \left(e_1^a(\vec{p})e_1^b(\vec{p}) - e_2^a(\vec{p})e_2^b(\vec{p})\right)\left(e_1^{c'}(\vec{p})e_1^{d'}(\vec{p}) - e_2^{c'}(\vec{p})e_2^{d'}(\vec{p})\right) \\ &= P^{ad'}P^{bc'} + P^{ac'}P^{bd'} - P^{ab}P^{c'd'} \end{aligned} \tag{16.57}$$

is the polarization tensor which can be written in terms of the projection tensor P^{ab} defined by

$$P^{ab} = \eta^{ab} + \delta_0^a\delta_0^b - \frac{p^a p^b}{\vec{p}^2}, \tag{16.58}$$

where $p^a = (0,\vec{p})$ in the coordinate system that we are using. The projection tensor P^{ab} satisfies

$$P^{ab}\eta_{ab} = 2, \qquad P^{ab}P^{cd}\eta_{bc} = P^{ad}. \tag{16.59}$$

From Eq. (16.59) we can calculate the lowest order CTP propagators for the graviton. They are given by

$$\bar{G}^{0AB}_{abc'd'}(\eta,\eta',\vec{p}) = \bar{G}^{0AB}(\eta,\eta',\vec{p})P_{abc'd'}(\vec{p}), \tag{16.60}$$

where the components of the 2×2 matrix $\bar{G}^0$ are

$$\begin{aligned} \bar{G}^{0++}(\eta,\eta',\vec{p}) &= -i\left[\Theta(\eta-\eta')f(\eta,\eta',|\vec{p}|) + \Theta(\eta'-\eta)f^*(\eta,\eta',|\vec{p}|)\right] \\ \bar{G}^{0+-}(\eta,\eta',\vec{p}) &= -if^*(\eta,\eta',|\vec{p}|) \\ \bar{G}^{0-+}(\eta,\eta',\vec{p}) &= -if(\eta,\eta',|\vec{p}|) \\ \bar{G}^{0--}(\eta,\eta',\vec{p}) &= -i\left[\Theta(\eta-\eta')f^*(\eta,\eta',|\vec{p}|) + \Theta(\eta'-\eta)f(\eta,\eta',|\vec{p}|)\right]. \end{aligned} \tag{16.61}$$

Since f goes like $|\vec{p}|^{-3}$ for small $|\vec{p}|$, this two-point function must be truncated at some small momentum $|\vec{p}| = \varepsilon$ to have a well-defined inverse Fourier transform. The two-point function in coordinate space diverges then as $\ln\varepsilon$ for $\varepsilon \to 0$, as was noted by a number of authors; see, e.g., [976, 568]. This was expected since the same problem occurs for a massless minimally coupled scalar field in de Sitter space [518, 519]. However, similarly to the stress-energy tensor

correlation function for a scalar field [520], gauge-invariant correlation functions of observables characterizing the local geometry in a finite-size region, such as the curvature tensor correlation functions, are finite and do not grow at large distances [970, 971, 977]. We will address this issue in Chapter 17.

16.3.2 The One-Loop Order

We can now use expression (9.66) for the two-point metric perturbations to calculate the one-loop order graviton propagators which are of order κ^2 in our perturbative scheme. As explained at the end of Section 9.3.3, this is easily accomplished by replacing the fields h_{ab} in the effective action (16.38) with the lowest order propagators. Fourier transforming the spatial coordinates, we get

$$\bar{G}^{AB}_{abc'd'}(\eta,\eta',\vec{p}) = \bar{G}^{0AB}_{abc'd'}(\eta,\eta',\vec{p}) + P_{abc'd'}(\vec{p})\kappa^2\bar{G}^{1AB}(\eta,\eta',\vec{p}) \tag{16.62}$$

where

$$\begin{aligned}
\bar{G}^{1AB}(\eta,\eta',\vec{p}) = {} & (5\alpha-2\beta)\vec{p}^{\,2}\int \tau^{-2}\bar{G}^0_{AM}(\eta,\tau,\vec{p})\epsilon_{MN}\bar{G}^0_{NB}(\tau,\eta',\vec{p})d\tau \\
& + (\alpha+2\beta)\int \tau^{-2}(\partial_\tau\bar{G}^0_{AM}(\eta,\tau,\vec{p}))\epsilon_{MN}(\partial_\tau\bar{G}^0_{NB}(\tau,\eta',\vec{p}))d\tau \\
& + 3\alpha\int \left[(\partial_\tau^2+\vec{p}^{\,2})\bar{G}^0_{AM}(\eta,\tau,\vec{p})\right]\epsilon_{MN}\left[(\partial_\tau^2+\vec{p}^{\,2})\bar{G}^0_{NB}(\tau,\eta',\vec{p})\right] \\
& \times \ln(-H\tau)d\tau \\
& - \frac{3}{2}\alpha\iint \left[(\partial_\tau^2+\vec{p}^{\,2})\bar{G}^0_{AM}(\eta,\tau,\vec{p})\right]V_{MN}(\tau,\tau',\vec{p}) \\
& \times \left[(\partial_{\tau'}^2+\vec{p}^{\,2})\bar{G}^0_{NB}(\tau',\eta',\vec{p})\right]d\tau d\tau'.
\end{aligned} \tag{16.63}$$

Here we have introduced the matrix $\epsilon_{MN} = \mathrm{diag}(1,-1)$ and

$$\begin{aligned}
V_{MN}(\tau,\tau',\vec{p}) = {} & \begin{pmatrix} \bar{L}(\tau-\tau',\vec{p};\bar{\mu}) & \bar{D}(\tau-\tau',\vec{p}) \\ -\bar{D}(\tau-\tau',\vec{p}) & -\bar{L}(\tau-\tau',\vec{p};\bar{\mu}) \end{pmatrix} \\
& + i\bar{N}(\tau-\tau',\vec{p})\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.
\end{aligned} \tag{16.64}$$

This is written in terms of the Fourier transforms of the following real-valued kernels, in terms of which the complex-valued kernels in (16.7) can be decomposed:

$$\begin{aligned}
L(x;\bar{\mu}) &= -\int \ln\left|\frac{p^2}{\bar{\mu}^2}\right| e^{ip\cdot x}\frac{d^4p}{(2\pi)^4} = L(-x;\bar{\mu}) \\
N(x) &= \pi\int \Theta(-p^2)e^{ip\cdot x}\frac{d^4p}{(2\pi)^4} = N(-x) \\
D(x) &= i\pi\int \Theta(-p^2)\,\mathrm{sgn}\,p^0\, e^{ip\cdot x}\frac{d^4p}{(2\pi)^4} = -D(-x).
\end{aligned} \tag{16.65}$$

The Fourier transforms of those kernels can be computed with the help of Section 16.7.2 and are given by

$$\begin{aligned}
\bar{L}(\eta-\eta',\vec{p}) &= \cos\left[|\vec{p}|(\eta-\eta')\right]\mathcal{P}'\frac{1}{|\eta-\eta'|}\\
\bar{D}(\eta-\eta',\vec{p}) &= \cos\left[|\vec{p}|(\eta-\eta')\right]\mathcal{P}\frac{1}{\eta-\eta'}\\
\bar{N}(\eta-\eta',\vec{p}) &= -\frac{\sin\left[|\vec{p}|(\eta-\eta')\right]}{\eta-\eta'}+\pi\delta(\eta-\eta').
\end{aligned} \tag{16.66}$$

After a long calculation which involves computing the integrals in Eq. (16.63) using the $i\epsilon$ prescription, as explained in some detail in Section 16.7.3, we get the two-point function of the metric perturbations

$$\begin{aligned}
\langle \mathrm{in}|\bar{\hat{h}}^{ab}(\eta,\vec{p})\bar{\hat{h}}^{m'n'}(\eta',\vec{q})|\mathrm{in}\rangle = (2\pi)^3\delta^3(\vec{p}+\vec{q})P^{abm'n'}\times\Bigg[&f(\eta,\eta',\vec{p})\\
&\times\left(1+6\alpha\kappa^2H^2\ln\left(\frac{\bar{\mu}}{H}\right)-(5\alpha-2\beta)\,\kappa^2H^2\right)\\
&+\frac{3}{2}\alpha\kappa^2H^4\Big(I_1(\eta,\eta',\vec{p})-I_2(\eta,\eta',\vec{p})-I_2^*(\eta',\eta,\vec{p})\Big)\\
&+\frac{3}{2}\alpha\kappa^2H^4\Big(I_3(\eta,\eta',\vec{p})-I_4(\eta,\eta',\vec{p})+I_5(\eta,\eta',\vec{p})\Big)\Bigg]\\
&+\mathcal{O}(\kappa^4),
\end{aligned} \tag{16.67}$$

where

$$\begin{aligned}
I_1(\eta,\eta',\vec{p}) &= 2|\vec{p}|^{-1}\eta\eta'e^{-i|\vec{p}|(\eta-\eta')}\\
I_2(\eta,\eta',\vec{p}) &= |\vec{p}|^{-3}e^{i|\vec{p}|(\eta+\eta')}\left(|\vec{p}|\eta+i\right)\left(|\vec{p}|\eta'+i\right)\times\\
&\quad\times\left[\mathrm{Ein}\left(-2i|\vec{p}|\eta\right)+\ln\left(2i|\vec{p}|\eta\right)+\gamma\right]\\
&\quad+|\vec{p}|^{-3}e^{-i|\vec{p}|(\eta-\eta')}\left(|\vec{p}|\eta-i\right)\left(|\vec{p}|\eta'+i\right)\ln\left(-2|\vec{p}|\eta\right)\\
I_3(\eta,\eta',\vec{p}) &= |\vec{p}|^{-3}e^{-i|\vec{p}|(\eta-\eta')}\left(|\vec{p}|\eta-i\right)\left(|\vec{p}|\eta'+i\right)\left[\ln\left(2i|\vec{p}|(\eta-\eta')\right)+\gamma\right]\\
I_4(\eta,\eta',\vec{p}) &= |\vec{p}|^{-3}e^{i|\vec{p}|(\eta-\eta')}\left(|\vec{p}|\eta+i\right)\left(|\vec{p}|\eta'-i\right)\times\\
&\quad\times\left[\mathrm{Ein}\left(-2i|\vec{p}|(\eta-\eta')\right)+\ln\left(2i|\vec{p}|(\eta-\eta')\right)+\gamma\right]\\
I_5(\eta,\eta',\vec{p}) &= \eta^2(\eta')^2\left[\bar{N}(\eta-\eta',\vec{p})-i\bar{D}(\eta-\eta',\vec{p})\right].
\end{aligned} \tag{16.68}$$

This is the exact result of the two-point function for tensor metric perturbations including one-loop corrections from conformal fields. It is given for arbitrary pairs of times, it is finite for non-coincident points and (apart from a global factor $|\vec{p}|^{-3}$) invariant under the simultaneous rescaling $\{\vec{p}\to\lambda^{-1}\vec{p};\eta,\eta'\to\lambda\eta,\lambda\eta'\}$, which is a necessary condition for de Sitter invariance.

Eq. (16.67) is valid for the particular choice of the renormalization scale $a_1^{\mathrm{R}}(\bar{\mu}) = 0$, as explained in Section 16.1. For arbitrary values of the renormalization scale one should make the following replacement in Eq. (16.67):

$$6\alpha\kappa^2 H^2 \ln\left(\frac{\bar{\mu}}{H}\right) \longrightarrow 6\kappa^2 H^2 \left[N a_1^{\mathrm{R}}(\mu) + \alpha \ln\left(\frac{\mu}{H}\right)\right]. \tag{16.69}$$

Note that this guarantees the invariance of the two-point function under the renormalization group, as follows from Eq. (16.8).

16.3.3 Equal Times and Power Spectrum

In cosmology, two-point correlations are often expressed in terms of the power spectrum. To calculate it, one needs to take the equal time limit $\eta' \to \eta$ of the expectation value (16.67). However, this expectation value is not a regular function but a distribution with singular support and the equal-time limit does not make sense in general. Therefore, the concept of a power spectrum is, strictly speaking, ill-defined at the one-loop level. See Fröb et al. [491] for a more detailed discussion of this point. In order to compare with other results in the literature, however, we can take the equal time limit of the regular part, which amounts to replace the kernels appearing in I_5 by

$$\begin{aligned} \bar{N}(\eta, \eta', \vec{p}) &\to -\frac{\sin\left[|\vec{p}|(\eta - \eta')\right]}{\eta - \eta'} \\ \bar{D}(\eta - \eta', \vec{p}) &\to 0. \end{aligned} \tag{16.70}$$

Following Section 18 of Mukhanov et al. [753], the power spectrum is defined to be a suitable normalization factor times the spatial Fourier transform of the contracted correlation function at equal times,

$$\begin{aligned} \delta^2(|\vec{p}|, \eta) &= \frac{\kappa^2}{4(2\pi)^3} |\vec{p}|^3 \eta_{am'} \eta_{bn'} \int \langle \mathrm{in}| h^{ab}(\eta, \vec{x}) h^{m'n'}(\eta, \vec{0}) |\mathrm{in}\rangle e^{-i\vec{p}\cdot\vec{x}} d^3x \\ &= \frac{\kappa^2}{32\pi^3} |\vec{p}|^3 \eta_{am'} \eta_{bn'} \int \langle \mathrm{in}| \bar{h}^{ab}(\eta, \vec{p}) \bar{h}^{m'n'}(\eta, \vec{q}) |\mathrm{in}\rangle \frac{d^3q}{(2\pi)^3}, \end{aligned} \tag{16.71}$$

which gives

$$\begin{aligned} \delta^2(|\vec{p}|, \eta) = \frac{\kappa^2 H^2}{32\pi^3} \Bigg[& 2\left(1 + \vec{p}^2\eta^2\right) \left(1 + 6\alpha\kappa^2 H^2 \ln\left(\frac{\bar{\mu}}{H}\right) - \left(5\alpha - 2\beta\right)\kappa^2 H^2\right) \\ & - 12\alpha\kappa^2 H^2 \Re\left[e^{2i|\vec{p}|\eta} \left(|\vec{p}|\eta + i\right)^2 \left[\mathrm{Ein}\left(-2i|\vec{p}|\eta\right) + \ln\left(2i|\vec{p}|\eta\right) + \gamma\right]\right] \\ & - 12\alpha\kappa^2 H^2 \left(1 + \vec{p}^2\eta^2\right) \ln\left(-2|\vec{p}|\eta\right) + 6\alpha\kappa^2 H^2 \vec{p}^2\eta^2 \left(2 - \vec{p}^2\eta^2\right) \Bigg] \\ & + \mathcal{O}(\kappa^6). \end{aligned} \tag{16.72}$$

Once again, for an arbitrary value of the renormalization scale μ one needs the replacement (16.69).

Let us consider two important limits of Eq. (16.72). It is useful to write it in terms of the physical momentum of the physical space-time $\tilde{g}_{ab}$, which is

$$\tilde{p} = \frac{|\vec{p}|}{a} = -H|\vec{p}|\eta. \tag{16.73}$$

The power spectrum depends on $|\vec{p}|$ and η only through the combination $\tilde{p}$, so that it is time-independent in the physical spacetime.

For *subhorizon* modes, i.e., modes with $\tilde{p} \gg H$, we should then recover flat-space behavior, as the curvature of spacetime should have no effect on those modes. Taking therefore the limit $H \to 0$, we get from Eq. (16.72)

$$\delta^2(\tilde{p}) \to \frac{\kappa^2}{16\pi^3}\tilde{p}^2\left(1 - 3\alpha\kappa^2\tilde{p}^2 + \mathcal{O}(\kappa^4)\right) + \mathcal{O}\left(\frac{H}{\tilde{p}}\right), \tag{16.74}$$

where we have used the asymptotic expansion of the Ein function from Section 16.7.1. We therefore have a power correction to the Minkowski spectrum.

On the other hand, for *superhorizon* modes, i.e., modes with $\tilde{p} \ll H$, we take the limit $\tilde{p} \to 0$ to find

$$\delta^2(\tilde{p}) \to \frac{\kappa^2 H^2}{16\pi^3}\left[1 + 6\alpha\kappa^2 H^2\left(\ln\left(\frac{\bar{\mu}}{H}\right) + \gamma\right) - (5\alpha - 2\beta)\kappa^2 H^2 + \mathcal{O}(\kappa^4)\right] + \mathcal{O}\left(\frac{\tilde{p}}{H}\right), \tag{16.75}$$

which corresponds to a small constant shift of the scale-invariant power spectrum for tensor perturbations in de Sitter obtained at tree level.

This result is in contrast with the logarithmic dependence on the comoving momentum $|\vec{p}|$ previously found for one-loop corrections to tensor [957] and scalar [427, 978] perturbations, but it is in agreement with the results on loop corrections for scalar perturbations due to minimally coupled fields obtained by Senatore and Zaldarriaga [567], where the origin of this discrepancy was identified.

16.4 Intrinsic/Induced Fluctuations and Secular Terms

In the one-loop two-point function of the tensor metric perturbations found, there is no dependence on the initial times $t_0^{\pm} = t_0(1 \mp i\epsilon)$ after taking the limit $t_0 \to -\infty$. The no dependence on t_0 is crucial for obtaining a result which is invariant (aside from a global factor $|\vec{p}|^{-3}$) under the simultaneous rescaling $\{\vec{p} \to \lambda^{-1}\vec{p};\, \eta, \eta' \to \lambda\eta, \lambda\eta'\}$, which is a necessary condition for the de Sitter invariance of the result.

It is worth mentioning that in the stochastic gravity framework described in Section 10.4, one rewrites the imaginary part of $S_{\rm eff}[h^{\pm}]$ using an auxiliary

stochastic field ξ_{ab} to arrive at a real *stochastic effective action* as in (10.42), from which manifestly real and causal equations of motion for the perturbation h_{ab} can be derived. An alternative expression for $\bar{G}_{AB}$ can then be given as a sum of two terms called *intrinsic* and *induced* fluctuations, where the induced fluctuations are a direct consequence of the stochastic source as in (10.16). It is then important to consider the contributions of both terms, since otherwise secular terms occur in the two-point function for the metric perturbations. However, in order to implement correctly the prescription for the adiabatic "in" vacuum in the interacting theory, it is more convenient to work directly with equation (9.66), which is derived from the generating functional, as we have done in Sections 16.1 and 16.3.

To illustrate this point let us consider the following decomposition for the anticommutator of the metric perturbations when the effects of matter loops are included

$$\begin{aligned}\frac{1}{2}\langle\{\hat{h}_{ab}(x),\hat{h}_{c'd'}(x')\}\rangle &= \langle h^{\rm h}_{ab}(x)h^{\rm h}_{c'd'}(x')\rangle_0 \\ &+\frac{\kappa^2}{4}\iint G^{\rm ret}_{ab\hat{m}\hat{n}}(x,y)N^{\hat{m}\hat{n}\hat{p}\hat{q}}(y,y') \\ &\times G^{\rm ret}_{\hat{p}\hat{q}c'd'}(y',x')\sqrt{-g}\sqrt{-g}d^4y'd^4y, \end{aligned} \tag{16.76}$$

where the noise kernel $N_{abc'd'}(x,y)$ corresponds to the symmetrized connected correlation function of the stress-energy tensor

$$N_{abc'd'}(x,x') = \frac{1}{2}\langle 0|\{\hat{T}_{ab}(x),\hat{T}_{c'd'}(x')\}|0\rangle - \langle 0|\hat{T}_{ab}(x)|0\rangle\langle 0|\hat{T}_{c'd'}(x')|0\rangle. \tag{16.77}$$

In addition, $h^{\rm h}_{ab}(x)$ correspond-to solutions of the linearized semiclassical Einstein equations, i.e., Eq. (10.14) with $\xi_{ab} = 0$, around a background metric g_{ab}, $\langle\ldots\rangle_0$ denotes an average over their initial conditions according to the initial quantum state of the metric perturbations, and $G^{\rm ret}_{abc'd'}$ is the retarded propagator associated with the linearized semiclassical equation.

Equation (16.76) can be derived using the stochastic gravity formalism as just mentioned, see (10.16), but also within the purely quantum field theoretical calculation of this chapter if we assume a factorized initial state with the metric perturbations and the matter fields uncorrelated. However, as we will see next, such kinds of initial states give rise to some pathologies; see Hu et al. [190] and Fleming et al. [979]. For this reason, in our study we have employed the method described in Section 9.3 to generate a properly dressed initial state for the interacting theory which implies correlations between the metric perturbations and the matter fields.

16.4.1 Secular Terms in a Simple Model

It turns out that the contributions from intrinsic and induced fluctuations exhibit *secular* terms when considered separately. The basic idea can be illustrated with the simple model of two bilinearly coupled harmonic oscillators by Calzetta et al. [688] and Fleming et al. [979], with action

$$S[Q,q] = \frac{1}{2}\int \left(M\dot{Q}^2 - M\Omega^2 Q^2 + m\dot{q}^2 - m\omega^2 q^2 - 2c\,Qq \right) d\eta, \tag{16.78}$$

where η denotes the time and overdots denote derivatives with respect to η. Here the oscillator q is the analogue of the matter fields, while the oscillator Q is the analogue of the metric perturbations.

The analogous of the anticommutator of the metric in this model is the correlation function $\frac{1}{2}\langle\{\hat{Q}(\eta),\hat{Q}(\eta')\}\rangle$ calculated perturbatively up to quadratic order in c. The free retarded propagator for the oscillator Q and the noise kernel characterizing the fluctuations of the oscillator q are given by

$$\begin{aligned} G^{\rm ret}(\eta-\eta') &= \frac{1}{M\Omega}\sin\left[\Omega(\eta-\eta')\right]\theta(\eta-\eta'), \\ N(\eta-\eta') &= \frac{1}{2m\omega}\cos\left[\omega(\eta-\eta')\right], \end{aligned} \tag{16.79}$$

where the noise kernel corresponds to the real part (the anticommutator) of the two-point function of the oscillator q in the interaction picture and we have assumed that it is initially in its ground state, so that $\langle\{\hat{q}(\eta),\hat{q}(\eta')\}\rangle = (2m\omega)^{-1}\exp[i\Omega(\eta-\eta')]$. In that case the term for the induced fluctuations is proportional to

$$\int_{t_0}^{\eta}\int_{t_0}^{\eta'} \sin\left[\Omega(\eta-\tau)\right]\cos\left[\omega(\tau-\tau')\right]\sin\left[\Omega(\eta'-\tau')\right] d\tau' d\tau. \tag{16.80}$$

The key point is that the sines and cosines involve exponentials with two opposite signs in the exponent. For resonant oscillators (with $\omega = \Omega$) this will give rise to terms for which the τ- or τ'-dependent exponents (or both) cancel out and one is left with terms that do not oscillate as a function of τ or τ'. Therefore, when integrating over these variables, one obtains *secular* terms proportional to $\eta - t_0$, $\eta' - t_0$ or both.

The structure of the induced fluctuations for the loop diagram associated with the cubic gravitational interaction between the metric perturbation and the matter field is completely analogous, with a sum over the spatial momentum running in the loop, and for massless fields there is always some momentum for which the resonance condition is fulfilled.

In fact, given a spatial Fourier mode $\vec{p}$ for the metric perturbations, the expression for the one-loop correction to its two-point function is the same as the contribution quadratic in the coupling constant c for a model consisting of an oscillator with $\Omega = |\vec{p}|$ interacting independently (and bilinearly) with a set of oscillators $\{q_j\}$ with frequencies $\{\omega_j\}$. The result is a simple generalization of

the two-oscillator model discussed above with an extra sum over the frequencies ω_j. More precisely, one needs to integrate over a continuous ω distribution, as dictated by the spatial Fourier transform of Eqs. (16.65), which involves just an integral over p^0 and whose result is given by Eqs. (16.66).

In addition, to mimic the gravitational case for metric perturbations around de Sitter, one needs to consider the coupling between the oscillators to be time-dependent with terms proportional to η. (One should also consider the frequency of the oscillator Q to be time dependent, but there the time-dependent terms are proportional to η^{-2}. Therefore, they are essentially irrelevant for the discussion about secular terms because they vanish as $\eta \to -\infty$.)

16.4.2 Absence of Secular Terms in the Correlator

Despite the existence of secular terms for the intrinsic and the induced fluctuations, they are absent in the result for the full correlation function because the secular terms from both contributions cancel out. Such cancellation can be inferred from our calculations and the final result in Section 16.3.2, but can also be understood when considering Feynman diagrams in the CTP formalism following the approach of Section 9.3. The basic idea is that for fixed arguments of the correlation function (associated with the external legs) and integrating over the times of internal vertices, all the propagators leaving a given vertex on the + branch are of positive frequency with respect to the time variable of that vertex at sufficiently early times (the analogous statement with negative frequencies holds for vertices on the − branch). In the gravitational case this is a consequence of $G^+(\eta_1, \eta)$ being the positive Wightman function of the free theory associated with the Bunch–Davies vacuum, which involves an exponential factor with exponent $-i\Omega(\eta_1 - \eta)$. In that case there are also factors involving powers of the conformal times, but the key role is still played by the oscillatory factors as far as the possible existence of secular terms is concerned.

Therefore, instead of an equation like (16.80), one gets a sum of terms of the following form (or analogous forms):

$$\int_{t_0^+}^{\eta} \int_{\tau}^{\eta'} e^{-i\Omega(\eta-\tau)}\, e^{-i\omega(\tau'-\tau)}\, e^{-i\Omega(\eta'-\tau')} d\tau' d\tau. \tag{16.81}$$

where in this particular case all the τ-dependent exponents have the same sign. This guarantees that there will be no secular terms and the limit $t_0 \to -\infty$ will converge. It should be emphasized that within this formulation the terms that naturally appear, such as (16.81), always contain a mixture of intrinsic and induced contributions.

Note also that it is important to distinguish the initial times t_0^+ and t_0^- for the two branches. Otherwise we would miss the exponential factor in terms proportional to $e^{i(\omega+\Omega)(t_0^+ - t_0^-)}$, or $t_0^{\pm} e^{i(\omega+\Omega)(t_0^+ - t_0^-)}$. This appears when one has a time-dependent coupling, like the cubic interaction terms in Eq. (16.1), quadratic in the matter fields and linear in the metric perturbations, for a nontrivial

FLRW background. Missing these exponential factors gives rise to spurious finite contributions or even divergences in the limit $t_0 \to -\infty$ for the second case.

An alternative way of understanding the cancellation of the secular terms is by noticing that the calculation of correlation functions using the method described in Section 9.3 gives the correlation function for the ground state of the interacting theory (or an adiabatic ground state for suitable time-dependent interactions). This is clearer for time-independent Hamiltonians (including the interactions). The ground state is then a stationary state and the time dependence of the correlation functions only involves the relative differences between their arguments. Therefore, there cannot be secular terms, i.e., depending on some particular initial time, in the exact result, which implies a complete cancellation of secular terms order by order in the perturbative calculation. The lack of dependence on an arbitrary initial time and the corresponding need for a cancellation of the secular terms can be extended to the case of time-dependent interactions provided that appropriate adiabaticity conditions are satisfied.

Note that in a similar calculation for flat space all the secular terms in the induced metric perturbations do cancel out. This is, however, an accident of such a special case: in general, other cubic interaction terms (different from the gravitational interaction) will not lead to such a cancellation even in flat space. The reason for the cancellation when considering the gravitational case in a Minkowski background (see Section 11.4 or [189]) is that the noise kernel for a massless field involves two global d'Alembertian operators which can be integrated by parts so that they act on the graviton propagator. The action of the Minkowski d'Alembertian on the graviton propagator gives a vanishing result in flat space, but this is not the case in a general FLRW spacetime.

16.5 Two-Point Functions for Metric Perturbations

Here we go back to the evaluation of the one-loop two-point functions for the metric perturbations. The two-point functions, which are true expectation values with respect to the *in* state, are obtained by functionally differentiating the generating functional $Z[J^{\pm}]$ with respect to the sources as in Eq. (9.62). The generating functional can be computed perturbatively in powers of κ, as one can see in Eq. (9.64), once the effective action has been written with this perturbative expansion up to order κ^2; see the Eqs. (9.56) or (16.28). Finally, functionally differentiating with respect to the sources $J^{\pm}$, we obtain Eq. (9.66) for the one-loop two-point functions of the metric perturbations up to order κ^2.

16.5.1 Two-Point Function for Vector and Scalar Parts

The scalar and vectorial parts of the lowest-order CTP effective action (9.60) are given by Eqs. (16.30) and (16.31), respectively. Let us now Fourier transform the scalar and vectorial perturbations with respect to the spatial arguments to obtain

$$S_0^{\rm S}+S_0^{\rm V}=\frac{1}{2}\int a^2\vec{p}^{\,2}\bar{v}_{\rm T}^m\bar{v}_m^{\rm T*}\frac{d^3p}{(2\pi)^3}d\eta-\frac{3}{2}H^2\int a^4\bar{\varphi}\bar{\varphi}^*\frac{d^3p}{(2\pi)^3}d\eta$$
$$-2\int Ha^3\vec{p}^{\,2}\bar{\varphi}\bar{\psi}^*\frac{d^3p}{(2\pi)^3}d\eta, \tag{16.82}$$

where all fields depend on η and $\vec{p}$ and we have used bars to denote Fourier-transformed quantities with respect to the spatial coordinates.

Introducing, as in Section 16.3.1, a right-handed orthonormal set of three vectors $\vec{e}_A(\vec{p})$ such that $\vec{p}=|\vec{p}|\vec{e}_3$ and choosing that under reflection they satisfy Eq. (16.42), we can form the two real scalars

$$\bar{v}_+=\frac{1}{2}(e_1^m+e_2^m)\bar{v}_m^{\rm T},\qquad \bar{v}_-=\frac{i}{2}(e_1^m-e_2^m)\bar{v}_m^{\rm T}. \tag{16.83}$$

Taking into account the transversality condition $p^m\bar{v}_m^{\rm T}=0$, the vector $\bar{v}_m^{\rm T}$ can be written as

$$\bar{v}_{\rm T}^m=\bar{v}_+\left(e_1^m+e_2^m\right)-i\bar{v}_-\left(e_1^m-e_2^m\right). \tag{16.84}$$

The action (16.82) then reads

$$S_0^{\rm S}+S_0^{\rm V}=-\frac{3}{2}H^2\int a^4\bar{\varphi}\bar{\varphi}^*\frac{d^3p}{(2\pi)^3}d\eta-2\int Ha^3\vec{p}^{\,2}\bar{\varphi}\bar{\psi}^*\frac{d^3p}{(2\pi)^3}d\eta$$
$$+\int a^2\vec{p}^{\,2}\left(\bar{v}_+\bar{v}_+^*+\bar{v}_-\bar{v}_-^*\right)\frac{d^3p}{(2\pi)^3}d\eta. \tag{16.85}$$

From this we see that the tree-level kinetic operator A^0 of Eq. (9.58) has the following non-vanishing components:

$$\bar{A}^{0++}_{v_+v_+}=\bar{A}^{0++}_{v_-v_-}=\frac{2}{H^2\eta^2}\vec{p}^{\,2},\quad \bar{A}^{0++}_{\varphi\varphi}=-\frac{3}{H^2\eta^4},$$
$$\bar{A}^{0++}_{\psi\psi}=0,\quad \bar{A}^{0++}_{\varphi\psi}=\bar{A}^0_{\psi\varphi}=\frac{2}{H^2\eta^3}\vec{p}^{\,2}, \tag{16.86}$$

with $\bar{A}^{0--}=-\bar{A}^{0++}$ in every case.

We see that the scalar and vectorial perturbations satisfy elliptic equations rather than hyperbolic ones, since these perturbations become purely algebraic in spatial Fourier space. This reflects the fact they are not true dynamical degrees of freedom but constrained ones. The CTP generating functional for the scalar and vectorial sector will still be given by Eq. (9.65), but with free "propagators" G^0_{AB} proportional to $\delta(\eta-\eta')$ that result from solving the algebraic equation

$$\begin{pmatrix}\bar{A}^{0++}(\eta,\vec{p}) & 0\\ 0 & -\bar{A}^{0++}(\eta,\vec{p})\end{pmatrix}\begin{pmatrix}\bar{G}^{0++}(\eta,\eta',\vec{p}) & \bar{G}^{0+-}(\eta,\eta',\vec{p})\\ \bar{G}^{0-+}(\eta,\eta',\vec{p}) & \bar{G}^{0--}(\eta,\eta',\vec{p})\end{pmatrix}=\mathbf{I}\delta(\eta-\eta'). \tag{16.87}$$

for their spatial Fourier transforms.

Inverting the first matrix on the left-hand side of Eq. (16.87), it follows that $\bar{G}^{0+-}(\eta,\eta',\vec{p}) = \bar{G}^{0-+}(\eta,\eta',\vec{p}) = 0$ as well as $\bar{G}^{0--}(\eta,\eta',\vec{p}) = -\bar{G}^{0++}(\eta,\eta',\vec{p})$, and for the latter "propagator" we have

$$\begin{aligned}
\bar{G}^{0++}_{v_+v_+} &= \bar{G}^{0++}_{v_-v_-} = \frac{H^2\eta^2}{2\vec{p}^2}\delta(\eta-\eta'),\\
\bar{G}^{0++}_{\varphi\psi} &= \bar{G}^{0++}_{\psi\varphi} = \frac{H^2\eta^3}{2\vec{p}^2}\delta(\eta-\eta'),\\
\bar{G}^{0++}_{\psi\psi} &= \frac{3H^2\eta^2}{4(\vec{p}^2)^2}\delta(\eta-\eta'),\\
\bar{G}^{0++}_{\varphi\varphi} &= 0.
\end{aligned} \tag{16.88}$$

Using the decomposition (16.26) for the metric perturbations, we readily obtain for the vector "propagator"

$$\begin{aligned}
\bar{G}^{0++\mathrm{V}}_{abc'd'}(\eta,\eta',\vec{p}) &= 4\delta^m_{(a}\delta^0_{b)}\delta^0_{(c'}\delta^n_{d')}\left(\vec{e}^{\,1}_m\vec{e}^{\,1}_n + \vec{e}^{\,2}_m\vec{e}^{\,2}_n\right)\left(\bar{G}^{0++}_{v_+v_+} + \bar{G}^{0++}_{v_-v_-}\right)\\
&= 4\delta^0_{(a}P_{b)(c'}\delta^0_{d')}\frac{H^2\eta^2}{\vec{p}^2}\delta(\eta-\eta'),
\end{aligned} \tag{16.89}$$

where the projection tensor P_{mn} is defined in Eq. (16.58) with $p^a = (0,\vec{p})$ in the spatially flat coordinate system that we have chosen.

Similarly, for the scalar "propagator" we obtain

$$\begin{aligned}
\bar{G}^{0++\mathrm{S}}_{abc'd'}(\eta,\eta',\vec{p}) &= 2i\left(\delta^0_{(a}p_{b)}\delta^0_{c'}\delta^0_{d'} - \delta^0_a\delta^0_b\delta^0_{(c'}p_{d')}\right)\bar{G}^{0++}_{\phi\psi}\\
&\quad + \delta^0_a\delta^0_b\delta^0_{c'}\delta^0_{d'}\bar{G}^{0++}_{\phi\phi} + 4\delta^0_{(a}p_{b)}\delta^0_{(c'}p_{d')}\bar{G}^{0++}_{\psi\psi}\\
&= \delta^0_{(a}\left[i\eta\left(p_{b)}\delta^0_{(c'} - \delta^0_{b)}p_{(c'}\right) + 3\frac{p_{b)}p_{(c'}}{\vec{p}^2}\right]\delta^0_{d')}\frac{H^2\eta^2}{\vec{p}^2}\delta(\eta-\eta').
\end{aligned} \tag{16.90}$$

Because they are proportional to $\delta(\eta-\eta')$, when substituting the propagators (16.89) and (16.90) into Eq. (9.66) to calculate the correlation function for the scalar and vectorial sector of the metric perturbations, the two time integrals can be trivially performed and the result is directly proportional to the corresponding component of the matrix V_{AB} defined in Eq. (9.59). This is equivalent to solving the constraints corresponding to the Einstein equations for the scalar and vectorial perturbations, which become proportional, in Fourier space, to the stress-energy tensor. Their correlator is then directly determined by the stress-energy tensor correlator of the matter fields, which essentially coincides with V_{AB}.

Furthermore, since the tree-level Wightman functions for the vectorial and scalar parts vanish, as seen above, the only parts of the matrix V_{MN} that contribute in this case to Eq. (9.66) are those with the index M equal to A and the index N equal to B. Therefore, calculating the Wightman function at

one-loop order, for which $AB = -+$, there will be no contribution from V_{++} nor V_{--}, which includes all of $S_2^{\rm g}$ in Eq. (16.29) as well as the terms with the two kernels $K^{\pm}$ in Eq. (16.5). Taking into account that $\bar{G}^{0--} = -\bar{G}^{0++}$, the vector contribution to the two-point function, the positive Wightman function, is then given by

$$\bar{G}^{-+{\rm V}}_{abc'd'}(\eta,\eta',\vec{p}) = \kappa^2 \iint \bar{G}^{0++{\rm V}}_{ab\hat{m}\hat{n}}(\eta,\tau,\vec{p})\bar{V}^{\hat{m}\hat{n}\bar{p}\bar{q}}_{-+}(\tau,\tau',\vec{p})\bar{G}^{0++{\rm V}}_{\bar{p}\bar{q}c'd'}(\tau',\eta',\vec{p})d\tau' d\tau, \tag{16.91}$$

where V_{-+} is obtained by functionally differentiating the terms involving the kernel K in Eq. (16.5).

After inserting the vector "propagator" (16.89) and performing the time integrals, the result reads

$$\bar{G}^{-+{\rm V}}_{abc'd'}(\eta,\eta',\vec{p}) = 6i\alpha\kappa^2\delta^0_{(a}P_{b)(c'}\delta^0_{d')}H^4\eta^2(\eta')^2\frac{1}{\vec{p}^2}\left(\partial_\eta^2+\vec{p}^2\right)\bar{S}(\eta-\eta',\vec{p}), \tag{16.92}$$

where we defined the kernel

$$\bar{S}(\eta-\eta',\vec{p}) = -ie^{-i|\vec{p}|(\eta-\eta')}\mathcal{P}\frac{1}{\eta-\eta'} + \pi\delta(\eta-\eta'). \tag{16.93}$$

This kernel is obtained (up to a proportionality factor) by the temporal Fourier transform of the kernel K in Eq. (16.7). Here $\mathcal{P}$ denotes Hadamard's finite-part distribution. Details on such Fourier transforms are given in Section 16.7.2.

Proceeding in the same way, one finds that the scalar contribution is given by

$$\begin{aligned}\bar{G}^{-+{\rm S}}_{abc'd'}(\eta,\eta',\vec{p}) = \frac{i}{2}\alpha\kappa^2H^4\Bigg[&9\delta^0_{(a}p_{b)}\delta^0_{(c'}p_{d')}\frac{\eta^2(\eta')^2}{(\vec{p}^2)^2}\partial_\eta^2 \\ &+ 3\delta^0_{(a}p_{b)}\delta^0_{(c'}p_{d')}\frac{\eta^2(\eta')^2}{\vec{p}^2}(\eta-\eta')\partial_\eta - \delta^0_{(a}p_{b)}\delta^0_{(c'}p_{d')}\eta^3(\eta')^3 \\ &- 3i\delta^0_a\delta^0_b\delta^0_{(c'}p_{d')}\frac{\eta^3(\eta')^2}{\vec{p}^2}\partial_\eta^2 + 3i\delta^0_{(a}p_{b)}\delta^0_{c'}\delta^0_{d'}\frac{\eta^2(\eta')^3}{\vec{p}^2}\partial_\eta^2 \\ &+ i\delta^0_{(a}\left(\delta^0_{b)}p_{(c'}+p_{b)}\delta^0_{(c'}\right)\delta^0_{d')}\eta^3(\eta')^3\partial_\eta \\ &+ \delta^0_a\delta^0_b\delta^0_{c'}\delta^0_{d'}\eta^3(\eta')^3\partial_\eta^2\Bigg]\bar{S}(\eta-\eta',\vec{p}).\end{aligned} \tag{16.94}$$

16.5.2 Two-Point function for tensor part

The two-point function for the tensor perturbations was derived in Section 16.3 and is given by Eq. (16.67). For the positive Wightman function it can be written (reintroducing the TT superscripts) as

$$G^{-+\mathrm{TT}}_{abc'd'}(x,x') = -i\langle \mathrm{in}|h^{\mathrm{TT}}_{ab}(x)h^{\mathrm{TT}}_{c'd'}(x')|\mathrm{in}\rangle = \int \bar{G}^{-+\mathrm{TT}}_{abc'd'}(\eta,\eta',\vec{p})e^{i\vec{p}\cdot(\vec{x}-\vec{x}')}\frac{d^3p}{(2\pi)^3}, \tag{16.95}$$

with

$$\bar{G}^{-+\mathrm{TT}}_{abc'd'}(\eta,\eta',\vec{p}) = -iP_{abc'd'}\Bigg[f(\eta,\eta',\vec{p})\Big(1+6\alpha\kappa^2H^2\ln\Big(\frac{\bar{\mu}}{H}\Big)-(5\alpha-2\beta)\,\kappa^2H^2\Big) + \frac{3}{2}\alpha\kappa^2H^4g(\eta,\eta',\vec{p})\Bigg]+\mathcal{O}(\kappa^4), \tag{16.96}$$

where the polarization tensor $P_{abc'd'}$ is given by the second equality of Eq. (16.57) in terms of the projection tensor P_{ab} defined in Eq. (16.58). The function $f(\eta,\eta',\vec{p})$ coincides with the tree-level result and is given by (16.56), and $g(\eta,\eta',\vec{p})$ is given by

$$g(\eta,\eta',\vec{p}) = I_1(\eta,\eta',\vec{p}) - I_2(\eta,\eta',\vec{p}) - I_2^*(\eta',\eta,\vec{p}) + I_3(\eta,\eta',\vec{p}) - I_4(\eta,\eta',\vec{p}) + I_5(\eta,\eta',\vec{p}), \tag{16.97}$$

where the terms I_k are defined in Eq. (16.68); note that an alternative expression for I_5 is $I_5(\eta,\eta',\vec{p}) = \eta^2(\eta')^2\bar{S}(\eta-\eta',\vec{p})$.

16.6 Effective Action for a General Conformal Field Theory

In this section we derive the effective action for a general conformal field theory (CFT), including interacting theories. In fact, as explained by Fröb et al. [639], see also [980], the result for a general conformal theory is essentially the same that is obtained for a free massless conformally coupled scalar field, except for the values of some parameters. To see this let us begin with the classical action (16.1), which can be written as $S[\tilde{g},\tilde{\phi}] = S_{\mathrm{g}}[\tilde{g}] + S_{\mathrm{m}}[\tilde{g},\tilde{\phi}]$, where S_{g} includes the bare Einstein–Hilbert action with a cosmological constant and the counterterms, and S_{m} is the matter field action

$$S_{\mathrm{m}}[\tilde{g},\tilde{\phi}] = -\frac{1}{2}\int\Big[\Big(\tilde{\nabla}^a\tilde{\phi}\Big)\Big(\tilde{\nabla}_a\tilde{\phi}\Big)+\xi(n)\tilde{R}\tilde{\phi}^2\Big]\sqrt{-\tilde{g}}d^nx, \tag{16.98}$$

where we have taken $N=1$.

For the metric (16.18), the explicit calculation of the influence action S_{IF} which results from functionally integrating out the scalar field (as has been done in the expression for the generating functional (9.50) to obtain (9.50)), leads to the following effective action for the gravitational field

$$S_{\mathrm{eff}}[\tilde{g}^{\pm}] = S_{\mathrm{g}}[\tilde{g}^{+}] - S_{\mathrm{g}}[\tilde{g}^{-}] + S_{\mathrm{IF}}[\tilde{g}^{\pm}]. \tag{16.99}$$

The influence action S_{IF} has ultraviolet divergencies associated with the matter loops which may be absorbed by the local counterterms that are included in the

bare gravitational action. In the case of conformal scalar fields, as one can see in Eq. (12.8), the necessary counterterms are given by

$$S_{\rm ct}^{\rm CV}[\tilde{g}] = \frac{\bar{\mu}^{n-4}}{2880\pi^2(n-4)} \int \left(\tilde{R}^{abcd}\tilde{R}_{abcd} - \tilde{R}^{ab}\tilde{R}_{ab} \right) \sqrt{-\tilde{g}} d^n x, \tag{16.100}$$

where $\bar{\mu}$ is the renormalization scale; the cosmological constant, the Einstein–Hilbert term and the square of the Ricci scalar do not get renormalized in dimensional regularization.

For a generic CFT the counterterms are given in the minimal subtraction (MS) scheme by Duff [422], see expression (2.60), as

$$S_{\rm ct}^{\rm MS}[\tilde{g}] = \frac{\bar{\mu}^{n-4}}{n-4} \int \left(b\tilde{C}^2 + b'\tilde{\mathcal{E}}_4 \right) \sqrt{-\tilde{g}} d^n x, \tag{16.101}$$

where $\tilde{C}^2 \equiv \tilde{C}^{abcd}\tilde{C}_{abcd}$ with the n-dimensional Weyl tensor $\tilde{C}_{abcd}$, and $\tilde{\mathcal{E}}_4 \equiv \tilde{R}^{abcd}\tilde{R}_{abcd} - 4\tilde{R}^{ab}\tilde{R}_{ab} + \tilde{R}^2$ is the Euler density; these scalars are called F and G, respectively, in Eq. (2.60). The constants b and b', which are called B and A in (2.60), are the coefficients that appear in the trace anomaly and depend on the theory under consideration. For the particular case of N_0 free massless conformally coupled scalar fields, $N_{1/2}$ free massless Dirac spinor fields and N_1 free vector fields we have [422]

$$b = \frac{N_0 + 6N_{1/2} + 12N_1}{1920\pi^2}, \tag{16.102a}$$

$$b' = -\frac{N_0 + 11N_{1/2} + 62N_1}{5760\pi^2}, \tag{16.102b}$$

in agreement with Eq. (2.63) and the definition of parameters that follow. In the literature, the symbols $a \equiv -(4\pi)^2 b'$ and $c \equiv (4\pi)^2 b$ are more common; however, to avoid confusion with the scale factor a we do not use that notation.

In the conformally flat case, the functional integral for the generating functional in (9.48) can be performed by rescaling the scalar matter as $\tilde{\phi} = a^{1-n/2}\phi$ and working in flat space. As one can see explicitly from Eqs. (12.2) and (12.4) the bare action is invariant under a conformal transformation of the metric, as in the first Eq. (16.18), and the rescaling of the scalar field,

$$S_{\rm m}\big[\tilde{g}, \tilde{\phi}\big] = S_{\rm m}[g, \phi]\,, \tag{16.103}$$

and, thus, the regulated bare influence action $S_{\rm IF}$ is also invariant under conformal transformations,

$$S_{\rm IF}\left[\tilde{g}^{\pm}\right] = S_{\rm IF}\left[g^{\pm}\right]. \tag{16.104}$$

However, the counterterms contained in $S_{\rm ct}$ are not invariant under such conformal transformations.

Thus, while in the sum

$$S_{\rm IF}\left[g^{\pm}\right] + S_{\rm ct}[g^{+}] - S_{\rm ct}[g^{-}]\,, \tag{16.105}$$

the counterterms cancel the divergences of $S_{\rm IF}$ as $n \to 4$ so that this limit is finite, we need to calculate this sum with the physical metric:

$$S_{\rm IF}\,[\tilde{g}^{\pm}] + S_{ct}\,[\tilde{g}^{+}] - S_{\rm ct}[\tilde{g}^{-}] = S_{\rm IF}\,[g^{\pm}] + S_{\rm ct}[g^{+}] - S_{\rm ct}[g^{-}] + (S_{\rm ct}[\tilde{g}^{+}] - S_{\rm ct}[g^{+}]) - (S_{\rm ct}[\tilde{g}^{-}] - S_{\rm ct}[g^{-}])\,. \tag{16.106}$$

Since also in flat space the counterterms cancel the divergences of $S_{\rm IF}$ when $n \to 4$, the terms in the last line do not diverge, but do not vanish either. They therefore give a finite contribution to the effective action, which can be calculated using the conformal transformations given in Section 12.5.2. This is the way how the correct trace anomaly arises in dimensional regularization; see, e.g., Mazur and Mottola [981].

For the counterterms in the MS scheme (16.101), we obtain

$$\lim_{n\to 4}\left(S^{\rm MS}_{\rm ct}[\tilde{g}] - S^{\rm MS}_{\rm ct}[g]\right) = \int\Big[(bC^2 + b'\mathcal{E}_4)\ln a - 4b'G^{mn}a^{-2}(\nabla_m a)(\nabla_n a) + 2b'a^{-4}\left[(\nabla^m a)(\nabla_m a) - 2a\nabla^m\nabla_m a\right] \times (\nabla^n a)(\nabla_n a)\Big]\sqrt{-g}d^4x. \tag{16.107}$$

For a single massless, conformally coupled scalar where, according to equation (16.102), $b = 1/(1920\pi^2)$ and $b' = -b/3$, we have

$$b\tilde{C}^2 + b'\tilde{\mathcal{E}}_4 = \frac{1}{2880\pi^2}\left(\tilde{R}^{abcd}\tilde{R}_{abcd} - \tilde{R}^{ab}\tilde{R}_{ab}\right) + \frac{n-4}{2880\pi^2}\left(\frac{3}{4}\tilde{C}^2 - \frac{3}{4}\tilde{\mathcal{E}}_4 + \frac{1}{12}\tilde{R}^2\right) + \mathcal{O}((n-4)^2), \tag{16.108}$$

so that the scheme used in Chapter 12 differs by a finite renormalization from the MS scheme as

$$S^{\rm MS}_{\rm ct} = S^{\rm CV}_{\rm ct} + \frac{1}{3840\pi^2}\int\left(\tilde{C}^2 - \tilde{\mathcal{E}}_4 + \frac{1}{9}\tilde{R}^2\right)\sqrt{-\tilde{g}}d^4x. \tag{16.109}$$

The integral of $\tilde{\mathcal{E}}_4$ is a topological invariant in four dimensions and thus does not contribute to the dynamics, and the finite renormalization of the square of the Weyl tensor can be absorbed simply into a redefinition of the renormalization scale $\bar{\mu}$ in the explicit result for the effective action $S_{\rm eff}$. However, the finite renormalization of the square of the Ricci scalar must be taken into account when we extend the result of $S_{\rm eff}$ to generic conformal field theories.

Including an arbitrary finite coefficient β for the square of the Ricci scalar, we obtain the effective action for general conformal theories by comparing formulas (16.99), (16.106), (16.107) and (16.109) with the results of Chapter 12. This means making use of Eqs. (12.45) and (12.46) for the effective action $S_{\rm eff}$,

with $\omega(\eta) = \ln a(\eta)$ and including now the cosmological constant Λ. The effective action is finally given by

$$S_{\rm eff}[\tilde{g}^{\pm}] = S_{\rm g}^{\rm R}[a, g^{+}] - S_{\rm g}^{\rm R}[a, g^{-}] + S_{\rm IF}^{\rm R}[g^{\pm}]\,, \tag{16.110}$$

where $S_{\rm g}^{\rm R}[a, g^{+}]$ is the renormalized, finite part of the local action given by

$$\begin{aligned} S_{\rm g}^{\rm R}[a, g] \equiv & \frac{1}{\kappa^2}\int \left(a^2 R - 6a\nabla^m\nabla_m a - 2\Lambda a^4\right)\sqrt{-g}d^4x \\ & + \int \left(bC^2 + b'\mathcal{E}_4\right)\ln a\sqrt{-g}d^4x \\ & - \frac{\beta}{12}\int \left(R - 6a^{-1}\nabla^2 a\right)^2\sqrt{-g}d^4x \\ & - 4b'\int G^{mn}a^{-2}(\nabla_m a)(\nabla_n a)\sqrt{-g}d^4x \\ & + 2b'\int a^{-4}\left[(\nabla^m a)(\nabla_m a) - 2a\nabla^m\nabla_m a\right] \\ & \times (\nabla^n a)(\nabla_n a)\sqrt{-g}d^4x, \end{aligned} \tag{16.111}$$

and $S_{\rm IF}^{\rm R}$ is the renormalized, finite part of the non-local term $S_{\rm IF}$, which reads

$$\begin{aligned} S_{\rm IF}^{\rm R}[g^{\pm}] \equiv & 2b\iint C^{+}_{abcd}(x)C^{-abcd}(y)K(x-y)d^4xd^4y \\ & + b\iint C^{+}_{abcd}(x)C^{+abcd}(y)K^{+}(x-y;\bar{\mu})d^4xd^4y \\ & - b\iint C^{-}_{abcd}(x)C^{-abcd}(y)K^{-}(x-y;\bar{\mu})d^4xd^4y, \end{aligned} \tag{16.112}$$

where the kernels K are given by (16.7). Notice that this effective action generalizes to any conformal field theory the result for N scalar fields given by Eqs. (16.5)–(16.6).

This result is valid in the MS renormalization scheme (16.101), where the renormalization scale $\bar{\mu}$ is chosen such that there is no term proportional to C^2 in the local part of the renormalized effective action $S_{\rm g}^{\rm R}$. That is, a value of the renormalization scale $\bar{\mu}$ has been chosen so that the renormalized coupling constant $a_1^{\rm R}(\bar{\mu}) = 0$, see Eq. (16.1). One can make this choice to shorten the intermediate expressions and then perform the changes needed to get a result valid for arbitrary μ. More specifically, we can use Eqs. (16.1) and (16.2) together with Eq. (16.112) and the expression (16.7) of the kernels, whose $\bar{\mu}$-dependent part is

$$-\frac{1}{2}\int \left(\ln\frac{1}{\bar{\mu}^2}\right)e^{ip\cdot x}\frac{d^4p}{(2\pi)^4} = \delta^4(x-x')\ln\bar{\mu}. \tag{16.113}$$

It then follows the relation between two different values of the renormalization scale μ_0 and μ:

$$Na_1^{\rm R}(\mu) = Na_1^{\rm R}(\mu_0) - \frac{2}{3}\, b\, \ln\left(\frac{\mu}{\mu_0}\right), \tag{16.114}$$

so that for an arbitrary value μ of the renormalization scale we need to make the following replacement

$$b \ln \bar{\mu} \to b \ln \mu + \frac{3}{2} N a_1^{\rm R}(\mu). \tag{16.115}$$

This guarantees that the effective action $S_{\rm eff}[\tilde{g}^{\pm}]$ is invariant under the action of the renormalization group.

Different CFTs will correspond to different values of the constants b and b'. For instance, $b = 3\alpha/2$ and $b' = -\alpha/2$ with $\alpha = N(2880\pi^2)^{-1}$ for the free scalar field model specified in equation (16.1) and β is a free parameter, but the expressions for $S_{\rm eff}$ are otherwise the same. For interacting fields this will include radiative corrections with an arbitrary number of matter loops but no graviton loops, which would give rise to contributions of higher order in $1/N$, or any internal graviton propagators connecting different matter loops, which would correspond to contributions of higher order in κ^2.

16.7 Mathematical Supplement

Here we collect some mathematical expressions that are useful for the calculations in this chapter.

16.7.1 Some Special Functions

We define the entire function $\mathrm{Ein}(z)$ by

$$\mathrm{Ein}(z) = \int_0^z \frac{e^t - 1}{t} dt = \sum_{k=1}^{\infty} \frac{z^k}{k\, k!}. \tag{16.116}$$

Its asymptotic expansion at infinity ($r \to \infty$) is given by

$$\mathrm{Ein}\,(\alpha r + \beta) \sim -\gamma - \ln\left(-(\alpha r + \beta)\right) + e^{\alpha r + \beta}\left[\frac{1}{\alpha r} + \frac{1-\beta}{\alpha^2 r^2} + \mathcal{O}(r^{-3})\right], \tag{16.117}$$

where γ is the Euler–Mascheroni constant.

The following useful integrals involving $\mathrm{Ein}(z)$ can be easily obtained by partial integration:

$$\int \frac{e^{at}}{bt+c} dt = \frac{1}{b} e^{-\frac{ac}{b}} \left[\mathrm{Ein}\left[\frac{a}{b}\,(bt+c)\right] + \ln\,(bt+c)\right]$$

$$\int e^{at} \ln(bt+c) dt = \frac{1}{a}\left[\left(e^{at} - e^{-\frac{ac}{b}}\right) \ln\,(bt+c) - e^{-\frac{ac}{b}}\, \mathrm{Ein}\left[\frac{a}{b}\,(bt+c)\right]\right]$$

$$\int e^{at}\mathrm{Ein}(bt+c)dt = \frac{1}{a}e^{at}\mathrm{Ein}(bt+c) + \frac{1}{a}e^{-\frac{ac}{b}}\left[\mathrm{Ein}\left(\frac{a}{b}(bt+c)\right)\right.$$
$$\left.-\mathrm{Ein}\left(\frac{a+b}{b}(bt+c)\right)\right]. \tag{16.118}$$

These integrals depend continuously on the parameters a and c, so that for instance the integral of $\mathrm{Ein}(bt+c)$ can be calculated by taking the limit $a \to 0$ on the right side of the last integral.

16.7.2 Distributions and Their Fourier Transforms

Here we analyze the main distributions which appear in Section 16.3.2. Good references for the general theory of distributions are Schwartz [731] and Gelfand and Shilov [982].

The principal value distribution, denoted by $\mathcal{P}\frac{1}{t}$, is defined as

$$\mathcal{P}\frac{1}{t} = \mathrm{d}\lim_{\epsilon\to 0}\frac{\Theta(t-\epsilon)+\Theta(-t-\epsilon)}{t}, \tag{16.119}$$

where $\mathrm{d}-\lim$ stands for "limit in the sense of distributions", i.e.

$$\int \mathcal{P}\frac{1}{t}f(t)dt = \lim_{\epsilon\to 0}\left[\int \frac{\Theta(t-\epsilon)+\Theta(-t-\epsilon)}{t}f(t)dt\right], \tag{16.120}$$

where $f(t)$ is an appropriate (fast decaying) test function. The Fourier transform can be calculated to give

$$\begin{aligned}
\int \mathcal{P}\frac{1}{t}e^{i\omega t}dt &= \lim_{\epsilon\to 0}\left[\left(\int_{-\infty}^{-\epsilon}+\int_{\epsilon}^{\infty}\right)\frac{e^{i\omega t}}{t}dt\right] \\
&= \lim_{\epsilon\to 0}\left[\lim_{r\to\infty}\int_{-r}^{-\epsilon}\frac{e^{i\omega t}}{t}dt + \lim_{r\to\infty}\int_{\epsilon}^{r}\frac{e^{i\omega t}}{t}dt\right] \\
&= \lim_{\epsilon\to 0}\left[\lim_{r\to\infty}\left[\mathrm{Ein}\,(i\omega t)+\ln|t|\right]_{-r}^{-\epsilon} + \lim_{r\to\infty}\left[\mathrm{Ein}\,(i\omega t)+\ln|t|\right]_{\epsilon}^{r}\right] \\
&= \lim_{\epsilon\to 0}\left[\mathrm{Ein}\,(-i\omega\epsilon)+\ln(i\omega)-\ln(-i\omega)-\mathrm{Ein}\,(i\omega\epsilon)\right] \\
&= i\arg(i\omega) - i\arg(-i\omega) = i\pi\,\mathrm{sgn}\,\omega.
\end{aligned} \tag{16.121}$$

Another distribution of interest is $\mathcal{P}'\frac{\Theta(t)}{t}$, which is defined as

$$\mathcal{P}'\frac{\Theta(t)}{t} = \mathrm{d}\lim_{\epsilon\to 0}\left[\frac{\Theta(t-\epsilon)}{t}+\delta(t)(\ln(\mu_0\epsilon)+\gamma)\right], \tag{16.122}$$

where $\mu_0 > 0$ is an arbitrary reference energy scale to make the definition dimensionally correct, and we use the symbol $\mathcal{P}'$ to distinguish from the previous

$\mathcal{P}$ which didn't include a δ distribution. Its Fourier transform can be calculated by proceeding as in the previous case, and is given by

$$\begin{aligned}\int \mathcal{P}'\frac{\Theta(t)}{t}e^{i\omega t}dt &= \lim_{\epsilon\to 0}\left[\int_\epsilon^\infty \frac{e^{i\omega t}}{t}dt + \ln(\mu_0\epsilon)+\gamma\right]\\ &= \lim_{\epsilon\to 0}\lim_{r\to\infty}\left[[-\gamma-\ln(-i\omega)-\ln r]+\ln r\right.\\ &\quad \left.-\mathrm{Ein}\,(i\omega\epsilon)-\ln\epsilon+\ln\epsilon+\gamma+\ln\mu_0\right]\\ &= \ln\left(\frac{\mu_0}{|\omega|}\right)+\frac{i\pi}{2}\mathrm{sgn}\omega. \end{aligned} \tag{16.123}$$

Using those two results we may write the following compact expression:

$$\begin{aligned}-\int \ln\left(\frac{\omega}{\mu_0}\right)^2 e^{-i\omega t}\frac{d\omega}{2\pi} &= 2\mathcal{P}'\frac{\Theta(t)}{t}-\mathcal{P}\frac{1}{t}\\ &= \mathrm{d}\lim_{\epsilon\to 0}\left[\frac{\Theta(|t|-\epsilon)}{|t|}+2\delta(t)(\ln(\mu_0\epsilon)+\gamma)\right], \end{aligned} \tag{16.124}$$

where we denote the distribution defined by the last line as $\mathcal{P}'|t|^{-1}$. Simply applying to this equation the formula for the inverse Fourier transform of a function with the argument shifted by a constant, one gets

$$-\int \ln\left(\frac{a+\omega}{\mu_0}\right)^2 e^{-i\omega t}\frac{d\omega}{2\pi} = e^{iat}\mathcal{P}'\frac{1}{|t|}. \tag{16.125}$$

We have now all the elements for calculating the temporal inverse Fourier transform of

$$\ln\left(\frac{p^2\pm i0}{\bar{\mu}^2}\right) = \ln\left|\frac{p^2}{\bar{\mu}^2}\right| \pm i\pi\Theta(-p^2). \tag{16.126}$$

First, from Eq. (16.125) we obtain

$$\begin{aligned}\int \ln|\frac{p^2}{\bar{\mu}^2}|e^{-ip^0t}\frac{dp^0}{2\pi} &= \int\left[\ln|p^0+|\vec{p}||+\ln|p^0-|\vec{p}||-2\ln\bar{\mu}\right]\\ &= -\cos\left(|\vec{p}|t\right)\mathcal{P}'\frac{1}{|t|}-\delta(t)\ln\left(\frac{\bar{\mu}^2}{\mu_0^2}\right). \end{aligned} \tag{16.127}$$

Second, writing $\Theta(-p^2)$ as $1-\Theta\big(|\vec{p}|^2-(p^0)^2\big)$, we can immediately calculate

$$\int \Theta(-p^2)e^{-ip^0t}\frac{dp^0}{2\pi} = \left[\delta(t)-\frac{\sin\left(|\vec{p}|t\right)}{\pi t}\right]. \tag{16.128}$$

Combining the two we finally get

$$\begin{aligned}\int \ln\left(\frac{p^2\pm i0}{\bar{\mu}^2}\right)e^{-ip^0t}\frac{dp^0}{2\pi} &= -\cos\left(|\vec{p}|t\right)\mathcal{P}'\frac{1}{|t|}-\delta(t)\ln\left(\frac{\bar{\mu}^2}{\mu_0^2}\right)\\ &\quad \pm i\pi\left[\delta(t)-\frac{\sin\left(|\vec{p}|t\right)}{\pi t}\right], \end{aligned} \tag{16.129}$$

which provides the kernels (16.66) which we need for our calculation.

16.7.3 One-Loop Graviton Propagator Integrals

Here we explain in some detail how to calculate the integrals in Eq. (16.62) using the iϵ prescription described in Section 9.3.

The first thing that we will do is to substitute the definitions of the distributions (16.66) from Section 16.7.2 into these integrals. We then note that the $\delta(t)\ln(\mu_0)$ part from Eq. (16.122) appearing in the kernel $\bar{L}$ can be used to integrate over τ', so that the logarithm can be combined with the one appearing in the third integral of Eq. (16.62). Setting $\mu_0 = \bar{\mu}$, this gives

$$\int \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0(\eta,\tau,\vec{p})\right] \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0(\tau,\eta',\vec{p})\right] \ln\left(\frac{\bar{\mu}}{H}\right) d\tau. \quad (16.130)$$

We will perform this integral explicitly, which will serve as an illustration of the computation of the other integrals since they all follow the same general pattern.

The two-point function that we will calculate is the imaginary unit times the positive Wightman function $\bar{G}^{-+}$ defined in (9.62). The contribution of the above integral is then given by $-3i\alpha\kappa^2 \ln\left(\frac{\bar{\mu}}{H}\right) J$, where

$$\begin{aligned} J = &\int_{t_0^+}^{T} \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+}(\eta,\tau,\vec{p})\right] \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{++}(\tau,\eta',\vec{p})\right] d\tau \\ &- \int_{t_0^-}^{T} \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{--}(\eta,\tau,\vec{p})\right] \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+}(\tau,\eta',\vec{p})\right] d\tau. \end{aligned} \quad (16.131)$$

We are integrating from the initial time $t_0^\pm$ to the final time T, according to whether the integration variable τ belongs to the upper or lower half of the complex contour.

For the Wightman functions we have

$$\begin{aligned} (\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+/+-}(\eta,\tau,\vec{p}) &= \frac{2}{\tau}\partial_\tau \bar{G}^0_{-+/+-}(\eta,\tau,\vec{p}) \\ (\partial_\tau^2 + \vec{p}^2)^2\bar{G}^0_{-+/+-}(\eta,\tau,\vec{p}) &= 0, \end{aligned} \quad (16.132)$$

since these functions are proportional to f or f^*, which are solutions of the equation of motion (16.50). To make sure that we do not get a contribution from the arbitrary final time T, we first write

$$\begin{aligned} \bar{G}^0_{++}(\eta,\eta',\vec{p}) &= \Theta(\eta-\eta')\left[\bar{G}^0_{-+}(\eta,\eta',\vec{p}) - \bar{G}^0_{+-}(\eta,\eta',\vec{p})\right] + \bar{G}^0_{+-}(\eta,\eta',\vec{p}) \\ &= \bar{G}^0_{\rm ret}(\eta,\eta',\vec{p}) + \bar{G}^0_{+-}(\eta,\eta',\vec{p}), \\ \bar{G}^0_{--}(\eta,\eta',\vec{p}) &= -\bar{G}^0_{\rm ret}(\eta,\eta',\vec{p}) + \bar{G}^0_{-+}(\eta,\eta',\vec{p}), \end{aligned} \quad (16.133)$$

which simplifies J to

$$\begin{aligned} J = &\int_{t_0^+}^{T} \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+}(\eta,\tau,\vec{p})\right]\left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{\rm ret}(\eta',\tau,\vec{p})\right] d\tau \\ &+ \int_{t_0^-}^{T} \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{\rm ret}(\eta,\tau,\vec{p})\right]\left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+}(\tau,\eta',\vec{p})\right] d\tau \\ &- \int_{t_0^-}^{t_0^+} \left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+}(\eta,\tau,\vec{p})\right]\left[(\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{-+}(\tau,\eta',\vec{p})\right] d\tau. \end{aligned} \tag{16.134}$$

We see that the result will not depend on T provided that $T > \eta, \eta'$, as expected. Next, we calculate

$$\begin{aligned} \partial_\tau \bar{G}^0_{\rm ret}(\eta,\tau,\vec{p}) &= \Theta(\eta-\tau)\left(\partial_\tau \bar{G}^0_{-+}(\eta,\tau,\vec{p}) - \partial_\tau \bar{G}^0_{+-}(\eta,\tau,\vec{p})\right) \\ (\partial_\tau^2 + \vec{p}^2)\bar{G}^0_{\rm ret}(\eta,\tau,\vec{p}) &= \frac{2}{\tau}\partial_\tau \bar{G}^0_{\rm ret}(\eta,\tau,\vec{p}) - H^2\eta^2\delta(\tau-\eta), \end{aligned} \tag{16.135}$$

and together with Eq. (16.132) this leads to

$$\begin{aligned} J = &-2H^2\eta\left[\partial_\eta \bar{G}^0_{-+}(\eta,\eta',\vec{p})\right] - 2H^2\eta'\left[\partial_{\eta'} \bar{G}^0_{-+}(\eta,\eta',\vec{p})\right] \\ &+ 4\int_{t_0^+}^{T} \frac{1}{\tau^2}\left[\partial_\tau \bar{G}^0_{-+}(\eta,\tau,\vec{p})\right]\left[\partial_\tau \bar{G}^0_{\rm ret}(\eta',\tau,\vec{p})\right] d\tau \\ &+ 4\int_{t_0^-}^{T} \frac{1}{\tau^2}\left[\partial_\tau \bar{G}^0_{\rm ret}(\eta,\tau,\vec{p})\right]\left[\partial_\tau \bar{G}^0_{-+}(\tau,\eta',\vec{p})\right] d\tau \\ &- 4\int_{t_0^-}^{t_0^+} \frac{1}{\tau^2}\left[\partial_\tau \bar{G}^0_{-+}(\eta,\tau,\vec{p})\right]\left[\partial_\tau \bar{G}^0_{-+}(\tau,\eta',\vec{p})\right] d\tau. \end{aligned} \tag{16.136}$$

Using the definition of the Wightman functions from Eq. (16.61), we get

$$\begin{aligned} J = &-\frac{H^4}{|\vec{p}|}(\eta')^2(|\vec{p}|\eta - i)e^{-i|\vec{p}|(\eta-\eta')} + \frac{H^4}{|\vec{p}|}\eta^2(|\vec{p}|\eta' + i)e^{-i|\vec{p}|(\eta-\eta')} \\ &- i\frac{H^4}{2|\vec{p}|^3}(|\vec{p}|\eta - i)(|\vec{p}|\eta' - i)e^{-i|\vec{p}|(\eta+\eta')}\left(e^{2i|\vec{p}|\eta'} - e^{2i|\vec{p}|t_0^+}\right) \\ &- i\frac{H^4}{2|\vec{p}|^3}(|\vec{p}|\eta + i)(|\vec{p}|\eta' + i)e^{i|\vec{p}|(\eta+\eta')}\left(e^{-2i|\vec{p}|\eta} - e^{-2i|\vec{p}|t_0^-}\right) \\ &+ \frac{H^4}{|\vec{p}|^2}(|\vec{p}|\eta - i)(|\vec{p}|\eta' + i)e^{-i|\vec{p}|(\eta-\eta')}(\eta' - \eta). \end{aligned} \tag{16.137}$$

Finally, we take the limit $t_0^\pm \to -\infty(1 \mp i\epsilon)$, which causes the exponentials containing $t_0^\pm$ vanish. After collecting terms we get

$$J = i\frac{H^4}{|\vec{p}|^3}(|\vec{p}|\eta - i)(|\vec{p}|\eta' + i)e^{-i|\vec{p}|(\eta-\eta')} = 2if(\eta,\eta',\vec{p}), \tag{16.138}$$

so that the contribution of the integral (16.130) is $6\alpha\kappa^2 \ln\left(\frac{\bar{\mu}}{H}\right) f(\eta, \eta', \vec{p})$, as given in the final result (16.67).

For the remaining integrals containing the kernels (16.65), we need the special function Ein, which is given in Section 16.7.1. To further facilitate the calculation, it is convenient to combine the integrals containing the kernels $\bar{L}$ and $\bar{N}$ as well as $\bar{D}$ and $\bar{N}$ to give the distributions $e^{\pm i|\vec{p}|(\tau-\tau')}\mathcal{P}\frac{1}{\tau-\tau'}$, which permits taking the limit $t_0^{\pm} \to -\infty(1 \mp i\epsilon)$ already in the result of the integral over τ' before integrating over τ.

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Riemann Tensor Correlator in de Sitter

The Riemann tensor correlation function exhibits a number of interesting features. First, with appropriately raised indices it is gauge-invariant even when including contributions from loops of matter fields but excluding graviton loops, which as we have seen before can be implemented systematically within a large N expansion for N matter fields. Second, it is compatible with de Sitter invariance, in contrast with other gauge-invariant quantities more commonly used in cosmology. In fact, whether de Sitter invariance is preserved can be manifestly seen by expressing the result in terms of maximally symmetric bitensors. Thirdly, the Riemann tensor provides a complete and suitable characterization of the local geometry: geometrical observables associated with a finite region of spacetime can be obtained from it. Thus, they are insensitive to potentially infrared divergent superhorizon contributions which cannot be probed by observables involving finite spacetime regions.

In this chapter the linearized Riemann tensor correlator on a de Sitter background including one-loop corrections from conformal fields is derived, following Fröb et al. [572]. In Section 17.1 we revisit the formalism described in Section 15.1 of de Sitter-invariant bitensors, and introduce an alternative set of bitensors which are useful for this computation. The main result of the chapter is obtained in Section 17.2: by taking suitable derivatives of the metric correlator function found in Section 16.5, the connected two-point correlator function of the Riemann tensor $R^{ab}{}_{cd}$ is computed. The result is written in a manifestly de Sitter-invariant form in terms of the bitensors introduced by Allen and Jacobson. Moreover, given the decomposition of the Riemann tensor in terms of Weyl and Ricci tensors, we write the explicit results for the Weyl and Ricci tensors correlators as well as the Weyl–Ricci tensors correlator and study both their subhorizon and superhorizon behavior. These results are extended to general conformal field theories in Section 17.3.

We also derive the Riemann tensor correlator in Minkowski spacetime in a manifestly Lorentz invariant form in Section 17.4, by carefully taking the flat-space limit of our result in de Sitter. To the best of our knowledge, this result for the Riemann tensor correlator in Minkowski spacetime including loop corrections from general conformal fields had not been obtained previously to the calculation in de Sitter by Fröb et al. [572]. There were results in the literature for the metric correlator by Tomboulis [686] and Martín and Verdaguer [189], and the correlator of the Einstein tensor by Osborn and Shore [204], which is closely related to the stress-energy tensor correlator, but not for the full Riemann tensor correlator.

17.1 De Sitter-Invariant Bitensors

We have seen in Section 15.1 that any de Sitter-invariant two-point correlation function can be expressed in terms of maximally symmetric bitensors. Here we adapt the formalism to our problem. The basic de Sitter-invariant biscalar $Z(x,x')$ was defined in (15.8) The geodesic distance $\mu(x,x')$ along the shortest geodesic connecting the points x and x' is related to the biscalar $Z(x,x')$ by Eq. (15.18), which we now write re-introducing the Hubble parameter H as

$$Z(x,x') = \cos\left(H\mu(x,x')\right). \tag{17.1}$$

For points such that $Z(x,x') < -1$, i.e., points not connected by a geodesic, μ can still be defined by a suitable analytic continuation from (17.1).

As we have seen in Section 15.1 a theorem by Allen and Jacobson [476] establishes that there exists a complete set of de Sitter-invariant bitensors from which all maximally symmetric bitensors can be constructed. This set is given by the metric $\hat{g}_{ab}(x)$, the geodesic distance $\mu(x,x')$, the unit tangent vectors to the geodesic at the points x and x', $n_a(x,x')$ and $n_{a'}(x,x')$, defined by Eq. (15.14), and the propagator $\hat{g}_{ab'}(x,x')$ which parallely transports a vector from x' to x along the geodesic connecting them. Here as in Chapter 16 we use a hat over a symbol to refer to quantities which are defined on the background de Sitter space, see Eq. (16.18).

By inverting Eq. (17.1), we can write the complete set of bitensors described above in terms of covariant derivatives of $Z(x,x')$ as follows:

$$\begin{gathered} n_a = -\frac{1}{H\sqrt{1-Z^2}}\hat{\nabla}_a Z, \qquad n_{a'} = -\frac{1}{H\sqrt{1-Z^2}}\hat{\nabla}_{a'} Z, \\ H^2\hat{g}_{ab'} = \hat{\nabla}_a\hat{\nabla}_{b'}Z - \frac{1}{1+Z}(\hat{\nabla}_a Z)(\hat{\nabla}_{b'}Z). \end{gathered} \tag{17.2}$$

In addition, as an illustration of the above theorem, see Eq. (15.17), we have that

$$\hat{\nabla}_a n_b = \frac{Z}{\sqrt{1-Z^2}} H\left(\hat{g}_{ab} - n_a n_b\right), \tag{17.3}$$

from which we can calculate the second derivative of Z at one point as

$$\hat{\nabla}_a \hat{\nabla}_b Z = -\sqrt{1-Z^2}\, H \hat{\nabla}_a n_b - Z H^2 n_a n_b = -Z H^2 \hat{g}_{ab}. \tag{17.4}$$

It turns out that in many calculations it is more convenient to use the set provided by covariant derivatives of Z, because the application of a covariant derivative directly gives another member of this set, with the only exception shown in (17.4) which has a very simple structure. In contrast, if one uses the Allen and Jacobson set, one encounters more or less complicated prefactors as in Eq. (17.3).

In the conformally flat coordinate system of the Poincaré patch, we can use Eq. (15.10) together with the embedding map $X^A(x)$ for these coordinates,

$$\begin{aligned} X^0 &= -\frac{1+\vec{x}^{\,2}}{2\,\eta} + \frac{\eta}{2}, \\ X^\alpha &= -\frac{x^\alpha}{\eta}, \quad \alpha = 1,2,3 \\ X^4 &= \frac{1-\vec{x}^{\,2}}{2\,\eta} + \frac{\eta}{2}, \end{aligned}$$

to obtain

$$Z(x,x') = 1 - \frac{(x-x')^2}{2\eta\eta'} = 1 - \frac{(\vec{x}-\vec{x}')^2 - (\eta-\eta')^2}{2\eta\eta'}, \tag{17.5}$$

from which we can derive

$$\begin{aligned} \partial_\eta Z &= \frac{1}{\eta'} - \frac{Z}{\eta}, & \partial_{\eta'} Z &= \frac{1}{\eta} - \frac{Z}{\eta'}, \\ \partial_\alpha Z &= -\frac{\eta_{\alpha\beta}(x^\beta - x'^\beta)}{\eta\eta'}, & \partial_{\alpha'} Z &= \frac{\eta_{\alpha\beta}(x^\beta - x'^\beta)}{\eta\eta'}. \end{aligned} \tag{17.6}$$

In the next section we need contractions of the derivatives of Z. To streamline the notation we use also a semicolon to denote the covariant derivative $\hat{\nabla}_a$ with respect to the background de Sitter metric, i.e., $Z_{;a} = \hat{\nabla}_a Z$. Using Eq. (17.2) as well as the normalization $n_a n^a = 1$ and $n_{a'} = -\hat{g}_{a'b} n^b$, which implies $Z_{;a'} = -\hat{g}_{a'b} Z^{;b}$, we get

$$\begin{aligned} Z_{;a} Z^{;a} &= H^2(1-Z^2), & Z_{;a'} Z^{;a'} &= H^2(1-Z^2), \\ Z_{;ab'} Z^{;a} &= -H^2 Z Z_{;b'}, & Z_{;ab'} Z^{;b'} &= -H^2 Z Z_{;a}, \\ Z_{;ab'} Z^{;ac'} &= H^4 \delta^{c'}_{b'} - H^2 Z_{;b'} Z^{;c'}, & Z_{;ab'} Z^{;cb'} &= H^4 \delta^c_a - H^2 Z_{;a} Z^{;c}. \end{aligned} \tag{17.7}$$

Of course, this can also be directly calculated from expression (17.5) for Z in the spatially flat coordinates.

17.2 Correlators of Curvature Tensors

We first compute the two-point correlator for the Riemann tensor and express it in terms of de Sitter-invariant bitensors. The Riemann tensor is then decomposed into its Ricci and Weyl tensor parts. The linearized Riemann tensor $\tilde{R}^{ab}{}_{cd}$ of the perturbed de Sitter spacetime,

$$\tilde{g}_{mn} = a^2(\eta)\left(\eta_{mn} + \kappa h_{mn}\right), \tag{17.8}$$

can be written in terms of the metric perturbations and its derivatives as

$$\begin{aligned}\tilde{R}^{ab}{}_{cd} &= 2H^2\delta^a_{[c}\delta^b_{d]}\left(1+\kappa h_{00}\right) + 2\kappa H^2\eta^2\eta^{m[a}\eta^{b]n}\partial_n\partial_{[c}h_{d]m} \\ &\quad + 2\kappa H^2\eta\delta^m_{[c}\delta^{[b}_{d]}\eta^{a]n}\left(2\partial_{(m}h_{n)0} - \partial_0 h_{mn}\right) + \mathcal{O}(\kappa^2) \\ &= 2H^2\delta^a_{[c}\delta^b_{d]} + \mathcal{T}^{ab}{}_{cd}{}^{mn}h_{mn} + \mathcal{O}(\kappa^2).\end{aligned} \tag{17.9}$$

In the first equality we have used Section 12.5, after replacing h_{mn} by κh_{mn}, for the perturbative expansion in κ. In the second equality we define the differential operator $\mathcal{T}$, which is of order κ.

One may check the gauge invariance of this expression using the gauge transformation (16.20). It is only with the given index position that gauge invariance holds, because then the Lie derivative of the unperturbed tensor with respect to an arbitrary vector field $\hat{\xi}^a$ vanishes;

$$\mathcal{L}_{\hat{\xi}}\hat{R}^{ab}{}_{cd} = \mathcal{L}_{\hat{\xi}}\left(2H^2\delta^a_{[c}\delta^b_{d]}\right) = 0, \tag{17.10}$$

which corresponds to the diffeomorphisms generated by $\hat{\xi}^a$ at linear order.

To calculate the Riemann tensor correlator, we apply the differential operator $\mathcal{T}$ defined in Eq. (17.9) to each argument of the two-point function of the metric perturbations given in Section 16.5. If we consider only the connected correlator

$$\langle A(x)B(x')\rangle_{\rm c} \equiv \langle {\rm in}|A(x)B(x')|{\rm in}\rangle - \langle {\rm in}|A(x)|{\rm in}\rangle\langle {\rm in}|B(x')|{\rm in}\rangle, \tag{17.11}$$

the terms of order κ^2 in the Riemann tensor, which would give a contribution when combined with the background value of the Riemann tensor, cancel out. Consequently, to lowest order we have

$$\langle\tilde{R}^{ab}{}_{cd}(x)\tilde{R}^{m'n'}{}_{p'q'}(x')\rangle_{\rm c} = 4\kappa^2H^4\hat{g}^{au}\hat{g}^{bv}\hat{g}^{m's'}\hat{g}^{n't'}U_{[uv][cd][s't'][p'q']}(x,x'), \tag{17.12}$$

where the bitensor U is defined in terms of its spatial Fourier transform $\bar{U}$, which is indicated by an overbar as in Eq. (16.25),

$$\begin{aligned}\bar{U}_{abcdm'n'p'q'}(\eta,\eta',\vec{p}) &= \frac{1}{H^8\eta^4(\eta')^4}\sum_{k,l=0}^{2}T^{(k)}{}_{abcd}{}^{rs}(\eta)T^{(l)*}{}_{m'n'p'q'}{}^{t'u'}(\eta') \\ &\quad\times\left(i\bar{G}^{-+}_{rst'u'}(\eta,\eta',\vec{p})\right),\end{aligned} \tag{17.13}$$

with the differential operators

$$
\begin{aligned}
T^{(0)}{}_{abcd}{}^{rs}(\eta) &= \left(\eta_{ac}\eta_{bd}\delta_0^r\delta_0^s + 2i\eta\eta_{ac}p_{(b}\delta_{d)}^r\delta_0^s + \eta^2 p_a p_c \delta_b^r\delta_d^s\right), \\
T^{(1)}{}_{abcd}{}^{rs}(\eta) &= \left(-\eta\eta_{ac}\delta_b^r\delta_d^s + 2\eta\eta_{ac}\delta^0_{(b}\delta^r_{d)}\delta_0^s - 2i\eta^2 p_{(a}\delta^0_{c)}\delta_b^r\delta_d^s\right)\partial_\eta, \\
T^{(2)}{}_{abcd}{}^{rs}(\eta) &= -\eta^2\delta^0_a\delta^0_c\delta^r_b\delta^s_d\,\partial_\eta^2,
\end{aligned}
\tag{17.14}
$$

where the superscript in parentheses indicates the number of time derivatives involved.

17.2.1 Riemann Tensor Correlator

The first step of the calculation is to compute the bitensor $\bar{U}$ defined in Eq. (17.13). After calculating $\bar{U}$ we have to invert the Fourier transform and write in terms of maximally symmetric bitensors the resulting tensor structure, which will consist of linear combinations of η_{ab}, δ^0_a and $r_a = \eta_{ab}r^b$, where $r^a = (0, \vec{r})$ with $\vec{r} = \vec{x} - \vec{x}'$.

The coefficient functions, which will be functions of η, η' and $\vec{r}$, also have to be written in terms of the de Sitter-invariant biscalar $Z(x, x')$. All the necessary substitutions can be deduced from expression (17.5) for $Z(x, x')$ and its derivatives (17.6) in our spatially flat coordinate system. Explicitly, we have

$$
\begin{aligned}
&\eta_{ab} \to H^2\eta^2\hat{g}_{ab}, && r_a \to -\eta\eta' Z_{;a} + \delta^0_a\left(\eta - \eta' Z\right), \\
&\eta_{a'b'} \to H^2(\eta')^2\hat{g}_{a'b'}, && r_{a'} \to \eta\eta' Z_{;a'} - \delta^0_{a'}\left(\eta' - \eta Z\right), \\
&\eta_{ab'} \to \eta\eta' Z_{;ab'} + \eta\delta^0_{b'}Z_{;a} + \eta'\delta^0_a Z_{;b'} + \delta^0_a\delta^0_{b'}(Z-1).
\end{aligned}
\tag{17.15}
$$

Technical details for the calculation of the Fourier transforms are provided in Section 17.5.

It turns out that after a long calculation, computing the Fourier transforms and implementing the substitutions described above, all the terms which are not de Sitter-invariant cancel out. For the very long tensor algebra the open source *xAct* tensor package by Martín-García [983] was used. We are then left with a de Sitter-invariant result. This may be expressed in terms of the complete set of Allen and Jacobson basis bitensors, which involve n_a, $n_{a'}$ and $\hat{g}_{ab'}$. Note that we do not write explicitly the dependence of those de Sitter-invariant bitensors on x and x'.

The final result is

$$
\begin{aligned}
\langle \tilde{R}^{ab}{}_{cd}(x)\tilde{R}^{m'n'}{}_{p'q'}(x')\rangle_{\mathrm{c}} = \frac{4\kappa^2H^6}{\pi^2}\sum_{k=1}^{9}{}^{(k)}\mathcal{N}^{[ab]}{}_{[cd]}{}^{[m'n']}{}_{[p'q']} \\
\times\,\mathrm{d}\lim_{\epsilon\to 0^+}\left[\frac{3}{2}\alpha\kappa^2H^2\mathcal{S}^{(1,k)}(Z(x,x') - i\epsilon^{\pm})\right.
\end{aligned}
$$

$$+\left(1+6\alpha\kappa^2H^2\left(\ln\left(\frac{\bar{\mu}}{H}\right)+\gamma\right)-(5\alpha-2\beta)\,\kappa^2H^2\right)$$
$$\times\,\mathcal{S}^{(0,k)}(Z(x,x')-i\epsilon^{\pm})\Bigg]. \tag{17.16}$$

Here $\epsilon^{\pm}=+\epsilon$ whenever x lies within the future light cone of x', while $\epsilon^{\pm}=-\epsilon$ when x lies within the past light cone of x'. If both points are spacelike separated, $Z(x,x')<1$, the sign of ϵ is irrelevant. The limit $\epsilon\to 0^+$ is to be taken in the sense of distributions, i.e., after smearing with a test function, which we indicate with the notation "d lim".

The ${}^{(k)}\mathcal{N}$ are a set of nine 8-index de Sitter-invariant bitensors constructed with the Allen and Jacobson basis, and which have also the appropriate symmetries when antisymmetrized in each pair of indices:

$$\begin{aligned}
{}^{(1)}\mathcal{N}_{abcdm'n'p'q'} &= \hat{g}_{ac}\,\hat{g}_{bd}\,\hat{g}_{m'p'}\,\hat{g}_{n'q'},\\
{}^{(2)}\mathcal{N}_{abcdm'n'p'q'} &= \hat{g}_{ac}\,\hat{g}_{m'p'}\left(\hat{g}_{bd}\,n_{n'}n_{q'}+n_bn_d\,\hat{g}_{n'q'}\right),\\
{}^{(3)}\mathcal{N}_{abcdm'n'p'q'} &= \hat{g}_{ac}\,\hat{g}_{m'p'}\,n_bn_dn_{n'}n_{q'},\\
{}^{(4)}\mathcal{N}_{abcdm'n'p'q'} &= 4\hat{g}_{ac}\,\hat{g}_{m'p'}\,n_{(b}\hat{g}_{d)(n'}n_{q')},\\
{}^{(5)}\mathcal{N}_{abcdm'n'p'q'} &= 2\hat{g}_{ac}\,\hat{g}_{m'p'}\,\hat{g}_{b(n'}\hat{g}_{q')d},\\
{}^{(6)}\mathcal{N}_{abcdm'n'p'q'} &= 2\left(\hat{g}_{ac}\,n_{m'}n_{p'}+n_an_c\,\hat{g}_{m'p'}\right)\hat{g}_{b(n'}\hat{g}_{q')d},\\
{}^{(7)}\mathcal{N}_{abcdm'n'p'q'} &= 2n_an_cn_{m'}n_{p'}\,\hat{g}_{b(n'}\hat{g}_{q')d},\\
{}^{(8)}\mathcal{N}_{abcdm'n'p'q'} &= 8n_{(a}\hat{g}_{c)(m'}n_{p')}\,\hat{g}_{b(n'}\hat{g}_{q')d},\\
{}^{(9)}\mathcal{N}_{abcdm'n'p'q'} &= 4\hat{g}_{a(m'}\hat{g}_{p')c}\,\hat{g}_{b(n'}\hat{g}_{q')d}.
\end{aligned} \tag{17.17}$$

The coefficients $\mathcal{S}^{(0,k)}(Z)$ and $\mathcal{S}^{(1,k)}(Z)$ are functions of Z and are, therefore, manifestly de Sitter-invariant biscalars. The explicit expressions for the tree-level coefficients $\mathcal{S}^{(0,k)}(Z)$ in Eq. (17.16) are given by

$$\begin{aligned}
&\mathcal{S}^{(0,1)}(Z)=\frac{1}{8}(1-3Z)(1-Z)^{-3}, && \mathcal{S}^{(0,2)}(Z)=\frac{3}{4}(1+Z)(1-Z)^{-3},\\
&\mathcal{S}^{(0,3)}(Z)=-12(1-Z)^{-3}, && \mathcal{S}^{(0,4)}(Z)=-\frac{3}{4}(3-Z)(1-Z)^{-3},\\
&\mathcal{S}^{(0,5)}(Z)=-\frac{3}{4}(1-Z)^{-2}, && \mathcal{S}^{(0,6)}(Z)=-\frac{3}{4}(1+Z)(1-Z)^{-3},\\
&\mathcal{S}^{(0,7)}(Z)=12(1-Z)^{-3}, && \mathcal{S}^{(0,8)}(Z)=\frac{1}{4}(7-Z)(1-Z)^{-3},\\
&\mathcal{S}^{(0,9)}(Z)=\frac{1}{8}(3-Z)(1-Z)^{-3}.
\end{aligned} \tag{17.18}$$

The explicit expressions for the matter loop corrections $\mathcal{S}^{(1,k)}$ are

$$\begin{aligned}
\mathcal{S}^{(1,1)}(Z) &= -4Z^3(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad -\frac{1}{6}(1-2Z-Z^2+26Z^3)(1+Z)^{-2}(1-Z)^{-4}, \\
\mathcal{S}^{(1,2)}(Z) &= 12Z(1+Z^2)(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad +Z(11+4Z+17Z^2)(1+Z)^{-2}(1-Z)^{-4}, \\
\mathcal{S}^{(1,3)}(Z) &= -24(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right]-10(3+5Z)(1-Z)^{-4}, \\
\mathcal{S}^{(1,4)}(Z) &= -6(1+Z+3Z^2-Z^3)(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad -\frac{1}{2}(11+21Z+45Z^2-13Z^3)(1+Z)^{-2}(1-Z)^{-4}, \\
\mathcal{S}^{(1,5)}(Z) &= -6Z(1+Z)^{-2}(1-Z)^{-2}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad -\frac{1}{2}Z(11-13Z)(1+Z)^{-1}(1-Z)^{-4}, \\
\mathcal{S}^{(1,6)}(Z) &= -12Z(1+Z^2)(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad -(2+17Z+10Z^2+19Z^3)(1+Z)^{-2}(1-Z)^{-4}, \\
\mathcal{S}^{(1,7)}(Z) &= 24(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right]+10(7+5Z)(1-Z)^{-4}, \\
\mathcal{S}^{(1,8)}(Z) &= 4(1+2Z+3Z^2)(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad +(9+20Z+19Z^2)(1+Z)^{-2}(1-Z)^{-4}, \\
\mathcal{S}^{(1,9)}(Z) &= 4Z(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad +(1+5Z)(1+Z)^{-2}(1-Z)^{-4}.
\end{aligned} \tag{17.19}$$

An important feature of those coefficients is that they are well-defined distributions. Regarded as functions, they are only singular as $Z \to 1$, i.e., when x and x' are null separated, which is also the case for the two-point function of the matter field ϕ. Hereafter the dependence of Z on x and x' should always be implicitly understood.

We have to emphasize that this is a rather remarkable and nontrivial result. De Sitter invariance has not been assumed but derived from the two-point metric correlations computed in the Poincaré patch, which are not only gauge-dependent

but also not manifestly de Sitter invariant. The only de Sitter-invariant input is the selection of a de Sitter-invariant initial vacuum. In this respect it is important to keep in mind that while the Bunch–Davies vacuum for a free theory is de Sitter invariant, it ceases to be de Sitter invariant when it is evolved by a Hamiltonian that includes interactions, a point which is often overlooked in the literature. A de Sitter-invariant state for the interacting theory is needed instead. The $i\epsilon$ prescription used selects as asymptotic initial state an adiabatic vacuum of the interacting theory which constitutes the appropriate generalization of the Bunch–Davies vacuum to the interacting case.

This result should also be seen as a nontrivial check of the contributions to the two-point correlation functions of the metric perturbations, given by (16.92), (16.94) and (16.96). In addition to the correct index symmetries, the two-point function of the Riemann tensor (17.16) obey both the first and second Bianchi identities as expected. For the Bianchi identities to be satisfied, the (distributional) kernel $\bar{S}$ of the I_5 term in the two-point function of the tensorial metric perturbations (16.96) is important.

Note that even though the correlation functions of the metric perturbations were calculated in the Poincaré patch (this was important in order to select a de Sitter-invariant vacuum for the interacting theory), when written in the de Sitter-invariant basis, the results for the Riemann tensor correlator can be extended to the full de Sitter spacetime.

In order to analyze the structure of the Riemann tensor correlator, it is better to decompose the Riemann tensor in terms of its Weyl and Ricci parts:

$$\begin{aligned}\langle \tilde{R}^{ab}{}_{cd}(x)\tilde{R}^{m'n'}{}_{p'q'}(x')\rangle_{\mathrm{c}} &= \langle \tilde{C}^{ab}{}_{cd}(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\rangle_{\mathrm{c}} \\ &\quad + 4\delta^{[a}_{[c}\langle \tilde{R}^{b]}{}_{d]}(x)\tilde{R}^{[m'}{}_{[p'}(x')\rangle_{\mathrm{c}}\delta^{n']}_{q']} \\ &\quad + 2\delta^{[a}_{[c}\langle \tilde{R}^{b]}{}_{d]}(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\rangle_{\mathrm{c}} \\ &\quad + 2\langle \tilde{C}^{ab}{}_{cd}(x)\tilde{R}^{[m'}{}_{[p'}(x')\rangle_{\mathrm{c}}\delta^{n']}_{q']}, \end{aligned} \tag{17.20}$$

where the correlation functions on the right-hand side are obtained from the Riemann tensor correlator by contracting indices (for the Ricci tensor) and by subtracting traces from the Riemann tensor correlator (for the Weyl tensor).

We discuss next, in more detail, the main properties of these correlators. From the previous decomposition it also follows that the scalar correlators defined as

$$\begin{aligned}\langle \tilde{\mathrm{Rie}}^2(x,x')\rangle_{\mathrm{c}} &\equiv \hat{g}_{am'}\hat{g}_{bn'}\hat{g}^{cp'}\hat{g}^{dq'}\langle \tilde{R}^{ab}{}_{cd}(x)\tilde{R}^{m'n'}{}_{p'q'}(x')\rangle_{\mathrm{c}}, \\ \langle \tilde{\mathrm{Weyl}}^2(x,x')\rangle_{\mathrm{c}} &\equiv \hat{g}_{am'}\hat{g}_{bn'}\hat{g}^{cp'}\hat{g}^{dq'}\langle \tilde{C}^{ab}{}_{cd}(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\rangle_{\mathrm{c}}, \\ \langle \tilde{\mathrm{Ric}}^2(x,x')\rangle_{\mathrm{c}} &\equiv \hat{g}_{bn'}\hat{g}^{dq'}\langle \tilde{R}^{b}{}_{d}(x)\tilde{R}^{n'}{}_{q'}(x')\rangle_{\mathrm{c}}, \end{aligned} \tag{17.21}$$

are related by the following identity:

$$\langle \tilde{\mathrm{Rie}}^2(x,x')\rangle_{\mathrm{c}} = \langle \tilde{\mathrm{Weyl}}^2(x,x')\rangle_{\mathrm{c}} + 2\langle \tilde{\mathrm{Ric}}^2(x,x')\rangle_{\mathrm{c}}. \tag{17.22}$$

17.2.2 Weyl Tensor Correlator

In four dimensions the Weyl tensor can be calculated by subtracting traces from the Riemann tensor as follows:

$$C^{ab}{}_{cd} = \left(\delta^a_k\delta^b_l\delta^s_c\delta^t_d + 2\delta^s_k\delta^{[a}_l\delta^{b]}_{[c}\delta^t_{d]} + \frac{1}{3}\delta^{[a}_{[c}\delta^{b]}_{d]}\delta^s_k\delta^t_l\right) R^{kl}{}_{st}. \tag{17.23}$$

We can then calculate its correlator very easily from the Riemann tensor correlator (17.16). However, because of the tracelessness of the Weyl tensor, the corresponding coefficients for the bitensor set (17.17) are not independent since those bitensors are not traceless.

In fact, there are only three combinations of invariant bitensors which have vanishing trace on any contraction. These are the traceless tensors ${}^{(k)}\mathcal{D}$ defined as

$$\begin{aligned}
{}^{(1)}\mathcal{D}_{abcdm'n'p'q'} &= \left[{}^{(1)}\mathcal{N} - 3\,{}^{(5)}\mathcal{N} + {}^{(9)}\mathcal{N}\right]_{abcdm'n'p'q'}\\
{}^{(2)}\mathcal{D}_{abcdm'n'p'q'} &= \left[-12\,{}^{(3)}\mathcal{N} - 3\,{}^{(4)}\mathcal{N} + 12\,{}^{(7)}\mathcal{N} + 2\,{}^{(8)}\mathcal{N}\right]_{abcdm'n'p'q'}\\
{}^{(3)}\mathcal{D}_{abcdm'n'p'q'} &= \Big[- 2\,{}^{(1)}\mathcal{N} + 6\,{}^{(2)}\mathcal{N} - 12\,{}^{(3)}\mathcal{N} + 3\,{}^{(4)}\mathcal{N}\\
&\quad + 3\,{}^{(5)}\mathcal{N} - 6\,{}^{(6)}\mathcal{N} + 12\,{}^{(7)}\mathcal{N}\Big]_{abcdm'n'p'q'},
\end{aligned} \tag{17.24}$$

where the basic bitensor set ${}^{(k)}\mathcal{N}$ was introduced in Eq. (17.17).

We, therefore, obtain the following result for the correlator of the Weyl tensor for perturbations around de Sitter spacetime including radiative corrections from conformal matter fields up to order κ^4:

$$\begin{aligned}
\left\langle \tilde{C}^{ab}{}_{cd}(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\right\rangle_{\rm c} &= \frac{4\kappa^2H^6}{\pi^2}\sum_{k=1}^{3}{}^{(k)}\mathcal{D}^{[ab]}{}_{[cd]}{}^{[m'n']}{}_{[p'q']}\mathrm{d}\\
&\quad\times\lim_{\epsilon\to0^+}\left[\frac{\alpha}{4}\kappa^2H^2\mathcal{D}^{(1,k)}(Z-i\epsilon^{\pm})\right.\\
&\quad+\left(1+6\alpha\kappa^2H^2\left(\ln\left(\frac{\bar{\mu}}{H}\right)+\gamma\right)-(5\alpha-2\beta)\,\kappa^2H^2\right)\\
&\quad\left.\times\,\mathcal{D}^{(0,k)}(Z-i\epsilon^{\pm})\right].
\end{aligned} \tag{17.25}$$

The coefficient functions $\mathcal{D}^{(0,k)}$ are given by

$$\begin{aligned}
\mathcal{D}^{(0,1)}(Z) &= \frac{1}{8}(3-Z)(1-Z)^{-3}\\
\mathcal{D}^{(0,2)}(Z) &= \frac{1}{8}(7-Z)(1-Z)^{-3}\\
\mathcal{D}^{(0,3)}(Z) &= \frac{1}{8}(1+Z)(1-Z)^{-3}.
\end{aligned} \tag{17.26}$$

The coefficient functions for the matter loop corrections $\mathcal{D}^{(1,k)}$ are given by

$$\begin{aligned}
\mathcal{D}^{(1,1)}(Z) &= 24Z(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad + 6(1+5Z)(1+Z)^{-2}(1-Z)^{-4} \\
\mathcal{D}^{(1,2)}(Z) &= 12(1+2Z+3Z^2)(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad + 3(9+20Z+19Z^2)(1+Z)^{-2}(1-Z)^{-4} \\
\mathcal{D}^{(1,3)}(Z) &= 12Z(1+Z^2)(1+Z)^{-3}(1-Z)^{-3}\ln\left[\frac{1}{2}(1-Z)\right] \\
&\quad + (8+35Z+28Z^2+25Z^3)(1+Z)^{-2}(1-Z)^{-4}.
\end{aligned} \tag{17.27}$$

Even though it seems that those coefficient functions are singular as $Z \to -1$ (which corresponds to antipodal points), this is only apparent. Expanding the logarithm around $Z = -1$ as

$$\ln\left[\frac{1}{2}(1-Z)\right] = -\sum_{k=1}^{\infty}\frac{(1+Z)^k}{2^k\,k}, \tag{17.28}$$

we get

$$\begin{aligned}
&\mathcal{D}^{(1,1)}(Z) = \frac{5}{16} + \mathcal{O}(Z+1), \quad \mathcal{D}^{(1,2)}(Z) = \frac{5}{8} + \mathcal{O}(Z+1), \\
&\mathcal{D}^{(1,3)}(Z) = \mathcal{O}(Z+1),
\end{aligned} \tag{17.29}$$

which are perfectly regular at $Z = -1$. This means that the Weyl tensor correlator is only singular when the two points x and x' in the arguments of the correlator are connected by a null geodesic, with $Z(x,x') = 1$.

The contracted Weyl tensor correlator defined in Eq. (17.21) reduces to

$$\begin{aligned}
\left\langle \tilde{\mathrm{Weyl}}^2(x,x')\right\rangle_{\mathrm{c}} &= -\frac{180\alpha\kappa^4H^8}{\pi^2}(1+Z)^{-3}\ln\left[\frac{1}{2}(1-Z+i\epsilon^{\pm})\right] \\
&\quad - \frac{15\alpha\kappa^4H^8}{\pi^2}(7-35Z+29Z^2-25Z^3) \\
&\quad \times (1+Z)^{-2}(1-Z+i\epsilon^{\pm})^{-4},
\end{aligned} \tag{17.30}$$

where again the singularity for $Z \to -1$ is only apparent, as can be seen by expanding the logarithm around $Z = -1$ using Eq. (17.28).

For a general metric g_{ab} the second Bianchi identity for the Riemann tensor (which also applies to its correlator) gives

$$\nabla^a C_{abcd} = \nabla_{[c}\left[R_{d]b} - \frac{1}{6}g_{d]b}R\right]. \tag{17.31}$$

For the background de Sitter space $\hat{R}_{ab} = \Lambda \hat{g}_{ab}$ and $\hat{C}_{abcd} = 0$, both sides of Eq. (17.31) vanish and the equation is trivially satisfied. Furthermore, without matter contributions $\hat{\nabla}_a \tilde{C}^{ab}{}_{cd} = 0$ and the Weyl tensor correlator at tree level, which corresponds to $\alpha = \beta = 0$, must fulfill this transversality condition as well. Setting $\alpha = \beta = 0$ in the explicit expression (17.25), one can check that this is indeed the case. On the other hand, when the matter contributions are included, the Bianchi identity (17.31) relates the Weyl and Ricci–Weyl tensors correlators at order κ^4 as

$$\hat{\nabla}_a \left\langle \tilde{C}^{ab}{}_{cd}(x) \tilde{C}^{m'n'}{}_{p'q'}(x') \right\rangle_c = \left(-\delta^b_s \delta^t_{[c} + \frac{1}{6} \delta^t_s \delta^b_{[c} \right) \hat{\nabla}_{d]} \left\langle \tilde{R}^s{}_t(x) \tilde{C}^{m'n'}{}_{p'q'}(x') \right\rangle_c . \tag{17.32}$$

which we have explicitly checked to be the case for our result.

The full Weyl tensor correlator at order κ^4, that is at one-loop order for free matter fields, encompasses some previously known results. In particular, the Weyl tensor correlator at tree level which was first calculated by Kouris [971] and later corrected [984]. Mora, Tsamis and Woodard have also calculated the Weyl tensor correlator at tree level [985, 986], in the latter article they state agreement with Kouris' result after the above corrections. Our result when restricted to the tree level, i.e., order κ^2, agrees with theirs; see Fröb et al. [572] for further details.

Behavior in Different Limits

Let us analyze the behavior of the Weyl tensor correlator (17.25) in various important limits. First, for $Z \to -1$ we have seen its regular behavior in (17.29). From Eq. (17.26) we see that in the light-cone limit $Z \to 1$, the components of the two-point Weyl tensor correlator diverge like

$$\begin{aligned} &\sim \kappa^2 H^6 (1 - Z + i\epsilon^{\pm})^{-3} + \kappa^4 H^8 (1 - Z + i\epsilon^{\pm})^{-4} \\ &\quad + \kappa^4 H^8 (1 - Z + i\epsilon^{\pm})^{-3} \ln \left[\frac{1}{2} (1 - Z + i\epsilon^{\pm}) \right] . \end{aligned} \tag{17.33}$$

The quantum corrections due to the interaction with matter fields are, therefore, more singular near the light cone than the tree-level result, which corresponds to the first term in (17.33). Throughout the rest of this section whenever writing the dominant terms for the large or the short-distance limits, which is indicated with the symbol $\sim$, we leave out the dimensionless coefficients independent of any dimensionful parameters.

In order to analyze the behavior for *large spacelike separations*, it is convenient to set the times η and η' of the two points x and x' equal by means of a de Sitter isometry, which is always possible for spacelike separated points. The physical distance $d(x, x')$ between those points on the spatial section is then given by $d^2 = a^2(\eta)(\vec{x} - \vec{x}')^2$. From the explicit expression of Z given in (17.5), this distance can be written as $d^2 = 2H^{-2}(1 - Z)$, which is equal to the squared Minkowski distance between those points in the embedding space, as seen from

Eq. (15.9). Hence, the limit $(\vec{x}-\vec{x}')^2 \to \infty$ for *large spacelike separations*, i.e., *superhorizon* scales, with $d \gg H^{-1}$, corresponds to $Z \to -\infty$, and we see that the components which decay more slowly go like

$$\sim \kappa^2 H^6 |Z|^{-2} + \kappa^2 H^6 |Z|^{-3} + \kappa^4 H^8 |Z|^{-2} + \kappa^4 H^8 |Z|^{-3} \ln|Z| + \cdots , \qquad (17.34)$$

which in terms of the proper physical distance d reads

$$\sim \kappa^2 H^2 d^{-4} + \kappa^2 d^{-6} + \kappa^4 H^4 d^{-4} + \kappa^4 H^2 d^{-6} \ln(Hd) + \ldots , \qquad (17.35)$$

where we recall that $\kappa = \sqrt{16\pi G_{\rm N}} = \sqrt{16\pi}\,\ell_{\rm P}$, where $\ell_{\rm P}$ is the Planck length (we use units in which $\hbar = c = 1$).

It is worth pointing out that the second and last terms in (17.35) can be combined in a way which suggests the following relationship to a power law with a perturbative correction to the exponent:

$$\kappa^2 d^{-6+\kappa^2 H^2} \sim \kappa^2 d^{-6} \Big[1 + \kappa^2 H^2 \ln(Hd) + \mathcal{O}(\kappa^2 H^2 \ln^2(Hd)) \Big]. \qquad (17.36)$$

This shows that even for $\kappa^2 H^2 \ll 1$ the perturbative expansion can break down for sufficiently large separations. Thus, in general one needs a nonperturbative calculation in $\kappa^2 H^2$ which resums the logarithms to capture correctly the behavior for large separations. The situation is analogous to the existence of secular terms in certain perturbative calculations when considering the evolution for a sufficiently long time; see Fröb et al. [639].

For the *short-distance* limit when the two points are spacelike separated, i.e., at *subhorizon* scales $d \ll H^{-1}$, one should recover the flat-space result. Indeed, using the previous expression connecting d and Z for points at equal times, one can see that $d \to 0$ implies $Z \to 1$, and that the logarithmic terms in (17.26) are subdominant in this limit:

$$\kappa^4 H^8 (1-Z)^{-3} \ln\left[\frac{1}{2}(1-Z)\right] \sim \kappa^4 H^2 d^{-6} \ln\left(\frac{1}{2} H^2 d^2\right) \ll \kappa^4 d^{-8} \sim \kappa^4 H^8 (1-Z)^{-4}. \qquad (17.37)$$

The dominant terms do not contain H, as one would expect, and we obtain

$$\sim \kappa^2 d^{-6} + \kappa^4 d^{-8} + \cdots . \qquad (17.38)$$

Instead, although the logarithmic term in (17.33) contributes to the divergent behavior in the limit $Z \to 1$, it does not contribute to the flat-space limit because it contains extra positive powers of H after writing the result in terms of the distance d. Further details on the flat-space limit and the full expressions of the curvature tensor correlators in Minkowski space are given in Section 17.4. Furthermore, we see that at large distances, or superhorizon scales, the Weyl tensor correlator in de Sitter falls off, as given by (17.35), more slowly than in flat space, characterized by (17.38). This was also found for the stress-energy tensor correlator for minimally coupled scalar fields, both massless and massive, in Chapter 15.

For *large timelike separations*, we can achieve $\vec{x} = \vec{x}'$ through a de Sitter isometry. The expression of Z in spatially flat coordinates (17.5) then shows that $|\eta - \eta'| \to \infty$ corresponds to $Z \to \infty$, and we get the result (17.34). The proper time elapsed along a geodesic connecting the two points is given by $\tau^2 = -\mu^2(x, x')$, where the relation between the geodesic distance μ and the biscalar Z is given by Eq. (17.1). In the limit $Z \to \infty$ it follows that $\tau = H^{-1} \ln Z$, and therefore the components of the two-point Weyl correlation function which fall off more slowly go like

$$\sim \kappa^2 H^6 e^{-2H\tau} + \kappa^4 H^8 e^{-2H\tau} + \kappa^4 H^9 \tau e^{-3H\tau} + \cdots , \tag{17.39}$$

i.e., they fall off exponentially with the proper-time separation.

17.2.3 Ricci and Ricci–Weyl Tensor Correlators

The correlator of the linearized Ricci tensor as well as the Ricci–Weyl tensors correlator can also be easily obtained from the Riemann tensor correlator (17.16). Suitably contracting indices, in the bitensor basis furnished by the normal vectors n_a and $n_{a'}$ and the parallel propagator $\hat{g}_{ab'}$, we get for the two-point Ricci tensor correlator

$$\begin{aligned}\left\langle \tilde{R}^b{}_d(x)\tilde{R}^{n'}{}_{q'}(x')\right\rangle_{\mathrm{c}} = \frac{15\alpha\kappa^4 H^8}{4\pi^2}\Big[&- \delta^b_d\delta^{n'}_{q'} + 16 n^b n_d n^{n'} n_{q'} + 2\left(\hat{g}^{bn'}\hat{g}_{dq'} + \hat{g}^b_{q'}\hat{g}^{n'}_d\right) \\ &+ 4\left(n^b\hat{g}^{n'}_d n_{q'} + n_d\hat{g}^{bn'}n_{q'} + n^b\hat{g}_{dq'}n^{n'} + n_d\hat{g}^b_{q'}n^{n'}\right)\Big] \\ &\times \mathrm{d}\lim_{\epsilon\to 0^+}(1 - Z + i\epsilon^{\pm})^{-4}\end{aligned} \tag{17.40}$$

On the other hand, using equation (17.23) we obtain for the Ricci–Weyl tensors correlator

$$\begin{aligned}\left\langle \tilde{R}^b{}_d(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\right\rangle_{\mathrm{c}} = \frac{3\alpha\kappa^4 H^8}{\pi^2}\Big[&2\left(\delta^b_d - n^b n_d\right)\delta^{[m'}_{[p'}\delta^{n']}_{q']} - 6\delta^b_d\delta^{[m'}_{[p'}n^{n']}n_{q']} \\ &- 6\delta^{[m'}_{[p'}\left(n_{q']}\hat{g}^{n']b}n_d + n^{n']}\hat{g}^b_{q']}n_d + n_{q']}\hat{g}^{n']}_d n^b\right. \\ &\left. + n^{n']}\hat{g}_{q']d}n^b\right) Z - 3\left(\delta^{[m'}_{[p'} - 2n^{[m'}n_{[p']}\right) \\ &\times\left(\hat{g}^b_{q']}\hat{g}^{n']}_d + \hat{g}^{n']b}\hat{g}_{q']d}\right)\Big](1+Z)\mathrm{d}\lim_{\epsilon\to 0^+}(1 - Z + i\epsilon^{\pm})^{-4}.\end{aligned} \tag{17.41}$$

Given that $Z(x, x') = Z(x', x)$, the manifestly de Sitter-invariant results (17.16) for the Riemann tensor correlator remain invariant if one swaps the two Riemann tensor operators. In doing so, $\epsilon^{\pm}$ will change sign, which is relevant when x and x' are timelike separated, but not for spacelike separations: in that case both

operators commute, in agreement with the microcausality principle. This implies that the result for the Weyl–Ricci tensors correlator $\left\langle \tilde{C}^{mn}{}_{pq}(x)\tilde{R}^{b'}{}_{d'}(x')\right\rangle_{\rm c}$ can be simply obtained from the right-hand side of (17.41) by replacing primed indices with unprimed ones and vice versa.

Notice that there are no tree-level contributions to connected correlators involving the Ricci tensor. This is understood by regarding the Einstein equation for the Ricci operator as a Heisenberg equation of motion. The cosmological constant source term is proportional to δ^a_b and its effect is entirely included in the background Ricci tensor, which is also the case for the expectation value of the stress-energy tensor operator evaluated on the background metric. Hence, they do not contribute to the connected correlator, which only gets contributions starting at order κ^2 and involving stress-energy tensor terms that give rise to matter loops when combined with similar terms from the other Ricci or Weyl operators. Note that the terms that contribute at tree level to the Weyl–Weyl tensor correlator cannot combine to form connected Feynman diagrams in this case. The first nonvanishing contributions to connected correlators involving at least one Ricci tensor are, therefore, of order κ^4 and one loop in the matter fields.

Note also that further contractions vanish, i.e., the two-point function of the Ricci scalar with any other curvature tensor is zero. This is a peculiarity of the conformal case which reflects the fact that the trace of the stress-energy tensor, which is entirely due to the trace anomaly, does not fluctuate, as had already been found in Minkowski spacetime [183]. For the contracted Ricci tensor correlator defined in Eq. (17.21) we obtain

$$\left\langle \tilde{\rm Ric}^2(x,x')\right\rangle_{\rm c} = \frac{45\alpha\kappa^4H^8}{\pi^2}{\rm d}\lim_{\epsilon\to 0^+}(1-Z+i\epsilon^{\pm})^{-4}. \tag{17.42}$$

Behavior in Different Limits

Both correlators are only singular as $Z\to 1$, where the most singular term goes like

$$\sim \kappa^4H^8(1-Z+i\epsilon^{\pm})^{-4}. \tag{17.43}$$

For *large time-* and *space-like separations*, the components which fall off more slowly go like

$$\sim \kappa^4H^8Z^{-4} \tag{17.44}$$

for the two-point Ricci tensor correlator and like

$$\sim \kappa^4H^8Z^{-2} \tag{17.45}$$

for the Ricci–Weyl tensors correlator. Similarly to the Weyl tensor correlator case and taking into account that for large spacelike separations $|Z|\sim H^2d^2$,

one can express this in terms of the proper physical distance d, while for large timelike separations the relation $Z \sim \exp(H\tau)$ allows the result to be expressed in terms of the proper time τ along a geodesic connecting x and x'.

17.2.4 Ricci and Stress-Energy Tensors Correlators

We can compare the Ricci tensor correlator obtained here to the stress-energy tensor correlator for conformal fields obtained by Osborn and Shore [204]. They use as a basic set of bitensors the geodesic distance μ, the normal vectors n_a and $n_{a'}$ and the parallel propagator $\hat{g}_{ab'}$, which they name θ (rescaled by H), $\hat{x}_a, \hat{y}_a$ and I_{ab}, respectively. Adapted to our notation, their result for the four-dimensional case is given by

$$\langle T_{ab} T_{c'd'} \rangle_{\mathrm{c}} = \frac{1}{48\pi^4} \frac{H^8}{(1-Z)^4} \left[\left(\hat{g}_{a(c'} + 2n_a n_{(c'} \right) \left(\hat{g}_{d')b} + 2n_{d')} n_b \right) - \frac{1}{4} \hat{g}_{ab} \hat{g}_{c'd'} \right], \tag{17.46}$$

where we should interpret their $1 - Z$ as $1 - Z + \mathrm{i}\epsilon^{\pm}$.

Since all connected correlators of the Ricci scalar vanish, as remarked before, up to the order at which we are working, the Einstein equation gives

$$\hat{g}_{as} \hat{g}_{c'm'} \left\langle \tilde{R}^{s}{}_{b} \tilde{R}^{m'}{}_{d'} \right\rangle_{\mathrm{c}} = \frac{1}{4} \kappa^4 \langle T_{ab} T_{c'd'} \rangle_{\mathrm{c}} , \tag{17.47}$$

which coincides exactly with equation (17.40) when we take $\alpha = (2880\pi^2)^{-1}$, corresponding to the single scalar field that Osborn and Shore considered; compare also with Eq. (15.47). Furthermore, taking into account the substitution $3\alpha/2 = b$ for the prefactor, as will be explained in Section 17.3, one can see that the result also agrees for the other examples of CFTs that they studied.

17.3 Riemann Tensor Correlator for General CFTs

In this section we explain how the results for the Riemann tensor correlator, and the related results for Weyl and Ricci tensor correlators, obtained above for free scalar fields can be extended to general conformal field theories (CFTs). In order to do so, we need to go back to equations (16.110)–(16.112) before particularizing the values of the constants b and b' to the case of free scalar fields.

We begin with some remarks about the different contributions to the Riemann tensor correlator which are useful for our reasoning. The sum of terms involving the biscalars $\mathcal{S}^{(0,k)}$ in Eq. (17.16) corresponds to the tree-level result (of order κ^2) except for the global coefficient, which includes corrections of order κ^4. Furthermore, from Eq. (16.111) we see that the terms proportional to the parameter b' in the effective action S_{eff} are either quadratic in h^{+}_{ab} or quadratic in h^{-}_{ab}, so that they do not contribute to the component V_{-+} of the matrix V_{AB} defined in Eq. (9.59). From Eq. (16.91) one can then conclude that there are no contributions

from scalar and vectorial metric perturbations to the Riemann tensor correlator which are proportional to b'. Exactly the same argument applies to the terms proportional to $\ln\bar{\mu}$.

Proceeding in the same way to derive Eq. (16.32) for the tensorial perturbations, one obtains the following result for arbitrary values of b and b':

$$S_2^{\mathrm{g,TT}}[h_{mn}] = -\frac{1}{2}(2b+b'-\beta)H^2\int a^2(\partial_s h_{mn})(\partial^s h^{mn})d^4x - bH^2\int a^2 h'_{mn}h'^{mn}d^4x + \frac{b}{2}\int(\Box h_{mn})(\Box h^{mn})\ln a d^4x. \quad (17.48)$$

When deriving this result, the terms proportional to b' give rise to a shift of the cosmological constant term (at all orders in h_{ab}) which is included in the corrected cosmological constant Λ_{eff} used to determine the background de Sitter metric as described in Section 16.1. On the other hand, the first integral on the right-hand side of Eq. (17.48) coincides with that in S_0^{TT} in equation (16.32), and it gives a contribution to the Riemann tensor correlator of the same form as the tree-level result.

Hence, comparing with Eq. (16.32), one can see that the factor $(5\alpha-2\beta)$ in equation (17.16) should be replaced with $(4b+2b'-2\beta)$. Since b' does not appear anywhere else and does not contribute to the scalar and vectorial correlators, one can use the substitution $3\alpha/2 = b$ for all the remaining instances of α, which also include the contributions from the nonlocal term $S_2^{\mathrm{K}}[h^{\pm}]$ in Eq. (16.29).

Finally the Riemann tensor correlator for the general case becomes

$$\left\langle \tilde{R}^{ab}{}_{cd}(x)\tilde{R}^{m'n'}{}_{p'q'}(x')\right\rangle_{\mathrm{c}} = \frac{4\kappa^2H^6}{\pi^2}\sum_{k=1}^{9}{}^{(k)}\mathcal{N}^{[ab]}{}_{[cd]}{}^{[m'n']}{}_{[p'q']} \times \mathrm{d}\lim_{\epsilon\to0^+}\Big[b\kappa^2H^2\mathcal{S}^{(1,k)}(Z(x,x')-\mathrm{i}\epsilon^{\pm}) + \Big(1+4b\kappa^2H^2\Big(\ln\Big(\frac{\bar{\mu}}{H}\Big)+\gamma\Big)-2\,(2b+b'-\beta)\,\kappa^2H^2\Big) \times \mathcal{S}^{(0,k)}(Z(x,x')-\mathrm{i}\epsilon^{\pm})\Big]. \quad (17.49)$$

Here the various objects are defined after equation (17.16) and for arbitrary values of the renormalization scale μ one needs to make the replacement (16.115), i.e., $b\ln\bar{\mu}\to b\ln\mu+(3/2)Na_1^{\mathrm{R}}(\mu)$.

Using the same kind of substitutions, one gets analogous results for equation (17.25) for the Weyl tensor correlator. Moreover, since the Ricci and Ricci–Weyl tensors correlators, obtained by suitable contractions of the Riemann tensor correlator, vanish for the tree-level result, they are entirely determined by the terms involving the biscalars $\mathcal{S}^{(1,k)}$. Therefore, one simply needs to make

the replacement $3\alpha/2 = b$ in the results for the Ricci and Ricci-Weyl tensors correlators (17.40) and (17.41), respectively.

The general result which is valid for any CFT corresponds for free theories to one-loop order in the matter fields, whereas for interacting theories it accounts for the effects of processes involving any number of matter loops but no internal graviton propagators.

17.3.1 Relevance of the Riemann Tensor Correlator

Besides the fact that it exhibits manifest de Sitter invariance when written in terms of maximally symmetric bitensors, the Riemann tensor correlator with appropriately raised indices has other appealing properties at the order at which we are working. It is gauge-invariant at order $1/N$ for a large N expansion in the number of matter fields, which excludes the contributions of graviton loops, of higher order in $1/N$. Furthermore, the Riemann tensor provides a suitable characterization of the local geometry and its correlator is an infrared-safe observable at that order, in the sense described in the cosmological literature [987, 988, 566, 573].

In contrast, there are gauge-invariant quantities which cannot be determined by the geometry within a region of fixed and finite proper size: they depend on global properties or on geometric properties associated with a region whose proper size grows with the cosmological expansion, e.g., a region of fixed comoving size. Classical no-hair theorems for de Sitter space do not apply to them and in general the correlators characterizing their quantum fluctuations exhibit infrared divergences as the number of past e-foldings tends to infinity, which is incompatible with de Sitter invariance. A simple example of such gauge-invariant objects is the tree-level correlator for the transverse-traceless metric perturbations in terms of the background spacetime coordinates; see Eqs. (16.61) and (16.56).

Thus, in the gravitational case de Sitter invariance can only hold at the quantum mechanical level for observables depending on the geometry within regions of arbitrary but finite proper size. The Riemann tensor and its correlators constitute a convenient example of such class of observables at leading order in $1/N$.

Of course it would be of great interest to extend the results presented here for the Riemann tensor correlator to include the effects of loop corrections from other kinds of fields besides conformal matter fields. Massless or sufficiently light minimally coupled fields are particularly interesting since they play an important role in inflationary cosmology and are the ones leading to stronger infrared effects in de Sitter.

More challenging is to include the effects of graviton loops. The necessary renormalization for dealing with the ultraviolet divergences that arise in this case when functionally integrating the metric perturbations should be performed

within a covariant treatment which requires in turn the use of DeWitt–Fadeev–Popov ghosts and BRST quantization [603], and is technically quite involved. Furthermore, once graviton loops are taken into account, the Riemann tensor correlator expressed in terms of background spacetime coordinates ceases to be gauge-invariant and infrared safe.

In fact, when going beyond a linear treatment of the metric perturbations, one needs in general to confront the difficulties of constructing diffeomorphism-invariant observables, which can be, at best, approximately local [565]. Nevertheless, for a perturbative calculation around the maximally symmetric de Sitter background, it may be possible to consider a generalized gauge-invariant Riemann tensor correlator by computing the parallel transport, with the perturbed metric, of the two Riemann tensors to a common point; see Fröb [989] for different possible approaches. At sufficiently higher order, however, new ultraviolet divergences will arise owing to the fact that the tensors are parallel transported along a one-dimensional geodesic and some kind of smearing around the curve should be introduced.

17.4 Riemann Tensor Correlator in Minkowski Spacetime

In this section we derive the Riemann tensor correlator in Minkowski spacetime as a byproduct of the correlators in de Sitter spacetime. For points separated by distances much smaller than the Hubble length H^{-1}, subhorizon scales, one expects that the two-point correlation function of the Riemann tensor should reproduce the Minkowski results. Thus, taking the curvature radius of de Sitter space H^{-1} to infinity, i.e., the Hubble parameter H to zero, we can obtain the two-point correlation functions of the curvature tensors in a Minkowski background. In the Poincaré patch of de Sitter spacetime, this can be done by choosing the spatially flat coordinate system

$$ds^2 = -dt^2 + \mathrm{e}^{2Ht} d\vec{x}^2, \tag{17.50}$$

expressing everything in terms of those coordinates and taking $H \to 0$ so that the metric goes over to the Minkowski metric.

However, the flat space limit can also be obtained easily in a way independent of the choice of a particular coordinate system by noting that the geodesic distance $\mu(x, x')$ between two points x and x' in the Minkowski spacetime is simply given by $\mu(x, x') = \sqrt{(x - x')^2}$. For the invariant biscalar Z given in Eq. (17.1) we then calculate

$$Z(x, x') = 1 - \frac{1}{2} H^2 (x - x')^2 + \mathcal{O}(H^3). \tag{17.51}$$

For the covariant derivatives of Z we have

$$\nabla_a Z \to -H^2 \eta_{ab} (x - x')^b + \mathcal{O}(H^3), \quad \nabla_a \nabla_{b'} Z \to H^2 \eta_{ab'} + \mathcal{O}(H^3), \tag{17.52}$$

and for the Allen and Jacobson bitensor set we obtain from equations (17.2)

$$n_a \to \frac{\eta_{ab}(x-x')^b}{\sqrt{(x-x')^2}} + \mathcal{O}(H), \qquad \hat{g}_{ab'} \to \eta_{ab'} + \mathcal{O}(H). \tag{17.53}$$

Thus, the Weyl tensor correlator (17.25) is in the flat-space limit given by

$$\begin{aligned}\left\langle \tilde{C}^{ab}{}_{cd}(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\right\rangle_{\rm c} &= \frac{8\kappa^2}{\pi^2}\left[{}^{(1)}\mathcal{W} + 3\,{}^{(2)}\mathcal{W} + {}^{(3)}\mathcal{W}\right]^{[ab]}{}_{[cd]}{}^{[m'n']}{}_{[p'q']} \\ &\quad \times \mathrm{d}\lim_{\epsilon\to 0^+}((x-x')^2 + \mathrm{i}\epsilon^{\pm})^{-3} \\ &\quad + \frac{48\alpha\kappa^4}{\pi^2}\left[3\,{}^{(1)}\mathcal{W} + 12\,{}^{(2)}\mathcal{W} + 8\,{}^{(3)}\mathcal{W}\right]^{[ab]}{}_{[cd]}{}^{[m'n']}{}_{[p'q']} \\ &\quad \times \mathrm{d}\lim_{\epsilon\to 0^+}((x-x')^2 + \mathrm{i}\epsilon^{\pm})^{-4}.\end{aligned} \tag{17.54}$$

Here the flat-space basis bitensors are

$$\begin{aligned}{}^{(1)}\mathcal{W}_{abcdm'n'p'q'} &= \eta_{ac}\eta_{bd}\eta_{m'p'}\eta_{n'q'} - 6\eta_{ac}\eta_{m'p'}\eta_{b(n'}\eta_{q')d} \\ &\quad + 4\eta_{a(m'}\eta_{p')c}\eta_{b(n'}\eta_{q')d} \\ {}^{(2)}\mathcal{W}_{abcdm'n'p'q'} &= -12\eta_{ac}\eta_{m'p'}y_b y_d y_{n'}y_{q'} - 12\eta_{ac}\eta_{m'p'}y_{(b}\eta_{d)(n'}y_{q')} \\ &\quad + 24y_a y_c y_{m'}y_{p'}\eta_{b(n'}\eta_{q')d} + 16y_{(a}\eta_{c)(m'}y_{p')}\eta_{b(n'}\eta_{q')d} \\ {}^{(3)}\mathcal{W}_{abcdm'n'p'q'} &= -2\eta_{ac}\eta_{bd}\eta_{m'p'}\eta_{n'q'} + 6\eta_{ac}\eta_{m'p'}\left(\eta_{bd}y_{n'}y_{q'} + y_b y_d \eta_{n'q'}\right) \\ &\quad - 12\eta_{ac}\eta_{m'p'}y_b y_d y_{n'}y_{q'} + 12\eta_{ac}\eta_{m'p'}y_{(b}\eta_{d)(n'}y_{q')} \\ &\quad + 6\eta_{ac}\eta_{m'p'}\eta_{b(n'}\eta_{q')d} + 24y_a y_c y_{m'}y_{p'}\eta_{b(n'}\eta_{q')d} \\ &\quad - 12\left(\eta_{ac}y_{m'}y_{p'} + y_a y_c \eta_{m'p'}\right)\eta_{b(n'}\eta_{q')d}.\end{aligned} \tag{17.55}$$

with unit vectors y_a defined as $y_a = \eta_{ab}(x^b - x'^b)((x-x')^2)^{-1/2}$. Note that the coefficient β as well as the renormalization scale μ do not appear, i.e., for conformally coupled scalar fields the flat-space limit is insensitive to the contributions from counterterms quadratic in the curvature tensors.

To compare with the de Sitter results, note that for points x and x' at equal time $x^0 = (x')^0$, the physical distance is $d^2 = (\vec{x} - \vec{x}')^2 = (x-x')^2$, and the two-point Weyl correlation function falls off like

$$\sim \kappa^2 d^{-6} + \kappa^4 d^{-8}, \tag{17.56}$$

which agrees with the short-distance limit for spacelike separations given by (17.38).

For the Ricci tensor correlator (17.40) we obtain

$$\begin{aligned}\left\langle \tilde{R}^{b}{}_{d}(x)\tilde{R}^{n'}{}_{q'}(x')\right\rangle_{\mathrm{c}} = \frac{60\alpha\kappa^4}{\pi^2}\Big[&- \delta^b_d\delta^{n'}_{q'} + 16y^b y_d y^{n'} y_{q'} + 2\left(\eta^{bn'}\eta_{dq'} + \eta^b_{q'}\eta^{n'}_d\right)\\ &+ 4\left(y^b\eta^{n'}_d y_{q'} + y_d\eta^{bn'}y_{q'} + y^b\eta_{dq'}y^{n'} + y_d\eta^b_{q'}y^{n'}\right)\Big]\\ &\times \mathrm{d}\lim_{\epsilon\to 0^+}((x-x')^2 + \mathrm{i}\epsilon^{\pm})^{-4},\end{aligned} \tag{17.57}$$

which is of order κ^4 and falls off like $\kappa^4 d^{-8}$.

Similarly, the Ricci–Weyl tensor correlation function (17.41) reads in the flat-space limit

$$\begin{aligned}\left\langle \tilde{R}^{b}{}_{d}(x)\tilde{C}^{m'n'}{}_{p'q'}(x')\right\rangle_{\mathrm{c}} = \frac{96\alpha\kappa^4}{\pi^2}\Big[&2\left(\delta^b_d - y^b y_d\right)\delta^{[m'}_{[p'}\delta^{n']}_{q']} - 6\delta^b_d\delta^{[m'}_{[p'}y^{n']}y_{q']}\\ &- 6\delta^{[m'}_{[p'}\Big(y_{q']}\eta^{n']b}y_d + y^{n']}\eta^b_{q']}y_d + y_{q']}\eta^{n']}_d y^b + y^{n']}\eta_{q']d}y^b\Big)\\ &- 3\left(\delta^{[m'}_{[p'} - 2y^{[m'}y_{[p'}\right)\left(\eta^b_{q']}\eta^{n']}_d + \eta^{n']b}\eta_{q']d}\right)\Big]\\ &\times \mathrm{d}\lim_{\epsilon\to 0^+}((x-x')^2 + \mathrm{i}\epsilon^{\pm})^{-4},\end{aligned} \tag{17.58}$$

which is also of order κ^4 and falls off like $\kappa^4 d^{-8}$.

As before, the Riemann tensor correlator can be calculated from the decomposition (17.20). This result is manifestly Lorentz-invariant and given in spacetime coordinates. Finally, the expressions for a general CFT can be easily obtained with the substitution $3\alpha/2 = b$ in all the results of this section, as discussed in Section 17.3.

17.5 Useful Fourier Transforms

For the calculation of the Riemann tensor correlator (17.12) we need Fourier transforms of the form

$$f_{i_1\ldots}(x,x') = \mathcal{F}(\vec{p},\vec{r})\left[\tilde{f}_{i_1\ldots}(\vec{p},\eta,\eta')\right] \equiv \int \bar{f}_{i_1\ldots}(\vec{p},\eta,\eta')\,\mathrm{e}^{\mathrm{i}\vec{p}\cdot\vec{r}}\frac{d^3p}{(2\pi)^3}, \tag{17.59}$$

where $\vec{r} = \vec{x} - \vec{x}'$. In order to compute this integrals, we first need to remove the tensorial structure. The tensors η_{ab} and δ^0_a can be simply factored out, whereas p_a factors can be extracted out of the Fourier transform as follows:

$$\mathcal{F}(\vec{p},\vec{r})\left[p_a\bar{f}_{i_1\ldots}(\vec{p},\eta,\eta')\right] = -\mathrm{i}\left(\partial_a - \delta^0_a\partial_\eta\right)\mathcal{F}(\vec{p},\vec{r})\left[\bar{f}_{i_1\ldots}(\vec{p},\eta,\eta')\right]. \tag{17.60}$$

The Fourier transform of the remaining scalar factor must then be understood in the sense of distributions, i.e., we calculate

$$f(x,x') = \mathrm{d}\lim_{\epsilon\to 0}\left[\frac{1}{2\pi^2|\vec{r}|}\int_0^\infty \mathrm{e}^{-\epsilon|\vec{p}|}\tilde{f}(|\vec{p}|,\eta,\eta')\sin\left(|\vec{p}||\vec{r}|\right)|\vec{p}|d|\vec{p}|\right], \tag{17.61}$$

where we have taken into account that because of rotational symmetry those scalar factors only depend on the absolute value of $\vec{p}$.

As a technical detail, we note that for the Fourier transform of, e.g., $p_a p_b|\vec{p}|^{-3}$ we have to introduce a lower bound ε for the integral in equation (17.61). After differentiating as indicated in Eq. (17.60), all dependence on ε disappears. This is because while the Fourier transform of $p_a p_b|\vec{p}|^{-3}$ is well defined (and is given by the result of this procedure), the Fourier transform of $|\vec{p}|^{-3}$ alone is not.

Employing the integrals for the Ein function given in Section 16.7 and using partial integration to reduce any powers of $|\vec{p}|$, most of the Fourier transforms can be readily calculated in this way. In addition, we need the indefinite integral $\int \ln(ap)\frac{1}{p}dp = \frac{1}{2}\ln^2(ap)$, and the definite integral

$$I = \int_0^\infty \mathrm{e}^{-\epsilon p}\mathrm{e}^{\mathrm{i}ap}\mathrm{Ein}(\mathrm{i}bp)\frac{1}{p}dp \tag{17.62}$$

for real parameters a and b.

Since it is absolutely convergent, to compute this integral we can insert the series expansion of the Ein function to get

$$\begin{aligned} I &= \sum_{k=1}^\infty \frac{(\mathrm{i}b)^k}{k\,k!}\int_0^\infty \mathrm{e}^{-\epsilon p}\mathrm{e}^{\mathrm{i}ap}p^{k-1}dp \\ &= \sum_{k=1}^\infty \frac{1}{k^2}\left(-\frac{b}{a+\mathrm{i}\epsilon}\right)^k = \mathrm{Li}_2\left(-\frac{b}{a+\mathrm{i}\epsilon}\right), \end{aligned} \tag{17.63}$$

where $\mathrm{Li}_2(z)$ is the dilogarithm function. However, this definite integral only occurs together with p_a, so that its derivative, $\mathrm{Li}_2'(z) = -\frac{\ln(1-z)}{z}$, which can also be inferred directly from the series expansion, is the only expression that we need for our computation.

18

Epilogue: Linkage with Quantum Gravity

Our book started with topics of quantum field theory *in* curved spacetime, and progressed to the backreaction effects of (the expectation values of the stress-energy tensor of) quantum matter fields *on* the background spacetime, culminating in the theory of *semiclassical gravity*, a mean field theory and, when their fluctuations or the stress tensor correlators are included, the theory of *stochastic gravity*. There was little mention of *quantum gravity*, even though historically many practitioners in this field, the authors of this book included, were motivated by the quest of it. Thus it is suitable to end this book with some description of the linkages of its contents with quantum gravity.

Details of the different veins of quantum gravity can be found in many excellent monographs and reviews mentioned in Chapter 1. We have presented one of these in the 'bottom (low energy)-up' approach, represented here by semiclassical and stochastic gravity. It is therefore necessary to bring in quantum gravity's dialectic dual, namely, *emergent gravity*, which, as we argued [483], is the imperative mission of all 'top (high energy)-down' theories, by way of strings, loops, simplices, causal sets, etc. To orient ourselves we first provide a general description of what emergent/quantum gravity is and is not.

18.1 A New Perspective and Two Different Routes

Beneath the diversity of theories and proposals, there is, after the last three decades, at least general agreement on the definition of quantum gravity: it is a theory for the *microscopic* structures of spacetime. Disagreement remains as to whether a theory for the microscopic structure of spacetime can be obtained by quantizing general relativity (GR), the singularly successful theory for the description of the *macroscopic* structure of spacetime, as much effort in the previous century can attest to. The alternative viewpoint is that GR is a low energy effective theory, and the metric and connection forms are the collective

or hydrodynamic variables of some to-be-discovered microscopic theory. These variables will lose their meaning at shorter wavelengths and higher energies. Classical gravity from this view is emergent from quantum gravity (definition above) but there is no guarantee that quantizing GR will lead to a theory for the microscopic structure of spacetime, namely, quantum gravity. To use the analogy of a crystal made of atoms, quantizing its vibrational modes yields phonons, not atoms. Finding the atomic structure of matter does not come from simply quantizing its collective degrees of freedom, but takes a very different path. In fact, when one gets down to the atomic scale, their collective excitations will no longer exist.

We should therefore distinguish the *two routes*: (1) 'bottom-up' ('up' here refers to energy scale) route of inferring the microscopic structure of spacetime from the given features of macroscopic spacetime described by GR; (2) 'top-down' route of deducing the known low energy macroscopic theory of spacetime from proposed candidate theories for the microscopic structures of spacetime. The former route defines the quest for quantum gravity, which has been in existence as a research discipline for the last 70 years. The latter route defines the task of emergent gravity, the challenge there is not just in proposing some attractive micro-constituent of spacetime, but in making explicit the processes and mechanisms of how the low energy phenomena emerge from it.

We can see that this *new perspective* on quantum gravity is very different from the old: one should be careful not to blindly follow the dogma that *quantizing* general relativity *naturally* yields a microscopic structure of spacetime, or accept the dictum that some micro-constituent is *the* theory which gives us *everything*.

We can also see that these two routes have very different emphasis, goals and methodology. In the bottom-up route, inferring the microscopic structure from the known macroscopic phenomena is a daunting task, as we have to face up to the unknown while painstakingly uncovering the hidden. But, if saying this offers some comfort, physics has progressed for centuries in this way, trying to infer the existence of atoms from the states of matter, quarks from nucleons, etc. For this purpose, concepts and methods from nonequilibrium statistical mechanics and examples from strongly correlated many-body systems will probably play an essential role hitherto largely ignored in quantum gravity. The recent effort to use tensor networks (e.g., [990, 991, 992, 993]) and entanglement entropy (e.g., [866, 870, 869, 868]) to see if any many-body quantum system can emerge in the large scale possessing properties of physical spacetime is encouraging.

On the top-down route one may think that deducing the macro theory from a *presumed* micro theory is easy. One can invoke familiar procedures such as taking the hydrodynamic or thermodynamic limits in classical physics or applying effective field theory in quantum physics. But as we shall see, it is not as straightforward as imagined. The calculation of string–string interactions via string field theory (which has not been fully constructed) or loop interactions via spin-foams, e.g., showing that the Minkowski space is the unique ground

state, is easier said than done. Even then, one would still need to show how specific collective variables suited for the description of macro physics can be constructed from these micro-variables. What remains uncertain is whether there could be different types of effective field theory: different coarse-graining measures giving rise to different emergent structures. Take, for example, nuclear physics: even with the known basic constituents and a certified theory of their interactions, deducing the different phases of nuclear matter from QCD, taking into account the complexity of nucleon interactions, is far from simple or straightforward.

18.2 Emergent versus Quantum Gravity

Let us examine the nature of the general missions and the specific tasks of these two routes.

18.2.1 'Top-Down' from Micro to Macro: How Spacetime Emerges

If spacetime is an emergent entity then its large-scale structures should be deducible from the underlying constituents and their interactions, like a crystal structure or a condensate constructed from atoms. Collective (macro) features such as elasticity (in gravity, metric elasticity [994]) associated with such a (micro) structure can lend themselves to a nice geometric depiction but these features will cease to make sense at scales close to or smaller than an atom. In these cases, the task of emergent gravity is similar to the derivation of hydrodynamic equations of motion or thermodynamic laws from molecular dynamics.

Spacetime is the resultant large-scale structure formed by the interaction of sub-level constituent particles. Even when the micro constituents are believed to be known – strings, loops, spin-foams, simplices, sets – one still needs to show the macro limit exists and how it is attained from their interactions. This is an obvious necessary criterion for a proposal to be qualified as a candidate theory of quantum gravity. For deductive emergent behavior, the path could be tortuous but in principle attainable.

As is known from many examples of condensed matter physics, it is often a nontrivial task to deduce mesoscopic behavior from micro-dynamics: one usually encounters nonlinear or even nonlocal interactions in strongly correlated systems, and some ingenuity is needed to identify collective variables at successive levels of structure to attain the simplest theory for an accurate enough description of the structure and its dynamics. Using the molecular to hydro-dynamics example, this intermediate scale corresponds to finding the appropriate kinetic variables and their associated dynamics (e.g., use of maximal entropy laws at successive stages of complexity).

In addition to nonlinearity, nonlocal properties can emerge from every level of structure and dynamics. This makes the transition to macroscopic limits much more involved. It requires not just a linear automatic progression of deductions from one single level to another, but the injection of new ideas at different levels in their constructions.

What is said above assumes that we are dealing with deductive-predictive theories. But we should be mindful that this macro manifestation could be via a non-deductive emergent behavior from the micro structures, in which case even micro- to macro-, sub- to super- structures would not be easy; some cases could even be impossible.

Thus for emergent theories, spelling out the instrumental mechanisms and the facilitating processes could be just as important as the knowledge of the basic constituents. This implies that, even if we don't know the underlying constituents, we can still learn from the characteristics of many emergent theories in the description of nature (abundant in biology, for example, and essential in condensed matter physics) to understand and formulate the basic rules of emergence. We could then apply the insights gained to describe the required conditions and necessary processes for the emergence of spacetime. These are the new tasks for emergent gravity in the 'top-down' direction.

18.2.2 'Bottom-Up' from Macro to Micro: Quest for Quantum Gravity

Compared to working out the macroscopic attributes of a presumed known microscopic theory, going the reverse way is always more difficult because it entails reconstructing lost or hidden information. One has only the large-scale features to go by but they are grossly coarse-grained from the microscopic. From the degraded information one hopes to find some hints to see what the micro structure could be like. To proceed from the bottom up, in the opinion of [483], focusing on two key elements may help:

(1) *Topological structures*, because they are more resilient to evolutionary or environmental changes. See, e.g., the works of Volovik [878] (He^3 analog, Fermi surface), Levin and Wen [995, 996] (string-nets, emergent light and fermions), and the growing literature on entanglement entropy and topological states in quantum many-body systems.
(2) *Noise and fluctuations*. Fluctuations could reveal some sub-structural contents and behavior, such as deducing molecular interaction from hydrodynamical fluctuations, or finding the universality classes of systems from their critical exponents at a phase transition.

This book focuses on themes pertaining to (2), but we wish to put on record that factor (1) is at least of equal, if not more, importance. Both features are needed in any 'bottom-up' expedition.

Common Features of Collective States Built from Different Constituents

In the depiction of the structure and properties of matter there are two almost orthogonal perspectives: one is by way of its constituents and interactions, the other according to its collective behavior. If we regard this chain of QED–QCD–GUT (grand-unification)–QG as a vertical progression depicting the hierarchy of basic constituents, there is also a horizontal progression in terms of the stochastic–statistical–kinetic–thermodynamic/hydrodynamic depiction of the collective states. There exist similarities between matters in the same collective state (e.g., hydrodynamics) but made from different constituents. Collective behavior of quark-gluon plasma is similar in many respects to that of electromagnetic plasma. Indeed, one talks about magneto-hydrodynamics from Maxwell's theory as well as magneto-chromo-hydrodynamics from QCD. In this long-wavelength, collision-dominated regime, their behavior can both be adequately depicted by the laws of hydrodynamics. The macroscopic, hydrodynamic equations and their conservation laws [997, 998, 999, 1000] are all based on the dynamical laws and conservation laws of microphysics but now in terms of a new set of collective (hydrodynamic) variables. This new macroscopic theory of the collective behavior of the basic constituents now emerges with a simple yet distinct life of its own. The underlying micro-theories could be different (molecular dynamics, electromagnetism, quantum chromodynamics), but the hydrodynamic behaviors of these constituents bear similarities.

18.3 Unraveling the Microstructures of Spacetime

Having provided the perspective of how different theories related to or pertaining to quantum gravity address the micro versus macro and quantum versus classical issues, let us now see how stochastic gravity is placed and linked to quantum gravity. Specifically, given the mean values and the fluctuations of the quantum matter stress-energy tensor, how do we come up with a sensible theory for the micro-structures of spacetime? This challenge is weighty, as all 'bottom-up' approaches are, because usually there is no simple or straightforward way to follow. If kinetic theory is useful as a guide in the pathway from stochastic gravity to quantum gravity, then the notion of *correlation dynamics* perhaps holds the key. In classical kinetic theory correlation dynamics refers to the evolution of correlation functions organized by the Boltzmann–BBGKY hierarchy (see, e.g., [1001]). The corresponding hierarchy for an interacting quantum field theory is the Schwinger–Dyson system of equations [1002, 1003]. To understand stochastic gravity in relation to quantum gravity let us first recall the theme presented in Sec. 2 of Chapter 1, namely, in many physical systems, between the quantum and the semiclassical regimes, a stochastic regime exists quite ubiquitously. We explained how these three levels of theoretical structures can be traversed by examining the interlocked issues of fluctuations, correlations and coherence.

Let us examine how fluctuations are related to correlations and how quantum correlations are linked to quantum coherence.

18.3.1 Noise and Fluctuations, Correlations and Coherence

For concreteness consider the example of gravitational perturbations (weak but deterministic) h_{ab} in a background spacetime with metric g_{ab} driven by the expectation value of the stress-energy tensor of a scalar field Φ, as well as its fluctuations $\hat{t}_{ab}(x)$. This is illustrated by the toy model of Section 10.2.3, which is useful for comparing the stochastic with the semiclassical and quantum equations of motion for the metric perturbation field.

We see first that in the semiclassical case the classical metric correlations are given by the product of the vacuum expectation value of the energy-momentum tensor, whereas in the quantum case it is given by the quantum average of the correlation of metric (operators) with respect to the fluctuations in both the matter and the gravitational fields. In the stochastic case the schematic form is closer to the quantum case than the semiclassical, except that the quantum average is replaced by the noise average, and the average of the stress-energy tensor is taken with respect only to the matter field.

The important improvement over semiclassical gravity is that stochastic gravity carries information about the correlations (and the related phase information) of the stress-energy tensor of quantum matter fields and their induced metric fluctuations which are absent in semiclassical gravity. Here we have invoked the innate relation between *fluctuations* and *correlations*. Stochastic gravity brings us closer than semiclassical gravity to quantum gravity in the sense that the correlations in the quantum matter field stress tensor and correlations in the induced geometric objects (such as the Ricci tensor correlator), fully present in quantum gravity, are partially retained in stochastic gravity. Because of the self-consistency condition required of the backreaction equations for the matter and spacetime sectors when solved simultaneously, the background spacetime has a way to 'tune in' to the correlations of the quantum matter fields registered in the noise terms which manifest through the induced metric fluctuations from solutions of the Einstein–Langevin equations. One can further link (but not equate) *correlations* in quantum field stress tensors to *coherence* in quantum gravity.[1]

Viewed in this broader light the Einstein–Langevin equation is only a partial (lowest correlation order) representation of the more complete theory for the micro-structures of spacetime and matter. There, the quantum coherence in the

[1] Here, we enter a deeper level of quantum structure involving entanglement: quantum correlations are insufficient to represent the full contents of entanglement [871, 1004, 1005]. Recent results on tensor product expansion and tensor networks in quantum information [1006, 1007, 990, 1008, 992] have advanced our understanding of this issue.

geometry sector is tethered to the quantum coherence in the matter field, as the quantum description of the combined matter and gravity sectors should be given by a completely coherent wave function of the closed quantum system. Semiclassical gravity forsakes all the quantum phase information in the gravity sector. Stochastic gravity captures only partial phase information or quantum coherence in the gravity sector induced by the correlations in the quantum matter fields. Since the degree of coherence can be represented by correlations, (putting aside the entanglement issues for the problem at hand), the strategy for the stochastic gravity program (in the broader sense) is to move up the correlation hierarchy from the second-order correlator (the variance) of the matter field stress-energy tensor to the higher-order correlations, and through their linkage with the gravity sector provided by the Einstein–Langevin equations for each order, retrieve the quantum attributes (partial coherence) of the quantum gravity theory. Thus, as remarked in [479], in this vein, focusing on the noise kernel, the stress-energy tensor two-point function, is only the first step (beyond the mean field semiclassical gravity theory) toward reconstructing the full theory of quantum gravity. This is the conceptual basis for the so-called 'kinetic theory approach to quantum gravity' [482].

If we view classical gravity as an effective theory, i.e., the metric or connection forms as collective variables of some basic constituents which make up spacetime in the large, and the theory of general relativity as the hydrodynamic limit, we can also ask if there is a midway station, like kinetic theory, in between molecular- and hydro-dynamics, from quantum micro–dynamics to classical hydrodynamics. This mesoscopic regime at the micro–macro interface is often where the quantum to classical transition takes place, depending on several key parameters of the system and its environment. Related to the unraveling of spacetime micro-dynamics via metric correlations and fluctuations we mention two conceptual proposals, one by way of the stochastic Boltzmann equation–BBGKY hierarchy analogy, the other by way of conductance and transport functions in mesoscopic physics.

18.3.2 Spacetime Conductance Fluctuations: Strongly Correlated Systems

Let us first examine the low energy hydrodynamics features, then consider ways to unravel the micro-structure. Notice what appears on the right-hand side of the Einstein–Langevin equation: the stress-energy two-point function is analogous to the conductance of electron transport which is given by the current-current two-point function [1009]. Physically this means that we are calculating the transport functions of the matter particles, here described by the quantum fields. Since conductivity and viscosity are transport functions in hydrodynamics (which can be obtained from kinetic theory) this is consistent with the view of general relativity as a hydrodynamics theory of spacetime [876]. The Einstein–Langevin

equation describes how these hydrodynamic transport functions in the quantum matter sector influence the spacetime dynamics, also at the hydrodynamics level. What we are after is how the correlation functions of geometric objects behave, such as the Einstein tensor correlation function induced by the matter field fluctuations [189] (described in Part III). Following Einstein's observation that spacetime dynamics is determined by (and also governs the motion of) the matter (stress-energy tensor), we expect that the transport function represented by the current-current correlation in the fluctuations of the matter energy density should also have a geometric counterpart. This relation between quantum matter correlators and classical geometric correlators persists at all orders, the higher-order correlators assuming an increasingly important role at higher energies or finer resolutions. When the interaction among the constituents gets stronger, or the probing scale gets shorter, effects associated with the higher correlation functions of the system will increase in importance. Studies in strongly correlated systems are revealing in this regard. For example, conductance fluctuations obtained from the four-point function of the current carry important information such as the sample specific signature and universality. Higher order correlators in geometric objects induced by higher moments in the matter stress-energy tensor [467, 468] may also begin to reveal fractal structures in spacetime [1010].

18.4 Relation to Quantum Gravity and Limitations of Stochastic Gravity

Continuing with this vein of reasoning we can probe beyond the geometro-hydrodynamics regime into the kinetic theory regime. The full BBGKY hierarchy of nth-order correlation functions ($n \to \infty$) constitutes a closed system. Coarse-graining (involving truncation and 'slaving' [2] – see [581] for its technical meaning) the hierarchy yields an effectively open system, for which Boltzmann's theory occupies the lowest order in that hierarchy. It describes the evolution of one particle distribution function driven by a two-particle collision integral under the assumption of a causal factorization initial condition, the molecular chaos assumption. Inclusion of a noise source representing the slaved contributions of the higher correlation functions yields the stochastic Boltzmann equation [1002]. The stochastic Boltzmann equation contains features which may enable us to make connections between spacetime micro-dynamics and stochastic gravity, specifically, how to unravel the higher-order correlations appearing as noise acting on the lower order correlations.

[2] Note that truncation without slaving will not produce an open system; the nth-order correlation functions constitute a smaller closed system. They obey an equation of the Vlasov type which is unitary.

18.4.1 The 'Kinetic Theory' Approach to Quantum Gravity

The kinetic theory approach to quantum gravity [482] proposes to unravel the microscopic structure of spacetime by examining how the correlation functions of the geometric objects providing an effective description of spacetime are driven by the correlation noises from the higher moments of the quantum matter stress-energy tensor. This paradigm is structured around the so-called *Boltzmann–Einstein Hierarchy* (B–E) of equations. For illustrative purpose, assuming that the micro-structure of spacetime can be represented by some interacting quantum field of micro-constituents, the Schwinger–Dyson hierarchy of n-point correlation functions would be the quantum parallel to the BBGKY hierarchy, the lowest order obeying the Boltzmann equation. For any nth-order correlation, if one can represent all the higher-order correlations as noise, called 'correlation noise' in [1002], this would give rise to a stochastic Boltzmann equation for the nth-order correlation. At the second correlation order, the Einstein tensor correlator with induced metric fluctuations [189] has been calculated for the Minkowski space; so have the Weyl and the Riemann tensors [980] for the de Sitter space and slow-roll Friedmann–Lemaître–Robertson–Walker spacetimes [989]. The higher-order correlations of these geometric objects with induced metric fluctuations can likewise be determined, albeit much harder, by solving the Einstein–Langevin equations with sources from the higher moments of the stress-energy tensor of the corresponding orders. The combined set of correlators of all orders of the geometric objects is given the name of a 'Boltzmann–Einstein hierarchy' in [482] because it has the structure of a BBGKY hierarchy whose lowest order is the Boltzmann equation while in the present spacetime context the lowest order in this hierarchy is the Einstein equation.[3]

Therefore, viewed in the light of mesoscopic physics, stochastic gravity is the theory which enables one to probe into the higher correlations of quantum matter and spacetime. From the excitations of the collective modes in geometro-hydrodynamics one tries to deduce the kinetic theory of spacetime meso-dynamics and eventually the full theory of quantum gravity for spacetime micro-dynamics.[4]

18.4.2 The Effective Theory Approach to Quantum Gravity

The conceptual scheme we depicted above shows the role of stochastic gravity in relation to semiclassical gravity and to quantum gravity, the latter is where the real challenge is. We can also take a more familiar homey route, that of the

[3] This is *not* the Einstein–Boltzmann equation in classical general relativity and relativistic kinetic theory where classical matter obeying the Boltzmann equation enters as a source of the Einstein equation. The Boltzmann–Einstein hierarchy refers to the spacetime sector alone.

[4] This paradigm was used by Mattingly [1011] as an example of emergence of general relativity from quantum gravity, mirroring the 'general relativity as geometro-hydrodynamics' perspective [876], by analogy with hydrodynamics being an emergent theory from molecular dynamics.

effective field theory approach. While it has been known for some time [194] that semiclassical gravity can be viewed as the leading-order theory in an $1/N$ expansion of quantum gravity coupled to a large number N of matter fields, here stochastic gravity is shown to be obtainable from a next-to-leading order $1/N$ expansion of such a theory. Thus in an effective field theory approach to quantum gravity, one can extend beyond, by going to the $(1/N)^2$ order, which entails the inclusion of graviton loops and internal graviton lines and dealing with the associated gauge problems.

18.4.3 Advantages and Limitations of Stochastic Gravity

Our understanding and presentation of a theory is incomplete if we only see its advantages and talk about its promises, but not its shortcomings and limitations. Stating what they are is perhaps the right way to close this book.

Recognizing that correlations and fluctuations are on the same footing, and for second-order correlations, that the Feynman–Vernon Gaussian functional identity enables us to translate quantum correlations to quantum noise, we can bring in useful concepts and techniques from nonequilibrium statistical physics to explore the many theoretical issues of stochastic gravity, and, through analog gravity, broaden the scope of its experimental realization, especially when quantum noise is involved. However, there are limitations on technical grounds. In stochastic gravity we do not quantize the metric perturbations; the stochastic correlations are shown to agree with the quantum correlations in the large N approximation. Going beyond the linear order in the matter fields, when computing the metric correlations including loops of matter fields, the separation of induced and intrinsic correlations is no longer useful because both the induced and the intrinsic parts diverge but their sum is finite. In this sense the 'in-in' effective action, where this book began, gives more transparent results (see discussions in Sec. 16.4). This shows that the advantage we see for one purpose could be the hindrance to another, perhaps higher, purpose. The quest continues, regardless.

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